SYNOPSYS®

DesignWare® Cores Multi-Protocol 16G PHY for TSMC 28 HPCP

Databook

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Revision History

PHY Version	Databook Version	Databook Date	Description
3.01a	1.00	May 2018	The changes since the previous version include the following: ■ Throughout the databook, updated the PCIe specification to: “PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG” ■ In the “General Product Description” section, added the third bullet to the note ■ In Table 1-4 (“Exceptions to Specification Allowed for PCIe 4.0”), updated the SRIS information ■ In Table 1-11 (“Exceptions to Specification Allowed for RXAUI/Double XAUI”), added the last two rows ■ In Table 1-12 (“Exceptions to Specification Allowed for 10GBASE-KR and 40GBASE-KR4”), added the last row ■ In Table 1-17 (“Primitive Devices Used”), updated the list of devices ■ Updated the “Raw PCS” section ■ Added the “RX CDR PPM Detector” section ■ Added the “PHY Power-Down” section ■ Updated the “RX Equalization and Adaptation” section ■ Updated the “CR Parallel interface” section ■ Updated Figure 5-15 (“CR Parallel Register Read Operation”) ■ Updated the “External SRAM Support” section ■ Updated the “Boundary Scan (ACJTAG)” section ■ Updated the “IDDQ” section ■ Updated the “Notes for Configuring the PHY” section ■ Updated Table 6-2 (“PHY Model Database Structure”) ■ Updated Table 6-3 (“Simulation Database Structure”) and the note below the table ■ Added the “PMA Gate-Level Model” and “Receiver Detection” sections

PHY Version	Databook Version	Databook Date	Description
3.01a	1.00	May 2018 (Continued...)	■ Updated the "Fast Simulation Mode" section ■ Updated Figure 7-1 ("Bump Map (x4)"), Figure 7-2 ("Bump Map (x2)"), and Figure 7-3 ("Bump Map (x1)") ■ Updated the "Bump Pad RDL and UBM Sizing" and "Assembly of PMA and Pads" sections ■ Added Figure 7-5 ("Origins for Abutment of PMA Hard Macro and Snap-On Pads for x2 Configuration"), and Figure 7-6 ("Origins for Abutment of PMA Hard Macro and Snap-On Pads for x1 Configuration") ■ Added the "Connectivity of PMA and Pads" and "PMA Routing Obstruction" sections ■ Updated the "PMA Abutment" section ■ In the "PMA Fill" section, added a note ■ In Table 7-4 ("Supported Power Supply Limits at the PMA Macro Bumps"), added footnotes ■ Added Figure 7-11 ("Back-to-Back Diodes in PHY") ■ Updated Table 7-8 ("Primitive Modules") ■ Updated the "Data Synchronizers" section ■ Revamped Chapter 8, "Board- and Package-Level Implementation" ■ Updated the "Reference Clock Requirements" section ■ Updated the "Reference Clock Implementation" section ■ In Chapter 10, "Electrical Specifications": - Updated the PCIe power dissipation values - Added the JESD204B power dissipation values ■ Updated Chapter 11, "Register Descriptions" ■ Updated Appendix A, "Support Configuration Settings" and Appendix B, "Lane Configuration Settings"
-	-	February 2017	The changes since the previous version include the following: ■ Throughout the databook, updated the PCIe 4.0 specification to: "PCI Express Base Specification, Revision 4.0 Version 0.7, October 2016, PCI-SIG" ■ In the "Customer Support" section, updated the URL of the global support centers to https://www.synopsys.com/support/global-support-centers.html ■ In the "Quick Reference" section, updated the infographic ■ In the "Features" section, added that the PHY supports a wide range of divided clock frequencies for external-to-PHY usage with optional spread-spectrum clock (SSC) capability ■ Updated Table 1-16 ("Process Information") ■ Updated Table 1-17 ("Primitive Devices Used") ■ In Table 3-1 ("PHY Signal Descriptions"), updated the descriptions of the following signals: - "ref_pad_clk_p" on page 110 - "ref_pad_clk_m" on page 110 - "txX_reset" on page 112 - "txX_disable" on page 112

PHY Version	Databook Version	Databook Date	Description
-	-	February 2017 (Continued...)	- "txX_req" on page 115 - "txX_detrx_req" on page 117 - "rxX_reset" on page 119 - "rxX_disable" on page 120 - "rxX_req" on page 126 - "sram_init_done" on page 156 ■ Updated the "RX Internal Eye Monitor" section ■ In the "PHY Initialization Sequence" section: - Updated that the maximum time from phy_reset de-assertion to de-assertion of txX_ack/rxX_ack can be as high as 15 ms in normal simulation mode - Updated Figure 5-1 ("PHY Initialization Sequence") on page 157 - Updated step 3 on page 159 to reflect that the phy_reset needs to be asserted for a minimum of 25 µs after power-supplies are ramped-up (power-on) instead of 15 µs ■ Updated the values in the following tables: - Table 5-1 ("Transmit Data-path latency") - Table 5-3 ("RX Data Path Latency") - Table 5-8 ("Power State Transition and Receiver Detection Times for PCIe (100 MHz Reference Clock)" ■ Updated the "Control Register Interface" section with respect to broadcast writes ■ In the "External SRAM Support" section, updated the note ■ In the "Burn-In" section, updated the note ■ Added the "PMA Fill" section ■ Updated Table 7-4 ("Supported Power Supply Limits at the PMA Macro Bumps") ■ Updated the "Board Requirements—External Components" section ■ Updated the values in the following tables: - Table 9-1 ("Reference Clock Input Requirements") - Table 9-2 ("Repeater Clock Output Specification") ■ Updated the "Power Dissipation Overview" section ■ Updated Chapter 11, "Register Descriptions"
-	-	December 2016	Preliminary version"Power Dissipation Overview"

Preface

Databook Organization

This databook is organized as follows:

- Chapter 1, "Product Overview", provides an introduction to the DesignWare Cores Multi-Protocol 16G PHY and its features.
- Chapter 2, "Deliverables", describes the deliverables included in the PHY full-package release.
- Chapter 3, "Signal Descriptions", describes the PHY top-level signals.
- Chapter 4, "Functional Description", provides a functional overview of the PHY and describes its test mode.
- Chapter 5, "PHY Usage and Configuration", explains the usage of PHY features and lists the PHY configuration settings for the protocols and data rates that are supported.
- Chapter 6, "Verilog Model Description", provides an overview of simulating the PHY Verilog model.
- Chapter 7, "Physical-Level Implementation", describes the physical implementation of the PHY.
- Chapter 8, "Board- and Package-Level Implementation", describes requirements and guidelines for implementing the PHY at the board and package level.
- Chapter 9, "Reference Clock Implementation", describes the reference clock that the PHY supports.
- Chapter 10, "Electrical Specifications", provides PHY electrical specifications for the TSMC28HPCP technology.
- Chapter 11, "Register Descriptions", describes the PHY control registers.

Web Resources

- DesignWare IP product information: http://www.designware.com
- Your custom DesignWare IP page: http://www.mydesignware.com
- Documentation through SolvNet: https://solvnet.synopsys.com (Synopsys password required)
- Synopsys Common Licensing (SCL): http://www.synopsys.com/keys

Reference Documentation

- PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG
- PCI Express Base Specification, Revision 3.1a, December 2015, PCI-SIG
- PCI Express Card Electromechanical Specification, Revision 3.0, July 2013, PCI-SIG
- PCI Express Mini Card Electromechanical Specification, Revision 2.0, April 2012, PCI-SIG
- IEEE Standard 802.3-2012, IEEE
- IEEE Standard 802.3ae-2002, IEEE
- IEEE standard 802.3ap-2007, IEEE
- IEEE standard 802.3az-2010, IEEE
- Serial ATA Revision 3.3, February 2016
- Clause 70 of the IEEE 802.3-2012 specification
- Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems
- Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems
- Clause 47 of the IEEE 802.3-2012 specification
- Clauses 48 and 71 of the IEEE 802.3-2012 specification
- Clauses 49 and 72 of the IEEE 802.3-2012 specification
- Clause 84 of IEEE 802.3-2012 specification
- RXAUI - Reduced Pin XAUI (2 x 6.25Gbps), Dune Networks, V2.1, January 2008
- Common Electrical I/O (CEI), Electrical and Jitter Interoperability agreements for 6G+ bps, 11G+ bps and 25G+ bps I/O, IA # OIF-CEI-03.1, February 2014, OIF
- Common Public Radio Interface (CPRI) specification V5.0, September 2011, CPRI Cooperation
- Open Base Station Architecture Initiative (OBSAI) Reference Point 3 Specification version 4.2, 2011, OBSAI
- Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converter JESD204B, July 2012
- IEEE Standard 1149.1-2001, IEEE
- IEEE Standard 1149.6-2003, IEEE
- DFT Compiler User Guide, Volume 1: Scan (XG Mode), Synopsys, Inc.
- RapidIO™ Interconnect Specification, Rev. 3.0, October 2013

Customer Support

To obtain support for your product, contact Support Center using one of the following methods:

- *For fastest response*, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.

  Go to http://solvnet.synopsys.com/EnterACall and click on the link to enter a call.
  Provide the requested information, including:

  - **Product:** DesignWare Cores
  - **Sub Product:** SERDES_16G_PHY
  - **Version:** <product_version_number>
  - **Problem Type:**
  - **Issue Severity:**
  - **Problem:** Provide a short summary of the issue or error message you have encountered.

    Also provide the full product name (Multi-Protocol 16G PHY for TSMC 28 HPCP) and a description of the problem. For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.

- Or, send an e-mail message to support_center@synopsys.com (your e-mail will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - Include the Product name, Sub Product name, Version number, and full product name (as identified above) in your e-mail so it can be routed correctly.
  - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - Attach any debug files you created.
- Or, telephone your local support center:
  - North America:
  - Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
  - All other countries:

    https://www.synopsys.com/support/global-support-centers.html

1 Product Overview

This chapter includes the following sections:

- "Quick Reference" on page 16
- "General Product Description" on page 17
- "Features" on page 18
- "Applications" on page 19
- "Standards Compliance" on page 20
- "Process Information" on page 31

1.1 Quick Reference

Multi-Protocol 16G PHY

Features:

Receiver: AFE containing CTLE and VGAs, and 5-tap DFE
Transmitter: 3-tap FFE, low power hybrid (voltage and current mode) driver
PLL: Two independent PLLs, based on LC-tank oscillators
Lane configurations: x1, x2, x4, x8, x16, with support for bifurcation
(x8 and x16 lane configurations are supported by aggregation)

Data Rates:

1.2 - 16.0 Gbps

EM:

100KPOH@110°C

Junction Temperature:

-40°C to 125°C

Process Technology:

TSMC 28 nm HKMG CLN28HPC+

Power Supplies:

VP = 0.9 V -7% to 0.9 V+10%
VPH = 1.5 V-7% to 1.8 V+10% (nominal = 1.5 or 1.8 V)

Test Modes Supported:

Boundary Scan (ACJTAG), Burn-In, IDDQ, Scan (At-Speed, Stuck-At), Diagnostics (serial and parallel loopback), and Normal Operation

ESD:

CDM = 6A peak discharge current
HBM = 2,000 V **MM** = 100 V

Metal Stacks and Package Configuration:

10M_5X2Y2Z + RDL
9M_6X2Z + RDL
8M_5X2Z + RDL
Flip-chip

Protocols/Specifications:

PCIe 4.0:	PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG
SATA:	Serial ATA Revision 3.3, February 2016
CEI-6G/CEI-11G:	Common Electrical I/O (CEI), Electrical and Jitter Interoperability agreements for 6G+ bps, 11G+ bps and 25G+ bps I/O, IA # OIF-CEI-03.1, February 2014, OIF
SGMII:	Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems
QSGMII:	Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems
1000BASE-KX:	Clause 70 of the IEEE 802.3-2012 specification
XAUI:/10GBASE-KX4:	Clause 47 of the IEEE 802.3-2012 specification
RXAUI/DoubleXAUI:	Clauses 48 and 71 of the IEEE 802.3-2012 specification
10GBASE-KR:/40GBASE-KR4:	Clauses 49 and 72 of the IEEE 802.3-2012 specification Clause 84 of IEEE 802.3-2012 specification
CPRI:	Common Public Radio Interface (CPRI) specification V5.0, September 2011, CPRI Cooperation
OBSAI:	Open Base Station Architecture Initiative (OBSAI) Reference Point 3 Specification version 4.2, 2011, OBSAI
JESD204B:	Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converter JESD204B, July 2012
SRIO 6.25G:	RapidIO™ Interconnect Specification, Rev. 3.0, October 2013

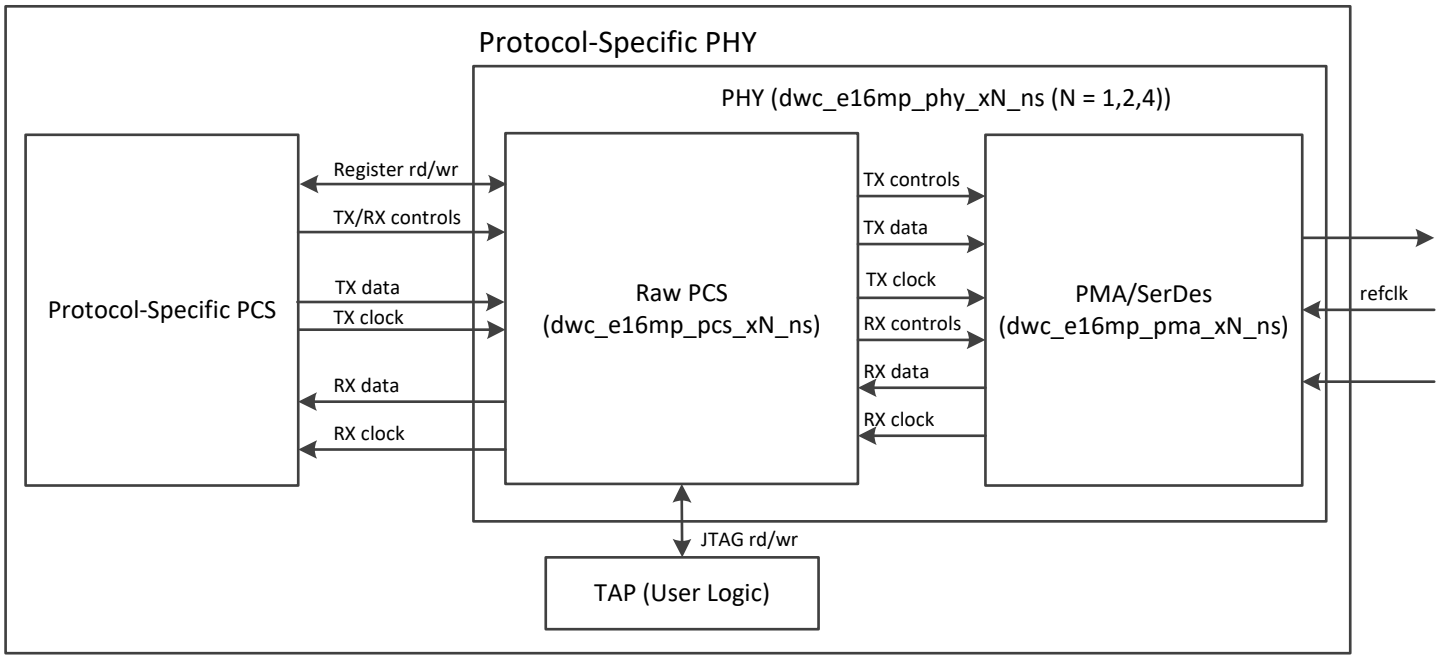
1.2 General Product Description

The DesignWare Cores Multi-Protocol 16G PHY (dwc_e16mp_phy_xN_ns; N = 1, 2, 4) provides full-duplex transceivers and consists of the following:

- A hard macro (physical medium attachment (PMA)/SerDes: (dwc_e16mp_pma_xN_ns; N = 1, 2, 4)
- A soft RTL module (raw physical coding sublayer (PCS): (dwc_e16mp_pcs_raw_xN_ns; N = 1, 2, 4)

Figure 1-1 illustrates the hierarchy of the PHY and its instantiation with a protocol-specific PCS to create a protocol-specific PHY. The PHY architecture supports aggregation of multiple PMA and raw PCS modules to create higher-lane configurations.

Figure 1-1 DesignWare Cores Multi-Protocol 16G PHY Usage and Integration Model



Note

- PHYs with number of lanes greater than N (shown in Figure 1-1) can be generated by aggregating multiple instances of (dwc_e16mp_phy_xN_ns; N = 1, 2, 4).
- Mandatory SRAM size: 16K single port SRAM, consisting of 16-bit locations
- Synopsys PHY does not use any special layers (esd, deep-nwell, and so on). It uses only the standard process kit.

The PHY supports a source-synchronous clocking interface on the transmit (TX) and receive (RX) data paths. The TX parallel data (per lane) is synchronous to the TX clock input of the corresponding lane. Similarly, the RX parallel data is output-synchronized (per lane) to an RX clock that is recovered from the serial data stream that corresponds to that lane.

The PHY supports different bit-width options for the TX and RX data path interfaces. Table 1-1 lists the general relation between baud rates and corresponding parallel data widths that are supported. For specific bit-widths supported for each protocol, see section 5.17 on page 207 ("Aggregation and Bifurcation Support").

Table 1-1 Data Widths Supported

Serial Bit Rate (Gbps)	TX/RX Interface Width
1.2 - 6.4	8-bit, 10-bit, 16-bit, 20-bit
6.4 - 16.0	16-bit, 20-bit

1.3 Features

The PHY provides the following features:

- Highly customizable physical medium attachment (PMA) configuration (controlled by PCS):
  - Configurable serial data:
    - 1.200-1.600 Gbps
    - 1.867-2.133 Gbps
    - 2.400-3.2004 Gbps
    - 3.733-4.266 Gbps
    - 4.800-6.400 Gbps
    - 7.467-8.533 Gbps
    - 9.600-12.800 Gbps
    - 14.932-16.000 Gbps
  - Single, dual and quad channel macros configurations
  - Extensive PMA debug capability via read/write and read-only registers in PCS
  - Register-based control of all PCS-to-PMA signals
  - Adaptive and configurable RX continuous time linear equalizer (CTLE) and decision feedback equalizer (DFE)
  - Adaptive and programmable TX equalization
- PCIe 4.0 aggregation and bifurcation up to x16 PHY configuration
- Supports PCIe L1-substate power managements and Separate Reference Clock Independent SSC Architecture (SRIS)
- IEEE 1149.1 and 1149.6 (AC JTAG) boundary scan
- Built-in Self-Test (BIST) features for production; at-speed testing on any digital tester
- Advanced, built-in diagnostics including on-chip sampling scope

- Visibility and controllability of hard macro functions through programmable registers in the design:
  - Overrides on all ASIC side functional control inputs for easy debug
  - Access register space through a parallel interface
  - Access register space through a JTAG port
- Configurable PCS with extensive debug options:
  - Independent TX and RX control per lane
  - Configurable TX and RX power modes
  - Pseudo-random bit sequence (PRBS) generation and checker (PRBS31, PRBS23, PRBS16, PRBS15, PRBS11, PRBS9, PRBS7)
  - Programmable 10-bit pattern generation with error injection capability
- A wide range of reference clock frequencies with optional fractional frequency correction capability
- A wide range of divided clock frequencies for external-to-PHY usage with optional spread-spectrum clock (SSC) capability
- Built-in, on-chip SSC generation and full configuration from -5000ppm to +5000ppm with a maximum range of 5000ppm



With fractional-N or internal SSC generation enabled, increased TX output jitter will result—contact Synopsys for more details.

- A reference clock repeater for other modules
- Supports flip-chip packaging

1.4 Applications

The DesignWare Cores Multi-Protocol 16G PHY is targeted for connectivity in chip-to-chip or card-to-card communication across a combination of printed circuit board, connectors, backplane wiring, or cable.

Designed for low power, efficient area, low test costs, and process centering for optimal yield and interoperability, the PHY enables designers to introduce competitive products using the latest standards.

1.5 Standards Compliance

The PHY is designed to be compliant with the standards listed within the subsections of this section with requisite exceptions listed. Table 1-2 below indicates the level of Simulation and Hardware Validation that Synopsys performs for each of the listed standards.

Table 1-2 Supported Level of Simulation and Hardware Validation

Protocol	Specification	Non Continuous Data Rate (Gbps)	Simulation Validation			Silicon Validation		
			Data Rate	Electrical	Synopsys Controller/ PHY Interoperation	Data Rate	Electrical	Synopsys Controller/ PHY Interoperation
PCIe 4.0:	PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG	2.5 - 16.0	Yes	Yes	Yes	Yes	Yes	Yes
SATA 3:	Serial ATA Revision 3.3, February 2016	1.5 - 6.0	Yes	Yes	Yes	Yes	Yes	Yes
40GBASE-KR4:	Clause 84 of IEEE 802.3-2012 specification	10.3125 (x4)	Yes	Yes	Yes	Yes	Yes	No[a]
10GBASE-KR:	Clauses 49 and 72 of the IEEE 802.3-2012 specification	10.3125	Yes	Yes	Yes	Yes	Yes	Yes
10GBASE-KX4:	Clause 47 of the IEEE 802.3-2012 specification	4 x 3.125	Yes	Yes	Yes	Yes	Yes	No
1000BASE-KX:	Clause 70 of the IEEE 802.3-2012 specification	1.25	Yes	Yes	Yes	Yes	Yes	Yes
CEI-6G:	OIF-CEI-03.1, February 2014	6.25	Yes	Yes	No[b]	No	No	No
CEI-11G:	OIF-CEI-03.1, February 2014	9.95 - 11.1	Yes	Yes	No	No	No	No
SGMII:	Clause 70 of the IEEE 802.3-2012 specification	1.25	Yes	Yes	No	No	No	No

Table 1-2 Supported Level of Simulation and Hardware Validation (Continued)

Protocol	Specification	Non Continuous Data Rate (Gbps)	Simulation Validation			Silicon Validation		
			Data Rate	Electrical	Synopsys Controller/ PHY Interoperation	Data Rate	Electrical	Synopsys Controller/ PHY Interoperation
QSGMII:	Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems	5.0	Yes	Yes	No	No	No	No
XAUI:	Clause 47 of the IEEE 802.3-2012 specification	4 x 3.125	Yes	Yes	No	No	No	No
RXAUI/Double XAUI:	Clauses 48 and 71 of the IEEE 802.3-2012 specification	2 x 6.25	Yes	Yes	No	No	No	No
CPRI:	Common Public Radio Interface (CPRI) specification V5.0, September 2011, CPRI Cooperation	1.2288 - 9.8304	Yes	Yes	No	No	No	No
OBSAI:	Open Base Station Architecture Initiative (OBSAI) Reference Point 3 Specification version 4.2, 2011, OBSAI	1.536 - 6.144	Yes	Yes	No	No	No	No
JESD204B:	Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converter JESD204B, July 2012	1.25 - 12.5	Yes	Yes	No	No	No	No
SRIO 6.25G:	RapidIO™ Interconnect Specification, Rev. 3.0, October 2013	1.25 - 6.25	Yes	Yes	No	No	No	No

a. Contact Synopsys for further details on Synopsys Controller/PHY Interoperation Testing
b. Contact Synopsys for further details on electrical characterization tests

1.5.1 PCIe 4.0

The PHY supports the PCIe 4.0 (16 Gbps) protocol and data rate, and is backward compatible with the PCIe 3.1 (8 Gbps), PCIe 2.1 (5 Gbps), and PCIe 1.1 (2.5 Gbps) protocol and data rates.

The PHY complies with the required features specified in the PCI Express Base Specification, Revision 4.0 Version 1.0, September 27, 2017, PCI-SIG.

Reference clock frequency supported

For the supported reference clocks, refer to "Support Configuration Settings" and "Lane Configuration Settings".

The PHY supports 100 MHz common reference clock (CC) configuration for both ends of the link. Additionally, separate reference clock independent spread (SRIS) and separate reference clock non spread (SRNS) architecture is supported for 100 MHz as well as alternate reference clock frequencies given in Table 1-3.

Table 1-3 Alternate Supported Reference Clock Frequencies in MHz

Protocol	Alternate supported reference clock frequencies in MHz								
PCIe 1.1:	20	25	40	50	62.5	78.125	125	156.25	200
PCIe 2.1:	20	25	40	50	62.5	78.125	125	156.25	200
PCIe 3.1:	20	25	40	50	62.5		125		200
PCIe 4.0:	20	25	40	50	62.5		125		200

Table 1-4 notes exceptions to the PCIe 4.0 specification.

Table 1-4 Exceptions to Specification Allowed for PCIe 4.0

Category	Exception Allowed
EI detection	Only inferred electrical idle detection is supported
SRIS	Currently, we support SRIS architecture with SSC on the reference clock source. If you would like to use internal SSC generation, contact Synopsys.

1.5.2 SATA 6G

The PHY supports the SATA 6G protocol and data rate and is backwards compatible to the SATA 3G and SATA 1.5G protocols and data rates.

The PHY complies with all of the required features of the Serial ATA Revision 3.3, February 2016. The PHY also supports Gen3i (6 Gbps), Gen2i and Gen2m (3 Gbps), Gen 1i and Gen1m (1.5 Gbps).

Reference clock frequencies supported

For the supported reference clocks, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-5)

Table 1-5 Exceptions to Specification Allowed for SATA 6G

Category	Exception Allowed
Reference clock	When 30 MHz is being used, OOB sampling window may not be met
TX rise/fall time	Faster than specification
No Gen 1x/2x	These have been removed from the SATA specification

1.5.3 CEI-6G-SR and CEI-6G-LR

The PHY supports the CEI-6G-SR and CEI-6G-LR (6.25 Gbps) data rate.

The PHY complies with the required features specified in the Common Electrical I/O (CEI), Electrical and Jitter Interoperability agreements for 6G+ bps, 11G+ bps and 25G+ bps I/O, IA # OIF-CEI-03.1, February 2014, OIF.

Reference clocks supported

For the reference clock supported at a data rate of 6.25 Gbps, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-6)

Table 1-6 Exceptions to Specification Allowed for CEI-6G-SR and CEI-6G-LR

Category	Exception Allowed
TX edge rate	Faster than specification
TX common mode noise	May exceed 5%*T_Vdif
Return loss	RX/TX return loss may violate specification
TX	■ Type 0 loads are supported ■ Type 1, 2, and 3 loads are not supported

Table 1-6 Exceptions to Specification Allowed for CEI-6G-SR and CEI-6G-LR (Continued)

Category	Exception Allowed
RX Termination	■ Can be configured as a Type 0 load ■ Type 0 DC-coupled link supported with common mode limited to 0-800 mV ■ Configuration as a Type 1, 2, or 3 loads is not supported

1.5.4 CEI-11G-SR

The PHY supports the CEI-11G-SR (9.95 Gbps-11.1 Gbps) data rate.

The PHY complies with the required features specified in the Common Electrical I/O (CEI), Electrical and Jitter Interoperability agreements for 6G+ bps, 11G+ bps and 25G+ bps I/O, IA # OIF-CEI-03.1, February 2014, OIF.

Reference clocks supported

For the reference clock supported at a data rate of 10.3125 Gbps, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-7)

Table 1-7 Exceptions to Specification Allowed for CEI-11G-SR

Category	Exception Allowed
TX edge rate	Faster than specification
TX common mode noise	May exceed 5%*T_Vdif
Return loss	RX/TX return loss may violate specification
TX	■ Type 0 loads are supported ■ Type 1, 2, and 3 loads are not supported
RX Termination	■ Can be configured as a Type 0 load ■ Type 0 DC-coupled link supported with common mode limited to 0-800 mV ■ Configuration as a Type 1, 2, or 3 load is not supported

1.5.5 1000BASE-KX/SGMII

The PHY supports Clause 70 of the IEEE 802.3-2012 specification and Serial Gigabit Media Independent Interface (SGMII). A digital interface that provides a 1.25 Gbps serial data rate data path between a 1000 Mbit/s PHY and a MAC sublayer, as defined in Serial Gigabit Media Independent Interface (SGMII) Specification, ENG-46158, Revision 1.8, November 2005, Cisco Systems.

The PHY meets the required features as stated in Cisco's SGMII Revision 1.8 electrical specification, November 2005.

Reference clocks supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-8)

Table 1-8 Exceptions to Specification Allowed for SGMII

Category	Exception Allowed
TX Vol, Vos	Lower than specification
TX rise/fall	Faster than specification
TX skew	May exceed limit
RX Vi	Input voltage range lower than specification
RX hysteresis	Not supported
Clock forwarding with data	The PHY only supports the recovered clock mode of link communication; the clock is not forwarded along with data

1.5.6 QSGMII

The PHY supports Quad Serial Gigabit Media Independent Interface (QSGMII) Specification, EDCS-540123, Revision 1.2, September 2007, Cisco Systems that provides four channels of 1.25 Gbps serial data rate datapath between four SGMII interfaces to be sent over a single MAC serial interface operating at 5 Gbps.

Reference clocks supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-9)

Table 1-9 Exceptions to Specification Allowed for QSGMII

Category	Exception Allowed
TX rise/fall	Faster than specification
TX common mode noise	May exceed 5% T_V dif
TX	■ Type 0 loads are supported ■ Type 1, 2, and 3 loads are not supported
RX Termination	■ Can be configured as a Type 0 load ■ Type 0 DC-coupled link supported with common mode limited to 0-800 mV ■ Configuration as a Type 1, 2, or 3 load is not supported

1.5.7 XAUI and 10GBASE-KX4

The PHY supports 10 Gigabit copper backplane 10GBASE-KX4 (3.125 Gbps) and 10 Gigabit Attachment Unit Interface (XAUI) (4x 3.125 Gbps). Serial data is striped over four serial interfaces operating at 3.125 Gbps.

The PHY complies with the required features of Clause 47 of the IEEE 802.3-2012 specification.

Reference clock frequencies supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-10)

Table 1-10 Exceptions to Specification Allowed for XAUI and 10GBASE-KX4

Category	Exception Allowed
TX rise/fall	Faster than specification
Lane-to-lane skew	Exceeds limit

1.5.8 RXAUI/Double XAUI

The PHY supports a 2x-scaled mode of XAUI, where all time-based portions of the XAUI electrical specification are scaled by a factor of 0.5. This mode is known as RXAUI (also Double XAUI) at twice the data rate. Serial data is striped over two serial interfaces operating at 6.25 Gbps. The PHY complies with Clauses 48 and 71 of the IEEE 802.3-2012 specification.

Reference clock frequencies supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-11)

Table 1-11 Exceptions to Specification Allowed for RXAUI/Double XAUI

Category	Exception Allowed
TX rise/fall	Faster than specification
Lane-to-lane skew	Exceeds limit
TX common mode noise	May exceed 5%*T_Vdif
TX	■ Type 0 loads are supported ■ Type 1, 2, and 3 loads are not supported
RX Termination	■ Can be configured as a Type 0 load ■ Type 0 DC-coupled link supported with common mode limited to 0-800 mV ■ Configuration as a Type 1, 2, or 3 load is not supported

1.5.9 10GBASE-KR and 40GBASE-KR4

For 10GBASE-KR, the PHY complies with Clauses 49 and 72 of the IEEE 802.3-2012 specification. For 40GBASE-KR4, the PHY complies with Clause 84 of IEEE 802.3-2012 specification. The PHY supports back channel adaptation, auto-negotiation, and forward error correction (FEC), as well as Energy-Efficient Ethernet (EEE).

Reference clock frequencies supported

For the reference clocks supported for 10GBASE-KR and 40GBASE-KR4, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-12)

Table 1-12 Exceptions to Specification Allowed for 10GBASE-KR and 40GBASE-KR4

Category	Exception Allowed
TX rise/fall edge rate control	Faster than specification
R_{PRE} for maximum C(0)	May exceed the preshoot limit for maximum C(0) setting

1.5.10 CPRI/OBSAI

The PHY supports the Common Public Radio Interface (CPRI) specification V5.0, September 2011, CPRI Cooperation, data rate at 1.2288 Gbps, 2.4576 Gbps, 3.0720 Gbps, 4.9152 Gbps, 6.144 Gbps, and 9.8304 Gbps.

The PHY also supports Open Base Station Architecture Initiative (OBSAI) Reference Point 3 Specification version 4.2, 2011, OBSAI, data rate at 1.536 Gbps, 3.072 Gbps, 6.144 Gbps.

Reference clock frequencies supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-13)

Table 1-13 Exceptions to Specification Allowed for CPRI/OBSAI

Category	Exception Allowed
TX rise/fall	Faster than specification
TX differential skew	May violate 15ps specification
Return loss	TX/RX return loss may violate specification
Lane to lane skew	May exceed limit
TX common mode noise	May exceed 5%*T_Vdiff
DC coupling	Limited to type 0 with 0 to 800mV input common mode voltage

1.5.11 JESD204B

The PHY supports Joint Electrical Device Engineering Council (JEDEC) Serial Interface for Data converter JESD204B, July 2012, which supports multiple speed grades leveraged from the following protocols:

- LV-OIF-SxI5: from 312.5 Mbps to 3.125 Gbps (adapted from OIF SxI-5, Ref. 3).
- LV-OIF-6G-SR: from 312.5 Mbps to 6.375 Gbps (adapted from CEI-6G-SR, Ref. 4).
- LV-OIF-11G-SR: from 312.5 Mbps to 12.5 Gbps (adapted from CEI-11G-SR, Ref. 4).

However, please note that the PHY supports 1.2288-2 Gbps, 2.45-4 Gbps, 4.9-8 Gbps, 9.8-12.5 Gbps data rates.

Reference clock frequencies supported

Use the following list to check the reference clocks supported:

- For JESD204B operating at 1.25 Gbps, refer to the configuration of 1000BASE-KX/SGMII/QSGMII on "Support Configuration Settings" and "Lane Configuration Settings"
- For JESD204B operating at 3.125 Gbps, refer to the configuration of XAUI/10GBASE-KX4 on "Support Configuration Settings" and "Lane Configuration Settings"
- For JESD204B operating at 5 Gbps, 6.25 Gbps, refer to the configuration of CEI-6G-SR and CEI-6G-LR on "Support Configuration Settings" and "Lane Configuration Settings"
- For JESD204B operating at 10.3125 Gpbs, refer to the configuration of CEI-11G-SR on "Support Configuration Settings" and "Lane Configuration Settings"

Exceptions to the specification allowed (see Table 1-14)

Table 1-14 Exceptions to Specification Allowed for JESD204B

Category	Exception Allowed
Data rate ranges	Support only for data rates above Gbps with the following discrete range: ■ 1.200-1.600 Gbps ■ 1.867-2.133 Gbps ■ 2.400-3.2004 Gbps ■ 3.733-4.266 Gbps ■ 4.800-6.400 Gbps ■ 7.467-8.533 Gbps ■ 9.600-12.800 Gbps ■ 14.932-16.000 Gbps
Unidirectional connection	Does not provide unidirectional PHY; either TX or RX may be turned OFF for a unidirectional option mode
TX rise/fall	Faster than specification

Table 1-14 Exceptions to Specification Allowed for JESD204B (Continued)

Category	Exception Allowed
TX common mode noise	May exceed 5% T_V dif
DC coupling	Support for DC floating termination with common mode voltage range is limited to 0-800 mV
Return loss	RX/TX differential and differential-to-common mode return loss may exceed specifications
Frame clock, multi-frame clock, and SYNC handling	Frame clock, multi-frame clock and SYNC handling are outside the scope of the macro

1.5.12 SRIO 6.25G

The PHY supports Level II SR/MR/LR (6.25 Gbps and 5 Gbps) data rate and is backwards compatible to Level I (3.125 Gbps, 2.5 Gbps, and 1.25 Gbps) data rate.

The PHY complies with the required features of the RapidIO™ Interconnect Specification, Rev. 3.0, October 2013.

Reference clocks supported

For the reference clocks supported, refer to "Support Configuration Settings" and "Lane Configuration Settings".

Exceptions to the specification allowed (see Table 1-15)

Table 1-15 Exceptions to Specification Allowed for SRIO 6.25G

Category	Exception Allowed
TX edge rate	Faster than specification
TX common mode noise	May exceed 5%*T_Vdif
TX lane to lane skew	Specified as 1000 ps for 4 lanes or less; 2UI+1000 ps for > 4 lanes; this limit may be exceeded when running at 1.25 Gbps for a part that has 4 or fewer lanes
TX differential skew	May exceed 25/20/15/15/15 ps limit for 1.25, 2.5, 3.125, 5, 6.25 Gbps operation
TX	■ Type 0 loads are supported ■ Type 1, 2, and 3 loads are not supported
RX Termination	■ Can be configured as a Type 0 load ■ Type 0 DC-coupled link supported with common mode limited to 0-800 mV ■ Configuration as a Type 1, 2, or 3 load is not supported

1.6 Process Information

Table 1-16 describes the process information for the PHY. For availability in alternate process nodes or variants, contact your sales representative.

Table 1-16 Process Information

Foundry	Process	Variant	Process Core Voltage	Process I/O Voltage	Metal Layers
TSMC	28 nm	CLN28HPC+	0.9 V	1.5 or 1.8 V	10M_5X2Y2Z + RDL 9M_6X2Z + RDL 8M_5X2Z + RDL

Note

- A list of GDSII layers used in the IP is included in a text file within the gds sub-directories of the release package. For more information, see Chapter 2, “Deliverables”.
- For PHY supply voltages, see “Power Supply Limits” on page 240.

Table 1-17 lists the primitive devices used by this PHY.

Table 1-17 Primitive Devices Used

Devices	Description
pdio_ll	Low Leakage P+/NW diode (core)
pdio_18ll	IO Low Leakage P+/NW diode
nmoscap_18	IO NMOS in N-Well Varactor
ndio	N+/PW Junction DIODE
nmoscap	NMOS in N-Well Varactor
rupolym_m	P+ Poly Non-Salicided resistor
nch_18_mac	Thick Oxide NMOS
pch_18_mac	Thick Oxide PMOS
nch_lvt_mac	Thin Oxide Low Voltage NMOS
pch_lvt_mac	Thin Oxide Low Voltage PMOS
nch_mac	CORE Standard VT NMOS
pch_mac	CORE Standard VT PMOS
pnp2_mis	Vertical pnp bipolar transistor (Emitter area 2.0um*2.0um)
ndio_hia_mac	CORE 2T ESD diode N+/PW

Table 1-17 Primitive Devices Used (Continued)

Devices	Description
rm1w	M1 resistor
rm2w	M2 resistor
rm3w	M3 resistor
rm4w	M4 resistor
rm5w	M5 resistor
rm7w	M7 resistor
rm8w	M8 resistor
rm9w	M9 resistor
rm10w	M10 resistor
moscap_rf	CORE RF MOSCAP Varactor wo DNW
moscap_rf18	IO RF MOSCAP Varactor wo DNW
rmap	Aluminum metal resistor

2 Deliverables

This chapter includes the following sections:

- "Introduction"
- "Front-End Views" on page 34
- "Back-End Views" on page 35
- "Signal-Integrity Views" on page 35
- "Test Views" on page 35

2.1 Introduction

The full-package for each DesignWare Cores Multi-Protocol 16G PHY release contains all of the design views and information required to instantiate the PHY in a top-level chip design. It also allows for the simulation of both the digital functionality of the PHY and the signal integrity of the link, and provides the information to enable production testing of the PHY.

For information about the PHY release directory structure, see the top-level Readme file included in the product deliverables.

The deliverables are grouped into four basic categories based on usage:

- Front-end view—Includes deliverables for system simulation and floor planning
- Back-end view—Includes deliverables for physical design and finishing
- Signal-integrity view—Includes deliverables for simulation of link performance at the analog level
- Test view—Includes deliverables for testing

2.2 Front-End Views

Front-end views are provided for both the hard macro portions of the PHY and the raw PCS soft-IP top-level wrapper.

A front-end view includes the following:

- Verilog test bench examples
- Digital simulation environment
- Synthesis timing models in .db and .lib formats
- Behavioral models for analog blocks
- Library Exchange Format (LEF) file with pin sizes and locations
- Protocol-specific test bench examples
- Synthesis constraints for the PCS
- IP-XACT XML file with register details
- DesignWare Cores Multi-Protocol 16G PHY Databook

2.3 Back-End Views

Back-end views are provided for the hard macro portions of the PHY. These deliverables tend to be metal stack dependent, so multiple files for each hard macro are provided. The top-level cell name inside all of the files is the same, but the content is not—you must select the appropriate file that matches your metal stack-up.

A back-end view includes the following:

- GDSII layout and layer map files
- Layout-Versus-Schematic (LVS) netlist in SPICE format, and LVS report
- LVS samples
- Design Rule Check (DRC) report
- DRC samples
- Generic technology (GTECH) netlists and models for simulation
- Gate-level Verilog
- Gate-level netlist and Standard Delay Format (SDF) timing file

2.4 Signal-Integrity Views

A signal-integrity view includes the following:

- S-Parameter models of RX and TX to facilitate package design
- IBIS-AMI models of RX and TX for signal integrity simulations
- Current waveforms, along with on-die power grid equivalent capacitance and resistance (Cdie and Rdie, respectively) for power delivery network simulations in the package

2.5 Test Views

A test view includes the following:

- Scan test environment with Automatic Test Pattern Generation (ATPG)
- Scan Verilog netlist

3 Signal Descriptions

This chapter details all possible I/O signals in the PHY IP.

Inputs are on the left of the signal diagrams; outputs are on the right.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

Voltage Range: Describes the voltage range expected on this pin. For the voltage range of the power supplies, see "Power Dissipation Overview".

Active State: Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).

Synchronous to: Indicates that the signal is asserted or de-asserted with respect to a clock edge.

Asynchronous: Indicates that the signal is not asserted or de-asserted with respect to a clock edge.

The I/O signals are grouped as follows:

- Power-Supply Signals on page 39
- Analog I/O Signals on page 41
- Boundary Scan Port Signals on page 44
- Parallel Control Register Port Signals on page 47
- Reset and Test Signals (Common Block Group) on page 49
- JTAG Port Signals on page 51
- Per-Transceiver Signals on page 53
- MPLL(A,B) Control Signals (Common Block Group) on page 54
- MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group) on page 64
- Power Gating Signals on page 68
- Reference Clock Control Signals (Common Block Group) on page 70
- Resistor Tune Signals (Common Block Group) on page 74

- Receiver Request Signals (Per-Lane Receive Interface Group) on page 76
- Receiver Adaptation Signals (Per-Lane Receive Interface Group) on page 83
- Receiver Datapath Signals (Per-Lane Receive Interface Group) on page 87
- Receiver Control Signals (Per-Lane Receive Interface Group) on page 90
- Receiver Test Signals on page 92
- RX LOS and TX Boost Level Control Signals (Common Block Group) on page 93
- Scan Interface Signals on page 95
- SRAM Control Signals on page 122
- Transmitter Request Signals (Per-Lane Transmit Interface Group) on page 124
- Transmitter Control Signals (Per-Lane Transmit Interface Group) on page 128
- Transmitter Datapath Signals (Per-Lane Transmit Interface Group) on page 130
- Transmitter Equalization Signals (Per-Lane Transmit Interface Group) on page 132
- Transmitter Test Signals on page 134

3.1 Power-Supply Signals

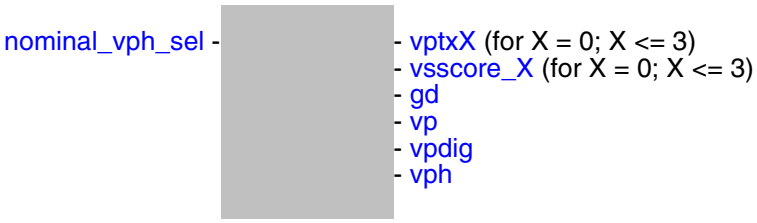


Table 3-1 Power-Supply Signals

Port Name	I/O	Description
vptxX (for X = 0; X <= 3)	IO	Analog transmitter termination voltage supply Off-chip analog power supply for TX lane X **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Active State:** N/A **Synchronous To:** N/A
vsscore_X (for X = 0; X <= 3)	IO	Ground; used by internal core logic Ground used by the core; this input is used for ESD protection **Voltage Range:** 0 V **Active State:** N/A **Synchronous To:** N/A
gd	IO	Ground This signal can be tied to chip ground plane in package, but must not be connected to ground distribution network on-die. **Voltage Range:** 0 V **Active State:** N/A **Synchronous To:** N/A
vp	IO	Low voltage supply Off-chip analog power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Active State:** N/A **Synchronous To:** N/A
vpdig	IO	Digital power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Active State:** N/A **Synchronous To:** N/A

Table 3-1 Power-Supply Signals (Continued)

Port Name	I/O	Description
vph	IO	I/O Voltage supply An off-chip analog high voltage power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Active State:** N/A **Synchronous To:** N/A
nominal_vph_sel[1:0]	I	VPH Nominal Selection (nominal_vph_sel) Indicates the VPH voltage level supplied: ■ 2'b01: Reserved ■ 2'b10 : 1.5 V ■ 2'b11: 1.8 V ■ 2'b00: Reserved **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.2 Analog I/O Signals

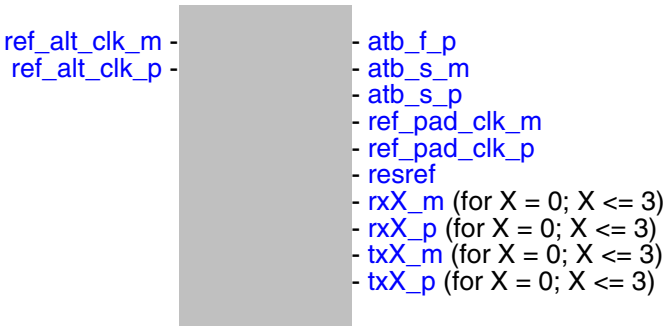


Table 3-2 Analog I/O Signals

Port Name	I/O	Description
atb_f_p	IO	Analog test bus force Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Active State:** N/A **Synchronous To:** N/A
atb_s_m	IO	Analog test bus sense (atb_s_m) Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Active State:** N/A **Synchronous To:** N/A
atb_s_p	IO	Analog test bus sense (atb_s_p) Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Active State:** N/A **Synchronous To:** N/A
ref_alt_clk_m	I	Low-Swing differential input clock pair internal pin (ref_alt_clk_m) Either ref_pad_clk_[p,m] or ref_alt_clk_[p,m] is available on a core, but both must not be active at the same time. The selection is based on whether the reference clock is provided to the core via an on-die distribution network or directly to the core from off- die. The pair that is selected depends on the value of the ref_use_pad input. **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A

Table 3-2 Analog I/O Signals (Continued)

Port Name	I/O	Description
ref_alt_clk_p	I	Low-Swing differential input clock pair internal pin (ref_alt_clk_p) Either ref_pad_clk_[p,m] or ref_alt_clk_[p,m] is available on a core, but both must not be active at the same time. The selection is based on whether the reference clock is provided to the core via an on-die distribution network or directly to the core from off- die. The pair that is selected depends on the value of the ref_use_pad input. **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A
ref_pad_clk_m	IO	Low-Swing differential input clock pair with pad (ref_pad_clk_m) Either ref_pad_clk_[p,m] or ref_alt_clk_[p,m] is available on a core, but both must not be active at the same time. The selection is based on whether the reference clock is provided to the core through an on-die distribution network or directly to the core from off-die. The pair that is selected depends on the value of the ref_use_pad input. **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A
ref_pad_clk_p	IO	Low-Swing differential input clock pair with pad (ref_pad_clk_p) **Voltage Range:** 0-vp Either ref_pad_clk_[p,m] or ref_alt_clk_[p,m] is available on a core, but both must not be active at the same time. The selection is based on whether the reference clock is provided to the core through an on-die distribution network or directly to the core from off-die. The pair that is selected depends on the value of the ref_use_pad input. **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A
resref	IO	Reference resistor connection Attach a 200 ohm 1% 100-ppm/C precision resistor-to- ground on the board. For information about resref, see the "Physical-Level Implementation" and "Board- and Package-Level Implementation" chapters. **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A
rxX_m (for X = 0; X <= 3)	IO	High-Speed differential receive pair (rxX_m) Receive differential pair for lane X **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A

Table 3-2 Analog I/O Signals (Continued)

Port Name	I/O	Description
rxX_p (for X = 0; X <= 3)	IO	High-Speed differential receive pair (rxX_p) Receive differential pair for lane X **Voltage Range:** 0-vp **Active State:** N/A **Synchronous To:** N/A
txX_m (for X = 0; X <= 3)	IO	High-Speed differential transmit pair (txX_m) Transmit differential pair for lane X **Voltage Range:** 0-vptxX **Active State:** N/A **Synchronous To:** N/A
txX_p (for X = 0; X <= 3)	IO	High-Speed differential transmit pair (txX_p) Transmit differential pair for lane X **Voltage Range:** 0-vptxX **Active State:** N/A **Synchronous To:** N/A

3.3 Boundary Scan Port Signals

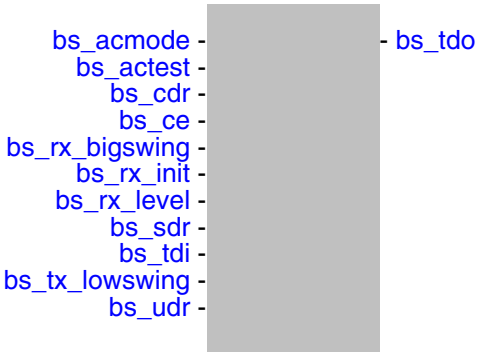


Table 3-3 Boundary Scan Port Signals

Port Name	I/O	Description
bs_acmode	I	Boundary Scan AC Mode (Figure 51, IEEE 1149.6) Enables Boundary Scan AC mode **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
bs_actest	I	Boundary Scan AC Test Signal (Figure 51, IEEE 1149.6) The AC Test Signal is generated by the user logic when AC EXTEST instruction is decoded. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
bs_cdr	I	Boundary Scan Clock-DR (IEEE 1149.1) Clocks the Boundary Scan capture registers, in either Scan or Capture modes **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
bs_ce	I	Boundary scan compliance enable signal Enables the Boundary Scan Transmitter and Receiver register cells on all lanes. This input should be driven from a sequential element and not from any combinatorial logic. Any glitch on this input causes the PHY to go into reset state. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-3 Boundary Scan Port Signals (Continued)

Port Name	I/O	Description
bs_rx_bigswing	I	RX Boundary Scan Big Swing Enables RX ACJTAG big-swing mode **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
bs_rx_init	I	RX Boundary Scan Initialize Signal This signal is similar to the 'Init Clk' referenced in test receiver diagrams in the ACJTAG specification (Figure 48, IEEE 1149.6). While this signal is high, the hysteresis in the receiver comparator is reset. The falling edge of this signal must occur at appropriate times, as specified in the ACJTAG specification. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
bs_rx_level[4:0]	I	ACJTAG Receiver Sensitivity Level Control Sets the internal thresholds for the ACJTAG receiver. The ACJTAG threshold level should be set to trigger for the expected amplitude for the given protocol. For threshold levels, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
bs_sdr	I	Boundary Scan Shift-DR (IEEE 1149.1) Enables shifting of the Boundary Scan capture registers **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** bs_cdr
bs_tdi	I	Boundary Scan Test Data In Boundary scan shift register input signal **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** bs_cdr
bs_tdo	O	Boundary Scan Test Data Out Boundary scan shift register output signal **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** bs_cdr

Table 3-3 Boundary Scan Port Signals (Continued)

Port Name	I/O	Description
bs_tx_lowswing	I	TX Boundary Scan Low Swing Enables TX JTAG low-swing mode (otherwise, it is in full- swing mode) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
bs_udr	I	Boundary Scan Update-DR (IEEE 1149.1) Clocks the Boundary Scan update registers **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

3.4 Parallel Control Register Port Signals

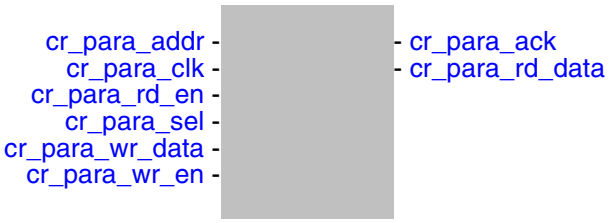


Table 3-4 Parallel Control Register Port Signals

Port Name	I/O	Description
cr_para_ack	O	CR Read/Write acknowledgement Acknowledge signal indicating completion of the CR parallel read/write access. For more information about operation during SRAM update, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** cr_para_clk
cr_para_addr[15:0]	I	CR address **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** cr_para_clk
cr_para_clk	I	CR clock **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
cr_para_rd_data[15:0]	O	CR read data **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** cr_para_clk
cr_para_rd_en	I	CR read enable Reads from the referenced address on cr_para_addr and provides the data on cr_para_rd_data **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** cr_para_clk

Table 3-4 Parallel Control Register Port Signals (Continued)

Port Name	I/O	Description
cr_para_sel	I	Control Register (CR) parallel interface select Controls selection between JTAG and CR interfaces: ■ 0: JTAG ■ 1: Control Register (CR) This input can only be changed when the cr_para_clk and jtag_tck clock inputs are disabled. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
cr_para_wr_data[15:0]	I	CR write data **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** cr_para_clk
cr_para_wr_en	I	CR write enable Writes the data on cr_para_wr_data to the referenced address on cr_para_addr **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** cr_para_clk

3.5 Reset and Test Signals (Common Block Group)

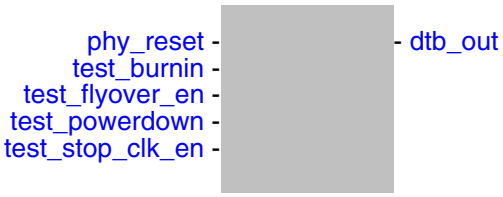


Table 3-5 Reset and Test Signals (Common Block Group)

Port Name	I/O	Description
dtb_out[1:0]	O	Digital test bus output Used in debug mode for probing the values of various internal signals in the PHY **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
phy_reset	I	PHY reset Asynchronously resets the core and all state machines. Asserting phy_reset resets all sequential elements in the design, including control registers. As a result, any register programming needs to be redone after phy_reset is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
test_burnin	I	All circuits activator Activates all circuitry in the PHY for burn-in testing. This signal must not be asserted while test_powerdown signal is set to 1'b1. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
test_flyover_en	I	Enable flyover test mode Flyover test mode enables direct access to the TX and RX bump pads, from the txX_flyover_data_p/m and rxX_flyover_data_p/m I/Os This mode can be enabled in either normal mode or scan mode. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-5 Reset and Test Signals (Common Block Group) (Continued)

Port Name	I/O	Description
test_powerdown	I	All circuits Power-Down control Powers down all circuitry in the PHY for IDDQ testing **Note:** The PHY is not functional in this mode and must be reset after this signal is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
test_stop_clk_en	I	Stop clock test mode enable Reserved **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.6 JTAG Port Signals



Table 3-6 JTAG Port Signals

Port Name	I/O	Description
jtag_tck	I	JTAG clock JTAG state machine clock signal **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
jtag_tdi	I	JTAG input port JTAG test data input signal **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** jtag_tck
jtag_tdo	O	JTAG output port JTAG test data output signal **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** jtag_tck
jtag_tdo_en	O	JTAG output enable JTAG test data output enable signal **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** jtag_tck
jtag_tms	I	JTAG state machine control JTAG test mode select signal **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** jtag_tck

Table 3-6 JTAG Port Signals (Continued)

Port Name	I/O	Description
jtag_trst_n	I	JTAG reset JTAG state machine reset signal. While this signal is held low, the state machine is held in test logic reset state. **Voltage Range:** 0-vpdig **Active State:** Low **Synchronous To:** Asynchronous

3.7 Per-Transceiver Signals (for X = 0; X <= 3)

laneX_rx2tx_par_lb_en -
laneX_tx2rx_ser_lb_en -



Table 3-7 Per-Transceiver Signals (for X = 0; X <= 3)

Port Name	I/O	Description
laneX_rx2tx_par_lb_en	I	Parallel (RX to TX) loopback enable When this signal is asserted, recovered parallel data from the receiver is looped back to the transmit serializer. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
laneX_tx2rx_ser_lb_en	I	Analog serial (TX to RX) loopback enable When this signal is asserted, data from the TX pre-driver is looped back to the RX analog front end (AFE); loss of signal (LOS) is bypassed and based on the txX_data_en input so that rxX_los = !txX_data_en. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.8 MPLL(A,B) Control Signals (Common Block Group)



Table 3-8 MPLL(A,B) Control Signals (Common Block Group)

Port Name	I/O	Description
mplla_bandwidth[15:0]	I	MPLLA bandwidth control Bandwidth control for MPLLA. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_div10_clk_en	I	MPLLA divide by 10 enable When asserted, the frequency of the mpll(a,b)_word_clk output clock is the MPLL(A,B) frequency divided by 10. This divide by 10 takes priority over mpll(a,b)_div8_clk_en. These inputs can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mplla_div16p5_clk	O	MPLLA divide by 16.5 clock Divide by 16.5 output clock derived from MPLL(A,B) **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mplla_div16p5_clk_en	I	MPLLA divide by 16.5 enable Enables the output clocks derived from MPLLA based on 16.5, 33 and 66 division ratios This input can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_div33_clk	O	MPLLA divide by 33 clock Divide by 33 output clock derived from MPLL(A,B) **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mplla_div66_clk	O	MPLLA divide by 66 clock Divide by 66 output clock derived from MPLL(A,B) **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mplla_div8_clk_en	I	MPLLA divide by 8 enable When asserted, the frequency of the mpll(a,b)_word_clk output clock is the MPLL(A,B) frequency divided by 8. If mpll(a,b)_div10_clk_en is also asserted, then divide by 10 takes priority. These inputs can only be changed when the txX_mpll_en input is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_div_clk	O	MPLLA divide output clock When enabled, the frequency of this clock is equal to the MPLL(A,B) frequency divided by mpll(a,b)_div_multiplier. If mpll(a,b)_ssc_en is asserted, then SSC is applied on this clock. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mplla_div_clk_en	I	MPLLA divide clock enable When asserted, the frequency of the mpll(a,b)_div_clk output clock is the MPLL(A,B) frequency divided by mpll(a,b)_div_multiplier. Fractional division and/or SSC is additionally applied to mpll(a,b)_div_clk depending on the settings of mpll(a,b)_ssc_en and mpll(a,b)_fracn_ctrl. These inputs can only be changed when the respective mpll(a,b)_ssc_en and txX_mpll_en inputs are de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_div_multiplier[6:0]	I	MPLLA output frequency multiplier control Frequency multiplication factor used to generate MPLL clock output from the reference clock input as seen by the MPLL (after ref_clk_div2_en and ref_clk_mplla_div2_en/ref_clk_mpllb_div2_en dividers). This input can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_dword_clk	O	Double-Rate word clock (mplla_dword_clk) Double-symbol rate (x2) clock derived from MPLLA **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mplla_force_ack	O	MPLLA Force Acknowledge Acknowledge output for mpll(a,b)_force_en. This output asserts when the MPLL is powered up in response to mpll(a,b)_force_en assertion. This output de-asserts as a response to mpll(a,b)_force_en de-assertion, and at that point the MPLL will be powereddown if txX_mpll_en of other lanes are de-asserted. **Note:** A four-way level handshake must be followed between mpll(a,b)_force_en and mpll(a,b)_force_ack. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mplla_force_en	I	MPLLA force enable When asserted, the corresponding MPLL is forced to be powered up, irrespective of the txX_mpll_en input. This input is used for applications where a free-running MPLL clock output is required. If MPLL is not powered up, Synopsys recommends that you follow the txX_mpll_en controls as described in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_init_cal_disable	I	MPLLA calibration disable control At power on, the internal PHY firmware will first calibrate both PLLs before powering down the unused one, unless mpll(a,b)_force_en is asserted to force it to remain powered up. The PLL configuration must be valid for calibration to complete in the specified time. ■ This is an asynchronous input tied high when the PLL is not used. ■ Asserting mpll[a/b]_init_cal_disable allows invalid configurations if the PLL is unused. ■ mplla_init_cal_disable = 1 disables mplla calibration ■ mpllb_init_cal_disable = 1 disables mpllb calibration **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_multiplier[7:0]	I	MPLLA frequency multiplier control Multiplies the reference clock to a frequency suitable for intended operating speed. For a list of values and corresponding frequencies, see the "Functional Description" chapter. The reference clocks used by the MPLLs are generated after applying ref_clk_div2_en and ref_clk_mplla_div2_en/ref_clk_mpllb_div2_en divisions on the ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m. These inputs can only be changed when phy_reset is asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_oword_clk	O	Octuple-Rate word clock (mplla_oword_clk) Octuple-symbol rate (x8) clock derived from MPLLA **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mplla_qword_clk	O	Quadruple-Rate word clock (mplla_qword_clk) Quadruple-symbol rate (x4) clock derived from MPLLA **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mplla_recal_bank_sel[1:0]	I	MPLLA Re-calibration Bank Select Selects the bank to store the MPLL calibration value **Note:** This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_state	O	MPLLA state indicator Indicates the state of MPLLA. This signal is asserted when MPLLA is powered up and phase-locked. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_tx_clk_div[1:0]	I	MPLLA TX Clock Divider Sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 2'b00 : div1 ■ 2'b01 : div2 ■ 2'b10 : div4 ■ 2'b11 : div1 (duplicate state) **Note:** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_word_clk	O	Word clock (mplla_word_clk) Symbol rate clock derived from MPLLA **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_bandwidth[15:0]	I	MPLLB bandwidth control Bandwidth control for MPLLB. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_div10_clk_en	I	MPLLB divide by 10 enable When asserted, the frequency of the mpll(a,b)_word_clk output clock is the MPLL(A,B) frequency divided by 10. This divide by 10 takes priority over mpll(a,b)_div8_clk_en. These inputs can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_div8_clk_en	I	MPLLB divide by 8 enable When asserted, the frequency of the mpll(a,b)_word_clk output clock is the MPLL(A,B) frequency divided by 8. If mpll(a,b)_div10_clk_en is also asserted, then divide by 10 takes priority. These inputs can only be changed when the txX_mpll_en input is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_div_clk	O	MPLLB divide output clock When enabled, the frequency of this clock is equal to the MPLL(A,B) frequency divided by mpll(a,b)_div_multiplier. If mpll(a,b)_ssc_en is asserted, then SSC is applied on this clock. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mpllb_div_clk_en	I	MPLLB divide clock enable When asserted, the frequency of the mpll(a,b)_div_clk output clock is the MPLL(A,B) frequency divided by mpll(a,b)_div_multiplier. Fractional division and/or SSC is additionally applied to mpll(a,b)_div_clk depending on the settings of mpll(a,b)_ssc_en and mpll(a,b)_fracn_ctrl. These inputs can only be changed when the respective mpll(a,b)_ssc_en and txX_mpll_en inputs are de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_div_multiplier[6:0]	I	MPLLB output frequency multiplier control Frequency multiplication factor used to generate MPLL clock output from the reference clock input as seen by the MPLL (after ref_clk_div2_en and ref_clk_mplla_div2_en/ref_clk_mpllb_div2_en dividers). This input can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_dword_clk	O	Double-Rate word clock (mpllb_dword_clk) Double-symbol rate (x2) clock derived from MPLLB **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mpllb_force_ack	O	MPLLB Force Acknowledge Acknowledge output for mpll(a,b)_force_en. This output asserts when the MPLL is powered up in response to mpll(a,b)_force_en assertion. This output de-asserts as a response to mpll(a,b)_force_en de-assertion, and at that point the MPLL will be powereddown if txX_mpll_en of other lanes are deasserted. **Note:** A four-way level handshake must be followed between mpll(a,b)_force_en and mpll(a,b)_force_ack. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_force_en	I	MPLLB force enable When asserted, the corresponding MPLL is forced to be powered up, irrespective of the txX_mpll_en input. This input is used for applications where a free-running MPLL clock output is required. If MPLL is not powered up, Synopsys recommends that you follow the txX_mpll_en controls as described in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_init_cal_disable	I	MPLLB calibration disable control At power on, the internal PHY firmware will first calibrate both PLLs before powering down the unused one, unless mpll(a,b)_force_en is asserted to force it to remain powered up. The PLL configuration must be valid for calibration to complete in the specified time. ■ This is an asynchronous input tied high when the PLL is not used. ■ Asserting mpll[a/b]_init_cal_disable allows invalid configurations if the PLL is unused. ■ mplla_init_cal_disable = 1 disables mplla calibration ■ mpllb_init_cal_disable = 1 disables mpllb calibration **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_multiplier[7:0]	I	MPLLB frequency multiplier control Multiplies the reference clock to a frequency suitable for intended operating speed. For a list of values and corresponding frequencies, see the "Functional Description" chapter. The reference clocks used by the MPLLs are generated after applying ref_clk_div2_en and ref_clk_mplla_div2_en/ref_clk_mpllb_div2_en divisions on the ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m. These inputs can only be changed when phy_reset is asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_oword_clk	O	Octuple-Rate word clock (mpllb_oword_clk) Octuple-symbol rate (x8) clock derived from MPLLBs **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
mpllb_qword_clk	O	Quadruple-Rate word clock (mpllb_qword_clk) Quadruple-symbol rate (x4) clock derived from MPLLB **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_recal_bank_sel[1:0]	I	MPLLB Re-calibration Bank Select Selects the bank to store the MPLL calibration value **Note:** This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_state	O	MPLLB state indicator Indicates the state of MPLLB. This signal is asserted when MPLLB is powered up and phase-locked. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_tx_clk_div[1:0]	I	MPLLB TX Clock Divider Sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 2'b00 : div1 ■ 2'b01 : div2 ■ 2'b10 : div4 ■ 2'b11 : div1 (duplicate state) **Note:** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_word_clk	O	Word clock (mpllb_word_clk) Symbol rate clock derived from MPLLB **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-8 MPLL(A,B) Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
sup_misc[7:0]	I	Support Miscellaneous Controls This input controls miscellaneous settings in the Support block. sup_misc[7:3] is reserved sup_misc[2:1] is used to indicate the nominal VP supply voltage with the following encoding: ■ 00: (0.8 V mode) ■ 01: (0.85 V mode) ■ 10: (0.9 V mode) ■ 11: (1.0 V mode) sup_misc[0] is used to get the required chop clock frequency based on the ref clock frequency. This input must be set to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.9 MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group)

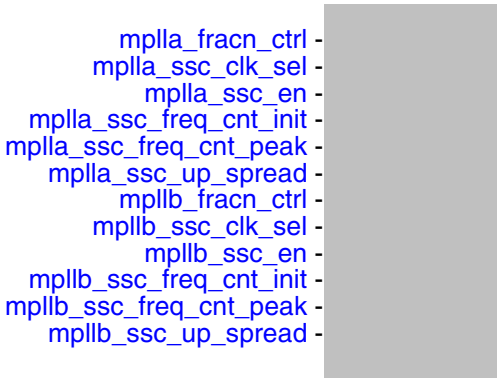


Table 3-9 MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group)

Port Name	I/O	Description
mplla_fracn_ctrl[10:0]	I	MPLLA fractional control MPLL(A,B) fractional control input to be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_ssc_clk_sel	I	MPLLA spread spectrum clock select MPLL(A,B) clock select for generating spread spectrum. This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-9 MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mplla_ssc_en	I	Spread spectrum enable (mplla_ssc_en) Enables spread-spectrum clock (SSC) generation on the mpll(a,b)_div_clk output. If the reference clock already has spread spectrum applied, mpll(a,b)_ssc_en must be de-asserted. These inputs can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mplla_ssc_freq_cnt_init[11:0]	I	MPLLA SSC Frequency Counter Initialization Sets the initialization value for the SSC frequency **Note:** ■ This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. ■ This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig counter. **Active State:** N/A **Synchronous To:** Asynchronous
mplla_ssc_freq_cnt_peak[7:0]	I	MPLLA SSC Frequency Counter Peak Sets the peak value for the SSC frequency counter.**Note:** ■ This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. ■ This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mplla_ssc_up_spread	I	MPLLA SSC Up Spread Enable When asserted, the SSC is applied in upwards direction (positive ppm) and when deasserted the SSC is applied downwards (negative ppm). **Note:**This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-9 MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_fracn_ctrl[10:0]	I	MPLLB fractional control MPLL(A,B) fractional control input to be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_ssc_clk_sel	I	MPLLB spread spectrum clock select MPLL(A,B) clock select for generating spread spectrum. This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_ssc_en	I	Spread spectrum enable (mpllb_ssc_en) Enables spread-spectrum clock (SSC) generation on the mpll(a,b)_div_clk output. If the reference clock already has spread spectrum applied, mpll(a,b)_ssc_en must be de-asserted. These inputs can only be changed when the txX_mpll_en inputs for all lanes are de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
mpllb_ssc_freq_cnt_init[11:0]	I	MPLLB SSC Frequency Counter Initialization Sets the initialization value for the SSC frequency **Note:** ■ This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. ■ This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig counter. **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-9 MPLL(A,B) Fractional Division and SSC Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
mpllb_ssc_freq_cnt_peak[7:0]	I	MPLLB SSC Frequency Counter Peak Sets the peak value for the SSC frequency counter. **Note:** ■ This input must be set as per the tables in the "Support Configuration Settings" appendix and the "Lane Configuration Settings" appendix. ■ This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
mpllb_ssc_up_spread	I	MPLLB SSC Up Spread Enable When asserted, the SSC is applied in upwards direction (positive ppm) and when de-asserted the SSC is applied downwards (negative ppm). **Note:**This input can be changed only when the MPLL is powered down. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.10 Power Gating Signals

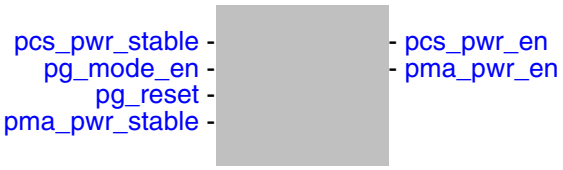


Table 3-10 Power Gating Signals

Port Name	I/O	Description
pcs_pwr_en	O	Power enable for Raw PCS power switches Enable signal for external switches to supply power to the power gated logic in Raw PCS. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
pcs_pwr_stable	I	Power stable for Raw PCS Status signal indicating that the power for the Raw PCS is stable. The pcs_pwr_stable signal should only be asserted if the supply is 90% of nominal or higher. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
pg_mode_en	I	Power gating support enable Control input to enable the power gating support. When de-asserted, the control inputs related to power gating are ignored. For power gating sequencing, see the "PHY Usage and Configuration" chapter. This input is meant to be a tie-off and can be transitioned only during the PHY reset sequence. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
pg_reset	I	Power gated reset Reset signal for the power gating domain, which controls the generation of the pcs_pwr_en and pma_pwr_en outputs. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-10 Power Gating Signals (Continued)

Port Name	I/O	Description
pma_pwr_en	O	Power enable for PMA power switch Enable signal for PMA power switch (external) to supply power to the PMA. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
pma_pwr_stable	I	Power stable for PMA Status signal indicating that the power for the PMA is stable. The pma_pwr_stable signal should only be asserted if the supply is 90% of nominal or higher. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.11 Reference Clock Control Signals (Common Block Group)



Table 3-11 Reference Clock Control Signals (Common Block Group)

Port Name	I/O	Description
phy_ref_dig_clk	O	Buffered version of input reference clock This is a buffered, single-ended version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of ref_use_pad. If ref_clk_div2_en is asserted, then phy_ref_dig_clk is the divided-by-2 version of the reference clock. If ref_range[2:0] is greater than three, then an additional divide-by-2 is applied. This is to ensure phy_ref_dig_clk has a maximum frequency of 100 MHz. **Notes:** ■ Do not use this signal as a precision clock source for other PHY instantiations. It is not appropriate for such use. ■ This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. ■ phy_ref_dig_clk will be running during phy_reset assertion, if ref_clk_en=1. To ensure clean start, customer should first deassert ref_clk_en and then assert it.**Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
ref_clk_div2_en	I	Input reference clock divider control The reference clock is divided by 2 when asserted. Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A

Table 3-11 Reference Clock Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
ref_clk_en	I	Reference clock enable Enables the reference clock for the PHY. The ref_clk_en signal must remain de-asserted until the reference clock is running at the appropriate frequency, at which point ref_clk_en can be asserted. For lower power states where the reference clock to the PHY is disabled, ref_clk_en may also be de-asserted. This signal must be negated after power-on because it is used as an asynchronous reset for some internal logic. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A
ref_clk_mplla_div2_en	I	MPLLA reference clock divider control The reference clock used for MPLLA calibration and locking can be divided by 2. This division is applied after ref_clk_div2_en. Hence the total division ratio (from the input reference clock) can be 1, 2 or 4. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A
ref_clk_mpllb_div2_en	I	MPLLB reference clock divider control The reference clock used for MPLLB calibration and locking can be divided by 2. This division is applied after ref_clk_div2_en. Hence the total division ratio (from the input reference clock) can be 1, 2 or 4. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A
ref_clk_req	O	Reference clock request Request for output logic to supply reference clock to the PHY. When this input is asserted, reference clock to the PHY cannot be disabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A
ref_clkdet_en	I	Enable Reference Clock Detection Enables detection of the reference clock on either the pads input or the alt input (depending on ref_use_pad). **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-11 Reference Clock Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
ref_clkdet_result	O	Reference Clock Detection result Indicates the presence of the reference clock on either the pads input or the alt input (depending on ref_use_pad). The result is only valid when ref_clkdet_en is asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
ref_dig_fr_clk	O	Free-running buffered version of input reference clock This is a buffered, single-ended version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of ref_use_pad. The frequency of this clock matches that of the input reference clock and is not affected by the setting of ref_clk_div2_en. **Notes:** ■ Do not use this signal as a precision clock source for other PHY instantiations. It is not appropriate for such use. ■ This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. ■ ref_dig_fr_clk will be running during phy_reset assertion, if ref_clk_en=1. To ensure clean start, customer should first deassert ref_clk_en and then assert it.**Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
ref_range[2:0]	I	Input reference clock frequency range Specifies the frequency range of the input reference clock (post ref_clk_div2_en division if any). The code mapping of ref_range is as follows: ■ 3'b000: 20 - 26 MHz ■ 3'b001: 26.1 - 52 MHz ■ 3'b010: 52.1 - 78 MHz ■ 3'b011: 78.1 - 104 MHz ■ 3'b100: 104.1 - 130 MHz ■ 3'b101: 130.1 - 156 MHz ■ 3'b110: 156.1 - 182 MHz ■ 3'b111: 182.1 - 200 MHz Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-11 Reference Clock Control Signals (Common Block Group) (Continued)

Port Name	I/O	Description
ref_repeat_clk_en	I	Repeat reference clock enable Enables the CML output clocks ref_repeat_clk_[p,m]. This pair of clocks can be used as reference clocks for other on-chip PHYs. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
ref_repeat_clk_m	O	Current-mode logic (CML) buffered version of input reference clock (ref_repeat_clk_m) This is a CML-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of ref_use_pad. This pair of clocks can be used as reference clocks for other PHYs on chip. **Note:**This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
ref_repeat_clk_p	O	Current-mode logic (CML) buffered version of input reference clock (ref_repeat_clk_p) This is a CML-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt_clk_p/ref_alt_clk_m, depending on the settings of ref_use_pad. This pair of clocks can be used as reference clocks for other PHYs on chip. **Note:**This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
ref_use_pad	I	Select reference clock connected to ref_pad_clk_p/ref_pad_clk_m Selects the external ref_pad_clk_p and ref_pad_clk_m inputs as the reference clock source when asserted. When de-asserted, ref_alt_clk_p and ref_alt_clk_m are the sources of the reference clock. Any change in this input must be followed by phy_reset assertion. **Note:** A transition on this input must be followed by the assertion of phy_reset. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** N/A

3.12 Resistor Tune Signals (Common Block Group)

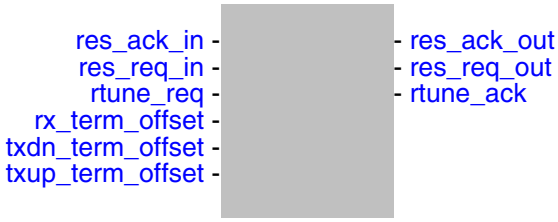


Table 3-12 Resistor Tune Signals (Common Block Group)

Port Name	I/O	Description
res_ack_in	I	Arbitration lines for sharing single reference resistor (res_ack_in) Arbitration lines used for sharing a single-reference resistor among multiple cores that are connected to a single set of pads **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
res_ack_out	O	Arbitration lines for sharing single reference resistor (res_ack_out) Arbitration lines used for sharing a single reference resistor among multiple cores that are connected to a single set of pads **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
res_req_in	I	Arbitration lines for sharing single reference resistor (res_req_in) Arbitration lines used for sharing a single-reference resistor among multiple cores that are connected to a single set of pads **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
res_req_out	O	Arbitration lines for sharing single reference resistor (res_req_out) Arbitration lines used for sharing a single reference resistor among multiple cores that are connected to a single set of pads **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rtune_ack	O	Resistor tune acknowledge Indicates that a resistor tune has completed **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-12 Resistor Tune Signals (Common Block Group) (Continued)

Port Name	I/O	Description
rtune_req	I	Resistor tune request Assertion triggers a resistor tune request (if one is not already in progress) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rx_term_offset[4:0]	I	Offset for RX Termination Specifies an additional fixed offset to calibrated RX termination value. This is a signed input with 2's complement encoding. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
txdn_term_offset[8:0]	I	Offset for TX Down Termination Specifies an additional fixed offset to calibrated TX down termination value. This is a signed input with 2's complement encoding. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
txup_term_offset[8:0]	I	Offset for TX Up Termination Specifies an additional fixed offset to calibrated TX up termination value. This is a signed input with 2's complement encoding. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

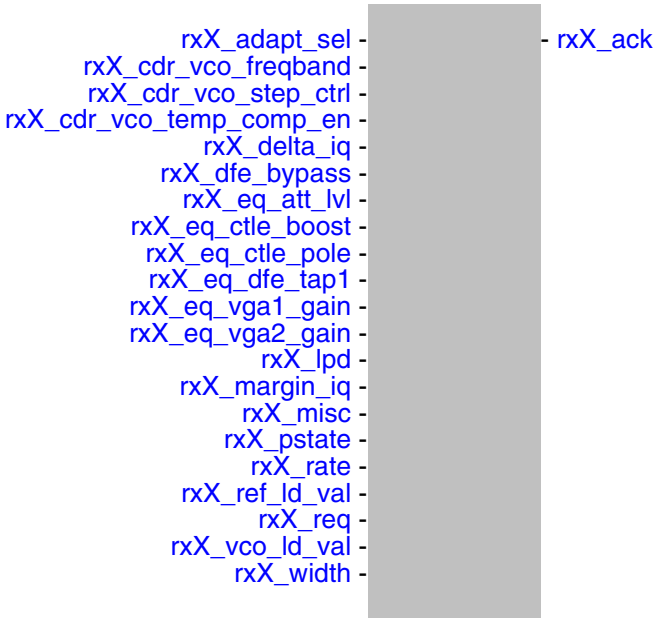


Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
rxX_ack	O	Receiver acknowledge The assertion of this signal indicates the requested receiver setting is complete. This signal stays asserted until the rxX_req input is de-asserted. Once asserted, rxX_ack is de-asserted when rxX_req is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_adapt_sel	I	Select Storage Bank for RX Adaptation The PHY supports two internal register storage banks for saving adaptation results?this input specifies which one is selected. The storage banks are used for restoring the adapted values when needed (for example, when returning from another rate). This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_cdr_vco_freqband[1:0]	I	RX CDR VCO Frequency Band Controls coarse-DAC step-size based on data rate and VPH level. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_cdr_vco_step_ctrl	I	RX CDR VCO Step Control Controls CDR VCO biasing based on data rate. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_cdr_vco_temp_comp_en	I	RX CDR Temperature Compensation Enable Control of CDR VCO temperature compensation circuit based on data rate. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_delta_iq[3:0]	I	IQ offset value Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This is an unsigned offset value and covers a range of 0UI to 0.2UI. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_dfe_bypass	I	RX DFE By-pass Enable Indicates by-passing of the decision feedback equalization (DFE) circuitry. The by-pass path is meant to save power for low data rates where the DFE is not required. The DFE is powered-off when in by-pass mode. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_eq_att_lvl[2:0]	I	RX equalization attenuation level Controls the AFE attenuation level from -2dB when set to 3'b000, to -6dB when set to 3'b111; binary encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_eq_ctle_boost[4:0]	I	RX equalization CTLE boost Controls the CTLE boost level; binary encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_eq_ctle_pole[1:0]	I	RX equalization CTLE pole Controls the continuous time linear equalizer (CTLE) boost pole location; binary encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_eq_dfe_tap1[7:0]	I	RX equalization DFE Tap1 Controls the value of DFE data Tap1. This is a signed input; two's complement encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_eq_vga1_gain[2:0]	I	RX equalization VGA gain 1 Controls the AFE first stage Variable Gain Amplifier gain; binary encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_eq_vga2_gain[2:0]	I	RX equalization VGA gain 2 Controls the AFE second stage Variable Gain Amplifier gain; binary encoded. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_lpd	I	Receiver lane power down Powers down the receiver into a power state equivalent to P1. This signal is useful for selectively powering down lanes in a link while maintaining the power state of the link as a whole (for example, link in P0). **Note:** If the rxX_pstate[1:0] input is set to P2, then that takes priority and the receiver is placed in P2. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_margin_iq[6:0]	I	Value for RX IQ margining This is a signed value for skewing the IQ separation for the purpose of RX margining. It covers a range from -0.5UI to 0.5UI. Each margin step corresponds to approximately 0.022UI phase shift in magnitude. For PCIe margining the recommended limit for RX margining is +/-9 codes corresponding to +/-0.2UI. This input requires an rxX_req/rxX_ack handshake, and is only valid when the RX DFE is enabled (rxX_dfe_bypass=0). **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_misc[7:0]	I	RX Miscellaneous Controls This input controls miscellaneous settings in the RX. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_pstate[1:0]	I	Receiver power state Sets the power state of the transmitter. The state mappings are as follows: ■ 2'b00: P0- Receiver is fully powered up and output receive clocks are active ■ 2'b01: P0S - RX voltage-controlled oscillator (VCO) is in continuous calibration mode, output receive clocks are not available. ■ 2'b10: P1- RX analog front-end (AFE) and voltage regulators are powered up, but RX VCO is in reset ■ 2'b11: P2- RX LOS detector is powered up and the rest of RX is powered down **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_rate[1:0]	I	RX data rate Select the baud rate for the receive path ■ 2'b00: baud ■ 2'b01: baud/2 ■ 2'b10: baud/4 ■ 2'b11: baud/8 **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_ref_ld_val[6:0]	I	RX VCO calibration reference load value This input is used to load internal calibration registers used to perform RX VCO calibration. This input must be set to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_req	I	Receiver request This signal indicates a new receiver setting request. This signal should stay high until the rxX_ack output is asserted. After rxX_ack is asserted, rxX_req should be de-asserted. **Notes:** The PHY captures the following input signals when rxX_req is asserted: ■ rxX_pstate ■ rxX_lpd ■ rxX_rate ■ rxX_width ■ rxX_dfe_bypass ■ rxX_eq_att_lvl ■ rxX_eq_vga1_gain ■ rxX_eq_vga2_gain ■ rxX_eq_ctle_pole ■ rxX_eq_ctle_boost ■ rxX_eq_dfe_tap1 ■ rxX_ref_ld_val ■ rxX_vco_ld_val ■ rxX_adapt_sel ■ rxX_cdr_vco_freqband ■ rxX_cdr_vco_step_ctrl ■ rxX_cdr_vco_temp_comp_en ■ rxX_delta_iq ■ rxX_margin_iq ■ rxX_misc These signals should be stable before of assertion of rxX_req and remain stable until rxX_req is de-asserted. Any combination of changes in these signals is valid for a single request, and the completion of all changes are indicated by the assertion of rxX_ack. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
rxX_vco_ld_val[12:0]	I	RX VCO calibration load value This input is used to load internal calibration registers used to perform RX VCO calibration. This input must be set to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-13 Receiver Request Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_width[1:0]	I	RX data width Select the width of the output receive data: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.14 Receiver Adaptation Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

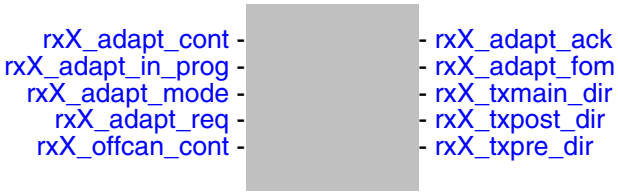


Table 3-14 Receiver Adaptation Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
rxX_adapt_ack	O	Receiver adaptation acknowledge The assertion of this output indicates completion of the receiver adaptation sequence and that values on rxX_adapt_fom, rxX_txmain_dir, rxX_txpre_dir, and rxX_txpost_dir are valid. **Note:** This signal remains asserted until rxX_adapt_req is de-asserted, after which rxX_adapt_ack de-asserts. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_adapt_cont	I	Receiver adaptation continuous operation This input should be asserted if continuous receiver adaptation is required. If this signal is de-asserted, the receiver adaptation stops when the adaptation acknowledge (rxX_adapt_ack) is asserted; otherwise, the receiver continues to adapt. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_adapt_fom[7:0]	O	Receiver figure of merit Output indicating the quality of the received data eye. A higher value of rxX_adapt_fom indicates better link equalization, with 8'd0 indicating worst equalization setting and 8'd255 indicating the best equalization setting. This output is valid only when rxX_adapt_ack is asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-14 Receiver Adaptation Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_adapt_in_prog	I	Setting this signal enables you to retain the selected bank in the current mode. For example, if the bank is in PCIe Gen4 mode, the selected bank continues to operate in that mode ■ rx_adapt_in_prog=0: When there is a reset (when rx_reset=1), reset all banks, rx_adpt_done_r (bank0) and rx_adpt_done_b1_r (bank1) ■ rx_adapt_in_prog=1: Only reset the selected bank (based on rx_adapt_sel); the other bank is not reset **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_adapt_mode[1:0]	I	RX Adaptation mode select Selects what RX adaptation mode will be performed. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_adapt_req	I	Receiver adaptation request This input indicates a request for the receiver to adapt to incoming data. This signal must remain asserted until rxX_adapt_ack is asserted, after which rxX_adapt_req can be de- asserted. **Note:** Aborting rxX_adapt_req by negating before rxX_adapt_ack is asserted or by asserting rxX_reset may leave the receiver with a less than optimal settings. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_offcan_cont	I	Receiver offset cancellation continuous operation If continuous receiver offset cancellation is required, then assert this signal. If this signal is de-asserted, then offset cancellation runs only when the receiver exits the P2 power state. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-14 Receiver Adaptation Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_txmain_dir[1:0]	O	Direction change for Far-End TX coefficients (rxX_txmain_dir) Indicates the recommended direction change for the far- end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir[1:0]. The INC/DEC definition for rxX_txpre_dir[1:0] and rxX_txpost_dir[1:0] may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_txpost_dir[1:0]	O	Direction change for Far-End TX coefficients (rxX_txpost_dir) Indicates the recommended direction change for the far- end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir[1:0]. The INC/DEC definition for rxX_txpre_dir[1:0] and rxX_txpost_dir[1:0] may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-14 Receiver Adaptation Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_txpre_dir[1:0]	O	Direction change for Far-End TX coefficients (rxX_txpre_dir) Indicates the recommended direction change for the far- end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir[1:0]. The INC/DEC definition for rxX_txpre_dir[1:0] and rxX_txpost_dir[1:0] may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.15 Receiver Datapath Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

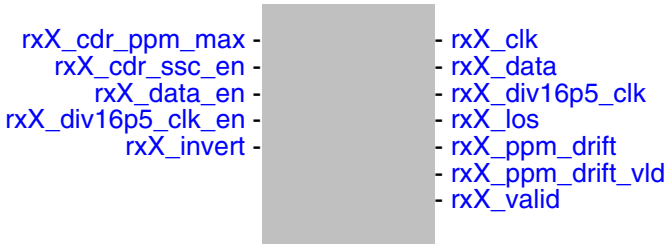


Table 3-15 Receiver Datapath Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
rxX_cdr_ppm_max[4:0]	I	Maximum Allowed PPM on the RX CDR Clock This input specifies the maximum PPM drift to be allowed on the RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. The recovery circuit forces the CDR to exit the data tracking mode even if rxX_data_en is asserted. Instead, the CDR enters the reference tracking mode and is gradually brought back to the ideal frequency before allowing the data tracking again (if rxX_data_en is still asserted). This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_cdr_ssc_en	I	Receiver CDR SSC-Mode enable Controls the CDR tracking gains and duration. It should be asserted when the receive data has a spread-spectrum clock (SSC), and de-asserted when the receive data does not have SSC. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_clk	O	Receiver output clock Received data clock **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-15 Receiver Datapath Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_data[19:0]	O	Receiver output data MSB is received first. This output must be qualified with rxX_valid. The valid range of the data is determined by the rxX_width[1:0] input as follows: - For rxX_width=2'b00, Valid Data Range=rxX_data[7:0] - For rxX_width=2'b01, Valid Data Range=rxX_data[9:0] - For rxX_width=2'b10, Valid Data Range=rxX_data[15:0] - For rxX_width=2'b11, Valid Data Range=rxX_data[19:0]**Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** rxX_clk
rxX_data_en	I	RX data enable Enables the receiver data output. When asserted in P0, the RX datapath is enabled and the CDR begins tracking the received data. This signal is only valid when the RX is in the P0 power state and should be set to 0 in the other power states. Note that when rx_data_en is de-asserted in P0, then the CDR tracks the reference clock in order to stay close to the desired frequency of operation. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_div16p5_clk	O	Receiver Div 16.5 output clock Recovered clock with frequency of RX VCO frequency divided by 16.5. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
rxX_div16p5_clk_en	I	Receiver Div 16.5 output clock enable Enables the divide-by-16.5 RX VCO recovered clock. This signal is only valid when the RX is in the P0 power state. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_invert	I	Invert logical sense of rxX_p and rxX_m inputs This signal has the effect of inverting the received data stream. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-15 Receiver Datapath Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_los	O	Receive LOS (Loss of Signal) output When asserted indicates that the receiver has lost signal. This signal is only valid for low frequency data on rxX_p/rxX_m. The behavior of the signal is as follows: ■ For rxX_p/m=Electrical Idle, rxX_los=1 ■ For rxX_p/m=Data (with pulses less than 400 ps), rxX_los=1/0 (unreliable) ■ For rxX_p/m=Data (with pulses greater than 400 ps), rxX_los=0 **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_ppm_drift[5:0]	O	RX CDR PPM Drift This value represents the amount of ppm on the rxX_clk with respect to the ideal desired frequency. For more information, see the "PHY Usage and Configuration" chapter. This output is valid when rxX_ppm_drift_vld is asserted. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_ppm_drift_vld	O	RX CDR PPM Drift Valid Indicates when the rxX_ppm_drift[5:0] output is valid and can be sampled. For more information, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_valid	O	Data on rxX_data valid indicator This signal is asserted when the RX clock and data recovery (CDR) is bit-locked. The rxX_valid will de-assert if CDR drift detector is enabled (determined by RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL.RX_CDR_DETECTOR_EN) and CDR has drifted beyond the specified maximum PPM (determined by rxX_cdr_ppm_max[4:0]). For more information, see the "PHY Usage and Configuration" chapter. **Note:** The rxX_los signal has no effect on the rxX_valid output. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** rxX_clk

3.16 Receiver Control Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

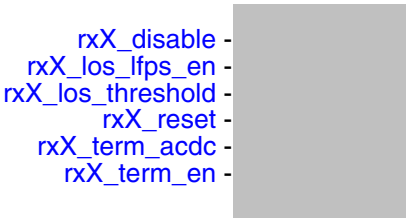


Table 3-16 Receiver Control Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
rxX_disable	I	Receiver disable Puts the receiver in a low-power mode where all the receiver blocks, including the LOS detector, are turned off. This signal should be asserted when in PCIe L1 Substates L1.1 and L1.2. The receiver should be in the P1 power state when using this signal. This input asynchronously disables RX terminations and does not require any clock to be supplied to the PHY. **Note:** The rxX_disable signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
rxX_los_lfps_en	I	Receiver LOS LFPS enable Enables the LFPS filter for USB-like LFPS detection. By default, this input should be set to 0. For more information, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-16 Receiver Control Signals (Per-Lane Receive Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
rxX_los_threshold[2:0]	I	Receiver LOS/Idle detection threshold Sets the LOS threshold level trip point. For the correct setting for a specific protocol, see the "PHY Usage and Configuration" chapter. ■ For rxX_los_threshold=3'b000, LOS threshold=Reserved ■ For rxX_los_threshold=3'b001, LOS threshold=90 mVpp ■ For rxX_los_threshold=3'b010, LOS threshold=120 mVpp ■ For rxX_los_threshold=3'b011, LOS threshold=150 mVpp ■ For rxX_los_threshold=3'b100, LOS threshold=180 mVpp ■ For rxX_los_threshold=3'b101, LOS threshold=210 mVpp ■ For rxX_los_threshold=3'b110, LOS threshold=240 mVpp ■ For rxX_los_threshold=3'b111, LOS threshold=270 mVpp **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
rxX_reset	I	Receiver reset This signal internally resets the RX datapath, all the receiver settings and state machines. Note: If rxX_clk and rxX_div16p5_clk are active at the time rxX_reset is asserted, then these output clock might experience a temporary glitch. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
rxX_term_acdc	I	Receiver termination control Controls the terminations of the RX. Selector for floating versus grounded RX termination: ■ 0 : Floating RX termination (DC coupled PHY) ■ 1 : Grounded RX termination (AC coupled PHY) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_term_en	I	Receiver termination enable When asserted HIGH, the RX is terminated with nominal 50 ohm resistance. When asserted LOW, the termination is in high impedance. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.17 Receiver Test Signals (for X = 0; X <= 3)



Table 3-17 Receiver Test Signals (for X = 0; X <= 3)

Port Name	I/O	Description
rxX_flyover_data_m	O	Receive Flyover Data Outputs (rxX_flyover_data_m) Outputs are directly connected to RX bump pads when test_flyover_en is enabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
rxX_flyover_data_p	O	Receive Flyover Data Outputs (rxX_flyover_data_p) Outputs are directly connected to RX bump pads when test_flyover_en is enabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.18 RX LOS and TX Boost Level Control Signals (Common Block Group)

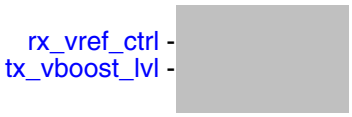
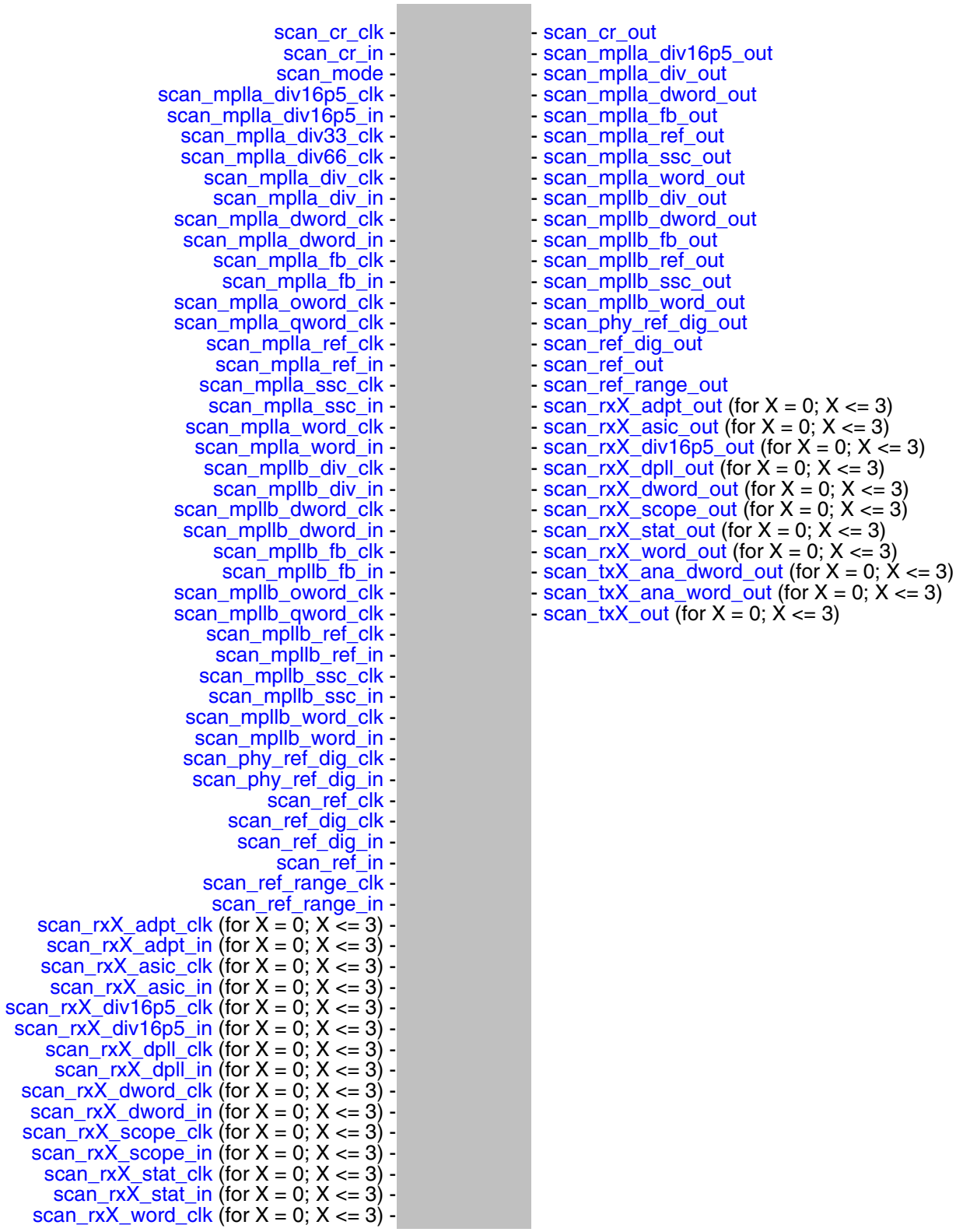


Table 3-18 RX LOS and TX Boost Level Control Signals (Common Block Group)

Port Name	I/O	Description
rx_vref_ctrl[4:0]	I	RX biasing current control Sets the RX biasing current for RX analog front end. The recommended default setting for this input is 5'b10001. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
tx_vboost_lvl[2:0]	I	TX voltage boost maximum level Sets the maximum achievable TX swing. The recommended default setting for this input is 3'b101 (tx_vboost_vref = 0.275 V). **Note:** This signal does not set the actual TX swing, it only sets the maximum limit. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.19 Scan Interface Signals



scan_rxX_word_in (for X = 0; X <= 3) -
scan_set_rst -
scan_shift -
scan_shift_cg -
scan_txX_ana_dword_clk (for X = 0; X <= 3) -
scan_txX_ana_dword_in (for X = 0; X <= 3) -
scan_txX_ana_word_clk (for X = 0; X <= 3) -
scan_txX_ana_word_in (for X = 0; X <= 3) -
scan_txX_in (for X = 0; X <= 3) -

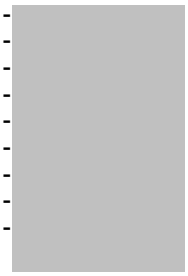


Table 3-19 Scan Interface Signals

Port Name	I/O	Description
scan_cr_clk	I	Scan clock per clock domain (scan_cr_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_cr_in[150:0]	I	Scan chain inputs per clock domain (scan_cr_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_cr_out[150:0]	O	Scan chain outputs per clock domain (scan_cr_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mode	I	Scan mode enable **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
scan_mplla_div16p5_clk	I	Scan clock per clock domain (scan_mplla_div16p5_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_div16p5_in	I	Scan chain inputs per clock domain (scan_mplla_div16p5_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_div16p5_out	O	Scan chain outputs per clock domain (scan_mplla_div16p5_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_div33_clk	I	Scan clock per clock domain (scan_mplla_div33_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_div66_clk	I	Scan clock per clock domain (scan_mplla_div66_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_div_clk	I	Scan clock per clock domain (scan_mplla_div_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_div_in	I	Scan chain inputs per clock domain (scan_mplla_div_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_div_out	O	Scan chain outputs per clock domain (scan_mplla_div_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mplla_dword_clk	I	Scan clock per clock domain (scan_mplla_dword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_dword_in	I	Scan chain inputs per clock domain (scan_mplla_dword_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_dword_out	O	Scan chain outputs per clock domain (scan_mplla_dword_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_fb_clk	I	Scan clock per clock domain (scan_mplla_fb_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_fb_in	I	Scan chain inputs per clock domain (scan_mplla_fb_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_fb_out	O	Scan chain outputs per clock domain (scan_mplla_fb_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mplla_oword_clk	I	Scan clock per clock domain (scan_mplla_oword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_qword_clk	I	Scan clock per clock domain (scan_mplla_qword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_ref_clk	I	Scan clock per clock domain (scan_mplla_ref_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_ref_in	I	Scan chain inputs per clock domain (scan_mplla_ref_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_ref_out	O	Scan chain outputs per clock domain (scan_mplla_ref_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_ssc_clk	I	Scan clock per clock domain (scan_mplla_ssc_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mplla_ssc_in[2:0]	I	Scan chain inputs per clock domain (scan_mplla_ssc_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_ssc_out[2:0]	O	Scan chain outputs per clock domain (scan_mplla_ssc_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mplla_word_clk	I	Scan clock per clock domain (scan_mplla_word_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mplla_word_in	I	Scan chain inputs per clock domain (scan_mplla_word_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mplla_word_out	O	Scan chain outputs per clock domain (scan_mplla_word_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_div_clk	I	Scan clock per clock domain (scan_mpllb_div_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_div_in	I	Scan chain inputs per clock domain (scan_mpllb_div_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_div_out	O	Scan chain outputs per clock domain (scan_mpllb_div_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_dword_clk	I	Scan clock per clock domain (scan_mpllb_dword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_dword_in	I	Scan chain inputs per clock domain (scan_mpllb_dword_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_dword_out	O	Scan chain outputs per clock domain (scan_mpllb_dword_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_fb_clk	I	Scan clock per clock domain (scan_mpllb_fb_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_fb_in	I	Scan chain inputs per clock domain (scan_mpllb_fb_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_fb_out	O	Scan chain outputs per clock domain (scan_mpllb_fb_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_oword_clk	I	Scan clock per clock domain (scan_mpllb_oword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mpllb_qword_clk	I	Scan clock per clock domain (scan_mpllb_qword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_ref_clk	I	Scan clock per clock domain (scan_mpllb_ref_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_mpllb_ref_in	I	Scan chain inputs per clock domain (scan_mpllb_ref_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_ref_out	O	Scan chain outputs per clock domain (scan_mpllb_ref_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_ssc_clk	I	Scan clock per clock domain (scan_mpllb_ssc_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_ssc_in[2:0]	I	Scan chain inputs per clock domain (scan_mpllb_ssc_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_ssc_out[2:0]	O	Scan chain outputs per clock domain (scan_mpllb_ssc_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_mpllb_word_clk	I	Scan clock per clock domain (scan_mpllb_word_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_mpllb_word_in	I	Scan chain inputs per clock domain (scan_mpllb_word_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_mpllb_word_out	O	Scan chain outputs per clock domain (scan_mpllb_word_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_phy_ref_dig_clk	I	Scan clock per clock domain (scan_phy_ref_dig_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_phy_ref_dig_in	I	Scan chain inputs per clock domain (scan_phy_ref_dig_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_phy_ref_dig_out	O	Scan chain outputs per clock domain (scan_phy_ref_dig_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_ref_clk	I	Scan clock per clock domain (scan_ref_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_ref_dig_clk	I	Scan clock per clock domain (scan_ref_dig_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_ref_dig_in[10:0]	I	Scan chain inputs per clock domain (scan_ref_dig_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_ref_dig_out[10:0]	O	Scan chain outputs per clock domain (scan_ref_dig_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_ref_in	I	Scan chain inputs per clock domain (scan_ref_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_ref_out	O	Scan chain outputs per clock domain (scan_ref_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_ref_range_clk	I	Scan clock per clock domain (scan_ref_range_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_ref_range_in[22:0]	I	Scan chain inputs per clock domain (scan_ref_range_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_ref_range_out[22:0]	O	Scan chain outputs per clock domain (scan_ref_range_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_adpt_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_adpt_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_adpt_in[5:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_adpt_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_adpt_out[5:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_adpt_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_asic_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_asic_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_asic_in (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_asic_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_asic_out (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_asic_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_div16p5_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_div16p5_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_div16p5_in (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_div16p5_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_div16p5_out (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_div16p5_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_dpll_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_dpll_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_dpll_in[1:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_dpll_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_dpll_out[1:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_dpll_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_dword_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_dword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_dword_in[9:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_dword_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_dword_out[9:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_dword_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_scope_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_scope_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_scope_in (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_scope_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_scope_out (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_scope_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_stat_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_stat_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_stat_in[8:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_stat_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_stat_out[8:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_stat_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_rxX_word_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_rxX_word_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_rxX_word_in (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_rxX_word_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_rxX_word_out (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_rxX_word_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_set_rst	I	Scan set reset **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk
scan_shift	I	Scan shift **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk
scan_shift_cg	I	Scan shift for clock gates **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_txX_ana_dword_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_txX_ana_dword_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
scan_txX_ana_dword_in[1:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_txX_ana_dword_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_txX_ana_dword_out[1:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_txX_ana_dword_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_txX_ana_word_clk (for X = 0; X <= 3)	I	Scan clock per clock domain (scan_txX_ana_word_clk) Test Clock muxed into the clock pin of registers during Scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

Table 3-19 Scan Interface Signals (Continued)

Port Name	I/O	Description
scan_txX_ana_word_in (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_txX_ana_word_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_txX_ana_word_out (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_txX_ana_word_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.
scan_txX_in[2:0] (for X = 0; X <= 3)	I	Scan chain inputs per clock domain (scan_txX_in) **Note:** The scan shift inputs and outputs reflect the scan chains for only the hard macro (PMA). For the PCS portion of the PHY, it is assumed that you insert scan and add the appropriate scan chains. The following scan clocks do not have associated scan chains in the PHY, but get muxed into functional clocks outputs: ■ scan_mplla_div33_clk ■ scan_mplla_div66_clk ■ scan_mplla_oword_clk ■ scan_mplla_oword_clk **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_in[*] is synchronous to txX_clk.
scan_txX_out[2:0] (for X = 0; X <= 3)	O	Scan chain outputs per clock domain (scan_txX_out) **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** scan_CLKDOMAIN_clk **Note:** scan_txX_out[*] is synchronous to txX_clk.

3.20 SRAM Control Signals

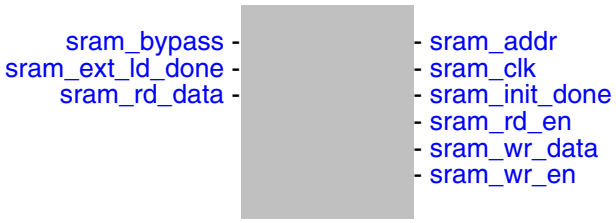


Table 3-20 SRAM Control Signals

Port Name	I/O	Description
sram_addr[13:0]	O	SRAM address bus Address port for the SRAM Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** sram_clk
sram_bypass	I	SRAM bypass Control signal when asserted, bypasses the SRAM interface. In this case, the adaptation and calibration algorithms are executed from the hard wired values within the Raw PCS. If SRAM is not bypassed, the internal algorithms are first loaded by Raw PCS into the SRAM at which point user can change the contents of the SRAM. The updated SRAM contents are used for the adaptation and calibration routines. This signal is meant to be used only for debugging purposes and must not change after phy_reset is negated. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
sram_clk	O	SRAM Clock Clock used to drive SRAM interface Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
sram_ext_ld_done	I	SRAM external load done Signal asserted by user after any updates to the SRAM have been loaded **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-20 SRAM Control Signals (Continued)

Port Name	I/O	Description
sram_init_done	O	SRAM Initialization done Signal indicating that the SRAM has been initialized by the boot loader in the Raw PCS **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
sram_rd_data[15:0]	I	SRAM read bus (sram_rd_data) Read data port for the SRAM Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** sram_clk
sram_rd_en	O	SRAM read enable Read enable control to read data from SRAM Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** sram_clk
sram_wr_data[15:0]	O	SRAM read bus (sram_wr_data) Write data port for the SRAM Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** sram_clk
sram_wr_en	O	SRAM write enable Write enable control to write data to SRAM Selected if macro *_SRAM_SUPPORT is defined **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** sram_clk

3.21 Transmitter Request Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)



Table 3-21 Transmitter Request Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
txX_ack	O	Transmitter acknowledge The assertion of this signal indicates the requested transmitter setting (txX_req) is complete or the requested RX detection operation is complete. This signal stays asserted until the request input (txX_req or txX_detrx_req) is de-asserted, after which txX_ack is de-asserted. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_detrx_req	I	Transmitter RX-Detection request This signal indicates an RX-detection request. When txX_ack is asserted then the detection result on txX_detrx_result is valid. **Notes:** ■ This signal should stay high until the txX_ack output is asserted, then txX_detrx_req should be de-asserted. ■ RX detection is only valid when the TX is in the P1 power state. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
txX_detrx_result	O	Transmitter RX-Detection result This signal is valid when txX_ack is asserted following a txX_detrx_req request. ■ 1'b0: Receiver not detected ■ 1'b1: Receiver detected **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-21 Transmitter Request Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_lpd	I	Transmitter lane power down Powers down the transmitter into a power state equivalent to P1. This signal is useful for selectively powering down lanes in a link while maintaining the power state of the link as a whole (for example, link in P0). **Note:** If the txX_pstate[1:0] input is set to P2, it takes priority over txX_lpd and the transmitter is placed in P2 state. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_master_mplla_state	I	Master MPLL power state per lane (txX_master_mplla_state) Indicates the state of MPLL(A,B) from the master PHY. Use when you want to aggregate more than one PHY into a single link. For single PHY usage, this input should be connected to the mpll(a,b)_state output of the PHY itself. For details about aggregating the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_master_mpllb_state	I	Master MPLL power state per lane (txX_master_mpllb_state) Indicates the state of MPLL(A,B) from the master PHY. Use when you want to aggregate more than one PHY into a single link. For single PHY usage, this input should be connected to the mpll(a,b)_state output of the PHY itself. For details about aggregating the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_mpll_en	I	TX MPLL enable Powers up or powers down the MPLLA or MPLLB (determined using txX_mpllb_sel). For normal operation, this input must be set to 1 when txX_pstate[1:0] is in P0 or P0s state. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
txX_mpllb_sel	I	TX MPLLB select Specifies the source of the TX analog clocks; if asserted, MPLLB is the source; otherwise, MPLLA is the source. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

Table 3-21 Transmitter Request Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_pstate[1:0]	I	Transmitter power state Sets the power state of the transmitter. For example, the settings for PCIe are: ■ 2'b00: P0-Transmitter is fully powered up. ■ 2'b01: P0S-Transmitter common mode is held, TX analog clocks are active, but TX serializer is off. ■ 2'b10: P1-Transmitter common mode is held, but TX analog clocks and TX serializer are off. ■ 2'b11: P2-Transmitter is powered down. For settings for other protocols, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
txX_rate[2:0]	I	TX data rate Select the baud rate for the transmit path. ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b111: baud/10 ■ 3'b100 - 3'b110: Not Supported **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-21 Transmitter Request Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_req	I	Transmitter request This signal indicates a new transmitter setting request. This signal should be asserted only when txX_ack is low and should stay high until the txX_ack output is asserted. After txX_ack is asserted, txX_req should be de-asserted. **Notes:** The following input signals are captured by the PHY when txX_req is asserted: ■ txX_pstate ■ txX_lpd ■ txX_rate ■ txX_width ■ txX_mpllb_sel ■ txX_mpll_en ■ txX_misc These signals should be stable before txX_req is asserted, and remain stable until txX_req is de-asserted. Any combination of changes in these signals is valid for a single request, and the completion of all changes is indicated by the assertion of txX_ack. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
txX_width[1:0]	I	TX data width Select the width of the input transmit data. ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

3.22 Transmitter Control Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

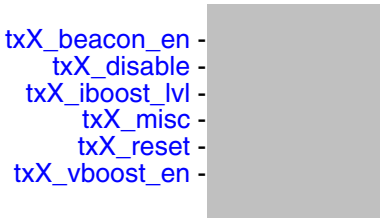


Table 3-22 Transmitter Control Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
txX_beacon_en	I	Transmitter beaconing enable This signal enables or disables transmitter beaconing. The period for transmit pulses is between 20-50 ns. When de- asserted, the Beacon oscillator completes its current cycle before being disabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_disable	I	Transmitter disable Puts the transmitter in a low-power mode where all the transmitter blocks, including TX common mode, are turned off. This signal should be asserted when in PCIe L1 Substate L1.2. The transmitter should be in the P1 power state when using this signal. This input asynchronously disables TX terminations and does not require any clock to be supplied to the PHY. **Note:** The txX_disable signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
txX_iboost_lvl[3:0]	I	Transmitter current boost level Controls the per lane TX swing level when TX amplitude boost mode is enabled using txX_vboost_en. The TX swing on all lanes is also affected by the tx_vboost_lvl[2:0] control. For more information about how amplitude is affected by these controls, see the "Functional Description" chapter. For per protocol settings for this control input, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous

Table 3-22 Transmitter Control Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_misc[7:0]	I	TX Miscellaneous Controls This input controls miscellaneous settings in the TX analog block. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** Asynchronous
txX_reset	I	Transmitter reset This signal internally resets the TX datapath, all the transmitter settings and state machines, including common-mode adjustment and receiver detection state machines. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous (must not glitch)
txX_vboost_en	I	Transmitter voltage boost enable When asserted, current mode TX swing boost is enabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

3.23 Transmitter Datapath Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

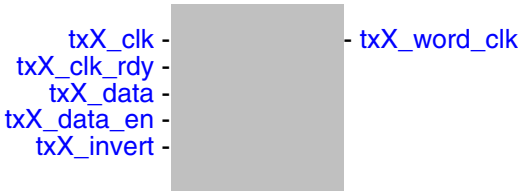


Table 3-23 Transmitter Datapath Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
txX_clk	I	Transmitter input clock Input data must be synchronous with this clock. If the lane is part of a multi-lane link and the link remains active (for example, in P0) then this clock should remain active (even if the lane is powered down or in reset). **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A
txX_clk_rdy	I	Transmitter input clock ready This signal is used for synchronizing TX lanes in a multi- lane link. When the txX_clk is active and stable, txX_clk_rdy should be asserted at the same time for all TX lanes and remain asserted as long as the link is active (for example, P0) even if some of the lanes in the link are powered down or in reset. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** txX_clk
txX_data[19:0]	I	Transmitter input parallel data MSB is transmitted first. The valid range of the data is determined by the txX_width[1:0] input as follows: ■ For txX_width=2'b00, Valid Data Range=txX_data[7:0] ■ For txX_width=2'b01, Valid Data Range=txX_data[9:0] ■ For txX_width=2'b10, Valid Data Range=txX_data[15:0] ■ For txX_width=2'b11, Valid Data Range=txX_data[19:0] **Note:** The invalid bits of txX_data are ignored by the PHY. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** txX_clk

Table 3-23 Transmitter Datapath Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_data_en	I	Transmitter data enable When asserted and the TX lane is in P0, then the TX output driver is enabled. If de-asserted or the TX lane is not in P0, then the TX output driver is disabled. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** txX_clk
txX_invert	I	Invert logical sense of txX_p and txX_m outputs This signal has the effect of inverting the output data stream. **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** txX_clk
txX_word_clk	O	Transmitter word clock output The TX word clock from the TX lane. This output is meant for debugging purposes only. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** N/A

3.24 Transmitter Equalization Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)



Table 3-24 Transmitter Equalization Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3)

Port Name	I/O	Description
txX_eq_main[5:0]	I	Transmitter amplitude adjustment control Control for setting the transmitter driver output amplitude (main coefficient) txX_eq_main[5:0]: Integer value (0 to 40)**Notes:** ■ If the fractional post is non-zero (in other words, txX_eq_post[1:0] != 2'b00) or the fractional pre is non-zero (in other words, txX_eq_pre[1:0] != 2'b00), then the txX_eq_main[5:0] input should be increased by 1. This is to complete the fractional portion given a fixed number of transmitter legs. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main[5:0] values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** txX_clk
txX_eq_post[5:0]	I	Transmitter Post-Emphasis level adjustment control Control for setting the transmitter driver output post- emphasis (post coefficient) ■ txX_eq_post[5:2]: Integer value (0 to 15) ■ txX_eq_post[1:0]: Fraction value (0, 0.25, 0.5, 0.75) **Notes:** ■ The maximum total post value is 15, therefore txX_eq_post[5:2]=15 is only allowed if the fractional part (txX_eq_post[1:0]) is zero. Otherwise txX_eq_post[5:2] should be less than or equal to 14. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main[5:0] values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** txX_clk

Table 3-24 Transmitter Equalization Signals (Per-Lane Transmit Interface Group) (for X = 0; X <= 3) (Continued)

Port Name	I/O	Description
txX_eq_pre[5:0]	I	Transmitter Pre-Emphasis level adjustment control Control for setting the transmitter driver output pre- emphasis (preshoot coefficient) ■ txX_eq_pre[5:2]: Integer value (0 to10) ■ txX_eq_pre[1:0]: Fraction value (0, 0.25, 0.5, 0.75) **Notes:** ■ The maximum total pre value is 10, therefore txX_eq_pre[5:2]=10 is only allowed if the fractional part (txX_eq_pre[1:0]) is zero; otherwise, txX_eq_pre[5:2] should be less than or equal to 9. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main[5:0] values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vpdig **Active State:** N/A **Synchronous To:** txX_clk

3.25 Transmitter Test Signals (for X = 0; X <= 3)

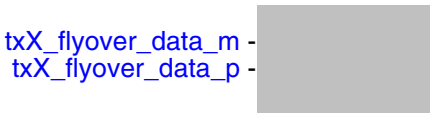


Table 3-25 Transmitter Test Signals (for X = 0; X <= 3)

Port Name	I/O	Description
txX_flyover_data_m	I	Transmit Flyover Data inputs (txX_flyover_data_m) Inputs are directly connected to TX bump pads when test_flyover_en is enabled **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous
txX_flyover_data_p	I	Transmit Flyover Data inputs (txX_flyover_data_p) Inputs are directly connected to TX bump pads when test_flyover_en is enabled **Voltage Range:** 0-vpdig **Active State:** High **Synchronous To:** Asynchronous

4 Functional Description

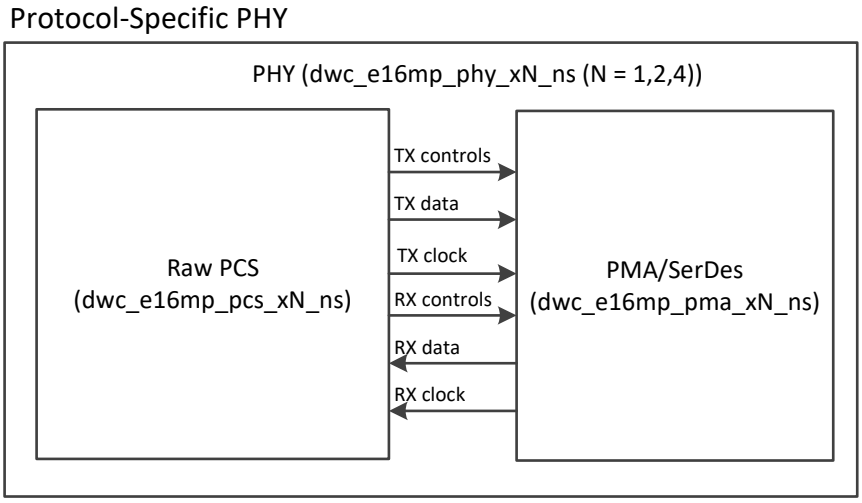
This chapter includes the following sections:

- "PHY" on page 136
- "PMA" on page 137
- "Raw PCS" on page 153

4.1 PHY

The DesignWare Cores Multi-Protocol 16G PHY consists of the hard macro (PMA) and the soft RTL macro (raw PCS). The major blocks are shown in Figure 4-1.

Figure 4-1 DesignWare Cores Multi-Protocol 16G PHY



The following sections provide the functional description of the PMA and the Raw PCS.

4.2 PMA

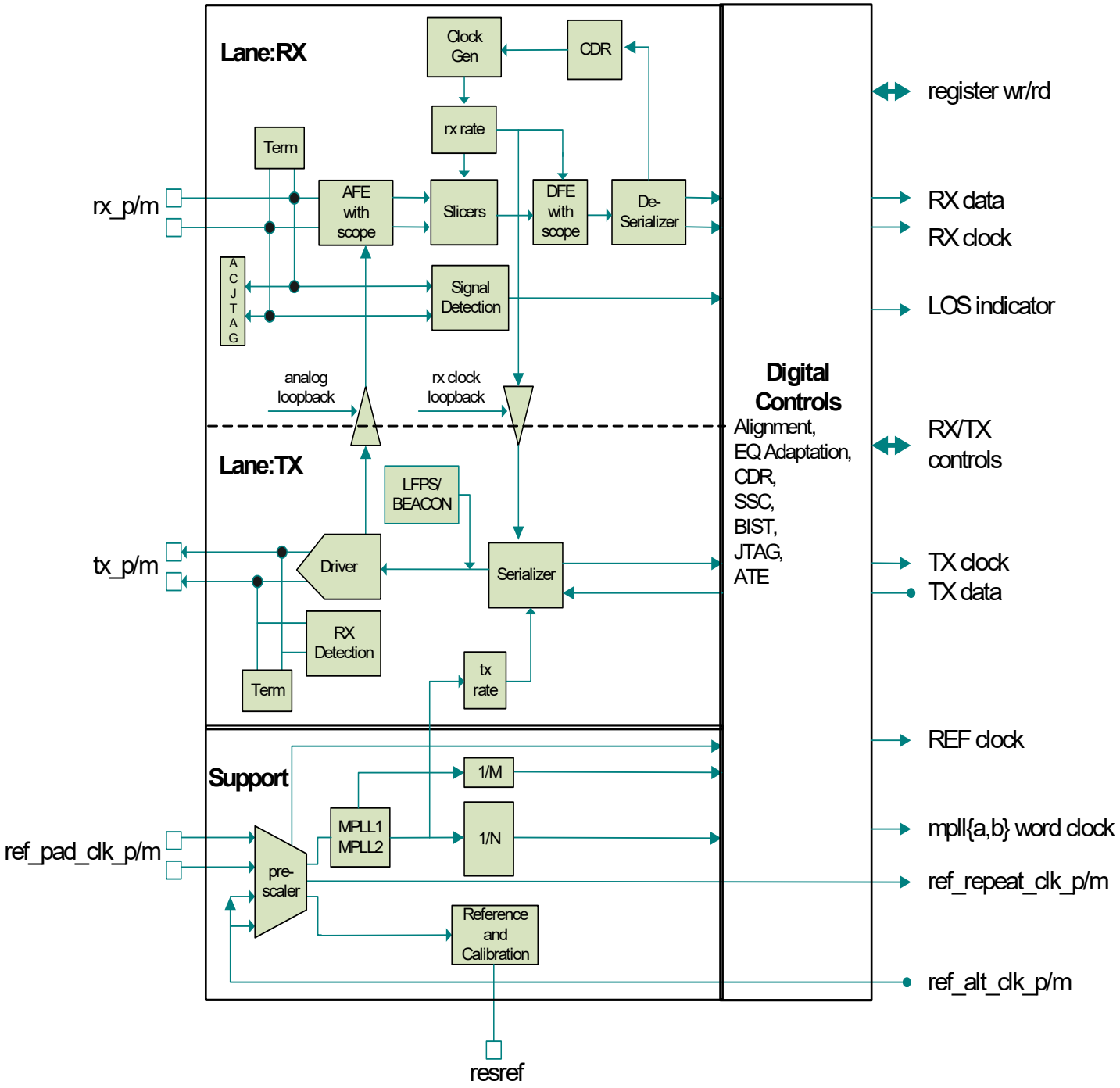
Each PMA comprises a full duplex transceiver for each lane. As shown in Figure 4-2 on page 138, the PMA comprises the following main blocks:

- Lane
  - Receiver (RX)
  - Transmitter (TX)
- Support
  - Reference clock
  - MPLLs
  - Termination Resistance Tuning
- Test Features

On the transmit (TX) side, the PMA receives parallel data from the relevant lane of the PCS. The data is serialized and sent to the TX driver, which is a high-speed, hybrid-mode (voltage and current), differential-output driver. TX driver characteristics, such as amplitude, pre-emphasis, and termination, are user-controllable. The TX driver output is sent to a pair of external pads.

On the receive (RX) side, the PMA receives differential serial data from a pair of external pads. Receiver characteristics, such as equalization and termination, are user-controllable. A continuous-time linear equalizer (CTLE) is first used to compensate for channel loss. A decision feedback equalizer (DFE) is also implemented after CTLE to optimize the data eye further for high-speed data that exceeds ~6.25 Gbps. The RX output signal is passed to the clock and data recovery (CDR) circuit, which recovers the embedded clock in the data. A dedicated VCO is designed as the source of the recovering clock. The CDR loop adjusts the VCO clock frequency and phase until the frequency matches incoming data and the phase is aligned with incoming data. This recovered clock is used to retime received data and send it to the deserializer, which produces parallel data and a parallel data clock for the relevant PCS lane.

Figure 4-2 PMA Architecture (Single Lane)



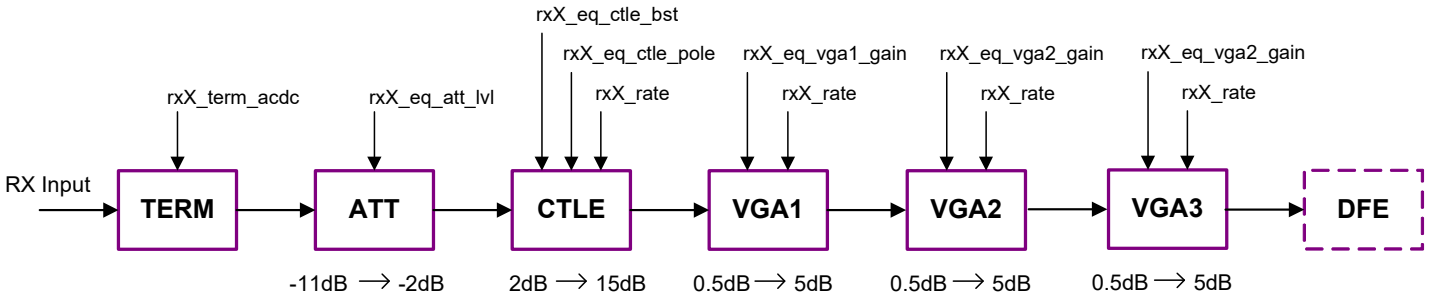
4.2.1 Lane: Receiver (RX)

This section describes the RX functionality.

4.2.1.1 RX Analog Front End

The analog front end (AFE) circuitry receives off-chip data and boosts high-frequency components to compensate part of the channel loss with programmable or adaptive CTLE.

Figure 4-3 RX AFE (Analog Front End) Diagram

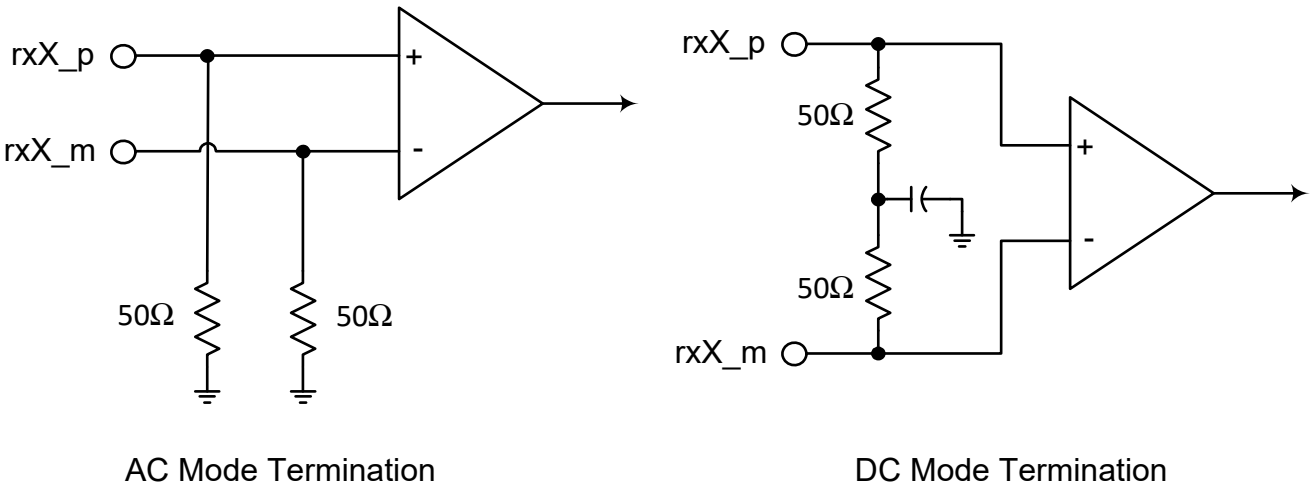


As shown in Figure 4-3, the AFE consists of the following blocks:

- Termination (TERM)

  The TERM block is tuned by the PHY using an off-chip reference resistor.

Figure 4-4 RX Termination



- Attenuator (ATT)

  The ATT block is responsible for attenuating the incoming signal to acceptable levels for the subsequent CTLE block to maintain linearity in the case of a short-reach link. Attenuation is controlled by rxX_eq_att_lvl.

- Continuous time-linear equalizer (CTLE)

  The CTLE block provides high-frequency boost range from 2dB to 15dB. The CTLE block has programmable boost level, boost pole frequency, and boost bandwidth (BW) control (provided by rxX_eq_ctle_boost, rxX_eq_ctle_pole, and rxX_rate, respectively).

- Voltage gain amplifiers (VGA1 and VGA2)

  The VGAs are implemented to maintain a constant signal swing. Analog gain control provided by rxX_eq_vga1_gain and rxX_eq_vga2_gain, and BW control provided by rxX_rate. There are two VGAs in series to improve the gain control range.

To achieve optimal link performance, the receiver equalization parameters are automatically tuned to their optimal values for protocols that are adapted. For more details about RX AFE equalization adaptation, see "RX Equalization and Adaptation" on page 142.

4.2.1.2 RX DFE (Decision Feedback Equalizer)

The post-VGA output is summed by a 5-tap, continuously adaptive Decision Feedback Equalizer (DFE). The output of the DFE is sampled by offset-compensated samplers, deserialized, and sent to the core along with the recovered clock.

The 5-tap DFE is a half-rate architecture with first tap unrolling. For more details about RX equalization adaptation, see "RX Equalization and Adaptation" on page 142.

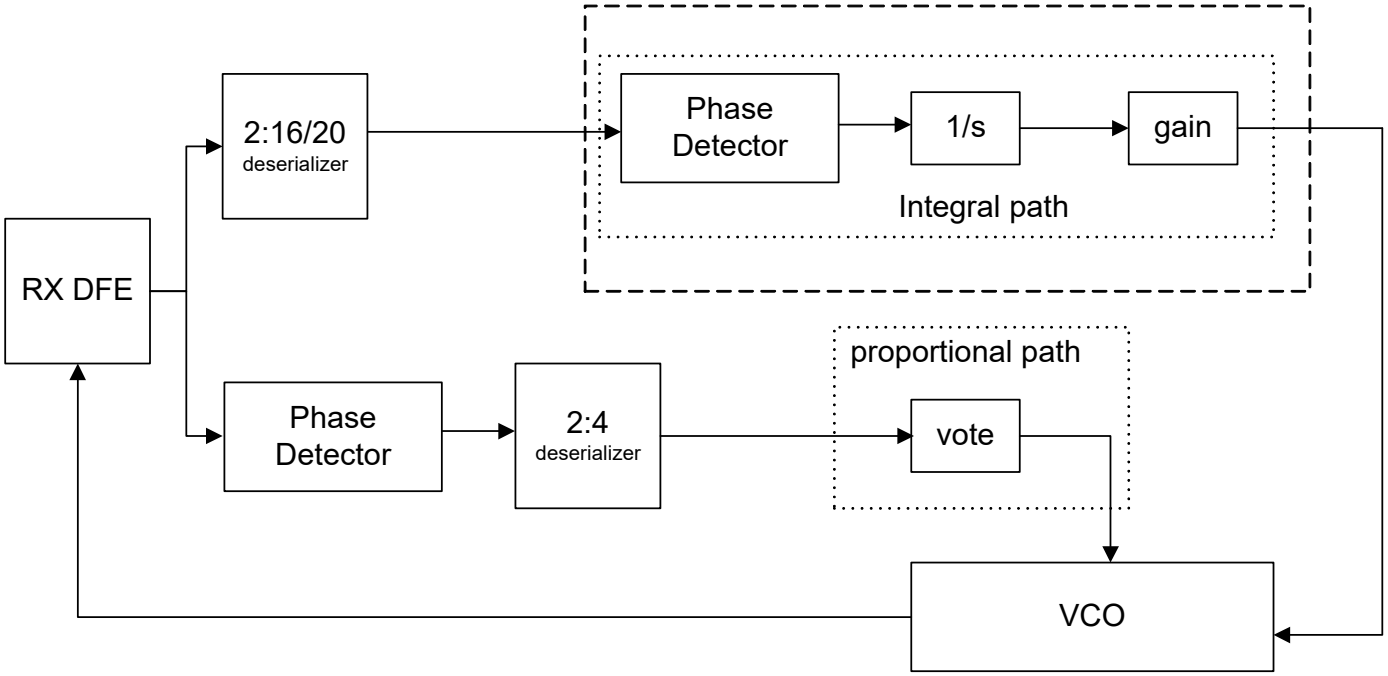
The DFE can be disabled (and DFE_BYPASS enabled) through top-level signal or register control:

- rxX_dfe_bypass
- LANEN_DIG_ANA_RX_CTL_OVRD_OUT.RX_ANA_DFE_BYPASS_EN register bit can be de-asserted to disable DFE and enable DFE_BYPASS
- The DFE and DFE_BYPASS can both be disabled through register LANEN_DIG_ANA_RX_CTL_OVRD_OUT.RX_ANA_DFE_TAPS_EN, which is used to reset tap2 to tap5, therefore virtually disabling the taps and the DFEs

4.2.1.3 RX Clock and Data Recovery

The clock and data recovery (CDR) circuit is implemented as a second-order digital loop. The loop includes a proportional path and an integral path. To reduce proportional path latency, the proportional path is implemented separately from the integral path, with much higher speed, as shown in Figure 4-5.

Figure 4-5 RX Clock and Data Recover (CDR)



The proportional path and integral path outputs control a dedicated CDR voltage-controlled oscillator (VCO), which generates an RX clock aligned with incoming data. To support spread-spectrum clocking (SSC) and to reduce CDR lock time, proportional path gain and integral path gain are dynamically set during the CDR locking process.

4.2.1.4 RX Deserialization

The deserializer converts serial data from RX and sends the data to the PCS as parallel data. The deserializer qualifies serial input data from the CDR using the high-speed serial recovered clock, and deserializes the data using the low-speed parallel recovered clock. The deserializer forwards the deserialized data to the receiver's PCS channel. The deserializer supports a 16-, or 20-bit interface. The deserializer also optionally supports an 8- or 10-bit interface for lower data-rates as specified in Table 1-1 on page 18.

4.2.1.5 RX Internal Eye Monitor

Provided with the PHY is an internal eye monitor, which can provide data to produce a plot of time and voltage axis using internal DFT hooks built into the AFE. The eye monitor is non-destructive and can be run in parallel with live traffic. This procedure can be used to characterize the input RX eye margin as seen after the ATT or at the even or odd input to the DFE.

4.2.1.6 RX Rate Control

The RX high-speed clock is generated by the CDR loop that controls and aligns the high-speed clock frequency and phase with data received by the RX. The CDR-controlled VCO provides a clock rate that can range from 4 GHz to 8 GHz. In addition, the RX has a flexible data rate control (configured by rxX_rate[1:0]) that allows the RX to reliably receive and process incoming data from 1.2 Gbps to a maximum of 16.0 Gbps.

For example, by setting rxX_rate[1:0] to 2'b10, the PHY operates at 2.5 Gbps (the RX clock rate is divided down from 5 GHz to 1.25 GHz). For detailed PMA configuration settings for all supported protocols, refer to Chapter 5, "PHY Usage and Configuration".

4.2.1.7 RX Calibration

The PHY has the ability to calibrate the RX AFE and DFE so as to minimize offset. This is run on start up out of reset and may also be run continuously during mission mode.

4.2.1.7.1 Startup Calibration

The PHY runs startup calibration after the RX is powered up and transitions from reset into the P0 power state. It calibrates the offsets of each stage of the AFE (ATT, CTLE, and VGA) in sequence, and then calibrates the offsets of the DFE data, phase, and error slicers.

4.2.1.7.2 Continuous Calibration

The PHY runs continuous calibration in mission-mode. The continuous calibration routine adjusts the offsets of each stage of the AFE (CTLE, VGA, and summer) in sequence. Continuous calibration runs in tandem with continuous adaptation when the latter is supported.

4.2.1.8 RX Equalization and Adaptation

For protocols that require link adaptation (such as, PCIe 3.1 and 10GBASE-KR), the PHY supports adaptation of the receiver equalization parameters, and requests for incrementing or decrementing the far-end transmitter coefficients. The PHY adapts the AFE equalization and gain, the DFE tap coefficients, and the remote TX equalization. It operates in two steps: startup adaptation and continuous adaptation.

4.2.1.8.1 Startup Adaptation

The PHY runs startup adaptation after startup calibration is complete, when the RX is receiving valid data and the CDR is locked. It first adapts the AFE, where the ATT level and VGA gain are adapted based on measurements of the average eye opening and the CTLE boost is adapted based on data bit correlation. It then adapts the DFE, where the tap coefficients are adapted based on the same correlation algorithm. It calculates a figure-of-merit based on collected statistics. Finally, it evaluates the TX pre-cursor and post-cursor and makes a decision on whether to adjust remote TX equalization.

4.2.1.8.2 Continuous Adaptation

The PHY runs continuous adaptation in mission-mode for protocols that support adaptation. The continuous adaptation routine adjusts the coefficients of DFE taps 1 to 5 and the VGA gain using an algorithm similar to the one used during startup adaptation. Continuous adaptation is run in tandem with continuous calibration.

4.2.2 Lane: Transmitter (TX)

This section describes the TX functionality.

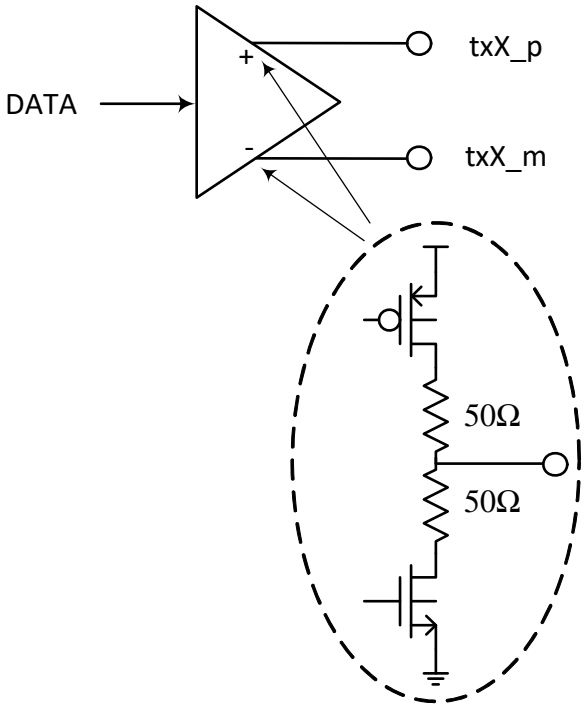
4.2.2.1 TX Serializer

Parallel data from the PCS is latched using a clock that is synchronous to the lane's PCS TX clock, and then serialized by the Parallel-to-Serial (P2S) converter. The serializer supports a 16- or 20-bit interface, or a 8- or 10-bit interface for lower data-rates as defined in Table 1-1 on page 18.

4.2.2.2 TX Termination

The TX is a voltage mode driver with 50Ω termination on both txX_p and txX_m pullup and pulldown, as shown in Figure 4-6. The termination is tuned by the PHY using an off-chip reference resistor.

Figure 4-6 TX Termination



4.2.2.3 TX Amplitude Control

The peak-to-peak output swing of the TX, Vswing, is nominally equal to the power supply, vptxX. However, the TX supports the boost function, which is enabled by the input signal of txX_vboost_en.

When the boost function is enabled and 4 * tx_vboost_vref > vptxX, then the swing is further increased to the amplitude given in the following equation:

Vswing (Vppd) = vptxX + 1/16 * (1 + txX_iboost_lvl[3:0]) * (4 * tx_vboost_vref - vptxX)

Where:

- tx_vboost_vref is an internal voltage that determines the maximum possible swing and is determined by tx_vboost_lvl[2:0] according to the following function:

  tx_vboost_vref = vbg * (tx_vboost_lvl[2:0] + 6) / 32

  where vbg = 0.8 V

  The default of tx_vboost_lvl[2:0] = 3'b101, which results in 274 mV

- txX_iboost_lvl[3:0] provides fine control of the boosted amplitude

Therefore, when txX_iboost_lvl[3:0] = '1111', Vswing is maximized and equal to max{vptxX, 4 * tx_vboost_vref}, as shown in Table 4-1.

Table 4-1 lists typical maximum TX swing values under the listed conditions.

Table 4-1 Typical Maximum TX Swing (Vswing) When txX_iboost_lvl[3:0]='1111'

Value of tx_vboost_lvl[2:0]	Amplitude (mVppd)
000	vptxX
001	vptxX
010	The higher value of {vptxX, vbg}
011	900
100	1000
101	1100
110	not recommended
111	not recommended

Table 4-2 lists the typical TX swing with the tx_vboost_vref default setting of “101” and vptxX = 0.9 V.

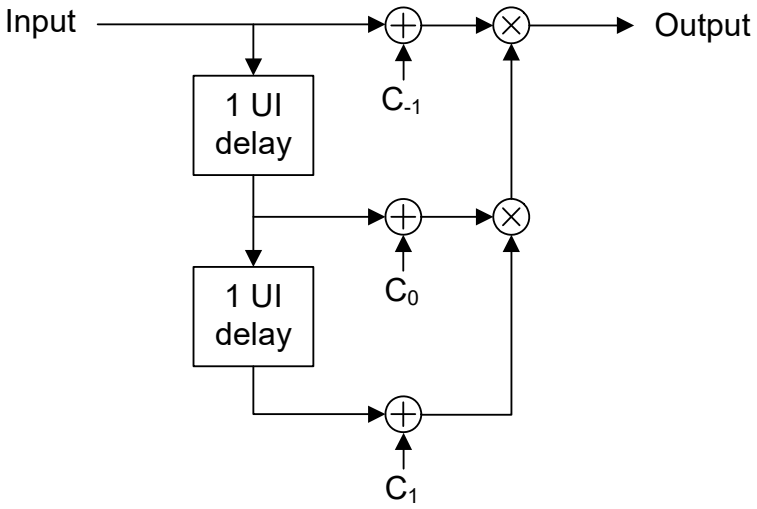
Table 4-2 Typical TX Launch Amplitude

Value of txX_iboost_lvl[3:0]	Amplitude (mVppd)
4'b0000	912.5
4'b0001	925
4'b0010	937.5
4'b0011	950
4'b0100	962.5
4'b0101	975
4'b0110	987.5
4'b0111	1000
4'b1000	1012.5
4'b1001	1025
4'b1010	1037.5
4'b1011	1050
4'b1100	1062.5
4'b1101	1075
4'b1110	1087.5
4'b1111	1100

4.2.2.4 TX Equalization

The TX driver is implemented with a 3-tap feed-forward equalization (FFE), as shown in Figure 4-7 on page 146.

Figure 4-7 TX Driver Architecture



Where:

- C_0, C_{-1}, and C_1 are TX driver main tap, pre-cursor tap, and post-cursor tap coefficients
- C_{-1} and C_1 are defined as negative values

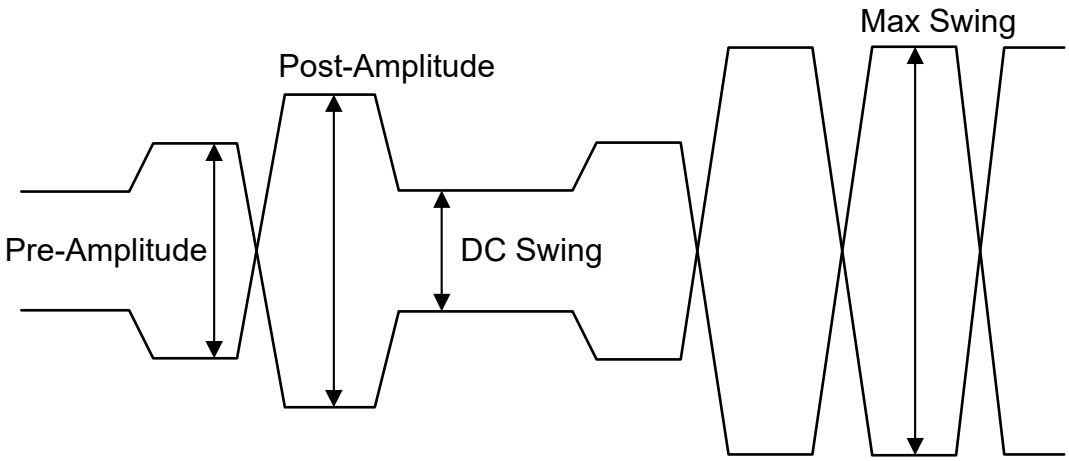
The FFE coefficients C_0, C_{-1}, C_1 are set by three control signals (txX_eq_main[5:0], txX_eq_pre[5:0], and txX_eq_post[5:0]) by the following equations:

- C_0 = txX_eq_main[5:0]
- $|C_{-1}|$ = txX_eq_pre[5:0]/4
- $|C_1|$ = txX_eq_post[5:0]/4

Note The equations above and below use the absolute values of C_{-1} and C_1.

TX Equalization with pre-amplitude, post-amplitude, and DC swing is illustrated in Figure 4-8.

Figure 4-8 TX Equalization



To calculate the differential pre-amplitude, post-amplitude, and DC swing, use the following equations:

- pre-amplitude = Vswing * (C_0 + |C_{-1}| - | C_1|)/40
- post_amplitude = Vswing * (C_0 - |C_{-1}| + |C_1|)/40
- DC_swing = Vswing * (C_0 - |C_{-1}| - |C_1|)/40
- Max_Swing = Vswing * (C_0 + |C_{-1}| + |C_1|)/40

To maximize the output swing, set it so that C_0 + |C_{-1}| + |C_{+1}| = 40, which results in:

Max_Swing = Vswing

To attenuate output swing, set C_0 + |C_{-1}| + |C_1| to be less than 40. For example, to get half swing output, set C_0 + |C_{-1}| + |C_1| = 20.

4.2.2.5 TX Rate control

The TX high-speed clock is generated by the common clock module in the Support block. As listed in "MPLL" on page 148, the Support block provides a clock rate that can range from 4.88 GHz to 8 GHz. In addition, the TX has a flexible data rate control (configured by controlling txX_rate) that allows the TX to operate from 1.2 Gbps to a maximum of 16.0 Gbps.

The TX high-speed clock is always divided down to be half of the baud rate frequency. For example, if the clock provided by the Support block is 5 GHz, by setting txX_rate to 3'b010, the clock is divided down by a factor of 4 to 1.25 GHz and the TX operates at 2.5 Gbps. For detailed PMA configuration settings for all supported protocols, refer to Chapter 5, "PHY Usage and Configuration".

4.2.3 Support Block

The Support block (see Figure 4-2 on page 138) provides all the common functions for the RX/TX link, TX clock generation, TX/RX termination calibration, biasing voltage, and reference clock buffering.

4.2.3.1 Reference Clock Input

The PHY requires a reference clock from either:

- An external pad input: ref_pad_clk_p or ref_pad_clk_m
- A core-side, on-chip input: ref_alt_clk_p or ref_alt_clk_m

The external or core-side input is selected by ref_use_pad. When ref_use_pad is asserted, the external reference clock is used; otherwise, the on-chip reference source is used.

For implementation details, see Chapter 9, "Reference Clock Implementation".

4.2.3.2 Reference Clock Output

The Support block includes a reference clock repeat function. A current-mode logic (CML) differential reference clock is driven on ref_repeat_clk_{p,m} when ref_repeat_clk_en is asserted. This clock can be used as the reference clock source for other PHYs. For details, refer to "Repeat Clock Specification" on page 268.

4.2.3.3 MPLL

The MPLL generates the high-frequency clocks used by the lane TX. There are two MPLLs, MPLL A and MPLL B, which can be either operated individually or combined to support lane bifurcation. The txX_mpllb_sel signal allows you to select which MPLL to use.

The following are the key features of the MPLLs:

- Total output frequency ranges from 4.8 GHz to 8 GHz
  - With MPLL A enabled, the frequency can range from 4.8 GHz to 6.4 GHz
  - With MPLL B enabled, the frequency can range from 7.47 GHz to 8.53 GHz
- BW requirement compliance for PCIe 1.1, PCIe 2.1, PCIe 3.1, and PCIe 4.0 with reference clock of 100 MHz
- Support for lane bifurcation (with two MPLLs enabled)
- Support for divide-by-fixed-16.5 in the output clock tree
- Support for data rates from 1.2 Gbps to 16.0 Gbps
- Support for Spread Spectrum generation with a range of -5000 ppm to 5000 ppm
- Support for fixed-frequency offset compensation up to 1600 ppm

4.2.3.3.1 Configuring the MPLL

You need to configure the MPLL so it runs at the correct frequency for your application and protocol. In particular, you need to set the feedback ratio (Feedback N).

To configure the MPLL, perform the following:

1. Enable the correct MPLL (MPLLA or MPLLB), according to the application and the frequency range. For more about enabling the correct MPLL, see "MPLL" on page 148.
2. Use the configuration settings in Chapter 5, "PHY Usage and Configuration" to determine the correct setting of mpll{a,b}_multiplier. The mapping of the mpll{a,b}_multiplier[6:0] input setting to the decimal multiplier ratio used by the MPLL is provided in Table 4-3. Assert mpll{a,b}_multiplier[7] for an additional divide-by-2 to enable division greater than 131. Note that, in addition to the programmable divider settings, there is already a fixed divide-by-2 in the MPLL feedback path.

Table 4-3 MPLL Feedback Divider Ratio (N) Configuration

mpll{a,b}_multiplier[6:0]	Feedback N Values (in Decimal Notation)			
7'h03 - 7'h00	131	130	129	128
7'h7F - 7'h7C	127	126	125	124
7'h7B - 7'h78	123	122	121	120
7'h77 - 7'h74	119	118	117	116
7'h73 - 7'h70	115	114	113	112
7'h6F - 7'h6C	111	110	109	108
7'h6B - 7'h68	107	106	105	104
7'h67 - 7'h64	103	102	101	100
7'h63 - 7'h60	99	98	97	96
7'h5F - 7'h5C	95	94	93	92
7'h5B - 7'h58	91	90	89	88
7'h57 - 7'h54	87	86	85	84
7'h53 - 7'h50	83	82	81	80
7'h4F - 7'h4C	79	78	77	76
7'h4B - 7'h48	75	74	73	72
7'h47 - 7'h44	71	70	69	68
7'h43 - 7'h40	67	66	65	64
7'h3F - 7'h3C	63	62	61	60
7'h3B - 7'h38	59	58	57	56
7'h37 - 7'h34	55	54	53	52

Table 4-3 MPLL Feedback Divider Ratio (N) Configuration (Continued)

mpll{a,b}_multiplier[6:0]	Feedback N Values (in Decimal Notation)			
7'h33 - 7'h30	51	50	49	48
7'h2F - 7'h2C	47	46	45	44
7'h2B - 7'h28	43	42	41	40
7'h27 - 7'h24	39	38	37	36
7'h23 - 7'h20	35	34	33	32
7'h1F - 7'h1C	31	30	29	28
7'h1B - 7'h18	27	26	25	24
7'h17 - 7'h14	23	22	21	20
7'h13 - 7'h10	19	18	17	16
7'h0F - 7'h0C	15	14	13	12
7'h0B - 7'h08	10	10	9	8
7'h07 - 7'h04	x	x	x	x

Note When mpll{a,b}_div_clk_en is enabled, the frequency of mpll{a,b}_div_clk clock is equal to the MPLL{A,B} frequency divided by mpll{a,b}_div_multiplier[6:0]. See Table 4-4 for mpll{a,b}_div_multiplier[6:0] values (in hexadecimal notation) and divider ratio values (in decimal notation). Divider ratio values between 0 to 7 (in decimal notation) are invalid and should not be used.

Table 4-4 MPLL Divide Output Clock configuration

mpll{a,b}_div_multiplier[6:0]	Divider Ratio (in Decimal Notation)			
7'h03 - 7'h00	131	130	129	128
7'h7F - 7'h7C	127	126	125	124
7'h7B - 7'h78	123	122	121	120
7'h77 - 7'h74	119	118	117	116
7'h73 - 7'h70	115	114	113	112
7'h6F - 7'h6C	111	110	109	108
7'h6B - 7'h68	107	106	105	104
7'h67 - 7'h64	103	102	101	100

Table 4-4 MPLL Divide Output Clock configuration (Continued)

mpll{a,b}_div_multiplier[6:0]	Divider Ratio (in Decimal Notation)			
7'h63 - 7'h60	99	98	97	96
7'h5F - 7'h5C	95	94	93	92
7'h5B - 7'h58	91	90	89	88
7'h57 - 7'h54	87	86	85	84
7'h53 - 7'h50	83	82	81	80
7'h4F - 7'h4C	79	78	77	76
7'h4B - 7'h48	75	74	73	72
7'h47 - 7'h44	71	70	69	68
7'h43 - 7'h40	67	66	65	64
7'h3F - 7'h3C	63	62	61	60
7'h3B - 7'h38	59	58	57	56
7'h37 - 7'h34	55	54	53	52
7'h33 - 7'h30	51	50	49	48
7'h2F - 7'h2C	47	46	45	44
7'h2B - 7'h28	43	42	41	40
7'h27 - 7'h24	39	38	37	36
7'h23 - 7'h20	35	34	33	32
7'h1F - 7'h1C	31	30	29	28
7'h1B - 7'h18	27	26	25	24
7'h17 - 7'h14	23	22	21	20
7'h13 - 7'h10	19	18	17	16
7'h0F - 7'h0C	15	14	13	12
7'h0B - 7'h08	10	10	9	8
7'h07 - 7'h04	x	5	x	4

Note: x refers to invalid divider setting and hence should not be used.

4.2.3.4 Termination Resistance Tuning

The PHY uses an external precision resistor to calibrate the termination impedances of the high-speed inputs and outputs of the PHY.

A termination calibration is performed any time the PHY is taken out of reset. In addition, you can request the PHY to perform a calibration by asserting rtune_req. When the calibration is completed, rtune_ack goes low.

The termination calibration is not continuous, so the PHY has the ability to share a single external resistor using a handshake protocol across several PHYs.

For implementation details, see Chapter 7, “Physical-Level Implementation”.

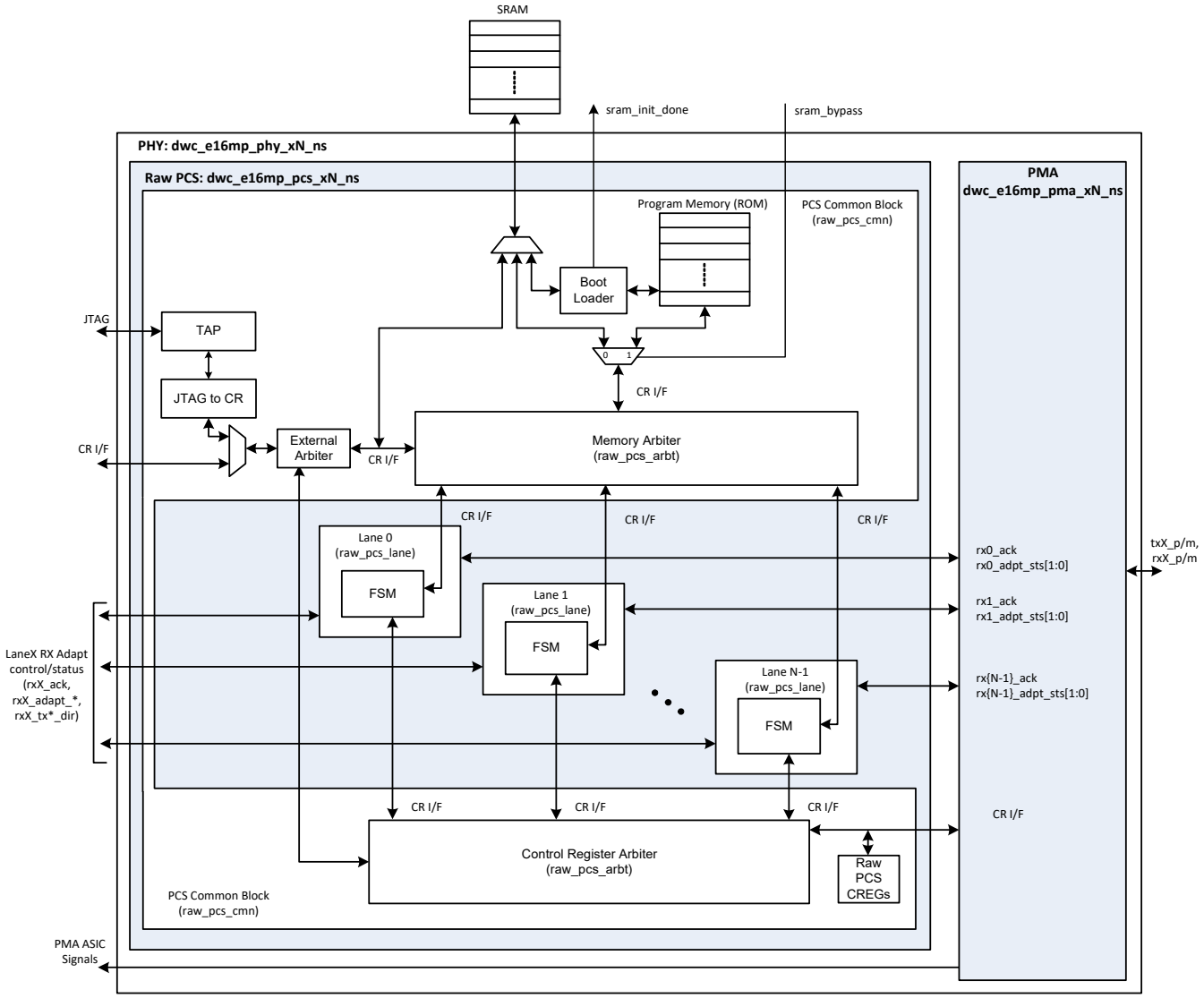
4.3 Raw PCS

The Raw PCS is a soft macro provided as Verilog RTL. The following features are implemented in the Raw PCS:

- Power-up calibration algorithms for the PMA analog front-end, for example, RX AFE offset cancellation, RX slicer calibration, MPLL calibrations, RX VCO calibrations
- Controls for the adaptation routines in the PMA for modes when adaptation is enabled
- MPLL power-up and power-down control based upon txX_pstate and txX_mpllb_sel inputs
- Programmable CDR unlock detector
- JTAG Tap controller for serial register access
- Register access arbitration between JTAG/CR Parallel access, and the internal power-up and adaptation routines. The selection between JTAG and CR Parallel interfaces is controlled using primary input, cr_para_sel (see section 5.14 on page 187 (“Control Register Interface”)), which means that both of these interfaces cannot operate simultaneously. A round-robin arbitration mechanism exists to provide equal priority to different sources requesting access to the shared internal register bus. These sources include multiple internal state machines as well as external register access interfaces (JTAG/CR-Parallel).

The block diagram of the Raw PCS is shown in Figure 4-9.

Figure 4-9 Raw PCS Block Diagram



Each lane module (raw_pcs_lane) contains finite state machines (FSM) to execute the power-up calibration and runtime adaptation algorithms. The algorithms are stored in a register bank located in a common module (raw_pcs_cmn). The design includes two register access arbiters. The Memory arbiter is used for accessing the register bank whereas the Control Register Arbiter is used to access all other registers in the Raw PCS and the PMA. This mechanism provides efficient use of the registers and state machines within each lane where each lane can independently execute a different portion of the algorithm while reducing the register access bottleneck.

All control registers in the Raw PCS and PMA can be externally accessed using JTAG or CR Parallel Interfaces, with only one interface at any given time.

5 PHY Usage and Configuration

This chapter explains the usage of PHY features and lists the PHY configuration settings for the protocols and data rates that are supported. It includes the following sections:

- "PHY Power-Up" on page 156
- "PHY Initialization Sequence" on page 156
- "Lane Configuration Update" on page 160
- "MPLL Configuration Update" on page 163
- "TX Data Path" on page 164
- "RX Data Path" on page 165
- "RX Loss of Signal (LOS) Detector" on page 167
- "RX CDR PPM Detector" on page 168
- "TX and RX Power States" on page 169
- "TX Equalization and Adaptation" on page 174
- "PHY Power-Down" on page 180
- "RX Equalization and Adaptation" on page 181
- "Receiver Detection" on page 186
- "Control Register Interface" on page 187
- "External SRAM Support" on page 192
- "Test Modes" on page 196
- "Aggregation and Bifurcation Support" on page 207
- "Power Gating" on page 210
- "Configuration for a Disabled PHY" on page 214
- "Notes for Configuring the PHY" on page 215
- "Configuring the PHY" on page 217

5.1 PHY Power-Up

Power up the vp* and vph power supplies in any order and then proceed with the PHY reset and initialization sequence (vp* indicates the grouping of all core supplies; vp, vpdig, and vptx0/1/2/3).

Note

- The state of the PHY is unknown after power up but before assertion of reset. The PHY may exhibit unwanted behavior such as TX enabled or RX terminations activated, for example.
- While the unknown state is undesirable, the PHY has been designed to prevent damage if it is left in this state.
- Supply ramp should be monotonic with a ramp time of no faster than 10 µs and a rate no faster than (supply voltage)/10 µs.

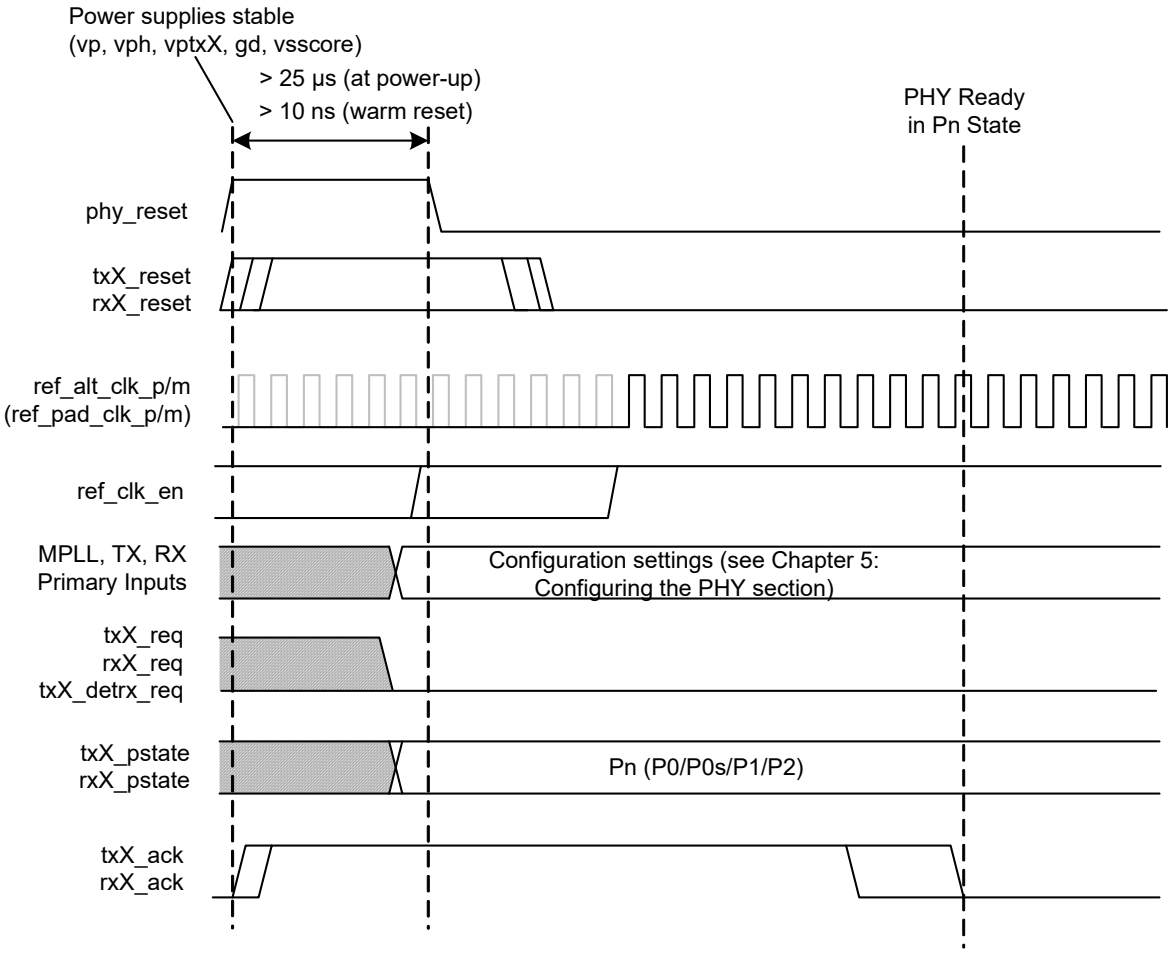
5.2 PHY Initialization Sequence

During PHY reset and initialization sequence, the PHY performs power up circuit calibrations and enters the configuration mode defined by the primary inputs. The PHY initialization timing diagram is shown in Figure 5-1 on page 157.

The PHY does not require any clocks to be supplied to enter the reset state (assertion of phy_reset). Exit from the reset state is indicated by de-assertion of txX_ack/rxX_ack, and requires a reference clock to be supplied and ref_clk_en input to be asserted. The maximum time from phy_reset de-assertion to de-assertion of txX_ack/rxX_ack can be as high as 5 ms in fast simulation mode and as high as 15 ms in normal simulation mode because offset calibration is not modeled.

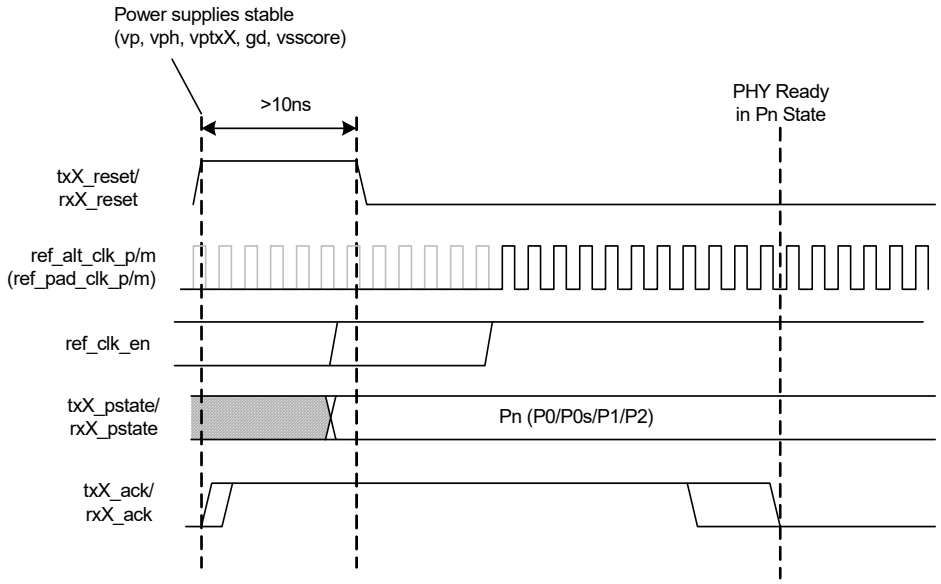
When phy_reset is asserted, the TX and RX terminations are disabled and txX_p/m are tri-stated for all lanes of the PHY. In addition to global phy_reset, txX_reset/rxX_reset inputs can be used to reset individual lanes in the PHY. When txX_reset is asserted, the txX_pstate is similar to P2, the TX terminations are disabled and txX_p/m are tri-stated for lane X. When rxX_reset is asserted, the rxX_pstate is similar to P2 except that the RX terminations are disabled for lane X. Asserting either txX_reset or rxX_reset during phy_reset assertion stalls the Raw_PCS FSM operation before internal registers are updated for lane X after phy_reset is de-asserted.

Figure 5-1 PHY Initialization Sequence



The PHY reset sequence is shown in Figure 5-2.

Figure 5-2 PHY Reset Lane X Sequence



txX_reset/rxX_reset usage

Note ref_clk_en must stay asserted, irrespective of txX_reset/rxX_reset, if any of the lanes require the reference clock input to be active.

The PHY reset/initialization steps are as follows:

1. Assert reset (phy_reset = 1).

   The lane-acknowledge outputs are asserted during reset (txX_ack = rxX_ack = 1).

2. Set the MPLL and lane (TX/RX) inputs to the desired configuration as defined in "Configuring the PHY" on page 217.

   Note that the TX and RX power state control inputs, txX_pstate[1:0] and rxX_pstate[1:0], can be set independently.

3. De-assert reset (phy_reset = 0). The phy_reset needs to be asserted for a minimum of 25 µs after power-supplies are ramped-up (power-on). For any subsequent phy_reset assertions (warm-reset), phy_reset needs to be asserted for a minimum of 10 ns.

4. Wait for the lane acknowledge outputs to de-assert (txX_ack = rxX_ack = 0). At this point, all the desired inputs set in Step 2 have been applied in the PHY.

Note Initialization will not complete until both enabled PLLs are calibrated and the reference resistor tuning is complete:

- Ensure that the resistor contention bus is connected as shown in Figure 7-13 on page 246 ("Reference Resistor Contention Bus - Single PHY") or Figure 7-14 on page 246 ("Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor")
- Ensure that both the MPLLs are configured properly or the mpll{a/b}_init_cal_disable signal is asserted

5. Assert txX_clk_rdy if needed. For usage of the txX_clk_rdy input, refer to "Lane Configuration Update" on page 160.

Note Warm reset will be 10 ns minimum if the power is up for more than 10 µs.

5.3 Lane Configuration Update

Each lane is configured using its TX and RX configuration inputs. Figure 5-3 and Figure 5-4 show the configuration change handshake for the TX and RX signals, respectively. Although the figures show all the inputs that are updated using the txX_req/txX_ack and rxX_req/rxX_ack handshake, it is allowed to change a subset of these inputs. These inputs must stay stable from assertion of txX_req/rxX_req until de-assertion of txX_ack/rxX_ack.

Figure 5-3 Lane TX Configuration Change Handshake

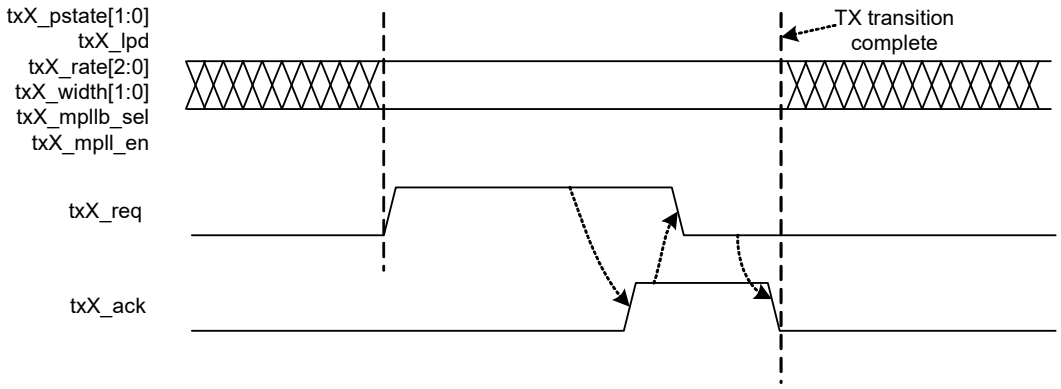
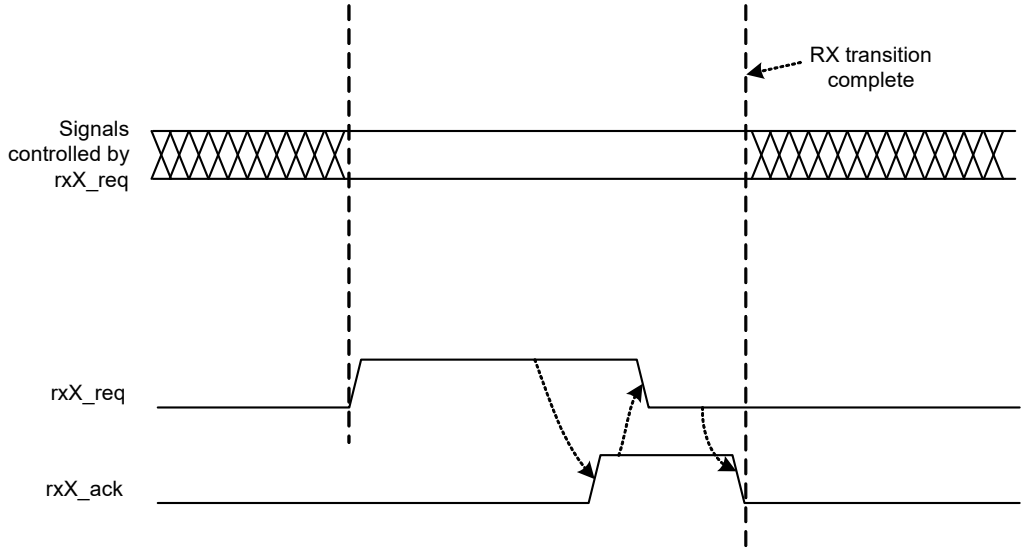


Figure 5-4 Lane RX Configuration Change Handshake

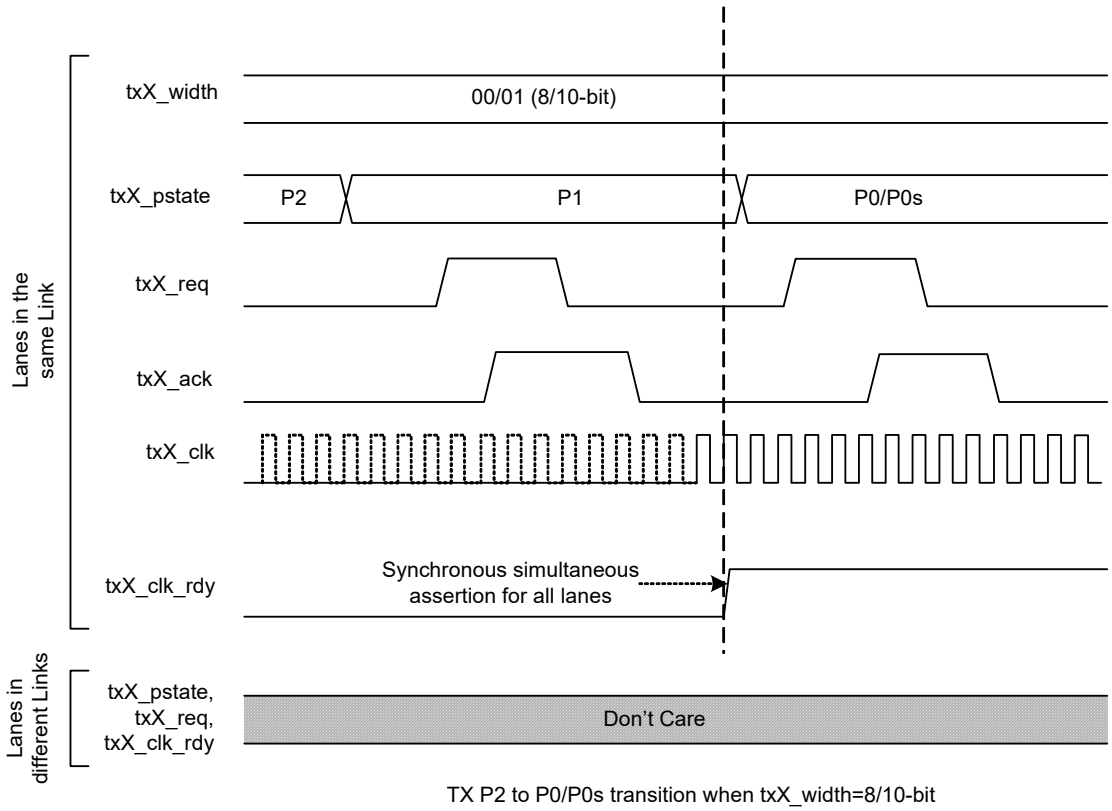


Lanes that are operating as part of one link must comply with the following requirements:

- TX configuration inputs txX_pstate, txXrate, txX_width, and txX_mpllb_sel must be set to the same value for all lanes.
- RX configuration inputs rxX_pstate, rxXrate, and rxX_width must be set to the same value for all lanes.
- A common clock must be supplied to txX_clk inputs for all lanes.
- txX_clk_rdy must be asserted prior to changing the txX_pstate to P0/P0s and must be asserted synchronously in the same clock cycle for all the lanes in a single link when txX_width=2'b00/01 (8/10-bit). Figure 5-5 shows the usage of txX_clk_rdy in 8/10-bit case. Figure 5-6 on page 162 shows the usage of txX_clk_rdy when txX_width=2'b10/11 (16/20-bit), in which case, txX_clk_rdy must be asserted before txX_clk is running and txX_req is asserted.

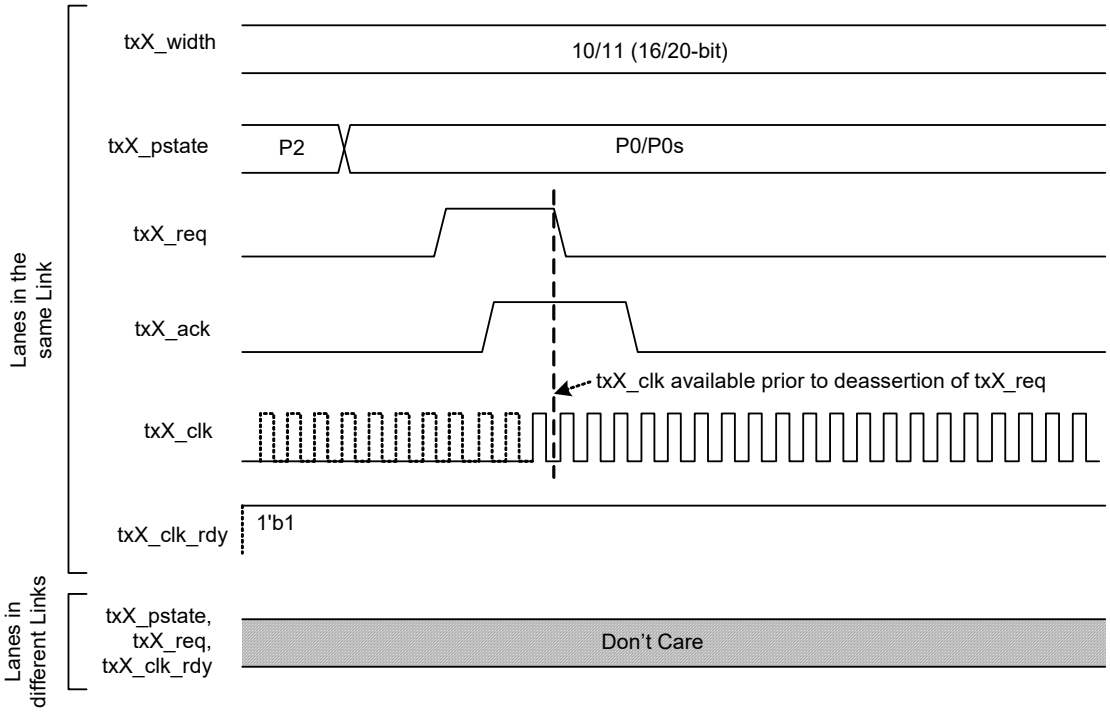
Note When txX_width=8/10-bit, txX_pstate transition from P2 (2'b11) to P0/P0s (2'b00/01) is not permitted. As shown in Figure 5-5, txX_pstate transitions from P2 to P1 to P0/P0s. This is because txX_clk_rdy input must be asserted synchronously on txX_clk prior to the internal transmit clock alignment sequence in the PHY when entering P0/P0s state.

Figure 5-5 TX P2 to P0/P0s Transition When txX_width = 8/10-bit



TX P2 to P0/P0s transition when txX_width=8/10-bit

Figure 5-6 TX P2 to P0/P0s Transition When txX_width = 16/20-bit

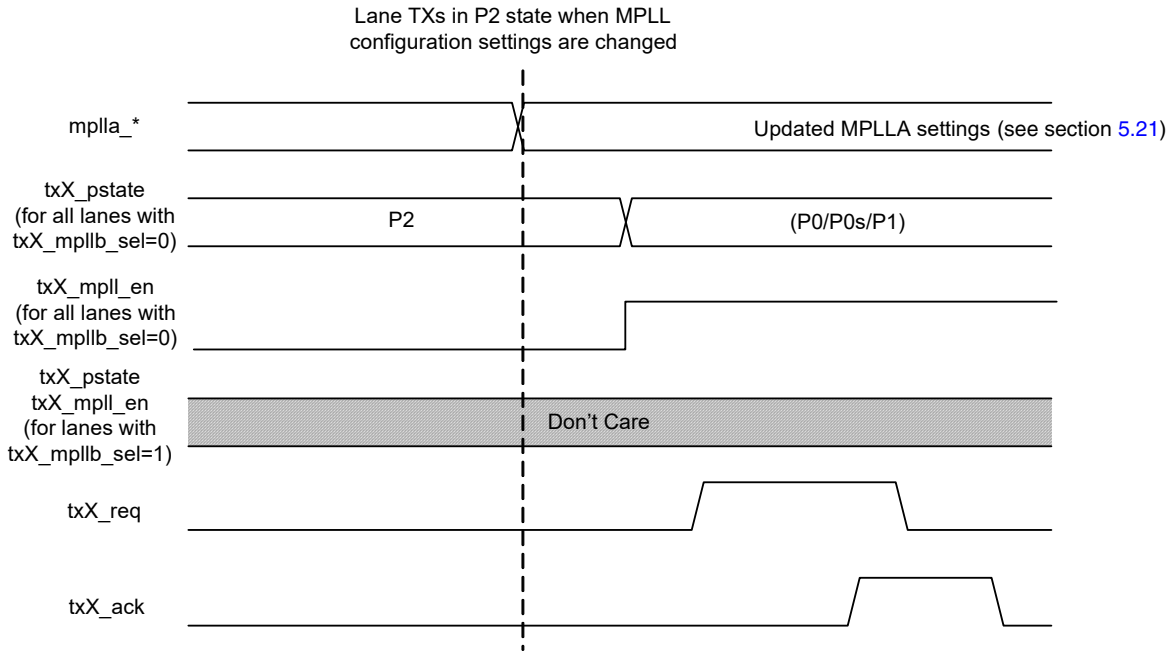


5.4 MPLL Configuration Update

The MPLLA and MPLLB in the PHY are powered up and powered down depending upon the txX_pstate and txX_mpllb_sel control inputs on each lane. With the default register settings, the MPLLA is powered down when mplla_force_en input is 0 and for each lane, txX_mpll_en=0 or txX_mpllb_sel=1. Likewise, MPLLB is powered down when mpllb_force_en input is 0 and, for each lane, txX_mpll_en=0 or txX_mpllb_sel=0. Both MPLLs are calibrated when phy_reset is de-asserted and mpll{a,b}_init_cal_disable inputs are de-asserted. Calibration of each MPLL can be skipped if the corresponding mpll{a,b}_init_cal_disable input is asserted when phy_reset is de-asserted.

The MPLL configuration settings can only be changed when the corresponding MPLL is powered down. As an example, Figure 5-7 shows the requirements for changing the MPLLA configuration settings.

Figure 5-7 MPLLA Configuration Settings Update Followed By a TX Power-state Change



5.5 TX Data Path

The TX data path is active when the TX lane is in P0 state (txX_pstate=2'b00) and txX_data_en is asserted.

The TX data path supports serialization of 8/10/16/20-bit data symbol determined by the setting of the input pin txX_width. The valid bits that are transmitted in the 8/10/16/20-bit mode are defined in the signal description of txX_data. The data symbols captured by the PHY on the rising edge of txX_clk when txX_data_en=1 are transmitted on txX_p/m serial lines. In addition, there are some dribble (invalid) bits transmitted before the first valid word and after the last valid word. The maximum number of dribble bits at the beginning and end of transmission is 4 and 2, respectively.

For lanes that are operating as part of one link, there is a clock alignment mechanism in the PHY to align the internal transmit clocks of each lane to the common tX_clk inputs. This alignment results in a maximum TX skew on txX_p/m across different lanes in the same link to within 2 unit-intervals (UI) + 500ps. If the txX_clk clocks are not aligned between lanes, then the serial skew can be up to 1 txX_clk (if 16/20b mode) or 2 txX_clk (if 8/10b mode).

The TX data-path latency for the PHY is measured as per Figure 5-8. The latency numbers are provided in Table 5-1.

Figure 5-8 Transmit Data Path Latency Measurement In the PHY

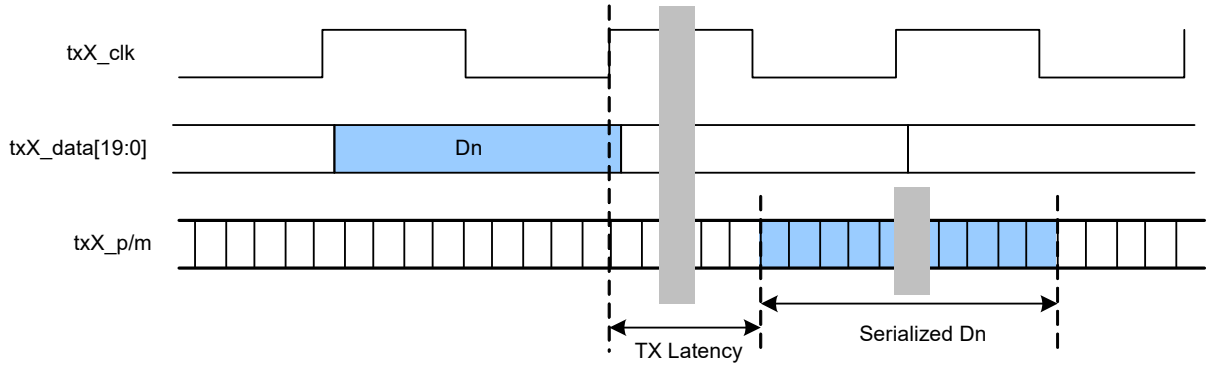


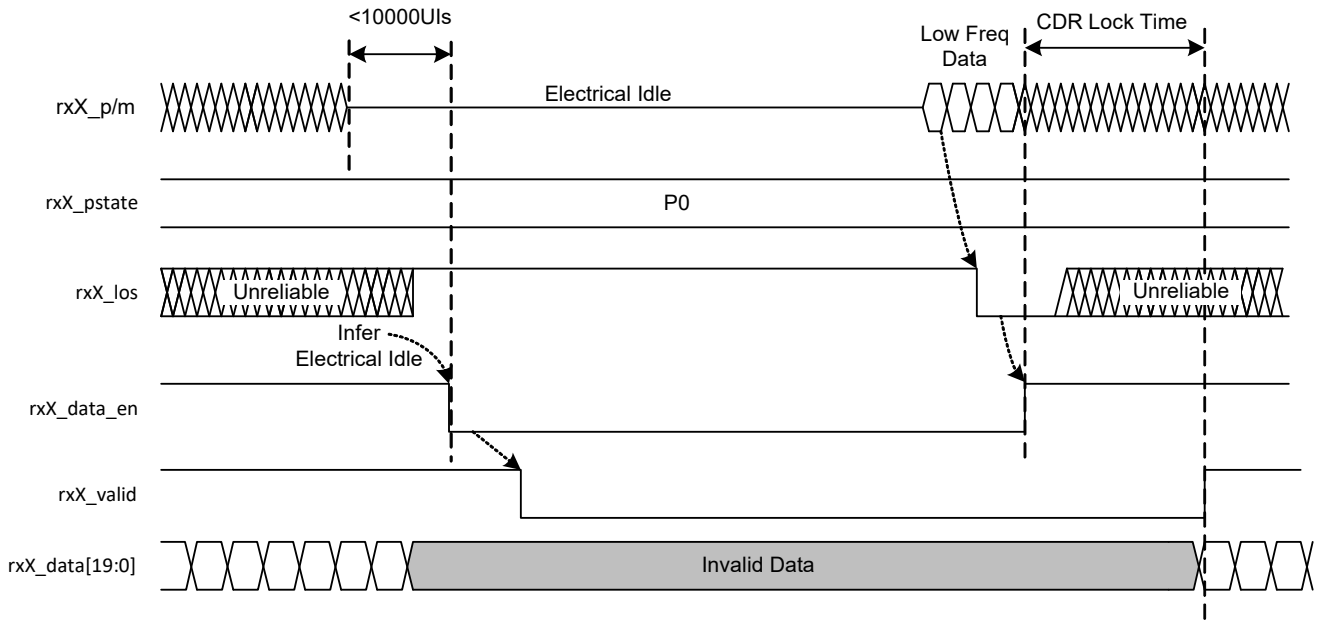
Table 5-1 Transmit Data-path latency

Lane TX Width (txX_width)	Min Latency with TX FIFO bypassed (UIs)	Max Latency with TX FIFO bypassed (UIs)	Min Latency with TX FIFO (UIs)	Max Latency with TX FIFO (UIs)
10-bit	58	61	77	81
16-bit	41	46	57	63
20-bit	47	52	67	71

5.6 RX Data Path

The RX data path is active when the RX lane is in P0 state (rxX_pstate=2'b00) and rxX_data_en is asserted. You should only assert rxX_data_en when the RX lane is in P0 state and the RX input signal is present (indicated by the de-assertion of the rxX_los output). If a loss of signal is detected, either by assertion of rxX_los for baud rates less-than-or-equal-to 2.5 Gbps or inferred using higher layer protocol-specific mechanisms, the clock-and-data-recovery (CDR) must be disabled by setting rxX_data_en=0 and re-enabled by setting rxX_data_en=1 after a signal is detected using rxX_los=0. A typical timing diagram of RX data path for baud rates greater than 2.5 Gbps is shown in Figure 5-9.

Figure 5-9 RX Data Path and Control Timing for Baud Rate Greater than 2.5 Gbps



The output rxX_valid is asserted, when a pre-determined bit-locking timer expires after assertion of rxX_data_en. In this case, rxX_valid is de-asserted, when rxX_data_en is de-asserted to 0 or when the CDR frequency drift goes beyond that set by the input rxX_cdr_ppm_max[4:0]. The default settings for rxX_cdr_ppm_max[4:0] values provided in this chapter set the drift threshold to 15000ppm. If the CDR frequency goes beyond this threshold, the rxX_data_en is internally overridden to enable the reference tracking mode until the CDR frequency returns to the ideal frequency. After which the locking to data will resume and rxX_valid will re-assert after the bit-locking timer expires.

When the input rxX_data_en is asserted, the CDR's tracking mode is enabled. When rxX_data_en=0, the RX VCO frequency is calibrated using the reference clock inputs (ref_alt_clk_p/m, ref_pad_clk_p/m) to ensure that the VCO frequency stays within the precision required for the CDR to lock to the received incoming data when rxX_data_en is re-asserted.

If rxX_data_en is enabled during invalid data, the fastest recovery will be to change the rxX_pstate from P0 to P1 to P0. This action takes approximately 30 µs and is much faster than negating rxX_data_en for an extended time.

The maximum CDR lock times as measured in Figure 5-9 are provided in Table 5-2 for both Spread-Spectrum-Clocking (SSC) and non-SSC modes.

Table 5-2 CDR Lock Times

CDR Mode (SSC/non-SSC)	Maximum CDR Lock Time (UIs)
Non-SSC (rxX_cdr_ssc_en=0)	6560
SSC (rxX_cdr_ssc_en=1)	36800

The RX data path supports de-serialization of 8/10/16/20-bit data symbol determined by the setting of the input pin rxX_width. The valid bits on rxX_data[19:0] in the 8/10/16/20-bit modes are defined in the signal description of rxX_data.

The RX data-path latency for the PHY is measured as per Figure 5-10. The latency numbers are provided in Table 5-3.

Figure 5-10 Receive Data Path Latency Measurement In the PHY

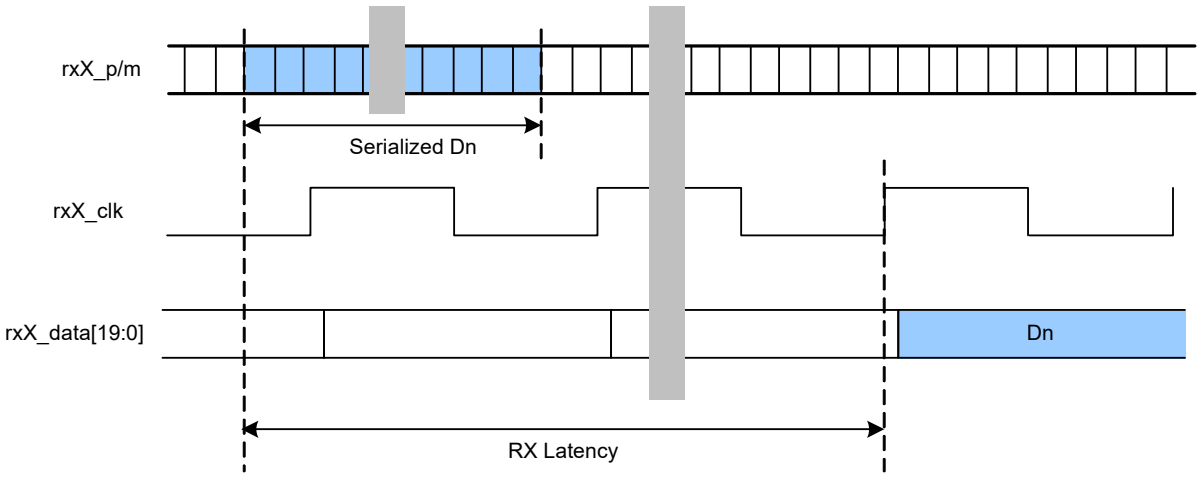


Table 5-3 RX Data Path Latency

Lane RX Width (rxX_width)	Min Latency (UIs)	Max Latency (UIs)
10-bit	50	51
16-bit	48	49
20-bit	60	61

5.7 RX Loss of Signal (LOS) Detector

The rxX_los output can be used to detect or infer electrical idle on the rxX_p/m inputs. The RX los detector is enabled when rxX_disable and rxX_reset inputs are de-asserted. The los detector can be configured to be used as a true LFPS detector for USB3-like applications when rxX_los_lfps_en input is asserted.

The rxX_los output is in an undetermined state when rxX_disable input is asserted. When rxX_disable is de-asserted, it may take up to 10 µs for the LOS detector to power up, during which time the rxX_los output is invalid and in an undetermined state.

The rxX_los output is in an undetermined state when RX terminations are disabled, that is, when rxX_term_en=0.

When rxX_los output is valid, the behavior of rxX_los is provided in Table 5-4.

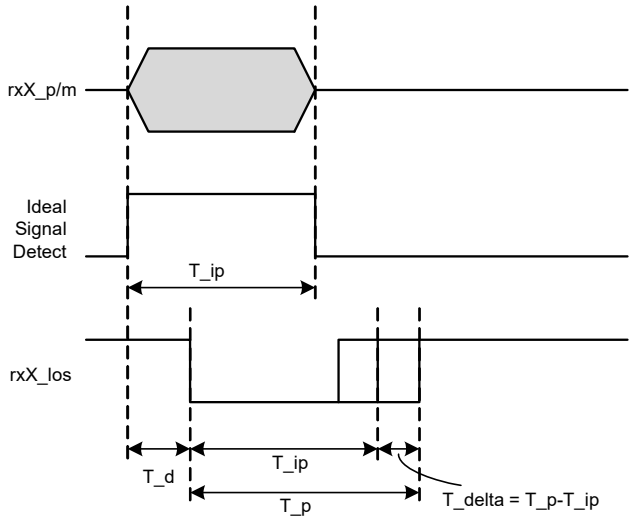
Table 5-4 rxX LOS Behavior

Signal on rxX_p/m	rxX_los
Pulses wider than 400ps	Stable 0
Pulses narrower than 400ps	Unreliable (0/1)
Electrical Idle	Stable 1

For cases where rxX_los output is valid, the response time and the distortion introduced by the LOS circuit are shown in Figure 5-11.

Note Use rxX_los only to detect exit from electrical idle for signals with pulse widths greater than 400 ps. It is recommended that you use an asynchronous pulse catching logic to capture the active low pulse on rxX_los to detect exit from electrical idle.

Figure 5-11 rxX LOS Response Time and Distortion Specification



The rxX_los specification is provided in Table 5-5.

Table 5-5 rxX LOS Specification

Parameter Description	Parameter	Specification
Response time from signals burst to rxX_los transition from 1 to 0	T_d	0ns<T_d<8ns
Pulse width distortion	T_delta	-8ns<T_delta<8ns

5.8 RX CDR PPM Detector

The PHY includes a CDR PPM detection circuit that contains the following features:

- Programmable PPM offset
- Current status of PPM offset relative to nominal frequency
- Ability to operate simply as a PPM monitor without intervening
- Ability to reset the CDR back to a nominal state when the clock drifts past the programmable PPM offset

The CDR PPM detector contains running counters and compares the incoming rxX_clk to a nominal clock based on the rxX_ref_ld_val and rxX_vco_ld_val inputs. If the difference exceeds the programmable PPM offset, the CDR unlock detector forces the rxX_data_en signal low to the hard-macro PMA and enables the PMA to track back to the nominal frequency.

The RX CDR PPM detector is enabled by default but can be disabled by setting the RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL.RX_CDR_DETECTOR_EN register to 1'b0. If you want to disable the correction mechanisms of the RX CDR PPM detector, you can also enable it in monitoring mode by setting the RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL.RX_CDR_PPM_MONITOR_MODE register to 1'b1.

The rxX_ppm_drift signal is the difference between the current running counter value relative to the programmed rxX_vco_ld_val. The output is a signed value that ranges from -64 (6'b100000) to +63 (6'b011111). This value is valid when rxX_ppm_drift_vld is asserted. The rxX_ppm_drift_vld signal is periodically asserted when the FSM has completed an iteration of comparing the ppm drift.

Note that the rxX_ppm_drift signal must not be used to control an ESTORE module in the protocol PCS.

This signal provides only a difference from the NOMINAL frequency and does not take into account the actual transmitter.

5.8.1 Configuring the RX CDR PPM Detector

The RX CDR PPM detector can be programmed by the changing the rxX_cdr_ppm_max[4:0] input signals.

This signal represents a difference from the nominal rxX_vco_ld_val that represents the appropriate PPM setting. This value is available in the PHY configuration section for 15,000 ppm (default setting). However, the value can be updated for each configuration based on the following equation (the example is based on PCIe Gen4 @ 100 MHz):

FLOOR(rxX_vco_ld_val * (ppm_value / 1e6)) - 2 (for margin) = (1,360 * 0.015) – 2 = 18

5.9 TX and RX Power States

The PHY supports independent power state controls for each lane and TX and RX within each lane. There are four power states supported for the TX and RX within each lane; P0, P0s, P1, and P2.

5.9.1 TX and RX Power State Description

The operational modes of the lane in each power state are provided in Table 5-6 and Table 5-7.

Table 5-6 TX Power State Descriptions

TX Power State	Description
P0	P0 is the fully operational power state for the TX. In the state: 1. txX_pstate is set to P0. 2. txX_reset, txX_disable inputs must be de-asserted. txX_lpd can be asserted to disable a lane in a link. 3. txX_mpll_en must be set to 1. 4. If tx_data_en is asserted, TX driver is enabled and serial data is transmitted. 5. txX_clk input must be valid and of correct frequency prior to de-asserting txX_req input when requesting entry into P0 state (shown in Figure 5-6 on page 162). 6. TX common-mode, appropriate MPLL clocks, TX clocks, and TX data-path are enabled. 7. MPLL is powered up. 8. When MPLLA is powered up (as described in "MPLL Configuration Update" on page 163), the following clocks are enabled and valid: - mplla_div16p5_clk, mplla_div33_clk, mplla_div66_clk, when mplla_div16p5_clk_en is asserted - mplla_div_clk, when mplla_div_clk_en is asserted - mplla_word_clk, mplla_dword_clk, mplla_qword_clk, mplla_oword_clk, when either or both of the signals mplla_div8_clk_en and mplla_div10_clk_en are asserted 9. When MPLLB is powered up (as described in "MPLL Configuration Update" on page 163), the following clocks are enabled and valid: - mpllb_div_clk, when mpllb_div_clk_en is asserted - mpllb_word_clk, mpllb_dword_clk, mpllb_qword_clk, mpllb_oword_clk, when either or both of the signals mpllb_div8_clk_en and mpllb_div10_clk_en are asserted
P0s	Same state as P0 with the following exceptions: 1. txX_pstate is set to P0s 2. tx_data_en must be de-asserted

Table 5-6 TX Power State Descriptions (Continued)

TX Power State	Description
P1	Same state as P0s with the following exceptions: 1. txX_pstate is set to P1. 2. txX_mpll_en/txX_disable may be set to 0/1, respectively. For example, in PCIe L1 state, txX_pstate=P1, txX_mpll_en=1, txX_disable=0. In PCIe L1 sub-states, txX_pstate=P1, txX_mpll_en=0, txX_disable=1. 3. MPLLs may be powered down if txX_mpll_en=0 and no other lane is requesting MPLLs to be powered up, for example, in PCIe L1 sub-states. In this case, MPLL clock outputs are disabled. 4. txX_clk may not be supplied.
P2	Same state as P1, with following exceptions: 1. txX_pstate is set to P2. 2. MPLL{A/B} is powered down if txX_mpll_en=0 and no other lane is requesting MPLLs to be powered up. In this case, MPLL clock outputs may be disabled. 3. MPLL{A/B} is powered up if txX_mpll_en=1. In this case, MPLL clock outputs are disabled.

Table 5-7 RX Power State Descriptions

RX Power State	Description
P0	All RX circuits are powered up. 1. When rxX_data_en are asserted, the RX data recovery and de-serialization is enabled. 2. rxX_reset and rxX_disable inputs must be de-asserted. rxX_lpd can be asserted for disabling lane while link in P0. rxX_term_en must be asserted. 3. rxX_clk is valid. **Note:** PCIe P0s state maps to P0 state on the PHY rxX_pstate input.
P0s	Same state as P0 with the following exceptions: 1. RX CDR and data-path are disabled. 2. rxX_reset, rx0_lpd, and rxX_disable inputs may be asserted. 3. rxX_data_en must be de-asserted. 4. rxX_clk is disabled.
P1	Same state as P0s with the following exceptions: 1. RX VCO is powered down. 2. rx0_lpd and rxX_disable inputs may be asserted. For example, in PCIE L1 sub-states, rxX_disable input is asserted.
P2	Same state as P1, with following exceptions: RX analog front end is powered down.

5.9.2 Power State Transition Times for TX and RX

The time taken for the PHY to assert txX_ack and rxX_ack in response to a txX_req and rxX_req corresponding to a txX_pstate and rxX_pstate transition is referred to as the power state transition time. The power state transition times are dependent upon the reference clock frequency, txX_clk/rxX_clk frequency, and ref_range input. In addition, the power state transition times are dependent upon a range of input and parameters. These include, but are not limited to: register settings, reference clock frequency, MPLL clock frequency, RX VCO clock frequency, lane divider settings.

The maximum power state transition times for PCIe mode with 100 MHz reference clock input and input settings from PCIe Gen1 ("Support Configuration Settings" and "Lane Configuration Settings") are provided in Table 5-8. To support faster simulation times, the PHY includes a scaled down mode that can be enabled by a sequence of register writes defined in "Fast Simulation Mode" on page 223. Power state transition times for scaled down mode are also listed in Table 5-8. The values assume the TX and RX power states are the same and transitioned at the same time.

Table 5-8 Power State Transition and Receiver Detection Times for PCIe (100 MHz Reference Clock)

Power State Transition	Non-Scaled-down Mode (µs)				Scaled-down Mode (µs)			
	PCIe 1.1	PCIe 2.1	PCIe 3.1	PCIe 4.1	PCIe 1.1	PCIe 2.1	PCIe 3.1	PCIe 4.1
RESET → P0	15000	15000	15000	15000	30	30	30	30
RESET → P1	15000	15000	15000	15000	30	30	30	30
RESET → P2	15000	15000	15000	15000	30	30	30	30
P0 → P0s	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
P0 → P1	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
P0 → P2	2	2	2	2	2	2	2	2
P0s → P0	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
P0s → P1	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
P0s → P2	2	2	2	2	2	2	2	2
P1 → P0	15.5	15.5	17.3	14.1	8	5	4	3.5
P1 → P2	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
P2 → P0	868	868	868	868	8.5	7	6.5	6
P2 → P1	850	850	850	850	2.5	2.5	2.5	2.5
P1.1 → P0	37	36	35	34	10	7.5	6.5	6

Table 5-8 Power State Transition and Receiver Detection Times for PCIe (100 MHz Reference Clock)

Power State Transition	Non-Scaled-down Mode (µs)				Scaled-down Mode (µs)			
	PCIe 1.1	PCIe 2.1	PCIe 3.1	PCIe 4.1	PCIe 1.1	PCIe 2.1	PCIe 3.1	PCIe 4.1
P1.2 → P0	57	56	55	54	10	7.7	6.7	6.3
P1.2 PG → P0[a]	78	77	76	75	29	29	28	27

Note:

- P1.2 values are preliminary and under review. For more information, contact Synopsys.
- Times shown are for single-link configurations. Timings for bifurcated links will vary slightly based on relative state of links.

a. The power switch turn-on time is not included

For power state transition times for other configurations of the PHY, contact Synopsys customer support.

Table 5-9 details the exit latencies for transition times between L1 substates power states.

Table 5-9 Substate Exit Latencies and Timing Information

Latency/Transition	Description	Value	Unit
T_COMMON_MODE	Worst-case time for the Common mode voltage to be reached on txX_{p,m} pins while transmitting in Recovery mode. This parameter applies to an exit from P1.2 through P1 to P0 (Recovery) and represents the extra time in which the MAC is required to send TS1 ordered sets.	60	µs

The protocol specific power state mappings are provided in Table 5-10.

Table 5-10 Protocol Power State Mapping

Protocol	Protocol Power State	PHY Input Settings					
		txX_pstate	rxX_pstate	txX_mpll_en	txX_disable	rxX_disable	ref_clk_en
PCIe	P0	P0	P0	1	0	0	1
	P0s	P0s	P0	1	0	0	1
	P1	P1	P1	1	0	0	1
	P1.1	P1	P1	0	0	1	0
	P1.2	P1	P1	0	1	1	0
	P1.CPM	P1	P1	0	0	0	0
	P2	P2	P2	0	0	0	1
	P2.CPM	P2	P2	0	0	0	0

Table 5-10 Protocol Power State Mapping (Continued)

Protocol	Protocol Power State	PHY Input Settings					
		txX_pstate	rxX_pstate	txX_mpll_en	txX_disable	rxX_disable	ref_clk_en
SATA	PHYRDY	P0	P0	1	0	0	1
	Partial	P0	P0	1	0	0	1
	Slumber	P2	P2	0	0	0	1
	DevSleep	P2	P2	0	0	0	1
Ethernet	Active	P0	P0	1	0	0	1
	Refresh	P0	P0	1	0	0	1
	Quiet	P0	P0s	1	0	0	1
	Disabled	P2	P2	0	0	0	1

Table 5-10 shows the settings of PHY inputs after the PHY has entered the corresponding power-state. While transitioning between power states, the PHY inputs require the appropriate sequencing in order to meet the following PHY requirements:

- txX_disable and rxX_disable must not be asserted before the txX_pstate[1:0] and rxX_pstate[1:0] transition into the P1 state is completed (indicated by the assertion of txX_ack and rxX_ack).
- ref_clk_en must not be de-asserted before txX_mpll_en is de-asserted and mpll{a,b}_state output is de-asserted.
- The TX data-path and RX CDR are enabled by asserting txX_data_en and rxX_data_en inputs. These can only be enabled in the active power state (PCIe: P0, SATA: PHYRDY, Ethernet: Active/Refresh) and must be de-asserted in all other power states.

5.10 TX Equalization and Adaptation

TX equalization (EQ) and transmit levels are controlled by the txX_eq_pre, txX_eq_post, and txX_eq_main input signals. These inputs are derived from the coefficients C_{-1}, C_1, and C_0 (defined in the PCIe Base specification) using the following equations.

Equation:

TX Equalization input mapping to specification parameters

txX_eq_pre = 4 x $|C_{-1}|$

txX_eq_post = 4 x $|C_1|$

txX_eq_main = C_0 + (txX_eq_pre != 0) + (txX_eq_post != 0)

The txX_eq_post, txX_eq_pre, and txX_eq_main inputs can be updated in a single txX_clk cycle to a known valid value. Alternately, the inputs can be sequenced to free up resources in the following order: Increment C_1, Increment C_{-1}, Decrement C_0, Decrement C_1, Decrement C_{-1}, then Increment C_0. An invalid value might cause TX serial data corruption in simulation.

5.10.1 TX EQ Initialization and Presets

The TX equalization input settings and TX Margining settings for PCIe are provided in Table 5-11.

Table 5-11 TX Equalization Settings and TX Margining Settings for PCIe

TxMargin	TxPreset	txX_eq_main	txX_eq_pre	txX_eq_post	TX de-emphasis (dB)	TX pre-shoot (dB)	TX max swing (Normalized)
0	0	30	0	40	-6.02	0.00	1
	1	33	0	28	-3.74	0.00	1
	2	32	0	32	-4.44	0.00	1
	3	35	0	20	-2.50	0.00	1
	4	40	0	0	0.00	0.00	1
	5	36	16	0	0.00	1.94	1
	6	35	20	0	0.00	2.50	1
	7	28	16	32	-6.02	3.52	1
	8	30	20	20	-3.52	3.52	1
	9	33	28	0	0.00	3.74	1
	10	27	0	52	-9.12	0.00	1

Table 5-11 TX Equalization Settings and TX Margining Settings for PCIe (Continued)

TxMargin	TxPreset	txX_eq_main	txX_eq_pre	txX_eq_post	TX de-emphasis (dB)	TX pre-shoot (dB)	TX max swing (Normalized)
1	0	30	0	40	-6.02	0.00	1
	1	33	0	28	-3.74	0.00	1
	2	32	0	32	-4.44	0.00	1
	3	35	0	20	-2.50	0.00	1
	4	40	0	0	0.00	0.00	1
	5	36	16	0	0.00	1.94	1
	6	35	20	0	0.00	2.50	1
	7	28	16	32	-6.02	3.52	1
	8	30	20	20	-3.52	3.52	1
	9	33	28	0	0.00	3.74	1
	10	27	0	52	-9.12	0.00	1
2	0	22	0	32	-6.62	0.00	0.75
	1	25	0	20	-3.52	0.00	0.75
	2	24	0	24	-4.44	0.00	0.75
	3	26	0	16	-2.69	0.00	0.75
	4	30	0	0	0.00	0.00	0.75
	5	27	12	0	0.00	1.94	0.75
	6	26	16	0	0.00	2.69	0.75
	7	21	12	24	-6.02	3.52	0.75
	8	22	16	16	-3.93	3.93	0.75
	9	25	20	0	0.00	3.52	0.75
	10	20	0	40	-9.54	0.00	0.75

Table 5-11 TX Equalization Settings and TX Margining Settings for PCIe (Continued)

TxMargin	TxPreset	txX_eq_main	txX_eq_pre	txX_eq_post	TX de-emphasis (dB)	TX pre-shoot (dB)	TX max swing (Normalized)
3	0	15	0	20	-6.02	0.00	0.5
	1	17	0	12	-3.10	0.00	0.5
	2	16	0	16	-4.44	0.00	0.5
	3	17	0	12	-3.10	0.00	0.5
	4	20	0	0	0.00	0.00	0.5
	5	18	8	0	0.00	1.94	0.5
	6	17	12	0	0.00	3.10	0.5
	7	14	8	16	-6.02	3.52	0.5
	8	16	8	8	-2.50	2.50	0.5
	9	17	12	0	0.00	3.10	0.5
	10	14	0	24	-7.96	0.00	0.5
4	0	7	0	12	-7.96	0.00	0.25
	1	8	0	8	-4.44	0.00	0.25
	2	8	0	8	-4.44	0.00	0.25
	3	9	0	4	-1.94	0.00	0.25
	4	10	0	0	0.00	0.00	0.25
	5	9	4	0	0.00	1.94	0.25
	6	9	4	0	0.00	1.94	0.25
	7	7	4	8	-6.02	3.52	0.25
	8	8	4	4	-2.50	2.50	0.25
	9	8	8	0	0.00	4.44	0.25
	10	7	0	12	-7.96	0.00	0.25
5,6,7	Not supported						

Note For full-swing mode (no attenuation): txX_eq_main[5:0] + txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4 = 40.

TX equalization input settings for 10GBASE-KR preset and initialize states are provided in Table 5-12.

Table 5-12 TX Equalization Settings for 10GBASE-KR

State	txX_eq_main[5:0] (decimal)	txX_eq_pre[5:0] (decimal)	txX_eq_post[5:0] (decimal)
PRESET	40	0	0
INITIALIZE	27	8	44

TX equalization input settings for SRIO-SR initialization are provided in Table 5-13.

Table 5-13 TX Equalization Settings for SRIO-SR

State	txX_eq_main[5:0] (decimal)	txX_eq_pre[5:0] (decimal)	txX_eq_post[5:0] (decimal)
Option 1: 0db de-emphasis	30	0	0
Option 2: 3.5db de-emphasis	25	0	20

5.10.2 TX EQ Adaptation

This section describes how the TX EQ inputs to the PHY should be updated when a coefficient update request is received from the link partner (far-end receiver) as part of link adaptation.

5.10.2.1 PCIe TX EQ Adaptation

As per the PCIe 3.1 specification, the link partner determines the coefficient updates needed for the local TX equalization and encapsulates the pre-cursor and post-cursor coefficients as part of its transmitted TS1 ordered sets. To do so, the link partner uses the local TX's preset-to-coefficient mapping table, as well as the FS and LF values. These parameters for the PHY TX are as follows:

- Preset to Coefficient Mapping (see Table 5-11 on page 174)
- FS: 40 (decimal)
- LF: 14 (decimal)

Note that the above parameters are outputs from the PCIe PCS to the MAC; for more, refer to the 16G PCS for the DesignWare® Cores Multi-Protocol 16G PHY Databook.

5.10.2.2 10GBASE-KR TX EQ Adaptation

As per the 10GBASE-KR specification, the link partner determines the increment/decrement/hold commands for the local TX coefficients and encapsulates these commands for each of the pre/main/post-cursors as part of its transmitted training frames. The algorithm to translate the increment/decrement/hold commands into the TX EQ coefficient inputs to the PHY is described below.

The number of TX driver legs used for attenuation is given by:

Natten = 40 - (txX_eq_main[5:0] + txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4)

For full-swing mode (no attenuation):

txX_eq_main[5:0] + txX_eq_pre[5:0]/4 + txX_eq_post[5:0]/4 = 40

The increment/decrement/hold commands from the link partner are translated into local variables ΔNatten, ΔNmain, ΔtxX_eq_pre, and ΔtxX_eq_post as per Table 5-14.

Table 5-14 Default TX Coefficient Update Sequence for 10GBASE-KR

Update Method	EQ Update Request from Link Partner's RX			Natten^k > 0?	TX Configuration			
	C_1	C_0	C_{-1}		ΔNatten [5:0] (decimal)	ΔNmain [7:0] (decimal)	ΔtxX_eq_pre [5:0] (decimal)	ΔtxX_eq_post [5:0] (decimal)
1	Increment	Hold	Hold	NA	+1	0	0	-4
2	Decrement	Hold	Hold	Yes	-1	0	0	+4
3[a]	Decrement	Hold	Hold	No	0	0	0	0
4	Hold	Increment	Hold	Yes	-1	+4	0	0
5	Hold	Increment	Hold	No	0	0	0	0
6	Hold	Decrement	Hold	NA	+1	-4	0	0
7	Hold	Hold	Increment	NA	+1	0	-4	0
8	Hold	Hold	Decrement	Yes	-1	0	+4	0
9[a]	Hold	Hold	Decrement	No	0	0	0	0

a. These update requests result in a MIN/MAX indication to the link partner as per the status response shown in Table 5-15 on page 179 (“Status Response to TX update requests”). An alternate algorithm for avoiding this limitation is provided in section 5.10.2.2.1 on page 180 (“Alternate TX Coefficient Update Algorithm for 10GBASE-KR”).

The procedure for updating the TX coefficients must be followed in the sequence below:

1. Natten $[5:0]^{k+1}$ = Natten$[5:0]^k$ + ΔNatten
2. Nmain$[7:0]^{k+1}$ = Nmain$[7:0]^k$ + ΔNmain
3. txX_eq_pre$[5:0]^{k+1}$ = txX_eq_pre$[5:0]^k$ + ΔtxX_eq_pre
4. txX_eq_post$[5:0]^{k+1}$ = txX_eq_post$[5:0]^k$ + ΔtxX_eq_post
5. txX_eq_main$[5:0]^{k+1}$ = 40 - Natten$[5:0]^{k+1}$ - txX_eq_pre$[5:2]^{k+1}$ - txX_eq_post$[5:2]^{k+1}$

   Where k is the current coefficient value, k+1 is the next update.

The starting values of Nmain are equal to 4*txX_eq_main[5:0], where txX_eq_main[5:0] is txX_eq_main[5:0] as defined in Table 5-12 on page 177.

If simultaneous increment/decrement requests are received (as defined in Table 45-55, 10GBASE-KR Specification), they should be processed in the following order:

1. Increment C_1
2. Increment C_{-1}
3. Decrement C_0
4. Decrement C_1
5. Decrement C_{-1}
6. Increment C_0

To ensure that the PHY meets the transmitter signaling requirements in the 10GBASE-KR specification, the following transmitter coefficient constraints must be met before updating the coefficient inputs defined in Table 5-14 on page 178.

- 22*4 <= Nmain[7:0] <= 40*4
- 0 <= txX_eq_post[5:0] <= 15*4
- 0 <= txX_eq_pre[5:0] <= 7*4
- 0 <= Natten[5:0] <= 18
- 0 <= (Nmain[7:0] + txX_eq_pre[5:0] + txX_eq_post[5:0]) <= 40*4
- 4*4 <= (Nmain[7:0] - txX_eq_pre[5:0] - txX_eq_post[5:0])

Checks should be performed for each coefficient update request to determine if the next k+1 set of coefficients is valid. If a coefficient update is within the above limits then the change is accepted and an UPDATED response can be sent to the link partner. If a coefficient update hits one of the above limits, the change is accepted and a MIN or MAX response can be sent to the link partner. If a coefficient update tries to go outside the above limits then the update is not performed and the MIN/MAX previous response is maintained

Table 5-15 summarizes system status response after coefficient update requests are received.

Table 5-15 Status Response to TX update requests

Coefficient	Status Response to increment/decrement (if rules are violated or limits are hit)		Status Response to increment/decrement (if within rules and limits are not hit)
	increment	decrement	
C_1	MAX	MIN	UPDATED
C_0	MAX	MIN	UPDATED
C_{-1}	MAX	MIN	UPDATED

5.10.2.2.1 Alternate TX Coefficient Update Algorithm for 10GBASE-KR

The default coefficient update algorithm meets the voltage update specification provided in Table 72-7, 10GBASE-KR Specification (IEEE Standard 802.3-2012, IEEE). However, this default method will not maintain the signal amplitude when only performing C_{+1}/C_{-1} INC/DEC commands. To maintain amplitude, an alternate update algorithm is shown in Table 5-16.

Table 5-16 Alternate TX Coefficient Update Algorithm for 10GBASE-KR

Coefficient Update Request						
C_{+1}	C_0	C_{-1}	Delta Natten	Delta L_{main} [7:0]	Delta L_{pre} [5:0]	Delta L_{post} [5:0]
Increment	Hold	Hold	0	+2	0	-2
Decrement	Hold	Hold	0	-2	0	+2
Hold	Increment	Hold	-1	+4	0	0
Hold	Decrement	Hold	+1	-4	0	0
Hold	Hold	Increment	0	+2	-2	0
Hold	Hold	Decrement	0	-2	+2	0

In addition to maintaining the amplitude for C_{+1}/C_{-1} INC/DEC updates, this algorithm also provides more flexibility in the link adaptation as it allows the link adaptation to proceed for the cases listed on rows 3 and 9 of Table 5-14 ("Default TX Coefficient Update Sequence for 10GBASE-KR" on page 178) where a MIN/MAX response would otherwise be generated. For example, when the transmitter is starting with the initial setting of PRESET (see Table 5-12 on page 177, "TX Equalization Settings for 10GBASE-KR") and getting a request from link partner to decrement C_1. Instead of sending a MIN response as per Table 5-15 ("Status Response to TX update requests" on page 179), the alternate TX update algorithm allows the link adaptation to continue searching for best setting.

Although this method allows the link adaptation to continue with the search for best settings, the coefficient updates using this method may marginally not meet the minimum voltage update specification provided in Table 72-7, 10GBASE-KR Specification (IEEE Standard 802.3-2012, IEEE).

Note It is recommended that the alternate TX coefficient update algorithm be implemented in the external logic as a programmable feature. By default, the alternate update algorithm should stay disabled.

5.11 PHY Power-Down

The PHY vp* and vph* power supplies may be powered down in any order.

For more information about power gating the PHY, see the "Power Gating" section.

See the "Configuration for a Disabled PHY" section, if you intend to power down the PHY but leave the rest of the SOC powered up.

5.12 RX Equalization and Adaptation

The PHY supports receiver equalization, adaptation of receiver parameters, and commands for incrementing or decrementing the transmitter coefficients of the link partner. For protocols that require DFE and link adaptation (PCIe 4.0, SRIO-SR, and 10GBASE-KR), the receiver adaptation is triggered by asserting the rxX_adapt_req input. Upon completion of the receiver adaptation, the PHY asserts rxX_adapt_ack and uses one of the following two mechanisms to adapt the transmitter coefficients of the link partner:

1. Figure of Merit (FOM)—An 8-bit rxX_adapt_fom[7:0] output indicating the quality of the received data eye. A higher value of rxX_adapt_fom [7:0] indicates better link equalization, with 8'd0 indicating worst equalization setting and 8'd255 indicating the best equalization setting.
2. Direction change—rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], and rxX_txpost_dir[1:0] outputs providing the increment/decrement/hold requests for the transmitter coefficients of the link partner. The outputs rxX_adapt_fom[7:0], rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], and rxX_txpost_dir[1:0] outputs are only valid when rxX_adapt_ack is asserted. The maximum time from assertion of rxX_adapt_req to rxX_adapt_ack is dependent on the baud rate and reference clock frequency. The maximum time is based on rxX_adapt_mode[1:0] and the frequency of the reference clock input. Table 5-17 below shows RX adaptation time for various configurations.

Table 5-17 RX Adaptation Time

rxX_adapt_mode[1:0]	Use Refclk Frequency	Adaptation Time
0	PCIe 4 100 MHz	1005.4 µs
1	PCIe 3 100 MHz	1107.45 µs
2	Short channel (PCIe 4) 100 MHz	590.18 µs
3	Long 156.2 MHz	3939.14 µs

Receiver adaptation is only supported when in P0 state, electrical idle exit is detected on the RX serial lines, rxX_data_en is asserted, and the clock and data recovery (CDR) circuitry is nominally locked (indicated by the assertion of rxX_valid output. When rxX_data_en is enabled during equalization phase, it is possible that for some equalization settings, CDR may drift and not able to recover. As a result, it is recommended to assert rxX_reset for at least 10 ns between performing RX adaptation requests.

The recommended sequence is as follows.

1. Assert rxX_reset to the PHY.
2. Wait for de-assertion of rxX_ack from the PHY.
3. Transition the PHY lane receiver into P0 state.
4. Wait for detection of electrical idle exit condition on rxX_los.
5. Assert rxX_data_en to the PHY.
6. Wait for assertion of pipe_rxX_valid from the PHY.

7. Perform an RX adaptation request and assert rxX_adapt_req.

   The PHY performs RX adaptation, then signals the completion by asserting rxX_adapt_ack.

8. De-assert rxX_adapt_req to the PHY.

Note

- rxX_adapt_req should not be asserted unless rxX_dfe_bypass = 0
- rxX_adapt_req input should only be asserted after rxX_valid output is asserted.
- Aborting rxX_adapt_req by negating before rxX_adapt_ack is asserted or by asserting rxX_reset may leave the receiver with a less than optimal settings.No requests should be sent to the PHY using rxX_req/rxX_ack while rxX_adapt_req is asserted.
- The internal RX CDR PPM detector is disabled during adaptation requests. Any external PPM detectors must also be turned off during the adaptation process.

5.12.1 RX Link Adaptation using FOM

The RX link adaptation can be performed using the Figure of Merit calculation in the PHY.

Although this mechanism can be used for all protocols, it is recommended for link adaptation when a wide range of TX Presets are available for the link partner's TX.

Note FOM mechanism is not recommended for 10GBASE-KR links.

Example of FOM Mechanism

An example of using the FOM mechanism for PCIe 4.0 link adaptation is described below.

The PCIe MAC iterates through the 10 predefined transmitter presets and selects the preset for which the PHY returns the best Figure of Merit. The terminating condition for PCIe 4.0 link adaptation is for the MAC to go through a total of 12 presets (11 presets + 1 preset with best Figure of Merit). The maximum time taken by the PHY for receiver evaluation for each preset is dependent on the baud rate and reference clock frequency. The maximum time in PCIe 4.0 mode for 100 MHz reference clock input is approximately 1 millisecond. Therefore, the iterations through 11 transmitter presets for the link partner complete deterministically within the timeout limit of 24 milliseconds, as specified in the PCIe base specification.

A detailed sequence of events for PCIe 4.0 link partner transmitter adaptation is as follows:

1. PHY lane receiver is transitioned into P0 state.
2. Electrical Idle Exit condition is detected on rxX_los.
3. The PCIe 4 PCS asserts rxX_data_en.
4. MAC sets the link partner preset index N to 0, best preset index N_best = 0, best figure of merit
   a. FOM_best = 0, TxPreset = Preset(0).
5. The MAC requests the link partner to choose TxPreset.
6. The MAC waits for assertion of pipe_rxX_valid from the PCIe 4.0 PCS.
7. The MAC requests receiver evaluation to PCIe 4.0 PCS and PCIe 4.0 PCS asserts rxX_adapt_req.
8. The PHY performs receiver evaluation, signals completion of receiver evaluation by asserting
   a. rxX_adapt_ack and generating rxX_adapt_fom[7:0].
9. PCIe 4.0 PCS de-asserts rxX_adapt_req to the PHY, passes on Figure of Merit (FOM) information to the MAC.
10. If FOM > FOM_best, MAC sets N_best = N, FOM_best = FOM.
11. If N < 11, set N = N + 1, TxPreset = Preset(N), MAC goes to Step 5, else go to Step 12.
12. For N = 11, set N = 12, TxPreset = Preset(N_best), MAC goes to Step 5. For N = 12, exits the loop and proceeds to the subsequent Equalization phases mentioned in the PCIe base specification.

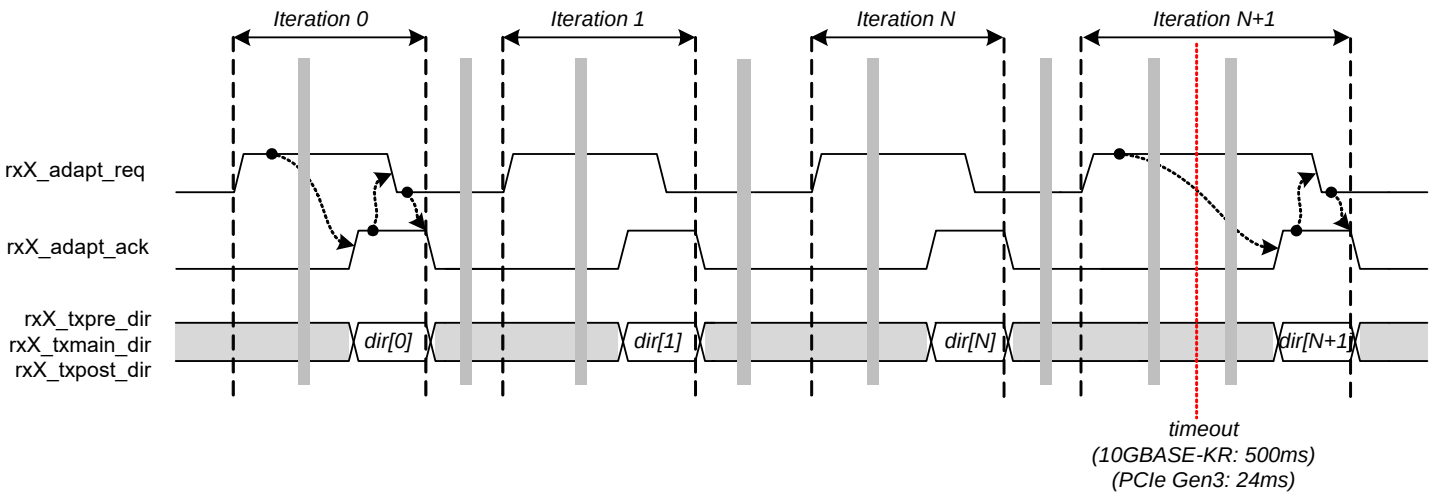
    When N= 12 and if there is time, many PCIe MACs will use the Preset(N_best) as a coarse remote tune and refine the adjustment by using RX Link Adaptation Using Direction Change (for more information, see section 5.12.2 on page 184).

Note During the phase when RX evaluation is in progress or when the link partner is changing its transmitter presets, the recovered data from the PHY can have significant bit errors.

5.12.2 RX Link Adaptation Using Direction Change

The RX link adaptation can also be performed using the Direction Change mechanism. This mechanism can be used for all protocols supporting link adaptation. The timing diagram for receiver evaluation using the direction change mechanism is shown in Figure 5-12.

Figure 5-12 Link Adaptation Using Direction Change Mechanism



The terminating condition for the link training algorithm should be a programmable time-out (for example, 500ms for 10GBASE-KR).

Direction change outputs (rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], rxX_txpost_dir[1:0]) are generated assuming the TX coefficients C_{-1} (pre-shoot) and C_1 (de-emphasis) are encoded as the absolute value of the actual FIR coefficient and thus incrementing or decrementing either value refers to the magnitude of the actual FIR coefficient. For example, incrementing de-emphasis (rxX_txpost_dir[1:0]) means increasing the amount of de-emphasis equalization on serial lines by sending a KR Decrement command. The main cursor (rxX_txmain_dir[1:0]) will never indicate an increment or decrement.

Note The PHY never generates an INC/DEC command for the TX main coefficient C_0; in other words, the PHY output rxX_txmain_dir[1:0] never generates 2'b01/2'b10.

It is recommended that you send the following direction updates for the pre-shoot (rxX_txpre_dir[1:0]=2'b01) and de-emphasis (rxX_txpost_dir[1:0]=2'b01) from the PHY to the link partner in a two-step sequence:

PHY Command	10GBASE-KR Link Command	
rxX_txpost_dir[1:0]=2'b01	Step 1	Decrement C_0
	Step 2	Decrement C_1
rxX_txpre_dir[1:0]=2'b01	Step 1	Decrement C_0
	Step 2	Decrement C_{-1}

This procedure can prevent scenarios where the link partner may not be able to accept equalization update commands and respond with a MIN/MAX status. For more information about the transmitter settings, see section 5.10.2.2.1 on page 180 ("Alternate TX Coefficient Update Algorithm for 10GBASE-KR").

Note It is recommended that the two-step sequence on the PHY RX commands be implemented in the external logic as a programmable feature. By default, this two-step sequence should be enabled.

Example of Direction Change Mechanism

An example of using the Direction Change mechanism for 10GBASE-KR link adaptation is described below. The sequence of events for 10GBASE-KR link partner transmitter adaptation is as follows:

1. PHY is transitioned into P0 state.
2. Electrical Idle Exit condition is detected on rxX_los.
3. The 10GBASE-KR PCS asserts rxX_data_en
4. The 10GBASE-KR PCS waits for assertion of rxX_valid output from the PHY.
5. The 10GBASE-KR PCS sets the link partner preset to the initial state for 10GBASE-KR training; in other words, TxPreset = INITIALIZE.
6. The 10GBASE-KR PCS resets link training iteration count i = 0, and starts the timer.
7. The 10GBASE-KR PCS performs link training iteration i, as follows:
   a. The 10GBASE-KR PCS requests receiver evaluation by asserting rxX_adapt_req.
   b. The Raw PCS performs receiver evaluation, and signals completion of receiver evaluation by asserting rxX_adapt_ack and generating rxX_txpre_dir[1:0], rxX_txmain_dir[1:0], rxX_txpost_dir[1:0]. This is collectively represented by dir[i] in Figure 5-12 on page 184.

Note FOM (rxX_adapt_fom[7:0]) is also valid when rxX_adapt_ack is asserted.

   c. The 10GBASE-KR PCS de-asserts rxX_adapt_req to the Raw PCS.
   d. At this point, the 10GBASE-KR PCS can optionally, after waiting for rxX_valid assertion, run a PRBS checker to check the status of the link.

e. If the timer has reached (or exceeded) the timeout condition (for example, 300ms) then go to Step 8 to exit the loop; else, continue below.

f. For each of pre, main, and post, if the corresponding result in dir[i] is INC or DEC, then the 10GBASE-KR PCS sends this request to the link partner for it to apply to its TX and go to Step g; else, if all of the coefficient dirs are HOLD (in other words, dir[i] = {HOLD, HOLD, HOLD}), then go to Step 8.

g. The 10GBASE-KR PCS waits for an acknowledgement response from the link partner for each coefficient request it sent (in other words, UPDATED, MAX or MIN).

h. Re-evaluate the link partner TX settings by performing a new iteration (for example, set i=i+1) and go to the top of Step 7.

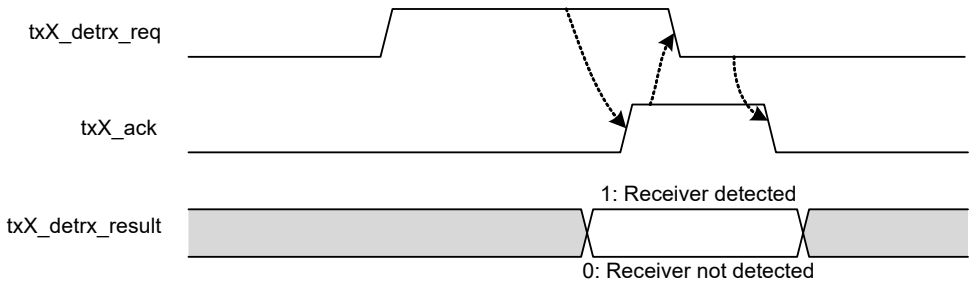
8. The link training algorithm has concluded and the link is considered “GOOD”. If the PRBS checker was used and it has errors from the last iteration then the link is declared “FAIL”.

Note that the time-out condition should only be checked after the rxX_adapt_ack has been asserted and before sending the resulting dir[i] to the link partner. This is needed to ensure a stable ending to the link training algorithm, with the local receiver and far-end transmitter in known and evaluated states.

5.13 Receiver Detection

Detection of a receiver (with terminations enabled) connected to the TX serial line txX_p/m on lane X in the PHY can be initiated in the TX state (txX_pstate) of P1 or P2 by asserting the input txX_detrx_req. The result of receiver detection is signaled on txX_detrx_result when txX__ack is asserted. The timing diagram of the 2-way handshake signals during receiver detection is shown in Figure 5-13.

Figure 5-13 Receiver Detection Handshake



During receiver detection, the common-mode voltage on txX_p/m is distorted and reverted back to nominal when txX_ack is asserted. The receiver detection enable signal txX_detrx_req must stay de-asserted in P0 and P0s states. The time taken by each lane to perform receiver detection (from txX_detrx_req assertion to txX_ack assertion) may vary across lanes to within one µs of other lanes.

5.14 Control Register Interface

The control registers in the PHY can be accessed through a Control Register (CR) Parallel register interface or a serial test interface (JTAG). These interfaces are defined in the following sections.

The PHY register space supports broadcast writes, as follows:

- If you set cr_para_addr[15:12] = 4'b1001, PHY broadcasts to all PMA lanes
- If you set cr_para_addr[15:12] = 4'b1010, PHY broadcasts to all RAW PCS lanes

For details, see Chapter 11, "Register Descriptions".

Refer to the section, 5.15 ("External SRAM Support" on page 192), for operation during SRAM update.

5.14.1 CR Parallel interface

The CR Parallel interface is a synchronous, 16-bit data/address parallel port provided for on-chip access to control registers inside the PHY. The CR Parallel interface is enabled when input cr_para_sel is asserted to 1. While access to these registers is not required for normal PHY operation, this interface is included for users that want to access some of the PHY's diagnostic features during normal operation or to override some of the PHY's control signals.



Wait a minimum of 10 phy_ref_dig_clk cycles if ref_clk_en = 1, or 10 cr_para_clk cycles if ref_clk_en = 0.

The CR Parallel interface consists of following PHY interface signals: cr_para_addr, cr_para_clk, cr_para_rd_data, cr_para_rd_en, cr_para_wr_data, cr_para_wr_en, and cr_para_ack; and is enabled when input cr_para_sel is set to 1.

All control registers can be accessed through either the CR interface or the internal firmware FSM operating on the cr_clk domain. A pulse-catching circuit captures the cr_para_clk read or write access in this clock domain. As a result, the cr_para_wr_en or cr_para_rd_en signal must be de-asserted for a minimum of seven phy_ref_dig_clk cycles between CR interface requests. Note that the CR interface cannot be accessed during assertion of phy_reset. However, any registers that need to be written before powering up the PHY can be done after de-asserting phy_reset but before de-asserting the rxX_reset/txX_reset signal.

A GLCM exists between the external cr_para_clk (or jtag_tck) and the internal reference clock for accessing registers inside the PHY. This GLCM is controlled by the ref_clk_en signal and always uses the internal reference clock when ref_clk_en is asserted. As a result, note that the CR interface latency might change when performing CR interface accesses while changing from one clock source to another. When power gating is enabled (pg_mode_en = 1'b1), the GLCM feature is disabled, and the cr_clk always operates on the internal reference clock.

When accessing the CR interface with the reference clock enabled, you must wait until at least 50 reference clock cycles have elapsed after phy_reset is de-asserted and ref_clk_en is asserted. Alternatively, you can wait for the phy_ref_dig_clk clock to start toggling. This waiting time is required for the clock to propagate through the internal PHY de-glitching circuit and to de-assert internal resets. When accessing the CR interface with the reference clock disabled, you must wait until at least 10 external CR clock cycles have elapsed (cr_para_clk or jtag_tck) after phy_reset has been de-asserted (or pg_reset has been de-asserted in the case of power-gating exit). In this case, the external clock needs to de-assert only internal resets.

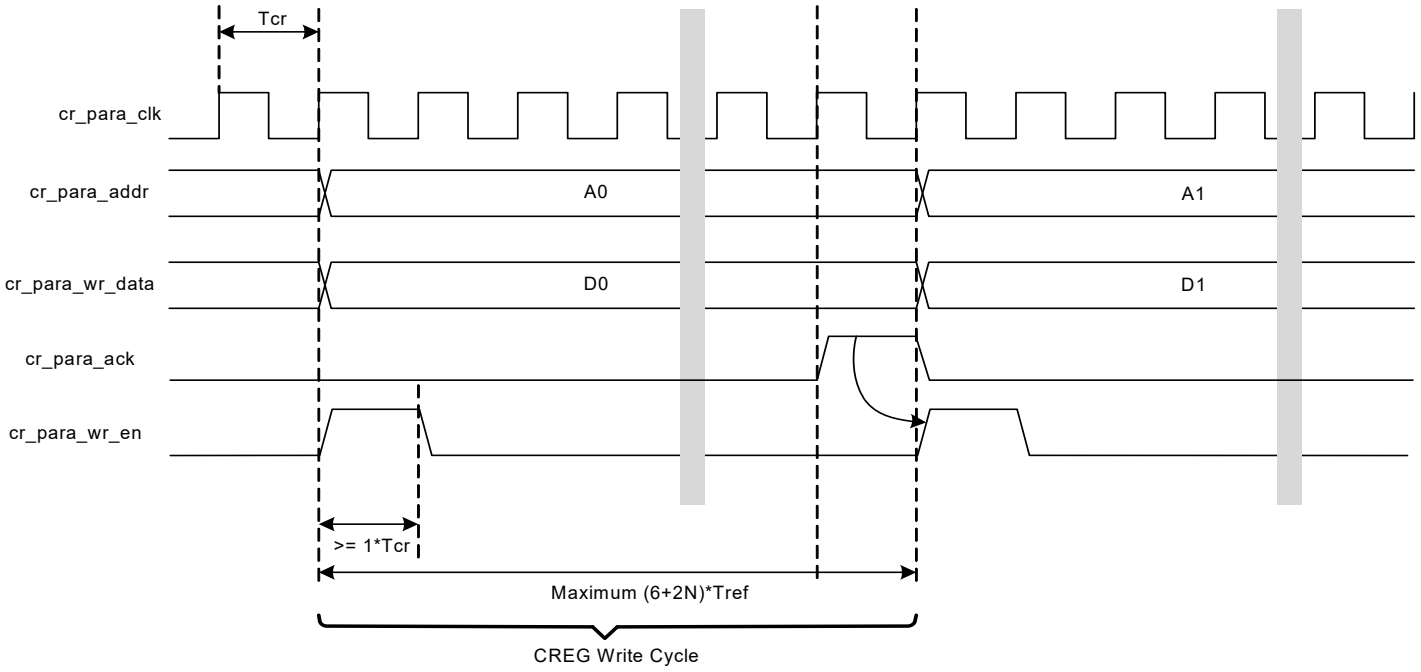
The timing diagrams of the CR Parallel register write and read operations are shown in Figure 5-14 and Figure 5-15, respectively. In these diagrams, “N” corresponds to number of lanes in the PHY; for example, for a 4-lane PHY, N=4. Parameters Tref and Tcr are defined below:

Tref: Time Period of phy_ref_dig_clk output clock, with a minimum value of 10ns.

Tcr: Time period of the cr_para_clk, with a minimum value of 10ns.

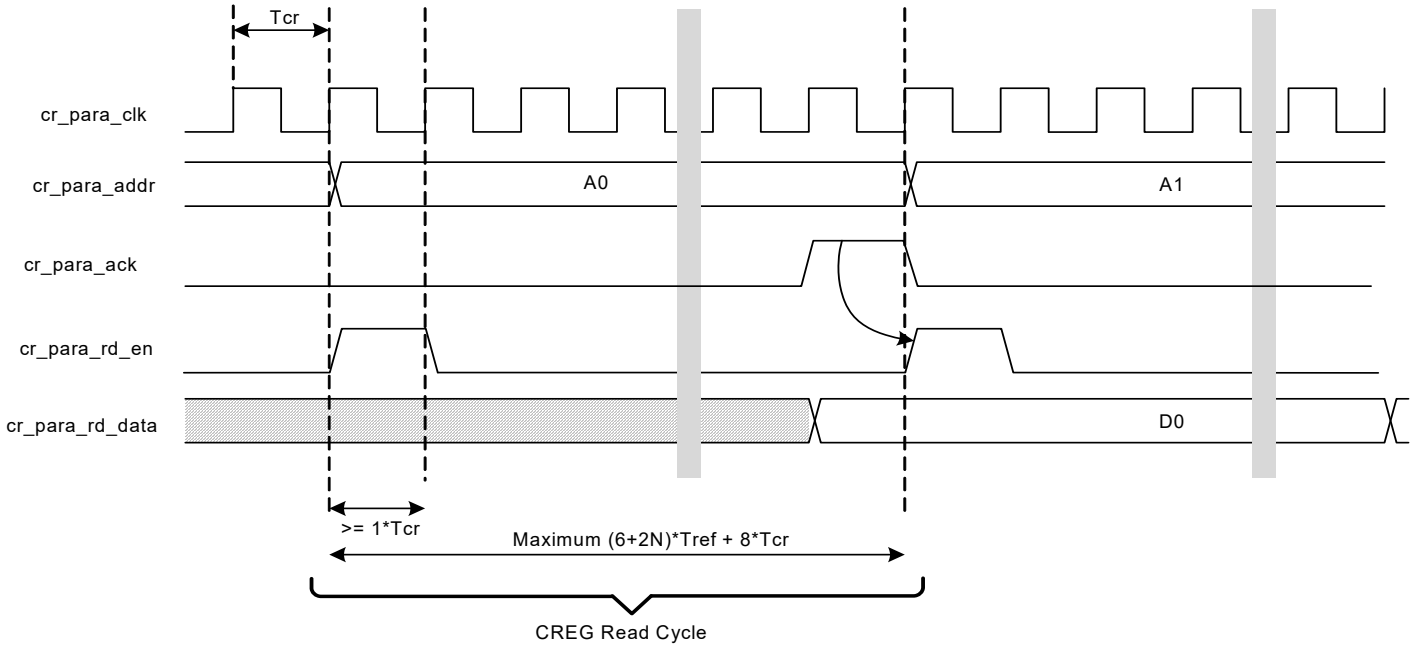
The completion of a write and read cycle is acknowledged by the assertion of the cr_para_ack output. The maximum times taken by write and read cycles are provided in Figure 5-14 and Figure 5-15.

Figure 5-14 CR Parallel Register Write Operation



The timing diagram of the CR Parallel register read operation is shown in Figure 5-15.

Figure 5-15 CR Parallel Register Read Operation



Note

The following registers require two back-to-back read operations to the same register in order to read the register value:

- LANEx_DIG_RX_DPLL_FREQ
- LANEx_DIG_RX_LBERT_ERR
- SUP_DIG_MPLLA_SSC_SS_FREQ_0
- SUP_DIG_MPLLA_SSC_SS_FREQ_1
- SUP_DIG_MPLLA_SSC_SS_PHASE
- SUP_DIG_MPLLB_SSC_SS_FREQ_0
- SUP_DIG_MPLLB_SSC_SS_PHASE
- SUP_DIG_MPLLB_SSC_SS_FREQ_1

5.14.2 JTAG interface

The JTAG interface in the PHY provides access to an internal TAP controller which implements the CRSEL instruction used for writing and reading registers in the PHY. The JTAG interface is enabled when input cr_para_sel is de-asserted to 0.

There is one JTAG interface per PHY, so for configurations where there are multiple PHYs aggregated (instantiated), there will be more than one JTAG interface. To control these with a single TAP controller, these multiple interfaces can be daisy chained externally.

5.14.2.1 IR Codes

Table 5-18 describes the supported JTAG instructions. The LSB of the IR and DR registers is the first bit shifted into jtag_tdi input.

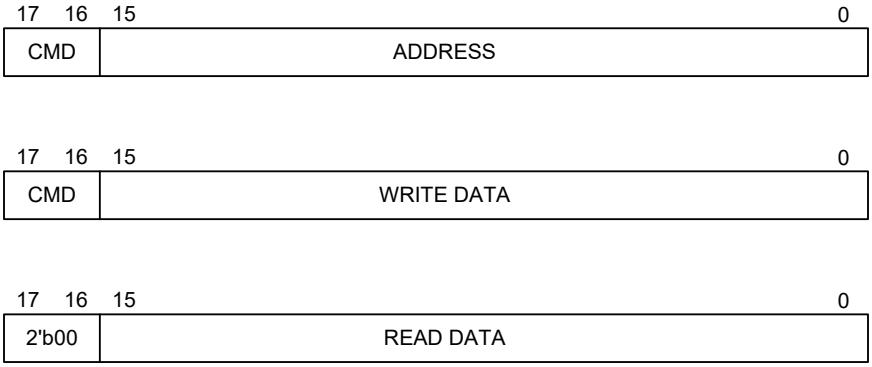
Table 5-18 Supported JTAG Instructions

Instruction	IR	DR	Operation
IDCODE	8'h01	32 bits	Shift out ID code. The 32-bit IDCODE can be found in the following registers: {SUP_DIG_IDCODE_HI, SUP_DIG_IDCODE_LO}.
CRSEL	8'h31	18 bits	Control register access
BYPASS	All others	1 bit	jtag_tdi to jtag_tdo bypass (bit is preloaded to 1)

5.14.2.2 CRSEL Instruction

When operating with the CRSEL instruction, internal registers can be accessed through the JTAG interface. When CRSEL instruction is selected, the shifted-in contents of the DR register are loaded into the CRSEL register during the UPDATE-DR phase of the JTAG state machine. The CRSEL register mapping when CRSEL instruction is selected as shown in Figure 5-16.

Figure 5-16 CRSEL Register Mapping



The CMD field determines what type of access will be performed. Table 5-19 shows the encodings which are defined for CMD field of the CRSEL register.

Table 5-19 Command Field Encodings for the CRSEL Register

CMD	Encoding	Description
ADDR	2'b00	Capture ADDRESS field for use in a subsequent WRITE command. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.
WRITE	2'b01	Write the value in the WRITE DATA field to the register selected by the address captured during the most recent ADDR or READ command. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.

Table 5-19 Command Field Encodings for the CRSEL Register (Continued)

CMD	Encoding	Description
DATA	2'b10	During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo. WRITE DATA value shifted in is unused.
READ	2'b11	Perform a read operation to the register selected by the ADDRESS field to make its current value available for subsequent CRSEL accesses. During subsequent SHIFT-DR state, shifts current value from most recently accessed register (via READ or WRITE) out on jtag_tdo.

To perform a control register read, shift a READ command into the CRSEL register along with the desired register address in the ADDRESS field. Any subsequent CRSEL access causes the data returned by the READ to be shifted out on jtag_tdo. This allows a limited form of pipelining of READ commands, as the second access may be shifting in the command and address for the subsequent READ while shifting out the data from the previous READ. To ensure reliable read operation, a minimum wait of 100 jtag_tck cycles in the run-test-idle state is needed. This wait is required for the register access to be transferred across clock domains and received by the JTAG controller successfully.

To perform a control register write, the ADDRESS of the register to be written must first be selected by performing either an ADDR or READ command and specifying the register address in the ADDRESS field. A second access using the WRITE command and specifying the data to be written in the WRITE DATA field causes the write to finally be performed to the desired register.

5.15 External SRAM Support

The Raw PCS stores the instruction code to execute the calibration and adaptation algorithms. In order to minimize the area of the Raw PCS, this pre-optimized and pre-validated code is stored in the form of hard-coded look up table. To provide flexibility of updating the code in silicon, an external SRAM interface is provided. Support of SRAM interface is mandatory.

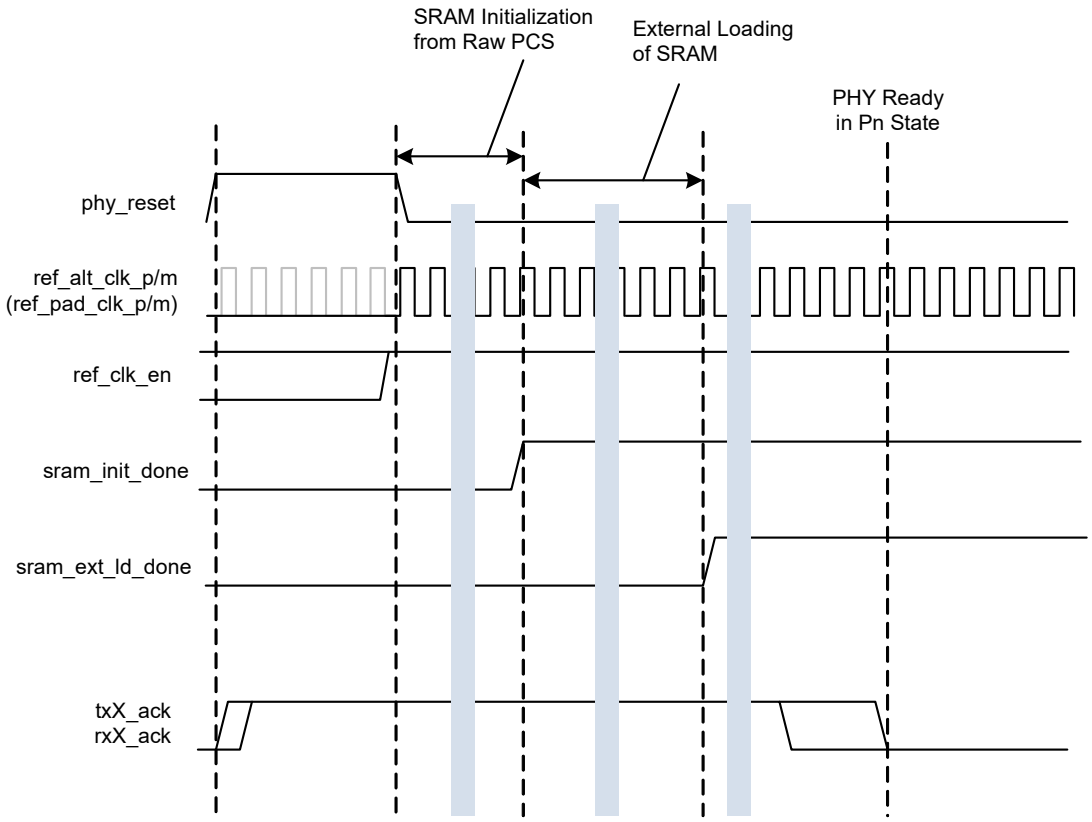
After de-assertion of phy_reset, the boot loader in the Raw PCS loads the code from the internal look up table to the external SRAM and asserts output signal, sram_init_done. Register access is not supported before assertion of sram_init_done. After initialization of SRAM, user can change the SRAM contents (or access any PHY register) using either the JTAG or the CR Parallel Interface by addressing the SRAM address space in the register map. After external access to the SRAM (or any other PHY register) is complete, input sram_ext_ld_done should be set high, allowing the FSMs in the Raw PCS to start executing the code from SRAM. For architecture diagram of the SRAM interface, refer to Figure 4-9 on page 154 (“Raw PCS Block Diagram”).



- SRAM address space is from “RAWMEM_DIG_RAM_CMN0_B0_R0” through “RAWMEM_DIG_RAM_CMN63_B7_R31”
- Mandatory SRAM size: 16K single port SRAM, consisting of 16-bit locations

Figure 5-17 shows the timing diagram of SRAM initialization and loading.

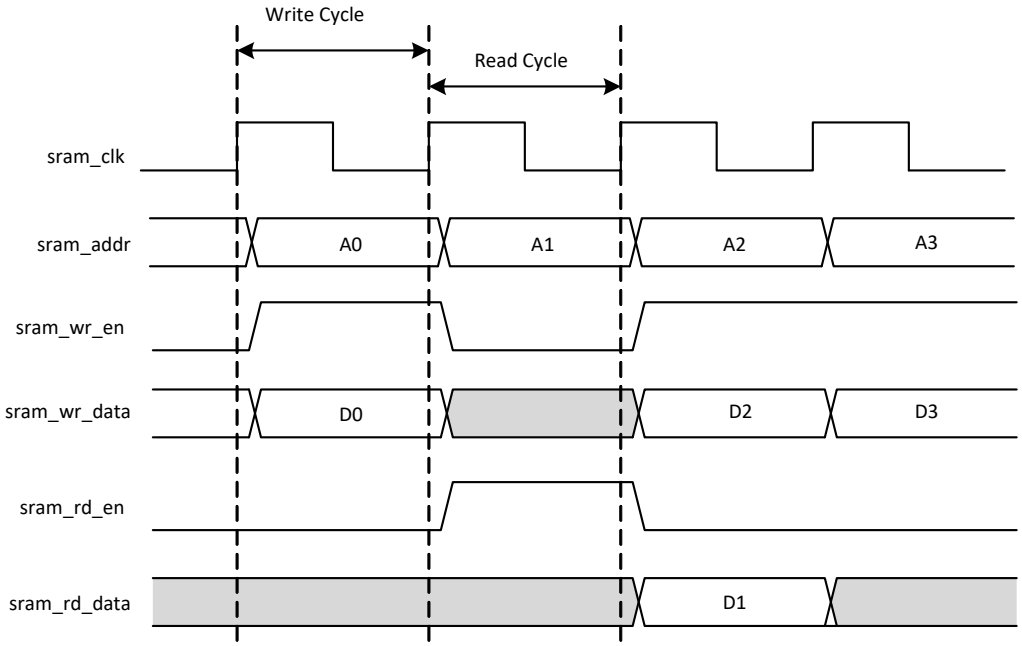
Figure 5-17 SRAM Initialization and Loading Sequence



When sram_bypass is asserted, the access to the SRAM is bypassed and the lane FSMs execute the code from the hard coded look up table in the Raw PCS. This signal is meant to be used only for debugging purposes and must not change after phy_reset is negated.

The timing diagram for the SRAM interface is shown in Figure 5-18.

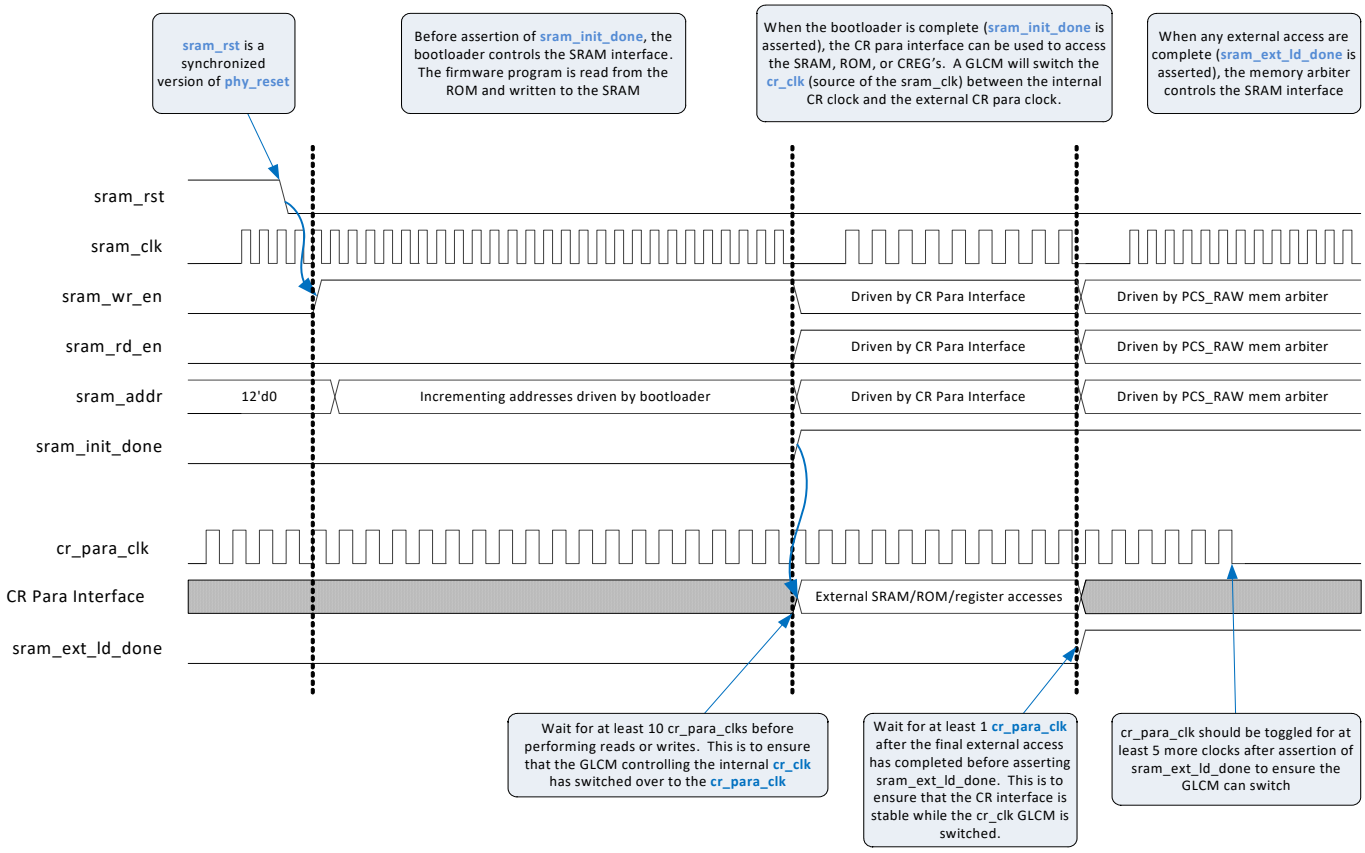
Figure 5-18 SRAM Read/Write Access Timings



Note Maximum clock speed requirement for the SRAM clock is 100 MHz.

The sequence of events on the SRAM interface as part of PHY power up are shown in Figure 5-19, if sram_bypass is negated. The SRAM contents after sram_init_done = 1 will be the same as in the file, `phy/include/*_pcs_raw_mem_rst.v`, used during synthesis.

Figure 5-19 Sequence of Events on the SRAM Interface



5.16 Test Modes

The test modes supported in the PHY and the input configuration settings for each mode are provided in Table 5-20.

Table 5-20 PHY Test Mode Selection

PHY Mode of Operation	bs_ce	test_burnin	test_powerdown	test_flyover_en	scan_mode	scan_shift scan_shift_cg
Boundary Scan (ACJTAG)	1	X	X	X	X	X
Burn-In	0	1	0	0	0	0
IDDQ	0	0	1	0	0	X
Flyover	0	0	0	1	0/1	0/1
Scan (At-Speed, Stuck-At)	0	0	0	0/1	1	0/1
Diagnostics (serial and parallel loopback)	0	0	0	0	0	0
Normal Operation	0	0	0	0	0	0

The following sections describe each test mode in more detail.

5.16.1 Boundary Scan (ACJTAG)

The PHY includes boundary scan registers per lane, which provide direct controllability and observability of the txX_p/m and rxX_p/m lines. The boundary scan instructions supported in the PHY are SAMPLE, PRELOAD, BYPASS, EXTEST_PULSE and EXTEST_TRAIN as defined in IEEE Std 1149.6-2003.

The states of all inputs of the PHY except bs_* and power supplies are ignored in boundary scan mode.

Figure 5-20 shows the logical structure of the boundary scan implementation in the N-lane PHY. The TX driver control logic is similar to the Output Driver Example defined in Sec 6.3.2. IEEE Std 1149.6-2003. The TX common mode is enabled in boundary scan mode. There is no single bit register bypass logic within the boundary scan logic of the PHY, but it can be added externally if required.

Table 5-21 on page 198 ("ACJTAG - Low Swing Mode Trip Levels") shows the low and big swing mode trip levels for the ACJTAG Input Receiver. This table provides the minimum and maximum trip levels over PVT corners for each setting. The low swing mode and big swing mode are intended for half swing (that is, 400-600 mVpp) and full swing (800-1200 mVpp) protocols, respectively. This table is only a reference, and the choice of trip level is determined based on customer's boundary scan implementation.



Note For ACJTAG DC Mode, it is recommended to use bs_rx_level[4:0]=5b'01100 (decimal 12) for a DC trip level between 85 mV and 115 mV.

Figure 5-20 Boundary Scan Structure

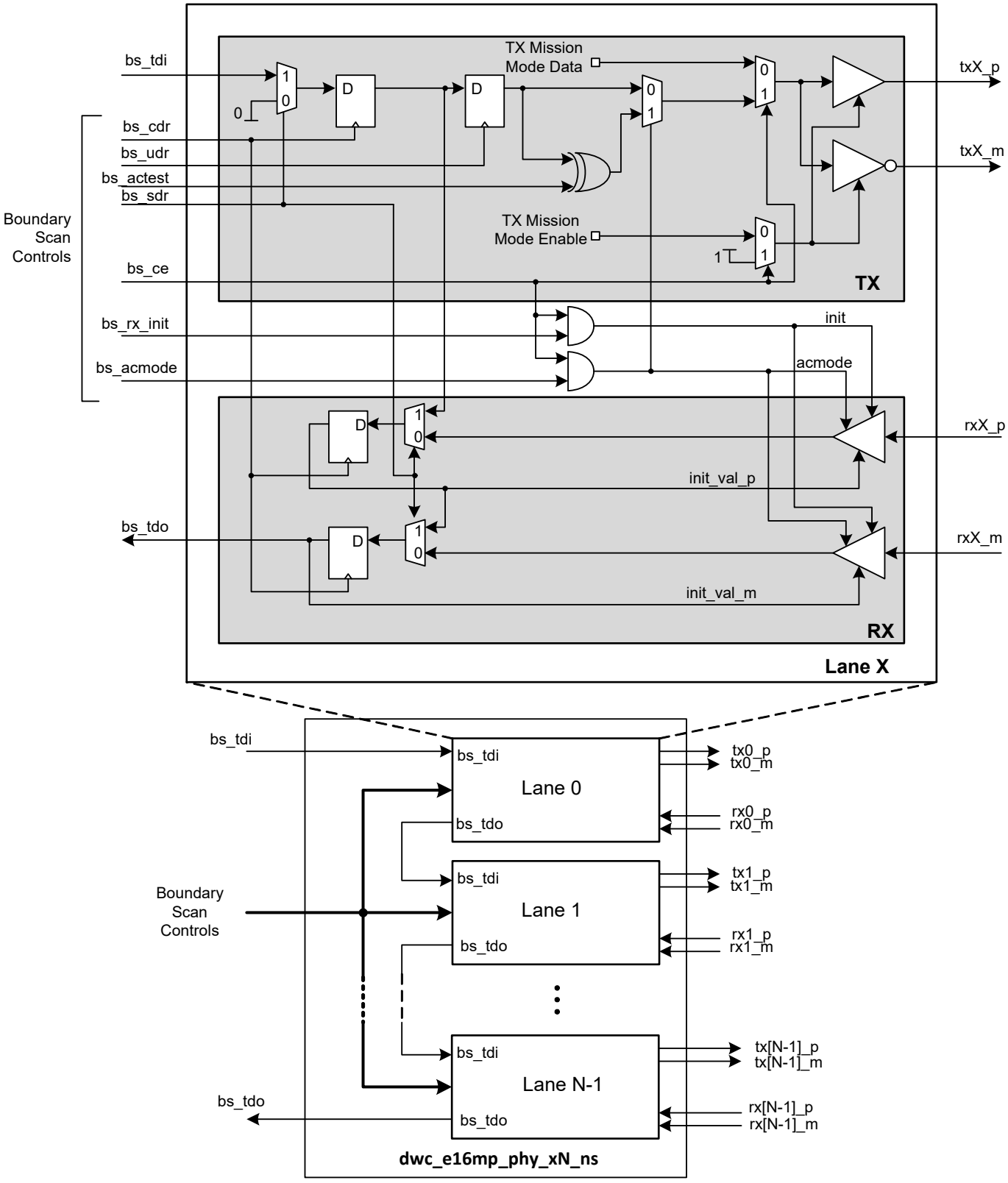


Table 5-21 ACJTAG - Low Swing Mode Trip Levels

bs_rx_level[4:0]	bs_rx_bigswing	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
00101	0	223	295
00110	0	242	328
00111	0	267	362
01000	0	283	396
01001	0	299	432
01010	0	314	468
01011	0	330	504
01100	0	355	541
01101	0	371	579
01110	0	396	617
01111	0	413	655

Table 5-22 ACJTAG - Big Swing Mode Trip Levels

bs_rx_level[4:0]	bs_rx_bigswing	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
00100	1	238	376
00101	1	269	413
00110	1	297	453
00111	1	323	490
01000	1	348	429
01001	1	370	570
01010	1	394	612
01011	1	416	654
01100	1	438	698
01101	1	461	742
01110	1	484	788

Table 5-22 ACJTAG - Big Swing Mode Trip Levels (Continued)

bs_rx_level[4:0]	bs_rx_bigswing	Minimum Trip Level (mVppd)	Maximum Trip Level (mVppd)
01111	1	506	836

5.16.2 Burn-In

In burn-in mode, all analog blocks are powered up, all clocks are enabled, the analog front end is put into TX-to-RX loopback mode, and the TX pattern generator and RX pattern matcher are enabled. This mode turns on an on-chip oscillator that is used as a reference clock for the MPLLs. This mode is intended for use during high-temperature operating life (HTOL) or Burn-in, and sets up the PHY to toggle as much circuitry as possible (for realistic hot carrier injection degradation - HCI), while unused circuitry is carefully biased in its power-down mode to ensure realistic degradation for NBTI (negative bias temperature instability).

Note

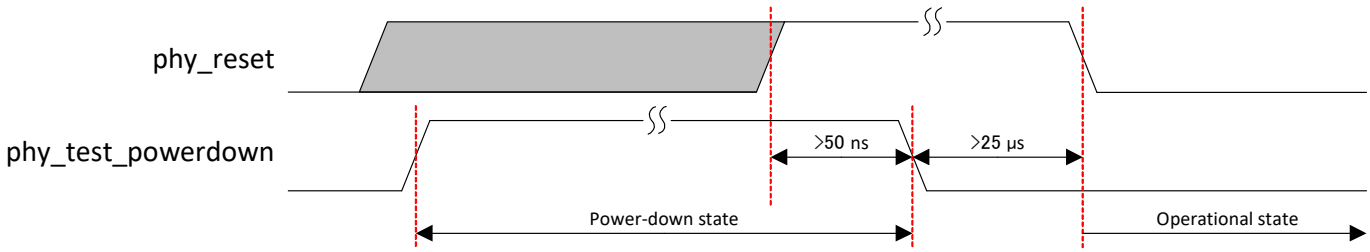
- Prior to enabling the burn-in mode by asserting test_burnin, phy_reset must be asserted and de-asserted to ensure the states of internal registers are reset.
- In burn-in mode:
  - vph_nominal[1:0] should be driven to 2'b11 to indicate that the VPH supply is elevated.
  - Synopsys recommends terminating the TX output with single ended AC coupled 50 Ω terminations to ground

5.16.3 IDDQ

In IDDQ test mode, all the analog blocks are powered down to allow leakage measurement. Input settings for enabling the IDDQ mode are provided in Table 5-20 on page 196. States of all other inputs are ignored.

It is recommended that phy_reset be asserted for at least 50 ns before test_powerdown is negated and remain asserted for at least 25 µs.

Figure 5-21 PHY Power-Down Sequence



5.16.4 Flyover

Settings to enable flyover mode are provided in Table 5-20 on page 196.

The flyover mode can be used for the following functions:

- Send scan data on and off die through the PHY TX and RX IO pins
- Output of high frequency clocks and data for observation and debug purposes

On the TX-side, the flyover mode allows you to asynchronously send an on-die signal off-chip from the txX_flyover_data_p and txX_flyover_data_m ASIC-side inputs to the txX_p and txX_m outputs. In this mode, a reference clock is not required.

On the RX-side, the flyover mode allows you to asynchronously send an off-chip signal on-die through the rxX_p and rxX_m inputs to the rxX_flyover_data_p and rxX_flyover_data_m ASIC-side outputs. In this mode, a reference clock is not required as the signal is received asynchronously using the test receiver.

The flyover mode supports up to 200 MHz single ended data on both the RX and the TX data paths. It also support up to 1 GHz TX output when driven with a complementary signal.

5.16.5 Scan (At-Speed, Stuck-At)

Settings to enable scan mode are provided in Table 5-20 on page 196. To achieve the coverage metrics provided below, all other inputs must be controllable and all outputs must be observable.



PMA .lib ETM modes:

- **NORMAL**—For Functional mode
- **SCAN**—For Stuck-at Shift and Capture mode
  In this mode, Shift and Capture happens at Scan Clock Frequency.
- **SCAN_ASST**—For At Speed Scan Testing

In this mode, Shift occurs at Scan Clock Frequency, whereas Capture occurs at Functional mode frequency.

Note the following scan features and requirements in the PHY:

- The PHY provides independent scan clock inputs per internal clock domain.
- Primary clock inputs to the PHY must have scan clocks supplied to them externally in scan mode.
- All clock outputs from the PHY, except ref_repeat_clk_p/m, have a scan clock input muxed on them.
- Synchronous inputs and outputs are at-speed scannable.
- All scan chains can be shifted in/out at the same time. By default, only one-hot scan clock capture is supported. Customers have the option to externally group the scan clocks, which are logically independent and share the same frequency. The ATPG flow in the deliverables provide an optional static clock grouping of the scan clocks. By default, no clock grouping is enabled for the vectors and scripts that are provided with the product. For more information on enabling clock grouping, see `./pma/atpg/<metal-stack>/readme_tmax.txt` file.

- All resets (primary input resets or internally generated resets) are over-ridden to primary scan reset input scan_set_rst in scan mode.
- Maximum scan shift frequency is 100 MHz.
- Maximum scan chain length is 100 flops.
- All scan chains are positive-edge triggered.
- All scan chains are terminated with lockup latches.
- All analog blocks are powered down in scan mode. To prevent propagation of logic X's in the scan chain, the outputs of all the analog blocks feeding the digital scannable logic are over-ridden to known states.
- Stuck-At-Fault (SAF) coverage of the PMA is greater than 98%.
- Transition-Delay-Fault (TDF) coverage of the PMA is greater than 85%.
- Coverage of the soft-macro (Raw PCS) is place-and-route implementation dependent.

5.16.6 Diagnostics

The PHY provides the following diagnostic features to enhance test bench characterization and ATE testing.

- Loopback functions (independent per-lane)
- TX-to-RX serial (pre-driver) loopback
- RX-to-TX parallel loopback
- BERT (independent per-lane)
- 7th, 9th, 11th, 15th, 16th, 23rd, and 31st order polynomial pattern generation and recognition
- Generation of simple test patterns
- Error insertion and detection

5.16.6.1 Loopback Functions

Each lane in the PHY supports a TX-to-RX and an RX-to-TX loopback mode. Only one loopback mode can be enabled at any given time. The following conditions must be met prior to enabling the loopback modes:

- Input settings must be set to values provided in Table 5-20 on page 196
- phy_reset must be asserted and then de-asserted
- Reference clock input to the PHY must be supplied
- MPLLA/B configuration settings must correspond to the required baud rate (for more information, see "Configuring the PHY" on page 217); this can be done by controlling primary inputs or accessing registers.
- Set the signals in Table 5-23 accordingly; this can be done by controlling primary inputs or accessing registers.

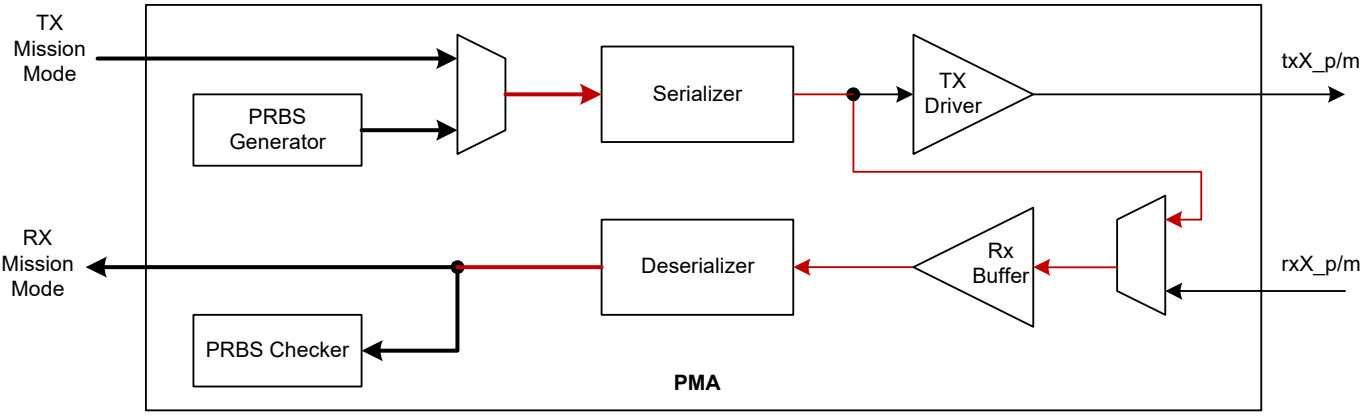
Table 5-23 Loopback Function Settings

Signal	Setting
txX_reset rxX_reset txX_lpd rxX_lpd	De-assert for the corresponding lane X
txX_pstate rxX_pstate	Set to P0
txX_rate[2:0] txX_width[1:0] txX_mpllb_sel rxX_rate[2:0] rxX_width[1:0]	Set to required baud rate (for more information, see “Configuring the PHY” on page 217

TX-to-RX Serial (Pre-Driver) Loopback

The TX-to-RX loopback path is shown in Figure 5-22.

Figure 5-22 TX-to-RX Serial (pre-driver) Loopback



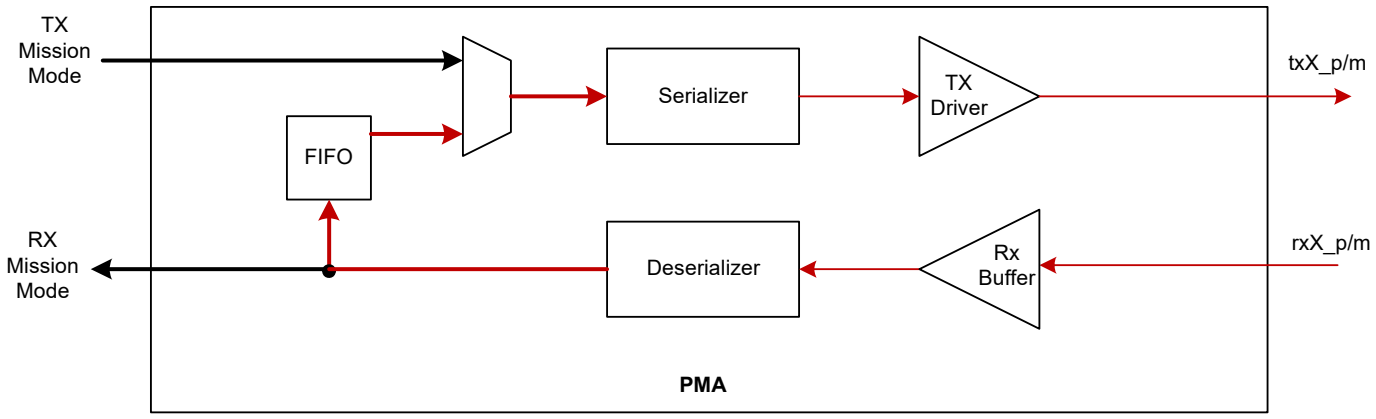
This mode uses either the user logic or the built-in PRBS Generator/Checker to generate and check the test data. The serial loopback path to the RX Buffer is after the TX pre-driver (before the TX driver). Using this mode, you can verify the operation for all enabled PCS and PMA functional blocks in the transmitter and receiver channel.

The TX-to-RX loopback mode is enabled by asserting the input laneX_tx2rx_ser_lb_en, and enabling the TX and RX data paths by asserting txX_data_en and rxX_data_en input (also accessible through over-ride registers LANEx_DIG_ASIC_LANE_OVRD_IN, LANEx_DIG_ASIC_TX_OVRD_IN_0, and LANEx_DIG_ASIC_RX_OVRD_IN_0).

RX-to-TX Parallel Loopback

The RX-to-TX parallel loopback path is shown in Figure 5-23.

Figure 5-23 RX-to-TX Parallel Loopback



The RX-to-TX parallel loopback mode is enabled by asserting the input laneX_rx2tx_par_lb_en, and enabling the TX and RX data paths by asserting txX_data_en and rxX_data_en (also accessible through override registers LANEx_DIG_ASIC_LANE_OVRD_IN, LANEx_DIG_ASIC_TX_OVRD_IN_0, and LANEx_DIG_ASIC_RX_OVRD_IN_0). In this loopback mode, the received recovered data rxX_p/m is transmitted back on the txX_p/m outputs.

To prevent errors that may get introduced in the RX to TX loopback data because of the frequency deviation between the local MPLLs and the external transmitter, the reference clock input to the PHY must be derived from the same source as the reference clock used by the external transmitter feeding data on rxX_p/m. For cases where txX_rate and rxX_rate are different (for example, SGMII in bifurcated mode ("Settings for 1000BASE-KX/SGMII (Bifurcated Usage)" in "Lane Configuration Settings"), txX_rate must be set equal to rxX_rate for RX to TX loopback to work. This is because in RX to TX loopback mode, RX recovered clock is used as transmit clock.

5.16.6.2 BERT

The Byte Error Rate Tester (BERT) comprises two independently programmed modules: a pattern generator and a pattern matcher/error counter.

5.16.6.2.1 BERT Pattern Generator

Table 5-24 provides a list of patterns that can be generated with the built-in pattern generator.

Table 5-24 Pattern Generator Mode Control

LANEx_DIG_TX_LBERT_CTL[3:0]	Description
4'b0000	Pattern generator is disabled
4'b0001	LFSR31: $x^{31} + x^{28} + 1$
4'b0010	LFSR23: $x^{23} + x^{18} + 1$
4'b0011	LFSR23: $x^{23} + x^{21} + x^{16} + x^{8} + x^{5} + x^{2} + 1$

Table 5-24 Pattern Generator Mode Control (Continued)

LANEx_DIG_TX_LBERT_CTL[3:0]	Description
4'b0100	LFSR16: $x^{16} + x^5 + x^4 + x^3 + 1$
4'b0101	LFSR15: $x^{15} + x^{14} + 1$
4'b0110	LFSR11: $x^{11} + x^9 + 1$
4'b0111	LFSR9: $x^9 + x^5 + 1$
4'b1000	LFSR7: $x^7 + x^6 + 1$
4'b1001	Fixed word (PAT0)
4'b1010	DC balanced word (PAT0, ~PAT0)
4'b1011	Fixed pattern (000, PAT0, 3ff, ~PAT0}

The desired 10-bit pattern PAT0 can be set by writing to the pattern generator control register LANEx_DIG_TX_LBERT_CTL[14:5]. Register bits LANEx_DIG_TX_LBERT_CTL[14:5] are used for 20-bit mode and bits LANEx_DIG_TX_LBERT_CTL[12:5] are used for 16-bit mode. The pattern generator only operates in either 20-bit or 16-bit mode. The mode selected depends on the transmit data width, which is selected using input txX_width or by writing into the control register LANEx_DIG_ASIC_TX_OVRD_IN_0. When the transmit data is set to 20-bit or 10-bit, then the pattern generator is set to 20-bit mode. When the transmit data is set to 16-bit or 8-bit, then the pattern generator is set to 16-bit mode.

Note The PAT0 data pattern has the following restriction:
Patterns of all 0's or all 1's should not be used for “Fixed word” and “Fixed pattern” modes

A single-bit error can be inserted into the LSB of a transmitted word by writing a 1 on the register LANEx_DIG_TX_LBERT_CTL[4].

5.16.6.2.2 BERT Pattern Matcher

The pattern matcher can detect and synchronize to errored bytes in multiple types of patterns. Errored symbols are counted in the error counter. The pattern matcher only operates in either 20-bit or 16-bit mode. The mode selected depends on the receive data width which is selected using input rxX_width or by writing into the control register LANEx_DIG_ASIC_RX_OVRD_IN_0. When the receive data is set to 20-bit or 10-bit, then the pattern matcher is set to 20-bit mode. When the receive data is set to 16-bit, then the pattern matcher is set to 16-bit mode.

For 20-bit mode, error detecting word size is 10 bits; bits 0 to 9 and bits 10 to 19. For 16-bit mode, the error detecting word size is 8 bits; bits 0 to 7 and bits 8 to 15.

Note that the controls for the pattern generator and pattern matcher within a lane are independent.

Table 5-25 describes the LANEx_DIG_RX_LBERT_CTL[3:0] field, which selects the expected pattern and operating mode.

Table 5-25 Pattern Matcher Mode Control

LANEx_DIG_RX_LBERT_CTL[3:0]	Description
4'b0000	Disabled
4'b0001	LFSR31: $x^{31} + x^{28} + 1$
4'b0010	LFSR23: $x^{23} + x^{18} + 1$
4'b0011	LFSR23: $x^{23} + x^{21} + x^{16} + x^8 + x^5 + x^2 + 1$
4'b0100	LFSR16: $x^{16} + x^5 + x^4 + x^3 + 1$
4'b0101	LFSR15: $x^{15} + x^{14} + 1$
4'b0110	LFSR11: $x^{11} + x^9 + 1$
4'b0111	LFSR9: $x^9 + x^5 + 1$
4'b1000	LFSR7: $x^7 + x^6 + 1$
4'b1001	d[n] = d[n-10]
4'b1010	d[n] = !d[n-10]
4'b1011	d[n] = !d[n-20]

The pattern matcher generates the expected pattern and synchronizes the generated pattern to the incoming pattern. Synchronization and error counting are initiated by asserting the sync bit (LANEx_DIG_RX_LBERT_CTL[4]).

The error counter in the pattern matcher is 22 bits wide. You can read the error count result from the LANEx_DIG_RX_LBERT_ERR register, the content of which is {lbert_err_ovbit14, lbert_err_cnt[14:0]}. The contents of LANEx_DIG_RX_LBERT_ERR are presented according to the following rules:

- When the error count is < 2^{15}, the bottom 15 bits are presented in the count field (lbert_err_cnt[14:0]) and the lbert_err_ovbit14 field is de-asserted.
- When the error count is < 2^{22} but >= 2^{15}, the top 15 bits are presented in the count field (lbert_err_cnt[14:0]) and the lbert_err_ovbit14 field is set. This result effectively scales the contents down by a factor of 2^7.
- When the 22-bit error counter overflows, the count field (lbert_err_cnt[14:0]) is set to 0x7FFF, and the lbert_err_ovbit14 field is asserted.

The pattern generator and pattern matchers are extensively used as part of ATE testing. Although all of the functional ATE tests are performed using register accesses, some inputs need to be driven to specific values as shown below:

- phy_reset = 0 (after subjecting the DUT though POR sequence)
- scan_mode = 1'b0
- rxX_adapt_req= 1'b0
- rxX_disable= 1'b0

- rxX_req=1'b0
- txX_disable= 1'b0
- txX_req=1'b0
- txX_detrx_req=1'b0
- res_req_in = 0
- phy*_bs* inputs = 0
- phy*_jtag_trst_n = 1
- phy_test_powerdown =0
- phy_test_burnin = 0

5.17 Aggregation and Bifurcation Support

Combining different instances of the PHY to operate as a common link is referred to as aggregation of PHY. Splitting different lanes in the same PHY across different links is referred to as bifurcation of PHY. The PHYs support aggregation and bifurcation, as well as a combination of both aggregation and bifurcation.

Ensure that the txX_master_mpll[a.b]_state inputs of lane X is connected to the mpll_[a,b]_state outputs of the same PHY from which the txX_clk input of the corresponding lane is derived from. For lanes belonging to the same link, ensure that the txX_clk inputs are derived from the same source and phase skew balanced to within 200ps.

Figure 5-24 and Figure 5-25 show connectivity examples of one x2 PHY that is configured to operate as one x2 link and two x1 links, respectively.

Figure 5-24 x2 PHY Configured as x2 Link

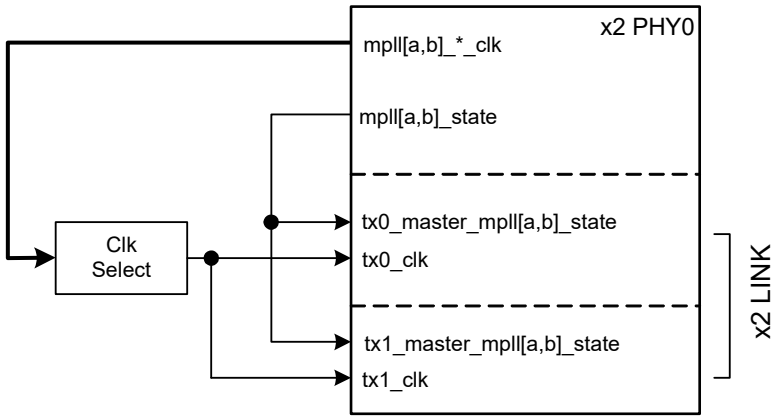


Figure 5-25 x2 PHY Configured as Two x1 Links

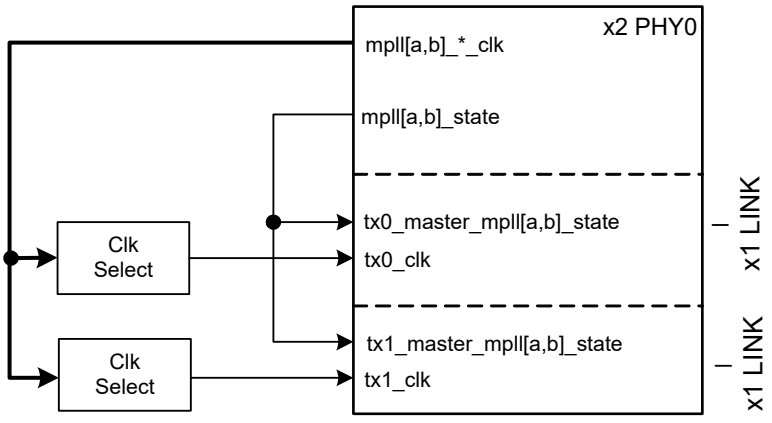


Figure 5-26 shows connectivity example of two x2 PHYs that are configured to operate as one x4 link.

Figure 5-26 Two x2 PHYs Configured as One x4 Link

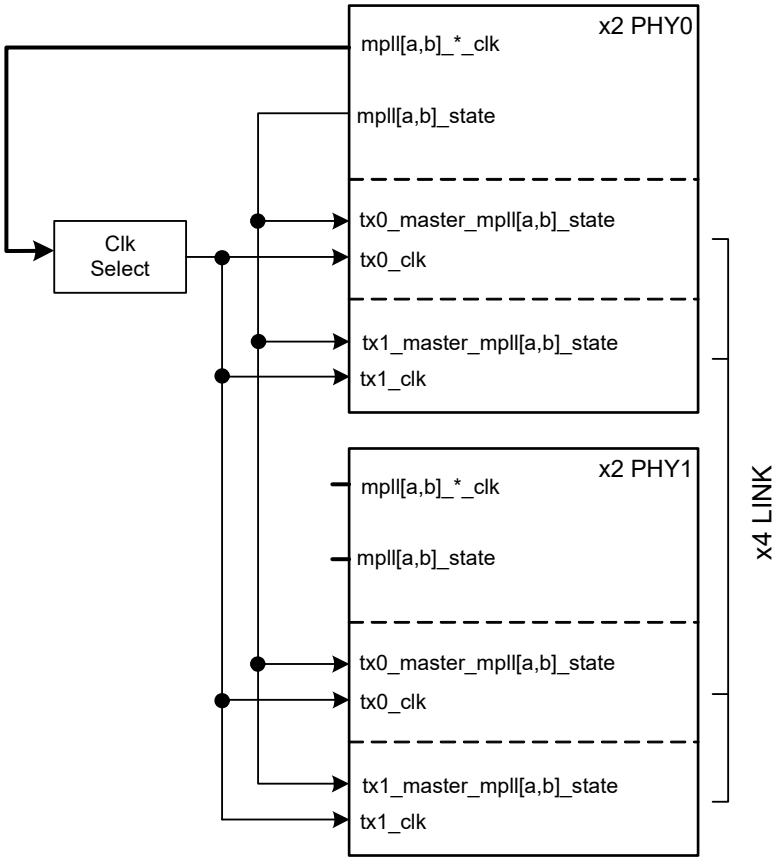
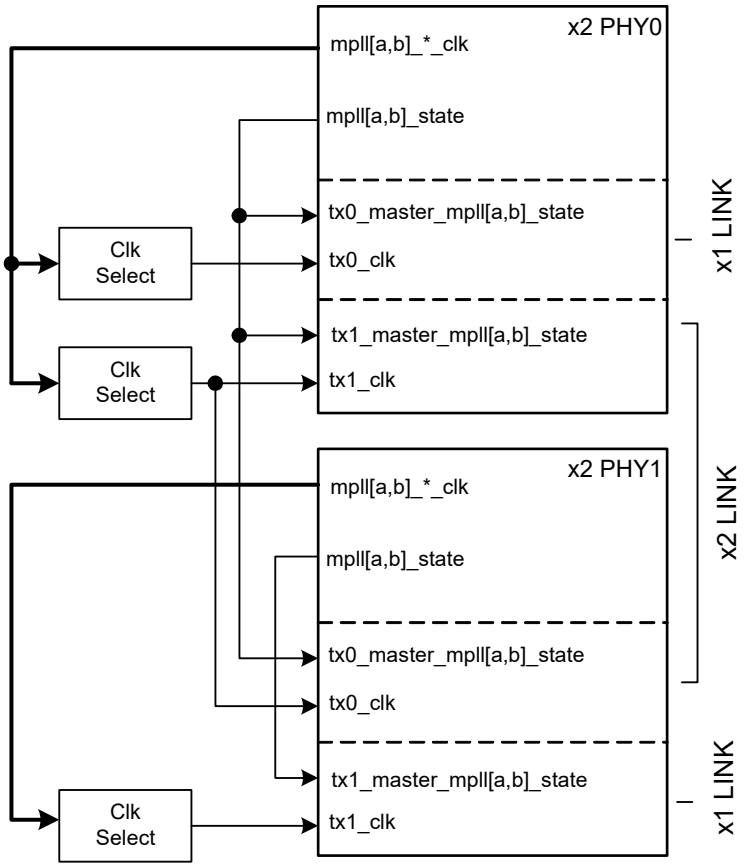


Figure 5-27 shows connectivity example of aggregation and bifurcation where two x2 PHYs are configured to operate as two x1 links and one common x2 link split across the two PHYs.

Figure 5-27 Two x2 PHYs Configured for Aggregation and Bifurcation (Two x1 Links and One x2 Link Split Across Two PHYs)



5.18 Power Gating

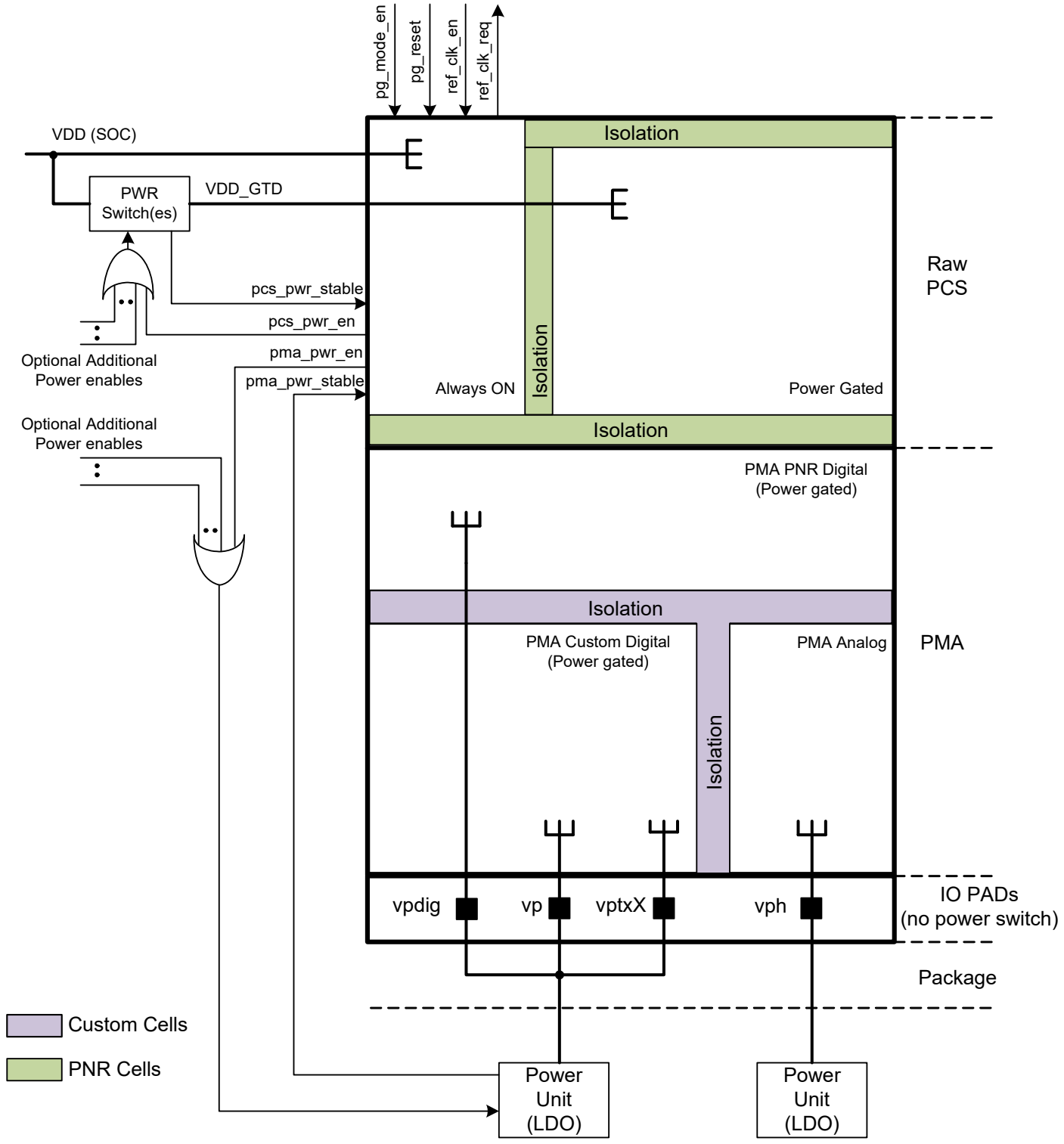
The PHY supports an optional power gating feature through control and status signals for external power switches for the Raw PCS logic and the PMA vp, vptxX, and vpdig supplies.

Note

- The PHY does not include internal power gating switches. Power gating is only supported using external power switches.
- Signals ref_repeat_clk_{p,m}, ref_dig_fr_clk, and phy_ref_dig_clk may be powered down when power gating is enabled. Do not use these signals for logic or PHYs meant to remain operational during power gating.

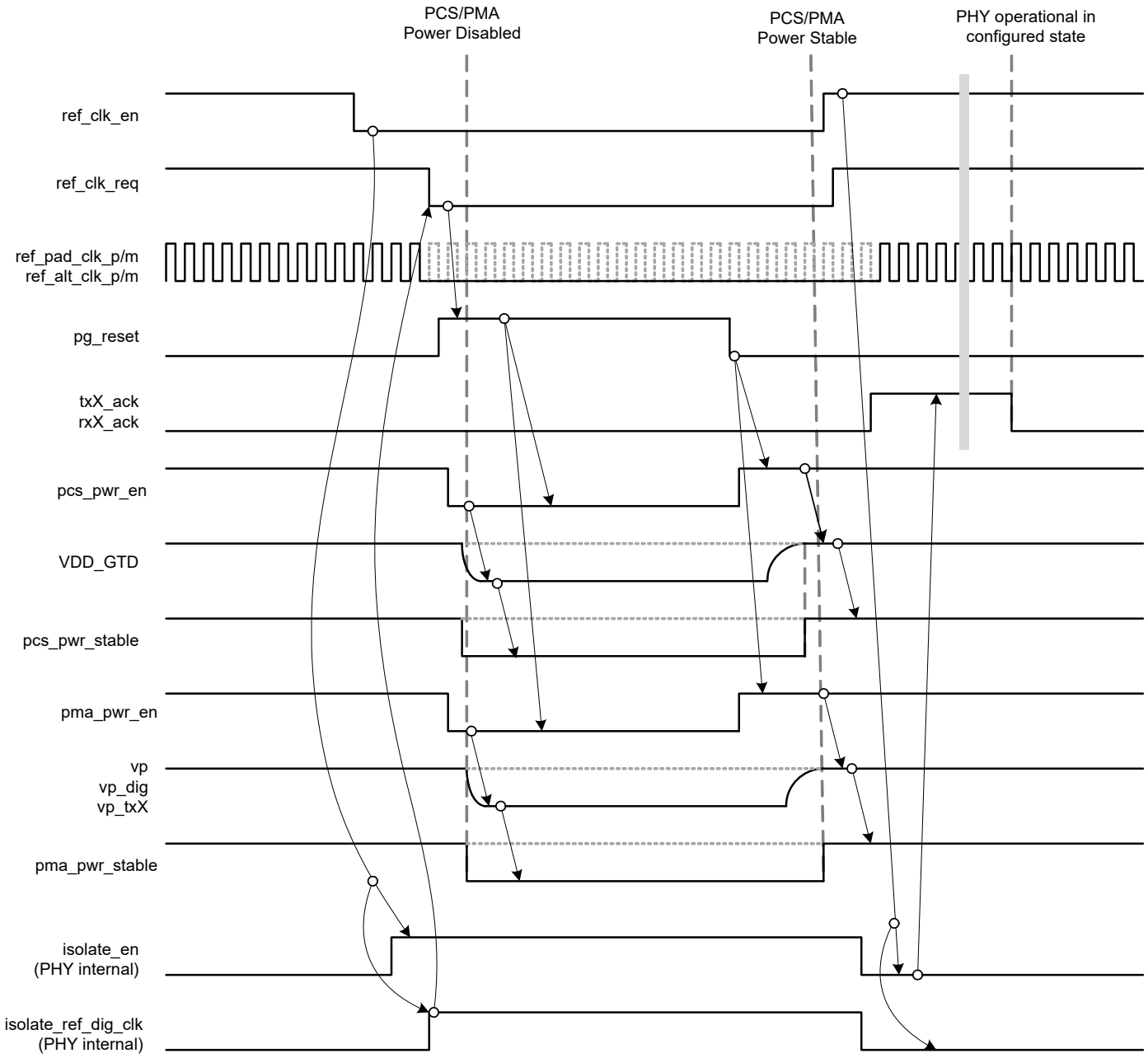
Figure 5-28 shows the control signal connectivity for power gating with external power switch for the PMA. The PHY generates power gating enables for the PCS switch (or multiple switches) as well as the PMA switch external to the PHY. The inputs pcs_pwr_stable and pma_pwr_stable must only be asserted after the supplies ramp up and are at 90% of nominal or higher. The power gating mode is enabled when pg_mode_en input is 1. When pg_mode_en is 0, the power gating control/status inputs to the PHY (except ref_clk_en, ref_clk_req, and txX/rxX_ack) shown in Figure 5-28 are ignored.

Figure 5-28 Power Gating Connectivity for External PMA Switch



The sequence of powering gating is shown in Figure 5-29.

Figure 5-29 Power-Gating Sequence for the PHY (Internal or External PMA Switch) with pg_mode_en=1



The entry into power gated state starts by the de-assertion of ref_clk_en. The PHY performs isolation sequencing in preparation of power gating by retaining states of some calibration registers and de-asserts the ref_clk_req. At this point, you can:

- Disable the reference clock inputs to the PHY.
- Assert the pg_reset input, which causes the pcs_pwr_en and pma_pwr_en signals to de-assert.
- Turn off the power supplies to PMA (vp, vp_dig, and vp_txX) and Raw PCS (VDD_GTD). However, it is allowed that these power supplies can stay ON because of any other SOC component requirement.

The exit from power gated state is initiated by de-assertion of pg_reset. The PHY requests power by assertion of pcs_pwr_en and pma_pwr_en. It is required that the ref_clk_en input be asserted after the PMA and Raw PCS power supplies are stable (pcs_pwr_stable/pma_pwr_stable=1/1). The PHY resets the internal power gated logic, disables the isolation on power gated signals, and asserts txX_ack/rxX_ack outputs. After the reference clock inputs are supplied, the PHY performs the fast power-up by recovering states of the retained calibration registers, and de-asserts txX_ack/rxX_ack outputs. At this point, the PHY is configured into the state defined by its primary configuration inputs.

5.19 Configuration for a Disabled PHY

There are several methods to disable a PHY, if the PHY is not being used in your SOC implementation (that is, SOC that may have several PHY instantiations where depending on an application, one or more PHYs are disabled). To disable a PHY, follow the methods outlined in Table 5-26.

Table 5-26 Signal Connections If the PHY is Not Used

Signal	Method 1: All Power Supplies (vp, vph, vptxX, vpdig, gd) at Nominal Voltage and IDDQ = 1	Method 2: All Power Supplies (vp, vph, vptxX, vpdig, gd) Connected to Ground	Method 3: All Power Supplies (vp, vph, vptxX, vpdig, gd) Left Unconnected (Floating)
PMA inputs from Raw PCS	Automatically engaged isolation cells	Tie all inputs to ground—recommended	Tie all inputs to ground—recommended
PMA outputs to Raw PCS	Insert isolation gate—recommended	Insert isolation gate	Insert isolation gate
rxX_p, rxX_m	Leave floating or tie low	Leave floating or tie low	Leave floating or tie low
txX_p, txX_m	Leave floating	Leave floating	Leave floating
resref	Leave floating	Leave floating	Leave floating
ref_pad_clk_p, ref_pad_clk_m	Tie low	Tie low	Tie low
ref_alt_clk_p, ref_alt_clk_m	Tie low	Tie low	Tie low

Note:

- For methods 2 and 3, to ensure that the outputs do not float high nor drive any active logic, you must use isolation gates on PHY outputs. Furthermore, to prevent the PHY circuitry from inadvertently toggling, it is good practice to tie all the inputs to the ground.
- If REXT is being shared with other PHYs, use only method 1. For REXT sharing, the request and ack handshake logic within each PHY must be powered and should allow the handshake signals from other PHYs to propagate through. For more information about resistor sharing, see Figure 7-14 on page 246 (“Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor”).
- If clock is being repeated, ensure that you do not power down the PMAs in the chain.
- For more information about IDDQ mode, see Table 5-20 on page 196 (“PHY Test Mode Selection”).

5.20 Notes for Configuring the PHY

For the configuration tables that follow, note the following important information:

- All values are in decimal
- The MPLL feedback divider ratios determined by the code word on primary inputs mplla_multiplier[7:0]/mpllb_multiplier[7:0] are provided in Table 4-3 on page 149.
- A change in the following inputs must be followed by a PHY reset (phy_reset) assertion and ref_clk_en de-assertion:
  - vph_nominal[1:0]
  - ref_range[2:0]
  - ref_clk_div2_en
  - ref_clk_mplla_div2_en/ref_clk_mpllb_div2_en
  - ref_use_pad
  - rx_vref_ctrl[4:0]
  - tx_vboost_lvl[2:0]
  - rxX_los_threshold[2:0]
  - rxX_los_lfps_en
  - mplla_multiplier[7:0]/mpllb_multiplier[7:0]
  - mplla_bandwidth[15:0]/mpllb_bandwidth[15:0]
  - rxX_term_acdc
  - pg_mode_en
  - cr_para_sel
  - sram_bypass
- The following inputs can be changed only when the corresponding MPLL is powered down:
  - mplla_div8_clk_en/mpllb_div8_clk_en
  - mplla_div10_clk_en/mpllb_div10_clk_en
  - mplla_tx_clk_div/mpllb_tx_clk_div
  - sup_misc[7:0]
  - mplla_ssc_en/mpllb_ssc_en
  - mplla_fracn_ctrl[10:0]/mpllb_fracn_ctrl[10:0]
  - mplla_ssc_clk_sel/mpllb_ssc_clk_sel
  - mplla_ssc_freq_cnt_init[11:0]/mpllb_ssc_freq_cnt_init[11:0]
  - mplla_ssc_freq_cnt_peak[7:0]/mpllb_ssc_freq_cnt_peak[7:0]

- mplla_ssc_up_spread/mpllb_ssc_up_spread
- mplla_div16p5_clk_en
- mplla_div_clk_en/mpllb_div_clk_en
- mplla_div_multiplier/mpllb_div_multiplier

- The following input can be changed only when the TX clock source is disabled (txX_pstate[1:0] = P1/P2, or txX_reset = 1):

  txX_disable

- The following input can be changed only when the RX clock source is disabled:

  rxX_disable

- The following inputs can be changed only when the RX clock source is disabled (rxX_pstate[1:0] = P1/P2; or rxX_reset = 1):
  - rxX_div16p5_clk_en
  - rxX_cdr_ssc_en

5.21 Configuring the PHY

- For support configuration values, see Appendix A, “Support Configuration Settings”.
- For lane configuration values, see Appendix B, “Lane Configuration Settings”.

6 Verilog Model Description

This chapter includes the following sections:

- "Overview" on page 220
- "Verilog Function" on page 220
- "Serial Line Modeling" on page 221
- "Verilog Model Files and Simulators" on page 221
- "Fast Simulation Mode" on page 223
- "PCIe Simulation Environment" on page 225

6.1 Overview

The product deliverables include Verilog RTL for the Raw PCS and a Verilog behavioral model of the PMA. Contrary to the Verilog RTL for the Raw PCS, which can be synthesized into a gate-level netlist by a synthesis tool, the Verilog behavioral model of the PMA includes a netlist of generic GTECH gates and a non-synthesizable behavioral model of the analog circuits.

In the set of deliverables, the Raw PCS Verilog RTL and the PMA Verilog behavioral model reside at:

- Raw PCS: ./Latest/phy/rtl
- PMA Verilog model: ./Latest/pma/behavior

Together, they form the Verilog model of the PHY (dwc_e16mp_phy_xN_ns.v; N=Number of lanes). The PHY model allows you to run functional verification of your logic along with the PHY.



This PHY model and test bench do not represent an example of exhaustive PHY testing.

6.2 Verilog Function

The Verilog model simulates using a four-state (0, 1, X, Z) simulator. Therefore, several features that require analog voltage and current values transmitted over wires cannot be simulated with this model.

Features that are not accurately modeled in the PHY model are as follows:

- Receive and transmit I/O terminations
- Termination resistor tuning and measurement
- Voltage-level setting for the Receiver Electrical Idle Detector
- AC signal characteristics of the ACJTAG receiver
- Voltage margining
- Analog test buses; the PHY pins atb_f_p, atb_s_m, atb_s_p are inaccurate during simulations
- Receiver detection

  The receiver detection status output from the PHY txX_detrx_result is inaccurate during simulations. To simulate the receiver detection feature, you should force the receiver detection status on txX_detrx_result when txX_ack is asserted in response to the assertion of txX_detrx_req, and release the force after txX_ack is de-asserted.
- Transmitter equalization settings (pre-shoot, swing, and post-cursor) on the transmit serial lines txX_p/m
- Receiver equalizer settings (AFE CTLE boost and VGA gain, and DFE taps)

  The RX Evaluation outputs rxX_adapt_fom, rxX_txmain_dir, rxX_txpost_dir, and rxX_txpre_dir (described in "RX Equalization and Adaptation" on page 181) are inaccurate during simulations. To simulate link adaptation, you should force the required settings at these PHY outputs when rxX_adapt_ack is asserted, and release the forces after rxX_adapt_ack is de-asserted.

6.3 Serial Line Modeling

The serial lines (txX_p/m, rxX_p/m) are fundamentally analog signals. Because each wire is limited to the simulator's four states, not all aspects of the analog signals can be reflected. Table 6-1 provides encodings of the analog transmitter model outputs.

Table 6-1 Transmitter Model Serial Encodings

Output	txX_p	txX_m
Differential 1	1	0
Differential 0	0	1
No signal, holding common mode	0	0
No signal, not holding common mode	Z	Z

The analog receiver model interprets only differential 0 or differential 1 encodings as valid signal levels.

6.4 Verilog Model Files and Simulators

The Verilog model is generated for simulations and can be simulated using Synopsys VCS or Cadence NCSim simulators.

Table 6-2 describes each portion of the Verilog model and indicates the product deliverables directory where the portion is stored.

Table 6-2 PHY Model Database Structure

Directory	Description
`./Latest/phy/include`	Contains macros used in the Raw PCS, PMA and simulation files.
`./Latest/upcs/include`	Contains macros used in the UPCS and simulation files.
`./Latest/upcs/rtl`	Contains Verilog RTL files for the UPCS, which includes the following: ■ dwc_e16mp_upcs_xN_ns_upcs_pcs_PxL.v: Top-level UPCS module (PxL = Number of total PCS lanes, P= Number of PIPE interfaces, L=Number of lanes per PIPE) For more information about these files, refer to the 16G PCS for the DesignWare® Cores Multi-Protocol 16G PHY Databook.
`./Latest/phy/rtl`	Contains Verilog RTL for the Raw PCS files ■ dwc_e16mp_phy_xN_ns_pcs_raw_xN.v: Top-level RTL file for the xN Raw PCS ■ dwc_e16mp_phy_xN_ns.v: PHY top-level wrapper instantiating PMA and Raw PCS
`./Latest/pma/behavior`	Contains Verilog RTL for the behavioral models for the PMA. ■ dwc_e16mp_pma_xN_ns.v: Top-level file for the xN PMA instantiating GTECH model of the digital and behavioral model of analog sub-blocks.

Instructions for the PHY simulations are provided in the TESTBENCH.README file under ./Latest/phy/testbench directory. The simulation database structure for PHY simulations is provided in Table 6-3.

Table 6-3 Simulation Database Structure

Directory	Description
`./Latest/phy/sim`	VCS and NC Verilog regression script for GTECH and gate-level simulations of the PHY, which includes the following: ■ run_cust_tb_regression.sh: Regression script for the PHY to run both on VCS and NC Verilog. This script has many command line arguments. See TESTBENCH.README for guidelines on how to run a test using this script.
`./Latest/phy/testbench`	Sample test-cases for the behavioral and gate-level simulations. ■ TESTBENCH.README: Run instructions for the PHY test bench.

Note For simulating the PHY model, define the ANI_SIM_MODE and ANI_RANDOM_SEED macros. For details on macros used in the Synopsys PHY testbench, refer to `./phy/testbench/TESTBENCH.README`.

6.4.1 PMA Gate-Level Model

For gate-level simulations, the DWC_E16MP_X4NS_SIM_GATES macro must be defined, and DWC_E16MP_X4NS_SIM_GTECH must not be defined. Refer to the `./phy/testbench/TESTBENCH.README` file for details.

The following issues exist with running PMA gate-level simulations:

- Due to non-trivial clocking structures within the PMA, a zero-time or unit-delay simulation of the gate-level model would be non-functional due to race conditions in the Verilog execution caused by simulator event ordering. Therefore, SDF back-annotation of the gate delays is required. The .sdf files are provided in the `./pma/timing/<metal>/sdf` directory and are automatically loaded at simulation startup.
- The .sdf files simply represent the timing of the hardware. These files are generated at worst-case and best-case process/temperature/voltage corners. However, they are not sufficient for replacing static timing. Synopsys has validated the PMA's internal timing and timing for the PMA interfaces are provided in the .lib files. Validate these timings using STA tools and not Verilog simulation.
- For proper operation of the PMA gate-level model with respect to standard cell library timing checks, the following switch is required:

  +notimingchecks

  This switch is required due to the invalid timing checks (caused by multiple clock domains, synchronizers, and so on) coupled with the standard cell library's behavior of inserting X's into the simulation when these checks fail.

- Each simulator loads the .sdf files in a different manner. Therefore, warnings during SDF annotation vary between simulators. These messages are innocuous and can be ignored. However, there must be no errors during SDF annotation. For example, the following warning is from VCS SDF annotation. VCS does not have a mechanism to suppress this warning.

```
Warning-[SDFCOM_TI] TIMINGCHECKs Ignored
SDF Warning: TIMINGCHECKs ignored, model compiled or invoked with
+notimingcheck option.
```

  This warning is due to the +notimingcheck switch and is expected.

6.4.2 Receiver Detection

True receiver detection is not possible in Verilog simulation, because the serial wires are unidirectional, and the receiver detection operation requires signals to propagate contrary to this direction. In the behavioral model of the transceiver analog circuitry, state bits (one per serial wire) are provided to control the result of receiver detection. These bits default to 0 (receiver not present) and can be forced to 1 (receiver present) as shown below, or negated, at any point prior to asserting txX_detrx_req.

```
force `PHY_PATH.pma.ana.lane<lane number>.tx.rxdetect.tx_rxdet_m_result = 1'b1;
force `PHY_PATH.pma.ana.lane<lane number>.tx.rxdetect.tx_rxdet_p_result = 1'b1;
```

6.5 Fast Simulation Mode

Some features, such as power-state transitions, receiver-detect, and RX adaptation, may take long simulation times in normal operating modes. These simulation times can be reduced for simulation purposes by either using fast simulation macros or by writing into the PHY control registers through the Parallel Control Register Interface or through JTAG (for information, see “Control Register Interface” on page 187). In addition, there are certain Verilog force statements that need to be included in the customer testbench to make simulations work.

The fast simulation mode can be enabled by defining the macro “DWC_E16MP_XnNS_SHORT_RESET.” The macro based fast simulation mode is only supported for the RTL model of the Raw PCS. The fast simulation mode is supported for both GTECH and gate-level models of the PMA. In real silicon or with synthesized Raw PCS, the following sequence can be used to modify the internal Raw PCS FSM operation:

1. Set phy_reset and all of rxX_reset and txX_reset to '1'.
2. Wait for 10 ns or a minimum of 10 µs after the power-supplies are ramped-up (at power-up).
3. Set phy_reset to '0'.
4. Wait a minimum of 10 phy_ref_dig_clk cycles to ensure that the Control Register interface is operational.
5. Write a 1'b1 to the registers fields described in Table 6-4.
6. Set all of rxX_reset and txX_reset to '0' for lanes, which are to be operational.

Table 6-4 Registers for Fast Simulation mode

Register	Register Field
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_OVRD	FAST_MPLL_LOCK
	FAST_MPLL_PWRUP
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_OVRD	FAST_MPLL_LOCK
	FAST_MPLL_PWRUP
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1	FAST_AFE_DFE_SETTLE
LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1	FAST_RX_VREG_EN
	FAST_RX_CLK_DCC_EN
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3	FAST_TX_RXDET
RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL	SKIP_RX_VGA_EXT_STARTUP
	SKIP_RX_REF_CAL_EXT_STARTUP
	SKIP_RX_DFE_ADAPT_STARTUP
	SKIP_RX_DFE_CAL_STARTUP
	SKIP_RX_DFE_CAL_EXT_STARTUP
	SKIP_RX_AFE_ADAPT_STARTUP
	SKIP_RX_AFE_CAL_STARTUP
	SKIP_RX_PHS_CAL_STARTUP
	SKIP_RX_PHS_CAL_EXT_STARTUP
	SKIP_RX_FOM_STARTUP
	SKIP_RX_TX_INC_DEC_STARTUP
	SKIP_RX_IQ_CAL_STARTUP
	SKIP_RX_IQ_DELTA_STARTUP
	SKIP_RX_IQ_ADAPT_STARTUP
	SKIP_RX_REF_CAL_STARTUP
RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL_1	SKIP_RX_VGA_SLICER_CAL_STARTUP
	SKIP_RX_BYPASS_CAL_STARTUP
	SKIP_RX_ERROR_CAL_STARTUP

Table 6-4 Registers for Fast Simulation mode (Continued)

Register	Register Field
RAWLANEN_DIG_AON_FAST_FLAGS	FAST_RX_VCO_CAL
	FAST_RX_VCO_WAIT
	FAST_RX_PWRUP
	FAST_TX_RXDET
	FAST_TX_CMN_MODE
	FAST_SUP
	FAST_RX_CONT_CAL_ADAPT
	FAST_RX_CONT_ADAPT
	FAST_RX_ADAPT
	FAST_RX_STARTUP_CAL

Note

- Asserting phy_reset resets the registers to default values. To preserve fast simulation mode, the registers enabling the fast simulation mode must be written after every phy_reset de-assertion.
- The DAC offset calibration takes approximately 15 ms. For digital simulations, to speed up runtime, it is recommended to define the FAST_RX_STARTUP_CAL macro or write "1" to the FAST_RX_STARTUP_CAL register field.
- Assert FAST_MPLL_LOCK, FAST_MPLL_PWRUP or FAST_SUP when writing to the FAST registers during gate level simulations to ensure that the MPLL clocks are at the proper frequency when enabled.
- Do not assert FAST_RX_VCO_CAL or FAST_RX_VCO_WAIT if DWC_*_RX_FORCE_FREQ_TUNE is not defined to speed up the RX VCO frequency tuning.
- If DWC_E16MP_XnNS_SHORT_RESET macro is used, write a zero to negate the FAST_* bits as desired.

6.6 PCIe Simulation Environment

The deliverables include a PCIe simulation environment including the xN PCIe PCS and the xN PHY as the Design Under Test (DUT). For details of the simulation environment, refer to Chapter 5, “Simulation”, in the 16G PCS for the DesignWare® Cores Multi-Protocol 16G PHY Databook.

7 Physical-Level Implementation

This chapter includes the following sections:

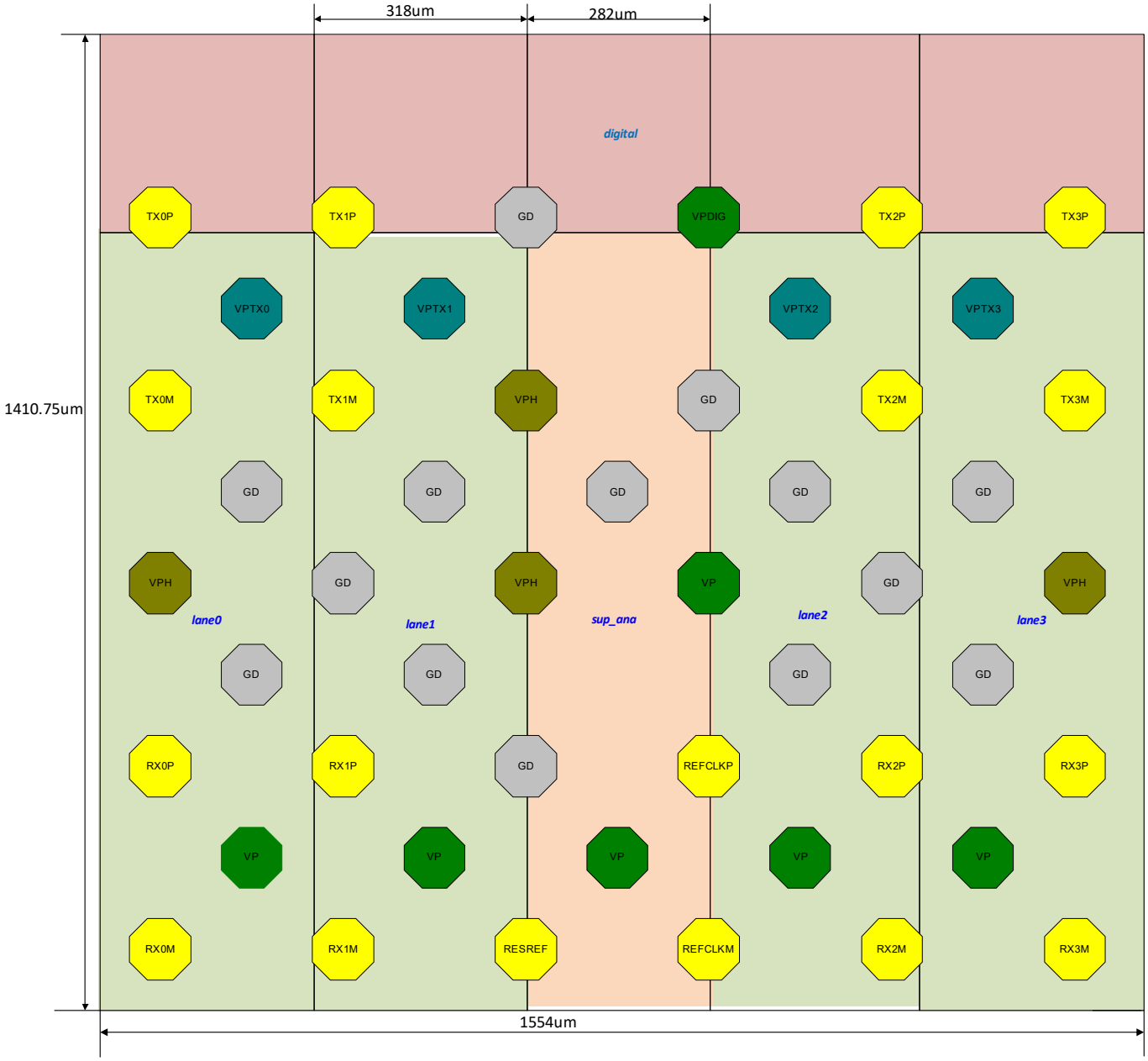
- "PMA Assembly" on page 228
- "PMA Placement" on page 236
- "PMA Fill" on page 239
- "Operating Conditions" on page 240
- "Power Supply Connectivity" on page 241
- "ESD" on page 242
- "Reference Clock" on page 245
- "Reference Resistor" on page 245
- "ATB and DTB Signals" on page 247
- "Raw PCS Synthesis" on page 247

7.1 PMA Assembly

7.1.1 Flip-Chip Pinout

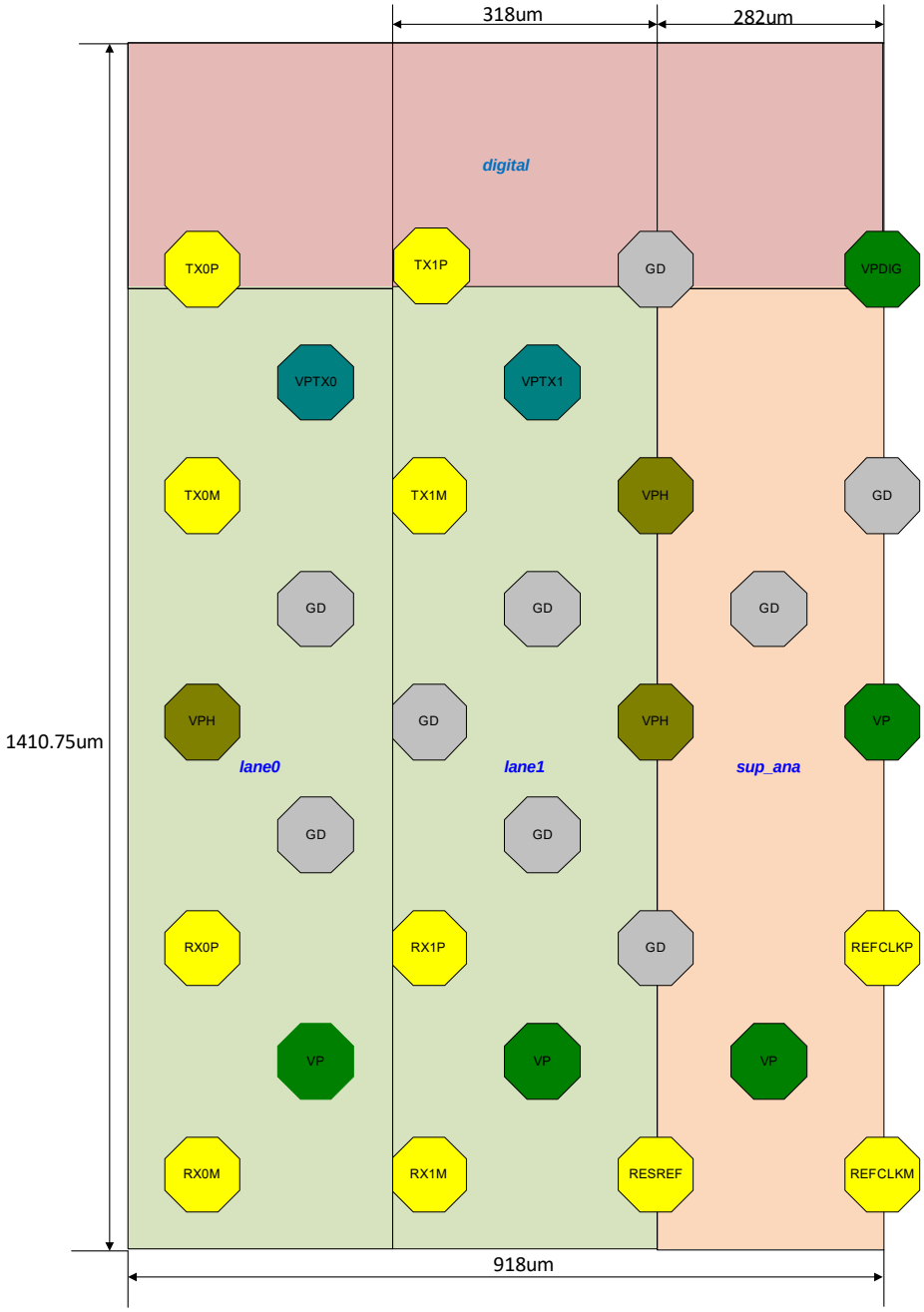
The assembled x4 PMA pinout is shown in Figure 7-1.

Figure 7-1 Bump Map (x4)



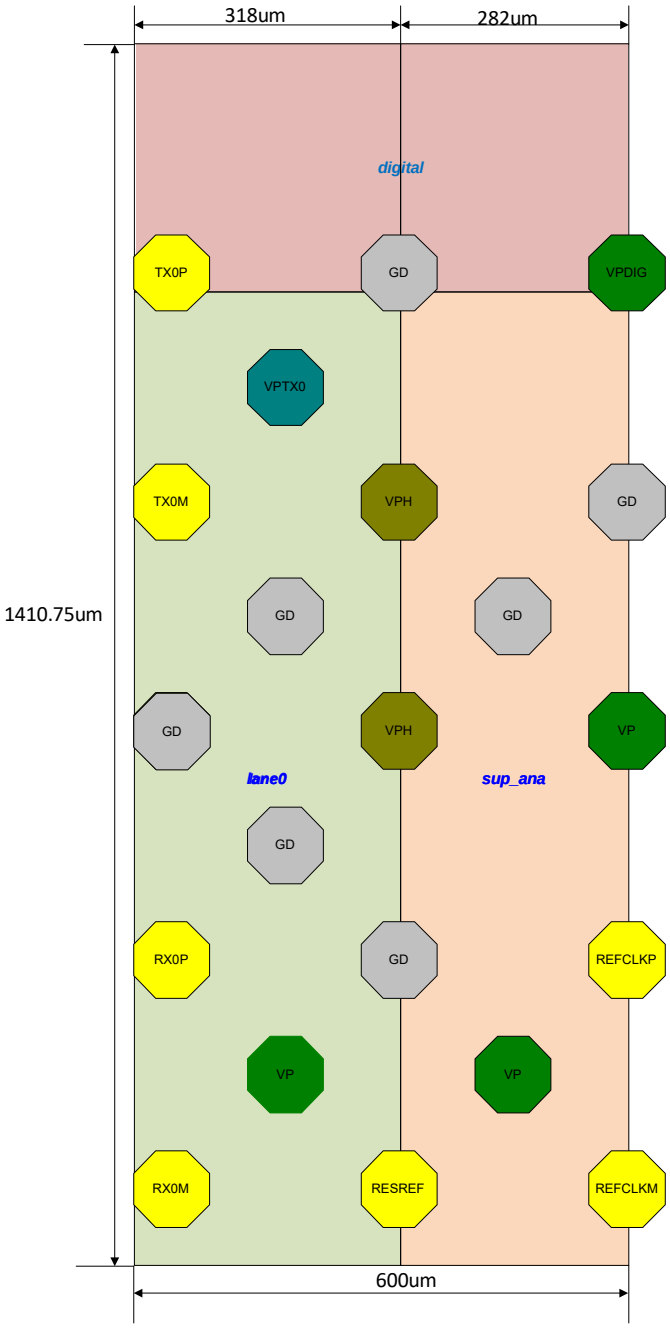
The assembled x2 PMA pinout is shown in Figure 7-2.

Figure 7-2 Bump Map (x2)



The assembled x1 PMA pinout is shown in Figure 7-3.

Figure 7-3 Bump Map (x1)



7.1.2 Bump Pad RDL and UBM Sizing

Foundry requirements state that pad RDL and UBM sizing is dictated by the minimum bump pitch found on the die.

The pads layout provided with the PMA have a bump pitch of 200 µm. The RDL and UBM sizing used is suitable for a minimum bump pitch between 178-200 µm. The bump RDL size is 98.9 µm and the UBM size is 94.4 µm.



Care should be taken to ensure that the under-bump metallization (UBM) is the correct size. Foundry and packaging requires that the UBM shape used for the bump is the same for all bumps within the same die. Furthermore, the size of the UBM shape is dictated by the minimum bump pitch in the die.

7.1.3 PMA Size

The size of the PMA is determined by the size of the following sub-components:

- PMA hard macro
- Flip-chip pads

The preliminary estimated size of the hard macro building blocks for X1, X2, and X4 PMA are shown in Table 7-1, Table 7-2, and Table 7-3.

Table 7-1 X1 PMA Size

Block	Drawn X (µm)	Drawn Y (µm)	Drawn Area (mm^2)	Shrink Factor	Actual Area (mm^2)
PMA	600	1410.75	0.85	0.81	0.69
Pads	700.25	1392	0.97	0.81	0.79

Table 7-2 X2 PMA Size

Block	Drawn X (µm)	Drawn Y (µm)	Drawn Area (mm^2)	Shrink Factor	Actual Area (mm^2)
PMA	918	1410.75	1.30	0.81	1.05
Pads	997.9	1392	1.39	0.81	1.13

Table 7-3 X4 PMA Size

Block	Drawn X (µm)	Drawn Y (µm)	Drawn Area (mm^2)	Shrink Factor	Actual Area (mm^2)
PMA	1554	1410.75	2.19	0.81	1.78
Pads	1554	1392	2.16	0.81	1.75

7.1.4 Assembly of PMA and Pads

The PMA hard macro and snap-on pads are connected by abutment with the same origin (0,0) as illustrated in Figure 7-5, with the same mirroring and rotation (if applicable).

Note

The flip-chip pads cell will extend beyond the extents of the PMA. It is allowed and expected that you will place and route core logic beneath these extensions--the only blockage in the pads cells extensions is on the RDL layer.

For x4 and x2 configurations, there are no offsets (Figure 7-5 and Figure 7-5 on page 233). However, for x1 configuration, there is an offset of (-20.35, 0) relative to origin (0, 0) (Figure 7-6 on page 234).

Figure 7-4 Origins for Abutment of PMA Hard Macro and Snap-On Pads for x4 Configuration

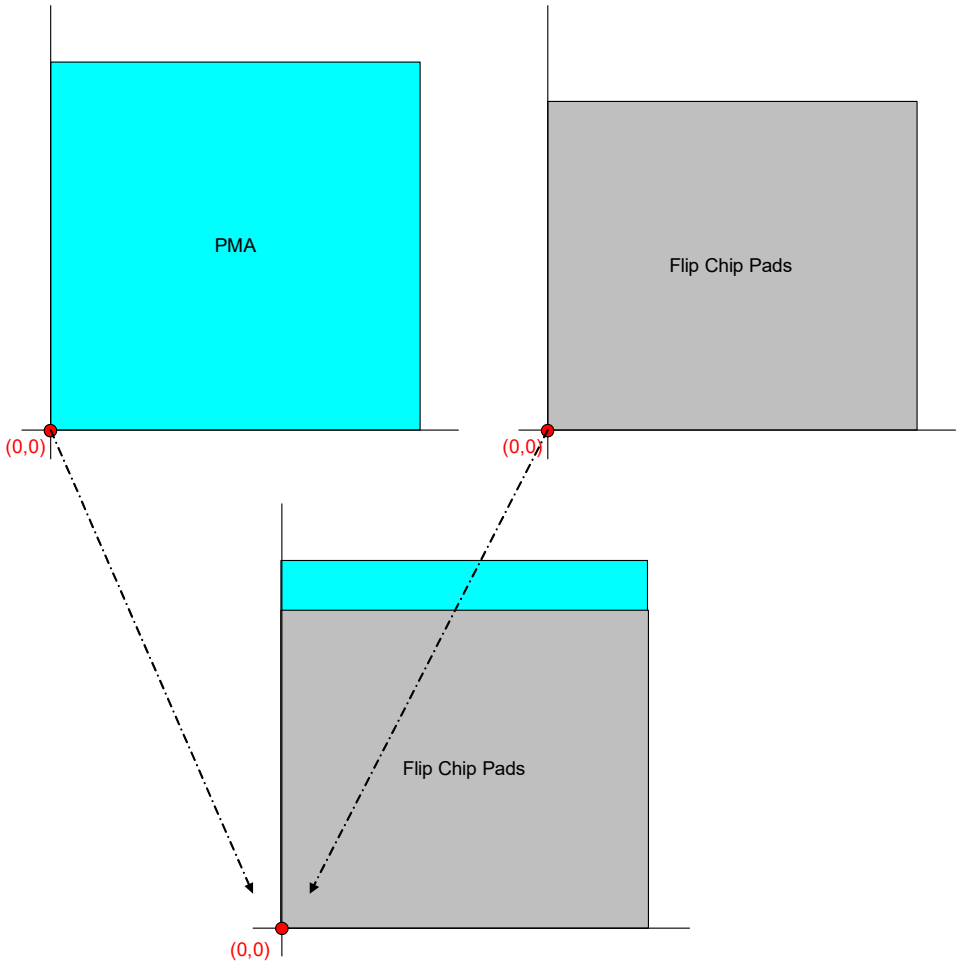


Figure 7-5 Origins for Abutment of PMA Hard Macro and Snap-On Pads for x2 Configuration

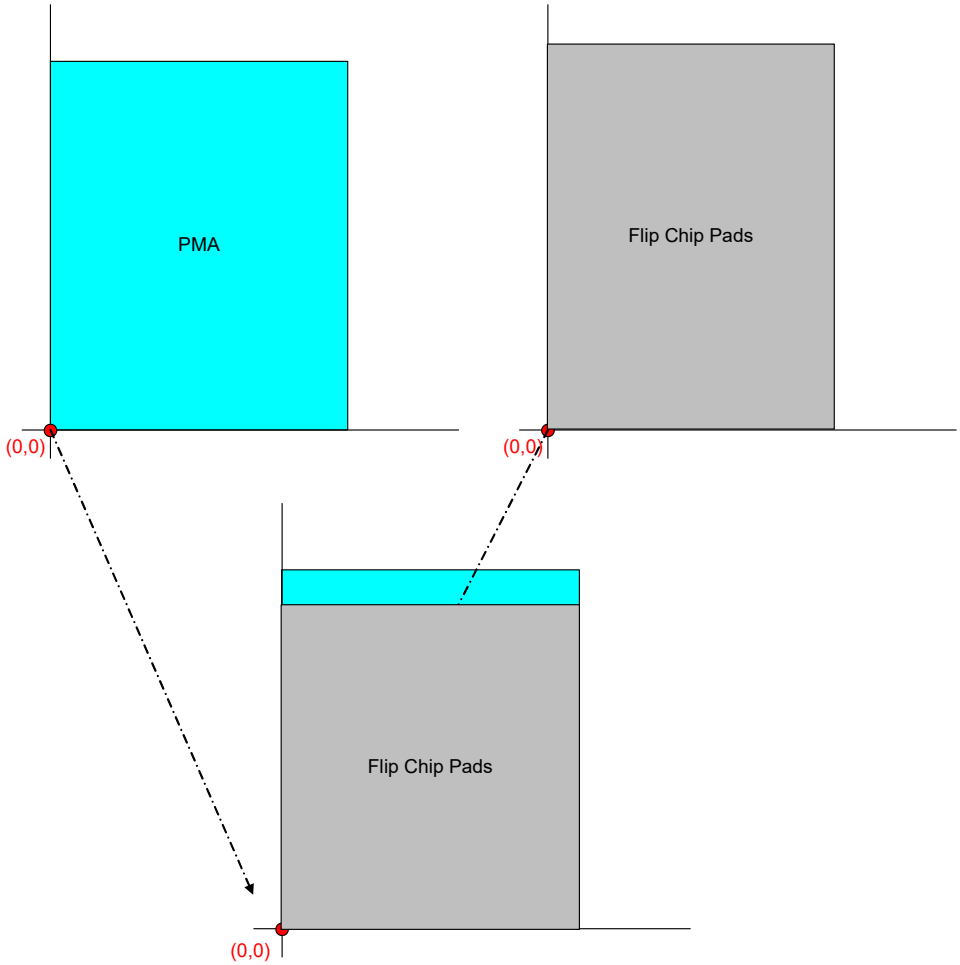
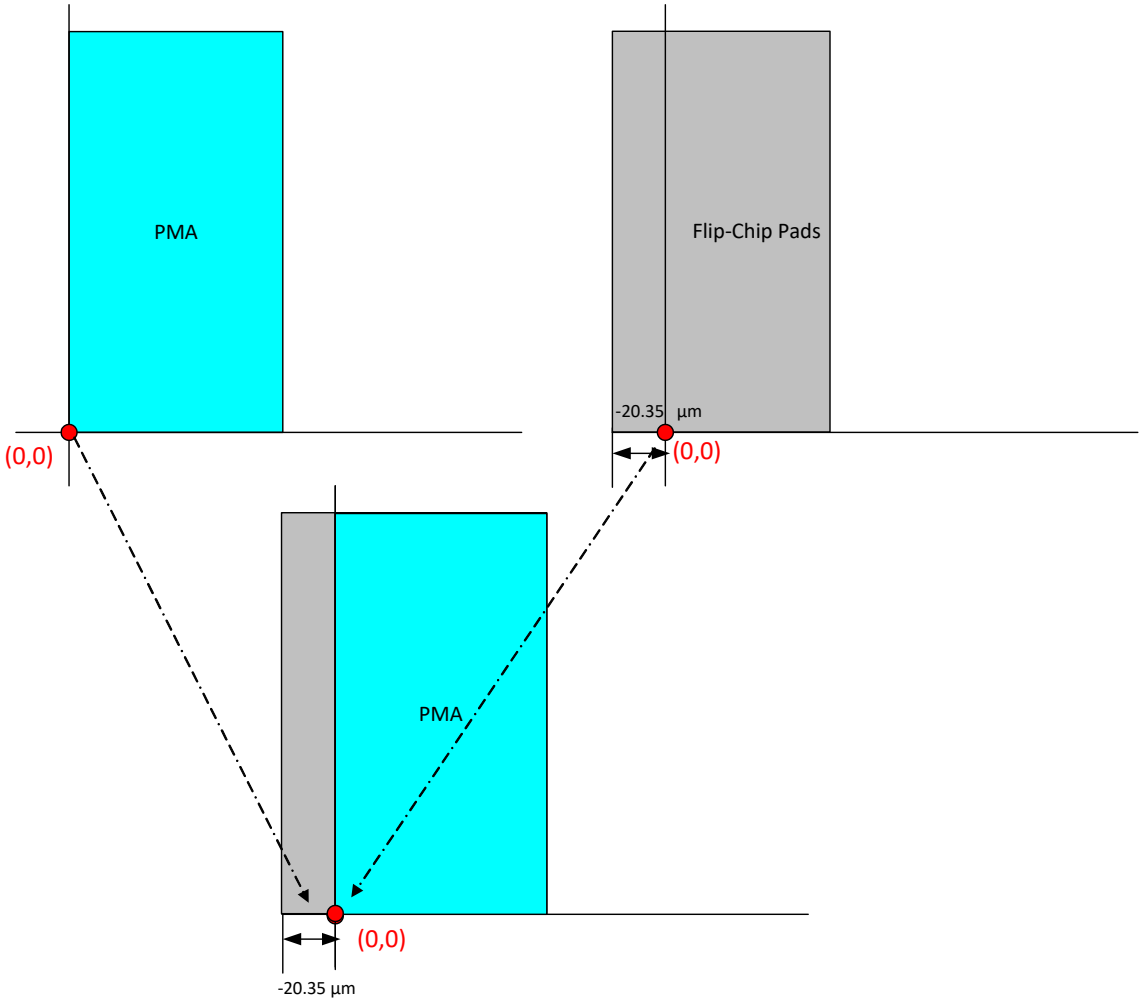


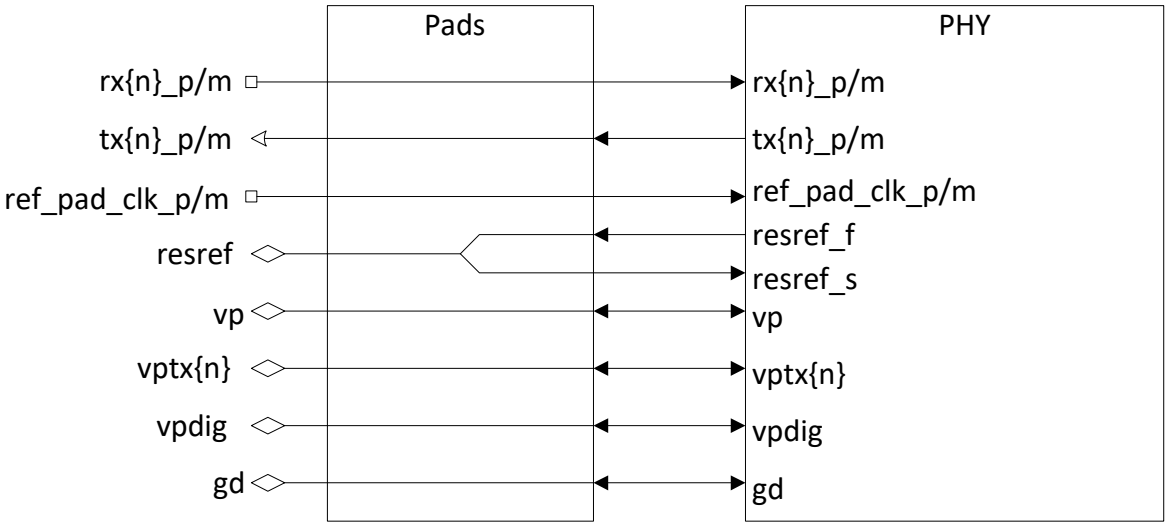
Figure 7-6 Origins for Abutment of PMA Hard Macro and Snap-On Pads for x1 Configuration



7.1.5 Connectivity of PMA and Pads

When PMA and PADs are instantiated together the resref_s and resref_f signals are shorted together as resref (see Figure 7-7).

Figure 7-7 Connectivity of PMA and PADs



7.1.6 PMA Routing Obstruction

The PMA LEF contains blockage for all metals not including RDL layer. The PADs LEF contains full blockage for all metals including RDL. When integrating the PHY you should ensure to include both LEFs with respect to top level routing blockage to keep SOC level routing out of the PHY.

7.2 PMA Placement

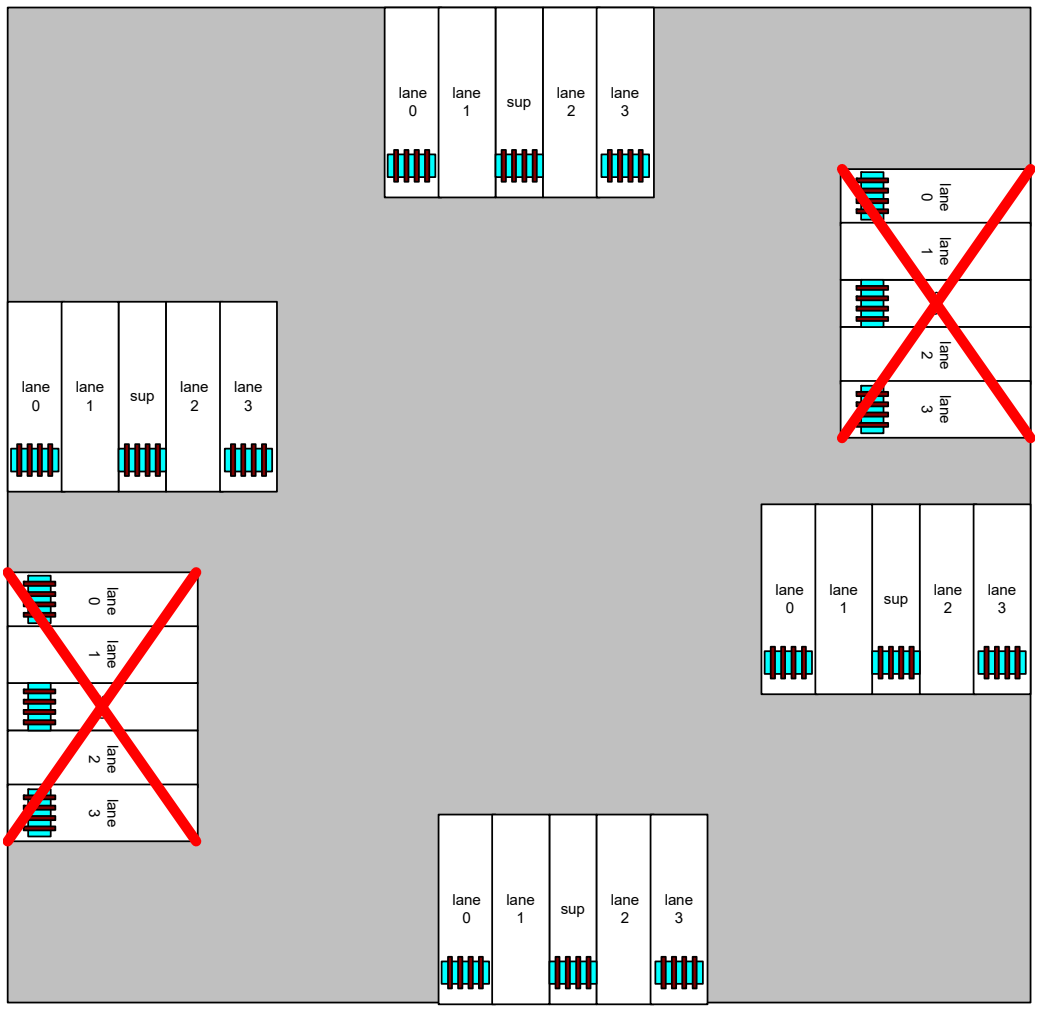
This section describes things to consider when placing the assembled PMA.

7.2.1 PMA Rotation and Mirroring

The FET POLY gates within the PMA must be vertical according to foundry rules.

The PMA hard macro is designed to be placed with preference for the north or south side of the die, as shown in Figure 7-8. The PMA hard macro may also be placed on the east or west (non-preferred) side of the die, as shown in Figure 7-8.

Figure 7-8 PMA Rotation and Mirroring



The PMA may be rotated by 180°, but not 90° or 270° due to the vertical POLY requirement.

The PMA may be mirrored in either the X or Y direction as this does not change POLY orientation.

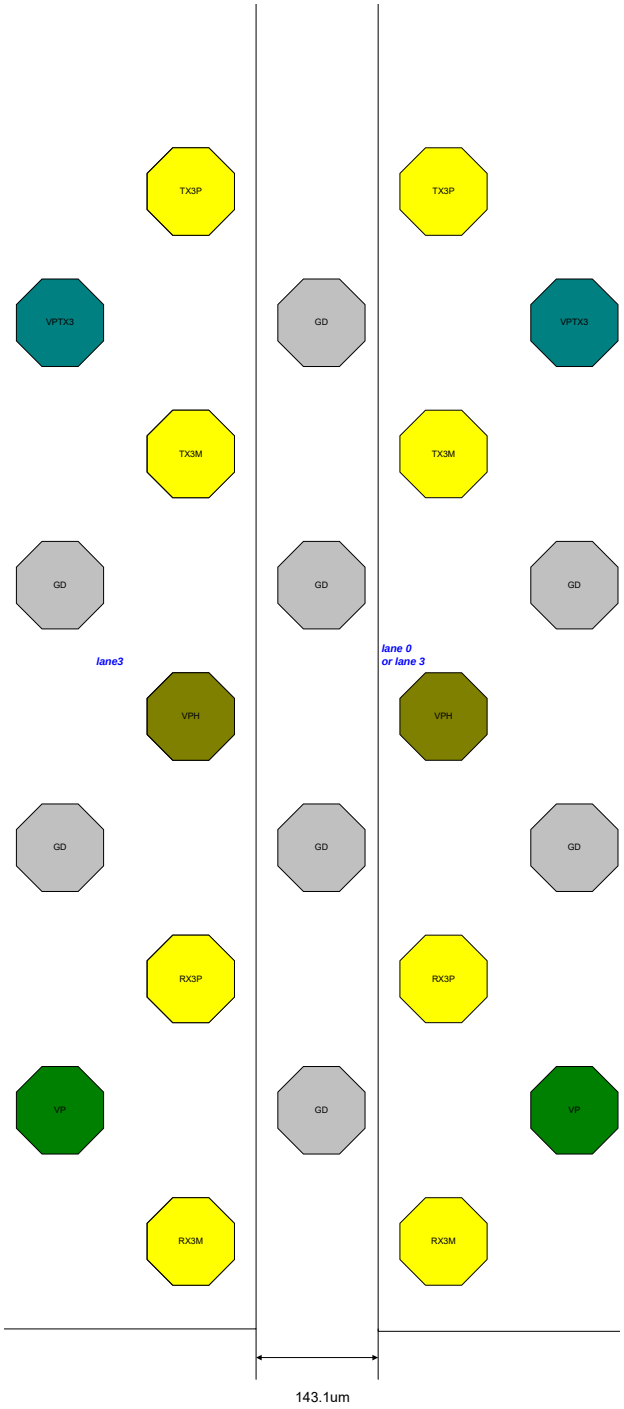
7.2.2 PMA Abutment

The following requirements must be met when placing the PMA adjacent to other circuitry:

- Spacing Requirements
  - PMA to PMA:
    - May be abutted with adjacent PMA on east and west edges with 143.1 µm spacing, as shown in Figure 7-9 on page 238
    - Do not place the x1 and x2 PMA such that the clock support block is adjacent to the clock support block of the next PMA (that is, next PMA is instantiated in a flip-X)
  - PMA to core logic:
    - The east, west, and south side of the PMA has a 10 µm spacing requirement
    - The north side of the PMA may abut with core logic
    - Special consideration for instantiation of a x1 or x2 - a 20 µm moat is required on the edge of the PMA that has the clock support block exposed to core logic-on the unflipped/rotated PMA, which is the east edge.
  - Moat requirements:
    - The "moat" may not contain any core logic circuitry including cells, signal routes, and power meshes
    - The "moat" may not contain any substrate taps

When multiple PMAs are being placed adjacent to each other, they may be placed as close as 10 µm from each other, as shown in Figure 7-9.

Figure 7-9 PMA Abutment



7.3 PMA Fill

When dummy fill is run on metal and base layers at the top level, fill shapes will be placed within the PMA. Synopsys does not provide a design with fill/dummy shapes. We place some obstructions on key sensitive circuitry, but there are other areas in the PMA itself that the foundry fill decks are expected to fill.

We check the design to ensure that adding dummy fill does not result in density violations at the SOC level. Before fill insertion, density errors are expected.



The PMA GDS does not come with fill added. You should not place any fill blockage over the PMA when integrating it. You should run the foundry fill deck and expect fill shapes to be placed within the PMA.

7.4 Operating Conditions

This section details the operating conditions of the PMA.

7.4.1 Power Supply Limits

The PMA supports the power supply voltages in Table 7-4.

Table 7-4 Supported Power Supply Limits at the PMA Macro Bumps

Supply	Description	Min	Typical	Max	Units
vph	High Voltage IO supply[a]	1.5-7%	1.5 or 1.8	1.8+10%	V
vp/vptxX/vpdig	Analog supply[b]	0.9-7%	0.9	0.9+10%	V

a. vph_nominal[1:0] should be set according to the nominal VPH supply voltage (see signal chapter for settings)
b. sup_misc[2:1] should be set according to the nominal VP supply voltage (for settings, see Chapter 3, “Signal Descriptions”)

Allowed vp supply variation (+10%/-7%) is less than the supply variation found in the PMA LIBs (+/-10%)

7.4.2 Temperature

The PMA supports an operating temperature range from -40°C to 125°C.

Temperature drift after initial startup adaptation and calibration can affect the performance of PMA, particularly the receiver. The amount of temperature drift that the PMA can withstand before violating the protocol bit error rate limit is dependent on the receiver, the data rate, and external channel impairments. If there is a temperature drift, it may be necessary to re-adapt and re-calibrate the receiver. Sensitivity to this effect can be reduced by implementation of continuous offset correction and adaptation (CCA).

This PHY implements continuous offset calibration and adaptation (CCA). This reduces but does not zero out the sensitivity of the receiver to changes in temperature. Changes in temperature may still result in reduced eye opening as seen by the receiver and will reduce margin for jitter tolerance.

7.4.3 Electromigration

The electromigration limit of the PMA is 100KPOH@110°C.


Operating at conditions that exceed the electromigration limits of PMA reduces the lifetime of the PMA. While the PMA can meet performance requirements at higher temperature, this is expected to be a short term condition, such as temporary failure in cooling, and so on.

7.5 Power Supply Connectivity

7.5.1 Power and Ground Isolation

Power supply connectivity must be designed accounting for all noise sources, decap, parasitics, package, and so on. You must perform a power integrity analysis.

To prevent power supply noise from inducing jitter on the transmitted and received waveforms, ensure that the PHY power supplies are isolated from the power supplies used on the rest of the ASIC. For more information about supply connectivity within the package and on the board, see section 8.3 ("Supplies Requirements" on page 257).

The PHY provides power pins for purpose of additional on-die decap, if needed, as determined by your power integrity analysis of the PHY + package + board. Due to jitter and isolation requirement these pins should not be used to connect the PHY to other on-die supplies.

The PHY ground, gd, is isolated on the die from the ASIC core ground vsscore_X. Back-to-back diodes between the ASIC ground (connected to vsscore_X pins) and the PHY ground, gd, are provided within the PHY macro as part of the ESD protection strategy (described in "ESD Protection Strategy" on page 242), but no direct metal-metal connection is permitted.

Isolation of all power and ground on the die implies the only connection between the PHY hard macro and the rest of the ASIC are for signal wires. The signal integrity of this interface is maintained using a fully clock-forwarded architecture where both clock and data cross the power domains together, so that subtle differences in supply voltages have no impact on timing.

7.5.2 On-Chip Supply Decoupling

If power integrity simulations indicate that additional PHY power supply decoupling is needed, one option is to include that capacitance on-chip. This may also be advantageous for utilizing space in between PHY instantiations or along the die-edge.

For this purpose, the vp, vph, and gd pins have been exposed on the last two metal layers.

Note Power integrity of the macro supplies must be considered early in the design process and can impact both on-die implementation (if additional on-die decap is needed) as well as decap design and board and package implementation. For more details on power integrity, including package and board requirements, see "Board Requirements—External Components" on page 264.

7.6 ESD

This section presents ESD considerations.

7.6.1 ESD Requirements

ESD performance is evaluated by the amount of voltage and current that the protection circuits can absorb during an ESD event while keeping voltages below damaging levels. The protection circuits are designed to meet the protection levels provided in Table 7-5.

Table 7-5 Minimum ESD Protection Levels

ESD Category	Description	Minimum Protection Level	JEDEC Class
Charged Device Model (CDM) [JEDEC JESD22-C101F]	High voltage High current Short duration	6A peak discharge current[a]	C2/C1 (500 V/250 V)[b]
Human Body Model (HBM) [JEDEC JS-001-2014]	High voltage Low current Very long duration	2,000 V	2
Machine Model (MM) [JEDEC JESD22-A115C]	Low voltage Medium current Long duration Bidirectional	100 V	N/A

a. The width of CDM pulse should be larger than 1 ns.
b. Support for both 250 V or 500 V CDM target level is dependent on maximum discharge current generated in final SOC/package implementation.

All high speed IOs (such as rxX_p/rxX_m and txX_p/txX_m) have 2,000 V human body model (HBM) protection for optimized performance at 8 Gbps and above.

7.6.2 ESD Protection Strategy

The PHY's ESD protection strategy provides the following features:

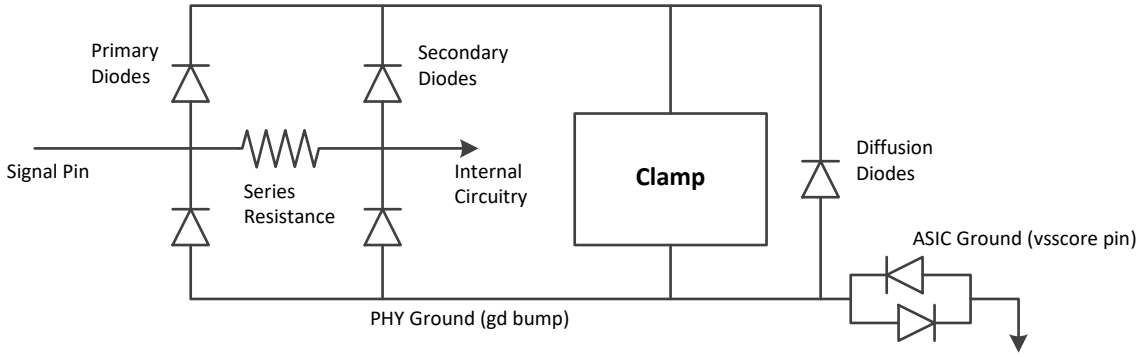
- Large primary diodes between signal pad and PHY power and ground supplies
- A series resistor and secondary diodes between signal pad and PHY power and ground supplies
- An active clamp between the PHY power and ground supplies

  The clamp and primary diodes are placed close to each other within the PHY, minimizing the resistance between them and maximizing the effectiveness of the ESD circuits.
- Back-to-back diodes between PHY ground, gd, and interfacing (ASIC) ground, vsscore_X
- The interface between the ASIC and hard macro is protected through the use of NAND/NOR CDM protection within the hard macro on all inputs and outputs

- Sufficient metal width in all ESD paths

Figure 7-10 shows the recommended ESD strategy to implement in the Multi-Protocol 16G PHY.

Figure 7-10 ESD Protection Strategy



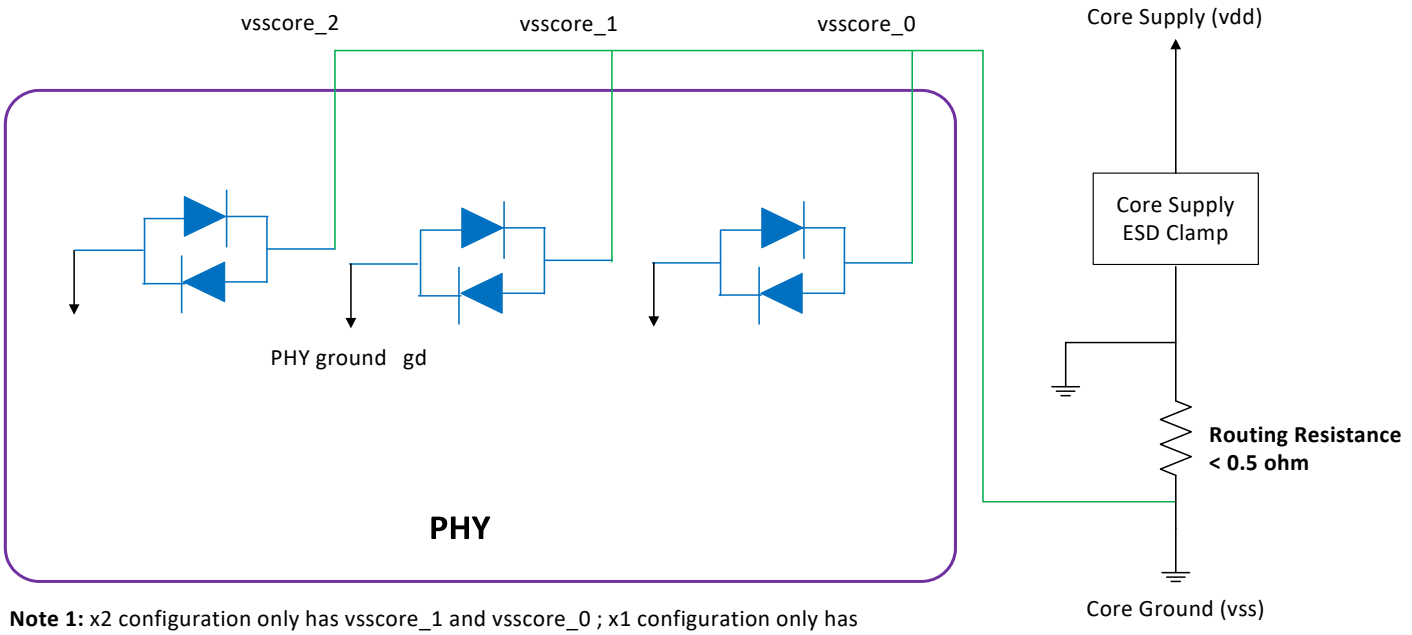
During ESD events, the primary diodes direct most of the current to the PHY power/ground buses and through the clamp. This path has minimal resistance, keeping the signal pad voltage and voltage across the PHY power buses low. Some current flows through the secondary diodes and develops a further voltage drop across the series resistor from the signal pad to the internal circuits. Back-to-back diodes enable current flow between ground domains-preventing large differences from developing between ground domains. The back-to-back diodes prevent damage between interfacing circuits due to power bus differences.

7.6.3 ESD Back-to-Back Diode Connection to Core VSS

Although vsscore and gd are to be connected together within the package, for ESD protection of the PHY/ASIC interface a back to back diode connection between vsscore and gd is provided within the PHY.

Each of the vsscore pins on the PHY must have a strong connection to the ASIC vsscore supply grid. The vsscore pins provided in the IP should be connected with a solid metal. There should be less than 0.5 Ohms of resistance between each vsscore pin on the PHY and the closest clamp within the ASIC (see Figure 7-11 on page 244).

Figure 7-11 Back-to-Back Diodes in PHY



Note 1: x2 configuration only has vsscore_1 and vsscore_0 ; x1 configuration only has vsscore_0
Note 2: Add a vdd-to-vss clamp close to the back-to-back diodes with no more than 0.5 ohm routing resistance from the diodes to the clamp

Additional back to back diodes outside of the PHY (or other vss coupling devices) can be used to connect ASIC ground and PHY gd for additional ESD protection if desired.

If non-ground supplies are connected together for ESD protection through back-to-back diodes, special care must be taken to check for, and guard against latch-up, power-up sequencing and leakage issues.

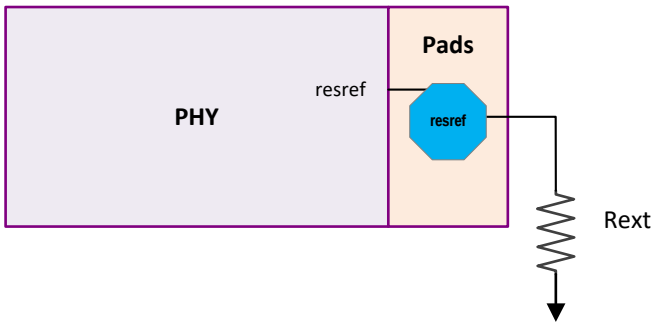
7.7 Reference Clock

For details about reference clock physical implementation, see Chapter 9, “Reference Clock Implementation”.

7.8 Reference Resistor

An external reference resistor is used for calibration of the RX and TX termination. The external resistor is automatically used for calibration when the PHY comes out of reset and when a manual tuning request is made. When a PHY is using the reference resistor, it forces current across the resistor. When a PHY is not using the reference resistor, the current is disabled.

Figure 7-12 Reference Resistor Physical Implementation



7.8.1 Reference Resistor Contention Bus

Each PHY requires access to a reference resistor. This access may be per PHY or it may be shared. A contention bus is used to control PHY access to the external resistor using res_req_in, res_req_out, res_ack_in, res_ack_out.

For a single-PHY implementation, connect the signals as shown in Figure 7-13 on page 246.

For sharing a reference resistor among several PHYs, connect the signals as shown in Figure 7-14 on page 246.

Note

- When sharing a reference resistor amongst several PHYs, it may take up to 30 µs per PHY to sequentially calibrate each PHY.
- Before choosing to share a reference resistor, make sure to be aware of how it will be connected in the package and on the board. For information on package and board implementation, see Chapter 8, “Board- and Package-Level Implementation”.

Figure 7-13 Reference Resistor Contention Bus - Single PHY

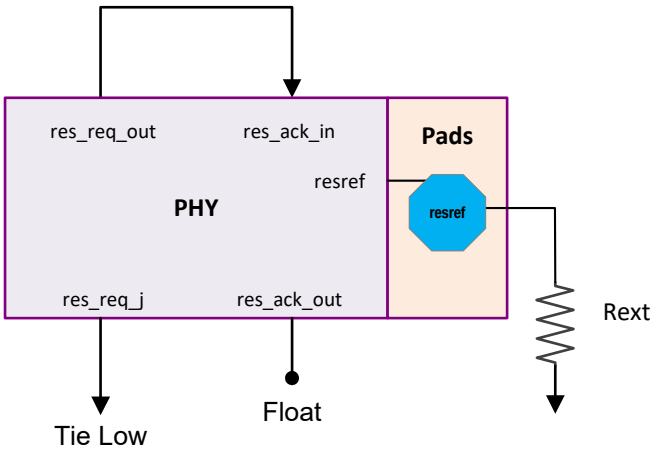
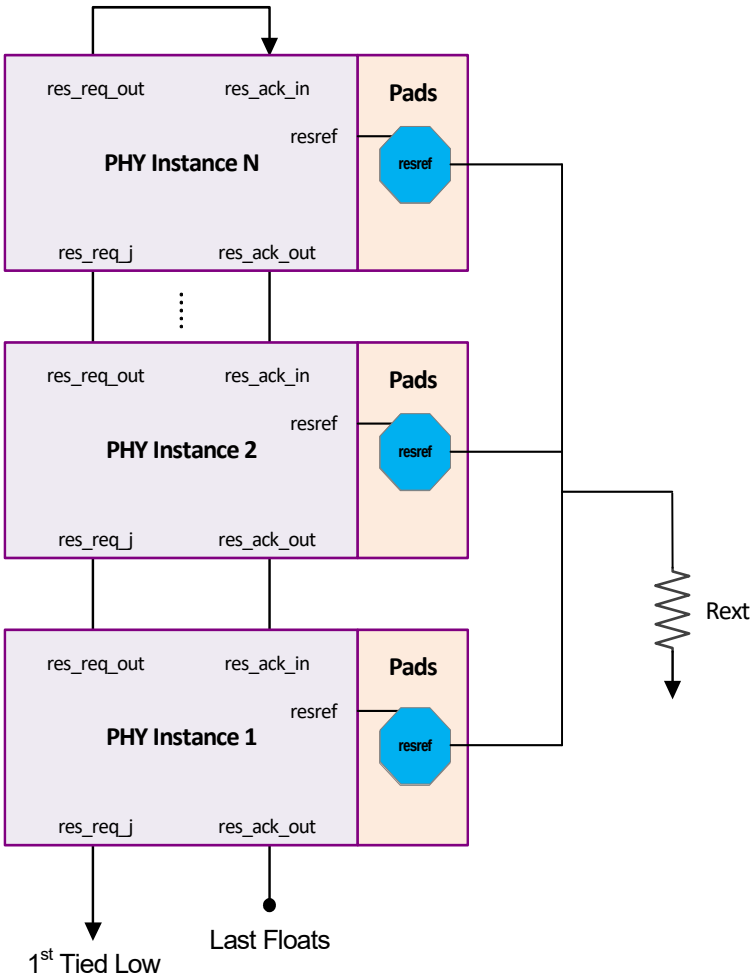


Figure 7-14 Reference Resistor Contention Bus - Multiple PHYs Sharing a Single Resistor



7.9 ATB and DTB Signals

The analog test bus (ATB) is reserved. These pins (atb_s_p, atb_s_m, and atb_f_p) should be left unconnected when integrating the PHY.

The digital test bus (DTB) is provided as a means of accessing internal digital signals for debug purposes. If not needed, these pins (dtb_out) may be left unconnected when integrating the PHY.

7.10 Raw PCS Synthesis

The Raw PCS is delivered as soft RTL, meaning that it will be hardened along with the rest of the RTL of the protocol-PCS controller and ASIC. Synthesizing the Raw PCS requires using a Verilog-2001 compatible synthesis tool.

7.10.1 Synthesis Database

The deliverables include the Design Compiler synthesis scripts in TCL format that can be used to synthesize the Raw PCS logic using the timing model of the PMA. The synthesis database structure is shown in Table 7-6 and the ATPG deliverables are listed in Table 7-7.

Table 7-6 Synthesis Database Structure for PHY

Directory	Description
./Latest/phy/synth	Synthesis directory for PHY ■ run_synth.sh: Shell script to run synthesis using Design Compiler TCL scripts. ■ run_formal.sh: Shell script to run logic equivalence checks between RTL and synthesized netlist of the Raw PCS using Formality. You need to update the variables in the scripts to match the installed technology library location (TECH_DIR, STD_DB. These scripts must be run in the directory ./Latest/phy/synth.
./Latest/phy/synth/scripts	Synthesis and Formal scripts ■ dwc_e16mp_phy_xN_ns_synth.tcl: Design Compiler Synthesis script to synthesize dwc_e16mp_phy_xN_ns_pcs_raw_xN using RTL files and libs for dwc_e16mp_phy_xN_ns PMA. ■ dwc_e16mp_phy_xN_ns_phy_con.tcl: Design Compiler synthesis constraint file for the Raw PCS in TCL format. ■ dwc_e16mp_phy_xN_ns_formal.tcl: Formality script for running logic equivalence check on the synthesized netlist of Raw PCS.

Table 7-7 ATPG Deliverables for the PHY

Directory	Description
./Latest/pma/atpg/<*metal_stack*>/ctl	CTL and SPF model
./Latest/pma/atpg/<*metal_stack*>/models	Verilog scan model of the PMA; only to be used by TetraMax (not to be used for simulations)
./Latest/pma/atpg/<*metal_stack*>/rpt	TetraMax log file
./Latest/pma/atpg/<*metal_stack*>/tetramax	run_tmax.sh: Top-level shell script to run TetraMax TCL script (tmax_atpg.tcl) for generating vectors
./Latest/pma/atpg/<*metal_stack*>/vectors	■ WGL vectors for tester (*_test.stuck.wgl and *_test.tdf.wgl) ■ Top-level Verilog ATPG serial test bench (*_test.serial.stuck.v and *_test.serial.tdf.v) ■ Top-level Verilog ATPG parallel test bench (*_test.para.tdf.v and *_test.para.stuck.v)



For information on Synthesis flow and database for PCIe PCS, refer to the 16G PCS for the DesignWare® Cores Multi-Protocol 16G PHY Databook.

7.10.2 Raw PHY (PCS & PMA) Constraints

The synthesis constraints are provided in the form of TCL script used by Synopsys Design Compiler. This file (dwc_e16mp_phy_xN_ns_con.tcl) describes the design intent and surrounding constraints for synthesis, clocking, timing, power, test, and environmental and operating conditions. The constraints include the input and output delays of interface pins, clock definitions, and any timing path exceptions in the design, such as false paths or multicycle paths.

In most cases, the Raw PCS is synthesized in conjunction with other logic that resides on the opposite side of the ASIC interface boundary. In this situation, timing constraints can be automatically inferred from the adjoining logic and from the PMA timing model to which the Raw PCS is connected. When synthesizing the Raw PCS as a standalone block, signals on the ASIC interface must be constrained so that sufficient time is budgeted for the adjoining logic to be connected later.

7.10.2.1 Hand-Placed Logic Constraints

There are generic, digital-cell, primitive modules within the Raw PCS that are used to instantiate a specific digital cell that must not be logically optimized during synthesis. These modules are typically used around asynchronous clock domain crossing (CDC) points within the design. It is essential that these modules are processed correctly, not just synthesized as usual with the Raw PCS logic. Failure to do so will likely create glitch-generating and reconvergent logic that can cause problems on the ASIC interface.

The modules are listed in Table 7-8.

Table 7-8 Primitive Modules

Module	Function	Hand-instantiated cell Required?
*_gen_sync	3-stage data signal synchronizer	No
*_gen_rst_sync	Asynchronous assert, synchronous de-assert reset synchronizer	No
*_ani_clk_gate	Clock-gating cell	Yes
*_ani_clk_and2	2 input clock AND gate	Yes
*_ani_clk_and3	3 input clock AND gate	Yes
*_ani_clk_or2	2 input clock OR gate	Yes
*_ani_clk_mux	2 input clock MUX	Yes
*_ani_mux	2 input MUX	Yes
*_ani_and2	2 input AND gate	Yes
*_ani_and3	3 input AND gate	Yes
*_ani_or2	2 input OR gate	Yes
*_gen_mux	2 input MUX wrapper – instantiates ani_mux module	No
*_gen_clk_mux	2 input clock MUX wrapper – instantiates ani_clk_mux module	No
*_gen_clk_gate	Clock-gating cell wrapper – instantiates ani_clk_gate module and gen_mux module for scan muxing	No
*_gen_clk_and2	2 input clock AND wrapper – instantiates ani_clk_and2 module	No
*_gen_clk_and3	3 input clock AND wrapper – instantiates ani_clk_and3 module	No
*_gen_clk_or2	2 input clock OR wrapper – instantiates ani_clk_or2 module	No
*_gen_pipe_delay	Pipeline delay module	No
*_gen_pcatch	Pulse catcher – instantiates a gen_sync module for synchronization and a gen_mux for scan muxing	No
*_pcs_raw_gen_glcm	Functional glitchless clock mux (GLCM) – instantiates the following modules: ■ gen_sync ■ gen_clk_and2 ■ gen_clk_gate ■ gen_clk_or2	No

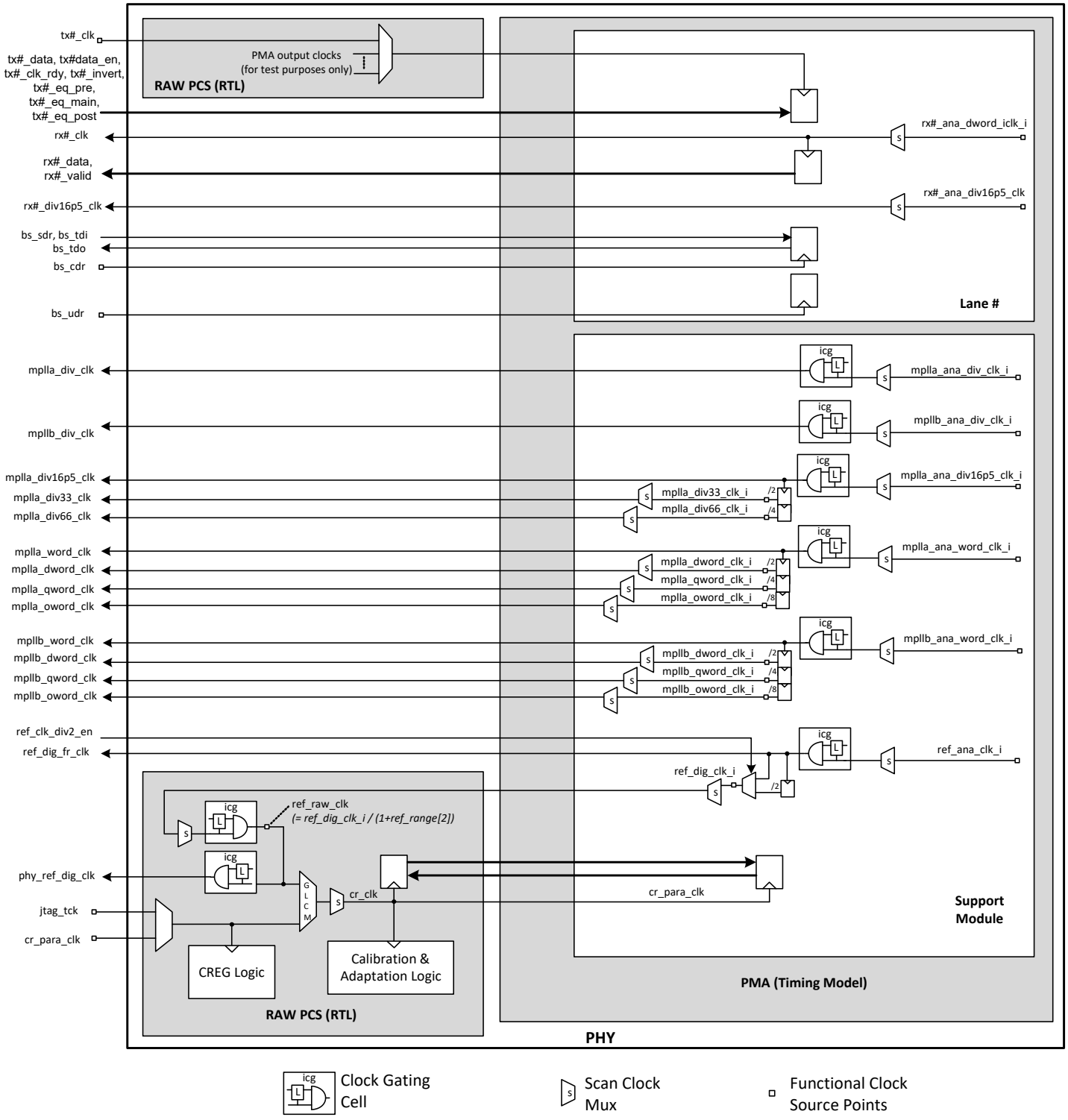
To prevent these modules from being optimized during synthesis, customers can either replace the contents of these modules with an instantiation of the appropriate standard cell from a digital library, or synthesize each module stand-alone with a restricted library, so that only the correct cell is selected by the tool. In both cases, a “set_size_only” attribute must be applied to prevent reoptimization of the instance during regular Raw PCS synthesis.

7.10.2.2 Clock Constraints

Customers need to add external scan clock muxes to txX_clk and jtag_tck, and not cr_para_clk. When scan_mode = 1, the internal cr_para_sel is overridden to zero, so jtag_tck is selected. Internal signal cr_ext_clk is from jtag_tck and cr_para_clk mux. Signal cr_ext_clk is then muxed with ref_raw_clk to produce the cr_clk and used to drive the PMA cr_para_clk input. Signal cr_clk is treated as a separate clock domain from cr_ext_clk, and hence has its own scan clock (scan_cr_clk). There is internal logic on both jtag_tck and cr_ext_clk (which is not on cr_clk), hence the need to externally mux a scan clock on jtag_tck. Also, there is a txX_clk scan clock mux internal to the PMA, which can source any of the MPLLA or MPLLB clocks for testing.

The clocking structure for functional clocks is shown in Figure 7-15 on page 251. The figure shows legends for clock sources (primary or generated clocks), clock gating cells, and scan clock muxes.

Figure 7-15 Functional Clock Definition Diagram for Raw PCS Synthesis



The clock definitions and associated false paths for the PHY are provided in the constraints file dwc_e16mp_phy_xN_ns_con.tcl delivered as part of the synthesis deliverables.

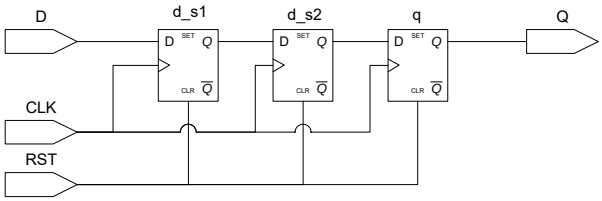
7.10.2.3 Synchronizers

Soft IP (Raw PCS) instantiates synchronizer modules used for clock domain crossing protection. There are two types of synchronizer modules used in the design.

7.10.2.3.1 Data Synchronizers

For single-bit signals that must be passed from one clock domain to another, an instance of the module(s) "dwc_e16mp_*_gen_sync" is used to synchronize the source signal into the receiving clock domain. The structure of a single-bit data synchronizer is shown in Figure 7-16.

Figure 7-16 dwc_e16mp_*_gen_sync module



The input to the first stage of the synchronizer can be considered completely asynchronous. Therefore, all paths to the d_s1 register (of the synchronizer only) can be considered false_paths. (Note, explicitly defining a false path to the first flop of the synchronizer is not required. The user can rely on clock domain separation via clock groups, or false_paths, in the synthesis constraints.)

The second and third registers in the dwc_e16mp _*_gen_sync module are d_s2 and q. To ensure stable synchronization, constrain the maximum delay to these cells, from d_s1/Q to d_s2/D and from d_s2/Q to q/Q to 20% of the sampling clock frequency.

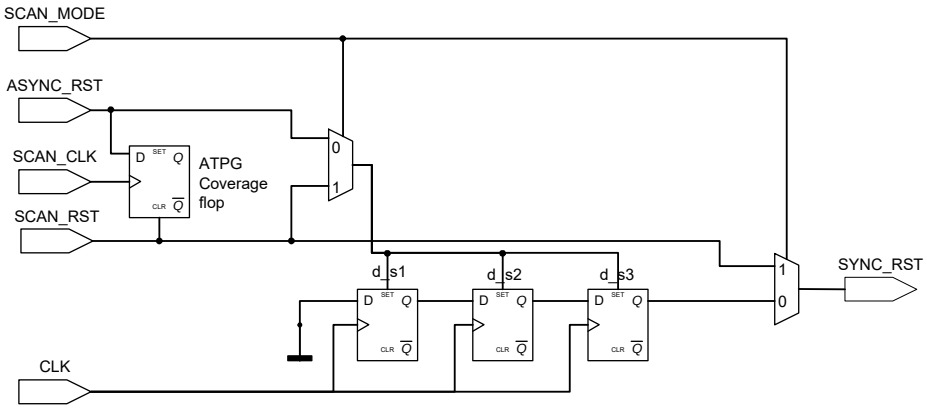
For ease of timing constraint definition, this delay can be implemented as 200 ps (20% of 1000 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instances. You can ignore the clock-tree delay if the paths are constrained explicitly from d_s1/Q to d_s2/D and from d_s2/Q to q/D.

7.10.2.3.2 Reset Synchronizers

For synchronizing the asynchronous reset signal, an instance of the modules, "dwc_e16mp_*_gen_rst_sync," is used. As shown in Figure 7-17 on page 253, any time an external or internal reset condition is passed from one clock domain to another domain, the reset condition must pass through a dwc_e16mp_*_gen_rst_sync module to do the following:

- Asynchronously capture the reset signal
- Stretch the reset pulse to be active for at least one full clock cycle in the receiving domain
- Synchronously de-assert the reset signal with respect to the receiving domain

Figure 7-17 dwc_e16mp_*_gen_rst_sync module



Because assertion of a reset will cause all flip-flops in the synchronizer module to capture the pulse, the de-assertion edge of the asynchronous reset input has no effect on the final flip-flop that drives the reset into the receiving domain. Therefore, all asynchronous paths through the d_s3 register of this module can be considered false.

The second and third registers in the dwc_e16mp_*_gen_rst_sync module are d_s2 and d_s3. To ensure stable synchronization, constrain the maximum delay to these cells, from d_s1/Q to d_s2/D and from d_s2/Q to d_s3/D to 20% of the sampling clock frequency.

For ease of timing constraint definition, this delay can be implemented as 200 ps (20% of 1000 MHz) plus the intended clock tree depth of the respective sampling clock for the associated module instance. You can ignore the clock-tree delay if the paths are constrained explicitly from d_s1/Q to d_s2/D and from d_s2/Q to d_s3/D.

Metastability behavior is modeled in the synchronizers and is activated by defining the macro ANI_SIM_MODE during simulations.

7.10.2.4 Unused Signals

Unused inputs to the raw PCS should be tied to ground potential.

Unused outputs from the raw PCS should be left floating.

8 Board- and Package-Level Implementation

This chapter includes the following sections:

- "Package Type" on page 256
- "RX and TX Differential Signal Guidelines" on page 256
- "Supplies Requirements" on page 257
- "External Reference Resistor" on page 260
- "Reference Clock" on page 264
- "Board Requirements—External Components" on page 264

8.1 Package Type

The PMA is designed to support flip-chip packaging technology. For PMA bump-map diagrams, see "Flip-Chip Pinout" on page 228.

8.2 RX and TX Differential Signal Guidelines

You must design and simulate each of the RX and TX differential routes to ensure that signal integrity requirements are met.



For more information, see the requirements listed in Chapter 2 of the DesignWare Cores SERDES PCB and Packaging Design Guide Application Note.

8.2.1 Package Guidelines—RX and TX Maximum Allowed Package Loss

In general, the maximum supported package insertion loss at Nyquist is 1.5 dB.

For most protocols, the TX eye requirements must be met at the package pins. Both TX output amplitude and equalization are affected by package loss. These considerations can limit the amount of package loss of certain protocols.

For SATA, the TX specifications are to be met at the SATA connector. Dependent on your package and board loss, both the amplitude and the de-emphasis level should be optimized to meet the host and device de-emphasis requirements given in Table 54 of the SATA specification.

For 10GBASE-KR, adjust both the amplitude and the de-emphasis level to account for package loss. Increase the de-emphasis level to account for the amount of package loss at Nyquist rate.

For PCIe, the transmit launch amplitude should be adjusted to account for package loss.

- For 2.5 Gbps and 5 Gbps, the PCIe 4.0 specification states that VTX-DIFF-PP should be measured as the average transition bit amplitude at the 0.5 UI location using across 500 repetitions of the compliance pattern.
- For 8 Gbps and 16 Gbps, VTX-DIFF-PP should be measured with the TX P4 equalization preset as the average amplitude of bits 57-62 of the 64-ones/64-zeros segment of the averaged across 500 repetitions of the compliance pattern.

8.2.2 Signal Integrity Analysis

To enable signal integrity (SI) analysis of the RX and TX routes, s-parameter models as well as an AMI model is provided with the PHY deliverables.

8.2.2.1 S-Parameter Model

To assist in package design, an S-parameter model of the RX input and TX output are provided. These models should be simulated along with your extracted package model to ensure that the protocol specific requirements for return loss and other signal integrity requirements are being met.

8.2.2.2 AMI Model

To assist link modeling, an IBIS Algorithmic Modeling Interface (IBIS-AMI) model of the PHY transmitter and receiver is provided along with a representative channel model.

The provided channel model is simply an example, and customers should measure the system board and connectors they are using to get a more accurate representation of the link performance.

8.3 Supplies Requirements

You must design and simulate the power distribution network (PDN) for each of the PHY supplies to ensure that power integrity requirements are met.

For more information, see the requirements listed in Chapter 3 of DesignWare Cores SERDES PCB and Packaging Design Guide Application Note.

8.3.1 Power Supplies Delivery to Pads

You must perform a power integrity (PI) analysis to ensure that the DC and AC noise supply requirements in Table 8-1 are met on-die.

Table 8-1 Power Supply Requirements

Pin	Supply Voltage at PHY Bumps	AC Noise
vph	DC: 1.5-7% to 1.8+10% (V)	< 3% Vpp of DC level
vp/vptxX/vpdig	DC: 0.9-7% to 0.9+10% (V)	< 5% Vpp of DC level
gd	0 V	N/A

Current waveforms along with Cdie and Rdie are provided with the PHY deliverables. You should simulate the supply noise using this PI enablement along with a model of your package and board to ensure that on-die supply requirements are being met.

The power supply requirements are on-die at the PHY pads.

Note

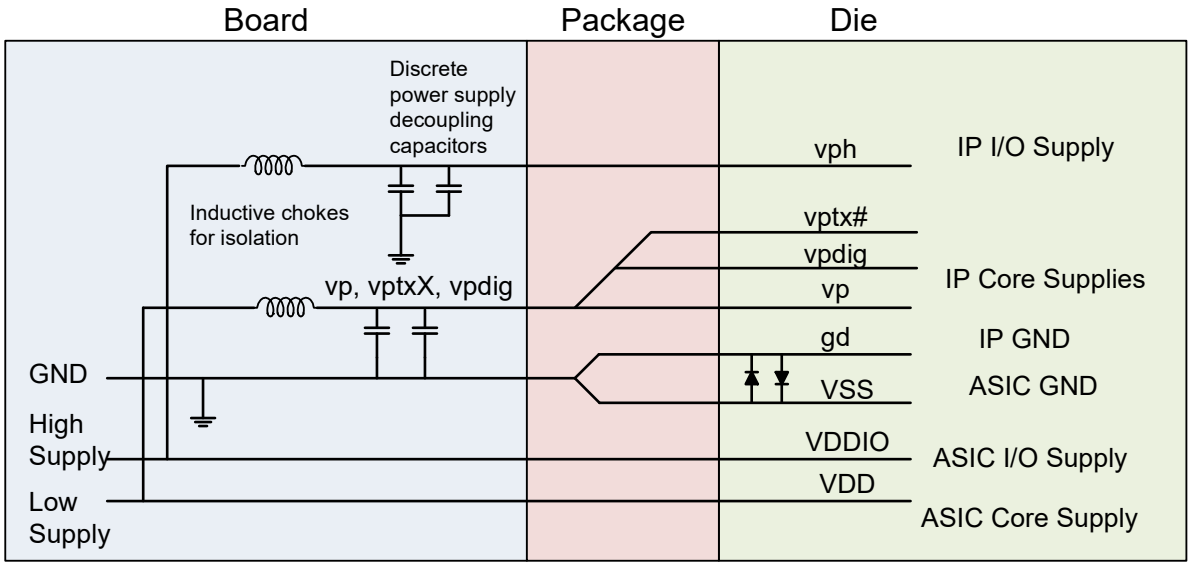
For the quiet VPH analog supply, an LDO is recommended in order to minimize supply noise. If a switching supply is used, there are the following additional recommendations:

- An inductor of ferrite bead should be used to isolate supply noise and filter edge rate at the package pins
- The filtered switching supply waveform should meet the following requirements:
  - For <10mV/us slew rate can accept up to 10mVpp step
  - For >10mV/us slew rate can accept up to 5mVpp step
- It is recommended to share power integrity waveforms with Synopsys for further analysis if the supply noise is close to maximum limits

8.3.2 Package and Board Guidelines—Power Supply Sharing

A diagram of the power supplies is shown in Figure 8-1.

Figure 8-1 Power Supply Connections



To ensure good jitter performance, the PHY is designed to completely isolate power and ground supplies from the rest of the ASIC on-die. Per the power supply connections diagram, you need to isolate the PHY power supplies from other ASIC supplies by using inductive chokes (such as, ferrite bead).

To ensure good ESD performance, the PHY requires a common ground in the package with the rest of the ASIC.

The following supplies may be connected/shorted together within the package:

- PHY core supplies (vptxX, vpdig, vp)
- PHY ground (gd) and ASIC ground

When multiple PHYs are instantiated, their vp and vph may be connected together within the package according to the following requirements:

- If you are using multiple reference clocks with different sources, particularly if they are SSC and each independently spread, then recommendation is not to share the VPH supply in order to mitigate the risk of PLL pulling.
- Per-PHY supply parasitics and bypass capacitance are maintained.
- Power integrity simulation has been performed to confirm AC supply noise requirements are observed for all PHYs.

8.3.3 Package Guidelines—Inductance

Recommended maximum supply inductance for each of vp and vph is given in Table 8-4 ("AC Coupling Capacitors - Target Protocols"). A power integrity simulation should be performed to ensure noise and low impedance requirements are met with the given package parasitics.

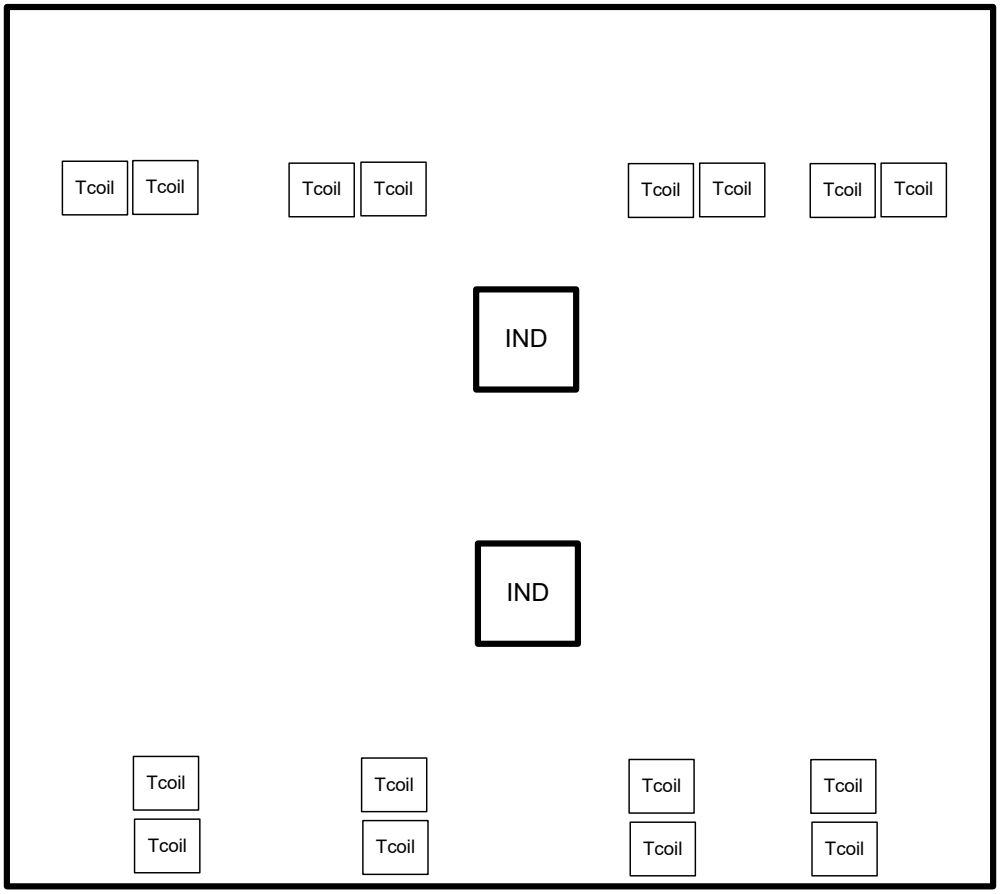
Table 8-2 Maximum Supply Inductance

Parameter	Target	Comments
Package inductance	<500pH	Inductance for each power plane from C4 bumps to BGA ball

8.3.4 Package Guidelines—Substrate Routing

There are on-die inductors implemented in the PHY: Tcoil inductors in transmit and receive paths, and in MPLL. In the silicon interposer or package layout, keep a minimum of 50 µm clearance between on chip inductors and the metal plane underneath. Synopsys recommends that you avoid any routing or metal fill in the area of interposer immediately beneath the on-chip inductors.

Figure 8-2 On-Die Inductors



8.4 External Reference Resistor

An external reference resistor is used for calibration of the RX and TX termination. The external resistor is automatically used for calibration when the PMA comes out of reset and when a manual tuning request is made. When a PMA is using the reference resistor, it forces current across the resistor. When a PMA is not using the reference resistor, the current is disabled. The requirements for the reference resistor are covered in Table 8-3.

For details on reference resistor implementation for a single PMA, see "Reference Resistor Implementation for a Single PMA" on page 262.

For details on reference resistor sharing between multiple PMAs, see "Sharing a Reference Resistor Between Multiple PMAs" on page 262.

Note For additional details on physical implementation and contention bus connectivity see "Reference Resistor" on page 245.

8.4.1 Package and Board Requirements—Reference Resistor

Reference resistor requirements are outlined in Table 8-3.

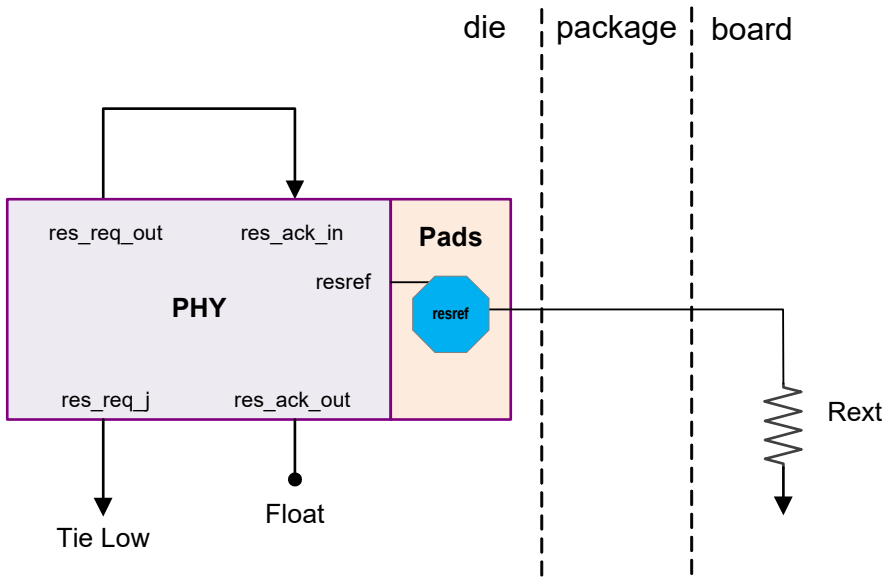
Table 8-3 Reference Resistor Requirements

Item	Value	Design Guidelines
External Reference Resistor Resistance	200 Ω precision resistor (+/-1%)	None
External Reference Resistor Maximum Current	2 mA	None
External Reference Resistor parasitic capacitance C_{load}	< 20 pF	■ The maximum parasitic capacitance includes external loading of package trace and PCB as well as PHY loading. ■ Each PHY connected to the resref pin adds an additional 1.5 pF loading. For example, a single PMA can support up to 18.5 pF (20 pF-1x1.5 pF) of external load. Sharing a single reference resistor between 4 PMAs can support up to 14 pF (20 pF -4x1.5 pF) of external load.
External Reference Resistor parasitic resistance through package and board R_{route}	1% of 200 Ω	Take care to minimize the resistance of the connection through the package to the external reference resistor as this will affect the accuracy of the termination tuning. This spec is for total resistance through the package and board to the board's external resistor. The ground used on the board to terminate the external resistor should be as close as possible to the PMA ground voltage. Noise adjacent signals (for example, clocks) should be spaced away from resref or should be shielded with ground, if possible.

8.4.2 Reference Resistor Implementation for a Single PMA

Package and board connectivity for a single PMA implementation is shown in Figure 8-3.

Figure 8-3 External Reference Resistor Connectivity for a Single PMA



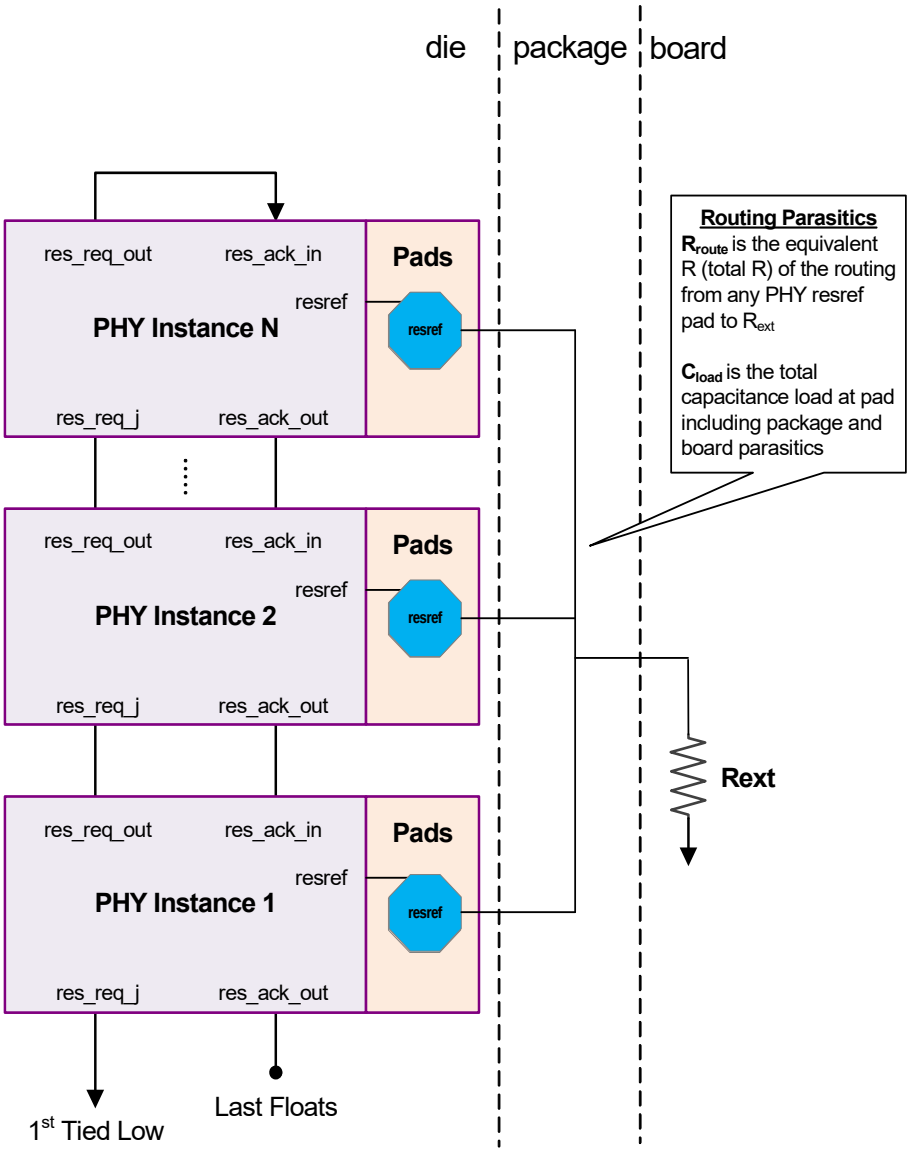
8.4.3 Sharing a Reference Resistor Between Multiple PMAs

The external reference resistor may be shared among a maximum of four instances of the Multi-Protocol 16G PHY for TSMC 28 HPCP PMA by connecting the resref signals together in the package substrate. Sharing an external reference resistor by tying the resref bus together in the package is shown in Figure 8-4.

Note

- If multiple PHYs are sharing a single reference resistor and one or more of the PHYs are being put in a disabled state, see the specific requirements provided in section 5.19 ("Configuration for a Disabled PHY" on page 214)
- If multiple PHYs are sharing a single reference resistor, do not use the VP supply power-gating on any of the PMAs because it will disable the contention bus
- Reference resistors may be shared with other instantiations of the same PMA, but should not be shared with different PMAs
- Take care to meet the parasitic resistance and capacitance requirements noted in "Package and Board Requirements—Reference Resistor" on page 261

Figure 8-4 External Resistor Sharing



8.5 Reference Clock

For details about reference clock board and package requirements, see Chapter 9, "Reference Clock Implementation".

8.6 Board Requirements—External Components

The PHY requires the following external components:

- Use AC coupling capacitors as specified in Table 8-4. For protocols that are unlisted in Table 8-4, use capacitors of 100 nF or as specified by the standard.
- Supply bypass capacitors selected to minimize supply noise
- Precision 200 Ω (± 1%) resistor used for biasing and termination impedance calibration; the precision resistor requirement is shown in Table 8-5.

Table 8-4 AC Coupling Capacitors - Target Protocols

Protocol	Min	Typ	Max	Units
PCIe 1.1 and PCIe 2.1	75	--	200	nF
PCIe 3.1 and PCIe 4.0	180	--	265	nF
SATA	--	--	12	nF
10GBASE-KR and 40GBASE-KR4	--	--	100	nF
10GBASE-KX4	--	--	4.7	nF
SGMII	--	--	4.7	nF
QSGMII	--	--	100	nF

Table 8-5 External Component Requirements

Component	Value
resref	200 Ω (± 1%)

9 Reference Clock Implementation

The PHY is designed to handle a wide range of input clock frequencies to support a wide range of applications.

This chapter includes the following sections:

- "Reference Clock Requirements" on page 266
- "Reference Clock Implementation" on page 269

9.1 Reference Clock Requirements

The PHY supports a differential reference clock source. The source may be driven through either external pads or internal pins. The reference clock must meet specific requirements for signal swing and jitter.

9.1.1 Reference Clock Requirements PCIe

The PCIe reference clock requirements (such as swing, edge rate, jitter, and so on) for all data rates are listed in the PCI Express Base Specification 4.0, in section 8.6 “Refclk Specifications.”



With respect to PCIe specifications, the PMA uses an Independent Refclk (IR) architecture, also known as a data clocked receiver architecture.

9.1.2 Generic Reference Clock Requirement for Protocols

Differential reference clock requirements for protocols other than PCIe are given in Table 9-1 for data rates below and above 8 Gbaud.

Table 9-1 Reference Clock Input Requirements

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	200	MHz	Not continuous; per protocol as noted in the databook
FREF_OFFSET	Reference clock frequency offset	-100	100	ppm	
		-350	350	ppm	SATA 1.5G/3G/6G
RMSJREF_CLK ≤ 8 Gbps	Reference clock random Jitter		1.6	ps_{rms}	Integrated RJ from 12 kHz to 20 MHz
			1.2		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK ≤ 8 Gbps	Reference clock deterministic jitter		2.8 5.6	ps, pp	0.75-10MHz (offset)[a] 0.2-50MHz (offset)[a]
RMSJREF_CLK > 8 Gbps	Reference clock random jitter		1.0	ps_{rms}	Integrated RJ from 12 kHz to 20 MHz
			0.8		Integrated Rj from 2 MHz to 20 MHz
DJREF_CLK > 8 Gbps	Reference clock deterministic jitter		1.7 3.4	ps, pp	0.75-10 MHz (offset)[a] 0.2-50 MHz (offset)[a]
DCREF_CLK	Duty cycle	40	60	%	
VCMREF_CLK	Common mode input level	0	VP	V	Differential inputs

Table 9-1 Reference Clock Input Requirements (Continued)

Symbol	Parameter	Min	Max	Units	Conditions
VDREF_CLK	Differential input swing	300	2*VP	mVpp	Differential inputs must meet both VDREF_CLK and VDSEREF_CLK
VDSEREF_CLK	Differential clock single ended voltage	0	VP		Differential input must meet both VDREF_CLK and VDSEREF_CLK
SKEWDIF_CLK	Differential clock input skew		100	ps	
SWREF_CLK	Differential clock input slew rate	0.6		V/ns	20–80%

Note:

For PCIe, follow the CEM specification requirements

a. DJ noise limits are for noise within an offset from the divided reference clock and harmonics of the divided reference clock.

9.1.3 Repeat Clock Specification

The PHY is able to repeat the input reference clocks through up to four (4) PHY macros. A reference clock repeater is provided to output the reference clock through ref_repeat_clk_p and ref_repeat_clk_m. The specifications for the repeater are provided in Table 9-2 on page 268.

Table 9-2 Repeater Clock Output Specification

Symbol	Parameter	Min	Max	Units	Conditions
FREF_CLK	Reference clock frequency	19.2	200	MHz	
DJREF_CLK	Increase in reference clock deterministic jitter for each repeat		0.3	ps	0.75-10 MHz (offset) Due to self noise from the PMA. Any additional external noise is additive.
DCREF_CLK	Duty cycle	40	60	%	No degradation is allowed
VCMREF_CLK	Common mode output	VP-0.2	VP-0.1	V	Differential CML output
VDREF_CLK	Differential output swing	400	800	mVpp	Recommended constant swing design, which allows for 400 mVpp swing all the time
SKEWDIF_CLK	Differential clock output skew		100	ps	No extra skew is allowed; instead, the PHY corrects the incoming clock skew (if needed)
CL_CLK	Single-ended output loading		1	pF	Specifies the maximum loading that the PHY can drive
BW_CLK	Clock repeater BW	500		MHz	Specifies rise/fall time degradation from inputs
LVL_OFF_CLK	Differential signal level when disabled		GD		Disabled clock repeater is pulled to 0 V
REPT_CLK	Maximum repeat times allowed		3		Repeat through three PMAs to the final fourth PMA (that is, for x16 aggregated PHY)

9.2 Reference Clock Implementation

There are two types of reference clock implementations:

- External off-chip reference clocks using ref_pad_clk_p/ref_pad_clk_m
- On-chip reference clocks using ref_alt_clk_p/ref_alt_clk_m

Both reference clock implementations support either a differential clock or a CMOS single-ended clock. The reference clock input is an unterminated, AC coupled input. If the reference clock is a single-ended CMOS clock, one of the clock inputs (such as ref_pad_clk_p) must be a full-swing CMOS clock, and the other input (such as ref_pad_clk_m) should be tied to either high or low. If the reference clock is differential, it may be CML or LVDS, or any differential specification that meets the requirements outlined in Table 9-1 on page 266.

Any reference clock inputs that are not being used should be tied to ground.

Note

You must keep the on-chip and off-chip inputs mutually exclusive:

- If an on-chip clock source is used, then you must either tie the off-chip source to ground or leave it floating.
- If an off-chip clock source is used, then you must either tie the on-chip source to ground or leave it floating.

Doing so guarantees that no noise couples through the MUX between the on-chip and off-chip selection.

9.2.1 External Off-Chip Clock Sources

When routing reference clocks in from off-chip, a signal integrity (SI) analysis should be performed using a model of the clock input, package, board traces, and clock source. This analysis should confirm that the reference clock input specification is being met across the external on-chip reference clock input (ref_pad_clk_p/ref_pad_clk_m), as outlined in "Reference Clock Requirements" on page 266.

A model of the reference clock input is provided in Figure 9-1 with parameters provided in Table 9-3.

Figure 9-1 Model of Reference Clock Pad Input Load

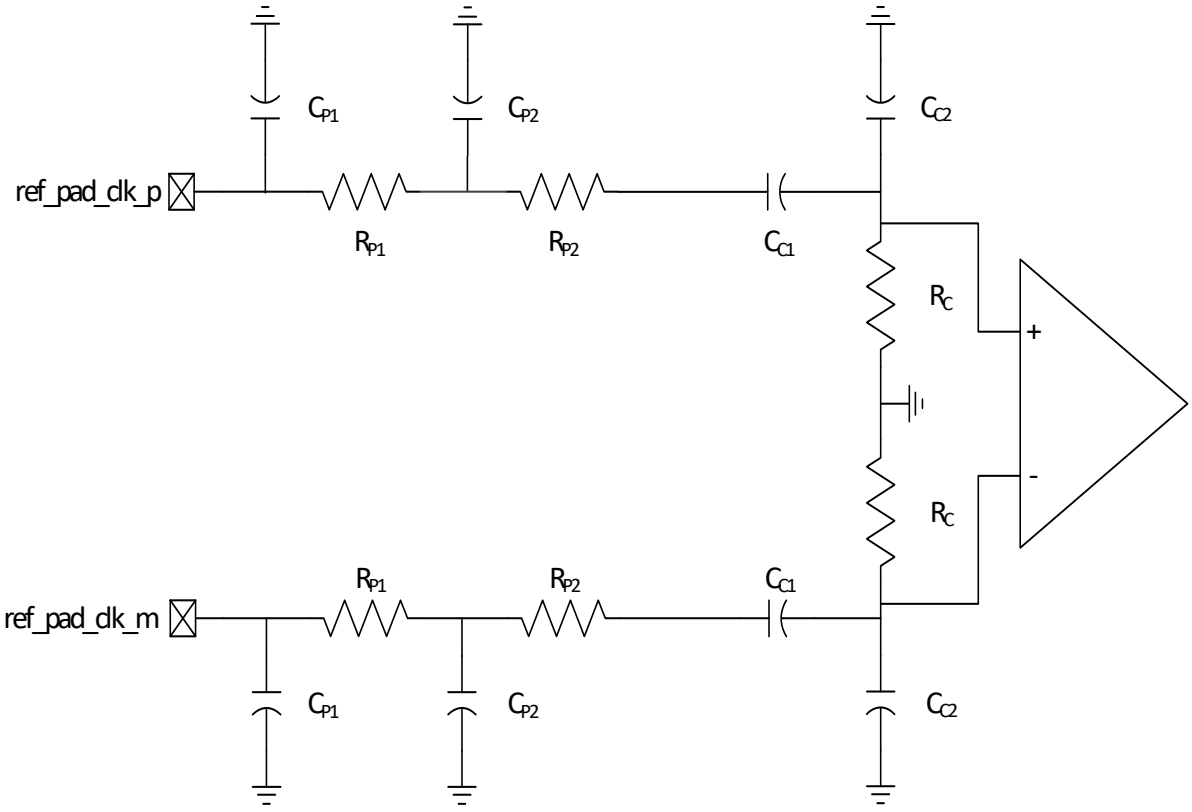


Table 9-3 Reference Clock External Pad Input Parasitics[a]

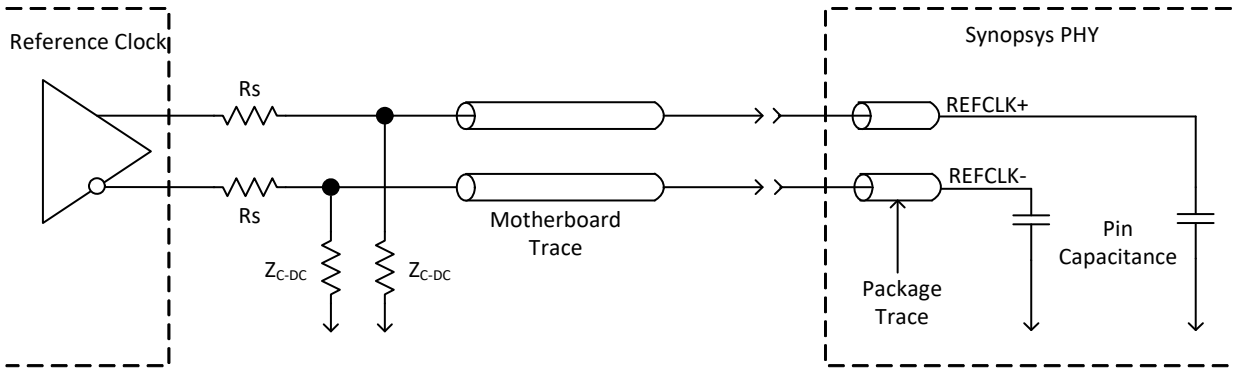
Parasitic Device	Value	Units
CP1	1.107	pF
CP2	0.0527	pF
RP1	750	Ω
RP2	360	Ω
CC1	0.532	pF
CC2	22	fF
RC	680	kΩ

a. All dimensions in these tables are preliminary and subject to change.

The reference clock inputs are not terminated. This is to allow for connection of reference clocks to multiple PHYs at the same time. If termination is used, it should be placed on your board as close to the chip as possible. For PCIe, see the PCIe CEM specification for detail on how to implement the on-board termination.

If you use off-chip reference clock (that is, ref_pad_clk_p or ref_pad_clk_m), you must terminate the off-chip clock source to suppress reflection. Also, make sure that the clock swing at the Synopsys PHY input is within reference clock specification. An example of this implementation is shown in Figure 9-2 on page 271.

Figure 9-2 Off-chip Clock Source Termination



9.2.2 On-Chip Clock Sources

Many applications require an on-chip reference clock. The challenge is getting the reference clock to the PHY without adding supply-noise-induced jitter to the clock. Check SolvNet for Synopsys application notes on techniques that have been used to successfully distribute clocks on-chip using isolated and filtered local supplies.

The following general guidelines should be followed when routing an on-chip reference clock:

- Design refclk routes as differential routes
- Route on higher, thicker metal to maximize bandwidth (lower level metals tend to be too resistive)
- Use shielding on both sides of differential routes
- Minimize any source of noise that may couple in to the refclk
- Balance any supplies/signals that may couple in so that noise is common mode
- Avoid using an automatic PNR flow to route the reference clock
- Simulate the reference clock routing (consult with Synopsys on an appropriate model)
- Ensure that the reference clock at the far end of the route at the input to the next PMA still meet signal swing requirements
- Share your results with Synopsys for review

A model of the reference clock input is provided in Figure 9-3 on page 272 with parameters provided in Table 9-4 on page 272.

When routing reference clocks on-chip, you should consider the parasitic loading of the clock distribution lines and the PHY clock input. This analysis should confirm that the reference clock input specification is being met across the on-chip reference clock input (ref_alt_clk_p/ref_alt_clk_m) as outlined in "Reference Clock Requirements" on page 266.

Figure 9-3 Model of Reference Clock ASIC Side ALT Input Loading

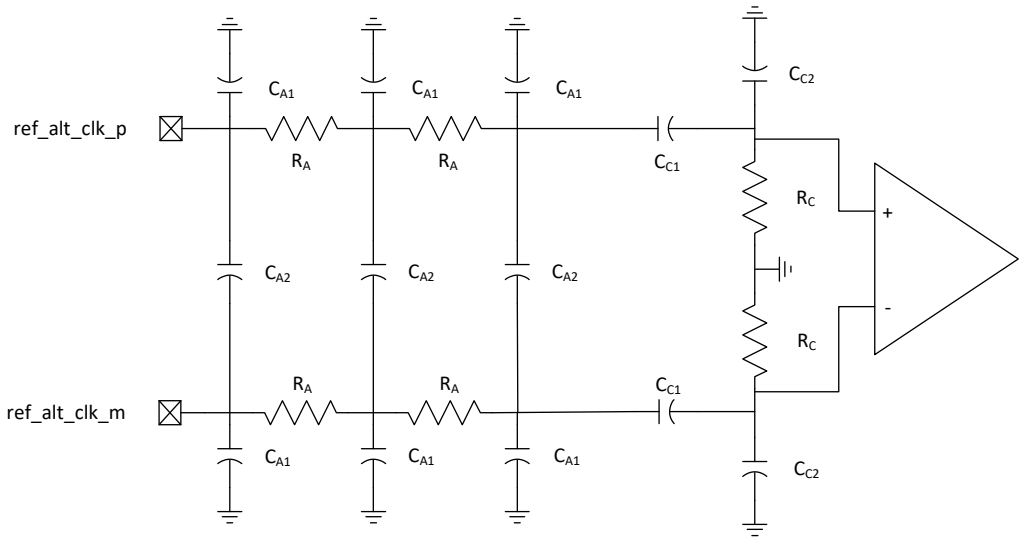


Table 9-4 Reference Clock ASIC side ALT Input Parasitics[a]

Parasitic Device	Value	Units
CA1	100	fF
CA2	20	fF
RA	40	Ω
CC1	0.532	pF
CC2	22	fF
RC	680	kΩ

a. All dimensions in these tables are preliminary and subject to change.

9.2.3 Reference Clock Repeater

There is a reference clock repeater output (ref_repeat_clk_{p,m}) that allows the PHY to buffer the reference clock and send it on to another PHY. This allows you to bring the reference clock in through one PHY and distribute it on-chip to other PHYs.

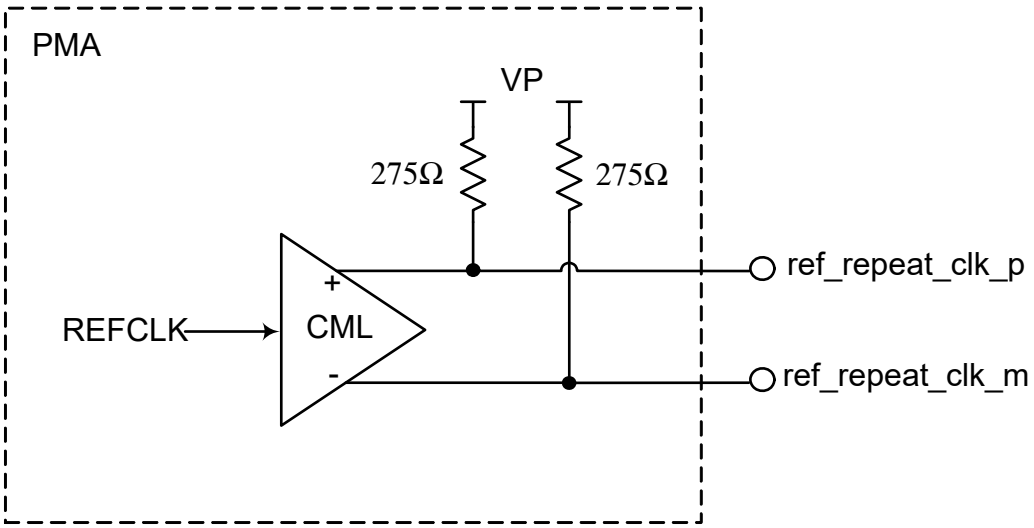
The driver is a VP supply referenced CML output clock driver, as shown in Figure 9-4.

The repeater is designed to meet the specifications given in "Repeat Clock Specification" on page 268.

When routing an on-chip clock source follow guidelines noted in section 9.2.2 ("On-Chip Clock Sources"). The repeater and routing should be simulated. Contact Synopsys for an appropriate model for the reference clock repeater.

Note Ref_clk repeat buffer and single ended buffers are not in an always on power domain. If power-gating is enabled, these outputs will be powered down. Do not use these signals for logic or PHYs meant to remain operational during power-gating.

Figure 9-4 Reference Clock Repeater



10 Electrical Specifications

This chapter includes the following sections:

- "Power Dissipation Overview" on page 276
- "Power Dissipation for PCIe" on page 277
- "Power Dissipation for Ethernet" on page 282
- "Power Dissipation for SATA" on page 284
- "Power Dissipation for JESD204B" on page 287

10.1 Power Dissipation Overview

The PHY supports the power supply voltages in Table 10-1. The values are supported across a temperature range of -40°C to 125°C.

Table 10-1 Supported Power Supply Limits at the PHY Macro Bumps

Supply	Description	Min	Typical	Max	Units
vph	High Voltage IO supply	1.5-7%	1.5 or 1.8	1.8+10%	V
vp/vptxX/vpdig	Analog supply	0.9-7%	0.9	0.9+10%	V

The following sections provide simulated power consumption values for the PMA hard macro.

- The typical (Typ) values are for TT, 25C, 0.9 V, 1.8 V.
- The worst-case (Wc) values are for FF, 125C, 0.99 V, 1.98 V.

Attention

- The power numbers in the following tables do not include PCS or raw PCS power consumption.
- Target power dissipation values indicated in the following tables are based on simulations and are preliminary.
- Power dissipation is subject to change based on PHY alignment with ratified PCIe 4.0 specification and silicon validation.

10.2 Power Dissipation for PCIe

Note

- Lower iboost setting can result in some corners being < 800mVpp full swing amplitude. Consult with Synopsys for this case.
- Bypass of the DFE may be suitable for shorter channels. Consult with Synopsys for this case.
- PCIe P1 mode power dissipation is provided for a data rate of 8 Gbps.

Table 10-2 PCIe x1 Power Dissipation

Mode			Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P0	tx_boost = 0	16Gbps Typ	100.6	53.4	154.0	170.6	186.7
		16Gbps Wc	184.7	69.0	253.7	296.7	319.4
	tx_boost = 8	16Gbps Typ	99.6	55.3	154.9	172.6	189.2
		16Gbps Wc	183.3	70.2	253.4	297.2	320.3
	tx_boost = 15	16Gbps Typ	98.6	57.3	156.0	174.8	192.0
		16Gbps Wc	181.7	71.4	253.1	297.6	321.2
		8Gbps Typ	62.5	44.5	106.9	122.9	136.3
		8Gbps Wc	133.5	58.0	191.5	227.9	247.1
		5Gbps Typ	44.0	35.4	79.5	92.8	103.4
		5Gbps Wc	109.3	46.2	155.6	184.5	199.8
		2.5Gbps Typ	33.1	33.1	66.2	79.4	89.4
		2.5Gbps Wc	95.1	43.3	138.4	165.5	179.8

Table 10-2 PCIe x1 Power Dissipation (Continued)

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P0s	16Gbps Typ	56.1	53.0	109.1	130.0	145.9
	16Gbps Wc	126.3	69.2	195.6	239.3	262.2
	8Gbps Typ	37.5	42.2	79.7	97.1	109.7
	8Gbps Wc	101.4	57.3	158.7	194.9	213.8
	5Gbps Typ	26.9	33.5	60.4	74.4	84.5
	5Gbps Wc	87.1	46.1	133.2	162.2	177.4
	2.5Gbps Typ	20.5	31.2	51.7	65.3	74.6
	2.5Gbps Wc	80.7	43.2	123.8	151.1	165.3
P1	Typ	6.0	14.0	20.0	26.4	30.5
	Wc	55.8	21.3	77.1	90.4	97.4
P1.1	Typ	1.1	0.9	2.0	2.4	2.7
	Wc	54.0	3.4	57.4	59.0	60.2
P1.2	Typ	0.80	0.09	0.9	0.9	0.9
	WC 125	53.4	3.00	56.4	57.8	58.8
P2	Typ	3.5	1.10	4.6	4.8	5.1
	Wc	53.7	3.2	56.9	58.4	59.4
Power Down with Power Gating (P1.2PG)	Typ	0.0	0.001	0.001	0.002	0.002
	Wc	0.00	0.213	0.213	0.351	0.421
IDDQ/Power Down	Typ	0.7	3.38E-02	0.8	0.7	0.7
	Wc	51.2	0.4	51.6	51.3	51.5

Table 10-3 PCIe x2 Power Dissipation

Mode			Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P0	tx_boost = 0	16Gbps Typ	197.9	94.7	292.6	320.2	348.6
		16Gbps Wc	358.0	120.4	478.4	553.1	592.9
	tx_boost = 8	16Gbps Typ	195.8	98.5	294.4	324.0	353.6
		16Gbps Wc	355.0	122.8	477.9	554.2	594.7
	tx_boost = 15	16Gbps Typ	193.9	102.6	296.5	328.4	359.2
		16Gbps Wc	352.0	125.2	477.2	555.1	596.4
		8Gbps Typ	121.7	76.9	198.6	224.9	248.0
		8Gbps Wc	255.5	98.6	354.1	415.7	448.2
		5Gbps Typ	85.5	59.4	144.9	166.0	183.8
		5Gbps Wc	208.0	76.5	284.4	332.1	357.3
		2.5Gbps Typ	63.6	54.7	118.3	139.3	155.7
		2.5Gbps Wc	179.4	70.6	250.1	294.2	317.5
P0s		16Gbps Typ	108.9	93.9	202.8	238.9	267.0
		16Gbps Wc	241.2	121.0	362.2	438.4	478.4
		8Gbps Typ	71.7	72.4	144.1	173.1	194.9
		8Gbps Wc	191.2	97.2	288.4	349.6	381.7
		5Gbps Typ	51.2	54.9	106.1	128.5	144.9
		5Gbps Wc	163.5	75.0	238.6	285.7	310.5
		2.5Gbps Typ	38.5	50.3	88.8	110.1	125.2
		2.5Gbps Wc	150.6	69.3	219.9	263.4	286.3
P1		Typ	8.9	15.8	24.7	31.7	36.5
		Wc	100.0	25.2	125.2	140.6	148.9
P1.1		Typ	1.9	1.9	3.8	4.6	5.1
		Wc	95.4	6.2	101.6	104.6	106.7

Table 10-3 PCIe x2 Power Dissipation (Continued)

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P1.2	Typ	1.3	0.2	1.5	1.4	1.5
	WC 125	94.3	4.54	98.8	100.8	102.3
P2	Typ	5.7	1.5	7.2	7.4	7.9
	Wc	97.2	5.1	102.3	104.7	106.4
Power Down with Power Gating (P1.2PG)	Typ	0.0	0.001	0.001	0.002	0.002
	Wc	0.00	0.238	0.238	0.393	0.472
IDDQ/Power Down	Typ	1.3	0.0	1.3	1.2	1.2
	Wc	93.2	0.5	93.7	93.1	93.3

Table 10-4 PCIe x4 Power Dissipation

Mode			Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P0	tx_boost = 0	16Gbps Typ	392.5	176.2	568.7	617.5	670.4
		16Gbps Wc	704.4	222.1	926.5	1063.9	1137.2
	tx_boost = 8	16Gbps Typ	388.4	183.8	572.2	625.3	680.4
		16Gbps Wc	698.6	226.9	925.5	1066.0	1140.5
	tx_boost = 15	16Gbps Typ	384.5	192.0	576.5	634.1	691.7
		16Gbps Wc	692.4	231.7	924.1	1067.8	1144.3
		8Gbps Typ	240.3	140.6	380.8	427.1	469.3
		8Gbps Wc	499.5	178.5	678.0	789.0	847.9
		5Gbps Typ	168.4	106.7	275.1	311.6	343.6
		5Gbps Wc	405.2	135.9	541.1	625.4	670.3
		2.5Gbps Typ	124.7	97.3	222.0	258.2	287.4
		2.5Gbps Wc	348.2	124.2	472.3	549.6	590.5

Table 10-4 PCIe x4 Power Dissipation (Continued)

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
P0s	16Gbps Typ	214.6	174.5	389.1	454.9	507.3
	16Gbps Wc	470.9	223.2	694.1	834.5	908.2
	8Gbps Typ	140.3	131.5	271.8	323.6	363.0
	8Gbps Wc	371.0	175.5	546.5	656.9	714.8
	5Gbps Typ	99.8	97.8	197.6	236.5	265.8
	5Gbps Wc	316.4	133.0	449.4	532.7	576.6
	2.5Gbps Typ	74.4	88.5	162.9	199.8	226.3
	2.5Gbps Wc	290.6	121.4	412.0	488.1	528.1
P1	Typ	14.6	18.4	33.0	40.7	46.3
	Wc	188.5	31.6	220.2	238.8	249.3
P1.1	Typ	3.6	3.7	7.3	8.8	10.0
	Wc	178.3	11.7	190.0	195.8	199.7
P1.2	Typ	2.4	0.3	2.7	2.6	2.7
	WC 125	176.1	7.61	183.7	186.9	189.4
P2	Typ	10.0	2.4	12.4	12.7	13.4
	Wc	184.2	9.1	193.3	197.4	200.4
Power Down with Power Gating (P1.2PG)	Typ	0.0	0.002	0.002	0.003	0.004
	Wc	0.00	0.289	0.289	0.477	0.573
IDDQ/Power Down	Typ	2.4	0.1	2.5	2.3	2.3
	Wc	177.2	0.8	177.9	176.7	176.9

10.3 Power Dissipation for Ethernet

Table 10-5 Ethernet x1 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	10.3Gbps Typ	69.0	42.8	111.8	126.2	139.1
	10.3Gbps Wc	180.9	55.8	236.8	271.2	289.7
	3.13Gbps Typ	33.1	36.0	69.1	83.8	94.6
	3.13Gbps Wc	134.0	47.2	181.3	210.6	226.2
	1.25Gbps Typ	25.5	32.4	57.9	71.5	81.2
	1.25Gbps Wc	124.1	42.7	166.7	193.2	207.3
Power Down	Typ	1.2	0.0	1.2	1.1	1.1
	Wc	89.5	0.37	89.9	89.2	89.4

Table 10-6 Ethernet x2 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	10.3Gbps Typ	135.7	73.6	209.3	232.5	254.5
	10.3Gbps Wc	347.3	94.7	442.0	500.0	531.3
	3.13Gbps Typ	63.9	60.9	124.8	148.9	167.1
	3.13Gbps Wc	253.3	78.1	331.4	379.6	405.4
	1.25Gbps Typ	48.7	53.2	101.9	123.7	139.7
	1.25Gbps Wc	233.1	69.4	302.4	345.2	368.1
Power Down	Typ	2.2	0.0	2.2	2.0	2.0
	Wc	165.3	0.50	165.8	164.5	164.7

Table 10-7 Ethernet x4 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	10.3Gbps Typ	269.1	134.0	403.2	443.3	483.5
	10.3Gbps Wc	679.9	170.3	850.2	954.1	1010.3
	3.13Gbps Typ	125.4	109.5	234.9	277.1	310.0
	3.13Gbps Wc	491.8	138.4	630.2	715.3	760.9
	1.25Gbps Typ	95.1	94.4	189.5	227.2	255.5
	1.25Gbps Wc	451.1	121.6	572.7	647.2	687.4
Power Down	Typ	4.1	0.1	4.2	3.8	3.8
	Wc	316.9	0.77	317.7	315.0	315.3

10.4 Power Dissipation for SATA

Table 10-8 SATA x1 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
PHYRDY	6Gbps Typ	46.0	35.8	81.8	95.1	105.9
	6Gbps Wc	149.3	49.0	198.3	228.7	244.8
	3Gbps Typ	41.8	34.1	75.9	88.7	99.0
	3Gbps Wc	143.3	46.9	190.2	219.2	234.7
	1.5Gbps Typ	41.4	33.3	63.1	87.3	97.3
	1.5Gbps Wc	141.3	45.9	83.8	215.7	230.8
Partial	Typ	33.6	35.8	69.4	84.0	94.8
	WC 125	135.0	49.4	184.4	215.1	231.4
Slumber	Typ	3.1	1.1	4.2	4.4	4.8
	Wc	91.7	3.2	94.8	96.0	97.0
Power Down	Typ	1.2	0.0	1.2	1.1	1.1
	WC 125	89.5	0.37	89.9	89.2	89.4
	WC 85	0.0	0.00	0.0	0.0	0.0
	WC 25	0.0	0.00	0.0	0.0	0.0

Table 10-9 SATA x2 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
PHYRDY	6Gbps Typ	89.8	59.6	149.4	170.2	188.1
	6Gbps Wc	283.9	80.4	364.3	413.7	440.3
	3Gbps Typ	81.3	56.2	137.5	157.5	174.3
	3Gbps Wc	271.9	76.2	348.1	394.9	420.0
	1.5Gbps Typ	80.6	54.7	115.3	154.5	170.9
	1.5Gbps Wc	267.8	74.3	150	387.7	412.2
Partial	Typ	64.9	59.7	124.6	148.0	165.9
	WC 125	255.3	81.1	336.4	386.6	413.4
Slumber	Typ	5.4	1.5	6.9	7.1	7.6
	Wc	169.5	5.1	174.6	176.3	178.0
Power Down	Typ	2.2	0.0	2.2	2.0	2.0
	WC 125	165.3	0.50	165.8	164.5	164.7
	WC 85	0.0	0.00	0.0	0.0	0.0
	WC 25	0.0	0.00	0.0	0.0	0.0

Table 10-10 SATA x4 Power Dissipation

Mode		Current from vp/vpdig/vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
PHYRDY	6Gbps Typ	177.3	106.6	283.9	319.5	351.4
	6Gbps Wc	553.2	142.4	695.6	782.6	829.6
	3Gbps Typ	160.4	99.7	260.1	293.9	323.8
	3Gbps Wc	529.1	134.0	663.1	744.9	789.1
	1.5Gbps Typ	159.0	97.0	221.3	288.6	317.7
	1.5Gbps Wc	521.0	130.1	284.7	730.5	773.5
Partial	Typ	127.5	106.8	234.3	274.9	307.0
	WC 125	495.9	143.9	639.8	728.4	775.9
Slumber	Typ	9.9	2.4	12.3	12.5	13.3
	Wc	325.1	9.1	334.3	336.9	339.9
Power Down	Typ	4.1	0.1	4.2	3.8	3.8
	WC 125	316.9	0.77	317.7	315.0	315.3
	WC 85	0.0	0.00	0.0	0.0	0.0
	WC 25	0.0	0.00	0.0	0.0	0.0

10.5 Power Dissipation for JESD204B

Table 10-11 JESD204B x1 Power Dissipation

Mode		Current from vp/vpdig (mA)	Current from vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	12.5Gbps Typ	81.7	40.6	122.3	134.4	146.6	81.7
	12.5Gbps Wc	158.5	55.2	213.6	247.9	266.1	158.5
Power Down	Typ	0.7	0.0	0.8	0.7	0.7	0.7
	Wc	51.2	0.4	51.5	51.3	51.4	51.2

Table 10-12 JESD204B x2 Power Dissipation

Mode		Current from vp/vpdig (mA)	Current from vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	12.5Gbps Typ	160.9	71.2	232.0	251.5	272.9	160.9
	12.5Gbps Wc	306.4	95.4	401.8	460.7	492.2	306.4
Power Down	Typ	1.3	0.0	1.3	1.2	1.2	1.3
	Wc	93.1	0.5	93.6	93.0	93.2	93.1

Table 10-13 JESD204B x4 Power Dissipation

Mode		Current from vp/vpdig (mA)	Current from vptx (mA)	Current from vph (mA)	Total Current (mA)	Total Power Consumption with 1.5 V vph (mW)	Total Power Consumption with 1.8 V vph (mW)
Active	12.5Gbps Typ	319.3	132.2	451.5	485.7	525.4	319.3
	12.5Gbps Wc	602.2	175.8	778.0	886.3	944.3	602.2
Power Down	Typ	2.4	0.1	2.5	2.3	2.3	2.4
	Wc	177.1	0.8	177.8	176.6	176.8	177.1

11 Register Descriptions

This chapter details all possible registers in the PHY and includes the following sections:

- "Memory Access Attributes" on page 290
- "PHY Registers" on page 293

11.1 Memory Access Attributes

The Memory Access attribute is defined as <ReadBehavior>/<WriteBehavior> which are defined in Table 11-1. Also, some register fields use the optional attributes listed in Table 11-3 on page 291.

Note

The PHY supports broadcast-write where a sub-set of addresses can be used to write into the same register across all lanes, concurrently. Address bits [15:12] can be set to 4'b1001 to access the broadcast feature for the PMA lanes, or set to 4'b1010 to access the broadcast feature for the Raw PCS lanes. For example, for a 4-lane PHY, a register write into address 0x9?AB performs simultaneous writes into addresses 0x10AB, 0x11AB, 0x12AB, 0x13AB.

When the firmware is running, there are register reads and writes using the same CR space. Therefore, certain registers may be overridden if the firmware is not halted.

Table 11-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field.
Wo	You can only write to this register once field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	Any write toggles this register field.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 11-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 11-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the core updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.

Table 11-3 Optional Attributes (Continued)

Attribute	Description
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

11.2 PHY Registers

In Table 11-4, follow the link for the register to see a detailed description of the register.

Table 11-4 Registers for Address Block: E16_mem_map

Register	Offset	Description
SUP_DIG_IDCODE_LO on page 324	0x0	**Description**: Low 16 bits of IDCODE
SUP_DIG_IDCODE_HI on page 325	0x1	**Description**: High 16 bits of IDCODE
SUP_DIG_REFCLK_OVRD_IN on page 326	0x2	**Description**: Override values for incoming REFCLK and RESET controls from ASIC
SUP_DIG_MPLLA_B_DIV_CLK_OVRD_IN on page 328	0x3	**Description**: Override values for incoming MPLLA_B_DIV_CLK controls from ASIC
SUP_DIG_MPLLA_OVRD_IN_0 on page 329	0x4	**Description**: Override values for incoming MPLLA controls from ASIC, register #0
SUP_DIG_MPLLA_OVRD_IN_1 on page 331	0x5	**Description**: Override values for incoming MPLLA controls from ASIC, register #1
SUP_DIG_MPLLA_OVRD_IN_2 on page 332	0x6	**Description**: Override values for incoming MPLLA controls from ASIC, register #2
SUP_DIG_MPLLA_OVRD_IN_3 on page 333	0x7	**Description**: Override values for incoming MPLLA controls from ASIC, register #3
SUP_DIG_MPLLA_OVRD_IN_4 on page 334	0x8	**Description**: Override values for incoming MPLLA controls from ASIC, register #4
SUP_DIG_MPLLB_OVRD_IN_0 on page 335	0x9	**Description**: Override values for incoming MPLLB controls from ASIC, register #0
SUP_DIG_MPLLB_OVRD_IN_1 on page 337	0xa	**Description**: Override values for incoming MPLLB controls from ASIC, register #1
SUP_DIG_MPLLB_OVRD_IN_2 on page 338	0xb	**Description**: Override values for incoming MPLLB controls from ASIC, register #2
SUP_DIG_MPLLB_OVRD_IN_3 on page 339	0xc	**Description**: Override values for incoming MPLLB controls from ASIC, register #3
SUP_DIG_MPLLB_OVRD_IN_4 on page 340	0xd	**Description**: Override values for incoming MPLLB controls from ASIC, register #4
SUP_DIG_SUP_OVRD_IN_0 on page 341	0xe	**Description**: Override values for support block ASIC inputs, register #0
SUP_DIG_SUP_OVRD_IN_1 on page 342	0xf	**Description**: Override values for support block ASIC inputs, register #1
SUP_DIG_SUP_OVRD_IN_2 on page 343	0x10	**Description**: Override values for support block ASIC inputs, register #2

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
SUP_DIG_SUP_OVRD_OUT on page 344	0x11	**Description**: Override values for support block ASIC outputs
SUP_DIG_LVL_OVRD_IN on page 346	0x12	**Description**: Override values for level settings
SUP_DIG_DEBUG on page 347	0x13	**Description**: Debug controls
SUP_DIG_MPLLA_ASIC_IN_0 on page 348	0x14	**Description**: Current values (pre-override) for incoming MPLLA controls from ASIC, register #0
SUP_DIG_MPLLA_ASIC_IN_1 on page 350	0x15	**Description**: Current values (pre-override) for incoming MPLLA controls from ASIC, register #1
SUP_DIG_MPLLA_ASIC_IN_2 on page 351	0x16	**Description**: Current values (pre-override) for incoming MPLLA controls from ASIC, register #2
SUP_DIG_MPLLB_ASIC_IN_0 on page 352	0x17	**Description**: Current values (pre-override) for incoming MPLLB controls from ASIC, register #0
SUP_DIG_MPLLB_ASIC_IN_1 on page 354	0x18	**Description**: Current values (pre-override) for incoming MPLLB controls from ASIC, register #1
SUP_DIG_MPLLB_ASIC_IN_2 on page 355	0x19	**Description**: Current values (pre-override) for incoming MPLLB controls from ASIC, register #2
SUP_DIG_MPLLA_B_DIV_CLK_ASIC_IN on page 356	0x1a	**Description**: Current values (pre-override) for incoming SUP controls from ASIC
SUP_DIG_ASIC_IN on page 357	0x1b	**Description**: Current values for incoming SUP control signals from/to ASIC
SUP_DIG_LVL_ASIC_IN on page 360	0x1c	**Description**: Current values (pre-override) for incoming SUP controls from ASIC
SUP_DIG_SUP_OVRD_MISC on page 362	0x1d	**Description**: Override values for incoming SUP MISC BUS from ASIC
SUP_DIG_MPLLA_FREQ_CNT_INIT_ASIC_IN on page 363	0x1e	**Description**: Current values (pre-override) for incoming SSC controls from ASIC
SUP_DIG_MPLLB_FREQ_CNT_INIT_ASIC_IN on page 364	0x1f	**Description**: Current values (pre-override) for incoming SSC controls from ASIC
SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL on page 365	0x20	**Description**: MPLL Calibration controls
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_OVRD on page 367	0x21	**Description**: MPLL override controls
SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT on page 369	0x22	**Description**: MPLL status register
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0 on page 372	0x23	**Description**: Thresholds for MPLL CAL Update timer and MPLL VCO Stabilization timer

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1 on page 373	0x24	**Description**: Thresholds for PCLK enable and MPLL VCO Clock Stabilization timer
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2 on page 374	0x25	**Description**: Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3 on page 375	0x26	**Description**: Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4 on page 376	0x27	**Description**: Thresholds for MPLL feedback clock enable and MPLL feedback digital clock disable timer
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL on page 377	0x28	**Description**: MPLL coarse_tune value register
SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE on page 378	0x29	**Description**: Value for MPLL coarse_tune when skipping calibration
SUP_DIG_MPLLA_SSC_SS_PHASE on page 379	0x2a	**Description**: Current MPLL phase selector value
SUP_DIG_MPLLA_SSC_SS_FREQ_0 on page 380	0x2b	**Description**: Frequency Control for Spread Spectrum #0
SUP_DIG_MPLLA_SSC_SS_FREQ_1 on page 381	0x2c	**Description**: Frequency Control for Spread Spectrum #1
SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL on page 382	0x2d	**Description**: MPLL Calibration controls
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_OVRD on page 384	0x2e	**Description**: MPLL override controls
SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT on page 386	0x2f	**Description**: MPLL status register
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0 on page 389	0x30	**Description**: Thresholds for MPLL CAL Update timer and MPLL VCO Stabilization timer
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1 on page 390	0x31	**Description**: Thresholds for PCLK enable and MPLL VCO Clock Stabilization timer
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2 on page 391	0x32	**Description**: Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3 on page 392	0x33	**Description**: Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4 on page 393	0x34	**Description**: Thresholds for MPLL feedback clock enable and MPLL feedback digital clock disable timer

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL on page 394	0x35	**Description**: MPLL coarse_tune value register
SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE on page 395	0x36	**Description**: Value for MPLL coarse_tune when skipping calibration
SUP_DIG_MPLLB_SSC_SS_PHASE on page 396	0x37	**Description**: Current MPLL phase selector value
SUP_DIG_MPLLB_SSC_SS_FREQ_0 on page 397	0x38	**Description**: Frequency Control for Spread Spectrum #0
SUP_DIG_MPLLB_SSC_SS_FREQ_1 on page 398	0x39	**Description**: Frequency Control for Spread Spectrum #1
SUP_DIG_CLK_RST_BG_PWRUP_TIME_0 on page 399	0x3a	**Description**: BG Power UP Time Register #0
SUP_DIG_CLK_RST_BG_PWRUP_TIME_1 on page 400	0x3b	**Description**: BG Power UP Time Register #1
SUP_ANA_MPLLA_LOOP_CTRL on page 401	0x40	**Description**: Configures MPLLA loop bandwidth. This register also enables you to control the current in PLL phase...
SUP_ANA_MPLLA_OVRD on page 403	0x41	**Description**: MPLLA Override Control Registers
SUP_ANA_MPLLA_ATB1 on page 405	0x42	**Description**: MPLLA analog test bus controls (1 of 3)
SUP_ANA_MPLLA_ATB2 on page 407	0x43	**Description**: MPLLA analog test bus controls (2 of 3)
SUP_ANA_MPLLA_ATB3 on page 410	0x44	**Description**: MPLLA analog test bus controls (3 of 3)
SUP_ANA_MPLLB_LOOP_CTRL on page 413	0x45	**Description**: Configure MPLLB loop bandwidth. This register also enables you to control the current in the phase...
SUP_ANA_MPLLB_OVRD on page 415	0x46	**Description**: MPLLB Override Control Registers
SUP_ANA_MPLLB_ATB1 on page 417	0x47	**Description**: MPLLB analog test bus controls (1 of 3)
SUP_ANA_MPLLB_ATB2 on page 419	0x48	**Description**: MPLLB analog test bus controls (2 of 3)
SUP_ANA_MPLLB_ATB3 on page 422	0x49	**Description**: MPLLB analog test bus controls (3 of 3)
SUP_ANA_RTUNE_CTRL on page 425	0x4a	**Description**: Resistor Tuning and Margin DAC control registers
SUP_ANA_SWITCH_PWR_MEAS on page 428	0x4b	**Description**: Local power supply measurement in SUP_ANA
SUP_ANA_SWITCH_MISC_MEAS on page 430	0x4c	**Description**: Miscellaneous analog test bus control and regulator reference voltage control

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
SUP_ANA_BG on page 432	0x4d	**Description**: BG
SUP_ANA_MPLLA_ATB4 on page 434	0x4e	**Description**: MPLLA_ATB4
SUP_ANA_MPLLB_ATB4 on page 436	0x4f	**Description**: MPLLB_ATB4
SUP_ANA_PRESCALER on page 438	0x50	**Description**: PRE
SUP_DIG_RTUNE_DEBUG on page 440	0x60	**Description**: Resistor tuning debug controls
SUP_DIG_RTUNE_CONFIG on page 442	0x61	**Description**: Configure Rtune Operation
SUP_DIG_RTUNE_STAT on page 443	0x62	**Description**: Resistor tuning register status
SUP_DIG_RTUNE_RX_SET_VAL on page 444	0x63	**Description**: Set value of RX resistor
SUP_DIG_RTUNE_TXDN_SET_VAL on page 445	0x64	**Description**: Set value of TX-DN Resistor
SUP_DIG_RTUNE_TXUP_SET_VAL on page 446	0x65	**Description**: Set value of TX-UP resistor
SUP_DIG_RTUNE_RX_STAT on page 447	0x66	**Description**: RX resistor tuning register status
SUP_DIG_RTUNE_TXDN_STAT on page 448	0x67	**Description**: TX-DN Resistor tuning register status
SUP_DIG_RTUNE_TXUP_STAT on page 449	0x68	**Description**: TX-UP resistor tuning register status
SUP_DIG_ANA_MPLLA_OVRD_OUT on page 450	0x80	**Description**: Override value for mplla signals going to ANA
SUP_DIG_ANA_MPLLB_OVRD_OUT on page 453	0x81	**Description**: Override value for mpllb signals going to ANA
SUP_DIG_ANA_RTUNE_OVRD_OUT on page 455	0x82	**Description**: Override value for RTUNE signals going to ANA
SUP_DIG_ANA_RX_TERM_OVRD_OUT on page 456	0x83	**Description**: Override value for RX termination signals going to ANA
SUP_DIG_ANA_STAT on page 457	0x84	**Description**: SUP input status register for SUP ANA outputs
SUP_DIG_ANA_BG_OVRD_OUT on page 458	0x85	**Description**: Override values for bandgap signals going to ANA
SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT on page 459	0x86	**Description**: Override value for mplla pmix signals going to ANA
SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT on page 460	0x87	**Description**: Override value for mpllb pmix signals going to ANA

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
SUP_DIG_FREQ_CNT_PEAK_ASIC_IN on page 461	0xa0	**Description**: Current values (pre-override) for incoming SSC controls from ASIC
SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN on page 462	0xa1	**Description**: Current values (pre-override) for incoming TXUP termination offset from ASIC
SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN on page 463	0xa2	**Description**: Current values (pre-override) for incoming TXDN termination offset from ASIC
SUP_DIG_MISC_ASIC_IN on page 464	0xa3	**Description**: Current values (pre-override) for incoming SUP controls from ASIC
LANEN_DIG_ASIC_LANE_OVRD_IN on page 465	0x1N00	**Description**: Override values for incoming LANE controls from ASIC
LANEN_DIG_ASIC_TX_OVRD_IN_0 on page 466	0x1N01	**Description**: Override values for incoming TX controls from ASIC, register #0
LANEN_DIG_ASIC_TX_OVRD_IN_1 on page 468	0x1N02	**Description**: Override values for incoming TX drive controls from ASIC, register #1
LANEN_DIG_ASIC_TX_OVRD_IN_2 on page 470	0x1N03	**Description**: Override values for incoming TX drive controls from ASIC, register #2
LANEN_DIG_ASIC_TX_OVRD_OUT on page 471	0x1N04	**Description**: Override values for outgoing TX controls to ASIC
LANEN_DIG_ASIC_RX_OVRD_IN_0 on page 472	0x1N05	**Description**: Override values for incoming RX controls from ASIC, register #0
LANEN_DIG_ASIC_RX_OVRD_IN_1 on page 474	0x1N06	**Description**: Override values for incoming RX controls from ASIC, register #1
LANEN_DIG_ASIC_RX_OVRD_IN_2 on page 475	0x1N07	**Description**: Override values for incoming RX controls from ASIC, register #2
LANEN_DIG_ASIC_RX_OVRD_IN_3 on page 476	0x1N08	**Description**: Override values for incoming RX controls from ASIC, register #3
LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0 on page 478	0x1N09	**Description**: Override values for incoming RX EQ controls from ASIC, register #0
LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1 on page 479	0x1N0a	**Description**: Override values for incoming RX EQ controls from ASIC, register #1
LANEN_DIG_ASIC_RX_OVRD_OUT_0 on page 480	0x1N0b	**Description**: Override values for outgoing RX controls to ASIC, register #0
LANEN_DIG_ASIC_LANE_ASIC_IN on page 481	0x1N0c	**Description**: Current values for incoming LANE controls from ASIC
LANEN_DIG_ASIC_TX_ASIC_IN_0 on page 482	0x1N0d	**Description**: Current values for incoming TX controls from ASIC, register #0

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_ASIC_TX_ASIC_IN_1 on page 485	0x1N0e	**Description**: Current values for incoming TX controls from ASIC, register #1
LANEN_DIG_ASIC_TX_ASIC_IN_2 on page 486	0x1N0f	**Description**: Current values for incoming TX controls from ASIC, register #2
LANEN_DIG_ASIC_TX_ASIC_OUT on page 487	0x1N10	**Description**: Current values for outgoing TX status controls from PHY
LANEN_DIG_ASIC_RX_ASIC_IN_0 on page 488	0x1N11	**Description**: Current values for incoming RX controls from ASIC, register #0
LANEN_DIG_ASIC_RX_ASIC_IN_1 on page 491	0x1N12	**Description**: Current values for incoming RX controls from ASIC, register #1
LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0 on page 493	0x1N13	**Description**: Current values for incoming RX EQ controls from ASIC, register #0
LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1 on page 494	0x1N14	**Description**: Current values for incoming RX EQ controls from ASIC, register #1
LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_0 on page 495	0x1N15	**Description**: Current values for incoming RX CDR VCO controls from ASIC, register #0
LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_1 on page 496	0x1N16	**Description**: Current values for incoming RX CDR VCO controls from ASIC, register #1
LANEN_DIG_ASIC_RX_ASIC_OUT_0 on page 497	0x1N17	**Description**: Current values for outgoing RX status controls from PHY, register #0
LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2 on page 498	0x1N18	**Description**: Override values for incoming RX EQ controls from ASIC, register #2
LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3 on page 499	0x1N19	**Description**: Override values for incoming RX EQ controls from ASIC, register #3
LANEN_DIG_ASIC_TX_OVRD_MISC on page 500	0x1N1a	**Description**: Override values for incoming TX MISC BUS from ASIC
LANEN_DIG_ASIC_RX_OVRD_MISC on page 501	0x1N1b	**Description**: Override values for incoming RX MISC BUS from ASIC
LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 on page 502	0x1N20	**Description**: TX Power State Control Register for P0
LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S on page 504	0x1N21	**Description**: TX Power State Control Register for P0S
LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 on page 506	0x1N22	**Description**: TX Power State Control Register for P1
LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 on page 508	0x1N23	**Description**: TX Power State Control Register for P2

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0 on page 510	0x1N24	**Description**: TX Power-up Time Register #0
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1 on page 511	0x1N25	**Description**: TX Power-up Time Register #1
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2 on page 512	0x1N26	**Description**: TX Power-up Time Register #2
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3 on page 513	0x1N27	**Description**: TX Power-up Time Register #3
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4 on page 514	0x1N28	**Description**: TX Power-up Time Register #4
LANEN_DIG_TX_PWRCTL_TX_FIFO_BYPASS on page 515	0x1N29	**Description**: TX fifo bypass Register
LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_5 on page 516	0x1N2a	**Description**: TX Power-up Time Register #4
LANEN_DIG_TX_LBERT_CTL on page 517	0x1N2b	**Description**: Pattern Generator controls
LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0 on page 518	0x1N2c	**Description**: TX Clock Alignment Control Register #0
LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1 on page 519	0x1N2d	**Description**: TX Clock Alignment Control Register #1
LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 on page 520	0x1N40	**Description**: RX Power State Control Register for P0
LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S on page 522	0x1N41	**Description**: RX Power State Control Register for P0S
LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 on page 524	0x1N42	**Description**: RX Power State Control Register for P1
LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 on page 526	0x1N43	**Description**: RX Power State Control Register for P2
LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_0 on page 528	0x1N44	**Description**: RX Power-up Time Register #0
LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1 on page 529	0x1N45	**Description**: RX Power-up Time Register #1
LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2 on page 530	0x1N46	**Description**: RX Power-up Time Register #2
LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_3 on page 531	0x1N47	**Description**: RX Power-up Control Register #0

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0 on page 532	0x1N48	**Description**: RX VCO calibration controls register #0
LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1 on page 533	0x1N49	**Description**: RX VCO calibration controls register #1
LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2 on page 535	0x1N4a	**Description**: RX VCO calibration controls register #2
LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_0 on page 536	0x1N4b	**Description**: RX Power UP Time Register #0
LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1 on page 537	0x1N4c	**Description**: RX Power UP Time Register #1
LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0 on page 538	0x1N4d	**Description**: RX VCO status register #0
LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1 on page 540	0x1N4e	**Description**: RX VCO status register #1
LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2 on page 542	0x1N4f	**Description**: RX VCO status register #2
LANEN_DIG_RX_RX_ALIGN_XAUI_COMM_MASK on page 543	0x1N50	**Description**: XAUI_COMMA Mask
LANEN_DIG_RX_LBERT_CTL on page 544	0x1N51	**Description**: Pattern Matcher controls
LANEN_DIG_RX_LBERT_ERR on page 545	0x1N52	**Description**: Pattern match error counter
LANEN_DIG_RX_CDR_CDR_CTL_0 on page 546	0x1N53	**Description**: Control bits for receiver in recovered domain
LANEN_DIG_RX_CDR_CDR_CTL_1 on page 548	0x1N54	**Description**: CDR Control Register #1
LANEN_DIG_RX_CDR_CDR_CTL_2 on page 549	0x1N55	**Description**: CDR Control Register #2
LANEN_DIG_RX_CDR_CDR_CTL_3 on page 550	0x1N56	**Description**: CDR Control Register #3
LANEN_DIG_RX_CDR_CDR_CTL_4 on page 552	0x1N57	**Description**: CDR Control Register #4
LANEN_DIG_RX_CDR_STAT on page 553	0x1N58	**Description**: Current output values to dpll (phug, frug)
LANEN_DIG_RX_DPLL_FREQ on page 554	0x1N59	**Description**: Current frequency integrator value.

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_RX_DPLL_FREQ_BOUND_0 on page 555	0x1N5a	**Description**: Frequency Bounds for incoming data stream #0
LANEN_DIG_RX_DPLL_FREQ_BOUND_1 on page 556	0x1N5b	**Description**: Frequency Bounds for incoming data stream #1
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0 on page 557	0x1N60	**Description**: Adaptation Configuration Register #0
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1 on page 558	0x1N61	**Description**: Adaptation Configuration Register #1
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2 on page 559	0x1N62	**Description**: Adaptation Configuration Register #2
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3 on page 560	0x1N63	**Description**: Adaptation Configuration Register #3
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4 on page 562	0x1N64	**Description**: Adaptation Configuration Register #4
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5 on page 563	0x1N65	**Description**: Adaptation Configuration Register #5
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6 on page 564	0x1N66	**Description**: Adaptation Configuration Register #6
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7 on page 565	0x1N67	**Description**: Adaptation Configuration Register #7
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8 on page 566	0x1N68	**Description**: Adaptation Configuration Register #8
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9 on page 567	0x1N69	**Description**: Adaptation Configuration Register #9
LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG on page 568	0x1N6a	**Description**: Reset Adaptation Configuration Register
LANEN_DIG_RX_ADPTCTL_ATT_STATUS on page 569	0x1N6b	**Description**: Value of ATT Adaptation code
LANEN_DIG_RX_ADPTCTL_VGA_STATUS on page 570	0x1N6c	**Description**: Value of VGA Adaptation code
LANEN_DIG_RX_ADPTCTL_CTLE_STATUS on page 571	0x1N6d	**Description**: Value of CTLE Adaptation code
LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS on page 572	0x1N6e	**Description**: Value of DFE Tap1 Adaptation code
LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS on page 573	0x1N6f	**Description**: Value of DFE Tap2 Adaptation code

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS on page 574	0x1N70	**Description**: Value of DFE Tap3 Adaptation code
LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS on page 575	0x1N71	**Description**: Value of DFE Tap4 Adaptation code
LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS on page 576	0x1N72	**Description**: Value of DFE Tap5 Adaptation code
LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_HIGH_VDAC_OFST on page 577	0x1N73	**Description**: Offset values for RX DFE Data Even High vDAC
LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_LOW_VDAC_OFST on page 578	0x1N74	**Description**: Offset values for RX DFE Data Even Low vDAC
LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_HIGH_VDAC_OFST on page 579	0x1N75	**Description**: Offset values for RX DFE Data Odd High vDAC
LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_LOW_VDAC_OFST on page 580	0x1N76	**Description**: Offset values for RX DFE Data Odd Low vDAC
LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_EVEN on page 581	0x1N77	**Description**: Sets values for RX SLICER CTRL EVEN signals going to ANA
LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_ODD on page 582	0x1N78	**Description**: Sets values for RX SLICER CTRL ODD signals going to ANA
LANEN_DIG_RX_ADPTCTL_DFE_ERROR_EVEN_VDAC_OFST on page 583	0x1N79	**Description**: Offset values for RX DFE Error Even vDAC
LANEN_DIG_RX_ADPTCTL_DFE_ERROR_ODD_VDAC_OFST on page 584	0x1N7a	**Description**: Offset values for RX DFE Error Odd vDAC
LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_LEVEL on page 585	0x1N7b	**Description**: Value of Error Slicer Level
LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST on page 586	0x1N7c	**Description**: Offset values for RX DFE By-Pass Even vDAC
LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST on page 587	0x1N7d	**Description**: Offset values for RX DFE By-Pass Odd vDAC
LANEN_DIG_RX_ADPTCTL_ADPT_RESET on page 588	0x1N7e	**Description**: Adaptation reset register
LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10 on page 589	0x1N7f	**Description**: Adaptation Configuration Register #10
LANEN_DIG_RX_STAT_LD_VAL_1 on page 590	0x1N80	**Description**: Stat load value for the sample counter #1
LANEN_DIG_RX_STAT_DATA_MSK on page 591	0x1N81	**Description**: Stat data mask bits [15:0]

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_RX_STAT_MATCH_CTL0on page 592	0x1N82	**Description**: Stat match controls register #0
LANEN_DIG_RX_STAT_MATCH_CTL1on page 593	0x1N83	**Description**: Stat match controls register #1
LANEN_DIG_RX_STAT_STAT_CTL0 on page 594	0x1N84	**Description**: Stat controls register #0
LANEN_DIG_RX_STAT_STAT_CTL1 on page 597	0x1N85	**Description**: Stat controls register #1
LANEN_DIG_RX_STAT_SMPL_CNT1 on page 599	0x1N86	**Description**: Sample Counter #1 Status
LANEN_DIG_RX_STAT_STAT_CNT_0on page 600	0x1N87	**Description**: Stat Counter 0 Status
LANEN_DIG_RX_STAT_STAT_CNT_1on page 601	0x1N88	**Description**: Stat Counter 1 Status
LANEN_DIG_RX_STAT_STAT_CNT_2on page 602	0x1N89	**Description**: Stat Counter 2 Status
LANEN_DIG_RX_STAT_STAT_CNT_3on page 603	0x1N8a	**Description**: Stat Counter 3 Status
LANEN_DIG_RX_STAT_STAT_CNT_4on page 604	0x1N8b	**Description**: Stat Counter 4 Status
LANEN_DIG_RX_STAT_STAT_CNT_5on page 605	0x1N8c	**Description**: Stat Counter 5 Status
LANEN_DIG_RX_STAT_STAT_CNT_6on page 606	0x1N8d	**Description**: Stat Counter 6 Status
LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL on page 607	0x1N8e	**Description**: Calibration Comparator Control
LANEN_DIG_RX_STAT_MATCH_CTL2on page 608	0x1N8f	**Description**: Stat match controls register #2
LANEN_DIG_RX_STAT_MATCH_CTL3on page 609	0x1N90	**Description**: Stat match controls register #3
LANEN_DIG_RX_STAT_MATCH_CTL4on page 610	0x1N91	**Description**: Stat match controls register #4
LANEN_DIG_RX_STAT_MATCH_CTL5on page 611	0x1N92	**Description**: Stat match controls register #5
LANEN_DIG_RX_STAT_STAT_CTL2 on page 612	0x1N93	**Description**: Stat controls register #2

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_RX_STAT_STAT_STOP on page 613	0x1N94	**Description**: Stat stop register
LANEN_DIG_ANA_TX_OVRD_OUT on page 614	0x1Na0	**Description**: Override values for TX signals going to ANA
LANEN_DIG_ANA_TX_TERM_UP_CODE_OVRD_OUT on page 617	0x1Na1	**Description**: Override value for TX termination code up going to ANA
LANEN_DIG_ANA_TX_TERM_DN_CODE_OVRD_OUT on page 618	0x1Na2	**Description**: Override value for TX termination down code going to ANA
LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0 on page 619	0x1Na3	**Description**: Override values for TX EQ signals going to ANA register #0
LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1 on page 620	0x1Na4	**Description**: Override values for TX EQ signals going to ANA register #1
LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2 on page 621	0x1Na5	**Description**: Override values for TX EQ signals going to ANA register #2
LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3 on page 622	0x1Na6	**Description**: Override values for TX EQ signals going to ANA register #3
LANEN_DIG_ANA_TX_EQ_OVRD_OUT_4 on page 623	0x1Na7	**Description**: Override values for TX EQ signals going to ANA register #4
LANEN_DIG_ANA_RX_CTL_OVRD_OUT on page 624	0x1Na8	**Description**: Override values for RX control signals going to ANA
LANEN_DIG_ANA_RX_PWR_OVRD_OUT on page 626	0x1Na9	**Description**: Override values for RX PWR UP/DN signals going to ANA
LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0 on page 628	0x1Naa	**Description**: Override values for RX VCO signals going to ANA #0
LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1 on page 630	0x1Nab	**Description**: Override values for RX VCO signals going to ANA #1
LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2 on page 632	0x1Nac	**Description**: Override values for RX VCO signals going to ANA #2
LANEN_DIG_ANA_RX_CAL on page 633	0x1Nad	**Description**: Sets values for RX CAL signals going to ANA register
LANEN_DIG_ANA_RX_VDAC_RANGE_SEL on page 635	0x1Nae	**Description**: Sets values for RX DAC CTRL value going to ANA
LANEN_DIG_ANA_RX_DAC_CTRL on page 636	0x1Naf	**Description**: Sets values for RX DAC CTRL value going to ANA
LANEN_DIG_ANA_RX_ANA_RTRIM on page 637	0x1Nb0	**Description**: Set value for RX RTRIM going to ANA

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_DIG_ANA_RX_DAC_CTRL_OVRD on page 638	0x1Nb1	**Description**: Overrides RX DAC CTRL bus (en/val/sel) going to ANA
LANEN_DIG_ANA_RX_DAC_CTRL_SEL on page 639	0x1Nb2	**Description**: Sets values for RX DAC CTRL Select signal going to ANA
LANEN_DIG_ANA_RX_AFE_ATT_VGA on page 640	0x1Nb3	**Description**: Value for RX AFE ATT & VGA signals going to ANA
LANEN_DIG_ANA_RX_AFE_CTLE on page 641	0x1Nb4	**Description**: Values for RX AFE CTLE signals going to ANA
LANEN_DIG_ANA_RX_SCOPE on page 642	0x1Nb5	**Description**: Values for RX SCOPE signals going to ANA
LANEN_DIG_ANA_RX_SLICER_CTRL on page 643	0x1Nb6	**Description**: Sets values for RX Slicer Ctrl signals going to ANA register
LANEN_DIG_ANA_RX_ANA_IQ_PHASE_ADJUST on page 644	0x1Nb7	**Description**: Sets values for RX ANA IQ PHASE Adjust signal going to ANA register
LANEN_DIG_ANA_RX_ANA_IQ_SENSE_EN on page 645	0x1Nb8	**Description**: Sets values for RX ANA IQ SENSE signal
LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN on page 646	0x1Nb9	**Description**: DAC CTRL enable signal
LANEN_DIG_ANA_RX_ANA_SIGNALS_CHANGES_ENABLE on page 647	0x1Nba	**Description**: Afe update enable signal
LANEN_DIG_ANA_RX_ANA_PHASE_ADJUST_CLK on page 648	0x1Nbb	**Description**: Phase adjust clock signal
LANEN_DIG_ANA_STATUS_0 on page 649	0x1Nbd	**Description**: Lane input status register #0
LANEN_DIG_ANA_STATUS_1 on page 651	0x1Nbe	**Description**: Lane input status register #1
LANEN_ANA_TX_OVRD_MEAS on page 652	0x1Nc0	**Description**: TX override control register
LANEN_ANA_TX_PWR_OVRD on page 654	0x1Nc1	**Description**: TX sub-block power control register
LANEN_ANA_TX_ALT_BUS on page 656	0x1Nc2	**Description**: TX oscillator and alternate data path control register
LANEN_ANA_TX_ATB1 on page 658	0x1Nc3	**Description**: TX analog test bus controls register (1 of 2)
LANEN_ANA_TX_ATB2 on page 661	0x1Nc4	**Description**: TX analog test bus controls register (2 of 2)

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_ANA_TX_VBOOST on page 663	0x1Nc5	**Description**: TX hybrid driver current-mode boosting control register
LANEN_ANA_TX_TERM_CODE_DN on page 666	0x1Nc6	**Description**: TX-DN termination analog code override register
LANEN_ANA_TX_TERM_CODE_UP on page 667	0x1Nc7	**Description**: TX-UP termination analog code override register
LANEN_ANA_TX_IBOOST_CODE on page 668	0x1Nc8	**Description**: TX hybrid driver current-mode boosting level control
LANEN_ANA_TX_OVRD_CLK on page 670	0x1Nc9	**Description**: TX clock control override register
LANEN_ANA_TX_MISC on page 672	0x1Nca	**Description**: Miscellaneous TX control register
LANEN_ANA_RX_ATB_IQSKEW on page 674	0x1Ncb	**Description**: RX analog test bus access for I/Q clock phase adjustment
LANEN_ANA_RX_DCC_OVRD on page 675	0x1Ncc	**Description**: RX Duty Cycle Correction and AFE data rate override register
LANEN_ANA_RX_PWR_CTRL1 on page 677	0x1Ncd	**Description**: RX sub-block power control register (1 of 2)
LANEN_ANA_RX_ATB_REGREF on page 679	0x1Nce	**Description**: RX analog test bus access to regulator reference levels
LANEN_ANA_RX_CDR_AFE on page 681	0x1Ncf	**Description**: RX clock data recovery mode control and AFE analog test bus access
LANEN_ANA_RX_PWR_CTRL2 on page 683	0x1Nd0	**Description**: RX sub-block power control (2 of 2) and loopback controls
LANEN_ANA_RX_MISC_OVRD on page 685	0x1Nd1	**Description**: Miscellaneous RX control register
LANEN_ANA_RX_CAL_MUXA on page 687	0x1Nd2	**Description**: RX calibration path A mux control register
LANEN_ANA_RX_ATB_MEAS1 on page 689	0x1Nd3	**Description**: RX analog test bus controls (1 of 2)
LANEN_ANA_RX_ATB_MEAS2 on page 691	0x1Nd4	**Description**: RX analog test bus controls (2 of 2)
LANEN_ANA_RX_CAL_MUXB on page 693	0x1Nd5	**Description**: RX calibration path B mux control register
LANEN_ANA_RX_TERM on page 695	0x1Nd6	**Description**: RX_TERM

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
LANEN_ANA_RX_SLC_CTRL on page 697	0x1Nd7	**Description**: RX_SLC_CTRL
LANEN_ANA_RX_ATB_VREG on page 698	0x1Nd8	**Description**: RX_ATB_VREG
LANEN_ANA_RX_CDR_VCO_CTRL on page 700	0x1Nd9	**Description**: Setting to control the voltage regulator for the RX_CDR_VCO block
RAWCMN_DIG_CMN_CTL on page 701	0x2000	**Description**: Common control register
RAWCMN_DIG_MPLLA_BW_OVRD_IN on page 702	0x2001	**Description**: Override values for incoming MPLLA bandwidth
RAWCMN_DIG_MPLLA_SSC_CTL_OVRD_IN on page 703	0x2002	**Description**: Override values for incoming MPLLA SSC control settings
RAWCMN_DIG_MPLLA_SSC_EN_OVRD_IN on page 704	0x2003	**Description**: Override values for incoming MPLLA SSC Input Signals
RAWCMN_DIG_MPLLA_MISC_OVRD_IN on page 705	0x2004	**Description**: Override values for incoming MPLLA-related Input Signals
RAWCMN_DIG_MPLLB_BW_OVRD_IN on page 706	0x2005	**Description**: Override values for incoming MPLLB bandwidth
RAWCMN_DIG_MPLLB_SSC_CTL_OVRD_IN on page 707	0x2006	**Description**: Override values for incoming MPLLB SSC control settings
RAWCMN_DIG_MPLLB_SSC_EN_OVRD_IN on page 708	0x2007	**Description**: Override values for incoming MPLLB SSC Input Signals
RAWCMN_DIG_MPLLB_MISC_OVRD_IN on page 709	0x2008	**Description**: Override values for incoming MPLLB-related Input Signals
RAWCMN_DIG_LANE_FSM_OP_XTND on page 710	0x2009	**Description**: Lane FSM OP XTND control register
RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0 on page 711	0x200a	**Description**: Override values for incoming TX term offset
RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1 on page 712	0x200b	**Description**: Override values for incoming TX/RX term offset
RAWCMN_DIG_CMN_CTL_1 on page 713	0x200c	**Description**: Common control register 1
RAWCMN_DIG_MPLL_OFF_TIME on page 714	0x200d	**Description**: Wait Time for turning MPLL off
RAWCMN_DIG_ATE_ALU_CTRL on page 715	0x200e	**Description**: ATE ALU module Control Register

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWCMN_DIG_ATE_ALU_ADDR on page 716	0x200f	**Description**: Address for Read and Write Command for ATE ALU Module
RAWCMN_DIG_ATE_ALU_DATA on page 717	0x2010	**Description**: Data for Write Command for ATE ALU Module
RAWCMN_DIG_ATE_ALU_FLAGS on page 718	0x2011	**Description**: Flags from ATE ALU Module
RAWCMN_DIG_ATE_ALU_ACCUM on page 719	0x2012	**Description**: Stores the ATE ALU Accumulator result
RAWCMN_DIG_MPLL_IN on page 720	0x2013	**Description**: Current values for incoming MPLL controls from PCS (pre-override Input Monitor)
RAWCMN_DIG_FW_PWRUP_DONE on page 721	0x2014	**Description**: Firmware Power-up Done Status
RAWCMN_DIG_MPLL_CLK_ASYNC_EN on page 722	0x2015	**Description**: Controls the MPLL Output Clcoks Asynchronously
RAWCMN_DIG_ATE_ALU_SPARE_0 on page 723	0x2016	**Description**: Spare register #1 for ATE ALU operations
RAWCMN_DIG_ATE_ALU_SPARE_1 on page 724	0x2017	**Description**: Spare register #2 for ATE ALU operations
RAWCMN_DIG_MPLL_STATE_OVRD_IN on page 725	0x2018	**Description**: Override for MPLLA/B State Outputs
RAWCMN_DIG_AON_SRAM_PGATE_BL_EN on page 726	0x2020	**Description**: Enable SRAM bootloader on power-gated exit
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_0 on page 727	0x2021	**Description**: MPLLA Coarse Tune Value for Bank 0
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_1 on page 728	0x2022	**Description**: MPLLA Coarse Tune Value for Bank 1
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_2 on page 729	0x2023	**Description**: MPLLA Coarse Tune Value for Bank 2
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_3 on page 730	0x2024	**Description**: MPLLA Coarse Tune Value for Bank 3
RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL on page 731	0x2025	**Description**: Value for MPLLA Bank Select
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_DONE on page 732	0x2026	**Description**: Indicates if the Selected MPLLA Calibration bank has the valid calibrated value
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_0 on page 733	0x2027	**Description**: MPLLB Coarse Tune Value for Bank 0

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_1 on page 734	0x2028	**Description**: MPLLB Coarse Tune Value for Bank 1
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_2 on page 735	0x2029	**Description**: MPLLB Coarse Tune Value for Bank 2
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_3 on page 736	0x202a	**Description**: MPLLB Coarse Tune Value for Bank 3
RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL on page 737	0x202b	**Description**: Value for MPLLB Bank Select
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_DONE on page 738	0x202c	**Description**: Indicates if the selected MPLLB Calibration bank has the valid calibrated value
RAWCMN_DIG_AON_MPLL_CAL_BANK_SEL_OVRD on page 739	0x202d	**Description**: Override the incoming mplla/b_recal_bank_sel input
RAWCMN_DIG_AON_MPLLA_COARSE_TUNE on page 740	0x202e	**Description**: Selected COARSE TUNE Value for MPLLA
RAWCMN_DIG_AON_MPLLB_COARSE_TUNE on page 741	0x202f	**Description**: Selected COARSE TUNE Value for MPLLB
RAWCMN_DIG_AON_MPLLA_IN_RECAL on page 742	0x2030	**Description**: Indicates if MPLLA is in Re-calibration
RAWCMN_DIG_AON_MPLLB_IN_RECAL on page 743	0x2031	**Description**: Indicates if MPLLB is in Re-calibration
RAWCMN_DIG_AON_PG_OVRD_IN on page 744	0x2032	**Description**: Override values for incoming power-gating signals
RAWCMN_DIG_AON_PG_OVRD_OUT on page 745	0x2033	**Description**: Override values for outgoing power-gating signals
RAWCMN_DIG_AON_SUP_OVRD_IN on page 746	0x2034	**Description**: Override Values for Incoming Support Block signals
RAWCMN_DIG_AON_SUP_OVRD_OUT on page 747	0x2035	**Description**: Override Values for Out-going Support Block signals
RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL on page 748	0x2036	**Description**: Current Bank selected for MPLLA Coarse Tune in PMA
RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL on page 749	0x2037	**Description**: Current Bank selected for MPLLB Coarse Tune in PMA
RAWCMN_DIG_AON_RTUNE_RX_VAL on page 750	0x2038	**Description**: Resistor Tune RX Value
RAWCMN_DIG_AON_RTUNE_TXDN_VAL on page 751	0x2039	**Description**: Resistor Tune TX Down Value

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWCMN_DIG_AON_RTUNE_TXUP_VAL on page 752	0x203a	**Description**: Resistor Tune TX Up Value
RAWCMN_DIG_AON_CMNCAL_STATUS on page 753	0x203b	**Description**: Common Calibration Status
RAWCMN_DIG_AON_SRAM_OVRD_IN on page 754	0x203c	**Description**: Override for incoming SRAM inputs
RAWCMN_DIG_AON_SRAM_IN on page 755	0x203d	**Description**: Monitor for SRAM inputs
RAWCMN_DIG_AON_SRAM_OUT on page 756	0x203e	**Description**: Monitor for SRAM outputs
RAWCMN_DIG_AON_MPLLA_BANK_SEL_DONE on page 757	0x2040	**Description**: Status for MPLLA Re-calibration or Switching
RAWCMN_DIG_AON_MPLLB_BANK_SEL_DONE on page 758	0x2041	**Description**: Status for MPLLB Re-calibration or Switching
RAWLANEN_DIG_PCS_XF_TX_OVRD_IN on page 759	0x3000	**Description**: Override values for incoming TX controls from PCS
RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 on page 761	0x3001	**Description**: Override values for incoming TX controls from PCS, register #1
RAWLANEN_DIG_PCS_XF_TX_PCS_IN on page 763	0x3002	**Description**: Current values for incoming TX controls from PCS (pre-override Input Monitor)
RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT on page 766	0x3003	**Description**: Override values for outgoing TX controls to PCS
RAWLANEN_DIG_PCS_XF_TX_PCS_OUT on page 767	0x3004	**Description**: Current values for outgoing TX status controls from Raw PCS (pre-override Output Monitor)
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN on page 768	0x3005	**Description**: Override values for incoming RX controls from PCS
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1 on page 770	0x3006	**Description**: Override values for incoming RX controls from PCS, register #1
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2 on page 771	0x3007	**Description**: Override values for incoming RX controls from PCS, register #2
RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2 on page 772	0x3008	**Description**: ATE Override input to control top-level inputs, register #3
RAWLANEN_DIG_PCS_XF_RX_PCS_IN on page 774	0x3009	**Description**: Current values for incoming RX controls from PCS (Monitor on post ATE override)
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1 on page 777	0x300a	**Description**: Current values for incoming RX controls from PCS, register #1 (Monitor on pre-override...

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2 on page 778	0x300b	**Description**: Current values for incoming RX controls from PCS, register #2 (Monitor on the pre-override...
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3 on page 779	0x300c	**Description**: Current values for incoming RX controls from PCS, register #3 ((Monitor on the pre-override...
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4 on page 780	0x300d	**Description**: Current values for incoming RX controls from PCS, register #4 ((Monitor on the pre-override...
RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT on page 781	0x300e	**Description**: Override values for outgoing RX controls to PCS
RAWLANEN_DIG_PCS_XF_RX_PCS_OUT on page 782	0x300f	**Description**: Current values for outgoing RX status controls from Raw PCS
RAWLANEN_DIG_PCS_XF_RX_ADAPT_ACK on page 783	0x3010	**Description**: RX Adaptation Acknowledge
RAWLANEN_DIG_PCS_XF_RX_ADAPT_FOM on page 784	0x3011	**Description**: RX Adaptation Figure of Merit
RAWLANEN_DIG_PCS_XF_RX_TXPRE_DIR on page 785	0x3012	**Description**: RX calculated direction for TX-pre
RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DIR on page 786	0x3013	**Description**: RX calculated direction for TX-Main
RAWLANEN_DIG_PCS_XF_RX_TXPOST_DIR on page 787	0x3014	**Description**: RX calculated direction for TX-Post
RAWLANEN_DIG_PCS_XF_LANE_NUMBER on page 788	0x3015	**Description**: Current lane number
RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN on page 789	0x3016	**Description**: ATE Override input to control top-level inputs, register #1
RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1 on page 791	0x3017	**Description**: Overrides the RX Output Clocks during the PHY Initialization, register #1
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_4 on page 792	0x3018	**Description**: Override values for incoming RX controls from PCS, register #4
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_5 on page 794	0x3019	**Description**: rx adaptation mode override register
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5 on page 796	0x301a	**Description**: Current values for incoming RX controls from PCS, register #5
RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6 on page 798	0x301b	**Description**: override register for rx IQ setting
RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1 on page 799	0x301c	**Description**: Override values for incoming RX EQ controls from PCS, register #1

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6 on page 801	0x301d	**Description**: Current values for incoming RX controls from PCS, register #6
RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1 on page 802	0x301e	**Description**: ATE Override input to control top-level inputs, register #2
RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_2 on page 804	0x301f	**Description**: Override values for incoming RX EQ controls from PCS, register #2
RAWLANEN_DIG_FSM_FSM_OVRD_CTL on page 805	0x3020	**Description**: FSM override control register
RAWLANEN_DIG_FSM_FSM_JMP_BANK on page 806	0x3021	**Description**: FSM Jump Bank Register
RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0 on page 807	0x3022	**Description**: FSM Breakpoint 0 on SRAM Address for Debugging
RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1 on page 808	0x3023	**Description**: FSM Breakpoint 1 on SRAM Address for Debugging
RAWLANEN_DIG_FSM_MEM_ADDR_MON on page 809	0x3025	**Description**: Memory Address Monitor
RAWLANEN_DIG_FSM_STATUS_MON on page 810	0x3026	**Description**: FSM Status Monitor
RAWLANEN_DIG_FSM_FAST_RX_STARTUP_CAL on page 812	0x3027	**Description**: Status of Fast RX Start Up Calibration
RAWLANEN_DIG_FSM_FAST_RX_ADAPT on page 813	0x3028	**Description**: Status of Fast RX Adaptation
RAWLANEN_DIG_FSM_FAST_SUP on page 814	0x3029	**Description**: Status of Fast Support block
RAWLANEN_DIG_FSM_FAST_TX_CMN_MODE on page 815	0x302a	**Description**: Status of Fast TX Common-mode Charge-up
RAWLANEN_DIG_FSM_FAST_TX_RXDET on page 816	0x302b	**Description**: Status of Fast TX detect RX
RAWLANEN_DIG_FSM_FAST_RX_PWRUP on page 817	0x302c	**Description**: Status of Fast RX Power-up
RAWLANEN_DIG_FSM_FAST_RX_VCO_WAIT on page 818	0x302d	**Description**: Status of Fast RX VCO Wait Times
RAWLANEN_DIG_FSM_FAST_RX_VCO_CAL on page 819	0x302e	**Description**: Status of Fast RX VCO Calibration
RAWLANEN_DIG_FSM_FAST_RX_CONT_CAL_ADAPT on page 820	0x302f	**Description**: Status of Fast RX Continuous Calibration/Adaptation

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_FSM_FAST_RX_CONT_ADAPT on page 821	0x3030	**Description**: Status of Fast RX Continuous Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_CAL on page 822	0x3031	**Description**: Status of RX AFE Startup Calibration
RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_CAL on page 823	0x3032	**Description**: Status of RX DFE Startup Calibration
RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT_STARTUP_CAL on page 824	0x3033	**Description**: Status of DFE extended start-up Calibration
RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELTA_STARTUP on page 825	0x3034	**Description**: Status of RX IQ Fixed Offset
RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_CAL on page 826	0x3035	**Description**: Status of RX IQ start-up calibration
RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT on page 827	0x3036	**Description**: Status of RX AFE start-up Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT on page 828	0x3037	**Description**: Status of RX DFE start-up Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT on page 829	0x3038	**Description**: Status of RX IO Start-up Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL on page 830	0x3039	**Description**: Status of RX Continuous Phase Calibration
RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL on page 831	0x303a	**Description**: Status of RX AFE Continuous Calibration
RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT on page 832	0x303b	**Description**: Status of RX Reference Level Continuous Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT on page 833	0x303c	**Description**: Status of RX VGA Continuous Adaptation
RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL on page 834	0x303d	**Description**: Status of RX Phase Start-up Calibration
RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL on page 835	0x303e	**Description**: Status of RX Phase Extended Start-up Calibration
RAWLANEN_DIG_FSM_FW_STATES_1 on page 836	0x303f	**Description**: Stores various states for Firmware - State #2
RAWLANEN_DIG_AON_AFE_ATT_IDAC_OFST on page 837	0x3040	**Description**: Offset value for RX AFE ATT iDAC
RAWLANEN_DIG_AON_AFE_CTLE_IDAC_OFST on page 838	0x3041	**Description**: Offset value for RX AFE CTLE iDAC

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_AON_AFE_VGA1_IDAC_OFST on page 839	0x3042	**Description**: Offset values for RX AFE VGA1 iDAC
RAWLANEN_DIG_AON_DFE_PHASE_EVEN_VDAC_OFST on page 840	0x3043	**Description**: Offset values for RX DFE Phase Even vDAC
RAWLANEN_DIG_AON_DFE_PHASE_ODD_VDAC_OFST on page 841	0x3044	**Description**: Offset values for RX DFE Phase Odd vDAC
RAWLANEN_DIG_AON_DFE_PHASE_EVEN_LOW_VDAC_OFST on page 842	0x3045	**Description**: Offset values for RX DFE Phase Even Low vDAC
RAWLANEN_DIG_AON_DFE_PHASE_ODD_LOW_VDAC_OFST on page 843	0x3046	**Description**: Offset values for RX DFE Phase Odd Low vDAC
RAWLANEN_DIG_AON_RX_PHSADJ_LIN on page 844	0x3047	**Description**: RX Phase Adjust Linear Value
RAWLANEN_DIG_AON_DFE_DATA_EVEN_HIGH_VDAC_OFST on page 845	0x3048	**Description**: Offset values for RX DFE Data Even High vDAC
RAWLANEN_DIG_AON_DFE_DATA_EVEN_LOW_VDAC_OFST on page 846	0x3049	**Description**: Offset values for RX DFE Data Even Low vDAC
RAWLANEN_DIG_AON_DFE_DATA_ODD_HIGH_VDAC_OFST on page 847	0x304a	**Description**: Offset values for RX DFE Data Odd High vDAC
RAWLANEN_DIG_AON_DFE_DATA_ODD_LOW_VDAC_OFST on page 848	0x304b	**Description**: Offset values for RX DFE Data Odd Low vDAC
RAWLANEN_DIG_AON_DFE_BYPASS_EVEN_VDAC_OFST on page 849	0x304c	**Description**: Offset values for RX DFE By-Pass Even vDAC
RAWLANEN_DIG_AON_DFE_BYPASS_ODD_VDAC_OFST on page 850	0x304d	**Description**: Offset values for RX DFE By-Pass Odd vDAC
RAWLANEN_DIG_AON_DFE_ERROR_EVEN_VDAC_OFST on page 851	0x304e	**Description**: Offset values for RX DFE Error Even vDAC
RAWLANEN_DIG_AON_DFE_ERROR_ODD_VDAC_OFST on page 852	0x304f	**Description**: Offset values for RX DFE Error Odd vDAC
RAWLANEN_DIG_AON_RX_CAL_IQ on page 853	0x3050	**Description**: Value for RX Calibrated IQ Phase
RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL on page 854	0x3051	**Description**: Set to Skip Firmware startup algorithms
RAWLANEN_DIG_AON_RX_CONT_ALGO_CTL on page 857	0x3052	**Description**: Set to Skip Firmware mission-mode algorithms
RAWLANEN_DIG_AON_INIT_PWRUP_DONE on page 859	0x3053	**Description**: Initial Power-Up Done Status

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_AON_RX_ADPT_ATT on page 860	0x3054	**Description**: RX Adapted value of ATT
RAWLANEN_DIG_AON_RX_ADPT_VGA on page 861	0x3055	**Description**: RX Adapted value of VGA
RAWLANEN_DIG_AON_RX_ADPT_CTLE on page 862	0x3056	**Description**: RX Adapted value of CTLE
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1 on page 863	0x3057	**Description**: RX Adapted value of DFE TAP1
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2 on page 864	0x3058	**Description**: RX Adapted value of DFE TAP2
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3 on page 865	0x3059	**Description**: RX Adapted value of DFE TAP3
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4 on page 866	0x305a	**Description**: RX Adapted value of DFE TAP4
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5 on page 867	0x305b	**Description**: RX Adapted value of DFE TAP5
RAWLANEN_DIG_AON_RX_ADPT_IQ on page 868	0x305c	**Description**: RX Adapted value of IQ for bank 0
RAWLANEN_DIG_AON_RX_ADPT_REF_ERR on page 869	0x305d	**Description**: RX Adapted value of Reference Level for Error Slicer for bank 0
RAWLANEN_DIG_AON_RX_ADAPT_DONE on page 870	0x305e	**Description**: RX Adaptation Done Status
RAWLANEN_DIG_AON_RX_ADPT_ATT_B1 on page 871	0x305f	**Description**: RX Adapted value of ATT for bank 1
RAWLANEN_DIG_AON_RX_ADPT_VGA_B1 on page 872	0x3060	**Description**: RX Adapted value of VGA for bank 1
RAWLANEN_DIG_AON_RX_ADPT_CTLE_B1 on page 873	0x3061	**Description**: RX Adapted value of CTLE for bank 1
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1_B1 on page 874	0x3062	**Description**: RX Adapted value of DFE TAP1 for bank 1
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2_B1 on page 875	0x3063	**Description**: RX Adapted value of DFE TAP2 for bank 1
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3_B1 on page 876	0x3064	**Description**: RX Adapted value of DFE TAP3 for bank 1
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4_B1 on page 877	0x3065	**Description**: RX Adapted value of DFE TAP4 for bank 1

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5_B1 on page 878	0x3066	**Description**: RX Adapted value of DFE TAP5 for bank 1
RAWLANEN_DIG_AON_RX_ADPT_IQ_B1 on page 879	0x3067	**Description**: RX Adapted value of IQ for bank 1
RAWLANEN_DIG_AON_RX_ADPT_REF_ERR_B1 on page 880	0x3068	**Description**: RX Adapted value of Reference Level for Error Slicer for bank 1
RAWLANEN_DIG_AON_RX_ADAPT_DONE_B1 on page 881	0x3069	**Description**: RX Adaptation Done Status for bank 1
RAWLANEN_DIG_AON_FAST_FLAGSon page 882	0x306a	**Description**: Fast flags for simulation only
RAWLANEN_DIG_AON_FW_STATES_0 on page 884	0x306b	**Description**: Stores various states for Firmware - State #1
RAWLANEN_DIG_AON_TXRX_OVRD_IN on page 885	0x306c	**Description**: Override values for incoming AON TX/RX controls from PCS
RAWLANEN_DIG_AON_TX_EQ_DIR_POLARITY_CTL on page 886	0x306d	**Description**: Controls the equalization direction polarity setting in PCS RAW
RAWLANEN_DIG_AON_TX_PRE_DIV on page 887	0x306e	**Description**: TX pre threshold div
RAWLANEN_DIG_AON_TX_MAIN_ATT_THRESHOLD on page 888	0x306f	**Description**: TX main ATT high and low threshold
RAWLANEN_DIG_AON_TX_MAIN_VGA_THRESHOLD on page 889	0x3070	**Description**: TX main vga high and low threshold
RAWLANEN_DIG_AON_TX_POST_BOOST_THRESHOLD on page 890	0x3071	**Description**: TX post boost high and low threshold
RAWLANEN_DIG_AON_TX_POST_TAP1_THRESHOLD on page 891	0x3072	**Description**: TX post tap1 high and low threshold
RAWLANEN_DIG_AON_RX_CAL_REF_EE_VDAC_OFST on page 892	0x3073	**Description**: Value for reference level error even slicer
RAWLANEN_DIG_AON_RX_CAL_REF_EO_VDAC_OFST on page 893	0x3074	**Description**: Value for reference level error odd slicer
RAWLANEN_DIG_AON_DFE_EE_VDAC_OFST on page 894	0x3075	**Description**: Value for DFE Error Even Low and High Sclier in Mission Mode
RAWLANEN_DIG_AON_DFE_EO_VDAC_OFST on page 895	0x3076	**Description**: Value for DFE Eerror Odd Low and High Sclier in Mission Mode
RAWLANEN_DIG_AON_RX_ADPT_ERR_SLC_MODE on page 896	0x3077	**Description**: RX Adaptation Error Slicer Mode for Reference-Level Calibration

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_AON_SETUP_REF_CTLE_IDAC_OFST on page 897	0x3078	**Description**: Offset value for AFE CTLE IDAC During Reference-Level Calibration
RAWLANEN_DIG_AON_SETUP_REF_VGA1_IDAC_OFST on page 898	0x3079	**Description**: Offset value for AFE VGA1 IDAC During Reference-Level Calibration
RAWLANEN_DIG_AON_SETUP_SLC_VGA1_IDAC_OFST on page 899	0x307a	**Description**: Offset Value for AFE VGA1 for Slicer Setup Calibration
RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL on page 900	0x307b	**Description**: RX CDR Detector Control
RAWLANEN_DIG_AON_RX_CDR_RECOVERY_TIME on page 901	0x307c	**Description**: RX CDR Recovery Time in reference clock for the intended PPM
RAWLANEN_DIG_AON_MEM_BREAKPOINT_2 on page 902	0x307d	**Description**: FSM Breakpoint 2 on SRAM Address for Debugging
RAWLANEN_DIG_AON_ADPT_CTL_0 on page 903	0x307e	**Description**: Adaptation Control register #0
RAWLANEN_DIG_AON_ADPT_CTL_1 on page 904	0x307f	**Description**: Adaptation Control register #1
RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ on page 905	0x3080	**Description**: Reset routine request
RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ on page 906	0x3081	**Description**: RX reset interrupt
RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ on page 907	0x3082	**Description**: RX request interrupt
RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ on page 908	0x3083	**Description**: RX rate change interrupt request
RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ on page 909	0x3084	**Description**: RX pstate change interrupt request
RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ on page 910	0x3085	**Description**: RX adaptation request interrupt
RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ on page 911	0x3086	**Description**: RX adaptation disable interrupt
RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR on page 912	0x3087	**Description**: RX reset interrupt clear
RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR on page 913	0x3088	**Description**: RX request interrupt clear
RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR on page 914	0x3089	**Description**: RX rate change interrupt clear

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR on page 915	0x308a	**Description**: RX pstate change interrupt clear
RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR on page 916	0x308b	**Description**: RX adaptation request interrupt clear
RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR on page 917	0x308c	**Description**: RX adaptation disable interrupt clear
RAWLANEN_DIG_IRQ_CTL_IRQ_MASK on page 918	0x308d	**Description**: Interrupt Mask
RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ on page 920	0x308e	**Description**: MPLLA/B Re-calibration interrupt
RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ_CLR on page 921	0x308f	**Description**: MPLL Re-calibration interrupt clear
RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ on page 922	0x3090	**Description**: RX IQ interrupt
RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ_CLR on page 923	0x3091	**Description**: RX IQ interrupt clear
RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN on page 924	0x30a0	**Description**: Override values for incoming LANE controls
RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT on page 925	0x30a1	**Description**: Override values for outgoing LANE controls
RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN on page 926	0x30a2	**Description**: Override values for incoming SUP controls from PMA
RAWLANEN_DIG_PMA_XF_SUP_PMA_IN on page 927	0x30a3	**Description**: Current values for coming SUP status controls from PMA
RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT on page 928	0x30a4	**Description**: Override values for outgoing TX controls to PMA
RAWLANEN_DIG_PMA_XF_TX_PMA_IN on page 929	0x30a5	**Description**: Current values for coming TX status controls from PMA
RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT on page 930	0x30a6	**Description**: Override values for outgoing RX controls to PMA
RAWLANEN_DIG_PMA_XF_RX_PMA_IN on page 931	0x30a7	**Description**: Current values for coming RX status controls from PMA
RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL on page 932	0x30a8	**Description**: Lane Rtune Controls
RAWLANEN_DIG_PMA_XF_RX_OVRD_IN on page 933	0x30a9	**Description**: Override values for incoming RX controls from PMA

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_PMA_XF_SRAM_REC_EN on page 934	0x30aa	**Description**: SRAM Record Enable
RAWLANEN_DIG_TX_CTL_TX_FSM_CTL on page 935	0x30c0	**Description**: TX FSM Control
RAWLANEN_DIG_TX_CTL_TX_CLK_CTL on page 936	0x30c1	**Description**: Select clock to act as TX input clock
RAWLANEN_DIG_RX_CTL_RX_FSM_CTL on page 937	0x30c2	**Description**: RX FSM control register
RAWLANEN_DIG_RX_CTL_RX_LOS_MASK_CTL on page 938	0x30c3	**Description**: RX LOS Mask Control
RAWLANEN_DIG_RX_CTL_RX_DATA_EN_OVRD_CTL on page 939	0x30c4	**Description**: RX Data Enable Override Control
RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS on page 940	0x30c5	**Description**: RX continuous offset cancellation status
RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS on page 941	0x30c6	**Description**: RX continuous adaptation status
RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE on page 942	0x30c7	**Description**: RX adaptation mode
RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL on page 943	0x30c8	**Description**: Select Between Two Banks Adaptation Settings for RX Adaptation
RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT on page 944	0x30c9	**Description**: RX CDR PPM Drift on rx clock
RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS on page 945	0x30ca	**Description**: RX CDR Detector Status Register
RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL on page 946	0x30cb	**Description**: Values for outgoing RX controls to PMA
RAWLANEN_DIG_RX_CTL_RX_PMA_EQ_IQ on page 947	0x30cc	**Description**: RX PMA Equalization IQ Phase value
RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN on page 948	0x30cd	**Description**: Enable RX Adapt Mode Override (using Custom Setting for RX Adaptation)
RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN on page 949	0x30ce	**Description**: Enable Using Four Different RX Adaptation Modes
RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM on page 950	0x30cf	**Description**: RX Adaptation Mission-Mode Figure of Merit
RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM on page 951	0x30d0	**Description**: RX Adaptation Start-up Figure of Merit (Base Mode or Extended Mode)

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN on page 952	0x30d1	**Description**: RX Adapted value of Reference Level for Error Even Slicer
RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD on page 953	0x30d2	**Description**: RX Adapted value of Reference Level for Error Odd Slicer
RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP on page 954	0x30d3	**Description**: RX Phase Adjust Mapped Value
RAWLANEN_DIG_RX_CTL_RX_DAC_IN_USE on page 955	0x30d4	**Description**: Indicates state of the DAC Interface
RAWLANEN_DIG_AON_ADPT_CTL_2 on page 956	0x30e0	**Description**: Adaptation Control register #2
RAWLANEN_DIG_AON_ADPT_CTL_3 on page 957	0x30e1	**Description**: Adaptation Control register #3
RAWLANEN_DIG_AON_ADPT_CTL_4 on page 958	0x30e2	**Description**: Adaptation Control register #4
RAWLANEN_DIG_AON_ADPT_CTL_5 on page 959	0x30e3	**Description**: Adaptation Control register #5
RAWLANEN_DIG_AON_ADPT_CTL_6 on page 960	0x30e4	**Description**: Adaptation Control register #6
RAWLANEN_DIG_AON_ADPT_CTL_7 on page 961	0x30e5	**Description**: Adaptation Control register #7
RAWLANEN_DIG_AON_ADPT_CTL_8 on page 962	0x30e6	**Description**: Adaptation Control register #8
RAWLANEN_DIG_AON_ADPT_CTL_9 on page 963	0x30e7	**Description**: Adaptation Control register #9
RAWLANEN_DIG_AON_ADPT_CTL_10 on page 964	0x30e8	**Description**: Adaptation Control register #10
RAWLANEN_DIG_AON_ADPT_CTL_11 on page 965	0x30e9	**Description**: Adaptation Control register #11
RAWLANEN_DIG_AON_ADPT_CTL_12 on page 966	0x30ea	**Description**: Adaptation Control register #12
RAWLANEN_DIG_AON_ADPT_CTL_13 on page 967	0x30eb	**Description**: Adaptation Control register #13
RAWLANEN_DIG_AON_ADPT_CTL_14 on page 968	0x30ec	**Description**: Adaptation Control register #14
RAWLANEN_DIG_AON_ADPT_CTL_15 on page 969	0x30ed	**Description**: Adaptation Control register #15

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWLANEN_DIG_AON_ADPT_CTL_16 on page 970	0x30ee	**Description**: Adaptation Control register #16
RAWLANEN_DIG_AON_ADPT_CTL_17 on page 971	0x30ef	**Description**: Adaptation Control register #17
RAWLANEN_DIG_AON_ADPT_CTL_18 on page 972	0x30f0	**Description**: Adaptation Control register #18
RAWLANEN_DIG_AON_ADPT_CTL_19 on page 973	0x30f1	**Description**: Adaptation Control register #19
RAWLANEN_DIG_AON_ADPT_CTL_20 on page 974	0x30f2	**Description**: Adaptation Control register #20
RAWLANEN_DIG_AON_ADPT_CTL_21 on page 975	0x30f3	**Description**: Adaptation Control register #21
RAWLANEN_DIG_AON_ADPT_CTL_22 on page 976	0x30f4	**Description**: Adaptation Control register #22
RAWLANEN_DIG_AON_ADPT_CTL_23 on page 977	0x30f5	**Description**: Adaptation Control register #23
RAWLANEN_DIG_AON_ADPT_CTL_24 on page 978	0x30f6	**Description**: Adaptation Control register #24
RAWLANEN_DIG_AON_ADPT_CTL_25 on page 979	0x30f7	**Description**: Adaptation Control register #25
RAWLANEN_DIG_AON_ADPT_CTL_26 on page 980	0x30f8	**Description**: Adaptation Control register #26
RAWLANEN_DIG_AON_SRAM_REC_CTRL on page 981	0x30f9	**Description**: SRAM Record Control
RAWLANEN_DIG_AON_SRAM_REC_ADDR on page 982	0x30fa	**Description**: Current SRAM Recording Address
RAWLANEN_DIG_AON_SRAM_REC_ITER on page 983	0x30fb	**Description**: Current Iteration Count for SRAM Recording
RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL_1 on page 984	0x30fc	**Description**: Set to Skip Firmware startup/rate-change algorithms
RAWLANEN_DIG_AON_RX_IQ_CTL on page 985	0x30fd	**Description**: RX IQ Adapatation Control Register #1
RAWLANEN_DIG_AON_RX_IQ_CTL_1 on page 986	0x30fe	**Description**: RX IQ Adapatation Control Register #2
RAWLANEN_DIG_AON_RX_ADPT_IQ_LIMIT on page 987	0x30ff	**Description**: RX IQ Adaptation Offset Limit

Table 11-4 Registers for Address Block: E16_mem_map (Continued)

Register	Offset	Description
RAWMEM_DIG_ROM_CMN0_B0_R0 on page 988	0x4000	**Description**: This is the first ROM memory address
RAWMEM_DIG_ROM_CMN63_B7_R31 on page 989	0x7fff	**Description**: This is the last ROM memory address
RAWMEM_DIG_RAM_CMN0_B0_R0 on page 990	0xc000	**Description**: This is the first RAM memory address
RAWMEM_DIG_RAM_CMN63_B7_R31 on page 991	0xffff	**Description**: This is the last RAM memory address

11.2.1 SUP_DIG_IDCODE_LO

- **Description:** Low 16 bits of IDCODE
- **Size:** 16 bits
- **Offset:** 0x0
- **Exists:** Always

Table 11-5 Fields for Register: SUP_DIG_IDCODE_LO

Bits	Name	Memory Access	Description
15:0	VAL	R	**Description**: The lower 16 bits of the IDCODE value It consists of the following fields 15 : Test Chip 14-12 : Site ID 11-0 : Synopsys Constant **Value After Reset**: 0x74cd **Exists**: Always **Reset Mask**: 0xffff

11.2.2 SUP_DIG_IDCODE_HI

- **Description:** High 16 bits of IDCODE
- **Size:** 16 bits
- **Offset:** 0x1
- **Exists:** Always

Table 11-6 Fields for Register: SUP_DIG_IDCODE_HI

Bits	Name	Memory Access	Description
15:0	VAL	R	**Description**: The upper 16 bits of the IDCODE value It consists of the following fields 31-28 : RTL Version 27-24 : Product 23-16 : Foundry/Size/Technology **Value After Reset**: 0x2b36 **Exists**: Always **Reset Mask**: 0xffff

11.2.3 SUP_DIG_REFCLK_OVRD_IN

- **Description:** Override values for incoming REFCLK and RESET controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x2
- **Exists:** Always

Table 11-7 Fields for Register: SUP_DIG_REFCLK_OVRD_IN

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	OVRD_EN	R/W	**Description**: Override enable for all inputs controlled by this register. Input signals overridden by this enable are: ref_clkdet_en bg_en nominal_vph_sel[1:0] ref_range[2:0] ref_repeat_clk_en ref_use_pad ref_clk_div2_en ref_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	REF_CLKDET_EN	R/W	**Description**: Override value for ref_clkdet_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	BG_EN	R/W	**Description**: Override value for bg_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	RESERVED	R/W	**Description**: Reserved bit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-7 Fields for Register: SUP_DIG_REFCLK_OVRD_IN (Continued)

Bits	Name	Memory Access	Description
8:7	VPH_NOMINAL	R/W	**Description**: Override value for nominal_vph_sel[1:0]. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
6:4	REF_CLK_RANGE	R/W	**Description**: Override value for ref_range[2:0]. The code mapping of ref_range is as follows: 3'b000 : 20.0 MHz to 26 MHz 3'b001 : 26.1 MHz to 52 MHz 3'b010 : 52.1 MHz to 78 MHz 3'b011 : 78.1 MHz to 104 MHz 3'b100 : 104.1 MHz to 130 MHz 3'b101 : 130.1 MHz to 156 MHz 3'b110 : 156.1 MHz to 182 MHz 3'b111 : 182.1 MHz to 200 MHz **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x7
3	REF_REPEAT_CLK_EN	R/W	**Description**: Override value for ref_repeat_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	REF_USE_PAD	R/W	**Description**: Override value for ref_use_pad. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	REF_CLK_DIV2_EN	R/W	**Description**: Override value for ref_clk_div2_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	REF_CLK_EN	R/W	**Description**: Override value for ref_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.4 SUP_DIG_MPLLA_B_DIV_CLK_OVRD_IN

- **Description:** Override values for incoming MPLLA_B_DIV_CLK controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x3
- **Exists:** Always

Table 11-8 Fields for Register: SUP_DIG_MPLLA_B_DIV_CLK_OVRD_IN

Bits	Name	Memory Access	Description
15:9	MPLLB_DIV_MULTIPLIER	R/W	**Description**: Override value for mpllb_div_multiplier. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
8	MPLLB_DIV_CLK_EN	R/W	**Description**: Override value for mpllb_div_clk_en. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:1	MPLLA_DIV_MULTIPLIER	R/W	**Description**: Override value for mplla_div_multiplier. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
0	MPLLA_DIV_CLK_EN	R/W	**Description**: Override value for mplla_div_clk_en. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.5 SUP_DIG_MPLLA_OVRD_IN_0

- **Description:** Override values for incoming MPLLA controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x4
- **Exists:** Always

Table 11-9 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_0

Bits	Name	Memory Access	Description
15	OVRD_EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: mplla_bandwidth[15:0] mplla_ssc_freq_cnt_peak[7:0] mplla_ssc_freq_cnt_init[11:0] mplla_ssc_clk_sel[2:0] mplla_fracn_ctrl[10:0] mplla_ssc_up_spread mplla_ssc_en mplla_div_multiplier[6:0] mplla_div_clk_en mplla_tx_clk_div[1:0] mplla_multiplier[7:0] mplla_div16p5_clk_en mplla_div10_clk_en mplla_div8_clk_en ref_clk_mplla_div2_en mplla_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:13	MPLLA_TX_CLK_DIV	R/W	**Description**: Override value for mplla_tx_clk_div[1:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
12:5	MPLLA_MULTIPLIER	R/W	**Description**: Override value for mplla_multiplier[7:0]. **Value After Reset**: 0x19 **Exists**: Always **Reset Mask**: 0xff

Table 11-9 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_0 (Continued)

Bits	Name	Memory Access	Description
4	MPLLA_DIV16P5_CLK_EN	R/W	**Description**: Override value for mplla_div16p5_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLA_DIV10_CLK_EN	R/W	**Description**: Override value for mplla_div10_clk_en. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	MPLLA_DIV8_CLK_EN	R/W	**Description**: Override value for mplla_div8_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	REF_CLK_MPLLA_DIV2_EN	R/W	**Description**: Override value for ref_clk_mplla_div2_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_EN	R/W	**Description**: Override value for mplla_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.6 SUP_DIG_MPLLA_OVRD_IN_1

- **Description:** Override values for incoming MPLLA controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x5
- **Exists:** Always

Table 11-10 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	MPLLA_SSC_CLK_SEL	R/W	**Description**: Override value for mplla_ssc_clk_sel[2:0]. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:2	MPLLA_FRACN_CTRL	R/W	**Description**: Override value for mplla_fracn_ctrl. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7ff
1	MPLLA_SSC_UP_SPREAD	R/W	**Description**: Override value for mplla_ssc_up_spread. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_SSC_EN	R/W	**Description**: Override value for mplla_ssc_en. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.7 SUP_DIG_MPLLA_OVRD_IN_2

- **Description:** Override values for incoming MPLLA controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x6
- **Exists:** Always

Table 11-11 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_2

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	MPLLA_SSC_FREQ_CNT_INIT	R/W	**Description**: Override value for mplla_ssc_freq_cnt_init[11:0]. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xfff

11.2.8 SUP_DIG_MPLLA_OVRD_IN_3

- **Description:** Override values for incoming MPLLA controls from ASIC, register #3
- **Size:** 16 bits
- **Offset:** 0x7
- **Exists:** Always

Table 11-12 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLLA_SSC_FREQ_CNT_PEAK	R/W	**Description**: Override value for mplla_ssc_freq_cnt_peak[7:0]. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.9 SUP_DIG_MPLLA_OVRD_IN_4

- **Description:** Override values for incoming MPLLA controls from ASIC, register #4
- **Size:** 16 bits
- **Offset:** 0x8
- **Exists:** Always

Table 11-13 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLA_BANDWIDTH	R/W	**Description**: Override value for mplla_bandwidth[15:0]. Its override enable is MPLLA_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0xffff

11.2.10 SUP_DIG_MPLLB_OVRD_IN_0

- **Description:** Override values for incoming MPLLB controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x9
- **Exists:** Always

Table 11-14 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	OVRD_EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: mpllb_bandwidth[15:0] mpllb_ssc_freq_cnt_peak[7:0] mpllb_ssc_freq_cnt_init[11:0] mpllb_ssc_clk_sel[2:0] mpllb_fracn_ctrl[10:0] mpllb_ssc_up_spread mpllb_ssc_en mpllb_div_multiplier[6:0] mpllb_div_clk_en mpllb_tx_clk_div[1:0] mpllb_multiplier[7:0] mpllb_div10_clk_en mpllb_div8_clk_en ref_clk_mpllb_div2_en mpllb_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:12	MPLLB_TX_CLK_DIV	R/W	**Description**: Override value for mpllb_tx_clk_div. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-14 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_0 (Continued)

Bits	Name	Memory Access	Description
11:4	MPLLB_MULTIPLIER	R/W	**Description**: Override value for mpllb_multiplier. **Value After Reset**: 0x19 **Exists**: Always **Reset Mask**: 0xff
3	MPLLB_DIV10_CLK_EN	R/W	**Description**: Override value for mpllb_div10_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_DIV8_CLK_EN	R/W	**Description**: Override value for mpllb_div8_clk_en. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	REF_CLK_MPLLB_DIV2_EN	R/W	**Description**: Override value for ref_clk_mpllb_div2_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLB_EN	R/W	**Description**: Override value for mpllb_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.11 SUP_DIG_MPLLB_OVRD_IN_1

- **Description:** Override values for incoming MPLLB controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0xa
- **Exists:** Always

Table 11-15 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	MPLLB_SSC_CLK_SEL	R/W	**Description**: Override value for mpllb_ssc_clk_sel[2:0]. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:2	MPLLB_FRACN_CTRL	R/W	**Description**: Override value for mpllb_fracn_ctrl. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7ff
1	MPLLB_SSC_UP_SPREAD	R/W	**Description**: Override value for mpllb_ssc_up_spread. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLB_SSC_EN	R/W	**Description**: Override value for mpllb_ssc_en. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.12 SUP_DIG_MPLLB_OVRD_IN_2

- **Description:** Override values for incoming MPLLB controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0xb
- **Exists:** Always

Table 11-16 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_2

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	MPLLB_SSC_FREQ_CNT_INIT	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_init[11:0]. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xfff

11.2.13 SUP_DIG_MPLLB_OVRD_IN_3

- **Description:** Override values for incoming MPLLB controls from ASIC, register #3
- **Size:** 16 bits
- **Offset:** 0xc
- **Exists:** Always

Table 11-17 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLLB_SSC_FREQ_CNT_PEAK	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_peak[7:0]. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.14 SUP_DIG_MPLLB_OVRD_IN_4

- **Description:** Override values for incoming MPLLB controls from ASIC, register #4
- **Size:** 16 bits
- **Offset:** 0xd
- **Exists:** Always

Table 11-18 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLB_BANDWIDTH	R/W	**Description**: Override value for mpllb_bandwidth[15:0]. Its override enable is MPLLB_OVRD_IN_0.OVRD_EN **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0xffff

11.2.15 SUP_DIG_SUP_OVRD_IN_0

- **Description:** Override values for support block ASIC inputs, register #0
- **Size:** 16 bits
- **Offset:** 0xe
- **Exists:** Always

Table 11-19 Fields for Register: SUP_DIG_SUP_OVRD_IN_0

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4	RES_OVRD_EN	R/W	**Description**: Override enable for res_req_in and res_ack_in Input signals overridden by this enable are: res_ack_in res_req_in **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RES_ACK_IN	R/W	**Description**: Override value for res_ack_in. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	RES_REQ_IN	R/W	**Description**: Override value for res_req_in. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RTUNE_OVRD_EN	R/W	**Description**: Override enable for rtune_req. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RTUNE_REQ	R/W	**Description**: Override value for rtune_req. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.16 SUP_DIG_SUP_OVRD_IN_1

- **Description:** Override values for support block ASIC inputs, register #1
- **Size:** 16 bits
- **Offset:** 0xf
- **Exists:** Always

Table 11-20 Fields for Register: SUP_DIG_SUP_OVRD_IN_1

Bits	Name	Memory Access	Description
15	RX_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override enable for RX termination Offset rx_term_offset[4:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:10	RX_TERM_OFFSET	R/W	**Description**: Override value for RX termination Offset rx_term_offset[4:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
9	TXUP_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override enable for TXUP termination offset txup_term_offset[8:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:0	TXUP_TERM_OFFSET	R/W	**Description**: Override value for TXUP termination offset txup_term_offset[8:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.17 SUP_DIG_SUP_OVRD_IN_2

- **Description:** Override values for support block ASIC inputs, register #2
- **Size:** 16 bits
- **Offset:** 0x10
- **Exists:** Always

Table 11-21 Fields for Register: SUP_DIG_SUP_OVRD_IN_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TXDN_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override enable for TXDN termination Offset txdn_term_offset[8:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:0	TXDN_TERM_OFFSET	R/W	**Description**: Override value for TXDN termination offset txdn_term_offset[8:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.18 SUP_DIG_SUP_OVRD_OUT

- **Description:** Override values for support block ASIC outputs
- **Size:** 16 bits
- **Offset:** 0x11
- **Exists:** Always

Table 11-22 Fields for Register: SUP_DIG_SUP_OVRD_OUT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	BG_SUP_OVRD_EN	R/W	**Description**: Override enable for bg_sup_state signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	BG_SUP_STATE_OVRD	R/W	**Description**: Override value for bg_sup_state signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	BG_LANE_OVRD_EN	R/W	**Description**: Override enable for bg_lane_state signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	BG_LANE_STATE_OVRD	R/W	**Description**: Override value for bg_lane_state signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	OVRD_EN	R/W	**Description**: Override enable for mplla_state/ack and mpllb_state/ack input signals overridden by this enable are: mpllb_state mplla_state res_ack_out res_req_out rtune_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-22 Fields for Register: SUP_DIG_SUP_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
4	MPLLB_STATE	R/W	**Description**: Override value for mpllb_state output. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLA_STATE	R/W	**Description**: Override value for mplla_state output. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RES_ACK_OUT	R/W	**Description**: Override value for res_ack_out output. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RES_REQ_OUT	R/W	**Description**: Override value for res_req_out output. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RTUNE_ACK	R/W	**Description**: Override value for rtune_ack output. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.19 SUP_DIG_LVL_OVRD_IN

- **Description:** Override values for level settings
- **Size:** 16 bits
- **Offset:** 0x12
- **Exists:** Always

Table 11-23 Fields for Register: SUP_DIG_LVL_OVRD_IN

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TX_VBOOST_LVL_EN	R/W	**Description**: Override enable for tx_vboost_lvl[2:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:6	TX_VBOOST_LVL	R/W	**Description**: Override value for tx_vboost_lvl[2:0]. **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x7
5	RX_VREF_CTRL_EN	R/W	**Description**: Override enable for rx_vref_ctrl[4:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4:0	RX_VREF_CTRL	R/W	**Description**: Override value for rx_vref_ctrl[4:0]. **Value After Reset**: 0x10 **Exists**: Always **Reset Mask**: 0x1f

11.2.20 SUP_DIG_DEBUG

- **Description:** Debug controls
- **Size:** 16 bits
- **Offset:** 0x13
- **Exists:** Always

Table 11-24 Fields for Register: SUP_DIG_DEBUG

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2:0	DTB_SEL	R/W	**Description**: The lane DTB's are OR'd together with the support DTB signals selected with the below encodings 0 - none 1 - mplla DTB output 2 - mpllb DTB output 3 - rtune DTB output **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.21 SUP_DIG_MPLLA_ASIC_IN_0

- **Description:** Current values (pre-override) for incoming MPLLA controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x14
- **Exists:** Always

Table 11-25 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:13	MPLLA_TX_CLK_DIV	R	**Description**: Value from ASIC for mplla_tx_clk_div (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:5	MPLLA_MULTIPLIER	R	**Description**: Value from ASIC for mplla_multiplier (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4	MPLLA_DIV16P5_CLK_EN	R	**Description**: Value from ASIC for mplla_div16p5_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3	MPLLA_DIV10_CLK_EN	R	**Description**: Value from ASIC for mplla_div10_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-25 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
2	MPLLA_DIV8_CLK_EN	R	**Description**: Value from ASIC for mplla_div8_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	REF_CLK_MPLLA_DIV2_EN	R	**Description**: Value from ASIC for ref_clk_mplla_div2_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLA_EN	R	**Description**: Value from ASIC for mplla_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.22 SUP_DIG_MPLLA_ASIC_IN_1

- **Description:** Current values (pre-override) for incoming MPLLA controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x15
- **Exists:** Always

Table 11-26 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	MPLLA_SSC_CLK_SEL	R	**Description**: Value from ASIC for mplla_ssc_clk_sel (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:2	MPLLA_FRACN_CTRL	R	**Description**: Value from ASIC for mplla_fracn_ctrl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	MPLLA_SSC_UP_SPREAD	R	**Description**: Value from ASIC for mplla_ssc_up_spread (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLA_SSC_EN	R	**Description**: Value from ASIC for mplla_ssc_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.23 SUP_DIG_MPLLA_ASIC_IN_2

- **Description:** Current values (pre-override) for incoming MPLLA controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x16
- **Exists:** Always

Table 11-27 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_2

Bits	Name	Memory Access	Description
15:0	MPLLA_BANDWIDTH	R	**Description**: Value from ASIC for mplla_bandwidth (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.24 SUP_DIG_MPLLB_ASIC_IN_0

- **Description:** Current values (pre-override) for incoming MPLLB controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x17
- **Exists:** Always

Table 11-28 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:12	MPLLB_TX_CLK_DIV	R	**Description**: Value from ASIC for mpllb_tx_clk_div (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:4	MPLLB_MULTIPLIER	R	**Description**: Value from ASIC for mpllb_multiplier (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3	MPLLB_DIV10_CLK_EN	R	**Description**: Value from ASIC for mpllb_div10_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	MPLLB_DIV8_CLK_EN	R	**Description**: Value from ASIC for mpllb_div8_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-28 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
1	REF_CLK_MPLLB_DIV2_EN	R	**Description**: Value from ASIC for ref_clk_mpllb_div2_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLB_EN	R	**Description**: Value from ASIC for mpllb_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.25 SUP_DIG_MPLLB_ASIC_IN_1

- **Description:** Current values (pre-override) for incoming MPLLB controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x18
- **Exists:** Always

Table 11-29 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_1

Bits	Name	Memory Access	Description
15:14	VPH_NOMINAL	R	**Description**: Value from ASIC for nominal_vph_sel (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MPLLB_SSC_CLK_SEL	R	**Description**: Value from ASIC for mpllb_ssc_clk_sel[2:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:2	MPLLB_FRACN_CTRL	R	**Description**: Value from ASIC for mpllb_fracn_ctrl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	MPLLB_SSC_UP_SPREAD	R	**Description**: Value from ASIC for mpllb_ssc_up_spread (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLB_SSC_EN	R	**Description**: Value from ASIC for mpllb_ssc_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.26 SUP_DIG_MPLLB_ASIC_IN_2

- **Description:** Current values (pre-override) for incoming MPLLB controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x19
- **Exists:** Always

Table 11-30 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_2

Bits	Name	Memory Access	Description
15:0	MPLLB_BANDWIDTH	R	**Description**: Value from ASIC for mpllb_bandwidth (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.27 SUP_DIG_MPLLA_B_DIV_CLK_ASIC_IN

- **Description:** Current values (pre-override) for incoming SUP controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1a
- **Exists:** Always

Table 11-31 Fields for Register: SUP_DIG_MPLLA_B_DIV_CLK_ASIC_IN

Bits	Name	Memory Access	Description
15:9	MPLLB_DIV_MULTIPLIER	R	**Description**: Value from ASIC for mpllb_div_multiplier (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	MPLLB_DIV_CLK_EN	R	**Description**: Value from ASIC for mpllb_div_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:1	MPLLA_DIV_MULTIPLIER	R	**Description**: Value from ASIC for mplla_div_multiplier (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLA_DIV_CLK_EN	R	**Description**: Value from ASIC for mplla_div_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.28 SUP_DIG_ASIC_IN

- **Description:** Current values for incoming SUP control signals from/to ASIC
- **Size:** 16 bits
- **Offset:** 0x1b
- **Exists:** Always

Table 11-32 Fields for Register: SUP_DIG_ASIC_IN

Bits	Name	Memory Access	Description
15	BG_EN	R	**Description**: Value from ASIC for bg_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14	MPLLB_STATE	R	**Description**: Value to ASIC for mpllb_state_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MPLLA_STATE	R	**Description**: Value to ASIC for mplla_state_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	RES_ACK_OUT	R	**Description**: Value to ASIC for res_ack_out_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	RES_ACK_IN	R	**Description**: Value from ASIC for res_ack_in (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-32 Fields for Register: SUP_DIG_ASIC_IN (Continued)

Bits	Name	Memory Access	Description
10	RES_REQ_OUT	R	**Description**: Value to ASIC for res_req_out_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9	RES_REQ_IN	R	**Description**: Value from ASIC for res_req_in (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	RTUNE_ACK	R	**Description**: Value to ASIC for rtune_ack_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	RTUNE_REQ	R	**Description**: Value from ASIC for rtune_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6	TEST_POWERDOWN	R	**Description**: Value from ASIC for test_powerdown (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	TEST_BURNIN	R	**Description**: Value from ASIC for test_burnin (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-32 Fields for Register: SUP_DIG_ASIC_IN (Continued)

Bits	Name	Memory Access	Description
4	REF_USE_PAD	R	**Description**: Value from ASIC for ref_use_pad (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3	REF_REPEAT_CLK_EN	R	**Description**: Value from ASIC for ref_repeat_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	REF_CLK_DIV2_EN	R	**Description**: Value from ASIC for ref_clk_div2_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	REF_CLK_EN	R	**Description**: Value from ASIC for ref_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	PHY_RESET	R	**Description**: Value from ASIC for phy_reset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.29 SUP_DIG_LVL_ASIC_IN

- **Description:** Current values (pre-override) for incoming SUP controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1c
- **Exists:** Always

Table 11-33 Fields for Register: SUP_DIG_LVL_ASIC_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:10	RX_TERM_OFFSET	R	**Description**: Value from ASIC for rx_term_offset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9	REF_CLKDET_EN	R	**Description**: Value from ASIC for ref_clkdet_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	REF_CLKDET_RESULT	R	**Description**: Value from ASIC for ref_clkdet_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:5	TX_VBOOST_LVL	R	**Description**: Value from ASIC for tx_vboost_lvl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-33 Fields for Register: SUP_DIG_LVL_ASIC_IN (Continued)

Bits	Name	Memory Access	Description
4:0	RX_VREF_CTRL	R	**Description**: Value from ASIC for rx_vref_ctrl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.30 SUP_DIG_SUP_OVRD_MISC

- **Description:** Override values for incoming SUP MISC BUS from ASIC
- **Size:** 16 bits
- **Offset:** 0x1d
- **Exists:** Always

Table 11-34 Fields for Register: SUP_DIG_SUP_OVRD_MISC

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	SUP_MISC_OVRD_EN	R/W	**Description**: Override enable for sup_misc[7:0] input signals overridden by this enable are: sup_misc[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	SUP_MISC_OVRD_VAL	R/W	**Description**: Override value for sup_misc[7:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.31 SUP_DIG_MPLLA_FREQ_CNT_INIT_ASIC_IN

- **Description:** Current values (pre-override) for incoming SSC controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1e
- **Exists:** Always

Table 11-35 Fields for Register: SUP_DIG_MPLLA_FREQ_CNT_INIT_ASIC_IN

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	MPLLA_SSC_FREQ_CNT_INIT	R	**Description**: Value from ASIC for mplla_ssc_freq_cnt_init (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.32 SUP_DIG_MPLLB_FREQ_CNT_INIT_ASIC_IN

- **Description:** Current values (pre-override) for incoming SSC controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1f
- **Exists:** Always

Table 11-36 Fields for Register: SUP_DIG_MPLLB_FREQ_CNT_INIT_ASIC_IN

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	MPLLB_SSC_FREQ_CNT_INIT	R	**Description**: Value from ASIC for mpllb_ssc_freq_cnt_init (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.33 SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL

- **Description:** MPLL Calibration controls
- **Size:** 16 bits
- **Offset:** 0x20
- **Exists:** Always

Table 11-37 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL

Bits	Name	Memory Access	Description
15	EXT_CAL_DONE	R/W	**Description**: Set the external calibration status to done **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:7	EXT_COARSE_TUNE	R/W	**Description**: Value of mpll_ana_coarse_tune_i[7:0] in external calibration mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
6	EXT_CHKFRQ_EN	R/W	**Description**: Check the frequency of the MPLL Only valid in external calibration mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	MPLL_EXTCAL	R/W	**Description**: Enable external calibration of MPLL **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	MPLL_SKIPCAL	R/W	**Description**: Skip automatic internal calibration of MPLL (and also skip external calibration if it is enabled) If skipcal is enabled, then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-37 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL (Continued)

Bits	Name	Memory Access	Description
3:0	LOAD_CNT	R/W	**Description**: MSBs for the CHKFRQ FSM ld_val[10:0] load value 0x0 - gives a ld_val of 0, no PPM difference can be detected 0x8 - gives a load value of 1024, 3000PPM resolution possible 0xA - gives a load value of 1280, 2343PPM resolution possible 0xB - gives a load value of 1408, 2130PPM resolution possible 0xC - gives a load value of 1536, 1953PPM resolution possible 0xF - gives a load value of 1920, 1600PPM resolution possible **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf

11.2.34 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_OVRD

- **Description:** MPLL override controls
- **Size:** 16 bits
- **Offset:** 0x21
- **Exists:** Always

Table 11-38 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_OVRD

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	DTB_SEL	R/W	**Description**: DTB select for MPLL dtb signals. Possible values are: 5'h01 => {mpll_fb_clk & ~scan_mode_i,ref_dig_clk & ~scan_mode_i} 5'h02 => {mpll_ana_output_en_i,mpll_pclk_en} 5'h04 => {mpll_cal_rdy,mpll_state} 5'h08 => {mpll_chkfrq_done,mpll_tooslow} 5'h10 => {curr_state[0], mpll_cnt[0]} **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
4	FAST_MPLL_LOCK	R/W	**Description**: Enable fast MPLL locking **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	FAST_MPLL_PWRUP	R/W	**Description**: Enable fast MPLL power-up **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLL_PCLK_EN	R/W	**Description**: Overrides the power FSM mpll_pclk_en signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-38 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_OVRD (Continued)

Bits	Name	Memory Access	Description
1	MPLL_FBDIGCLK_EN	R/W	**Description**: Overrides the power FSM mpll_fb_dig_clk_en signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	OVRD_SEL	R/W	**Description**: Override enable bit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.35 SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT

- **Description:** MPLL status register
- **Size:** 16 bits
- **Offset:** 0x22
- **Exists:** Always

Table 11-39 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	MPLL_ANA_EN	R	**Description**: Current value of mpll_ana_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MPLL_ANA_RST	R	**Description**: Current value of mpll_ana_rst_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	MPLL_ANA_CAL	R	**Description**: Current value of mpll_ana_cal_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	MPLL_ANA_FB_CLK_EN	R	**Description**: Current value of mpll_ana_fb_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-39 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT (Continued)

Bits	Name	Memory Access	Description
10	MPLL_ANA_OUTPUT_EN	R	**Description**: Current value of mpll_ana_output_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9	MPLL_PCLK_EN	R	**Description**: Current value of mpll_pclk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	MPLL_L_LANES	R	**Description**: Current value of lane_mpll_l (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	MPLL_R_LANES	R	**Description**: Current value of lane_mpll_r (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6	MPLL_CAL_RDY	R	**Description**: Current value of mpll_cal_rdy (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	CHKFRQ_DONE	R	**Description**: Current value of mpll_chkfrq_done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-39 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT (Continued)

Bits	Name	Memory Access	Description
4	MPLL_TOOSLOW	R	**Description**: Current value of mpll_tooslow (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3:0	FSM_STATE	R	**Description**: Current value of the power FSM state register (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.36 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

- **Description:** Thresholds for MPLL CAL Update timer and MPLL VCO Stabilization timer
- **Size:** 16 bits
- **Offset:** 0x23
- **Exists:** Always

Table 11-40 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:9	MPLL_CAL_UPDATE_TIME	R/W	**Description**: Threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf
8:0	VCO_STABILIZATION_TIME	R/W	**Description**: Threshold for the VCO stabilization timer in terms of number of reference clock cycles Here, the reference clock means the one that is fed to the phase detector of the corresponding PLL and not the raw reference clock. **Value After Reset**: 0x100 **Exists**: Always **Reset Mask**: 0x1ff

11.2.37 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

- **Description:** Thresholds for PCLK enable and MPLL VCO Clock Stabilization timer
- **Size:** 16 bits
- **Offset:** 0x24
- **Exists:** Always

Table 11-41 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:11	PCLK_EN_TIME	R/W	**Description**: Threshold for the PCLK enable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f
10:0	VCO_CLK_STABILIZATION_TIME	R/W	**Description**: Threshold for the VCO clock stabilization timer in terms of number of reference clock cycles Here, the reference clock means the one that is fed to the phase detector of the corresponding PLL and not the raw reference clock. **Value After Reset**: 0x400 **Exists**: Always **Reset Mask**: 0x7ff

11.2.38 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

- **Description:** Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
- **Size:** 16 bits
- **Offset:** 0x25
- **Exists:** Always

Table 11-42 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	MPLL_VCO_PWRDN_TIME	R/W	**Description**: Threshold for the MPLL VCO power-down timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f
4:0	PCLK_DIS_TIME	R/W	**Description**: Threshold for the PCLK disable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f

11.2.39 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

- **Description:** Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
- **Size:** 16 bits
- **Offset:** 0x26
- **Exists:** Always

Table 11-43 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	MPLL_ANA_PWRUP_TIME	R/W	**Description**: Threshold for the MPLL analog power-up timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0x7f

11.2.40 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

- **Description:** Thresholds for MPLL feedback clock enable and MPLL feedback digital clock disable timer
- **Size:** 16 bits
- **Offset:** 0x27
- **Exists:** Always

Table 11-44 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	MPLL_FBDIGCLK_DIS_TIME	R/W	**Description**: Threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf
3:0	MPLL_FBCLK_EN_TIME	R/W	**Description**: Threshold for the MPLL feedback clock enable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf

11.2.41 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL

- **Description:** MPLL coarse_tune value register
- **Size:** 16 bits
- **Offset:** 0x28
- **Exists:** Always

Table 11-45 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLL_COARSE_TUNE_VAL	R	**Description**: Current value of mpll_ana_coarse_tune_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.42 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE

- **Description:** Value for MPLL coarse_tune when skipping calibration
- **Size:** 16 bits
- **Offset:** 0x29
- **Exists:** Always

Table 11-46 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLL_SKIPCAL_COARSE_TUNE	R/W	**Description**: Value for MPLL coarse_tune when calibration is skipped **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.43 SUP_DIG_MPLLA_SSC_SS_PHASE

- **Description:** Current MPLL phase selector value
- **Size:** 16 bits
- **Offset:** 0x2a
- **Exists:** Always

Table 11-47 Fields for Register: SUP_DIG_MPLLA_SSC_SS_PHASE

Bits	Name	Memory Access	Description
15:12	FRACN_CTRL_DIV	R/W	**Description**: FracN Ctrl Divisor to get better resolution and lower PPM errors, when synthesizing from a non-integer reference clock. NOTE: Default value is 8. One is added in hardware to give a range of 1 to 16. (volatile and 2 reads needed to read value) **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf **Volatile**: true
11	ZERO_FREQ	R/W	**Description**: Zero frequency register. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
10:2	VAL	R/W	**Description**: Phase value from zero reference (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff **Volatile**: true
1:0	DTHR	R/W	**Description**: Bits below the useful resolution (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3 **Volatile**: true

11.2.44 SUP_DIG_MPLLA_SSC_SS_FREQ_0

- **Description:** Frequency Control for Spread Spectrum #0
- **Size:** 16 bits
- **Offset:** 0x2b
- **Exists:** Always

Table 11-48 Fields for Register: SUP_DIG_MPLLA_SSC_SS_FREQ_0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	FREQ_0_OVRD	R/W	**Description**: Frequency register override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
11:0	FREQ_CNT_INIT	R/W	**Description**: Initial Frequency Counter Value (volatile and 2 reads needed to read value) **Value After Reset**: 0x25 **Exists**: Always **Reset Mask**: 0xfff **Volatile**: true

11.2.45 SUP_DIG_MPLLA_SSC_SS_FREQ_1

- **Description:** Frequency Control for Spread Spectrum #1
- **Size:** 16 bits
- **Offset:** 0x2c
- **Exists:** Always

Table 11-49 Fields for Register: SUP_DIG_MPLLA_SSC_SS_FREQ_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	FREQ_1_OVRD	R/W	**Description**: Frequency register override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
7:0	FREQ_PEAK	R/W	**Description**: Peak Frequency Value (for changing direction) (volatile and 2 reads needed to read value) **Value After Reset**: 0x66 **Exists**: Always **Reset Mask**: 0xff **Volatile**: true

11.2.46 SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL

- **Description:** MPLL Calibration controls
- **Size:** 16 bits
- **Offset:** 0x2d
- **Exists:** Always

Table 11-50 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL

Bits	Name	Memory Access	Description
15	EXT_CAL_DONE	R/W	**Description**: Set the external calibration status to done **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:7	EXT_COARSE_TUNE	R/W	**Description**: Value of mpll_ana_coarse_tune_i[7:0] in external calibration mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
6	EXT_CHKFRQ_EN	R/W	**Description**: Check the frequency of the MPLL Only valid in external calibration mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	MPLL_EXTCAL	R/W	**Description**: Enable external calibration of MPLL **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	MPLL_SKIPCAL	R/W	**Description**: Skip automatic internal calibration of MPLL (and also skip external calibration if it is enabled) If skipcal is enabled, then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-50 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL (Continued)

Bits	Name	Memory Access	Description
3:0	LOAD_CNT	R/W	**Description**: MSBs for the CHKFRQ FSM ld_val[10:0] load value 0x0 - gives a ld_val of 0, no PPM difference can be detected 0x8 - gives a load value of 1024, 3000PPM resolution possible 0xA - gives a load value of 1280, 2343PPM resolution possible 0xB - gives a load value of 1408, 2130PPM resolution possible 0xC - gives a load value of 1536, 1953PPM resolution possible 0xF - gives a load value of 1920, 1600PPM resolution possible **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf

11.2.47 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_OVRD

- **Description:** MPLL override controls
- **Size:** 16 bits
- **Offset:** 0x2e
- **Exists:** Always

Table 11-51 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_OVRD

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	DTB_SEL	R/W	**Description**: DTB select for MPLL dtb signals. Possible values are: 5'h01 => {mpll_fb_clk & ~scan_mode_i,ref_dig_clk & ~scan_mode_i} 5'h02 => {mpll_ana_output_en_i,mpll_pclk_en} 5'h04 => {mpll_cal_rdy,mpll_state} 5'h08 => {mpll_chkfrq_done,mpll_tooslow} 5'h10 => {curr_state[0], mpll_cnt[0]} **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
4	FAST_MPLL_LOCK	R/W	**Description**: Enable fast MPLL locking **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	FAST_MPLL_PWRUP	R/W	**Description**: Enable fast MPLL power-up **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLL_PCLK_EN	R/W	**Description**: Overrides the power FSM mpll_pclk_en signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-51 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_OVRD (Continued)

Bits	Name	Memory Access	Description
1	MPLL_FBDIGCLK_EN	R/W	**Description**: Overrides the power FSM mpll_fb_dig_clk_en signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	OVRD_SEL	R/W	**Description**: Override enable bit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.48 SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT

- **Description:** MPLL status register
- **Size:** 16 bits
- **Offset:** 0x2f
- **Exists:** Always

Table 11-52 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	MPLL_ANA_EN	R	**Description**: Current value of mpll_ana_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MPLL_ANA_RST	R	**Description**: Current value of mpll_ana_rst_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	MPLL_ANA_CAL	R	**Description**: Current value of mpll_ana_cal_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	MPLL_ANA_FB_CLK_EN	R	**Description**: Current value of mpll_ana_fb_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-52 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT (Continued)

Bits	Name	Memory Access	Description
10	MPLL_ANA_OUTPUT_EN	R	**Description**: Current value of mpll_ana_output_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9	MPLL_PCLK_EN	R	**Description**: Current value of mpll_pclk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	MPLL_L_LANES	R	**Description**: Current value of lane_mpll_l (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	MPLL_R_LANES	R	**Description**: Current value of lane_mpll_r (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6	MPLL_CAL_RDY	R	**Description**: Current value of mpll_cal_rdy (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	CHKFRQ_DONE	R	**Description**: Current value of mpll_chkfrq_done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-52 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT (Continued)

Bits	Name	Memory Access	Description
4	MPLL_TOOSLOW	R	**Description**: Current value of mpll_tooslow (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3:0	FSM_STATE	R	**Description**: Current value of the power FSM state register (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.49 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

- **Description:** Thresholds for MPLL CAL Update timer and MPLL VCO Stabilization timer
- **Size:** 16 bits
- **Offset:** 0x30
- **Exists:** Always

Table 11-53 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:9	MPLL_CAL_UPDATE_TIME	R/W	**Description**: Threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf
8:0	VCO_STABILIZATION_TIME	R/W	**Description**: Threshold for the VCO stabilization timer in terms of number of reference clock cycles Here, the reference clock means the one that is fed to the phase detector of the corresponding PLL and not the raw reference clock. **Value After Reset**: 0x100 **Exists**: Always **Reset Mask**: 0x1ff

11.2.50 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

- **Description:** Thresholds for PCLK enable and MPLL VCO Clock Stabilization timer
- **Size:** 16 bits
- **Offset:** 0x31
- **Exists:** Always

Table 11-54 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:11	PCLK_EN_TIME	R/W	**Description**: Threshold for the PCLK enable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f
10:0	VCO_CLK_STABILIZATION_TIME	R/W	**Description**: Threshold for the VCO clock stabilization timer in terms of number of reference clock cycles Here, the reference clock means the one that is fed to the phase detector of the corresponding PLL and not the raw reference clock. **Value After Reset**: 0x400 **Exists**: Always **Reset Mask**: 0x7ff

11.2.51 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

- **Description:** Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
- **Size:** 16 bits
- **Offset:** 0x32
- **Exists:** Always

Table 11-55 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	MPLL_VCO_PWRDN_TIME	R/W	**Description**: Threshold for the MPLL VCO power-down timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f
4:0	PCLK_DIS_TIME	R/W	**Description**: Threshold for the PCLK disable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0x1f

11.2.52 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

- **Description:** Thresholds for PCLK disable and MPLL VCO POWER DOWN and MPLL ANA POWER UP timer
- **Size:** 16 bits
- **Offset:** 0x33
- **Exists:** Always

Table 11-56 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	MPLL_ANA_PWRUP_TIME	R/W	**Description**: Threshold for the MPLL analog power-up timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0x7f

11.2.53 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

- **Description:** Thresholds for MPLL feedback clock enable and MPLL feedback digital clock disable timer
- **Size:** 16 bits
- **Offset:** 0x34
- **Exists:** Always

Table 11-57 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	MPLL_FBDIGCLK_DIS_TIME	R/W	**Description**: Threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf
3:0	MPLL_FBCLK_EN_TIME	R/W	**Description**: Threshold for the MPLL feedback clock enable timer in terms of number of ref_range_clk cycles **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf

11.2.54 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL

- **Description:** MPLL coarse_tune value register
- **Size:** 16 bits
- **Offset:** 0x35
- **Exists:** Always

Table 11-58 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_COARSE_TUNE_VAL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLL_COARSE_TUNE_VAL	R	**Description**: Current value of mpll_ana_coarse_tune_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.55 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE

- **Description:** Value for MPLL coarse_tune when skipping calibration
- **Size:** 16 bits
- **Offset:** 0x36
- **Exists:** Always

Table 11-59 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_COARSE_TUNE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLL_SKIPCAL_COARSE_TUNE	R/W	**Description**: Value for MPLL coarse_tune when calibration is skipped **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.56 SUP_DIG_MPLLB_SSC_SS_PHASE

- **Description:** Current MPLL phase selector value
- **Size:** 16 bits
- **Offset:** 0x37
- **Exists:** Always

Table 11-60 Fields for Register: SUP_DIG_MPLLB_SSC_SS_PHASE

Bits	Name	Memory Access	Description
15:12	FRACN_CTRL_DIV	R/W	**Description**: FracN Ctrl Divisor to get better resolution and lower PPM errors, when synthesizing from a non-integer reference clock. NOTE: Default value is 8. One is added in hardware to give a range of 1 to 16. (volatile and 2 reads needed to read value) **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf **Volatile**: true
11	ZERO_FREQ	R/W	**Description**: Zero frequency register. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
10:2	VAL	R/W	**Description**: Phase value from zero reference (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff **Volatile**: true
1:0	DTHR	R/W	**Description**: Bits below the useful resolution (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3 **Volatile**: true

11.2.57 SUP_DIG_MPLLB_SSC_SS_FREQ_0

- **Description:** Frequency Control for Spread Spectrum #0
- **Size:** 16 bits
- **Offset:** 0x38
- **Exists:** Always

Table 11-61 Fields for Register: SUP_DIG_MPLLB_SSC_SS_FREQ_0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	FREQ_0_OVRD	R/W	**Description**: Frequency register override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
11:0	FREQ_CNT_INIT	R/W	**Description**: Initial Frequency Counter Value (volatile and 2 reads needed to read value) **Value After Reset**: 0x25 **Exists**: Always **Reset Mask**: 0xfff **Volatile**: true

11.2.58 SUP_DIG_MPLLB_SSC_SS_FREQ_1

- **Description:** Frequency Control for Spread Spectrum #1
- **Size:** 16 bits
- **Offset:** 0x39
- **Exists:** Always

Table 11-62 Fields for Register: SUP_DIG_MPLLB_SSC_SS_FREQ_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	FREQ_1_OVRD	R/W	**Description**: Frequency register override. NOTE: Must be set for PHASE writes to stick (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
7:0	FREQ_PEAK	R/W	**Description**: Peak Frequency Value (for changing direction) (volatile and 2 reads needed to read value) **Value After Reset**: 0x66 **Exists**: Always **Reset Mask**: 0xff **Volatile**: true

11.2.59 SUP_DIG_CLK_RST_BG_PWRUP_TIME_0

- **Description:** BG Power UP Time Register #0
- **Size:** 16 bits
- **Offset:** 0x3a
- **Exists:** Always

Table 11-63 Fields for Register: SUP_DIG_CLK_RST_BG_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	FAST_BG_WAIT	R/W	**Description**: Enable fast BG times (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:0	BG_SUP_EN_TIME	R/W	**Description**: Power up time (in ref_range cycles) for bandgap in SUP (spec >=25us expandible to 50us) **Value After Reset**: 0x28a **Exists**: Always **Reset Mask**: 0x7ff

11.2.60 SUP_DIG_CLK_RST_BG_PWRUP_TIME_1

- **Description:** BG Power UP Time Register #1
- **Size:** 16 bits
- **Offset:** 0x3b
- **Exists:** Always

Table 11-64 Fields for Register: SUP_DIG_CLK_RST_BG_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10:0	BG_LANE_EN_TIME	R/W	**Description**: Power up time (in ref_range cycles) for bandgap in LANE (spec >=0us) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7ff

11.2.61 SUP_ANA_MPLLA_LOOP_CTRL

- **Description:** Configures MPLLA loop bandwidth. This register also enables you to control the current in PLL phase mixer
- **Size:** 16 bits
- **Offset:** 0x40
- **Exists:** Always

Table 11-65 Fields for Register: SUP_ANA_MPLLA_LOOP_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:6	mpll_reset_width_sel_1_0	R/W	**Description**: Reset pulse width control programming of PFD **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
5	lpn_vreg	R/W	**Description**: Bit is not used/connected **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4	meas_iv_23	R/W	**Description**: VCO control buffer bypass (active high) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	pr_bypass	R/W	**Description**: Set local pr bypass control to ON **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	mode_old_ssc	R/W	**Description**: Bit is not used/connected **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	ovrd_pr_bypass	R/W	**Description**: Enable local control of PR bypass signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-65 Fields for Register: SUP_ANA_MPLLA_LOOP_CTRL (Continued)

Bits	Name	Memory Access	Description
0	NC0	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.62 SUP_ANA_MPLLA_OVRD

- **Description:** MPLLA Override Control Registers.
- **Size:** 16 bits
- **Offset:** 0x41
- **Exists:** Always

Table 11-66 Fields for Register: SUP_ANA_MPLLA_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	reset_reg	R/W	**Description**: Set local reset control to ON **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_reset	R/W	**Description**: Enable local control of reset signal (mpll_rst) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	fb_clk_en_reg	R/W	**Description**: If bit 4 (ovrd_fb_clk_en) = 1, this bit controls the feedback clock control signal 1: Feedback clock is turned on 0: Feedback clock is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	ovrd_fb_clk_en	R/W	**Description**: If this bit = 1, it allows bit 5 (fb_clk_en_reg) to control of feedback clock enabling signal (mpll_fb_clk_en) 1: One can use bit 5 (fb_clk_en_reg) to control feedback clock enabling function 0: Instead of analog register, internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-66 Fields for Register: SUP_ANA_MPLLA_OVRD (Continued)

Bits	Name	Memory Access	Description
3	cal_reg	R/W	**Description**: If bit 2 (ovrd_cal) = 1, this bit sets local calibration control to ON/OFF 1: Local calibration is turned on 0: Local calibration is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_cal	R/W	**Description**: If this bit = 1, it allows analog register bit 3(cal_reg) to control of calibration signal (mpll_cal) 1: One can use bit 3 (cal_reg) to turn ON/OFF the calibration of MPLLA 0: Instead of analog registers, internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	enable_reg	R/W	**Description**: If bit 0(ovrd_enable) = 1, this bit controls the MPLLA enable signal 1: MPLLA is turned on 0: MPLLA is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_enable	R/W	**Description**: If this bit = 1, it allows bit 1 (enable_reg) to control enable signal (mplla_en) of MPLLA 1: Allows bit 1 (enable_reg) to control enable signal (mplla_en) of MPLLA 0: Instead of analog register, control this functionality externally using mplla_en pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.63 SUP_ANA_MPLLA_ATB1

- **Description:** MPLLA analog test bus controls (1 of 3)
- **Size:** 16 bits
- **Offset:** 0x42
- **Exists:** Always

Table 11-67 Fields for Register: SUP_ANA_MPLLA_ATB1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_7	R/W	**Description**: Measures the vreg_left output voltage on the atb_s_p pin 1: Connects the vreg_left regulator output node to the atb_s_p pin for measurement 0: Disconnects the vreg_left regulator output node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_iv_6	R/W	**Description**: Measures the MPLLA charge pump output (vdd_h/2) voltage on the atb_s_p pin 1: Connects the MPLLA charge pump output (vdd_h/2) node to the atb_s_p pin for measurement 0: Disconnects MPLLA charge pump output (vdd_h/2) node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_5	R/W	**Description**: Measures the MPLLA vreg_vco voltage on the atb_s_p pin 1: Connects the vreg_vco internal node to the atb_s_p pin for measurement 0: Disconnects the vreg_vco internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-67 Fields for Register: SUP_ANA_MPLLA_ATB1 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_4	R/W	**Description**: Override vreg_cp to be 13/32X of Vbg voltage on atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_3	R/W	**Description**: Sets the MPLLA vreg_vp voltage to be 3 times the voltage applied on atb_s_m pin 1: vreg_vp voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_vp voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_2	R/W	**Description**: Sets MPLLA vreg_left voltage to be 3 time the voltage applied on atb_s_m pin 1: vreg_left voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_left voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_1	R/W	**Description**: Sets MPLLA vreg_right voltage to be 3 times the voltage applied on atb_s_m pin 1: vreg_right voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_right voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	meas_iv_0	R/W	**Description**: Sets MPLLA vreg_vco voltage to be 3 times the voltage applied on atb_s_m pin 1: vreg_vco voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_vco voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.64 SUP_ANA_MPLLA_ATB2

- **Description:** MPLLA analog test bus controls (2 of 3)
- **Size:** 16 bits
- **Offset:** 0x43
- **Exists:** Always

Table 11-68 Fields for Register: SUP_ANA_MPLLA_ATB2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_15	R/W	**Description**: Measures the reference voltage of VCO magnitude control amplifier on atb_s_p pin 1: Connects the VCO magnitude control amplifier reference voltage node to the atb_s_p pin for measurement 0: Disconnects the VCO magnitude control amplifier reference voltage node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_iv_14	R/W	**Description**: Measures the VCO magnitude control signal vdrive on the atb_s_p pin 1: Connects the VCO magnitude control signal vdrive internal node to the atb_s_p pin for measurement 0: Disconnects the VCO magnitude control signal vdrive internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_13	R/W	**Description**: Measures the buffered version of ctrl_fine_int voltage on the atb_s_p pin 1: Connects the ctrl_fine_int internal node to the atb_s_p pin for measurement 0: Disconnects the ctrl_fine_int internal node to the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-68 Fields for Register: SUP_ANA_MPLLA_ATB2 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_12	R/W	**Description**: Measures the MPLLA internal analog ground gd voltage on the atb_s_p pin 1: Connects the MPLLA internal analog ground gd net to the atb_s_p pin for measurement 0: Disconnects the MPLLA internal analog ground gd net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_11	R/W	**Description**: Measures MPLLA internal analog low supply vp voltage on the atb_s_p pin 1: Connects the MPLLA internal analog low supply vp net to the atb_s_p pin for measurement 0: Disconnects the MPLLA internal analog low supply vp net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_10	R/W	**Description**: Measures the MPLLA charge pump regulator voltage (vreg_cp) on the atb_s_p pin 1: Connects the MPLLA vreg_cp internal node to the atb_s_p pin for measurement 0: Disconnects the vreg_cp internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_9	R/W	**Description**: Measures the MPLLA vp tracking regulator output (vreg_vp) voltage on the atb_s_p pin 1: Connects the vreg_vp internal node to the atb_s_p pin for measurement 0: Disconnects the vreg_vp internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-68 Fields for Register: SUP_ANA_MPLLA_ATB2 (Continued)

Bits	Name	Memory Access	Description
0	meas_iv_8	R/W	**Description**: Measures the MPLLA vreg_right voltage on the atb_s_p pin 1: Connects the vreg_right internal node to the atb_s_p pin for measurement 0: Disconnects the vreg_right internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.65 SUP_ANA_MPLLA_ATB3

- **Description:** MPLLA analog test bus controls (3 of 3)
- **Size:** 16 bits
- **Offset:** 0x44
- **Exists:** Always

Table 11-69 Fields for Register: SUP_ANA_MPLLA_ATB3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_22	R/W	**Description**: Measure CP vsense on atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	atb_select	R/W	**Description**: This bit connects/disconnects the internal ATB_S signals to external ATB_S signals 1: Connects the signals ATB_S signals to external ATB_S signals 0: Disconnects the signals **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_21	R/W	**Description**: Measure the vbias voltage inside phase mixer (pmix) on the atb_s_p pin 1: Connects the vbias internal node in pmix of MPLLA to atb_s_p pin for measurement 0: Disconnects the vbias internal node in pmix from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-69 Fields for Register: SUP_ANA_MPLLA_ATB3 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_20	R/W	**Description**: Override the reference voltage for VCO amplitude in MPLLA to a lower value 1: VCO amplitude reference voltage = 3/5th of the voltage that is on the vreg_vco node 0: VCO amplitude reference voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_19	R/W	**Description**: Override the reference voltage for VCO amplitude in MPLLA to a higher value 1: VCO amplitude reference voltage = 4/5th of the voltage that is on the vreg_vco node 0: VCO amplitude reference voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_18	R/W	**Description**: Set the voltage to be applied on the VCO amplitude reference node in MPLLA 1: VCO amplitude reference voltage in MPLLA = voltage that is forced on the atb_s_m pin 0: VCO amplitude reference voltage in MPLLA is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_17	R/W	**Description**: Set the fine control voltage in MPLLA to a lower value 1: Fine control voltage in MPLLA = 10% of vreg_cp in calibration mode (minimum value) 0: Release fine control voltage in MPLLA; that is, the voltage is controlled by the internal circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-69 Fields for Register: SUP_ANA_MPLLA_ATB3 (Continued)

Bits	Name	Memory Access	Description
0	meas_iv_16	R/W	**Description**: Set the fine control voltage in MPLLA to a higher value 1: Fine control voltage in MPLLA = 60% of vreg_cp in calibration mode (maximum value) 0: Release fine control voltage in MPLLA; that is, the voltage is controlled by the internal circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.66 SUP_ANA_MPLLB_LOOP_CTRL

- **Description:** Configure MPLLB loop bandwidth. This register also enables you to control the current in the phase mixer
- **Size:** 16 bits
- **Offset:** 0x45
- **Exists:** Always

Table 11-70 Fields for Register: SUP_ANA_MPLLB_LOOP_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:6	mpll_reset_width_sel_1_0	R/W	**Description**: Reset pulse width control programming of PFD **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
5	lpn_vreg	R/W	**Description**: Bit is not used/connected **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4	meas_iv_23	R/W	**Description**: VCO control buffer bypass (active high) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	pr_bypass	R/W	**Description**: Set local pr bypass control to ON **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	mode_old_ssc	R/W	**Description**: Bit is not used/connected **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	ovrd_pr_bypass	R/W	**Description**: Enable local control of PR bypass signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-70 Fields for Register: SUP_ANA_MPLLB_LOOP_CTRL (Continued)

Bits	Name	Memory Access	Description
0	NC0	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.67 SUP_ANA_MPLLB_OVRD

- **Description:** MPLLB Override Control Registers.
- **Size:** 16 bits
- **Offset:** 0x46
- **Exists:** Always

Table 11-71 Fields for Register: SUP_ANA_MPLLB_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	reset_reg	R/W	**Description**: Set local reset control to ON **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_reset	R/W	**Description**: Enable local control of reset signal (mpll_rst) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	fb_clk_en_reg	R/W	**Description**: Set local feedback clock control to ON **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	ovrd_fb_clk_en	R/W	**Description**: If this bit = 1, it allows analog register bit 5 (fb_clk_en_reg) to control feedback clock 1: Use analog register bit 5 (fb_clk_en_reg) to control feedback clock 0: Instead of analog registers, internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-71 Fields for Register: SUP_ANA_MPLLB_OVRD (Continued)

Bits	Name	Memory Access	Description
3	cal_reg	R/W	**Description**: If bit 2 (ovrd_cal) = 1, use this analog registers to turn on/off the local calibration in MPLLB 1: Local calibration is turned on 0: Local calibration is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_cal	R/W	**Description**: If this bit = 1, it allows bit 3 (cal_reg) to turn on/off the local calibration in MPLLB 1: Allows bit 3 (cal_reg) to turn on/off the local calibration in MPLLB 0: Instead of analog registers, the internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	enable_reg	R/W	**Description**: If bit 0 (ovrd_enable) = 1, this bit enables/disables the MPLLB 1: MPLLB is turned on 0: MPLLB is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_enable	R/W	**Description**: If this bit = 1, it allows analog register bit 1(enable_reg) to turn on/off MPLLB 1: Allows analog register bit 1(enable_reg) to turn on/off MPLLB 0: Instead of analog registers, control this functionality externally using mpllb_en pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.68 SUP_ANA_MPLLB_ATB1

- **Description:** MPLLB analog test bus controls (1 of 3)
- **Size:** 16 bits
- **Offset:** 0x47
- **Exists:** Always

Table 11-72 Fields for Register: SUP_ANA_MPLLB_ATB1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_7	R/W	**Description**: Measure MPLLB regulator voltage vreg_left on the atb_s_p pin 1: Connects the vreg_left internal node of MPLLB to the atb_s_p pin for measurement 0: Disconnects the vreg_left internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_iv_6	R/W	**Description**: Measures the charge pump output (vdd_h/2) node voltage in MPLLB on atb_s_p pin 1: Connects the charge pump output (vdd_h/2) in MPLLB to the atb_s_p pin for measurement 0: Disconnects the charge pump output (vdd_h/2) in MPLLB from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_5	R/W	**Description**: Measures MPLLB vco regulator output vreg_vco on atb_s_p pin 1: Connects the vreg_vco internal node in MPLLB to the atb_s_p pin for measurement 0: Disconnects the vreg_vco internal node in MPLLB from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-72 Fields for Register: SUP_ANA_MPLLB_ATB1 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_4	R/W	**Description**: Override vreg_cp to be 13/32X of Vbg voltage on atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_3	R/W	**Description**: Overrides vp tracking regulator voltage vreg_vp in MPLLB such that vreg_vp to be 3 times of voltage applied on atb_s_m 1: vreg_vp voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_vp voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_2	R/W	**Description**: Overrides the vreg_left voltage in MPLLB to be 3 times the voltage applied on atb_s_m 1: vreg_left voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_left voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_1	R/W	**Description**: Overrides the vreg_right voltage in MPLLB to be 3 times the voltage applied on atb_s_m 1: vreg_right voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_right voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	meas_iv_0	R/W	**Description**: Override MPLLB vco regulator voltage vreg_vco to be 3 times the voltage applied on atb_s_m 1: vreg_vco voltage = 3 times the voltage that is applied on the atb_s_m pin 0: vreg_vco voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.69 SUP_ANA_MPLLB_ATB2

- **Description:** MPLLB analog test bus controls (2 of 3)
- **Size:** 16 bits
- **Offset:** 0x48
- **Exists:** Always

Table 11-73 Fields for Register: SUP_ANA_MPLLB_ATB2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_15	R/W	**Description**: Measures the reference voltage for MPLLB VCO magnitude control amplifier on the atb_s_p pin 1: Connects the MPLLB VCO magnitude control amplifier internal reference node to the atb_s_p pin for measurement 0: Disconnects the VCO magnitude control amplifier internal reference node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_iv_14	R/W	**Description**: Measures the MPLLB VCO magnitude control signal vdrive on the atb_s_p pin 1: Connects the MPLLB VCO magnitude control signal vdrive internal node to the atb_s_p pin for measurement 0: Disconnects the VCO magnitude control signal vdrive internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_13	R/W	**Description**: Measures the buffered version of ctrl_fine_int in MPLLB on atb_s_p pin 1: Connects the ctrl_fine_int internal node in MPLLB to the atb_s_p pin for measurement 0: Disconnects the ctrl_fine_int internal node in MPLLB from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-73 Fields for Register: SUP_ANA_MPLLB_ATB2 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_12	R/W	**Description**: Measures the MPLLB internal analog ground gd voltage on the atb_s_p pin 1: Connects the MPLLB internal analog ground gd net to the atb_s_p pin for measurement 0: Disconnects the MPLLB internal analog ground gd net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_11	R/W	**Description**: Measures MPLLB internal analog low supply vp on the atb_s_p pin 1: Connects the MPLLB internal analog low supply vp voltage net to the atb_s_p pin for measurement 0: Disconnects the MPLLB internal analog low supply vp voltage net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_10	R/W	**Description**: Measures MPLLB charge pump regulator output vreg_cp on atb_s_p pin 1: Connects the MPLLB vreg_cp internal node to the atb_s_p pin for measurement 0: Disconnects the MPLLB vreg_cp internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_9	R/W	**Description**: Measures MPLLB vp tracking regulator output vreg_vp on atb_s_p pin 1: Connects the MPLLB vreg_vp internal node to the atb_s_p pin for measurement 0: Disconnects the MPLLB vreg_vp internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-73 Fields for Register: SUP_ANA_MPLLB_ATB2 (Continued)

Bits	Name	Memory Access	Description
0	meas_iv_8	R/W	**Description**: Measures the MPLLB vreg_right internal voltage node on atb_s_p pin 1: Connects the MPLLB vreg_right internal node to the atb_s_p pin for measurement 0: Disconnects the MPLLB vreg_right internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.70 SUP_ANA_MPLLB_ATB3

- **Description:** MPLLB analog test bus controls (3 of 3)
- **Size:** 16 bits
- **Offset:** 0x49
- **Exists:** Always

Table 11-74 Fields for Register: SUP_ANA_MPLLB_ATB3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_iv_22	R/W	**Description**: Measure CP vsense on atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	atb_select	R/W	**Description**: Connects MPLLB internal ATB_S signals to external ATB_S signals 1: Connects the MPLLB internal ATB_S signals to external ATB_S signals 0: Disconnects the signals **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_iv_21	R/W	**Description**: Measures the pmix vbias internal node voltage in MPLLB on atb_s_p pin 1: Connects the pmix vbias internal node in MPLLB to the atb_s_p pin for measurement 0: Disconnects the pmix vbias internal node in MPLLB from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-74 Fields for Register: SUP_ANA_MPLLB_ATB3 (Continued)

Bits	Name	Memory Access	Description
4	meas_iv_20	R/W	**Description**: Overrides VCO amp reference to a lower value 1: VCO amplitude reference voltage = 3/5th of the voltage that is on the vreg_vco node in MPLLB 0: VCO amplitude reference voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_19	R/W	**Description**: Overrides VCO amp reference to a higher value 1: VCO amplitude reference voltage = 4/5th of the voltage that is on the vreg_vco node in MPLLB 0: VCO amplitude reference voltage is internally generated in MPLLB **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_iv_18	R/W	**Description**: Override VCO amp reference in MPLLB to be the voltage applied on atb_s_m 1: VCO amplitude reference voltage = voltage that is applied on the atb_s_m pin 0: VCO amplitude reference voltage is internally generated **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_iv_17	R/W	**Description**: Forces the fine control voltage in MPLLB to be 10% of vreg_cp in cal mode 1: Fine control voltage in MPLLB = 10% of vreg_cp in calibration mode (minimum value) 0: Release fine control voltage; that is, the voltage is controlled by the internal circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-74 Fields for Register: SUP_ANA_MPLLB_ATB3 (Continued)

Bits	Name	Memory Access	Description
0	meas_iv_16	R/W	**Description**: Forces fine control voltage in MPLLB to be 60% of vreg_cp in cal mode 1: Fine control voltage in MPLLB = 60% of vreg_cp in calibration mode (maximum value) 0: Release fine control voltage; that is, the voltage is controlled by the internal circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.71 SUP_ANA_RTUNE_CTRL

- **Description:** Resistor Tuning and Margin DAC control registers
- **Size:** 16 bits
- **Offset:** 0x4a
- **Exists:** Always

Table 11-75 Fields for Register: SUP_ANA_RTUNE_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	rt_en_frcon	R/W	**Description**: Internal analog register enable control for resistor tuning block RTUNE block 1: Forces resistor tuning control block to turn ON 0: Enable normal resistor tuning block ON/OFF operation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	x4_frcoff	R/W	**Description**: Increases calibration current in RTUNE block to 4x by default 1: Increase the RTUNE calibration current to 4x times from the default value 0: Turn off this feature **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5:4	rt_dac_mode	R/W	**Description**: Margin DAC mode control bits; for more details, see the description for the rt_dac_chop field **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-75 Fields for Register: SUP_ANA_RTUNE_CTRL (Continued)

Bits	Name	Memory Access	Description
3	rt_dac_chop	R/W	**Description**: Use the rt_dac_mode<5:4> and rt_dac_chop (bit 3) bits to perform the following functions 000: Margin DAC power down 001: Margin DAC single ended output drives atb_s_m 010: Margin Dac differential output (1/3VDAC+2/3VCM) drives atb_s_m, (4/3VCM-1/3VDAC) drives atb_s_p 011: Illegal state 100: Margin DAC power down 101: Margin DAC single ended output drives atb_s_p 110: Margin DAC differential output (1/3VDAC+2/3VCM) drives atb_s_p, (4/3VCM-1/3VDAC) drives atb_s_m 111: Illegal state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	rt_atb	R/W	**Description**: RTUNE ATB mode control. Combine this bit (MSB) with rt_ana_mode<1:0> (LSB) to perform the following functions: For example, 010 means rt_atb = 0 and rt_ana_mode<1:0> = 10. 000: Power-down mode 001: CAL RX replica 010: CAL TX down replica 011: CAL TX up replica 100: ADC function 101: CAL RX or through ATB 110: NOP 111: Rref Detect **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	rt_sel_atbp	R/W	**Description**: atb_s_p/m selection bit for resistor tuning block RTUNE 1: Selects atb_s_p 0: Selects atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-75 Fields for Register: SUP_ANA_RTUNE_CTRL (Continued)

Bits	Name	Memory Access	Description
0	rt_sel_atbf	R/W	**Description**: GD or atb_s_p/m selection bit for the resistor tuning block RTUNE 1: Selects gd as input 0: Selects atb_s_p/m as input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.72 SUP_ANA_SWITCH_PWR_MEAS

- **Description:** Local power supply measurement in SUP_ANA
- **Size:** 16 bits
- **Offset:** 0x4b
- **Exists:** Always

Table 11-76 Fields for Register: SUP_ANA_SWITCH_PWR_MEAS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	temp_meas	R/W	**Description**: Measure bandgap's PNP diode voltage (temperature) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	atb_switch_6	R/W	**Description**: Connect atb_s_p to atb_s_m if asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	atb_switch_5	R/W	**Description**: Connect atb_s_p to vph/2 if asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	atb_switch_4	R/W	**Description**: Connect atb_s_m to gd if asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	atb_switch_3	R/W	**Description**: Connect atb_s_p to vp if asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	atb_switch_2	R/W	**Description**: Connect atb_s_p to fastvreg_vref if asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-76 Fields for Register: SUP_ANA_SWITCH_PWR_MEAS (Continued)

Bits	Name	Memory Access	Description
1	atb_switch_1	R/W	**Description**: If bit 5 (atb_sw_half_vph) = 0, this bit connects/disconnects atb_s_p pin to internal analog high supply vph of SUP_ANA 1: If bit 5 (atb_sw_half_vph) = 0, connects vph to atb_s_p pin. NOTE: If bit 5 (atb_sw_half_vph) = 1, then connects vph/2 to atb_s_p pin (not recommended). 0: Disconnects vph from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	atb_switch_0	R/W	**Description**: This bit connects/disconnects slow regulator reference voltage node slowvreg_vref in SUP_ANA to atb_s_p pin 1: Connects the atb_s_p pin to the slowvreg_vref internal node for measurement 0: Disconnects the atb_s_p pin connection from slowvreg_vref internal node **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.73 SUP_ANA_SWITCH_MISC_MEAS

- **Description:** Miscellaneous analog test bus control and regulator reference voltage control
- **Size:** 16 bits
- **Offset:** 0x4c
- **Exists:** Always

Table 11-77 Fields for Register: SUP_ANA_SWITCH_MISC_MEAS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:5	vref_sel_slowreg_vph1p2	R/W	**Description**: Sets value of slowreg_vref for VPH=1.2V 000: 425mV 001: 450mV 010: 475mV 011: 500mV 100: 538mV (default) 101: 550mV 110: 575mV 111: 600mV **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
4:2	vref_sel_slowreg_vph1p5	R/W	**Description**: Sets value of slowreg_vref for VPH=1.5 or 1.8V 000: 425mV 001: 450mV 010: 475mV 011: 500mV 100: 538mV (default) 101: 550mV 110: 575mV 111: 600mV **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7

Table 11-77 Fields for Register: SUP_ANA_SWITCH_MISC_MEAS (Continued)

Bits	Name	Memory Access	Description
1:0	hyst_ref	R/W	**Description**: Sets prescaler input buffer hysteresis 00: No hysteresis 01: 18mVpp hysteresis 10: 35mVpp hysteresis (default) 11: 50mVpp hysteresis **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x3

11.2.74 SUP_ANA_BG

- **Description:** BG
- **Size:** 16 bits
- **Offset:** 0x4d
- **Exists:** Always

Table 11-78 Fields for Register: SUP_ANA_BG

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	NC7	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_temp_cntrl	R/W	**Description**: Monitor the temp_cntrl signal of temperature sensor comparator through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	chop_en_int	R/W	**Description**: Enable chopper clock for bandgap **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4:3	vref_sel_fastreg_vph1p2	R/W	**Description**: Sets value of fastreg_vref for VPH=1.2V 00: 300mV 01: 325mV (default) 10: 350mV 11: 375mV **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3

Table 11-78 Fields for Register: SUP_ANA_BG (Continued)

Bits	Name	Memory Access	Description
2:1	vref_sel_fastreg_vph1p5	R/W	**Description**: Sets value of fastreg_vref for VPH=1.5 or 1.8V 00: 300mV 01: 325mV (default) 10: 350mV 11: 375mV **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
0	bypass_bg	R/W	**Description**: Bypass bandgap with VP **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.75 SUP_ANA_MPLLA_ATB4

- **Description:** MPLLA_ATB4
- **Size:** 16 bits
- **Offset:** 0x4e
- **Exists:** Always

Table 11-79 Fields for Register: SUP_ANA_MPLLA_ATB4

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:6	vreg_ring_ctr	R/W	**Description**: vreg_ring_ctr<1:0> Function: 00: CP/OSC current 1.8uA 01: CP/OSC current 3.6uA 10: CP/OSC current 5.4uA 11: CP/OSC current 7.2uA **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
5	temp_sense	R/W	**Description**: Temperature sense control to reduce the varactor temp variation. By asserting this bit temperature sense is enabled **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	meas_iv_25	R/W	**Description**: Override VCO amp reference to be the voltage on atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_24	R/W	**Description**: Force fine control voltage to be ~45mV lower than mid value in cal mode (min value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-79 Fields for Register: SUP_ANA_MPLLA_ATB4 (Continued)

Bits	Name	Memory Access	Description
2	trim_reg_2	R/W	**Description**: Control the bias current in Phase Interpolator **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	trim_reg_1	R/W	**Description**: Control the bias current in Phase interpolator **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	trim_reg_0	R/W	**Description**: Control the bias current in Phase interpolator **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.76 SUP_ANA_MPLLB_ATB4

- **Description:** MPLLB_ATB4
- **Size:** 16 bits
- **Offset:** 0x4f
- **Exists:** Always

Table 11-80 Fields for Register: SUP_ANA_MPLLB_ATB4

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:6	vreg_ring_ctr	R/W	**Description**: vreg_ring_ctr<1:0> Function: 00: CP/OSC current 1.8uA 01: CP/OSC current 3.6uA 10: CP/OSC current 5.4uA 11: CP/OSC current 7.2uA **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
5	temp_sense	R/W	**Description**: Temperature sense control to reduce the varactor temp variation. By asserting this bit temperature sense is enabled **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	meas_iv_25	R/W	**Description**: Override VCO amp reference to be the voltage on atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_iv_24	R/W	**Description**: Force fine control voltage to be ~45mV lower than mid value in cal mode (min value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-80 Fields for Register: SUP_ANA_MPLLB_ATB4 (Continued)

Bits	Name	Memory Access	Description
2	trim_reg_2	R/W	**Description**: Control the bias current in Phase Interpolator **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	trim_reg_1	R/W	**Description**: Control the bias current in Phase interpolator **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	trim_reg_0	R/W	**Description**: Control the bias current in Phase interpolator **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.77 SUP_ANA_PRESCALER

- **Description:** PRE
- **Size:** 16 bits
- **Offset:** 0x50
- **Exists:** Always

Table 11-81 Fields for Register: SUP_ANA_PRESCALER

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	NC7	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	vp_enable	R/W	**Description**: If asserted Prescaler regulator is bypassed with Vp supply **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5:4	vreg_ring_ctr	R/W	**Description**: vreg_ring_ctr<1:0> Function: 00: CP/OSC current 1.8uA 01: CP/OSC current 3.6uA 10: CP/OSC current 5.4uA 11: CP/OSC current 7.2uA **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
3	meas_atb_vreg_suppre	R/W	**Description**: Measure Regulator output voltage in SUP PRE regulator on atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-81 Fields for Register: SUP_ANA_PRESCALER (Continued)

Bits	Name	Memory Access	Description
2	override_regref_suppre	R/W	**Description**: Override vp_lcl to be vp/3 of voltage on atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_atb_vreg_suppre_cp	R/W	**Description**: Measure Presclaer CP output voltage on atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	atb_select	R/W	**Description**: Connect internal ATB_S signals to external ATB_S signals **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.78 SUP_DIG_RTUNE_DEBUG

- **Description:** Resistor tuning debug controls
- **Size:** 16 bits
- **Offset:** 0x60
- **Exists:** Always

Table 11-82 Fields for Register: SUP_DIG_RTUNE_DEBUG

Bits	Name	Memory Access	Description
15	TXUP_GO	R/W	**Description**: Enable TxUP tune to continue in manual tune mode when TYPE is TxUP tune. When in non TxUP manual tune mode, this bit must be 0. When in TxUP manual tune mode and after TxUP manual tune is triggered, if the read only register SUP.RTUNE_STAT.DTB_RTUNE equals to 2'b01, it is required to first release the rtune related asynchronous override on analog/digital interface then turn this bit TXUP_GO from low to high to continue TxUP manual tune which also indicates that this bit needs to be low when triggering TxUP manual tune. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:5	VALUE	R/W	**Description**: Value to use when triggering SET_VAL field only the 6 LSB's are used when setting rx or tx cal values **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff
4:3	TYPE	R/W	**Description**: Type of manual tuning or register read/write to execute 0 - ADC, or read/write rt_value 1 - Rx tune, or read/write rx_cal_val (only 6 bits) 2 - TxDN tune, or read/write txdn_cal_val (10 bits) 3 - TxUP tune, or read/write txup_cal_val (10 bits) or Resref detect (no affect when triggering SET_VAL field) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-82 Fields for Register: SUP_DIG_RTUNE_DEBUG (Continued)

Bits	Name	Memory Access	Description
2	SET_VAL	R/W	**Description**: Set value Write to a 1 to manually write the register specified by the TYPE field to the value in the VALUE field **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MAN_TUNE	R/W	**Description**: Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know what the part is doing (w.r.t. resistor tuning) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	FLIP_COMP	R/W	**Description**: Invert analog comparator output **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.79 SUP_DIG_RTUNE_CONFIG

- **Description:** Configure Rtune Operation
- **Size:** 16 bits
- **Offset:** 0x61
- **Exists:** Always

Table 11-83 Fields for Register: SUP_DIG_RTUNE_CONFIG

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_CAL	R/W	**Description**: Skip calibration of RX resistor **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.80 SUP_DIG_RTUNE_STAT

- **Description:** Resistor tuning register status
- **Size:** 16 bits
- **Offset:** 0x62
- **Exists:** Always

Table 11-84 Fields for Register: SUP_DIG_RTUNE_STAT

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:10	DTB_RTUNE	R	**Description**: DTB sampling for rtune where dtb_rtune = {(rtune_state[2]^rtune_state[1]),rtune_state[0]} and rtune_state[2:0] equals to RTUNE_IDLE = 3'b000; RTUNE_REXT = 3'b001; RTUNE_PWRUP = 3'b011; RTUNE_COMP_RST = 3'b010; RTUNE_COMP_EVAL = 3'b110; RTUNE_DONE = 3'b100; RTUNE_TXDN2UP = 3'b101; (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9:0	STAT	R	**Description**: Current value of the register specifed by the DEBUG.TYPE field (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.81 SUP_DIG_RTUNE_RX_SET_VAL

- **Description:** Set value of RX resistor
- **Size:** 16 bits
- **Offset:** 0x63
- **Exists:** Always

Table 11-85 Fields for Register: SUP_DIG_RTUNE_RX_SET_VAL

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_SET_VAL	R/W	**Description**: Set value of RX resistor Writing a value to this register will set the RX resistor value. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

11.2.82 SUP_DIG_RTUNE_TXDN_SET_VAL

- **Description:** Set value of TX-DN Resistor
- **Size:** 16 bits
- **Offset:** 0x64
- **Exists:** Always

Table 11-86 Fields for Register: SUP_DIG_RTUNE_TXDN_SET_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	TXDN_SET_VAL	R/W	**Description**: Set value of TX-DN resistor Writing a value to this register will set the TX-DN resistor value. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.83 SUP_DIG_RTUNE_TXUP_SET_VAL

- **Description:** Set value of TX-UP resistor
- **Size:** 16 bits
- **Offset:** 0x65
- **Exists:** Always

Table 11-87 Fields for Register: SUP_DIG_RTUNE_TXUP_SET_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	TXUP_SET_VAL	R/W	**Description**: Set value of TX-UP resistor Writing a value to this register will set the TX-UP resistor value. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.84 SUP_DIG_RTUNE_RX_STAT

- **Description:** RX resistor tuning register status
- **Size:** 16 bits
- **Offset:** 0x66
- **Exists:** Always

Table 11-88 Fields for Register: SUP_DIG_RTUNE_RX_STAT

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_STAT	R	**Description**: Current value of the RX resistor tuning register (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.85 SUP_DIG_RTUNE_TXDN_STAT

- **Description:** TX-DN Resistor tuning register status
- **Size:** 16 bits
- **Offset:** 0x67
- **Exists:** Always

Table 11-89 Fields for Register: SUP_DIG_RTUNE_TXDN_STAT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	TXDN_STAT	R	**Description**: Current value of the TX-DN resistor tuning register (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.86 SUP_DIG_RTUNE_TXUP_STAT

- **Description:** TX-UP resistor tuning register status
- **Size:** 16 bits
- **Offset:** 0x68
- **Exists:** Always

Table 11-90 Fields for Register: SUP_DIG_RTUNE_TXUP_STAT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	TXUP_STAT	R	**Description**: Current value of the TX-UP resistor tuning register (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.87 SUP_DIG_ANA_MPLLA_OVRD_OUT

- **Description:** Override value for mplla signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x80
- **Exists:** Always

Table 11-91 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	OVRD_SEL	R/W	**Description**: Override enable for mplla_ana outputs Output signals overridden by this enable are: mplla_ana_div_clk_en mplla_ana_div16p5_clk_en mplla_fb_clk_en mplla_ana_div10_clk_en mplla_ana_div8_clk_en mplla_ana_output_r_en mplla_ana_output_l_en mplla_ana_output_en mplla_ana_cal mplla_ana_rst mplla_ana_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	MPLLA_DIV_CLK_EN	R/W	**Description**: Override value for mplla_ana_div_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	MPLLA_DIV16P5_CLK_EN	R/W	**Description**: Override value for mplla_ana_div16p5_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-91 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
8	MPLLA_FBCLK_EN	R/W	**Description**: Override value for mplla_fb_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	MPLLA_DIV10_CLK_EN	R/W	**Description**: Override value for mplla_ana_div10_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	MPLLA_DIV8_CLK_EN	R/W	**Description**: Override value for mplla_ana_div8_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	MPLLA_OUTPUT_R_EN	R/W	**Description**: Override value for mplla_ana_output_r_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	MPLLA_OUTPUT_L_EN	R/W	**Description**: Override value for mplla_ana_output_l_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLA_OUTPUT_EN	R/W	**Description**: Override value for mplla_ana_output_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLA_CAL	R/W	**Description**: Override value for mplla_ana_cal signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLA_RST	R/W	**Description**: Override value for mplla_ana_rst signal. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-91 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
0	MPLLA_ANA_EN	R/W	**Description**: Override value for mplla_ana_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.88 SUP_DIG_ANA_MPLLB_OVRD_OUT

- **Description:** Override value for mpllb signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x81
- **Exists:** Always

Table 11-92 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	OVRD_SEL	R/W	**Description**: Override enable for mpllb_ana outputs Output signals overridden by this enable are: mpllb_ana_div_clk_en mpllb_fb_clk_en mpllb_ana_div10_clk_en mpllb_ana_div8_clk_en mpllb_ana_output_r_en mpllb_ana_output_l_en mpllb_ana_output_en mpllb_ana_cal mpllb_ana_rst mpllb_ana_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	MPLLB_DIV_CLK_EN	R/W	**Description**: Override value for mpllb_ana_div_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	MPLLB_FBCLK_EN	R/W	**Description**: Override value for mpllb_fb_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-92 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
7	MPLLB_DIV10_CLK_EN	R/W	**Description**: Override value for mpllb_ana_div10_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	MPLLB_DIV8_CLK_EN	R/W	**Description**: Override value for mpllb_ana_div8_clk_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	MPLLB_OUTPUT_R_EN	R/W	**Description**: Override value for mpllb_ana_output_r_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	MPLLB_OUTPUT_L_EN	R/W	**Description**: Override value for mpllb_ana_output_l_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLB_OUTPUT_EN	R/W	**Description**: Override value for mpllb_ana_output_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_CAL	R/W	**Description**: Override value for mpllb_ana_cal signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLB_RST	R/W	**Description**: Override value for mpllb_ana_rst signal. **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	MPLLB_ANA_EN	R/W	**Description**: Override value for mpllb_ana_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.89 SUP_DIG_ANA_RTUNE_OVRD_OUT

- **Description:** Override value for RTUNE signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x82
- **Exists:** Always

Table 11-93 Fields for Register: SUP_DIG_ANA_RTUNE_OVRD_OUT

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	RTUNE_OVRD_EN	R/W	**Description**: Override enable for rtune outputs. Output signals overridden by this enable are: rt_ana_value[9:0] rt_ana_en rt_ana_mode[1:0] rt_ana_comp_rst rx_ana_term_val[5:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:4	RTUNE_VALUE	R/W	**Description**: Override value for rt_ana_value[9:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff
3	RTUNE_EN	R/W	**Description**: Override value for rt_ana_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2:1	RTUNE_MODE	R/W	**Description**: Override value for rt_ana_mode[1:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
0	RTUNE_COMP_RST	R/W	**Description**: Override value for rt_ana_comp_rst signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.90 SUP_DIG_ANA_RX_TERM_OVRD_OUT

- **Description:** Override value for RX termination signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x83
- **Exists:** Always

Table 11-94 Fields for Register: SUP_DIG_ANA_RX_TERM_OVRD_OUT

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_TERM_VAL	R/W	**Description**: Override value for rx_ana_term_val[5:0] signal. Its override enable is RTUNE_OVRD_OUT.RTUNE_OVRD_EN. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

11.2.91 SUP_DIG_ANA_STAT

- **Description:** SUP input status register for SUP ANA outputs
- **Size:** 16 bits
- **Offset:** 0x84
- **Exists:** Always

Table 11-95 Fields for Register: SUP_DIG_ANA_STAT

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	REF_ANA_CLKDET_RESULT	R	**Description**: Value from ANA for ref_ana_clkdet_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	RT_ANA_COMP_RESULT	R	**Description**: Value from ANA for rt_ana_comp_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.92 SUP_DIG_ANA_BG_OVRD_OUT

- **Description:** Override values for bandgap signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x85
- **Exists:** Always

Table 11-96 Fields for Register: SUP_DIG_ANA_BG_OVRD_OUT

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	BG_OVRD_EN	R/W	**Description**: Override enable for bandgap outputs Output signals overridden by this enable are: bg_ana_en bg_ana_vref_fast_start **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	BG_EN	R/W	**Description**: Override value for bg_ana_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	BG_FAST_START	R/W	**Description**: Override value for bg_ana_vref_fast_start signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.93 SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT

- **Description:** Override value for mplla pmix signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x86
- **Exists:** Always

Table 11-97 Fields for Register: SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	MPLLA_PMIX_OVRD_EN	R/W	**Description**: Override enable for mplla_ana_pmix_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	MPLLA_PMIX_EN	R/W	**Description**: Override value for mplla_ana_pmix_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	MPLLA_PMIX_SEL_OVRD_EN	R/W	**Description**: Override enable for mplla_ana_pmix_sel[7:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_PMIX_SEL	R/W	**Description**: Override value for mplla_ana_pmix_sel[7:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.94 SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT

- **Description:** Override value for mpllb pmix signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x87
- **Exists:** Always

Table 11-98 Fields for Register: SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	MPLLB_PMIX_OVRD_EN	R/W	**Description**: Override enable for mpllb_ana_pmix_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	MPLLB_PMIX__EN	R/W	**Description**: Override value for mpllb_ana_pmix_en signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	MPLLB_PMIX_SEL_OVRD_EN	R/W	**Description**: Override enable for mpllb_ana_pmix_sel[7:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_PMIX_SEL	R/W	**Description**: Override value for mpllb_ana_pmix_sel[7:0] signal. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.95 SUP_DIG_FREQ_CNT_PEAK_ASIC_IN

- **Description:** Current values (pre-override) for incoming SSC controls from ASIC
- **Size:** 16 bits
- **Offset:** 0xa0
- **Exists:** Always

Table 11-99 Fields for Register: SUP_DIG_FREQ_CNT_PEAK_ASIC_IN

Bits	Name	Memory Access	Description
15:8	MPLLA_SSC_FREQ_CNT_PEAK	R	**Description**: Value from ASIC for mplla_ssc_freq_cnt_peak (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:0	MPLLB_SSC_FREQ_CNT_PEAK	R	**Description**: Value from ASIC for mpllb_ssc_freq_cnt_peak (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.96 SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN

- **Description:** Current values (pre-override) for incoming TXUP termination offset from ASIC
- **Size:** 16 bits
- **Offset:** 0xa1
- **Exists:** Always

Table 11-100 Fields for Register: SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	TXUP_TERM_OFFSET	R	**Description**: Value from ASIC for txup_term_offset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.97 SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN

- **Description:** Current values (pre-override) for incoming TXDN termination offset from ASIC
- **Size:** 16 bits
- **Offset:** 0xa2
- **Exists:** Always

Table 11-101 Fields for Register: SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	TXDN_TERM_OFFSET	R	**Description**: Value from ASIC for txup_term_offset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.98 SUP_DIG_MISC_ASIC_IN

- **Description:** Current values (pre-override) for incoming SUP controls from ASIC
- **Size:** 16 bits
- **Offset:** 0xa3
- **Exists:** Always

Table 11-102 Fields for Register: SUP_DIG_MISC_ASIC_IN

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TEST_FLYOVER_EN	R	**Description**: Value from ASIC for test_flyover_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	TEST_STOP_CLK_EN	R	**Description**: Value from ASIC for test_stop_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:0	SUP_MISC	R	**Description**: Value from ASIC for sup_misc (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.99 LANEN_DIG_ASIC_LANE_OVRD_IN

- **Description:** Override values for incoming LANE controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N00
- **Exists:** Always

Table 11-103 Fields for Register: LANEN_DIG_ASIC_LANE_OVRD_IN

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: lane_rx2tx_par_lb_en lane_tx2rx_ser_lb_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	LANE_RX2TX_PAR_LB	R/W	**Description**: Override value for lane_rx2tx_par_lb_en_r. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	LANE_TX2RX_SER_LB	R/W	**Description**: Override value for lane_tx2rx_ser_lb_en_r. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.100 LANEN_DIG_ASIC_TX_OVRD_IN_0

- **Description:** Override values for incoming TX controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N01
- **Exists:** Always

Table 11-104 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_0

Bits	Name	Memory Access	Description
15	EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: tx_detrx_req tx_mpllb_sel tx_width[1:0] tx_rate[2:0] tx_pstate[1:0] tx_lpd tx_req tx_data_en tx_invert tx_reset tx_clk_rdy **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	DETECT_RX_REQ	R/W	**Description**: Override value for tx_detrx_req. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	MPLLB_SEL	R/W	**Description**: Override value for tx_mpllb_sel. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:11	WIDTH	R/W	**Description**: Override value for tx_width[1:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-104 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_0 (Continued)

Bits	Name	Memory Access	Description
10:8	RATE	R/W	**Description**: Override value for tx_rate[2:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
7:6	PSTATE	R/W	**Description**: Override value for tx_pstate[1:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
5	LPD	R/W	**Description**: Override value for tx_lpd. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	REQ	R/W	**Description**: Override value for tx_req. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	DATA_EN	R/W	**Description**: Override value for tx_data_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	INVERT	R/W	**Description**: Override value for tx_invert. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RESET	R/W	**Description**: Override value for tx_reset. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	CLK_RDY	R/W	**Description**: Override value for tx_clk_rdy. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.101 LANEN_DIG_ASIC_TX_OVRD_IN_1

- **Description:** Override values for incoming TX drive controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N02
- **Exists:** Always

Table 11-105 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_1

Bits	Name	Memory Access	Description
15	MAIN_OVRD_EN	R/W	**Description**: Override enable for tx_eq_main input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:9	TX_MAIN_CURSOR	R/W	**Description**: Override value for tx_eq_main. **Value After Reset**: 0x28 **Exists**: Always **Reset Mask**: 0x3f
8	EN	R/W	**Description**: Override enable for inputs below controlled by this register Input signals overridden by this enable are: tx_vboost_en tx_iboost_lvl tx_beacon_en tx_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	VBOOST_EN	R/W	**Description**: Override value for tx_vboost_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6:3	IBOOST_LVL	R/W	**Description**: Override value for tx_iboost_lvl. **Value After Reset**: 0xf **Exists**: Always **Reset Mask**: 0xf
2	BEACON_EN	R/W	**Description**: Override value for tx_beacon_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-105 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_1 (Continued)

Bits	Name	Memory Access	Description
1	DISABLE	R/W	**Description**: Override value for tx_disable. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	NYQUIST_DATA	R/W	**Description**: Override incoming data to nyquist **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.102 LANEN_DIG_ASIC_TX_OVRD_IN_2

- **Description:** Override values for incoming TX drive controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x1N03
- **Exists:** Always

Table 11-106 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	POST_OVRD_EN	R/W	**Description**: Override enable for tx_eq_post input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:7	TX_POST_CURSOR	R/W	**Description**: Override value for tx_eq_post. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f
6	PRE_OVRD_EN	R/W	**Description**: Override enable for tx_eq_pre input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5:0	TX_PRE_CURSOR	R/W	**Description**: Override value for tx_eq_pre. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

11.2.103 LANEN_DIG_ASIC_TX_OVRD_OUT

- **Description:** Override values for outgoing TX controls to ASIC
- **Size:** 16 bits
- **Offset:** 0x1N04
- **Exists:** Always

Table 11-107 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_OUT

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	EN_DETRX_RESULT	R/W	**Description**: Override enable for tx_detrx_result **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	DETRX_RESULT	R/W	**Description**: Override value for tx_detrx_result. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	EN_TX_ACK	R/W	**Description**: Override enable for tx_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	TX_ACK	R/W	**Description**: Override value for tx_ack. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.104 LANEN_DIG_ASIC_RX_OVRD_IN_0

- **Description:** Override values for incoming RX controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N05
- **Exists:** Always

Table 11-108 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: rx_dfe_bypass rx_div16p5_clk_e rx_width rx_rate rx_pstate rx_lpd rx_req rx_data_en rx_invert rx_reset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	RX_DFE_BYPASS	R/W	**Description**: Override value for rx_dfe_bypass. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	DIV16P5_CLK_EN	R/W	**Description**: Override value for rx_div16p5_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-108 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_0 (Continued)

Bits	Name	Memory Access	Description
10:9	WIDTH	R/W	**Description**: Override value for rx_width. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
8:7	RATE	R/W	**Description**: Override value for rx_rate. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
6:5	PSTATE	R/W	**Description**: Override value for rx_pstate. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
4	LPD	R/W	**Description**: Override value for rx_lpd. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	REQ	R/W	**Description**: Override value for rx_req. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	DATA_EN	R/W	**Description**: Override value for rx_data_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	INVERT	R/W	**Description**: Override value for rx_invert. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RESET	R/W	**Description**: Override value for rx_reset. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.105 LANEN_DIG_ASIC_RX_OVRD_IN_1

- **Description:** Override values for incoming RX controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N06
- **Exists:** Always

Table 11-109 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	RX_CDR_VCO_STEP_CTRL	R/W	**Description**: override value for rx_cdr_vco_step_ctrl. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_CDR_VCO_TEMP_COMP_EN	R/W	**Description**: override value for rx_cdr_vco_temp_comp_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	EN	R/W	**Description**: Override enable for all inputs controlled by this register input signals overridden by this enable are: rx_cdr_vco_step_ctrl rx_cdr_vco_temp_comp_en rx_cdr_vco_freqband[1:0] rx_ref_ld_val[6:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:7	RX_CDR_VCO_FREQBAND_1_0	R/W	**Description**: Override value for rx_cdr_vco_freqband. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
6:0	RX_REF_LD_VAL	R/W	**Description**: Override value for rx_ref_ld_val[6:0]. **Value After Reset**: 0x14 **Exists**: Always **Reset Mask**: 0x7f

11.2.106 LANEN_DIG_ASIC_RX_OVRD_IN_2

- **Description:** Override values for incoming RX controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x1N07
- **Exists:** Always

Table 11-110 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signal overridden by this enable is: rx_vco_ld_val **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:0	RX_VCO_LD_VAL	R/W	**Description**: Override value for rx_vco_ld_val. **Value After Reset**: 0x3e8 **Exists**: Always **Reset Mask**: 0x1fff

11.2.107 LANEN_DIG_ASIC_RX_OVRD_IN_3

- **Description:** Override values for incoming RX controls from ASIC, register #3
- **Size:** 16 bits
- **Offset:** 0x1N08
- **Exists:** Always

Table 11-111 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_3

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	EN	R/W	**Description**: Override enable for all inputs controlled by this register Input signals overridden by this enable are: rx_term_acdc rx_term_en rx_los_lfps_en rx_los_threshold rx_disable rx_clk_shift rx_align_en rx_cdr_ssc_en rx_cdr_track_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	TERM_ACDC	R/W	**Description**: Override value for rx_term_acdc. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	TERM_EN	R/W	**Description**: Override value for rx_term_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	LOS_LPFS_EN	R/W	**Description**: Override value for rx_los_lfps_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-111 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_3 (Continued)

Bits	Name	Memory Access	Description
7:5	LOS_THRSHLD	R/W	**Description**: Override value for rx_los_threshold. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
4	DISABLE	R/W	**Description**: Override value for rx_disable. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	CLK_SHIFT	R/W	**Description**: Override value for rx_clk_shift. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ALIGN_EN	R/W	**Description**: Override value for rx_align_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	CDR_SSC_EN	R/W	**Description**: Override value for rx_cdr_ssc_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	CDR_TRACK_EN	R/W	**Description**: Override value for rx_cdr_track_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.108 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0

- **Description:** Override values for incoming RX EQ controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N09
- **Exists:** Always

Table 11-112 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:9	EQ_CTLE_BOOST	R/W	**Description**: Override value for rx_eq_ctle_boost. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x1f
8:6	EQ_VGA2_GAIN	R/W	**Description**: Override value for rx_eq_vga2_gain. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x7
5:3	EQ_VGA1_GAIN	R/W	**Description**: Override value for rx_eq_vga1_gain. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x7
2:0	EQ_ATT_LVL	R/W	**Description**: Override value for rx_eq_att_lvl. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.109 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1

- **Description:** Override values for incoming RX EQ controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N0a
- **Exists:** Always

Table 11-113 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	EQ_OVRD_EN	R/W	**Description**: Override enable for RX EQ inputs Input signals overridden by this enable are: rx_eq_dfe_tap1 rx_eq_ctle_pole rx_eq_ctle_boost rx_eq_vga2_gain rx_eq_vga1_gain rx_eq_att_lvl **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:2	EQ_DFE_TAP1	R/W	**Description**: Override value for rx_eq_dfe_tap1. **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff
1:0	EQ_CTLE_POLE	R/W	**Description**: Override value for rx_eq_ctle_pole. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.110 LANEN_DIG_ASIC_RX_OVRD_OUT_0

- **Description:** Override values for outgoing RX controls to ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N0b
- **Exists:** Always

Table 11-114 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_OUT_0

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4	EN_CTL	R/W	**Description**: Override enable for all control outputs of this register Input signals overridden by this enable are: rx_adapt_sts rx_los rx_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:2	ADAPT_STS	R/W	**Description**: Override value for rx_adapt_sts. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
1	LOS	R/W	**Description**: Override value for rx_los. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ACK	R/W	**Description**: Override value for rx_ack. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.111 LANEN_DIG_ASIC_LANE_ASIC_IN

- **Description:** Current values for incoming LANE controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N0c
- **Exists:** Always

Table 11-115 Fields for Register: LANEN_DIG_ASIC_LANE_ASIC_IN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	LANE_RX2TX_PAR_LB	R	**Description**: Value from ASIC for lane_rx2tx_par_lb_en_r (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	LANE_TX2RX_SER_LB	R	**Description**: Value from ASIC for lane_tx2rx_ser_lb_en_r (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.112 LANEN_DIG_ASIC_TX_ASIC_IN_0

- **Description:** Current values for incoming TX controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N0d
- **Exists:** Always

Table 11-116 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0

Bits	Name	Memory Access	Description
15	DISABLE	R	**Description**: Value from ASIC for tx_disable (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14	DETECT_RX_REQ	R	**Description**: Value from ASIC for tx_detrx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MPLLB_SEL	R	**Description**: Value from ASIC for tx_mpllb_sel (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:11	WIDTH	R	**Description**: Value from ASIC for tx_width[1:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
10:8	RATE	R	**Description**: Value from ASIC for tx_rate[2:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-116 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
7:6	PSTATE	R	**Description**: Value from ASIC for tx_pstate[1:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	LPD	R	**Description**: Value from ASIC for tx_lpd (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4	REQ	R	**Description**: Value from ASIC for tx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3	DATA_EN	R	**Description**: Value from ASIC for tx_data_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	INVERT	R	**Description**: Value from ASIC for tx_invert (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	RESET	R	**Description**: Value from ASIC for tx_reset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-116 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
0	CLK_RDY	R	**Description**: Value from ASIC for tx_clk_rdy (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.113 LANEN_DIG_ASIC_TX_ASIC_IN_1

- **Description:** Current values for incoming TX controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N0e
- **Exists:** Always

Table 11-117 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:6	TX_MAIN_CURSOR	R	**Description**: Value from ASIC for tx_eq_main (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	VBOOST_EN	R	**Description**: Value from ASIC for tx_vboost_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4:1	IBOOST_LVL	R	**Description**: Value from ASIC for tx_iboost_lvl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	BEACON_EN	R	**Description**: Value from ASIC for tx_beacon_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.114 LANEN_DIG_ASIC_TX_ASIC_IN_2

- **Description:** Current values for incoming TX controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x1N0f
- **Exists:** Always

Table 11-118 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_2

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:6	TX_POST_CURSOR	R	**Description**: Value from ASIC for tx_eq_post (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5:0	TX_PRE_CURSOR	R	**Description**: Value from ASIC for tx_eq_pre (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.115 LANEN_DIG_ASIC_TX_ASIC_OUT

- **Description:** Current values for outgoing TX status controls from PHY
- **Size:** 16 bits
- **Offset:** 0x1N10
- **Exists:** Always

Table 11-119 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_OUT

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	DETRX_RESULT	R	**Description**: Value from PHY for tx_detrx_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	TX_ACK	R	**Description**: Value from PHY for tx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.116 LANEN_DIG_ASIC_RX_ASIC_IN_0

- **Description:** Current values for incoming RX controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N11
- **Exists:** Always

Table 11-120 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	CDR_TRACK_EN	R	**Description**: Value from ASIC for rx_cdr_track_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	RX_DFE_BYPASS	R	**Description**: Value from ASIC for rx_dfe_bypass (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	DIV16P5_CLK_EN	R	**Description**: Value from ASIC for rx_div16p5_clk_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
10:9	WIDTH	R	**Description**: Value from ASIC for rx_width (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-120 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
8:7	RATE	R	**Description**: Value from ASIC for rx_rate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6:5	PSTATE	R	**Description**: Value from ASIC for rx_pstate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4	LPD	R	**Description**: Value from ASIC for rx_lpd (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3	REQ	R	**Description**: Value from ASIC for rx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	DATA_EN	R	**Description**: Value from ASIC for rx_data_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	INVERT	R	**Description**: Value from ASIC for rx_invert (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-120 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0 (Continued)

Bits	Name	Memory Access	Description
0	RESET	R	**Description**: Value from ASIC for rx_reset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.117 LANEN_DIG_ASIC_RX_ASIC_IN_1

- **Description:** Current values for incoming RX controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N12
- **Exists:** Always

Table 11-121 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	RX_TERM_ACDC	R	**Description**: Value from ASIC for rx_term_acdc (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	RX_TERM_EN	R	**Description**: Value from ASIC for rx_term_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	LOS_LPFS_EN	R	**Description**: Value from ASIC for rx_los_lfps_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6:4	LOS_THRSHLD	R	**Description**: Value from ASIC for rx_los_threshold (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-121 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_1 (Continued)

Bits	Name	Memory Access	Description
3	DISABLE	R	**Description**: Value from ASIC for rx_disable (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	CLK_SHIFT	R	**Description**: Value from ASIC for rx_clk_shift (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	ALIGN_EN	R	**Description**: Value from ASIC for rx_align_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	CDR_SSC_EN	R	**Description**: Value from ASIC for rx_cdr_ssc_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.118 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0

- **Description:** Current values for incoming RX EQ controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N13
- **Exists:** Always

Table 11-122 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:9	EQ_CTLE_BOOST	R	**Description**: Value from ASIC for rx_eq_ctle_boost (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8:6	EQ_VGA2_GAIN	R	**Description**: Value from ASIC for rx_eq_vga2_gain (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5:3	EQ_VGA1_GAIN	R	**Description**: Value from ASIC for rx_eq_vga1_gain (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2:0	EQ_ATT_LVL	R	**Description**: Value from ASIC for rx_eq_att_lvl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.119 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1

- **Description:** Current values for incoming RX EQ controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N14
- **Exists:** Always

Table 11-123 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:2	EQ_DFE_TAP1	R	**Description**: Value from ASIC for rx_eq_dfe_tap1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1:0	EQ_CTLE_POLE	R	**Description**: Value from ASIC for rx_eq_ctle_pole (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.120 LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_0

- **Description:** Current values for incoming RX CDR VCO controls from ASIC, register #0
- **Size:** 16 bits
- **Offset:** 0x1N15
- **Exists:** Always

Table 11-124 Fields for Register: LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_0

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10:4	RX_REF_LD_VAL	R	**Description**: Value from ASIC for rx_ref_ld_val (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3:2	RX_CDR_VCO_FREQBAND	R	**Description**: Value from ASIC for rx_cdr_vco_freqband (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	RX_CDR_VCO_STEP_CTRL	R	**Description**: Value from ASIC for rx_cdr_vco_step_ctrl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	RX_CDR_VCO_TEMP_COMP_EN	R	**Description**: Value from ASIC for rx_cdr_vco_temp_comp_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.121 LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_1

- **Description:** Current values for incoming RX CDR VCO controls from ASIC, register #1
- **Size:** 16 bits
- **Offset:** 0x1N16
- **Exists:** Always

Table 11-125 Fields for Register: LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN_1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:0	RX_VCO_LD_VAL	R	**Description**: Value from ASIC for rx_vco_ld_val (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.122 LANEN_DIG_ASIC_RX_ASIC_OUT_0

- **Description:** Current values for outgoing RX status controls from PHY, register #0
- **Size:** 16 bits
- **Offset:** 0x1N17
- **Exists:** Always

Table 11-126 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_OUT_0

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4:3	ADAPT_STS	R	**Description**: Value from PHY for rx_adapt_sts (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	VALID	R	**Description**: Value from PHY for rx_valid (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	LOS	R	**Description**: Value from PHY for rx_los (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	ACK	R	**Description**: Value from PHY for rx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.123 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2

- **Description:** Override values for incoming RX EQ controls from ASIC, register #2
- **Size:** 16 bits
- **Offset:** 0x1N18
- **Exists:** Always

Table 11-127 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:7	EQ_DFE_TAP3	R/W	**Description**: Override value for rx_eq_dfe_tap3. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x40 **Exists**: Always **Reset Mask**: 0x7f
6:0	EQ_DFE_TAP2	R/W	**Description**: Override value for rx_eq_dfe_tap2. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x40 **Exists**: Always **Reset Mask**: 0x7f

11.2.124 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3

- **Description:** Override values for incoming RX EQ controls from ASIC, register #3
- **Size:** 16 bits
- **Offset:** 0x1N19
- **Exists:** Always

Table 11-128 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:7	EQ_DFE_TAP5	R/W	**Description**: Override value for rx_eq_dfe_tap5. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x40 **Exists**: Always **Reset Mask**: 0x7f
6:0	EQ_DFE_TAP4	R/W	**Description**: Override value for rx_eq_dfe_tap4. Its override enable is RX_OVRD_EQ_IN_1.EQ_OVRD_EN **Value After Reset**: 0x40 **Exists**: Always **Reset Mask**: 0x7f

11.2.125 LANEN_DIG_ASIC_TX_OVRD_MISC

- **Description:** Override values for incoming TX MISC BUS from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N1a
- **Exists:** Always

Table 11-129 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_MISC

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	TX_MISC_OVRD_EN	R/W	**Description**: Override enable for tx_misc[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	TX_MISC_OVRD_VAL	R/W	**Description**: Override value for tx_misc[7:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.126 LANEN_DIG_ASIC_RX_OVRD_MISC

- **Description:** Override values for incoming RX MISC BUS from ASIC
- **Size:** 16 bits
- **Offset:** 0x1N1b
- **Exists:** Always

Table 11-130 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_MISC

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	RX_MISC_OVRD_EN	R/W	**Description**: Override enable for rx_misc[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	RX_MISC_OVRD_VAL	R/W	**Description**: Override value for rx_misc[7:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.127 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0

- **Description:** TX Power State Control Register for P0
- **Size:** 16 bits
- **Offset:** 0x1N20
- **Exists:** Always

Table 11-131 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TX_P0_ALLOW_VBOOST	R/W	**Description**: If asserted, then vboost is allowed in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
8	TX_P0_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	TX_P0_DATA_EN	R/W	**Description**: This is ANDed with top-level tx_data_en asic input **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
6	TX_P0_DIG_CLK_EN	R/W	**Description**: Enable/Disable TX digital clocks in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
5	TX_P0_ANA_SERIAL_EN	R/W	**Description**: Value of TX ana serial_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4	TX_P0_ANA_RESET	R/W	**Description**: Value of TX ana reset in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-131 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 (Continued)

Bits	Name	Memory Access	Description
3	TX_P0_ANA_WORD_CLK_EN	R/W	**Description**: Value of TX ana word_clk_en in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_P0_ANA_CLK_EN	R/W	**Description**: Value of TX ana clk_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	TX_P0_ANA_VCM_HOLD	R/W	**Description**: Value of TX ana vcm_hold in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	TX_P0_ANA_REFGEN_EN	R/W	**Description**: Value of TX ana refgen_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.128 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S

- **Description:** TX Power State Control Register for P0S
- **Size:** 16 bits
- **Offset:** 0x1N21
- **Exists:** Always

Table 11-132 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TX_P0s_ALLOW_VBOOST	R/W	**Description**: If asserted, then vboost is allowed in P0s **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
8	TX_P0S_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	TX_P0S_DATA_EN	R/W	**Description**: This is ANDed with top-level tx_data_en asic input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	TX_P0S_DIG_CLK_EN	R/W	**Description**: Enable/Disable TX digital clocks in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
5	TX_P0S_ANA_SERIAL_EN	R/W	**Description**: Value of TX ana serial_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-132 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S (Continued)

Bits	Name	Memory Access	Description
4	TX_P0S_ANA_RESET	R/W	**Description**: Value of TX ana reset in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	TX_P0S_ANA_WORD_CLK_EN	R/W	**Description**: Value of TX ana word_clk_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_P0S_ANA_CLK_EN	R/W	**Description**: Value of TX ana clk_en in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	TX_P0S_ANA_VCM_HOLD	R/W	**Description**: Value of TX ana vcm_hold in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	TX_P0S_ANA_REFGEN_EN	R/W	**Description**: Value of TX ana refgen_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.129 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1

- **Description:** TX Power State Control Register for P1
- **Size:** 16 bits
- **Offset:** 0x1N22
- **Exists:** Always

Table 11-133 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TX_P1_ALLOW_VBOOST	R/W	**Description**: If asserted, then vboost is allowed in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
8	TX_P1_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
7	TX_P1_DATA_EN	R/W	**Description**: This is ANDed with top-level tx_data_en asic input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	TX_P1_DIG_CLK_EN	R/W	**Description**: Enable/Disable TX digital clocks in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	TX_P1_ANA_SERIAL_EN	R/W	**Description**: Value of TX ana serial_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	TX_P1_ANA_RESET	R/W	**Description**: Value of TX ana reset in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-133 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 (Continued)

Bits	Name	Memory Access	Description
3	TX_P1_ANA_WORD_CLK_EN	R/W	**Description**: Value of TX ana word_clk_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_P1_ANA_CLK_EN	R/W	**Description**: Value of TX ana clk_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TX_P1_ANA_VCM_HOLD	R/W	**Description**: Value of TX ana vcm_hold in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	TX_P1_ANA_REFGEN_EN	R/W	**Description**: Value of TX ana refgen_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.130 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2

- **Description:** TX Power State Control Register for P2
- **Size:** 16 bits
- **Offset:** 0x1N23
- **Exists:** Always

Table 11-134 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TX_P2_ALLOW_VBOOST	R/W	**Description**: If asserted, then vboost is allowed in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	TX_P2_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
7	TX_P2_DATA_EN	R/W	**Description**: This is ANDed with top-level tx_data_en asic input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	TX_P2_DIG_CLK_EN	R/W	**Description**: Enable/Disable TX digital clocks in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	TX_P2_ANA_SERIAL_EN	R/W	**Description**: Value of TX ana serial_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	TX_P2_ANA_RESET	R/W	**Description**: Value of TX ana reset in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-134 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 (Continued)

Bits	Name	Memory Access	Description
3	TX_P2_ANA_WORD_CLK_EN	R/W	**Description**: Value of TX ana word_clk_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_P2_ANA_CLK_EN	R/W	**Description**: Value of TX ana clk_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TX_P2_ANA_VCM_HOLD	R/W	**Description**: Value of TX ana vcm_hold in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	TX_P2_ANA_REFGEN_EN	R/W	**Description**: Value of TX ana refgen_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.131 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0

- **Description:** TX Power-up Time Register #0
- **Size:** 16 bits
- **Offset:** 0x1N24
- **Exists:** Always

Table 11-135 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:8	TX_CLK_EN	R/W	**Description**: Power-up time (in ref_range cycles) for TX ana clock enable (spec: >=1us) **Value After Reset**: 0x1a **Exists**: Always **Reset Mask**: 0xff
7:0	TX_REFGEN_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for TX ana refgen enable (spec: >=1us) **Value After Reset**: 0x1a **Exists**: Always **Reset Mask**: 0xff

11.2.132 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1

- **Description:** TX Power-up Time Register #1
- **Size:** 16 bits
- **Offset:** 0x1N25
- **Exists:** Always

Table 11-136 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15	SKIP_TX_VCM_HOLD_WAIT	R/W	**Description**: Skip wait for TX common mode hold Power-up **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
14:0	TX_VCM_HOLD_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for TX ana vreg enable (spec: >=800us) **Value After Reset**: 0x515 **Exists**: Always **Reset Mask**: 0x7fff

11.2.133 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2

- **Description:** TX Power-up Time Register #2
- **Size:** 16 bits
- **Offset:** 0x1N26
- **Exists:** Always

Table 11-137 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15:13	DTB_SEL	R/W	**Description**: Selects data to drive on DTB 0 - disabled 1 - tx_ack and tx_pwrsm_state[0] 2 - tx_ana_rxdetp_result_i, tx_ana_rxdetm_result_i 3 - tx_ana_reset_i, tx_ana_clk_en_i 4 - analog/asic clocks 5 - asic early signal / clock aligner shift 6 - tx_clk_state counter / lbert strobe 7 - ref_dig_rst/tx_dig_rst **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
12:0	TX_VBOOST_DIS_TIME	R/W	**Description**: TX vboost disable time (in ref_range cycles) (spec: >=160us) **Value After Reset**: 0x1040 **Exists**: Always **Reset Mask**: 0x1fff

11.2.134 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3

- **Description:** TX Power-up Time Register #3
- **Size:** 16 bits
- **Offset:** 0x1N27
- **Exists:** Always

Table 11-138 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:11	TX_RESET_TIME	R/W	**Description**: TX Reset de-assertion time (in ref_range_clk cycles) (spec: >=50ns) **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
10	FAST_TX_RXDET	R/W	**Description**: Enable fast TX RX-detection (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:0	TX_RXDET_TIME	R/W	**Description**: RX detect up time (in ref_range_clk cycles), starting from asserting rxdet_en (spec: 12us < time < 26us) **Value After Reset**: 0x172 **Exists**: Always **Reset Mask**: 0x3ff

11.2.135 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4

- **Description:** TX Power-up Time Register #4
- **Size:** 16 bits
- **Offset:** 0x1N28
- **Exists:** Always

Table 11-139 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4:0	TX_SERIAL_EN_TIME	R/W	**Description**: Power-up time in (ref_range_clk) cycles for TX ana serial enable (spec: 500ns < time < 1us) **Value After Reset**: 0xd **Exists**: Always **Reset Mask**: 0x1f

11.2.136 LANEN_DIG_TX_PWRCTL_TX_FIFO_BYPASS

- **Description:** TX fifo bypass Register
- **Size:** 16 bits
- **Offset:** 0x1N29
- **Exists:** Always

Table 11-140 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_FIFO_BYPASS

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	TX_FIFO_BYPASS	R/W	**Description**: By-pass TX datapath FIFO **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.137 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_5

- **Description:** TX Power-up Time Register #4
- **Size:** 16 bits
- **Offset:** 0x1N2a
- **Exists:** Always

Table 11-141 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_5

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:0	TX_VCM_HOLD_GS_TIME	R/W	**Description**: TX common mode gear-shift time (in ref range cycles) (spec: >=400us) **Value After Reset**: 0x28a0 **Exists**: Always **Reset Mask**: 0x7fff

11.2.138 LANEN_DIG_TX_LBERT_CTL

- **Description:** Pattern Generator controls
- **Size:** 16 bits
- **Offset:** 0x1N2b
- **Exists:** Always

Table 11-142 Fields for Register: LANEN_DIG_TX_LBERT_CTL

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:5	PAT0	R/W	**Description**: Pattern for modes 3-5 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff
4	TRIGGER_ERR	R/W	**Description**: Insert a single error into a lsb Any write of a 1 to this bit will insert an error **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:0	MODE	R/W	**Description**: Pattern to generate When changing modes, you must change to disabled first 0 - Disabled 1 - lfsr31: X^31 + X^28 + 1 2 - lfsr23: X^23 + X^18 + 1 3 - lfsr23: x^23 + x^21 + x^16 + x^8 + x^5 + x^2 + 1 4 - lfsr16: x^16 + x^5 + x^4 + x^3 + 1 5 - lfsr15: X^15 + X^14 + 1 6 - lfsr11: X^11 + X^9 + 1 7 - lfsr9: X^9 + X^5 + 1 8 - lfsr7: X^7 + X^6 + 1 9 - Fixed word (PAT0) 10 - DC balanced word (PAT0, ~PAT0) 11 - Fixed pattern: (000, PAT0, 3ff, ~PAT0) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.139 LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0

- **Description:** TX Clock Alignment Control Register #0
- **Size:** 16 bits
- **Offset:** 0x1N2c
- **Exists:** Always

Table 11-143 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	TX_NUM_2UI_SHIFTS_20B_MODE	R/W	**Description**: Number of UI shift in 20 Bit mode **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0xf
3:0	TX_NUM_2UI_SHIFTS_16B_MODE	R/W	**Description**: Number of UI shift in 16 Bit mode **Value After Reset**: 0x5 **Exists**: Always **Reset Mask**: 0xf

11.2.140 LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1

- **Description:** TX Clock Alignment Control Register #1
- **Size:** 16 bits
- **Offset:** 0x1N2d
- **Exists:** Always

Table 11-144 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	TX_STARTUP_DELAY_R	R/W	**Description**: TX startup delay value **Value After Reset**: 0x11f **Exists**: Always **Reset Mask**: 0x1ff

11.2.141 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0

- **Description:** RX Power State Control Register for P0
- **Size:** 16 bits
- **Offset:** 0x1N40
- **Exists:** Always

Table 11-145 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	RX_P0_FORCE_DFE_BYPASS	R/W	**Description**: Enable/disable RX DFE in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	RX_P0_DIG_CLK_EN	R/W	**Description**: Enable/disable RX digital clocks in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
10	RX_P0_VCO_CONTCAL_EN	R/W	**Description**: Enable/disable continuous calibration of the RX VCO in P0 If RX_P0_DIG_CLK_EN and the top-level rx_data_en are both asserted, then contiuous calibration is turned off and this value is ignored **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
9	RX_P0_VCO_CAL_RST	R/W	**Description**: Enable/disable resetting the RX VCO in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	RX_P0_VCO_FREQ_RST	R/W	**Description**: Enable/disable resetting the RX VCO frequency in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-145 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 (Continued)

Bits	Name	Memory Access	Description
7	RX_P0_ANA_CDR_EN	R/W	**Description**: Value of RX ana cdr_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
6	RX_P0_ANA_DESER_EN	R/W	**Description**: Value of RX ana deserial_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
5	RX_P0_ANA_CLK_DCC_EN	R/W	**Description**: Value of RX ana CLK_DCC_EN in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4	RX_P0_ANA_CLK_EN	R/W	**Description**: Value of RX ana clk_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
3	RX_P0_ANA_DIV16P5_CLK_EN	R/W	**Description**: Value of RX ana div16p5_clk_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	RX_P0_ANA_CLK_VREG_EN	R/W	**Description**: Value of RX ana clk_vreg_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RX_P0_ANA_AFE_EN	R/W	**Description**: Value of RX ana afe_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RX_P0_ANA_LOS_EN	R/W	**Description**: Value of RX ana los_en in P0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.142 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S

- **Description:** RX Power State Control Register for P0S
- **Size:** 16 bits
- **Offset:** 0x1N41
- **Exists:** Always

Table 11-146 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	RX_P0s_FORCE_DFE_BYPASS	R/W	**Description**: Enable/disable RX DFE in P0s **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	RX_P0S_DIG_CLK_EN	R/W	**Description**: Enable/disable RX digital clocks in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_P0S_VCO_CONTCAL_EN	R/W	**Description**: Enable/disable continuous calibration of the RX VCO in P0S If RX_P0S_DIG_CLK_EN and the top-level rx_data_en are both asserted, then contiuous calibration is turned off and this value is ignored **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
9	RX_P0S_VCO_CAL_RST	R/W	**Description**: Enable/disable resetting the RX VCO in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	RX_P0S_VCO_FREQ_RST	R/W	**Description**: Enable/disable resetting the RX VCO frequency in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-146 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S (Continued)

Bits	Name	Memory Access	Description
7	RX_P0S_ANA_CDR_EN	R/W	**Description**: Value of RX ana cdr_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_P0S_ANA_DESER_EN	R/W	**Description**: Value of RX ana deserial_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_P0S_ANA_CLK_DCC_EN	R/W	**Description**: Value of RX ana CLK_DCC_EN in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
4	RX_P0S_ANA_CLK_EN	R/W	**Description**: Value of RX ana clk_en in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
3	RX_P0S_ANA_DIV16P5_CLK_EN	R/W	**Description**: Value of RX ana div16p5_clk_en in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_P0S_ANA_CLK_VREG_EN	R/W	**Description**: Value of RX ana clk_vreg_en in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RX_P0S_ANA_AFE_EN	R/W	**Description**: Value of RX ana afe_en in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RX_P0S_ANA_LOS_EN	R/W	**Description**: Value of RX ana los_en in P0S **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.143 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1

- **Description:** RX Power State Control Register for P1
- **Size:** 16 bits
- **Offset:** 0x1N42
- **Exists:** Always

Table 11-147 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	RX_P1_FORCE_DFE_BYPASS	R/W	**Description**: Enable/disable RX DFE in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
11	RX_P1_DIG_CLK_EN	R/W	**Description**: Enable/disable RX digital clocks in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_P1_VCO_CONTCAL_EN	R/W	**Description**: Enable/disable continuous calibration of the RX VCO in P1 If RX_P1_DIG_CLK_EN and the top-level rx_data_en are both asserted, then contiuous calibration is turned off and this value is ignored **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	RX_P1_VCO_CAL_RST	R/W	**Description**: Enable/disable resetting the RX VCO in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
8	RX_P1_VCO_FREQ_RST	R/W	**Description**: Enable/disable resetting the RX VCO frequency in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-147 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 (Continued)

Bits	Name	Memory Access	Description
7	RX_P1_ANA_CDR_EN	R/W	**Description**: Value of RX ana cdr_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_P1_ANA_DESER_EN	R/W	**Description**: Value of RX ana deserial_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_P1_ANA_CLK_DCC_EN	R/W	**Description**: Value of RX ana CLK_DCC_EN in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_P1_ANA_CLK_EN	R/W	**Description**: Value of RX ana clk_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RX_P1_ANA_DIV16P5_CLK_EN	R/W	**Description**: Value of RX ana div16p5_clk_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_P1_ANA_CLK_VREG_EN	R/W	**Description**: Value of RX ana clk_vreg_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_P1_ANA_AFE_EN	R/W	**Description**: Value of RX ana afe_en in P1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_P1_ANA_LOS_EN	R/W	**Description**: Value of RX ana los_en in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.144 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2

- **Description:** RX Power State Control Register for P2
- **Size:** 16 bits
- **Offset:** 0x1N43
- **Exists:** Always

Table 11-148 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	RX_P2_FORCE_DFE_BYPASS	R/W	**Description**: Enable/disable RX DFE in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
11	RX_P2_DIG_CLK_EN	R/W	**Description**: Enable/disable RX digital clocks in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_P2_VCO_CONTCAL_EN	R/W	**Description**: Enable/disable continuous calibration of the RX VCO in P2 If RX_P2_DIG_CLK_EN and the top-level rx_data_en are both asserted, then contiuous calibration is turned off and this value is ignored **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	RX_P2_VCO_CAL_RST	R/W	**Description**: Enable/disable resetting the RX VCO in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
8	RX_P2_VCO_FREQ_RST	R/W	**Description**: Enable/disable resetting the RX VCO frequency in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

Table 11-148 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 (Continued)

Bits	Name	Memory Access	Description
7	RX_P2_ANA_CDR_EN	R/W	**Description**: Value of RX ana cdr_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_P2_ANA_DESER_EN	R/W	**Description**: Value of RX ana deserial_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_P2_ANA_CLK_DCC_EN	R/W	**Description**: Value of RX ana CLK_DCC_EN in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_P2_ANA_CLK_EN	R/W	**Description**: Value of RX ana clk_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RX_P2_ANA_DIV16P5_CLK_EN	R/W	**Description**: Value of RX ana div16p5_clk_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_P2_ANA_CLK_VREG_EN	R/W	**Description**: Value of RX ana clk_vreg_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_P2_ANA_AFE_EN	R/W	**Description**: Value of RX ana afe_en in P2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_P2_ANA_LOS_EN	R/W	**Description**: Value of RX ana los_en in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.145 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_0

- **Description:** RX Power-up Time Register #0
- **Size:** 16 bits
- **Offset:** 0x1N44
- **Exists:** Always

Table 11-149 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	FAST_RX_LOS_EN_WAIT	R/W	**Description**: Enable fast wait for RX LOS enable (simulation Only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:0	RX_LOS_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for RX ana los enable (spec >=1.1us) **Value After Reset**: 0x1d **Exists**: Always **Reset Mask**: 0x3ff

11.2.146 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1

- **Description:** RX Power-up Time Register #1
- **Size:** 16 bits
- **Offset:** 0x1N45
- **Exists:** Always

Table 11-150 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	FAST_RX_VREG_EN	R/W	**Description**: Enable fast RX VREG enable (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:2	RX_VREG_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles divided by 2) for RX ana vreg enable (spec >2us) **Value After Reset**: 0x1a **Exists**: Always **Reset Mask**: 0x7f
1:0	RX_RATE_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for RX ana rate or width change **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.147 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2

- **Description:** RX Power-up Time Register #2
- **Size:** 16 bits
- **Offset:** 0x1N46
- **Exists:** Always

Table 11-151 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:7	RX_DESER_DIS_TIME	R/W	**Description**: Power down time in (ref_range cycles) for RX ana deserial enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
6:5	RX_DESER_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for RX ana deserial enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
4:0	RX_CDR_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for RX ana cdr (or dfe/dfe_taps) enable (spec 200ns) **Value After Reset**: 0x5 **Exists**: Always **Reset Mask**: 0x1f

11.2.148 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_3

- **Description:** RX Power-up Control Register #0
- **Size:** 16 bits
- **Offset:** 0x1N47
- **Exists:** Always

Table 11-152 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	FAST_RX_CLK_DCC_EN	R/W	**Description**: Enable fast RX clock dcc enable (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:0	RX_CLK_DCC_EN_TIME	R/W	**Description**: Power-up time (in ref_range cycles) for RX ana clk dcc enable (spec >1us) **Value After Reset**: 0x1a **Exists**: Always **Reset Mask**: 0x3ff

11.2.149 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0

- **Description:** RX VCO calibration controls register #0
- **Size:** 16 bits
- **Offset:** 0x1N48
- **Exists:** Always

Table 11-153 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2:0	INT_GAIN_CAL_BOUNCE_CNT	R/W	**Description**: Number of bounces (i.e. direction changes) on the int_gain code before indicating that the RX VCO calibration is done **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x7

11.2.150 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1

- **Description:** RX VCO calibration controls register #1
- **Size:** 16 bits
- **Offset:** 0x1N49
- **Exists:** Always

Table 11-154 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1

Bits	Name	Memory Access	Description
15:9	DTB_SEL	R/W	**Description**: DTB select for RX VCO dtb signals 7'h01 - {chkfrq_en, ref_dig_clk} 7'h02 - {rx_ana_cdr_vco_en_i, rx_ana_cdr_startup_i} 7'h04 - {rx_vco_up, dpll_freq_rst} 7'h08 - {rx_vco_contcal_en, rx_vco_cal_rst} 7'h10 - {chkfrq_done, vcoclk_too_fast} 7'h20 - {cal_dir, rx_vco_cal_done} 7'h40 - {curr_state[0], rx_vco_cnt[0]} **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
8:5	DPLL_CAL_UG	R/W	**Description**: DPLL calibration update on int_gain code 3'h0 - 0 Else - (1/16)*2^(DPLL_CAL_UG-1) LSB/update Maximum DPLL_CAL_UG=10, i.e. 32 LSB/update **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf
4	DISABLE_INT_CAL_MODE	R/W	**Description**: When asserted, then the DPLL frequency register is never modified by the RX VCO calibration FSM (even if DPLL_CAL_UG is non-zero). In this case, the calibration will always be performed on the VCO freq_tune code. This allows disabling of integral calibration feature, and hence only using freq_tune calibration. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-154 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1 (Continued)

Bits	Name	Memory Access	Description
3	RX_VCO_CONTCAL_EN	R/W	**Description**: Override value for the continuous calibration enable from the RX PWRSM **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_VCO_CAL_RST	R/W	**Description**: Override value for the calibration reset from the RX PWRSM **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_VCO_FREQ_RST	R/W	**Description**: Override value for the frequency reset from the RX PWRSM **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_VCO_OVRD_SEL	R/W	**Description**: Override the calibration controls from the RX PWRSM **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.151 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2

- **Description:** RX VCO calibration controls register #2
- **Size:** 16 bits
- **Offset:** 0x1N4a
- **Exists:** Always

Table 11-155 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:10	FREG_TUNE_CAL_STEPS	R/W	**Description**: Number of cal steps of freq tune **Value After Reset**: 0x9 **Exists**: Always **Reset Mask**: 0xf
9:0	FREG_TUNE_START_VAL	R/W	**Description**: Starting value of freq tune code **Value After Reset**: 0x200 **Exists**: Always **Reset Mask**: 0x3ff

11.2.152 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_0

- **Description:** RX Power UP Time Register #0
- **Size:** 16 bits
- **Offset:** 0x1N4b
- **Exists:** Always

Table 11-156 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_0

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	FAST_RX_VCO_WAIT	R/W	**Description**: Enable fast RX VCO wait times (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:4	RX_VCO_CNTR_PWRUP_TIME	R/W	**Description**: Power up time for Rx ana vco cnter (spec: >2us) **Value After Reset**: 0x34 **Exists**: Always **Reset Mask**: 0xff
3:0	RX_VCO_UPDATE_TIME	R/W	**Description**: Settle time in (ref_range cycles) for RX ana vco update (freq_tune or int_gain) (spec >350ns) **Value After Reset**: 0x9 **Exists**: Always **Reset Mask**: 0xf

11.2.153 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1

- **Description:** RX Power UP Time Register #1
- **Size:** 16 bits
- **Offset:** 0x1N4c
- **Exists:** Always

Table 11-157 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10:4	RX_VCO_STARTUP_TIME	R/W	**Description**: Power up time in ref_range cycles for RX ana vco startup (spec >250ns) **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x7f
3	FAST_RX_VCO_CAL	R/W	**Description**: Enable fast RX VCO calibration (skips calibration) (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2:0	RX_VCO_CNTR_SETTLE_TIME	R/W	**Description**: RX VCO counter value settling time in (ref_dig_clk cycles) (spec: 1 ref_dig_clk cycle) **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7

11.2.154 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0

- **Description:** RX VCO status register #0
- **Size:** 16 bits
- **Offset:** 0x1N4d
- **Exists:** Always

Table 11-158 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	RX_ANA_CLK_EN	R	**Description**: Current value of rx_ana_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	RX_ANA_CDR_VCO_EN	R	**Description**: Current value of rx_ana_cdr_vco_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	RX_ANA_CDR_STARTUP	R	**Description**: Current value of rx_ana_cdr_startup_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	RX_ANA_VCO_CNTR_EN	R	**Description**: Current value of rx_ana_vco_cntr_en_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-158 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0 (Continued)

Bits	Name	Memory Access	Description
10	RX_ANA_VCO_CNTR_PD	R	**Description**: Current value of rx_ana_vco_cntr_pd_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9:0	RX_ANA_CDR_FREQ_TUNE	R	**Description**: Current value of rx_ana_cdr_freq_tune_i (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.155 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1

- **Description:** RX VCO status register #1
- **Size:** 16 bits
- **Offset:** 0x1N4e
- **Exists:** Always

Table 11-159 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	DPLL_FREQ_RST	R	**Description**: Indicates that the RX integral frequency is reset or not (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	RX_VCO_CAL_DONE	R	**Description**: Indicates that the RX VCO has completed calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6	RX_VCO_CONTCAL_EN	R	**Description**: Value of the continuous calibration enable from the RX PWRSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	RX_VCO_CAL_RST	R	**Description**: Value of the calibration reset from the RX PWRSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-159 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1 (Continued)

Bits	Name	Memory Access	Description
4	RX_VCO_FREQ_RST	R	**Description**: Value of the RX VCO frequency reset from the RX PWRSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3:0	RX_VCO_FSM_STATE	R	**Description**: Value of the RX VCO CAL FSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.156 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2

- **Description:** RX VCO status register #2
- **Size:** 16 bits
- **Offset:** 0x1N4f
- **Exists:** Always

Table 11-160 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2

Bits	Name	Memory Access	Description
15	RX_VCO_UP	R	**Description**: Indicates that the RX VCO is ready (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14	RX_VCO_CORRECT	R	**Description**: Indicates that the RX VCO clock has the correct frequency (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	VCOCLK_TOO_FAST	R	**Description**: Indicates that the RX VCO clock frequency is too fast (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:0	VCO_CNTR_FINAL	R	**Description**: Value of Rx VCO counter when refclk counter expired (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.157 LANEN_DIG_RX_RX_ALIGN_XAUI_COMM_MASK

- **Description:** XAUI_COMMA Mask
- **Size:** 16 bits
- **Offset:** 0x1N50
- **Exists:** Always

Table 11-161 Fields for Register: LANEN_DIG_RX_RX_ALIGN_XAUI_COMM_MASK

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	XAUI_COMM_MASK	R/W	**Description**: XAUI_COMMA Mask. For 10-bit COMMA set the mask to 0x3FF and for 7-bit COMMA set the mask to 0x3F8 **Value After Reset**: 0x3ff **Exists**: Always **Reset Mask**: 0x3ff

11.2.158 LANEN_DIG_RX_LBERT_CTL

- **Description:** Pattern Matcher controls
- **Size:** 16 bits
- **Offset:** 0x1N51
- **Exists:** Always

Table 11-162 Fields for Register: LANEN_DIG_RX_LBERT_CTL

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4	SYNC	R/W	**Description**: Synchronize pattern matcher LFSR with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. There is no need to write this back to zero to run normally. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:0	MODE	R/W	**Description**: Pattern to match When changing modes, you must change to disabled first 0 - Disabled 1 - lfsr31: X^31 + X^28 + 1 2 - lfsr23: X^23 + X^18 + 1 3 - lfsr23: x^23 + x^21 + x^16 + x^8 + x^5 + x^2 + 1 4 - lfsr16: x^16 + x^5 + x^4 + x^3 + 1 5 - lfsr15: X^15 + X^14 + 1 6 - lfsr11: X^11 + X^9 + 1 7 - lfsr9: X^9 + X^5 + 1 8 - lfsr7: X^7 + X^6 + 1 9 - d[n] = d[n-10] 10 - d[n] = !d[n-10] 11 - d[n] = !d[n-20] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.159 LANEN_DIG_RX_LBERT_ERR

- **Description:** Pattern match error counter
- **Size:** 16 bits
- **Offset:** 0x1N52
- **Exists:** Always

Table 11-163 Fields for Register: LANEN_DIG_RX_LBERT_ERR

Bits	Name	Memory Access	Description
15	OV14	R/W	**Description**: If active, multiply COUNT by 128. If OV14=1 and COUNT=2^15-1, signals overflow of counter (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
14:0	COUNT	R/W	**Description**: A read of this register, or a sync of the PM resets the error count. Current error count If OV14 field is active, then multiply count by 128 (volatile and 2 reads needed to read value) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7fff **Volatile**: true

11.2.160 LANEN_DIG_RX_CDR_CDR_CTL_0

- **Description:** Control bits for receiver in recovered domain
- **Size:** 16 bits
- **Offset:** 0x1N53
- **Exists:** Always

Table 11-164 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_0

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10:7	DTB_SEL	R/W	**Description**: Select to drive various signals onto the dtb 0 - disabled 1 - rx_pr_stable, rx_afe_stable from rx_ana_ctl 2 - com_good, com_bad from rx_align 3 - shift_in_prog, ana_odd_data from rx_align 4 - 2 MSB's of XAUI align FSM state 5 - 2 LSB's of XAUI align FSM state 6 - error_high, low from lbert_pm 7 - ana_los, los_filter from los block 8 - eios_state[0], eios_det from los block 9 - cdr_valid, MSB of FSM state from cdr_ctl A - 2 LSB's of FSM state from cdr_ctl B - rx_dig_rst,rx_dig_en C - rx_ana_word_clk_i, rx_ana_dword_clk_i **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
6	ALWAYS_REALIGN	R/W	**Description**: Realign on any misaligned comma **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	PHDET_EN_PR_MODE	R/W	**Description**: Enable partial response phase detector mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-164 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_0 (Continued)

Bits	Name	Memory Access	Description
4	PHDET_POL	R/W	**Description**: Reverse polarity of phase error **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:2	PHDET_EDGE	R/W	**Description**: Edges to use for phase detection. 10 - Use both edges 01 - Use rising edges only 11 - Use falling edges only 00 - Ignore all edges **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3
1:0	PHDET_EN	R/W	**Description**: Enable phase detector. top bit is odd slicers, bottom is even **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3

11.2.161 LANEN_DIG_RX_CDR_CDR_CTL_1

- **Description:** CDR Control Register #1
- **Size:** 16 bits
- **Offset:** 0x1N54
- **Exists:** Always

Table 11-165 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_1

Bits	Name	Memory Access	Description
15:10	SSC_OFF_CNT1	R/W	**Description**: When SSC mode is disabled, the 12-bit word count in gain stage 1 is: (SSC_OFF_CNT1 * 4) in 20b mode (SSC_OFF_CNT1 * 5) in 16b mode **Value After Reset**: 0x13 **Exists**: Always **Reset Mask**: 0x3f
9:0	SSC_OFF_CNT0	R/W	**Description**: When SSC mode is disabled, the 12-bit word count in gain stage 0 is: (SSC_OFF_CNT0 * 4) in 20b mode (SSC_OFF_CNT0 * 5) in 16b mode **Value After Reset**: 0x39 **Exists**: Always **Reset Mask**: 0x3ff

11.2.162 LANEN_DIG_RX_CDR_CDR_CTL_2

- **Description:** CDR Control Register #2
- **Size:** 16 bits
- **Offset:** 0x1N55
- **Exists:** Always

Table 11-166 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_2

Bits	Name	Memory Access	Description
15:9	SSC_ON_CNT1	R/W	**Description**: When SSC mode is enabled, the 12-bit word count in gain stage 1 is: (SSC_ON_CNT1 * 8) in 20b mode (SSC_ON_CNT1 * 10) in 16b mode **Value After Reset**: 0x25 **Exists**: Always **Reset Mask**: 0x7f
8:0	SSC_ON_CNT0	R/W	**Description**: When SSC mode is enabled, the 12-bit word count in gain stage 0 is: (SSC_ON_CNT0 * 8) in 20b mode (SSC_ON_CNT0 * 10) in 16b mode **Value After Reset**: 0xbb **Exists**: Always **Reset Mask**: 0x1ff

11.2.163 LANEN_DIG_RX_CDR_CDR_CTL_3

- **Description:** CDR Control Register #3
- **Size:** 16 bits
- **Offset:** 0x1N56
- **Exists:** Always

Table 11-167 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_3

Bits	Name	Memory Access	Description
15:13	FRUG_OVRD_VALUE	R/W	**Description**: Override value for FRUG (frequency update gain) 3'h0 - 0 3'h1 - 1/16 LSB/update 3'h2 - 1/8 LSB/update 3'h3 - 1/4 LSB/update 3'h4 - 1/2 LSB/update 3'h5 - 1 LSB/update 3'h6 - 2 LSB/update 3'h7 - 4 LSB/update **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
12:10	PHUG_OVRD_VALUE	R/W	**Description**: Override value for PHUG (phase update gain) 3'h0 - 0 3'h1 - 1000 ppm 3'h2 - 2000 ppm 3'h3 - 3000 ppm 3'h4 - 4000 ppm 3'h5 - 5000 ppm 3'h6 - 6000 ppm 3'h7 - 7000 ppm **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
9	OVRD_DPLL_GAIN	R/W	**Description**: Override PHUG and FRUG values **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-167 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_3 (Continued)

Bits	Name	Memory Access	Description
8:6	SSC_OFF_FRUG0	R/W	**Description**: When SSC mode is disabled, the frug value in gain stage 0 is SSC_OFF_FRUG0 **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7
5:3	SSC_OFF_PHUG1	R/W	**Description**: When SSC mode is disabled, the phug value in gain stage 1 is SSC_OFF_PHUG1 **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x7
2:0	SSC_OFF_PHUG0	R/W	**Description**: When SSC mode is disabled, the phug value in gain stage 0 is SSC_OFF_PHUG0 **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x7

11.2.164 LANEN_DIG_RX_CDR_CDR_CTL_4

- **Description:** CDR Control Register #4
- **Size:** 16 bits
- **Offset:** 0x1N57
- **Exists:** Always

Table 11-168 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_4

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:12	SSC_ON_PHUG1	R/W	**Description**: When SSC mode is enabled, the phug value in gain stage 1 is SSC_ON_PHUG1 **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x7
11:9	SSC_ON_PHUG0	R/W	**Description**: When SSC mode is enabled, the phug value in gain stage 0 is SSC_ON_PHUG0 **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x7
8:6	SSC_ON_FRUG1	R/W	**Description**: When SSC mode is enabled, the frug value in gain stage 1 is SSC_ON_FRUG1 **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7
5:3	SSC_ON_FRUG0	R/W	**Description**: When SSC mode is enabled, the frug value in gain stage 0 is SSC_ON_FRUG0 **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7
2:0	SSC_OFF_FRUG1	R/W	**Description**: When SSC mode is disabled, the frug value in gain stage 1 is SSC_OFF_FRUG1 **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7

11.2.165 LANEN_DIG_RX_CDR_STAT

- **Description:** Current output values to dpll (phug, frug)
- **Size:** 16 bits
- **Offset:** 0x1N58
- **Exists:** Always

Table 11-169 Fields for Register: LANEN_DIG_RX_CDR_STAT

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:3	FRUG_VALUE	R	**Description**: Current value for dpll_frug[2:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2:0	PHUG_VALUE	R	**Description**: Current value for dpll_phug[2:0] (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.166 LANEN_DIG_RX_DPLL_FREQ

- **Description:** Current frequency integrator value.
- **Size:** 16 bits
- **Offset:** 0x1N59
- **Exists:** Always

Table 11-170 Fields for Register: LANEN_DIG_RX_DPLL_FREQ

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:0	VAL	R/W	**Description**: Freq is 125*VAL ppm from the reference (volatile and 2 reads needed to read value) **Value After Reset**: 0x2000 **Exists**: Always **Reset Mask**: 0x3fff **Volatile**: true

11.2.167 LANEN_DIG_RX_DPLL_FREQ_BOUND_0

- **Description:** Frequency Bounds for incoming data stream #0
- **Size:** 16 bits
- **Offset:** 0x1N5a
- **Exists:** Always

Table 11-171 Fields for Register: LANEN_DIG_RX_DPLL_FREQ_BOUND_0

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10:1	UPPER_FREQ_BOUND	R/W	**Description**: Upper frequency bound in terms of LSBs of the integral control code **Value After Reset**: 0x264 **Exists**: Always **Reset Mask**: 0x3ff
0	FREQ_BOUND_EN	R/W	**Description**: Enable the frequency bounds feature **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.168 LANEN_DIG_RX_DPLL_FREQ_BOUND_1

- **Description:** Frequency Bounds for incoming data stream #1
- **Size:** 16 bits
- **Offset:** 0x1N5b
- **Exists:** Always

Table 11-172 Fields for Register: LANEN_DIG_RX_DPLL_FREQ_BOUND_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	LOWER_FREQ_BOUND	R/W	**Description**: Lower frequency bound in terms of LSBs of the integral control code **Value After Reset**: 0x19c **Exists**: Always **Reset Mask**: 0x3ff

11.2.169 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0

- **Description:** Adaptation Configuration Register #0
- **Size:** 16 bits
- **Offset:** 0x1N60
- **Exists:** Always

Table 11-173 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0

Bits	Name	Memory Access	Description
15	ADPT_CLK_DIV4_EN	R/W	**Description**: Set the adaptation clock to be divided by 4 (default is div2) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	START_ASM1	R/W	**Description**: Start adaptation state machine #1 (VGA, CTLE, DFE, EYEH) This register-bit is self-clearing **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:10	N_TGG_ASM1	R/W	**Description**: Number of toggle loop iterations for ASM1 **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0xf
9:0	N_TOP_ASM1	R/W	**Description**: Number of top level loop iterations for ASM1 **Value After Reset**: 0x10 **Exists**: Always **Reset Mask**: 0x3ff

11.2.170 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1

- **Description:** Adaptation Configuration Register #1
- **Size:** 16 bits
- **Offset:** 0x1N61
- **Exists:** Always

Table 11-174 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	DFE_T1_ANA_DIS	R/W	**Description**: Disable updating DFE tap1 analog values during adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	CTLE_POLE_OVRD_EN	R/W	**Description**: Override CTLE pole value (only valid if adaptation is run) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:8	CTLE_POLE_OVRD_VAL	R/W	**Description**: CTLE Pole override value to load at start of adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
7	FAST_AFE_DFE_SETTLE	R/W	**Description**: Enable fast AFE and DFE settling time (simulation only) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6:0	N_WAIT_ASM1	R/W	**Description**: Number of wait cycles for Adaptation SM #1 **Value After Reset**: 0x9 **Exists**: Always **Reset Mask**: 0x7f

11.2.171 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2

- **Description:** Adaptation Configuration Register #2
- **Size:** 16 bits
- **Offset:** 0x1N62
- **Exists:** Always

Table 11-175 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	TGG_PTTRN_1	R/W	**Description**: Pattern for the second toggle loop Error slicer is moved upward by Data tap1 if this pattern is matched **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x1f
4:0	TGG_PTTRN_0	R/W	**Description**: Pattern for the first toggle loop Error slicer is moved downward by Data tap1 if this pattern is matched **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x1f

11.2.172 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3

- **Description:** Adaptation Configuration Register #3
- **Size:** 16 bits
- **Offset:** 0x1N63
- **Exists:** Always

Table 11-176 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3

Bits	Name	Memory Access	Description
15	ESL_TWICE_DSL	R/W	**Description**: Assert if error slicer has twice the voltage range as the data slicer (for the same 8 bits). **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	TGG_EN	R/W	**Description**: Enable toggling of the error slicer **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	EYEHO_EN	R/W	**Description**: Enable eye height measurement using odd error slicer **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	EYEHE_EN	R/W	**Description**: Enable eye height measurement using even error slicer **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:7	DFE_EN	R/W	**Description**: Enable DFE adaptation for taps 5-1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
6	ATT_EN	R/W	**Description**: Enable ATT adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-176 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3 (Continued)

Bits	Name	Memory Access	Description
5	VGA_EN	R/W	**Description**: Enable VGA adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4:0	CTLE_EN	R/W	**Description**: Enable CTLE boost adaptation The five bits determine which correlators are used to adapt the CTLE **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

11.2.173 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4

- **Description:** Adaptation Configuration Register #4
- **Size:** 16 bits
- **Offset:** 0x1N64
- **Exists:** Always

Table 11-177 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4

Bits	Name	Memory Access	Description
15:12	DFE2_TH	R/W	**Description**: DFE Tap2 correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
11:8	DFE1_TH	R/W	**Description**: DFE Tap1 correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
7:4	VGA_TH	R/W	**Description**: VGA correlation decision threshold (2^N-1) During eye height measurement, the VGA_TH is repurposed for error slicer updates. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
3:0	CTLE_TH	R/W	**Description**: CTLE correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.174 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5

- **Description:** Adaptation Configuration Register #5
- **Size:** 16 bits
- **Offset:** 0x1N65
- **Exists:** Always

Table 11-178 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5

Bits	Name	Memory Access	Description
15:12	TH_OFFSET	R/W	**Description**: Apply an offset to the decision threshold **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
11:8	DFE5_TH	R/W	**Description**: DFE Tap5 correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
7:4	DFE4_TH	R/W	**Description**: DFE Tap4 correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
3:0	DFE3_TH	R/W	**Description**: DFE Tap3 correlation decision threshold (2^N-1) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.175 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6

- **Description:** Adaptation Configuration Register #6
- **Size:** 16 bits
- **Offset:** 0x1N66
- **Exists:** Always

Table 11-179 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6

Bits	Name	Memory Access	Description
15:13	ATT_LOW_TH	R/W	**Description**: ATT low threshold **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7
12	VGA_SAT_CNT_STICKY	R/W	**Description**: If de-asserted, then VGA saturation counts must be consecutive to change ATT **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
11:9	VGA_SAT_CNT	R/W	**Description**: VGA saturation count **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
8:6	ATT_MU	R/W	**Description**: ATT gain code update gain (2^N) **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
5:3	VGA_MU	R/W	**Description**: VGA gain code update gain (2^N) During eye height measurement, the VGA_MU is repurposed for error slicer updates. **Value After Reset**: 0x5 **Exists**: Always **Reset Mask**: 0x7
2:0	CTLE_MU	R/W	**Description**: CTLE Boost code update gain (2^N) **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7

11.2.176 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7

- **Description:** Adaptation Configuration Register #7
- **Size:** 16 bits
- **Offset:** 0x1N67
- **Exists:** Always

Table 11-180 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:8	VGA_LEV_LOW	R/W	**Description**: VGA level low saturation limit **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0xf
7:4	VGA_LEV_HIGH	R/W	**Description**: VGA level high saturation limit **Value After Reset**: 0xa **Exists**: Always **Reset Mask**: 0xf
3:0	VGA_MIN_SAT	R/W	**Description**: VGA minimum saturation limit **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0xf

11.2.177 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8

- **Description:** Adaptation Configuration Register #8
- **Size:** 16 bits
- **Offset:** 0x1N68
- **Exists:** Always

Table 11-181 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:12	DFE5_MU	R/W	**Description**: DFE tap5 code update gain (2^N) **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
11:9	DFE4_MU	R/W	**Description**: DFE tap4 code update gain (2^N) **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
8:6	DFE3_MU	R/W	**Description**: DFE tap3 code update gain (2^N) **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
5:3	DFE2_MU	R/W	**Description**: DFE tap2 code update gain (2^N) **Value After Reset**: 0x4 **Exists**: Always **Reset Mask**: 0x7
2:0	DFE1_MU	R/W	**Description**: DFE tap1 code update gain (2^N) **Value After Reset**: 0x5 **Exists**: Always **Reset Mask**: 0x7

11.2.178 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9

- **Description:** Adaptation Configuration Register #9
- **Size:** 16 bits
- **Offset:** 0x1N69
- **Exists:** Always

Table 11-182 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9

Bits	Name	Memory Access	Description
15:8	ERR_SLO_ADPT_INIT	R/W	**Description**: The error odd slicer is initialized to this value at the start of a new adaptation request. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
7:0	ERR_SLE_ADPT_INIT	R/W	**Description**: The error even slicer is initialized to this value at the start of a new adaptation request. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.179 LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG

- **Description:** Reset Adaptation Configuration Register
- **Size:** 16 bits
- **Offset:** 0x1N6a
- **Exists:** Always

Table 11-183 Fields for Register: LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4	RST_ADPT_TAP1	R/W	**Description**: Reset Data Tap1 when turning off DFE adaptation (Taps 2-5 are always turned off when DFE adaptation is turned off) **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
3	RST_ADPT_CTLE_POLE	R/W	**Description**: Reset CTLE Pole when turning off AFE adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	RST_ADPT_CTLE_BOOST	R/W	**Description**: Reset CTLE Boost when turning off AFE adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RST_ADPT_VGA	R/W	**Description**: Reset VGA when turning off AFE adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RST_ADPT_ATT	R/W	**Description**: Reset ATT when turning off AFE adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.180 LANEN_DIG_RX_ADPTCTL_ATT_STATUS

- **Description:** Value of ATT Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N6b
- **Exists:** Always

Table 11-184 Fields for Register: LANEN_DIG_RX_ADPTCTL_ATT_STATUS

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	ASM1_DON	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:0	ATT_ADPT_CODE	R	**Description**: Value of ATT adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.181 LANEN_DIG_RX_ADPTCTL_VGA_STATUS

- **Description:** Value of VGA Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N6c
- **Exists:** Always

Table 11-185 Fields for Register: LANEN_DIG_RX_ADPTCTL_VGA_STATUS

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8:0	VGA_ADPT_CODE	R	**Description**: Value of VGA adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.182 LANEN_DIG_RX_ADPTCTL_CTLE_STATUS

- **Description:** Value of CTLE Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N6d
- **Exists:** Always

Table 11-186 Fields for Register: LANEN_DIG_RX_ADPTCTL_CTLE_STATUS

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:10	CTLE_POLE_ADPT_CODE	R	**Description**: Value of CTLE Pole adaptation code (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9:0	CTLE_BOOST_ADPT_CODE	R	**Description**: Value of CTLE Boost adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.183 LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS

- **Description:** Value of DFE Tap1 Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N6e
- **Exists:** Always

Table 11-187 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12:0	DFE_TAP1_ADPT_CODE	R	**Description**: Value of DFE tap1 adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.184 LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS

- **Description:** Value of DFE Tap2 Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N6f
- **Exists:** Always

Table 11-188 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:0	DFE_TAP2_ADPT_CODE	R	**Description**: Value of DFE tap2 adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.185 LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS

- **Description:** Value of DFE Tap3 Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N70
- **Exists:** Always

Table 11-189 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:0	DFE_TAP3_ADPT_CODE	R	**Description**: Value of DFE tap3 adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.186 LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS

- **Description:** Value of DFE Tap4 Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N71
- **Exists:** Always

Table 11-190 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:0	DFE_TAP4_ADPT_CODE	R	**Description**: Value of DFE tap4 adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.187 LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS

- **Description:** Value of DFE Tap5 Adaptation code
- **Size:** 16 bits
- **Offset:** 0x1N72
- **Exists:** Always

Table 11-191 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	ASM1_DONE	R	**Description**: Asserts when Adaptation state machine #1 is done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11:0	DFE_TAP5_ADPT_CODE	R	**Description**: Value of DFE tap5 adaptation code The lower 5 bits are used to smooth out value transition (dither bits). (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.188 LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_HIGH_VDAC_OFST

- **Description:** Offset values for RX DFE Data Even High vDAC
- **Size:** 16 bits
- **Offset:** 0x1N73
- **Exists:** Always

Table 11-192 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_HIGH_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_EVEN_HIGH_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Even High vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.189 LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Data Even Low vDAC
- **Size:** 16 bits
- **Offset:** 0x1N74
- **Exists:** Always

Table 11-193 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA_EVEN_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_EVEN_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Even Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.190 LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_HIGH_VDAC_OFST

- **Description:** Offset values for RX DFE Data Odd High vDAC
- **Size:** 16 bits
- **Offset:** 0x1N75
- **Exists:** Always

Table 11-194 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_HIGH_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_ODD_HIGH_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Odd High vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.191 LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Data Odd Low vDAC
- **Size:** 16 bits
- **Offset:** 0x1N76
- **Exists:** Always

Table 11-195 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA_ODD_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_ODD_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Odd Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.192 LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_EVEN

- **Description:** Sets values for RX SLICER CTRL EVEN signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1N77
- **Exists:** Always

Table 11-196 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_EVEN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3:0	RX_ANA_SLICER_CTRL_E	R/W	**Description**: Value for rx_ana_slicer_ctrl_e[3:0] **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0xf

11.2.193 LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_ODD

- **Description:** Sets values for RX SLICER CTRL ODD signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1N78
- **Exists:** Always

Table 11-197 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_ODD

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3:0	RX_ANA_SLICER_CTRL_O	R/W	**Description**: Value for rx_ana_slicer_ctrl_o[3:0] **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0xf

11.2.194 LANEN_DIG_RX_ADPTCTL_DFE_ERROR_EVEN_VDAC_OFST

- **Description:** Offset values for RX DFE Error Even vDAC
- **Size:** 16 bits
- **Offset:** 0x1N79
- **Exists:** Always

Table 11-198 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_ERROR_EVEN_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_ERROR_EVEN_VDAC_OFST	R/W	**Description**: Offset value for DFE Error Even vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.195 LANEN_DIG_RX_ADPTCTL_DFE_ERROR_ODD_VDAC_OFST

- **Description:** Offset values for RX DFE Error Odd vDAC
- **Size:** 16 bits
- **Offset:** 0x1N7a
- **Exists:** Always

Table 11-199 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_ERROR_ODD_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_ERROR_ODD_VDAC_OFST	R/W	**Description**: Offset value for DFE Error Odd vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.196 LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_LEVEL

- **Description:** Value of Error Slicer Level
- **Size:** 16 bits
- **Offset:** 0x1N7b
- **Exists:** Always

Table 11-200 Fields for Register: LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_LEVEL

Bits	Name	Memory Access	Description
15:8	E_SLE_LVL	R	**Description**: Even Error Slicer Level (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7:0	E_SLO_LVL	R	**Description**: Odd Error Slicer Level (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.197 LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST

- **Description:** Offset values for RX DFE By-Pass Even vDAC
- **Size:** 16 bits
- **Offset:** 0x1N7c
- **Exists:** Always

Table 11-201 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_BYPASS_EVEN_VDAC_OFST	R/W	**Description**: Offset value for DFE By-Pass Even vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.198 LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST

- **Description:** Offset values for RX DFE By-Pass Odd vDAC
- **Size:** 16 bits
- **Offset:** 0x1N7d
- **Exists:** Always

Table 11-202 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_BYPASS_ODD_VDAC_OFST	R/W	**Description**: Offset value for DFE By-Pass Odd vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.199 LANEN_DIG_RX_ADPTCTL_ADPT_RESET

- **Description:** Adaptation reset register
- **Size:** 16 bits
- **Offset:** 0x1N7e
- **Exists:** Always

Table 11-203 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_RESET

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RESET_ASM1	R/W	**Description**: Resets adaptation state machine (ASM1) as well as the stats capture block. This is a self-clearing bit and requires restart of ASM1. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.200 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10

- **Description:** Adaptation Configuration Register #10
- **Size:** 16 bits
- **Offset:** 0x1N7f
- **Exists:** Always

Table 11-204 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2:0	CTLE_T1_WT	R/W	**Description**: When calculating CTLE correlation, tap1 correlation will be right shifted by CTLE_T1_WT before being used. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.201 LANEN_DIG_RX_STAT_LD_VAL_1

- **Description:** Stat load value for the sample counter #1
- **Size:** 16 bits
- **Offset:** 0x1N80
- **Exists:** Always

Table 11-205 Fields for Register: LANEN_DIG_RX_STAT_LD_VAL_1

Bits	Name	Memory Access	Description
15	SC1_START	R/W	**Description**: Start sample counter #1 This is a self-clearing bit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:0	SC1_LD_VAL	R/W	**Description**: Sample counter #1 load value **Value After Reset**: 0x40 **Exists**: Always **Reset Mask**: 0x7fff

11.2.202 LANEN_DIG_RX_STAT_DATA_MSK

- **Description:** Stat data mask bits [15:0]
- **Size:** 16 bits
- **Offset:** 0x1N81
- **Exists:** Always

Table 11-206 Fields for Register: LANEN_DIG_RX_STAT_DATA_MSK

Bits	Name	Memory Access	Description
15:0	DATA_MSK_15_0	R/W	**Description**: Value of data_msk_r[15:0] **Value After Reset**: 0xffff **Exists**: Always **Reset Mask**: 0xffff

11.2.203 LANEN_DIG_RX_STAT_MATCH_CTL0

- **Description:** Stat match controls register #0
- **Size:** 16 bits
- **Offset:** 0x1N82
- **Exists:** Always

Table 11-207 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL0

Bits	Name	Memory Access	Description
15:14	SCOPE_DLY	R/W	**Description**: Number of clock cycle delays on scope_data_rx_clk An additional MSB is added in SCOPE_DLY_2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
13:10	DATA_MSK_19_16	R/W	**Description**: Value of data_msk_r[19:16] **Value After Reset**: 0xf **Exists**: Always **Reset Mask**: 0xf
9:5	PTTRN_CR1A_4_0	R/W	**Description**: Value of pattern A for 1st correlator (bits 4:0) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
4:0	PTTRN_MSK_CR1A_4_0	R/W	**Description**: Value of pattern A mask for 1st correlator (bits 4:0) **Value After Reset**: 0x6 **Exists**: Always **Reset Mask**: 0x1f

11.2.204 LANEN_DIG_RX_STAT_MATCH_CTL1

- **Description:** Stat match controls register #1
- **Size:** 16 bits
- **Offset:** 0x1N83
- **Exists:** Always

Table 11-208 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	PTTRN_CR1A_ADPT_EN	R/W	**Description**: Enable ORing of adaptation pattern with pattern CR1A **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
10:6	PTTRN_CR1B_4_0	R/W	**Description**: Value of pattern B for 1st correlator (bits 4:0) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
5:1	PTTRN_MSK_CR1B_4_0	R/W	**Description**: Value of pattern B mask for 1st correlator (bits 4:0) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
0	PTTRN_CR1B_EN	R/W	**Description**: Enable pattern B matching for 1st correlator **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.205 LANEN_DIG_RX_STAT_STAT_CTL0

- **Description:** Stat controls register #0
- **Size:** 16 bits
- **Offset:** 0x1N84
- **Exists:** Always

Table 11-209 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0

Bits	Name	Memory Access	Description
15	SKIP_EN	R/W	**Description**: Value of skip_en_r **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	SC_TIMER_MODE	R/W	**Description**: Sample counter operation mode 0x0 - counts number of matched samples 0x1 - counts clock cycles (i.e. a timer) **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
13	STAT_RXCLK_SEL	R/W	**Description**: Select stat clock 0x0 - ref_range_clk 0x1 - rx_dig_clk (i.e. rx dword clk) Before changing stat_rxclk_sel_r from 1->0, the rx_dig_clk must be active (i.e. enabled) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:10	STAT_SRC_SEL	R/W	**Description**: Select stat source input 0x0 - {20{rx_cal_result}} 0x1 - {{20{scope_data_rxclk_dly_s01}},{20{scope_data_rxclk_dly}}} 0x2 - rx_phase[39:0] 0x3 - rx_error[39:0] 0x4 - rx_data[39:0] 0x5 - rx_phdir[39:0] 0x6 - 40'hFF_FFFF_FFFF **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

Table 11-209 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0 (Continued)

Bits	Name	Memory Access	Description
9:6	STAT_SHFT_SEL	R/W	**Description**: Select stat source shift value 0x0 - Correlate N-1 -> N+3 (use N for offset calibration) 0x1 - Correlate N+1 -> N+5 (for taps1-5) 0x2 - Correlate N+6 -> N+10 0x3 - Correlate N+11 -> N+15 0x4 - Correlate N+16 -> N+20 0x5 - Correlate N+21 -> N+25 0x6 - Correlate N+26 -> N+30 0x7 - Correlate N+31 -> N+35 0x8 - Correlate N+36 -> N+39 Setting 0x8 is only used in 20b mode (for checking corr on bits 36-39) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
5	CORR_MODE_EN	R/W	**Description**: Enable correlation mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4:3	CORR_SRC_SEL	R/W	**Description**: Select correlation input source 0x0 - rx_error[39:0] 0x1 - rx_phase[39:0] 0x2 - {{20{scope_data_rxclk_dly_s01}},{20{scope_data_rxclk_dly}}} 0x3 - No correlation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
2	CORR_SHFT_SEL	R/W	**Description**: Select shift for phase. 0:none, 1:>>1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	CORR_SHFT_SEL_VGA	R/W	**Description**: Select shift for error going to VGA. 0:none, 1:>>1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-209 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0 (Continued)

Bits	Name	Memory Access	Description
0	RESERVED_0	R/W	**Description**: Reserved bit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.206 LANEN_DIG_RX_STAT_STAT_CTL1

- **Description:** Stat controls register #1
- **Size:** 16 bits
- **Offset:** 0x1N85
- **Exists:** Always

Table 11-210 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL1

Bits	Name	Memory Access	Description
15:14	VLD_CTL	R/W	**Description**: Gating configuration of stats collection 0x0 - ignore both cdr_valid and rx_valid 0x1 - gate stats collection with cdr_valid 0x2 - gate stats collection with rx_valid 0x3 - ignore both cdr_valid and rx_valid **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
13	VLD_LOSS_CLR	R/W	**Description**: Clearing of stats collection upon loss of valid 0x0 - hold sample and stat counters 0x1 - clear sample and stat counters **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:11	DATA_DLY_SEL	R/W	**Description**: # of clock cycle delays on rx_data[19:0] An additional MSB is added in DATA_DLY_SEL_2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
10	STAT_CLK_EN	R/W	**Description**: Clock gate enable for stat clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	SC_PAUSE	R/W	**Description**: Pause the sample counter and stat counters **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:7	RESERVED_8_7	R/W	**Description**: Reserved bits **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-210 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL1 (Continued)

Bits	Name	Memory Access	Description
6	STAT_CNT_6_EN	R/W	**Description**: Enable for stat counter 6 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	STAT_CNT_5_EN	R/W	**Description**: Enable for stat counter 5 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	STAT_CNT_4_EN	R/W	**Description**: Enable for stat counter 4 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	STAT_CNT_3_EN	R/W	**Description**: Enable for stat counter 3 Only counter to be enabled by default, since used for offset calibration **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	STAT_CNT_2_EN	R/W	**Description**: Enable for stat counter 2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	STAT_CNT_1_EN	R/W	**Description**: Enable for stat counter 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	STAT_CNT_0_EN	R/W	**Description**: Enable for stat counter 0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.207 LANEN_DIG_RX_STAT_SMPL_CNT1

- **Description:** Sample Counter #1 Status
- **Size:** 16 bits
- **Offset:** 0x1N86
- **Exists:** Always

Table 11-211 Fields for Register: LANEN_DIG_RX_STAT_SMPL_CNT1

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	SMPL_CNT1	R	**Description**: Current value of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.208 LANEN_DIG_RX_STAT_STAT_CNT_0

- **Description:** Stat Counter 0 Status
- **Size:** 16 bits
- **Offset:** 0x1N87
- **Exists:** Always

Table 11-212 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_0

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_0	R	**Description**: Current value of stat counter #0 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.209 LANEN_DIG_RX_STAT_STAT_CNT_1

- **Description:** Stat Counter 1 Status
- **Size:** 16 bits
- **Offset:** 0x1N88
- **Exists:** Always

Table 11-213 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_1

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_1	R	**Description**: Current value of stat counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.210 LANEN_DIG_RX_STAT_STAT_CNT_2

- **Description:** Stat Counter 2 Status
- **Size:** 16 bits
- **Offset:** 0x1N89
- **Exists:** Always

Table 11-214 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_2

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_2	R	**Description**: Current value of stat counter #2 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.211 LANEN_DIG_RX_STAT_STAT_CNT_3

- **Description:** Stat Counter 3 Status
- **Size:** 16 bits
- **Offset:** 0x1N8a
- **Exists:** Always

Table 11-215 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_3

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_3	R	**Description**: Current value of stat counter #3 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.212 LANEN_DIG_RX_STAT_STAT_CNT_4

- **Description:** Stat Counter 4 Status
- **Size:** 16 bits
- **Offset:** 0x1N8b
- **Exists:** Always

Table 11-216 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_4

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_4	R	**Description**: Current value of stat counter #4 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.213 LANEN_DIG_RX_STAT_STAT_CNT_5

- **Description:** Stat Counter 5 Status
- **Size:** 16 bits
- **Offset:** 0x1N8c
- **Exists:** Always

Table 11-217 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_5

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_5	R	**Description**: Current value of stat counter #5 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.214 LANEN_DIG_RX_STAT_STAT_CNT_6

- **Description:** Stat Counter 6 Status
- **Size:** 16 bits
- **Offset:** 0x1N8d
- **Exists:** Always

Table 11-218 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_6

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	**Description**: Status of sample counter #1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
14:0	STAT_CNT_6	R	**Description**: Current value of stat counter #6 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.215 LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL

- **Description:** Calibration Comparator Control
- **Size:** 16 bits
- **Offset:** 0x1N8e
- **Exists:** Always

Table 11-219 Fields for Register: LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:3	REF_DIV_CNT	R/W	**Description**: Ref range clock count (e.g. 5'd3 = 4 ref_range cycles) **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x7
2:0	PRECHRGE_CNT	R/W	**Description**: Precharge Count (e.g. 5'd1 = 2 ref_range cycles) **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x7

11.2.216 LANEN_DIG_RX_STAT_MATCH_CTL2

- **Description:** Stat match controls register #2
- **Size:** 16 bits
- **Offset:** 0x1N8f
- **Exists:** Always

Table 11-220 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:0	PTTRN_CR1A_19_5	R/W	**Description**: Value of pattern A for 1st correlator (bits 19:5) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7fff

11.2.217 LANEN_DIG_RX_STAT_MATCH_CTL3

- **Description:** Stat match controls register #3
- **Size:** 16 bits
- **Offset:** 0x1N90
- **Exists:** Always

Table 11-221 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL3

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:0	PTTRN_MSK_CR1A_19_5	R/W	**Description**: Value of pattern A mask for 1st correlator (bits 19:5) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7fff

11.2.218 LANEN_DIG_RX_STAT_MATCH_CTL4

- **Description:** Stat match controls register #4
- **Size:** 16 bits
- **Offset:** 0x1N91
- **Exists:** Always

Table 11-222 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL4

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:0	PTTRN_CR1B_19_5	R/W	**Description**: Value of pattern B for 1st correlator (bits 19:5) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7fff

11.2.219 LANEN_DIG_RX_STAT_MATCH_CTL5

- **Description:** Stat match controls register #5
- **Size:** 16 bits
- **Offset:** 0x1N92
- **Exists:** Always

Table 11-223 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL5

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:0	PTTRN_MSK_CR1B_19_5	R/W	**Description**: Value of pattern B mask for 1st correlator (bits 19:5) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7fff

11.2.220 LANEN_DIG_RX_STAT_STAT_CTL2

- **Description:** Stat controls register #2
- **Size:** 16 bits
- **Offset:** 0x1N93
- **Exists:** Always

Table 11-224 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL2

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	SCOPE_DLY_2	R/W	**Description**: Additional MSB bit for SCOPE_DLY to extend the delay range to 0->7 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	DATA_DLY_SEL_2	R/W	**Description**: Additional MSB bit for DATA_DLY_SEL to extend the delay range to 0->7 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.221 LANEN_DIG_RX_STAT_STAT_STOP

- **Description:** Stat stop register
- **Size:** 16 bits
- **Offset:** 0x1N94
- **Exists:** Always

Table 11-225 Fields for Register: LANEN_DIG_RX_STAT_STAT_STOP

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SC1_STOP	R/W	**Description**: Stop sample counters #1 and associated stat counters. This is a self-clearing bit, and requires restart of sample counter #1. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.222 LANEN_DIG_ANA_TX_OVRD_OUT

- **Description:** Override values for TX signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Na0
- **Exists:** Always

Table 11-226 Fields for Register: LANEN_DIG_ANA_TX_OVRD_OUT

Bits	Name	Memory Access	Description
15	TX_OVRD_EN	R/W	**Description**: Override enable for all outputs controlled by this register Output signals overridden by this signal are : tx_ana_rxdet_en tx_ana_div4_en tx_ana_data_rate tx_ana_serial_en tx_ana_reset tx_ana_mpllb_clk_en tx_ana_mplla_clk_en tx_ana_word_clk_en tx_ana_clk_en tx_ana_vcm_hold tx_ana_refgen_en tx_ana_data_en tx_ana_clk_shift **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	TX_RXDET_EN	R/W	**Description**: Override value for tx_ana_rxdet_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	TX_DIV4_EN	R/W	**Description**: Override value for tx_ana_div4_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:10	TX_ANA_DATA_RATE	R/W	**Description**: Override value for tx_ana_data_rate. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

Table 11-226 Fields for Register: LANEN_DIG_ANA_TX_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
9	TX_ANA_SERIAL_EN	R/W	**Description**: Override value for tx_ana_serial_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	TX_ANA_RESET	R/W	**Description**: Override value for tx_ana_reset. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	TX_ANA_MPLLB_CLK_EN	R/W	**Description**: Override value for tx_ana_mpllb_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	TX_ANA_MPLLA_CLK_EN	R/W	**Description**: Override value for tx_ana_mplla_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	TX_ANA_WORD_CLK_EN	R/W	**Description**: Override value for tx_ana_word_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	TX_ANA_CLK_EN	R/W	**Description**: Override value for tx_ana_clk_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	TX_ANA_VCM_HOLD	R/W	**Description**: Override value for tx_ana_vcm_hold. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_ANA_REFGEN_EN	R/W	**Description**: Override value for tx_ana_refgen_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TX_ANA_DATA_EN	R/W	**Description**: Override value for tx_ana_data_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-226 Fields for Register: LANEN_DIG_ANA_TX_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
0	TX_ANA_CLK_SHIFT	R/W	**Description**: Override value for tx_ana_clk_shift. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.223 LANEN_DIG_ANA_TX_TERM_UP_CODE_OVRD_OUT

- **Description:** Override value for TX termination code up going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Na1
- **Exists:** Always

Table 11-227 Fields for Register: LANEN_DIG_ANA_TX_TERM_UP_CODE_OVRD_OUT

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	TX_ANA_VCM_HOLD_GS	R/W	**Description**: Override value for tx_ana_vcm_hold_gs Its override enable is TX_OVRD_OUT.TX_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	TX_CLK_LB_EN	R/W	**Description**: Override value for tx_ana_clk_lb_en Its override enable is TX_OVRD_OUT.TX_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	TX_TERM_UP_OVRD_EN	R/W	**Description**: Overrides enable for tx_ana_term_code_up[9:0] output signals overridden by this enable are: tx_ana_term_code_up **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:0	TX_TERM_UP_CODEP	R/W	**Description**: Override value for tx_ana_term_code_up[9:0] Its override enable is TX_TERM_UP_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.224 LANEN_DIG_ANA_TX_TERM_DN_CODE_OVRD_OUT

- **Description:** Override value for TX termination down code going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Na2
- **Exists:** Always

Table 11-228 Fields for Register: LANEN_DIG_ANA_TX_TERM_DN_CODE_OVRD_OUT

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	TX_TERM_DN_OVRD_EN	R/W	**Description**: Override enable for tx_ana_term_code_dn[9:0] signal **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:0	TX_TERM_DN_CODE	R/W	**Description**: Override value for tx_ana_term_code_dn[9:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.225 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0

- **Description:** Override values for TX EQ signals going to ANA register #0
- **Size:** 16 bits
- **Offset:** 0x1Na3
- **Exists:** Always

Table 11-229 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0

Bits	Name	Memory Access	Description
15	TX_EQ_OVRD_EN	R/W	**Description**: Override enable for tx eq signals output signals overridden by this enable are: tx_ana_load_clk tx_ana_ctrl_atten tx_ana_ctrl_half_up tx_ana_ctrl_half_dn tx_ana_ctrl_pre tx_ana_ctrl_pre_half tx_ana_ctrl_pre_qtr tx_ana_ctrl_post tx_ana_ctrl_post_half tx_ana_ctrl_post_qtr **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:1	TX_ANA_CTRL_ATTEN_13_0	R/W	**Description**: Override value for tx_ana_ctrl_atten[13:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3fff
0	TX_ANA_LOAD_CLK	R/W	**Description**: Override value for tx_ana_load_clk **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.226 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1

- **Description:** Override values for TX EQ signals going to ANA register #1
- **Size:** 16 bits
- **Offset:** 0x1Na4
- **Exists:** Always

Table 11-230 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1

Bits	Name	Memory Access	Description
15:0	TX_ANA_CTRL_ATTEN_29_14	R/W	**Description**: Override value for tx_ana_ctrl_atten[29:14] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.227 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2

- **Description:** Override values for TX EQ signals going to ANA register #2
- **Size:** 16 bits
- **Offset:** 0x1Na5
- **Exists:** Always

Table 11-231 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2

Bits	Name	Memory Access	Description
15:7	TX_ANA_CTRL_PRE	R/W	**Description**: Override value for tx_ana_ctrl_pre[8:0] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff
6	TX_ANA_CTRL_HALF_DN	R/W	**Description**: Override value for tx_ana_ctrl_half_dn Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	TX_ANA_CTRL_HALF_UP	R/W	**Description**: Override value for tx_ana_ctrl_half_up Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4:0	TX_ANA_CTRL_ATTEN_34_30	R/W	**Description**: Override value for tx_ana_ctrl_atten[34:30] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

11.2.228 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3

- **Description:** Override values for TX EQ signals going to ANA register #3
- **Size:** 16 bits
- **Offset:** 0x1Na6
- **Exists:** Always

Table 11-232 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:4	TX_ANA_CTRL_POST_QTR	R/W	**Description**: Override value for tx_ana_ctrl_post_qtr[1:0] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
3	TX_ANA_CTRL_POST_HALF	R/W	**Description**: Override value for tx_ana_ctrl_post_half Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2:1	TX_ANA_CTRL_PRE_QTR	R/W	**Description**: Override value for tx_ana_ctrl_pre_qtr[1:0] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
0	TX_ANA_CTRL_PRE_HALF	R/W	**Description**: Override value for tx_ana_ctrl_pre_half Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.229 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_4

- **Description:** Override values for TX EQ signals going to ANA register #4
- **Size:** 16 bits
- **Offset:** 0x1Na7
- **Exists:** Always

Table 11-233 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_4

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:0	TX_ANA_CTRL_POST	R/W	**Description**: Override value for tx_ana_ctrl_post[13:0] Its override enable is TX_EQ_OVRD_OUT_0.TX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3fff

11.2.230 LANEN_DIG_ANA_RX_CTL_OVRD_OUT

- **Description:** Override values for RX control signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Na8
- **Exists:** Always

Table 11-234 Fields for Register: LANEN_DIG_ANA_RX_CTL_OVRD_OUT

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	RX_CTL_OVRD_EN	R/W	**Description**: override Enable for outputs [7-0] below output signals overridden by this enable are: rx_ana_div16p5_clk_en rx_ana_data_rate rx_ana_word_clk_en rx_ana_div4_en rx_ana_dfe_taps_en rx_ana_dfe_bypass_n rx_ana_loopback_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	RX_LBK_CLK_EN	R/W	**Description**: Override value for rx_ana_loopback_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_ANA_DFE_BYPASS_N	R/W	**Description**: Override value for rx_ana_dfe_bypass_n **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_ANA_DFE_TAPS_EN	R/W	**Description**: Override value for rx_ana_dfe_taps_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-234 Fields for Register: LANEN_DIG_ANA_RX_CTL_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
4	RX_ANA_DIV4_EN	R/W	**Description**: Override value for rx_ana_div4_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RX_ANA_WORD_CLK_EN	R/W	**Description**: Override value for rx_ana_word_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2:1	RX_ANA_DATA_RATE	R/W	**Description**: Override value for rx_ana_data_rate **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
0	RX_ANA_DIV16P5_CLK_EN	R/W	**Description**: Override value for rx_ana_div16p5_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.231 LANEN_DIG_ANA_RX_PWR_OVRD_OUT

- **Description:** Override values for RX PWR UP/DN signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Na9
- **Exists:** Always

Table 11-235 Fields for Register: LANEN_DIG_ANA_RX_PWR_OVRD_OUT

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	RX_PWR_OVRD_EN	R/W	**Description**: Override enable for all outputs controlled by this register output signals overridden by this enable are: rx_ana_deserial_en rx_ana_cdr_en rx_ana_clk_en rx_ana_clk_dcc_en rx_ana_clk_vreg_en rx_ana_afe_en rx_ana_los_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_ANA_DESERIAL_EN	R/W	**Description**: Override value for rx_ana_deserial_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_ANA_CDR_EN	R/W	**Description**: Override value for rx_ana_cdr_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_ANA_CLK_EN	R/W	**Description**: Override value for rx_ana_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-235 Fields for Register: LANEN_DIG_ANA_RX_PWR_OVRD_OUT (Continued)

Bits	Name	Memory Access	Description
3	RX_ANA_CLK_DCC_EN	R/W	**Description**: Override value for rx_ana_clk_dcc_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_ANA_CLK_VREG_EN	R/W	**Description**: Override value for rx_ana_clk_vreg_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_ANA_AFE_EN	R/W	**Description**: Override value for rx_ana_afe_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_LOS_EN	R/W	**Description**: Override value for rx_ana_los_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.232 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0

- **Description:** Override values for RX VCO signals going to ANA #0
- **Size:** 16 bits
- **Offset:** 0x1Naa
- **Exists:** Always

Table 11-236 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0

Bits	Name	Memory Access	Description
15	RX_CDR_FREQ_TUNE_OVRD_EN	R/W	**Description**: override Enable for rx_ana_cdr_freq_tune output signals overridden by this enable are: rx_ana_cdr_freq_tune rx_ana_cdr_freq_tune_clk **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14	RX_ANA_VCO_CNTR_CLK	R/W	**Description**: Override value for rx_ana_vco_cntr_clk Its override enable is RX_VCO_OVRD_OUT_1.RX_ANA_VCO_CNTR_OVER_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	RX_ANA_VCO_CNTR_EN	R/W	**Description**: Override value for rx_ana_vco_cntr_en Its override enable is RX_VCO_OVRD_OUT_1.RX_ANA_VCO_CNTR_OVER_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:3	RX_ANA_CDR_FREQ_TUNE	R/W	**Description**: Override value for rx_ana_cdr_freq_tune Its override enable is RX_CDR_FREQ_TUNE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

Table 11-236 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0 (Continued)

Bits	Name	Memory Access	Description
2	RX_VCO_CDR_OVRD_EN	R/W	**Description**: Override enable for cdr_vco_en and cdr_startup output signals overridden by this enable are: rx_ana_cdr_vco_en rx_ana_cdr_startup **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_ANA_CDR_STARTUP	R/W	**Description**: Override value for rx_ana_cdr_startup **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_CDR_VCO_EN	R/W	**Description**: Override value for rx_ana_cdr_vco_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.233 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1

- **Description:** Override values for RX VCO signals going to ANA #1
- **Size:** 16 bits
- **Offset:** 0x1Nab
- **Exists:** Always

Table 11-237 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	RX_ANA_CDR_VCO_STEP_CTRL_OVRD_EN	R/W	**Description**: Override enable for rx_ana_cdr_vco_step_ctrl **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	RX_ANA_CDR_VCO_STEP_CTRL	R/W	**Description**: Override value for rx_ana_cdr_vco_step_ctrl **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	RX_ANA_CDR_VCO_TEMP_COMP_OVRD_EN	R/W	**Description**: Override enable for rx_ana_cdr_vco_temp_comp_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_ANA_CDR_VCO_TEMP_COMP_EN	R/W	**Description**: Override value for rx_ana_cdr_vco_temp_comp **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_ANA_VCO_CNTR_OVER_EN	R/W	**Description**: Override enable for rx_ana_vco_cntr_pd output signals overridden by this enable are: rx_ana_vco_cntr_pd rx_ana_vco_cntr_clk rx_ana_vco_cntr_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-237 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1 (Continued)

Bits	Name	Memory Access	Description
3	RX_ANA_VCO_CNTR_PD	R/W	**Description**: Override value for rx_ana_vco_cntr_pd **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
2	RX_ANA_CDR_VCO_FREQBAND_OVRD_EN	R/W	**Description**: Override enable for rx_ana_cdr_vco_freqband **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	RX_ANA_CDR_VCO_FREQBAND	R/W	**Description**: Override value for rx_ana_cdr_vco_freqband **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.234 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2

- **Description:** Override values for RX VCO signals going to ANA #2
- **Size:** 16 bits
- **Offset:** 0x1Nac
- **Exists:** Always

Table 11-238 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ANA_CDR_FREQ_TUNE_CLK	R/W	**Description**: Override value for rx_ana_cdr_freq_tune_clk Its override enable is RX_VCO_OVRD_OUT_0.RX_CDR_FREQ_TUNE_OVRD_EN Self-clearing to generate a pulse 1 cr_clk wide **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.235 LANEN_DIG_ANA_RX_CAL

- **Description:** Sets values for RX CAL signals going to ANA register
- **Size:** 16 bits
- **Offset:** 0x1Nad
- **Exists:** Always

Table 11-239 Fields for Register: LANEN_DIG_ANA_RX_CAL

Bits	Name	Memory Access	Description
15	RX_ANA_CAL_COMP_EN	R/W	**Description**: Value for rx_ana_cal_comp_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:13	RX_ANA_CAL_MODE	R/W	**Description**: Value for rx_ana_cal_mode[1:0] 00 - Differential comparison 01 - Single-ended, positive node to positive node 10 - Single-ended, negative node to negative node 11 - Common mode comparison **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
12	RX_ANA_SLICER_CAL_EN	R/W	**Description**: Value for rx_ana_slicer_cal_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	RX_ANA_CAL_SHORT_EN	R/W	**Description**: Value for rx_ana_cal_short_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_ANA_CAL_LPFBYP_EN	R/W	**Description**: Value for rx_ana_cal_lpfbyp_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:5	RX_ANA_CAL_MUXB_SEL	R/W	**Description**: Value for rx_ana_cal_muxb_sel[4:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

Table 11-239 Fields for Register: LANEN_DIG_ANA_RX_CAL (Continued)

Bits	Name	Memory Access	Description
4:0	RX_ANA_CAL_MUXA_SEL	R/W	**Description**: Value for rx_ana_cal_muxa_sel[4:0]. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

11.2.236 LANEN_DIG_ANA_RX_VDAC_RANGE_SEL

- **Description:** Sets values for RX DAC CTRL value going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nae
- **Exists:** Always

Table 11-240 Fields for Register: LANEN_DIG_ANA_RX_VDAC_RANGE_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RX_ANA_CAL_VDAC_DERANGE_SEL	R/W	**Description**: Value for rx_ana_cal_vdac_derange **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_CAL_VDAC_PRANGE_SEL	R/W	**Description**: Value for rx_ana_cal_vdac_prange **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.237 LANEN_DIG_ANA_RX_DAC_CTRL

- **Description:** Sets values for RX DAC CTRL value going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Naf
- **Exists:** Always

Table 11-241 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ANA_CAL_DAC_CTRL	R/W	**Description**: Override value for rx_ana_cal_dac_ctrl[7:0] Its override enable is RX_DAC_CTRL_OVRD.RX_CAL_DAC_CTRL_OVRD **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.238 LANEN_DIG_ANA_RX_ANA_RTRIM

- **Description:** Set value for RX RTRIM going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb0
- **Exists:** Always

Table 11-242 Fields for Register: LANEN_DIG_ANA_RX_ANA_RTRIM

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	RX_AFE_TRIM_EN_R	R/W	**Description**: enable for signal rx_afe_trim_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:2	RX_ANA_AFE_TRIM_THRESH_R	R/W	**Description**: Threshold value for rx_ana_afe_trim **Value After Reset**: 0x1a **Exists**: Always **Reset Mask**: 0x3f
1	RX_ANA_AFE_TRIM_OVRD_EN	R/W	**Description**: Override enable rx_ana_afe_trim_ovrd_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_AFE_TRIM_OVRD_VAL	R/W	**Description**: Override value for rx_ana_afe_trim_ovrd_val **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.239 LANEN_DIG_ANA_RX_DAC_CTRL_OVRD

- **Description:** Overrides RX DAC CTRL bus (en/val/sel) going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb1
- **Exists:** Always

Table 11-243 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL_OVRD

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_CAL_DAC_CTRL_OVRD	R/W	**Description**: Override enable for Cal DAC control output signals overridden by this enable are: rx_ana_cal_dac_ctrl_en rx_ana_cal_dac_ctrl rx_ana_cal_dac_ctrl_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.240 LANEN_DIG_ANA_RX_DAC_CTRL_SEL

- **Description:** Sets values for RX DAC CTRL Select signal going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb2
- **Exists:** Always

Table 11-244 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL_SEL

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4:0	RX_ANA_CAL_DAC_CTRL_SEL	R/W	**Description**: Value for rx_ana_cal_dac_ctrl_sel[4:0] Its override enable is RX_DAC_CTRL_OVRD.RX_CAL_DAC_CTRL_OVRD **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

11.2.241 LANEN_DIG_ANA_RX_AFE_ATT_VGA

- **Description:** Value for RX AFE ATT & VGA signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb3
- **Exists:** Always

Table 11-245 Fields for Register: LANEN_DIG_ANA_RX_AFE_ATT_VGA

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	RX_AFE_OVRD_EN	R/W	**Description**: Override enable for AFE control output signals overridden by this enable are: rx_ana_afe_update_en rx_ana_afe_att_lvl :: rx_ana_afe_vga1_gain rx_ana_afe_vga2_gain rx_ana_afe_ctle_boost rx_ana_afe_ctle_pole **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:6	RX_ANA_AFE_VGA2_GAIN	R/W	**Description**: Value for rx_ana_afe_vga2_gain[2:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
5:3	RX_ANA_AFE_VGA1_GAIN	R/W	**Description**: Value for rx_ana_afe_vga1_gain[2:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
2:0	RX_ANA_AFE_ATT_LVL	R/W	**Description**: Value for rx_ana_afe_att_lvl[2:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.242 LANEN_DIG_ANA_RX_AFE_CTLE

- **Description:** Values for RX AFE CTLE signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb4
- **Exists:** Always

Table 11-246 Fields for Register: LANEN_DIG_ANA_RX_AFE_CTLE

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:7	RX_ANA_AFE_BIAS_MT	R/W	**Description**: Value for rx_ana_afe_bias_mt[1:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
6:2	RX_ANA_AFE_CTLE_BOOST	R/W	**Description**: Override value for rx_ana_afe_ctle_boost[30:0] Its override enable is RX_AFE_ATT_VGA.RX_AFE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
1:0	RX_ANA_AFE_CTLE_POLE	R/W	**Description**: Override value for rx_ana_afe_ctle_pole[14:0] Its override enable is RX_AFE_ATT_VGA.RX_AFE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.243 LANEN_DIG_ANA_RX_SCOPE

- **Description:** Values for RX SCOPE signals going to ANA
- **Size:** 16 bits
- **Offset:** 0x1Nb5
- **Exists:** Always

Table 11-247 Fields for Register: LANEN_DIG_ANA_RX_SCOPE

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	RX_ANA_SCOPE_CLK_EN	R/W	**Description**: Enable the scope clocks going to the scope slicer and the lane digital part **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:3	RX_ANA_SCOPE_PHASE	R/W	**Description**: Sets value for rx_ana_scope_phase[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
2	RX_ANA_SCOPE_PH_CLK	R/W	**Description**: Sets value for rx_ana_scope_ph_clk This bit is self-clearing (i.e. only asserts for one cr_clk cycle) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_ANA_SCOPE_SEL	R/W	**Description**: Sets value for rx_ana_scope_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_SCOPE_EN	R/W	**Description**: Sets value for rx_ana_scope_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.244 LANEN_DIG_ANA_RX_SLICER_CTRL

- **Description:** Sets values for RX Slicer Ctrl signals going to ANA register
- **Size:** 16 bits
- **Offset:** 0x1Nb6
- **Exists:** Always

Table 11-248 Fields for Register: LANEN_DIG_ANA_RX_SLICER_CTRL

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	RX_ANA_SLICER_CTRL_OVRD_EN	R/W	**Description**: Override enable for Rx ANA Slicer Ctrl output signals overridden by this enable are: rx_ana_slicer_ctrl_o rx_ana_slicer_ctrl_e **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:4	RX_ANA_SLICER_CTRL_O	R/W	**Description**: Value for rx_ana_slicer_ctrl_o[3:0] **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0xf
3:0	RX_ANA_SLICER_CTRL_E	R/W	**Description**: Value for rx_ana_slicer_ctrl_e[3:0] **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0xf

11.2.245 LANEN_DIG_ANA_RX_ANA_IQ_PHASE_ADJUST

- **Description:** Sets values for RX ANA IQ PHASE Adjust signal going to ANA register
- **Size:** 16 bits
- **Offset:** 0x1Nb7
- **Exists:** Always

Table 11-249 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQ_PHASE_ADJUST

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	RX_ANA_IQ_PHASE_ADJUST	R/W	**Description**: Value for rx_ana_iq_phase_adjust[6:0] **Value After Reset**: 0x17 **Exists**: Always **Reset Mask**: 0x7f

11.2.246 LANEN_DIG_ANA_RX_ANA_IQ_SENSE_EN

- **Description:** Sets values for RX ANA IQ SENSE signal
- **Size:** 16 bits
- **Offset:** 0x1Nb8
- **Exists:** Always

Table 11-250 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQ_SENSE_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ANA_IQ_SENSE_EN	R/W	**Description**: Value for rx_ana_iq_sense_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.247 LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN

- **Description:** DAC CTRL enable signal
- **Size:** 16 bits
- **Offset:** 0x1Nb9
- **Exists:** Always

Table 11-251 Fields for Register: LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	DAC_CTRL_SELF_CLEAR_DISABLE	R/W	**Description**: Disable self-clearing for the rx_ana_cal_dac_ctrl_en register **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_CAL_DAC_CTRL_EN	R/W	**Description**: Value for rx_ana_cal_dac_ctrl_en Its override enable is RX_DAC_CTRL_OVRD.RX_CAL_DAC_CTRL_OVRD If DAC_CTRL_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.248 LANEN_DIG_ANA_RX_ANA_SIGNALS_CHANGES_ENABLE

- **Description:** Afe update enable signal
- **Size:** 16 bits
- **Offset:** 0x1Nba
- **Exists:** Always

Table 11-252 Fields for Register: LANEN_DIG_ANA_RX_ANA_SIGNALS_CHANGES_ENABLE

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	AFE_UPDATE_SELF_CLEAR_DISABLE	R/W	**Description**: Disable self-clearing for the rx_ana_afe_update_en register **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_AFE_UPDATE_EN	R/W	**Description**: Override value for rx_ana_afe_update_en. Its override enable is RX_AFE_ATT_VGA.RX_AFE_OVRD_EN If AFE_UPDATE_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.249 LANEN_DIG_ANA_RX_ANA_PHASE_ADJUST_CLK

- **Description:** Phase adjust clock signal
- **Size:** 16 bits
- **Offset:** 0x1Nbb
- **Exists:** Always

Table 11-253 Fields for Register: LANEN_DIG_ANA_RX_ANA_PHASE_ADJUST_CLK

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	PHASE_ADJUST_SELF_CLEAR_DISABLE	R/W	**Description**: Disable self-clearing for the rx_ana_iq_phase_adjust_clk register **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ANA_IQ_PHASE_ADJUST_CLK	R/W	**Description**: Value for rx_ana_iq_phase_adjust_clk If PHASE_ADJUST_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.250 LANEN_DIG_ANA_STATUS_0

- **Description:** Lane input status register #0
- **Size:** 16 bits
- **Offset:** 0x1Nbd
- **Exists:** Always

Table 11-254 Fields for Register: LANEN_DIG_ANA_STATUS_0

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	RX_ANA_SCOPE_DATA	R	**Description**: Value from ANA for rx_ana_scope_data (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6	RX_ANA_CAL_RESULT	R	**Description**: Value from ANA for rx_ana_cal_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	RX_ANA_LOS	R	**Description**: Value from ANA for rx_ana_los (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4	TX_ANA_CLK_LB_EN	R	**Description**: Value of tx_ana_clk_lb_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-254 Fields for Register: LANEN_DIG_ANA_STATUS_0 (Continued)

Bits	Name	Memory Access	Description
3	TX_ANA_LOOPBACK_EN	R	**Description**: Value of tx_ana_loopback_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	TX_ANA_RXDETM_RESULT	R	**Description**: Value from ANA for tx_ana_rxdetm_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	TX_ANA_RXDETP_RESULT	R	**Description**: Value from ANA for tx_ana_rxdetp_result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	TX_ANA_CLK_SHIFT_ACK	R	**Description**: Value from ANA for tx_ana_clk_shift_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.251 LANEN_DIG_ANA_STATUS_1

- **Description:** Lane input status register #1
- **Size:** 16 bits
- **Offset:** 0x1Nbe
- **Exists:** Always

Table 11-255 Fields for Register: LANEN_DIG_ANA_STATUS_1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:0	RX_ANA_VCO_CNTR	R	**Description**: Value from ANA for rx_ana_vco_cntr (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.252 LANEN_ANA_TX_OVRD_MEAS

- **Description:** TX override control register
- **Size:** 16 bits
- **Offset:** 0x1Nc0
- **Exists:** Always

Table 11-256 Fields for Register: LANEN_ANA_TX_OVRD_MEAS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	pull_dn_reg	R/W	**Description**: Use this register bit to Pull down TX output 1: Pull down the TX output to zero 0: Pull down function is disabled **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	pull_up_reg	R/W	**Description**: Use this register bit to Pull up TX output 1: Pulls up the TX output to high 0: Pulls up function is disabled **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	vcm_hold_reg	R/W	**Description**: If bit 4 (ovrd_vcm_hold) = 1, this bit is used to set TX differential output to a common mode 1: Enables TX common mode hold function 0: Disables TX common mode hold function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-256 Fields for Register: LANEN_ANA_TX_OVRD_MEAS (Continued)

Bits	Name	Memory Access	Description
4	ovrd_vcm_hold	R/W	**Description**: If this bit = 1, it allows register bit 5 (vcm_hold_reg) to control TX common mode hold function 1: Allows internal analog register bit 5 (vcm_hold_reg) to control TX common mode hold function 0: Instead of analog register, internal state machine controls TX common mode hold function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_samp_p	R/W	**Description**: Measures the clock p duty cycle on clock psample through atb_s_p 1: Connects the dcc_fb_clk_p to the atb_s_p pin for measurement 0: Disconnects the dcc_fb_clk_p from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_samp_m	R/W	**Description**: Measures the clock m duty cycle on clock msample through atb_s_p 1: Connects the dcc_fb_clk_m to the atb_s_p pin for measurement 0: Disconnects the dcc_fb_clk_m from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	clk_shift_reg	R/W	**Description**: If bit 0 (ovrd_clk_shift) = 1, this bit controls TX clock shift function 1: TX Clock shift is performed 0: TX Clock shift function is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_clk_shift	R/W	**Description**: If this bit = 1, it allows analog register bit 1 (clk_shift_reg) to control TX clock shift function 1: Allows bit 1 (clk_shift_reg) to enable/disable the TX clock shift 0: Disallows bit 1 (clk_shift_reg) to enable/disable the TX clock shift. Instead, it is controlled by internal state machines **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.253 LANEN_ANA_TX_PWR_OVRD

- **Description:** TX sub-block power control register
- **Size:** 16 bits
- **Offset:** 0x1Nc1
- **Exists:** Always

Table 11-257 Fields for Register: LANEN_ANA_TX_PWR_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_en	R/W	**Description**: If this bit = 1, it enables analog registers to take control of TX power state 1: Analog registers take control of the TX power state 0: External pin, txX_pstate[1:0], controls the TX power state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	serial_en_reg	R/W	**Description**: If bit 7 (ovrd_en) = 1, this bit turns ON/OFF TX serializer 1: TX serializer is turned on 0: TX serializer is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	clk_en_reg	R/W	**Description**: If bit 7 (ovrd_en) = 1, this bit enables/disables TX clock 1: TX clock is turned on 0: TX clock is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-257 Fields for Register: LANEN_ANA_TX_PWR_OVRD (Continued)

Bits	Name	Memory Access	Description
4	data_en_reg_int	R/W	**Description**: If bit 7 (ovrd_en) = 1, this bit enables/disables TX driver data path 1: Enables TX driver data path 0: Disables TX driver data path **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	clk_div_en_reg	R/W	**Description**: If bit 7 (ovrd_en) = 1, this bit enables/disables TX divider, it overrides !tx_reset pin function 1: TX divider function is turned on irrespective of !tx_reset 0: TX divider function is turned off irrespective of !tx_reset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	refgen_en_reg	R/W	**Description**: If bit 7 (ovrd_en) = 1, this bit enables/disables TX biasing 1: TX biasing block is turned on 0: TX biasing block is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	loopback_en_reg	R/W	**Description**: If bit 0 (ovrd_tx_loopback) = 1, this bit enables/disables TX loopback path to RX 1: Enables TX loopback path to RX 0: Disables TX loopback path to RX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_tx_loopback	R/W	**Description**: If this bit = 1, it allows to turn on/off TX loopback mode using analog register bit 1 (loopback_en_reg) 1: Allows TX loopback mode to be controlled by analog register bit 1 (loopback_en_reg) 0: Disallows TX loopback mode to be controlled by analog register bit 1 (loopback_en_reg) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.254 LANEN_ANA_TX_ALT_BUS

- **Description:** TX oscillator and alternate data path control register
- **Size:** 16 bits
- **Offset:** 0x1Nc2
- **Exists:** Always

Table 11-258 Fields for Register: LANEN_ANA_TX_ALT_BUS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	jtag_data_reg	R/W	**Description**: If bit 2 (ovrd_alt_bus) = 1, this bit becomes one of the sources of TX data. TX transmits data from this register bit instead of JTAG data from digital **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	osc_vp	R/W	**Description**: This bit selects/deselects the vp powered svt device ring oscillator output clock as input data for TX 1: TX selects its data input from vp powered svt device ring oscillator output clock 0: Deselects TX data input from vp powered svt device ring oscillator output clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	osc_vp_lvt	R/W	**Description**: This bit selects/deselects the vp powered lvt device ring oscillator output clock as input data for TX 1: TX selects its data input from vp powered lvt device ring oscillator output clock 0: Deselects TX data input from vp powered lvt device ring oscillator output clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-258 Fields for Register: LANEN_ANA_TX_ALT_BUS (Continued)

Bits	Name	Memory Access	Description
4	osc_vptx	R/W	**Description**: This bit selects/deselects the vptx powered svt device ring oscillator output clock as input data for TX 1: TX selects its data input from vptx powered svt device ring oscillator output clock 0: Deselects TX data input from vptx powered svt device ring oscillator output clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	osc_vph	R/W	**Description**: This bit selects/deselects the vph powered ring oscillator output clock as input data for TX 1: TX selects its data input from vph powered ring oscillator output clock 0: Deselects TX data input from vph powered ring oscillator output clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_alt_bus	R/W	**Description**: If this bit = 1, it allows you to control the TX data source selection using analog registers drv_source_reg<1:0> 1: Analog registers drv_source_reg<1:0> control the data source for TX 0: Internal state machine controls this function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	drv_source_reg	R/W	**Description**: If bit 2 (ovrd_alt_bus) = 1, use these analog registers to control for TX data source while 00: Alt is disabled, local ringo or data is transmitted depending on osc enabled or tx data enabled 01: Differential ACJTAG data on alt_bus, full amplitude 10: LFPS/beacon oscillator, differential on alt bus 11: Differential ACJTAG data on alt_bus, half_amplitude **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.255 LANEN_ANA_TX_ATB1

- **Description:** TX analog test bus controls register (1 of 2)
- **Size:** 16 bits
- **Offset:** 0x1Nc3
- **Exists:** Always

Table 11-259 Fields for Register: LANEN_ANA_TX_ATB1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	atb_vreg_s	R/W	**Description**: This bit connects/disconnects the TX regulator charge pump output divided by 2 (vdd_h/2) to the atb_s_p pin 1: Connects the TX regulator charge pump output divided by 2 (vdd_h/2) to the atb_s_p pin for measurement 0: Disconnects TX regulator charge pump output divided by 2 (vdd_h/2) from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	atb_iboost	R/W	**Description**: put iboost output on atb_f_p when asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	atb_vreg0	R/W	**Description**: This bit connects/disconnects the tx regulator output vreg_tx net to the atb_s_p pin 1: Connects the tx regulator output vreg_tx net to the atb_s_p pin for measurement 0: Disconnects the tx regulator output vreg_tx net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-259 Fields for Register: LANEN_ANA_TX_ATB1 (Continued)

Bits	Name	Memory Access	Description
4	override_regref_0	R/W	**Description**: This bit is used to override the reference voltage of TX regulator through atb_s_m pin 1: TX regulator reference voltage = voltage applied on the atb_s_m pin 0: TX regulator reference voltage = internal reference voltage generated by diving vptx **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	atb_vptx	R/W	**Description**: This bit connects/disconnects TX driver internal supply vptx to atb_s_p pin 1: Connects the TX driver internal vptx voltage to the atb_s_p pin for measurement 0: Disconnects TX driver internal vptx voltage from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	atb_vdccp	R/W	**Description**: This bit connects/disconnects the positive side of TX DCC control voltage vddcp to atb_s_p pin 1: Connects the positive side of the TX DCC control voltage vdccp to atb_s_p pin for measurement 0: Disconnects the positive side of TX DCC control voltage vdccp from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	atb_vdccm	R/W	**Description**: This bit connects/disconnects the negative side of TX DCC control voltage vddcm to atb_s_p pin 1: Connects the negative side of the TX DCC control voltage vdccm to atb_s_p pin for measurement 0: Disconnects the negative side of TX DCC control voltage vdccm from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-259 Fields for Register: LANEN_ANA_TX_ATB1 (Continued)

Bits	Name	Memory Access	Description
0	atb_gd	R/W	**Description**: This bit connects/disconnects the TX internal analog ground gd to atb_s_p 1: Connects the TX internal analog ground gd to the atb_s_p pin for measurement 0: Disconnects the TX internal analog ground gd from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.256 LANEN_ANA_TX_ATB2

- **Description:** TX analog test bus controls register (2 of 2)
- **Size:** 16 bits
- **Offset:** 0x1Nc4
- **Exists:** Always

Table 11-260 Fields for Register: LANEN_ANA_TX_ATB2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	atb_pbias	R/W	**Description**: This bit connects/disconnects TX pbias node to atb_s_p 1: Connects TX pbias node to the atb_s_p pin for measurement 0: Disconnects TX pbias node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	atb_nbias	R/W	**Description**: This bit connects/disconnects TX nbias node to atb_s_p 1: Connects TX nbias node to the atb_s_p pin for measurement 0: Disconnects TX nbias node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	atb_rxdetref	R/W	**Description**: This bit connects/disconnects TX RX-detection reference voltage to atb_s_p 1: Connects TX RX-detection reference voltage to atb_s_p pin for measurement 0: Disconnects TX RX-detection reference voltage from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-260 Fields for Register: LANEN_ANA_TX_ATB2 (Continued)

Bits	Name	Memory Access	Description
4	atb_txfp	R/W	**Description**: Using this bit one can overdrive tx_p pin by applying voltage on atb_f_p 1: Voltage on the tx_p pin = voltage that is applied on the atb_f_p pin 0: Disconnects the atb_f_p pin from the tx_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	atb_txfm	R/W	**Description**: Using this bit one can overdrive tx_m pin by applying voltage on atb_f_p 1: Voltage on the tx_m pin = voltage that is applied on the atb_f_p pin 0: Disconnects the atb_f_p pin from the tx_m pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	atb_txsp	R/W	**Description**: This bit connects/disconnects tx_p to atb_s_p pin 1: Connects the tx_p internal net to the atb_s_p pin for measurement 0: Disconnects the tx_p internal net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	atb_txsm	R/W	**Description**: This bit connects/disconnects tx_m to atb_s_p pin 1: Connects the tx_m internal net to the atb_s_p pin for measurement 0: Disconnects the tx_m internal net from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	atb_vcm	R/W	**Description**: This bit connects/disconnects the TX vcm internal node to atb_s_p 1: Connects the TX vcm internal node to the atb_s_p pin for measurement 0: Disconnects the TX vcm internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.257 LANEN_ANA_TX_VBOOST

- **Description:** TX hybrid driver current-mode boosting control register
- **Size:** 16 bits
- **Offset:** 0x1Nc5
- **Exists:** Always

Table 11-261 Fields for Register: LANEN_ANA_TX_VBOOST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_vboost_en	R/W	**Description**: When this bit = 1, it allows us to enable/disable TX boost mode using bit 6 (vboost_en_reg) 1: Allows TX boost mode enable/disable controlled by bit 6 (vboost_en_reg) 0: Disallows TX boost mode enable/disable using bit 6 (vboost_en_reg) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	vboost_en_reg	R/W	**Description**: If bit 7 (ovrd_vboost_en) = 1, this analog register takes control of TX vboost enable/disable 1: TX boost is turned on 0: TX boost is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	boost_vptx_mode_n	R/W	**Description**: This bit controls TX boost mode 1: Enables TX direct boost mode, the tx differential voltage is according to the following equation: Vdiffpp=vptx+0.05*(2.5*(1+iboost_code)*(2*tx_vboost_vref)in default TX boost mode 0: TX direct boost mode is disabled and tx differential voltage is according to the following equation: Vdiffpp=vptx+0.05*(2.5*(1+iboost_code)*((2*tx_vboost_vref) - 0.5*vptx) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-261 Fields for Register: LANEN_ANA_TX_VBOOST (Continued)

Bits	Name	Memory Access	Description
4	atb_vboost	R/W	**Description**: When TX boost is enabled (controlled by bit 6 and bit 7), this bit connects/disconnects vptx/2 to atb_s_p pin 1: Connects vptx/2 to the atb_s_p pin for measurement 0: Disconnects vptx/2 from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	atb_vboost_vref	R/W	**Description**: This bit connects/disconnects TX boost reference internal node voltage to atb_s_p pin 1: Connects the TX boost reference internal node voltage to the atb_s_p pin for measurement 0: Disconnects the TX boost reference internal node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	override_vref_boost_ref	R/W	**Description**: If this bit = 1, it allows the TX boost reference voltage (tx_vboost_vref) be forced through atb_s_m pin 1: TX boost reference voltage = voltage applied on atb_s_m pin voltage 0: TX boost reference voltage comes from internal bandgap voltage **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	atb_s_enable	R/W	**Description**: Connects/disconnects TX internal atb bus atb_s_p/m_int to atb_s_p/m to make it visible externally 1: Connects TX internal atb bus atb_s_p/m_int to atb_s_p/m pins for measurements 0: Disconnects TX internal atb bus atb_s_p/m_int from atb_s_p/m pins **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-261 Fields for Register: LANEN_ANA_TX_VBOOST (Continued)

Bits	Name	Memory Access	Description
0	meas_atb_vph_half	R/W	**Description**: Connects/disconnects tx internal analog high voltage vph/2 to atb_s_p 1: Connects tx internal analog high voltage vph/2 to the atb_s_p pin for measurement 0: Disconnects tx internal analog high voltage vph/2 from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.258 LANEN_ANA_TX_TERM_CODE_DN

- **Description:** TX-DN termination analog code override register
- **Size:** 16 bits
- **Offset:** 0x1Nc6
- **Exists:** Always

Table 11-262 Fields for Register: LANEN_ANA_TX_TERM_CODE_DN

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:1	term_code_dn_reg	R/W	**Description**: TX nmos leg biasing registers (7 MSBs out of 10 bits); this overrides the term_code_dn<9:3> when bit 0 (term_code_dn_ovrd) is set to 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
0	term_code_dn_ovrd	R/W	**Description**: If this bit = 1, it allows analog register bits term_code_dn_reg<7:1> to control TX nmos leg biasing 1: Allows analog register bits term_code_dn_reg<7:1> to control of TX nmos leg biasing codes term_code_dn<9:3> 0: The codes term_code_dn<9:3> comes from digital state machine and control of TX nmos leg biasing **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.259 LANEN_ANA_TX_TERM_CODE_UP

- **Description:** TX-UP termination analog code override register
- **Size:** 16 bits
- **Offset:** 0x1Nc7
- **Exists:** Always

Table 11-263 Fields for Register: LANEN_ANA_TX_TERM_CODE_UP

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:1	term_code_up_reg	R/W	**Description**: TX pmos leg biasing registers (7 MSBs out of 10 bits); this overrides the term_code_up<9:3> when bit 0 (term_code_up_ovrd) is set to 1 **Value After Reset**: 0x7f **Exists**: Always **Reset Mask**: 0x7f
0	term_code_up_ovrd	R/W	**Description**: If this bit = 1, it allows analog register bits term_code_up_reg<7:1> to control TX pmos leg biasing 1: Allows analog registers term_code_up_reg<7:1> to control of TX pmos leg biasing codes term_code_up<9:3> 0: The codes term_code_up<9:3> comes from digital state machine **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.260 LANEN_ANA_TX_IBOOST_CODE

- **Description:** TX hybrid driver current-mode boosting level control
- **Size:** 16 bits
- **Offset:** 0x1Nc8
- **Exists:** Always

Table 11-264 Fields for Register: LANEN_ANA_TX_IBOOST_CODE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	iboost_code_reg	R/W	**Description**: When TX boost is enabled and bit 3 (iboost_code_ovrd) = 1, these 4 bits (iboost_code<3:0>) control of TX boost amount, Vdiffpp = vptx + 0.05 * (2.5 * (1 + iboost_code) * (2 * tx_vboost_vref - vptx / 2) in default TX boost mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
3	iboost_code_ovrd	R/W	**Description**: If this bit = 1, It allows analog register bits iboost_code<7:4> to control TX boost amount 1: TX boost amount is controlled by internal analog registers iboost_code<7:4> 0: TX boost amount is controlled externally using the txX_iboost_lvl[3:0] pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	term_code_dn_reg_2	R/W	**Description**: TX nmos leg biasing register bit 2, this overrides the term_code_dn_reg<2> when analog register term_code_dn_ovrd is set to 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-264 Fields for Register: LANEN_ANA_TX_IBOOST_CODE (Continued)

Bits	Name	Memory Access	Description
1	term_code_up_reg_2	R/W	**Description**: TX pmos leg biasing register bit 2, this overrides the term_code_up_reg<2> when analog register term_code_up_ovrd is set to 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	lfps_high_priority	R/W	**Description**: If this bit = 1, lfps/beacon enable has higher priority than data enable 1: LFPS/beacon data is transmitted first 0: TX data is transmitted first instead of LFPS/beacon data **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.261 LANEN_ANA_TX_OVRD_CLK

- **Description:** TX clock control override register
- **Size:** 16 bits
- **Offset:** 0x1Nc9
- **Exists:** Always

Table 11-265 Fields for Register: LANEN_ANA_TX_OVRD_CLK

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_word_clk_en	R/W	**Description**: If this bit = 1, it allows bit 6 (word_clk_en_reg) to override TX word clock enable 1: Allows bit 6 (word_clk_en_reg) to switch on or switch off the TX word clock 0: Instead of this register, internal state machine switch on/off the TX word clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	word_clk_en_reg	R/W	**Description**: If bit 7 (ovrd_word_clk_en) = 1, this bit enables/disables TX word clock 1: Switches on the TX word clock 0: Switches off the TX word clock **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	ovrd_mpllab_en	R/W	**Description**: If this bit = 1, it allows bit 3 (mpllb_clk_en_reg) and 4 (mplla_clk_en_reg) to take control of selecting MPLLB/MPLLA clocks 1: Allows bit 3 (mpllb_clk_en_reg) and 4 (mplla_clk_en_reg) to take control of selecting MPLLB/MPLLA clocks 0: Disallows bit 3 (mpllb_clk_en_reg) and 4 (mplla_clk_en_reg) to take control of selecting MPLLB/MPLLA clocks **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-265 Fields for Register: LANEN_ANA_TX_OVRD_CLK (Continued)

Bits	Name	Memory Access	Description
4	mplla_clk_en_reg	R/W	**Description**: When bit 5 (ovrd_mpllab_en) = 1, this bit selects/deselects MPLLA clock as TX data output clock 1: MPLLA clock is the TX data output clock 0: MPLLA clock is deselected as the TX data output clock Note: Only one bit out of bits 1, 3, 4 can be asserted at a time **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	mpllb_clk_en_reg	R/W	**Description**: When bit 5 (ovrd_mpllab_en) = 1, this bit selects/deselects MPLLB clock as TX data output clock 1: MPLLB clock is the TX data output clock 0: MPLLB clock is deselected as the TX data output clock Note: Only one bit out of bits 1, 3, 4 can be asserted at a time **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_lb_en	R/W	**Description**: If this bit = 1, it allows bit 1 (clk_lb_en_reg) to take control of RX clock loopback to TX 1: Allows bit 1 (clk_lb_en_reg) to take control of RX clock loopback to TX 0: Disallows bit 1 (clk_lb_en_reg) to take control of RX clock loopback to TX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	clk_lb_en_reg	R/W	**Description**: When bit 2 (ovrd_lb_en) = 1, this bit enables/disables the RX clock loopback to TX 1: Enables RX to TX loopback 0: Disables RX to TX loopback **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	override_regref_1	R/W	**Description**: Bit is not used(NC) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.262 LANEN_ANA_TX_MISC

- **Description:** Miscellaneous TX control register
- **Size:** 16 bits
- **Offset:** 0x1Nca
- **Exists:** Always

Table 11-266 Fields for Register: LANEN_ANA_TX_MISC

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	override_rxdetref	R/W	**Description**: If this bit = 1, it allows to override RX detection reference voltage through atb_s_m pin 1: RX detection reference voltage = the voltage applied on the atb_s_m pin 0: Disconnects the RX detection reference voltage node from the atb_s_m pin. When disconnected from the atb_s_m pin, the RX detection reference voltage comes from the internal bandgap **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_atb_bias_vptx	R/W	**Description**: This bit connects/disconnects TX bias local voltage vptx to atb_s_p for measurement 1: Connects TX bias internal voltage vptx node to the atb_s_p pin for measurement 0: Disconnects TX bias local voltage vptx node from the atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	osc_nmos	R/W	**Description**: The nmos device ring oscillator ouputs from TX when asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-266 Fields for Register: LANEN_ANA_TX_MISC (Continued)

Bits	Name	Memory Access	Description
4	osc_pmos	R/W	**Description**: The pmos device ring oscillator ouputs from TX when asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:2	vreg_ring_ctr	R/W	**Description**: vreg_ring_ctr<1:0> Function: 00: CP/OSC current 1.8uA 01: CP/OSC current 3.6uA 10: CP/OSC current 5.4uA 11: CP/OSC current 7.2uA **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3
1:0	NC10	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.263 LANEN_ANA_RX_ATB_IQSKEW

- **Description:** RX analog test bus access for I/Q clock phase adjustment
- **Size:** 16 bits
- **Offset:** 0x1Ncb
- **Exists:** Always

Table 11-267 Fields for Register: LANEN_ANA_RX_ATB_IQSKEW

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	master_atb_en	R/W	**Description**: Connects/disconnects RX internal ATB sensing bus atb_s_p/m_int to atb_s_p/m pins to make it externally visible 1: Connects RX internal ATB sensing bus atb_s_p/m_int to atb_s_p/m pins to measure various internal nodes of RX 0: Disconnects RX internal ATB sensing bus atb_s_p/m_int from the atb_s_p/m pins Note: Bit 5 does not require this bit to be 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_atb_vp	R/W	**Description**: If this bit = 1, enables scope linearity characterization through atb_s_p/m measurement **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5:0	iq_phase_adjust_reg	R/W	**Description**: The iq phase adjust reg is valid only when ovrd_iq_phase_adjust is asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

11.2.264 LANEN_ANA_RX_DCC_OVRD

- **Description:** RX Duty Cycle Correction and AFE data rate override register
- **Size:** 16 bits
- **Offset:** 0x1Ncc
- **Exists:** Always

Table 11-268 Fields for Register: LANEN_ANA_RX_DCC_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	dcc_en_reg	R/W	**Description**: This bit enables/disables RX Duty Cycle Corrector (DCC) block when bit 6 (ovrd_dccandafe_en) = 1 1: RX DCC block is turned on 0: RX DCC block is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_dccandafe_en	R/W	**Description**: This bit enables you to control the RX DCC block and AFE data rate through register bit 7 (dcc_en_reg) and bit<1:0> (data_rate_reg), respectively 1: Control dcc and rate functions using register bit 7 (dcc_en_reg) and bit<1:0> (data_rate_reg), respectively 0: Instead of registers, internal state machines control these functions **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	word_clk_en_reg	R/W	**Description**: If asserted, it allows bit 4 to override rx word clock enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-268 Fields for Register: LANEN_ANA_RX_DCC_OVRD (Continued)

Bits	Name	Memory Access	Description
4	ovrd_word_clk_en	R/W	**Description**: The rx word clock enable/disable when bit 5 is asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:2	meas_atb_vdcc	R/W	**Description**: meas_atb_vdcc<3:2> bits enable you to measure the voltages of various internal nodes inside RX DCC block through the atb_s_p/m pins 00: Disables atb measurement 01: Connects vdcc_i_p/m nodes to atb_s_p/m pins for measurement 10: Connects vdcc_q_p/m nodes to atb_s_p/m pins for measurement 11: Connects vdcc_i/q nodes to atb_s_p/m pins for measurement **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
1:0	afe_rate_reg	R/W	**Description**: When bit 6 is asserted, the two bits (afe_rate_reg<1:0>) override RX AFE rate which controls CTLE boost frequency **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.265 LANEN_ANA_RX_PWR_CTRL1

- **Description:** RX sub-block power control register (1 of 2)
- **Size:** 16 bits
- **Offset:** 0x1Ncd
- **Exists:** Always

Table 11-269 Fields for Register: LANEN_ANA_RX_PWR_CTRL1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	pwron_reg	R/W	**Description**: This bit enables/disables AFE when bit 6 (ovrd_afe_en) = 1 1: AFE is turned on 0: AFE is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_pwron	R/W	**Description**: If this bit = 1, it allows bit 7 (afe_en_reg) to take control of AFE enable/disable 1: Allows bit 7 (afe_en_reg) to take control of AFE enable/disable 0: Internal state machine controls this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	los_en_reg	R/W	**Description**: This bit enables/disables RX LOS block when bit 4 (ovrd_los_en) = 1 1: RX LOS is turned on 0: RX LOS is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-269 Fields for Register: LANEN_ANA_RX_PWR_CTRL1 (Continued)

Bits	Name	Memory Access	Description
4	ovrd_los_en	R/W	**Description**: If this bit = 1, it allows bit 5 (los_en_reg) to enable/disable RX LOS detection block 1: Allows bit 5 (los_en_reg) to enable/disable RX LOS detection block 0: Internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	clk_en_reg	R/W	**Description**: This bit enables/disables RX clock when bit 2 (ovrd_clk_en) = 1 1: RX clock is turned on 0: RX clock is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_clk_en	R/W	**Description**: This bit enables you to control RX clock through register bit 3 (clk_en_reg) 1: Allows you to enable/disable RX clock using register bit 3 (lclk_en_reg) 0: Internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	acjt_en_reg	R/W	**Description**: Can be used to enable/disable ACJTAG when bit 0 (ovrd_acjt_en) = 1 1: ACJTAG is turned on 0: ACJTAG is turned off **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_acjt_en	R/W	**Description**: This bit enables you to turn on/off ACJTAG using register bit 1 (acjt_en_reg) 1: Allows you to turn on/off ACJTAG using register bit 1 (acjt_en_reg) 0: Internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.266 LANEN_ANA_RX_ATB_REGREF

- **Description:** RX analog test bus access to regulator reference levels
- **Size:** 16 bits
- **Offset:** 0x1Nce
- **Exists:** Always

Table 11-270 Fields for Register: LANEN_ANA_RX_ATB_REGREF

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:5	meas_atb_cal_mux	R/W	**Description**: This is the meas_atb_cal_mux<2:0> bit and used to measure the internal node voltages of RX calibration block 100: That is Meas_atb_cal_mux<2> = 1, RX cal comparator first stage differential outputs are measured through atb_s_p/m pins. 010: That is Meas_atb_cal_mux<1> = 1, RX cal mux B differential outputs are measured through atb_s_p/m pins. 001: That is Meas_atb_cal_mux<0> = 1, RX cal mux A differential outputs are measured through atb_s_p/m pins **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
4	override_regref_clk	R/W	**Description**: If set to 1, RX clock regulator reference can be applied through atb_s_m pin 1: RX clock regulator reference voltage = voltage applied on the atb_s_m pin 0: The RX clock regulator reference voltage comes from internal bandgap reference circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-270 Fields for Register: LANEN_ANA_RX_ATB_REGREF (Continued)

Bits	Name	Memory Access	Description
3	override_regref_scope	R/W	**Description**: If set to 1, RX scope regulator reference can be applied through atb_s_m pin 1: RX scope regulator reference voltage = voltage applied on the atb_s_m pin 0: The RX scope regulator reference voltage comes from internal bandgap reference circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	cdr_vco_dac_filt_bypass_reg	R/W	**Description**: Bypass ibias active filter in mission mode when asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	meas_atb_samp	R/W	**Description**: meas_atb_samp<1:0> bits control the measurement of various input voltages of RX DCC AMP block 00: Disables atb measurement 01: Connects clk_i_p/m sampling nodes to atb_s_p/m pins for measurement 10: Connects clk_q_p/m sampling nodes to atb_s_p/m pins for measurement 11: Connects clk_i/q sampling nodes to atb_s_p/m pins for measurement **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.267 LANEN_ANA_RX_CDR_AFE

- **Description:** RX clock data recovery mode control and AFE analog test bus access
- **Size:** 16 bits
- **Offset:** 0x1Ncf
- **Exists:** Always

Table 11-271 Fields for Register: LANEN_ANA_RX_CDR_AFE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	phdet_even_reg	R/W	**Description**: If asserted, CDR phase detector uses only even data path 1: RX CDR phase detector uses only even data path 0: RX CDR phase detector uses both odd and even data paths if bit 6 (phdet_odd_reg) is also 0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
6	phdet_odd_reg	R/W	**Description**: If asserted, CDR phase detector uses only odd data path 1: RX CDR phase detector only uses odd data path 0: RX CDR phase detector uses both odd and even data paths if bit 7 (phdet_even_reg) is also 0 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
5	meas_sum_ereg	R/W	**Description**: If asserted, place even summer outp/m nodes on atb_s_p|m_int **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	meas_sum_oreg	R/W	**Description**: If asserted, place odd summer outp/m nodes on atb_s_p|m_int **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-271 Fields for Register: LANEN_ANA_RX_CDR_AFE (Continued)

Bits	Name	Memory Access	Description
3:0	meas_atb_rx	R/W	**Description**: This controls meas_atb_rx<9:6> bits to measure various internal voltages in the RX AFE block. 1000: That is meas_atb_rx<9> = 1, AFE biasing vbp node voltage is connected to atb_s_p pin for measurement 0100: That is meas_atb_rx<8> = 1, AFE biasing vbn node voltage is connected to atb_s_m pin for measurement 0010: That is meas_atb_rx<7> = 1, rx_p node voltage is connected to atb_s_p pin for measurement 0001: That is meas_atb_rx<6> = 1, rx_p node voltage can be driven through atb_f_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.268 LANEN_ANA_RX_PWR_CTRL2

- **Description:** RX sub-block power control (2 of 2) and loopback controls
- **Size:** 16 bits
- **Offset:** 0x1Nd0
- **Exists:** Always

Table 11-272 Fields for Register: LANEN_ANA_RX_PWR_CTRL2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	loopback_en_reg	R/W	**Description**: Enables/disables TX to RX data loopback when bit 6 (ovrd_loopback_en) = 1 1: Enables TX to RX data loopback 0: Disbales TX to RX data loopback **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_loopback_en	R/W	**Description**: If this bit = 1, bit 7 (loopback_en_reg) can be used to enable/disable TX to RX data loopback 1: Allows bit 7 (loopback_en_reg) to enable/disable the TX to RX data loopback 0: Internal state machine controls this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	deserial_en_reg	R/W	**Description**: If bit 4 (ovrd_deserial_en) = 1, this bit enables/disables RX deserializer 1: Enables RX deserializer 0: Disables RX deserializer **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-272 Fields for Register: LANEN_ANA_RX_PWR_CTRL2 (Continued)

Bits	Name	Memory Access	Description
4	ovrd_des_en	R/W	**Description**: If this bit = 1, bit 5 (deserial_en_reg) takes control of RX desrializer enable/disable 1: Allows bit 5 (deserial_en_reg) to enable/disable RX desrializer 0: Internal state machine controls this function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	dfe_en_reg	R/W	**Description**: If bit 2 (ovrd_dfe_en) = 1, this bit enables/disables RX DFE 1: Enables RX DFE 0: Disables RX DFE **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	ovrd_dfe_en	R/W	**Description**: If this bit = 1, it allows bit 3 (dfe_en_reg) to control of RX DFE enable/disable 1: Allows bit 3 (dfe_en_reg) to control of RX DFE enable/disable 0: Internal state machine control this function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	override_regref_vco	R/W	**Description**: If asserted, RX CDR VCO regulator reference is overridden through atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	afe_idac_vga_range_en_reg	R/W	**Description**: If asserted afe idac vga range will be decreased by half. Default mode is to cover the total range **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.269 LANEN_ANA_RX_MISC_OVRD

- **Description:** Miscellaneous RX control register
- **Size:** 16 bits
- **Offset:** 0x1Nd1
- **Exists:** Always

Table 11-273 Fields for Register: LANEN_ANA_RX_MISC_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_short_en	R/W	**Description**: If this bit = 1, it allows bit 6 (short_en_reg) to enable/disable RX front short function 1: Allows bit 6 to enable/disable RX front short function 0: Disallows bit 6 to enable/disable RX front short function, internal state machine controls this function instead **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	shor_en_reg	R/W	**Description**: If bit 7 (ovrd_short_en) = 1, this bit enables/disables RX front short function 1: Enables RX front short function: 0: Disables RX front short function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	ovrd_rx_los_lfps_en	R/W	**Description**: If this bit = 1, it allows bit 4 (rx_los_lfps_en_reg) to enable/disable RX true LFPS detection 1: Allows bit 4 (rx_los_lfps_en_reg) to enable/disable RX true LFPS detection 0: Disallows bit 4 (rx_los_lfps_en_reg) to control this function, rxN_los_lfps_en pin controls this function instead **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-273 Fields for Register: LANEN_ANA_RX_MISC_OVRD (Continued)

Bits	Name	Memory Access	Description
4	rx_los_lfps_en_reg	R/W	**Description**: If bit 5 (ovrd_los_lfps_en) = 1, this bit enables/disables true LFPS detection in rx 1: Enables true LFPS detection in rx 0: Disables true LFPS detection in rx **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	vreg_ring_ctrl	R/W	**Description**: The vreg_ring_ctrl<1> controls CP OSC current **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	loopback_clk_sel_reg	R/W	**Description**: selects b/w iand qv phase for loopback clocks **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	rx_loopback_clk_en_reg	R/W	**Description**: The loopback clk en is valid only when ovrd_rx_loopback_clk_en is asserted **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_rx_loopback_clk_en	R/W	**Description**: The overides loopback en is from digital and selects loopback_clk_en from regs **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.270 LANEN_ANA_RX_CAL_MUXA

- **Description:** RX calibration path A mux control register
- **Size:** 16 bits
- **Offset:** 0x1Nd2
- **Exists:** Always

Table 11-274 Fields for Register: LANEN_ANA_RX_CAL_MUXA

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_cal_muxa_sel	R/W	**Description**: If this bit = 1, it allows register bits cal_muxa_sel_reg<6:2> to act as mux select in RX calibration mux path A 1: Allows register bits cal_muxa_sel_reg<6:2> to act as mux select in RX calibration mux path A 0: You cannot control RX calibration mux path A using this register. Instead, internal state machines control this functionality **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6:2	cal_muxa_sel_reg	R/W	**Description**: If bit 7 (ovrd_cal_muxa_sel) = 1, these analog registers act as mux select signals in RX calibration mux path A **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
1	meas_atb_vibias_vco	R/W	**Description**: This bit connects/disconnects RX CDR VCO ibias_cdr_vco net to atb_s_p pin for measurement 1: Connects RX CDR VCO ibias_cdr_vco net to atb_s_p pin for measurement 0: Disconnects RX CDR VCO ibias_cdr_vco net from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-274 Fields for Register: LANEN_ANA_RX_CAL_MUXA (Continued)

Bits	Name	Memory Access	Description
0	meas_atv_vco_200u	R/W	**Description**: This bit connects/disconnects CDR VCO iref net to atb_s_m pin 1: Connects CDR VCO iref net to atb_s_m pin for measurement 0: Disconnects CDR VCO iref net from atb_s_m pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.271 LANEN_ANA_RX_ATB_MEAS1

- **Description:** RX analog test bus controls (1 of 2)
- **Size:** 16 bits
- **Offset:** 0x1Nd3
- **Exists:** Always

Table 11-275 Fields for Register: LANEN_ANA_RX_ATB_MEAS1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	atb_frc_vlos	R/W	**Description**: This bit can be used to force rx los detection reference voltage through atb_s_m pin 1: RX los detection reference voltage = the voltage applied on atb_s_m pin 0: RX los detection reference voltage is from internal bandgap circuit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6:1	meas_atb_rx	R/W	**Description**: These registers bits can be used to control the following functions 100000: That is meas_atb_rx<5> = 1, rx_m can be driven by applying a voltage on atb_f_p pin 010000: That is meas_atb_rx<4> = 1, rx_m voltage can be sensed through atb_s_m pin 001000: That is meas_atb_rx<3> = 1, RX LOS detection threshold voltage can be measured on atb_s_p pin 000100: That is meas_atb_rx<2> = 1, RX bias reference voltage can be measured on atb_s_p pin 000010: That is meas_atb_rx<1> = 1, RX bias pmos biasing voltage bp1 can be measured on atb_s_p pin 000001: That is meas_atb_rx<0> = 1, RX bias cascode pmos biasing voltage bp1_c can be measured on atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

Table 11-275 Fields for Register: LANEN_ANA_RX_ATB_MEAS1 (Continued)

Bits	Name	Memory Access	Description
0	vreg_ring_ctrl	R/W	**Description**: The vreg_ring_ctrl<0> controls CP OSC current **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.272 LANEN_ANA_RX_ATB_MEAS2

- **Description:** RX analog test bus controls (2 of 2)
- **Size:** 16 bits
- **Offset:** 0x1Nd4
- **Exists:** Always

Table 11-276 Fields for Register: LANEN_ANA_RX_ATB_MEAS2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	meas_gd	R/W	**Description**: This bit is used to connect/disconnect RX regulator internal analog ground gd to atb_s_p pin 1: Connects RX regulator internal analog ground gd to atb_s_p pin for measurement 0: Disconnects RX regulator internal analog ground gd from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	meas_vreg_s	R/W	**Description**: This bit is used to connect/disconnect RX regulator charge pump output divided by 2 (vdd_h/2) to atb_s_p pin 1: Connects RX regulator charge pump output divided by 2 (vdd_h/2) to atb_s_p pin for measurement 0: Disconnects RX regulator charge pump output divided by 2 (vdd_h/2) from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_vreg_clk	R/W	**Description**: This bit is used to connect/disconnect RX clock regulator output vreg_clk to atb_s_p pin 1: Connects RX clock regulator output vreg_clk to atb_s_p pin for measurement 0: Disconnects RX clock regulator output vreg_clk from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-276 Fields for Register: LANEN_ANA_RX_ATB_MEAS2 (Continued)

Bits	Name	Memory Access	Description
4	afe_att_cm_sel_reg	R/W	**Description**: When asserted Vbg/R based CM is enabled. By default, it is enabled with constant gm based circuitry **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_vreg_scope	R/W	**Description**: This bit is used to connect/disconnect RX scope regulator output vreg_scope to atb_s_p pin 1: Connects RX scope regulator output vreg_scope to atb_s_p pin for measurement 0: Disconnects RX scope regulator output vreg_scope from atb_s_p pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	meas_vreg_vco	R/W	**Description**: If asserted, measure RX VCO regulator output voltage through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	meas_atb_vco_gd	R/W	**Description**: This bit connects/disconnects RX CDR VCO internal analog ground gd to atb_s_m pin 1: Connects RX CDR VCO internal analog ground gd to atb_s_m pin for measurement 0: Disconnects RX CDR VCO internal analog ground gd from atb_s_m pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	meas_atb_vco_vosc	R/W	**Description**: This bit connects/disconnects RX CDR VCO oscillator voltage to atb_s_m pin for measurement 1: Connects RX CDR VCO oscillator voltage to atb_s_m pin for measurement 0: Disconnects RX CDR VCO oscillator voltage from atb_s_m pin **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.273 LANEN_ANA_RX_CAL_MUXB

- **Description:** RX calibration path B mux control register
- **Size:** 16 bits
- **Offset:** 0x1Nd5
- **Exists:** Always

Table 11-277 Fields for Register: LANEN_ANA_RX_CAL_MUXB

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_cal_muxb_sel	R/W	**Description**: If this bit = 1, it allows analog register cal_muxb_sel_reg<6:2> to act as mux select for RX calibration mux path B 1: Allows analog register cal_muxb_sel_reg<6:2> to act as mux select of RX calibration mux path B 0: Disallows analog register cal_muxb_sel_reg<6:2> to act as mux select of RX calibration mux path B, rather internal state machine controls the function **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6:2	cal_muxb_sel_reg	R/W	**Description**: These analog registers act as mux select in RX calibration mux path B, if bit 7 (ovrd_cal_muxb_sel) = 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
1	ovrd_pwron_taps	R/W	**Description**: If this bit = 1, it allows bit 0 (dfe_taps_en_reg) to enable/disable rx dfe tap2 to tap5 1: Allows bit 0 (dfe_taps_en_reg) to enable/disable rx dfe tap2 to tap5 0: Disallows bit 0 (dfe_taps_en_reg) to enable/disable rx dfe tap2 to tap5, the internal state machine controls this instead **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-277 Fields for Register: LANEN_ANA_RX_CAL_MUXB (Continued)

Bits	Name	Memory Access	Description
0	dfe_taps_en_reg	R/W	**Description**: If bit 1 (ovrd_dfe_taps_en) = 1, this bit enables/disables RX DFE tap2 to tap5 1: Enables RX DFE tap2 to tap5 0: Disables RX DFE tap2 to tap5 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.274 LANEN_ANA_RX_TERM

- **Description:** RX_TERM
- **Size:** 16 bits
- **Offset:** 0x1Nd6
- **Exists:** Always

Table 11-278 Fields for Register: LANEN_ANA_RX_TERM

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_rx_term_dc_en	R/W	**Description**: If asserted the DC termination enable value is controlled via registers **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	rx_term_dc_en_reg	R/W	**Description**: If termination override is asserted, controls the DC termination enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	ovrd_rx_term_gd_en	R/W	**Description**: If asserted the ground termination enable value is controlled via registers **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	rx_term_gd_en_reg	R/W	**Description**: If termination override is asserted, controls the ground termination enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	ovrd_iq_phase_adjust	R/W	**Description**: If asserted the iq_phase_adjust value is controlled via registers **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-278 Fields for Register: LANEN_ANA_RX_TERM (Continued)

Bits	Name	Memory Access	Description
2:0	NC20	R/W	**Description**: Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.275 LANEN_ANA_RX_SLC_CTRL

- **Description:** RX_SLC_CTRL
- **Size:** 16 bits
- **Offset:** 0x1Nd7
- **Exists:** Always

Table 11-279 Fields for Register: LANEN_ANA_RX_SLC_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	rx_slicer_ctrl_e_reg	R/W	**Description**: If ovrd_rx_slicer_ctrl_reg is asserted, controls the even slicer configuration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
3:0	rx_slicer_ctrl_o_reg	R/W	**Description**: If ovrd_rx_slicer_ctrl_reg is asserted, controls the odd slicer configuration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.276 LANEN_ANA_RX_ATB_VREG

- **Description:** RX_ATB_VREG
- **Size:** 16 bits
- **Offset:** 0x1Nd8
- **Exists:** Always

Table 11-280 Fields for Register: LANEN_ANA_RX_ATB_VREG

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	ovrd_regref_iqc	R/W	**Description**: If asserted, main PMIX regulator reference is overridden through atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ovrd_regref_iqc_scope	R/W	**Description**: If asserted, scope PMIX regulator reference is overridden through atb_s_m **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	meas_atb_vreg_dfe	R/W	**Description**: If asserted, measure RX fast DFE regulator output voltage through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	meas_atb_vreg_dfe_byp	R/W	**Description**: If asserted, measure RX fast DFE bypass regulator output voltage through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	meas_atb_vreg_iqc	R/W	**Description**: If asserted, measure RX fast main PMIX regulator output voltage through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-280 Fields for Register: LANEN_ANA_RX_ATB_VREG (Continued)

Bits	Name	Memory Access	Description
2	meas_atb_vreg_iqc_scope	R/W	**Description**: If asserted, measure RX fast scope PMIX regulator output voltage through atb_s_p **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	ovrd_iqc_vref_sel	R/W	**Description**: If asserted the vref on the iqc regulator is controlled via registers **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ovrd_rx_slicer_ctrl_reg	R/W	**Description**: If asserted the slicer configuration value is controlled via registers **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.277 LANEN_ANA_RX_CDR_VCO_CTRL

- **Description:** Setting to control the voltage regulator for the RX_CDR_VCO block
- **Size:** 16 bits
- **Offset:** 0x1Nd9
- **Exists:** Always

Table 11-281 Fields for Register: LANEN_ANA_RX_CDR_VCO_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	creg_cdr_vco_misc_ctrl	R/W	**Description**: Enable/disable functions: bit 3: Bit is not used. bit 2: Bit is not used. bit 1: Bit is not used bit 0: Turns off cascode in ibias main branch when asserted **Value After Reset**: 0xe **Exists**: Always **Reset Mask**: 0xf
3:2	creg_cdr_vco_lowfreq	R/W	**Description**: Description: bit 1: Shuts off part of ibias current mirrors when asserted bit 0: Engages varactor in oscillator when asserted **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3
1:0	afe_att_cm_reg_1_0	R/W	**Description**: afe_att_cm_reg(1:0) Function: 00: 770mV 01: 730mV (Default) 10: 688mV 11: 646mV **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x3

11.2.278 RAWCMN_DIG_CMN_CTL

- **Description:** Common control register
- **Size:** 16 bits
- **Offset:** 0x2000
- **Exists:** Always

Table 11-282 Fields for Register: RAWCMN_DIG_CMN_CTL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	PHY_FUNC_RST	R/W	**Description**: Resets the PHY except registers in the Raw PCS common and always-on registers. Useful for resetting the PHY after reloading the Memory and without resetting the Memory. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.279 RAWCMN_DIG_MPLLA_BW_OVRD_IN

- **Description:** Override values for incoming MPLLA bandwidth
- **Size:** 16 bits
- **Offset:** 0x2001
- **Exists:** Always

Table 11-283 Fields for Register: RAWCMN_DIG_MPLLA_BW_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_OVRD_VAL	R/W	**Description**: Override value for mplla_bandwidth[15:0] **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0xffff

11.2.280 RAWCMN_DIG_MPLLA_SSC_CTL_OVRD_IN

- **Description:** Override values for incoming MPLLA SSC control settings
- **Size:** 16 bits
- **Offset:** 0x2002
- **Exists:** Always

Table 11-284 Fields for Register: RAWCMN_DIG_MPLLA_SSC_CTL_OVRD_IN

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	MPLLA_SSC_CTL_OVRD_EN	R/W	**Description**: Override value for MPLLA SSC control settings **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	MPLLA_SSC_CLK_SEL	R/W	**Description**: Override value for mplla_ssc_clk_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	MPLLA_SSC_UP_SPREAD	R/W	**Description**: Override value for mplla_ssc_up_spread **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:0	MPLLA_FRACN_CTRL	R/W	**Description**: Override value for mplla_fracn_ctrl[10:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7ff

11.2.281 RAWCMN_DIG_MPLLA_SSC_EN_OVRD_IN

- **Description:** Override values for incoming MPLLA SSC Input Signals
- **Size:** 16 bits
- **Offset:** 0x2003
- **Exists:** Always

Table 11-285 Fields for Register: RAWCMN_DIG_MPLLA_SSC_EN_OVRD_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	MPLLA_SSC_EN_OVRD_EN	R/W	**Description**: Override enable for MPLLA SSC enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	MPLLA_SSC_EN_OVRD_VAL	R/W	**Description**: Override value for MPLLA SSC enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	MPLLA_SSC_FREQ_CNT_INIT_EN	R/W	**Description**: Override value for mplla_ssc_freq_cnt_init **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:0	MPLLA_SSC_FREQ_CNT_INIT	R/W	**Description**: Override value for mplla_ssc_freq_cnt_init **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xfff

11.2.282 RAWCMN_DIG_MPLLA_MISC_OVRD_IN

- **Description:** Override values for incoming MPLLA-related Input Signals
- **Size:** 16 bits
- **Offset:** 0x2004
- **Exists:** Always

Table 11-286 Fields for Register: RAWCMN_DIG_MPLLA_MISC_OVRD_IN

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	MPLLA_OVRD_EN	R/W	**Description**: Override enable for mplla_bandwidth and mplla_tx_clk_div **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:9	MPLLA_TX_CLK_DIV_OVRD_VAL	R/W	**Description**: Override value for MPLLA SSC enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
8	MPLLA_SSC_FREQ_CNT_PEAK_EN	R/W	**Description**: Override value for mplla_ssc_freq_cnt_peak **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_SSC_FREQ_CNT_PEAK	R/W	**Description**: Override value for mplla_ssc_freq_cnt_peak[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.283 RAWCMN_DIG_MPLLB_BW_OVRD_IN

- **Description:** Override values for incoming MPLLB bandwidth
- **Size:** 16 bits
- **Offset:** 0x2005
- **Exists:** Always

Table 11-287 Fields for Register: RAWCMN_DIG_MPLLB_BW_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_OVRD_VAL	R/W	**Description**: Override value for mpllb_bandwidth[15:0] **Value After Reset**: 0x43 **Exists**: Always **Reset Mask**: 0xffff

11.2.284 RAWCMN_DIG_MPLLB_SSC_CTL_OVRD_IN

- **Description:** Override values for incoming MPLLB SSC control settings
- **Size:** 16 bits
- **Offset:** 0x2006
- **Exists:** Always

Table 11-288 Fields for Register: RAWCMN_DIG_MPLLB_SSC_CTL_OVRD_IN

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	MPLLB_SSC_CTL_OVRD_EN	R/W	**Description**: Override value for MPLLB SSC control settings **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	MPLLB_SSC_CLK_SEL	R/W	**Description**: Override value for mpllb_ssc_clk_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	MPLLB_SSC_UP_SPREAD	R/W	**Description**: Override value for mpllb_ssc_up_spread **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:0	MPLLB_FRACN_CTRL	R/W	**Description**: Override value for mpllb_fracn_ctrl[10:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7ff

11.2.285 RAWCMN_DIG_MPLLB_SSC_EN_OVRD_IN

- **Description:** Override values for incoming MPLLB SSC Input Signals
- **Size:** 16 bits
- **Offset:** 0x2007
- **Exists:** Always

Table 11-289 Fields for Register: RAWCMN_DIG_MPLLB_SSC_EN_OVRD_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	MPLLB_SSC_EN_OVRD_EN	R/W	**Description**: Override enable for MPLLB SSC enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	MPLLB_SSC_EN_OVRD_VAL	R/W	**Description**: Override value for MPLLB SSC enable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	MPLLB_SSC_FREQ_CNT_INIT_EN	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_init **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:0	MPLLB_SSC_FREQ_CNT_INIT	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_init **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xfff

11.2.286 RAWCMN_DIG_MPLLB_MISC_OVRD_IN

- **Description:** Override values for incoming MPLLB-related Input Signals
- **Size:** 16 bits
- **Offset:** 0x2008
- **Exists:** Always

Table 11-290 Fields for Register: RAWCMN_DIG_MPLLB_MISC_OVRD_IN

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	MPLLB_OVRD_EN	R/W	**Description**: Override enable for mpllb_bandwidth and mpllb_tx_clk_div **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10:9	MPLLB_TX_CLK_DIV_OVRD_VAL	R/W	**Description**: Override value for mpllb_tx_clk_div **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
8	MPLLB_SSC_FREQ_CNT_PEAK_EN	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_peak **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_SSC_FREQ_CNT_PEAK	R/W	**Description**: Override value for mpllb_ssc_freq_cnt_peak[7:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.287 RAWCMN_DIG_LANE_FSM_OP_XTND

- **Description:** Lane FSM OP XTND control register
- **Size:** 16 bits
- **Offset:** 0x2009
- **Exists:** Always

Table 11-291 Fields for Register: RAWCMN_DIG_LANE_FSM_OP_XTND

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	VAL	R/W	**Description**: Required when accessing the analog CREGs in firmware to prevent timing violations across ANA-DIG interface. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.288 RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0

- **Description:** Override values for incoming TX term offset
- **Size:** 16 bits
- **Offset:** 0x200a
- **Exists:** Always

Table 11-292 Fields for Register: RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	TXUP_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override Enable for txup_term_offset[8:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:0	TXUP_TERM_OFFSET_OVRD_VAL	R/W	**Description**: Override value for txup_term_offset[8:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.289 RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1

- **Description:** Override values for incoming TX/RX term offset
- **Size:** 16 bits
- **Offset:** 0x200b
- **Exists:** Always

Table 11-293 Fields for Register: RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1

Bits	Name	Memory Access	Description
15	RX_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override enable for rx_term_offset[4:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:10	RX_TERM_OFFSET_OVRD_VAL	R/W	**Description**: Override value for rx_term_offset[4:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
9	TXDN_TERM_OFFSET_OVRD_EN	R/W	**Description**: Override enable for txdn_term_offset[8:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:0	TXDN_TERM_OFFSET_OVRD_VAL	R/W	**Description**: Override value for txdn_term_offset[8:0] **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.290 RAWCMN_DIG_CMN_CTL_1

- **Description:** Common control register 1
- **Size:** 16 bits
- **Offset:** 0x200c
- **Exists:** Always

Table 11-294 Fields for Register: RAWCMN_DIG_CMN_CTL_1

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5	RTUNE_REQ_OVRD_EN	R/W	**Description**: Override enable for rtune_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RTUNE_REQ_OVRD_VAL	R/W	**Description**: Override value for rtune_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLB_INIT_CAL_DISABLE_OVRD_EN	R/W	**Description**: Override enable for mpllb_init_cal_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_INIT_CAL_DISABLE_OVRD_VAL	R/W	**Description**: Override value for mpllb_init_cal_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLA_INIT_CAL_DISABLE_OVRD_EN	R/W	**Description**: Override enable for mplla_init_cal_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_INIT_CAL_DISABLE_OVRD_VAL	R/W	**Description**: Override value for mplla_init_cal_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.291 RAWCMN_DIG_MPLL_OFF_TIME

- **Description:** Wait Time for turning MPLL off
- **Size:** 16 bits
- **Offset:** 0x200d
- **Exists:** Always

Table 11-295 Fields for Register: RAWCMN_DIG_MPLL_OFF_TIME

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	MPLLA_OFF_TIME	R/W	**Description**: Wait Counter for turning off MPLL **Value After Reset**: 0x1e **Exists**: Always **Reset Mask**: 0x3f

11.2.292 RAWCMN_DIG_ATE_ALU_CTRL

- **Description:** ATE ALU module Control Register
- **Size:** 16 bits
- **Offset:** 0x200e
- **Exists:** Always

Table 11-296 Fields for Register: RAWCMN_DIG_ATE_ALU_CTRL

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3:0	ATE_ALU_OPCODE	R/W	**Description**: OPCode for ATE ALU Module Commands Encoding ZERO 4'b0000 ADD_RD_DATA 4'b0001 ADD_DATA 4'b0010 SHL1 4'b0011 SHR1 4'b0100 ASHR1 4'b0101 INV 4'b0110 NEG 4'b0111 ABS 4'b1000 READ 4'b1110 WRITE 4'b1111 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.293 RAWCMN_DIG_ATE_ALU_ADDR

- **Description:** Address for Read and Write Command for ATE ALU Module
- **Size:** 16 bits
- **Offset:** 0x200f
- **Exists:** Always

Table 11-297 Fields for Register: RAWCMN_DIG_ATE_ALU_ADDR

Bits	Name	Memory Access	Description
15:0	ADDR	R/W	**Description**: Address for read and write command for ATE ALU module **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.294 RAWCMN_DIG_ATE_ALU_DATA

- **Description:** Data for Write Command for ATE ALU Module
- **Size:** 16 bits
- **Offset:** 0x2010
- **Exists:** Always

Table 11-298 Fields for Register: RAWCMN_DIG_ATE_ALU_DATA

Bits	Name	Memory Access	Description
15:0	DATA	R/W	**Description**: Data for read and write command for ATE ALU module **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.295 RAWCMN_DIG_ATE_ALU_FLAGS

- **Description:** Flags from ATE ALU Module
- **Size:** 16 bits
- **Offset:** 0x2011
- **Exists:** Always

Table 11-299 Fields for Register: RAWCMN_DIG_ATE_ALU_FLAGS

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ATE_ALU_ACCUM_SIGN	R	**Description**: Indicate the sign of the accumulator in ATE ALU (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.296 RAWCMN_DIG_ATE_ALU_ACCUM

- **Description:** Stores the ATE ALU Accumulator result
- **Size:** 16 bits
- **Offset:** 0x2012
- **Exists:** Always

Table 11-300 Fields for Register: RAWCMN_DIG_ATE_ALU_ACCUM

Bits	Name	Memory Access	Description
15:0	VAL	R	**Description**: Stores the ATE ALU Accumulator result (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.297 RAWCMN_DIG_MPLL_IN

- **Description:** Current values for incoming MPLL controls from PCS (pre-override Input Monitor)
- **Size:** 16 bits
- **Offset:** 0x2013
- **Exists:** Always

Table 11-301 Fields for Register: RAWCMN_DIG_MPLL_IN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	MPLLB_INIT_CAL_DISABLE	R	**Description**: Value from PCS for mpllb_init_cal_disable (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLA_INIT_CAL_DISABLE	R	**Description**: Value from PCS for mplla_init_cal_disable (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.298 RAWCMN_DIG_FW_PWRUP_DONE

- **Description:** Firmware Power-up Done Status
- **Size:** 16 bits
- **Offset:** 0x2014
- **Exists:** Always

Table 11-302 Fields for Register: RAWCMN_DIG_FW_PWRUP_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	DONE	R/W	**Description**: Firmware Power-up Status for phy_reset and pg_reset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.299 RAWCMN_DIG_MPLL_CLK_ASYNC_EN

- **Description:** Controls the MPLL Output Clcoks Asynchronously
- **Size:** 16 bits
- **Offset:** 0x2015
- **Exists:** Always

Table 11-303 Fields for Register: RAWCMN_DIG_MPLL_CLK_ASYNC_EN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	MPLLB_CLK_EN	R/W	**Description**: Asynchronously enables the MPLLB output clocks. This can only be changed if the MPLLB are off to avoid the glitch on the MPLLB output clocks **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_CLK_EN	R/W	**Description**: Asynchronously enables the MPLLA output clocks This can only be changed if the MPLLA is off to avoid the glitch on the MPLLA output clocks **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.300 RAWCMN_DIG_ATE_ALU_SPARE_0

- **Description:** Spare register #1 for ATE ALU operations
- **Size:** 16 bits
- **Offset:** 0x2016
- **Exists:** Always

Table 11-304 Fields for Register: RAWCMN_DIG_ATE_ALU_SPARE_0

Bits	Name	Memory Access	Description
15:0	DATA	R/W	**Description**: Spare for ATE ALU operations **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.301 RAWCMN_DIG_ATE_ALU_SPARE_1

- **Description:** Spare register #2 for ATE ALU operations
- **Size:** 16 bits
- **Offset:** 0x2017
- **Exists:** Always

Table 11-305 Fields for Register: RAWCMN_DIG_ATE_ALU_SPARE_1

Bits	Name	Memory Access	Description
15:0	DATA	R/W	**Description**: Spare for ATE ALU operations **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.302 RAWCMN_DIG_MPLL_STATE_OVRD_IN

- **Description:** Override for MPLLA/B State Outputs
- **Size:** 16 bits
- **Offset:** 0x2018
- **Exists:** Always

Table 11-306 Fields for Register: RAWCMN_DIG_MPLL_STATE_OVRD_IN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	MPLLB_STATE_OVRD_EN	R/W	**Description**: Override enable for mpllb_state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_STATE_OVRD_VAL	R/W	**Description**: Override value for mpllb_state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLA_STATE_OVRD_EN	R/W	**Description**: Override enable for mplla_state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_STATE_OVRD_VAL	R/W	**Description**: Override value for mplla_state **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.303 RAWCMN_DIG_AON_SRAM_PGATE_BL_EN

- **Description:** Enable SRAM bootloader on power-gated exit
- **Size:** 16 bits
- **Offset:** 0x2020
- **Exists:** Always

Table 11-307 Fields for Register: RAWCMN_DIG_AON_SRAM_PGATE_BL_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SRAM_PGATE_BL_EN	R/W	**Description**: Enable SRAM bootloader on power-gated exit. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.304 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_0

- **Description:** MPLLA Coarse Tune Value for Bank 0
- **Size:** 16 bits
- **Offset:** 0x2021
- **Exists:** Always

Table 11-308 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLA_COARSE_TUNE_DONE_BANK_0	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_COARSE_TUNE_BANK_0	R/W	**Description**: Stored bank0 coarse tune value for MPLLA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.305 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_1

- **Description:** MPLLA Coarse Tune Value for Bank 1
- **Size:** 16 bits
- **Offset:** 0x2022
- **Exists:** Always

Table 11-309 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLA_COARSE_TUNE_DONE_BANK_1	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_COARSE_TUNE_BANK_1	R/W	**Description**: Stored bank 1 coarse tune value for MPLLA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.306 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_2

- **Description:** MPLLA Coarse Tune Value for Bank 2
- **Size:** 16 bits
- **Offset:** 0x2023
- **Exists:** Always

Table 11-310 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLA_COARSE_TUNE_DONE_BANK_2	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_COARSE_TUNE_BANK_2	R/W	**Description**: Stored bank 2 coarse tune value for MPLLA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.307 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_3

- **Description:** MPLLA Coarse Tune Value for Bank 3
- **Size:** 16 bits
- **Offset:** 0x2024
- **Exists:** Always

Table 11-311 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_BANK_3

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLA_COARSE_TUNE_DONE_BANK_3	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLA_COARSE_TUNE_BANK_3	R/W	**Description**: Stored bank 3 coarse tune value for MPLLA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.308 RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL

- **Description:** Value for MPLLA Bank Select
- **Size:** 16 bits
- **Offset:** 0x2025
- **Exists:** Always

Table 11-312 Fields for Register: RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	MPLLA_CAL_BANK_SEL	R	**Description**: Value for MPLLA Bank select (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.309 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_DONE

- **Description:** Indicates if the Selected MPLLA Calibration bank has the valid calibrated value
- **Size:** 16 bits
- **Offset:** 0x2026
- **Exists:** Always

Table 11-313 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLLA_COARSE_TUNE_DONE	R	**Description**: Indicates if the selected MPLLA Calibration bank has the valid calibrated value (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.310 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_0

- **Description:** MPLLB Coarse Tune Value for Bank 0
- **Size:** 16 bits
- **Offset:** 0x2027
- **Exists:** Always

Table 11-314 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLB_COARSE_TUNE_DONE_BANK_0	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_COARSE_TUNE_BANK_0	R/W	**Description**: Stored bank0 coarse tune value for MPLLB **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.311 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_1

- **Description:** MPLLB Coarse Tune Value for Bank 1
- **Size:** 16 bits
- **Offset:** 0x2028
- **Exists:** Always

Table 11-315 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLB_COARSE_TUNE_DONE_BANK_1	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_COARSE_TUNE_BANK_1	R/W	**Description**: Stored bank 1 coarse tune value for MPLLB **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.312 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_2

- **Description:** MPLLB Coarse Tune Value for Bank 2
- **Size:** 16 bits
- **Offset:** 0x2029
- **Exists:** Always

Table 11-316 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLB_COARSE_TUNE_DONE_BANK_2	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_COARSE_TUNE_BANK_2	R/W	**Description**: Stored bank 2 coarse tune value for MPLLB **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.313 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_3

- **Description:** MPLLB Coarse Tune Value for Bank 3
- **Size:** 16 bits
- **Offset:** 0x202a
- **Exists:** Always

Table 11-317 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_BANK_3

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	MPLLB_COARSE_TUNE_DONE_BANK_3	R/W	**Description**: Indicates if the coarse tune value is calibrated value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:0	MPLLB_COARSE_TUNE_BANK_3	R/W	**Description**: Stored bank 3 coarse tune value for MPLLB **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.314 RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL

- **Description:** Value for MPLLB Bank Select
- **Size:** 16 bits
- **Offset:** 0x202b
- **Exists:** Always

Table 11-318 Fields for Register: RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	MPLLB_CAL_BANK_SEL	R	**Description**: Value for MPLL B Bank select (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.315 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_DONE

- **Description:** Indicates if the selected MPLLB Calibration bank has the valid calibrated value
- **Size:** 16 bits
- **Offset:** 0x202c
- **Exists:** Always

Table 11-319 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLLB_COARSE_TUNE_DONE	R	**Description**: Indicates if the selected MPLLB Calibration bank has the valid calibrated value (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.316 RAWCMN_DIG_AON_MPLL_CAL_BANK_SEL_OVRD

- **Description:** Override the incoming mplla/b_recal_bank_sel input
- **Size:** 16 bits
- **Offset:** 0x202d
- **Exists:** Always

Table 11-320 Fields for Register: RAWCMN_DIG_AON_MPLL_CAL_BANK_SEL_OVRD

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5	MPLLB_CAL_BANK_SEL_OVRD_EN	R/W	**Description**: Enable override for mplla_cal_bank_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4:3	MPLLB_CAL_BANK_SEL_OVRD_VAL	R/W	**Description**: Override value for incoming input mpllb_recal_bank_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
2	MPLLA_CAL_BANK_SEL_OVRD_EN	R/W	**Description**: Enable override for mplla_recal_bank_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	MPLLA_CAL_BANK_SEL_OVRD_VAL	R/W	**Description**: Override value for incoming input mplla_recal_bank_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.317 RAWCMN_DIG_AON_MPLLA_COARSE_TUNE

- **Description:** Selected COARSE TUNE Value for MPLLA
- **Size:** 16 bits
- **Offset:** 0x202e
- **Exists:** Always

Table 11-321 Fields for Register: RAWCMN_DIG_AON_MPLLA_COARSE_TUNE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLLA_COARSE_TUNE	R	**Description**: Selected coarse tune value for MPLLA (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.318 RAWCMN_DIG_AON_MPLLB_COARSE_TUNE

- **Description:** Selected COARSE TUNE Value for MPLLB
- **Size:** 16 bits
- **Offset:** 0x202f
- **Exists:** Always

Table 11-322 Fields for Register: RAWCMN_DIG_AON_MPLLB_COARSE_TUNE

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	MPLLB_COARSE_TUNE	R	**Description**: Selected coarse tune value for MPLLB (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.319 RAWCMN_DIG_AON_MPLLA_IN_RECAL

- **Description:** Indicates if MPLLA is in Re-calibration
- **Size:** 16 bits
- **Offset:** 0x2030
- **Exists:** Always

Table 11-323 Fields for Register: RAWCMN_DIG_AON_MPLLA_IN_RECAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLLA_IN_RECAL	R/W	**Description**: Indicates if MPLLA is in re-calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.320 RAWCMN_DIG_AON_MPLLB_IN_RECAL

- **Description:** Indicates if MPLLB is in Re-calibration
- **Size:** 16 bits
- **Offset:** 0x2031
- **Exists:** Always

Table 11-324 Fields for Register: RAWCMN_DIG_AON_MPLLB_IN_RECAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLLB_IN_RECAL	R/W	**Description**: Indicates if MPLLB is in re-calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.321 RAWCMN_DIG_AON_PG_OVRD_IN

- **Description:** Override values for incoming power-gating signals
- **Size:** 16 bits
- **Offset:** 0x2032
- **Exists:** Always

Table 11-325 Fields for Register: RAWCMN_DIG_AON_PG_OVRD_IN

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5	PG_MODE_EN_OVRD_EN	R/W	**Description**: Override enable for pg_mode_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	PG_MODE_EN_OVRD_VAL	R/W	**Description**: Override value for pg_mode_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	PG_RESET_OVRD_EN	R/W	**Description**: Override enable for pg_reset. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	PG_RESET_OVRD_VAL	R/W	**Description**: Override value for pg_reset. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	PCS_PWR_STABLE_OVRD	R/W	**Description**: Enable overriding value for pcs_pwr_stable. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	PMA_PWR_STABLE_OVRD	R/W	**Description**: Enable overriding value for pma_pwr_stable. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.322 RAWCMN_DIG_AON_PG_OVRD_OUT

- **Description:** Override values for outgoing power-gating signals
- **Size:** 16 bits
- **Offset:** 0x2033
- **Exists:** Always

Table 11-326 Fields for Register: RAWCMN_DIG_AON_PG_OVRD_OUT

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	PCS_PWR_EN_OVRD	R/W	**Description**: Enable overriding value for pcs_pwr_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	PMA_PWR_EN_OVRD	R/W	**Description**: Enable overriding value for pma_pwr_en. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.323 RAWCMN_DIG_AON_SUP_OVRD_IN

- **Description:** Override Values for Incoming Support Block signals
- **Size:** 16 bits
- **Offset:** 0x2034
- **Exists:** Always

Table 11-327 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_IN

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5	REF_CLK_EN_OVRD_EN	R/W	**Description**: Override enable for ref_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	REF_CLK_EN_OVRD_VAL	R/W	**Description**: Override value for ref_clk_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	MPLLB_FORCE_EN_OVRD_EN	R/W	**Description**: Override enable for mpllb_force_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_FORCE_EN_OVRD_VAL	R/W	**Description**: Override value for mpllb_force_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLA_FORCE_EN_OVRD_EN	R/W	**Description**: Override enable for mplla_force_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_FORCE_EN_OVRD_VAL	R/W	**Description**: Override value for mplla_force_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.324 RAWCMN_DIG_AON_SUP_OVRD_OUT

- **Description:** Override Values for Out-going Support Block signals
- **Size:** 16 bits
- **Offset:** 0x2035
- **Exists:** Always

Table 11-328 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_OUT

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	MPLLB_FORCE_ACK_OVRD_EN	R/W	**Description**: Override enable for output mpllb_force_ack. The output(s) overridden by it: mpllb_force_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	MPLLB_FORCE_ACK_OVRD_VAL	R/W	**Description**: Override value for output mpllb_force_ack. Its override enable is: SUP_OVRD_OUT.MPLLB_FORCE_ACK_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLA_FORCE_ACK_OVRD_EN	R/W	**Description**: Override enable for output mplla_force_ack. The output(s) overridden by it: mplla_force_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_FORCE_ACK_OVRD_VAL	R/W	**Description**: Override value for output mplla_force_ack. Its override enable is: SUP_OVRD_OUT.MPLLA_FORCE_ACK_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.325 RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL

- **Description:** Current Bank selected for MPLLA Coarse Tune in PMA
- **Size:** 16 bits
- **Offset:** 0x2036
- **Exists:** Always

Table 11-329 Fields for Register: RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	PMA_MPLLA_RECAL_BANK_SEL	R/W	**Description**: Current Bank selected for MPLLA coarse tune in PMA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.326 RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL

- **Description:** Current Bank selected for MPLLB Coarse Tune in PMA
- **Size:** 16 bits
- **Offset:** 0x2037
- **Exists:** Always

Table 11-330 Fields for Register: RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	PMA_MPLLB_RECAL_BANK_SEL	R/W	**Description**: Current Bank selected for MPLLB coarse tune in PMA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.327 RAWCMN_DIG_AON_RTUNE_RX_VAL

- **Description:** Resistor Tune RX Value
- **Size:** 16 bits
- **Offset:** 0x2038
- **Exists:** Always

Table 11-331 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RTUNE_RX_VAL	R/W	**Description**: Stored resister tune RX value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3f

11.2.328 RAWCMN_DIG_AON_RTUNE_TXDN_VAL

- **Description:** Resistor Tune TX Down Value
- **Size:** 16 bits
- **Offset:** 0x2039
- **Exists:** Always

Table 11-332 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	RTUNE_TXDN_VAL	R/W	**Description**: Stored resister tune TX down value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.329 RAWCMN_DIG_AON_RTUNE_TXUP_VAL

- **Description:** Resistor Tune TX Up Value
- **Size:** 16 bits
- **Offset:** 0x203a
- **Exists:** Always

Table 11-333 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	RTUNE_TXUP_VAL	R/W	**Description**: Stored resister tune TX up value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.330 RAWCMN_DIG_AON_CMNCAL_STATUS

- **Description:** Common Calibration Status
- **Size:** 16 bits
- **Offset:** 0x203b
- **Exists:** Always

Table 11-334 Fields for Register: RAWCMN_DIG_AON_CMNCAL_STATUS

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	CMNCAL_DONE	R/W	**Description**: Indicates whether common calibration (mpll and rtune) has been completed by this lane or not. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	CMNCAL_INIT	R/W	**Description**: Indicates whether common calibration (mpll and rtune) has been started by this lane or not. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.331 RAWCMN_DIG_AON_SRAM_OVRD_IN

- **Description:** Override for incoming SRAM inputs
- **Size:** 16 bits
- **Offset:** 0x203c
- **Exists:** Always

Table 11-335 Fields for Register: RAWCMN_DIG_AON_SRAM_OVRD_IN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	SRAM_BYPASS_OVRD_EN	R/W	**Description**: Override enable for input sram_bypass **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	SRAM_BYPASS_OVRD_VAL	R/W	**Description**: Override value for input sram_bypass **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	SRAM_EXT_LD_DONE_OVRD_EN	R/W	**Description**: Override enable for input sram_ext_ld_done **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	SRAM_EXT_LD_DONE_OVRD_VAL	R/W	**Description**: Override value for input sram_ext_ld_done **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.332 RAWCMN_DIG_AON_SRAM_IN

- **Description:** Monitor for SRAM inputs
- **Size:** 16 bits
- **Offset:** 0x203d
- **Exists:** Always

Table 11-336 Fields for Register: RAWCMN_DIG_AON_SRAM_IN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	SRAM_EXT_LD_DONE	R	**Description**: Value from RAW PCS for sram_ext_ld_done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	SRAM_BYPASS	R	**Description**: Value from RAW PCS for sram_bypass (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.333 RAWCMN_DIG_AON_SRAM_OUT

- **Description:** Monitor for SRAM outputs
- **Size:** 16 bits
- **Offset:** 0x203e
- **Exists:** Always

Table 11-337 Fields for Register: RAWCMN_DIG_AON_SRAM_OUT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SRAM_INIT_DONE	R	**Description**: Value from RAW PCS for sram_init_done (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.334 RAWCMN_DIG_AON_MPLLA_BANK_SEL_DONE

- **Description:** Status for MPLLA Re-calibration or Switching
- **Size:** 16 bits
- **Offset:** 0x2040
- **Exists:** Always

Table 11-338 Fields for Register: RAWCMN_DIG_AON_MPLLA_BANK_SEL_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	DONE	R	**Description**: Status for MPLLA re-calibration or Switching (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.335 RAWCMN_DIG_AON_MPLLB_BANK_SEL_DONE

- **Description:** Status for MPLLB Re-calibration or Switching
- **Size:** 16 bits
- **Offset:** 0x2041
- **Exists:** Always

Table 11-339 Fields for Register: RAWCMN_DIG_AON_MPLLB_BANK_SEL_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	DONE	R	**Description**: Status for MPLLB re-calibration or Switching (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.336 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN

- **Description:** Override values for incoming TX controls from PCS
- **Size:** 16 bits
- **Offset:** 0x3000
- **Exists:** Always

Table 11-340 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	OVRD_EN	R/W	**Description**: Override enable for all input signals below Input signals overridden by it: tx_master_mpllb_state tx_master_mplla_state tx_mpll_en tx_mpllb_sel tx_rate tx_width tx_lpd tx_pstate **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	MSTR_MPLLB_STATE	R/W	**Description**: Override value for tx_master_mpllb_state. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	MSTR_MPLLA_STATE	R/W	**Description**: Override value for tx_master_mplla_state. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-340 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN (Continued)

Bits	Name	Memory Access	Description
9	MPLL_EN	R/W	**Description**: Override value for tx_mpll_en. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	MPLLB_SEL	R/W	**Description**: Override value for tx_mpllb_sel. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7:5	RATE	R/W	**Description**: Override value for tx_rate. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x7
4:3	WIDTH	R/W	**Description**: Override value for tx_width. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3
2	LPD	R/W	**Description**: Override value for tx_lpd. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	PSTATE	R/W	**Description**: Override value for tx_pstate. Its override enable is: TX_OVRD_IN.OVRD_EN **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3

11.2.337 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1

- **Description:** Override values for incoming TX controls from PCS, register #1
- **Size:** 16 bits
- **Offset:** 0x3001
- **Exists:** Always

Table 11-341 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	IBOOST_LVL_OVRD_EN	R/W	**Description**: Override enable for tx_iboost_lvl[3:0] Input signal overridden by it: tx_iboost_lvl **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:8	IBOOST_LVL_OVRD_VAL	R/W	**Description**: Override value for tx_iboost_lvl[3:0]. Its override enable is: TX_OVRD_IN_1.IBOOST_LVL_OVRD_EN **Value After Reset**: 0xf **Exists**: Always **Reset Mask**: 0xf
7	VBOOST_EN_OVRD_EN	R/W	**Description**: Override enable for tx_vboost_en. Input signal overridden by it: tx_vboost_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	VBOOST_EN_OVRD_VAL	R/W	**Description**: Override value for tx_vboost_en. Its override enable is: TX_OVRD_IN_1.VBOOST_EN_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-341 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 (Continued)

Bits	Name	Memory Access	Description
5	DETRX_REQ_OVRD_EN	R/W	**Description**: Override enable for tx_detrx_req Input signal overridden by it: tx_detrx_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	DETRX_REQ_OVRD_VAL	R/W	**Description**: Override value for tx_detrx_req. Its override enable is: TX_OVRD_IN_1.DETRX_REQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	REQ_OVRD_EN	R/W	**Description**: Override enable for tx_req. Input signal overridden by it: tx_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	REQ_OVRD_VAL	R/W	**Description**: Override value for tx_req. Its override enable is: TX_OVRD_IN_1.REQ_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RESET_OVRD_EN	R/W	**Description**: Override enable for tx_reset Input signal overridden by it: tx_reset **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RESET_OVRD_VAL	R/W	**Description**: Override value for tx_reset. Its override enable is: TX_OVRD_IN_1.RESET_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.338 RAWLANEN_DIG_PCS_XF_TX_PCS_IN

- **Description:** Current values for incoming TX controls from PCS (pre-override Input Monitor)
- **Size:** 16 bits
- **Offset:** 0x3002
- **Exists:** Always

Table 11-342 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	DETRX_REQ	R	**Description**: Value from PCS for tx_detrx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	MSTR_MPLLA_STATE	R	**Description**: Value from PCS for tx_master_mplla_state (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	MSTR_MPLLB_STATE	R	**Description**: Value from PCS for tx_master_mpllb_state (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	MPLL_EN	R	**Description**: Value from PCS for tx_mpll_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-342 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN (Continued)

Bits	Name	Memory Access	Description
10	MPLLB_SEL	R	**Description**: Value from PCS for tx_mpllb_sel (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9:7	RATE	R	**Description**: Value from PCS for tx_rate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6:5	WIDTH	R	**Description**: Value from PCS for tx_width (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4	LPD	R	**Description**: Value from PCS for tx_lpd (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
3:2	PSTATE	R	**Description**: Value from PCS for tx_pstate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	REQ	R	**Description**: Value from PCS for tx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-342 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN (Continued)

Bits	Name	Memory Access	Description
0	RESET	R	**Description**: Value from PCS for tx_reset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.339 RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT

- **Description:** Override values for outgoing TX controls to PCS
- **Size:** 16 bits
- **Offset:** 0x3003
- **Exists:** Always

Table 11-343 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	EN_CTL	R/W	**Description**: Enable override values for all control outputs of this register Output Signal overridden by it: tx_detrx_result tx_ack **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	DETRX_RESULT	R/W	**Description**: Override value for tx_detrx_result. Its override enable is: TX_OVRD_OUT.EN_CTL **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	ACK	R/W	**Description**: Override value for tx_ack. Its override enable is: TX_OVRD_OUT.EN_CTL **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.340 RAWLANEN_DIG_PCS_XF_TX_PCS_OUT

- **Description:** Current values for outgoing TX status controls from Raw PCS (pre-override Output Monitor)
- **Size:** 16 bits
- **Offset:** 0x3004
- **Exists:** Always

Table 11-344 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_OUT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ACK	R	**Description**: Value from Raw PCS for tx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.341 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN

- **Description:** Override values for incoming RX controls from PCS
- **Size:** 16 bits
- **Offset:** 0x3005
- **Exists:** Always

Table 11-345 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	DFE_BYPASS	R/W	**Description**: Override value for rx_dfe_bypass. Its override enable is: RX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	OVRD_EN	R/W	**Description**: Enable override values for all fields in this register Input signals overridden by it: rx_dfe_bypass rx_lpd rx_pstate rx_width rx_rate **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
6	LPD	R/W	**Description**: Override value for rx_lpd. Its override enable is: RX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-345 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN (Continued)

Bits	Name	Memory Access	Description
5:4	PSTATE	R/W	**Description**: Override value for rx_pstate. Its override enable is: RX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
3:2	WIDTH	R/W	**Description**: Override value for rx_width. Its override enable is: RX_OVRD_IN.OVRD_EN **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0x3
1:0	RATE	R/W	**Description**: Override value for rx_rate. Its override enable is: RX_OVRD_IN.OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.342 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1

- **Description:** Override values for incoming RX controls from PCS, register #1
- **Size:** 16 bits
- **Offset:** 0x3006
- **Exists:** Always

Table 11-346 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	REQ_OVRD_EN	R/W	**Description**: Override enable for rx_req Input signals overridden by it: rx_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	REQ_OVRD_VAL	R/W	**Description**: Override value for rx_req. Its override enable is: RX_OVRD_IN_1.REQ_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RESET_OVRD_EN	R/W	**Description**: Override enable for rx_reset Input signals overridden by it: rx_reset **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RESET_OVRD_VAL	R/W	**Description**: Override value for rx_reset. Its override enable is: RX_OVRD_IN_1.RESET_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.343 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2

- **Description:** Override values for incoming RX controls from PCS, register #2
- **Size:** 16 bits
- **Offset:** 0x3007
- **Exists:** Always

Table 11-347 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13	VCO_LD_VAL_OVRD_EN	R/W	**Description**: Enable override for rx_vco_ld_val Input signals overridden by it: rx_vco_ld_val **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12:0	VCO_LD_VAL_OVRD	R/W	**Description**: Override value for rx_vco_ld_val. Its override enable is: RX_OVRD_IN_2.VCO_LD_VAL_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1fff

11.2.344 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2

- **Description:** ATE Override input to control top-level inputs, register #3
- **Size:** 16 bits
- **Offset:** 0x3008
- **Exists:** Always

Table 11-348 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	CONT_OVRD_EN	R/W	**Description**: Enable override values for rx_adapt_cont and rx_offcan_cont Input signals overridden by it: rx_adapt_cont rx_offcan_cont **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	OFFCAN_CONT	R/W	**Description**: Override value for rx_offcan_cont. Its override enable is: ATE_OVRD_IN_2.CONT_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	ADAPT_CONT	R/W	**Description**: Override value for rx_adapt_cont. Its override enable is: ATE_OVRD_IN_2.CONT_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	ADAPT_REQ_OVRD_EN	R/W	**Description**: Enable override values for rx_adapt_req Input signals overridden by it: rx_adapt_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-348 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2 (Continued)

Bits	Name	Memory Access	Description
4	ADAPT_REQ	R/W	**Description**: Override value for rx_adapt_req. Its override enable is: ATE_OVRD_IN_2.ADAPT_REQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RX_LOS_THRSHLD_OVRD_EN	R/W	**Description**: Enable override for rx_los_threshold. Input signals overridden by it: rx_los_threshold **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2:0	RX_LOS_THRSHLD_OVRD_VAL	R/W	**Description**: Override value for rx_los_threshold. Its override enable is: ATE_OVRD_IN_2.RX_LOS_THRSHLD_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.345 RAWLANEN_DIG_PCS_XF_RX_PCS_IN

- **Description:** Current values for incoming RX controls from PCS (Monitor on post ATE override)
- **Size:** 16 bits
- **Offset:** 0x3009
- **Exists:** Always

Table 11-349 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	RESET	R	**Description**: Value from PCS for rx_reset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13	OFFCAN_CONT	R	**Description**: Value from PCS for rx_offcan_cont (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
12	ADAPT_CONT	R	**Description**: Value from PCS for rx_adapt_cont (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
11	ADAPT_REQ	R	**Description**: Value from PCS for rx_adapt_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-349 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN (Continued)

Bits	Name	Memory Access	Description
10	RX_DFE_BYPASS	R	**Description**: Value from PCS for rx_dfe_bypass (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9:8	CDR_VCO_FREQBAND	R	**Description**: Value from PCS for rx_cdr_vco_freqband (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	LPD	R	**Description**: Value from PCS for rx_lpd (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6:5	PSTATE	R	**Description**: Value from PCS for rx_pstate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4:3	WIDTH	R	**Description**: Value from PCS for rx_width (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2:1	RATE	R	**Description**: Value from PCS for rx_rate (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-349 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN (Continued)

Bits	Name	Memory Access	Description
0	REQ	R	**Description**: Value from PCS for rx_req (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.346 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1

- **Description:** Current values for incoming RX controls from PCS, register #1 (Monitor on pre-override signals)
- **Size:** 16 bits
- **Offset:** 0x300a
- **Exists:** Always

Table 11-350 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	REF_LD_VAL	R	**Description**: Value from PCS for rx_ref_ld_val (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.347 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2

- **Description:** Current values for incoming RX controls from PCS, register #2 (Monitor on the pre-override signals)
- **Size:** 16 bits
- **Offset:** 0x300b
- **Exists:** Always

Table 11-351 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:0	VCO_LD_VAL	R	**Description**: Value from PCS for rx_vco_ld_val (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.348 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3

- **Description:** Current values for incoming RX controls from PCS, register #3 ((Monitor on the pre-override signals)
- **Size:** 16 bits
- **Offset:** 0x300c
- **Exists:** Always

Table 11-352 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
13:9	EQ_CTLE_BOOST	R	**Description**: Value from ASIC for rx_eq_ctle_boost (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8:6	EQ_VGA2_GAIN	R	**Description**: Value from ASIC for rx_eq_vga2_gain (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5:3	EQ_VGA1_GAIN	R	**Description**: Value from ASIC for rx_eq_vga1_gain (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2:0	EQ_ATT_LVL	R	**Description**: Value from ASIC for rx_eq_att_lvl (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.349 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4

- **Description:** Current values for incoming RX controls from PCS, register #4 ((Monitor on the pre-override signals)
- **Size:** 16 bits
- **Offset:** 0x300d
- **Exists:** Always

Table 11-353 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:2	EQ_DFE_TAP1	R	**Description**: Value from ASIC for rx_eq_dfe_tap1 (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1:0	EQ_CTLE_POLE	R	**Description**: Value from ASIC for rx_eq_ctle_pole (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.350 RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT

- **Description:** Override values for outgoing RX controls to PCS
- **Size:** 16 bits
- **Offset:** 0x300e
- **Exists:** Always

Table 11-354 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	EN_CTL	R/W	**Description**: Enable override values for all control outputs of this register Output Signal overridden by it rx_ack **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	ACK	R/W	**Description**: Override value for rx_ack Its override enable is RX_OVRD_OUT.EN_CTL **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.351 RAWLANEN_DIG_PCS_XF_RX_PCS_OUT

- **Description:** Current values for outgoing RX status controls from Raw PCS
- **Size:** 16 bits
- **Offset:** 0x300f
- **Exists:** Always

Table 11-355 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_OUT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ACK	R	**Description**: Value from Raw PCS for rx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.352 RAWLANEN_DIG_PCS_XF_RX_ADAPT_ACK

- **Description:** RX Adaptation Acknowledge
- **Size:** 16 bits
- **Offset:** 0x3010
- **Exists:** Always

Table 11-356 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_ADAPT_ACK

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_ACK	R/W	**Description**: RX Adaptation Acknowledge **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.353 RAWLANEN_DIG_PCS_XF_RX_ADAPT_FOM

- **Description:** RX Adaptation Figure of Merit
- **Size:** 16 bits
- **Offset:** 0x3011
- **Exists:** Always

Table 11-357 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_ADAPT_FOM

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ADAPT_FOM	R/W	**Description**: RX Adaptation Figure of Merit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.354 RAWLANEN_DIG_PCS_XF_RX_TXPRE_DIR

- **Description:** RX calculated direction for TX-pre
- **Size:** 16 bits
- **Offset:** 0x3012
- **Exists:** Always

Table 11-358 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXPRE_DIR

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	RX_TXPRE_DIR	R/W	**Description**: RX calculated direction for TX-pre 00 - No change 01 - Increment 10 - Decrement 11 - Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.355 RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DIR

- **Description:** RX calculated direction for TX-Main
- **Size:** 16 bits
- **Offset:** 0x3013
- **Exists:** Always

Table 11-359 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DIR

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	RX_TXMAIN_DIR	R/W	**Description**: RX calculated direction for TX-Main 00 - No change 01 - Increment 10 - Decrement 11 - Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.356 RAWLANEN_DIG_PCS_XF_RX_TXPOST_DIR

- **Description:** RX calculated direction for TX-Post
- **Size:** 16 bits
- **Offset:** 0x3014
- **Exists:** Always

Table 11-360 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXPOST_DIR

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	RX_TXPOST_DIR	R/W	**Description**: RX calculated direction for TX-Post 00 - No change 01 - Increment 10 - Decrement 11 - Reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.357 RAWLANEN_DIG_PCS_XF_LANE_NUMBER

- **Description:** Current lane number
- **Size:** 16 bits
- **Offset:** 0x3015
- **Exists:** Always

Table 11-361 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_NUMBER

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3:0	LANE_NUMBER	R	**Description**: Current lane number (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0

11.2.358 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN

- **Description:** ATE Override input to control top-level inputs, register #1
- **Size:** 16 bits
- **Offset:** 0x3016
- **Exists:** Always

Table 11-362 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN

Bits	Name	Memory Access	Description
15	RX_WIDTH_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_width input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
14:13	RX_WIDTH_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for rx_width input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
12	RX_RATE_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_rate input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:10	RX_RATE_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for rx_rate input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
9	RX_DFE_BYPASS_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_dfe_bypass input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	RX_DFE_BYPASS_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for rx_dfe_bypass input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	TX_REQ_ATE_OVRD_EN	R/W	**Description**: Enable ATE override vale for tx_req input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-362 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (Continued)

Bits	Name	Memory Access	Description
6	TX_REQ_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level tx_req input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_REQ_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_req input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_REQ_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_req input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	TX_RESET_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for tx_reset input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_RESET_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level tx_reset input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_RESET_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_reset input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_RESET_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_reset input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.359 RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1

- **Description:** Overrides the RX Output Clocks during the PHY Initialization, register #1
- **Size:** 16 bits
- **Offset:** 0x3017
- **Exists:** Always

Table 11-363 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RX_DIV16P5_CLK_EN	R/W	**Description**: Enable the outgoing rx_div16p5_clk **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	RX_CLK_EN	R/W	**Description**: Enable the outgoing rx_clk **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.360 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_4

- **Description:** Override values for incoming RX controls from PCS, register #4
- **Size:** 16 bits
- **Offset:** 0x3018
- **Exists:** Always

Table 11-364 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_4

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	REF_LD_VAL_OVRD_EN	R/W	**Description**: Enable override for rx_ref_ld_val Input signals overridden by it: rx_ref_ld_val **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:7	REF_LD_VAL_OVRD	R/W	**Description**: Override value for rx_ref_ld_val. Its override enable is: RX_OVRD_IN_4.REF_LD_VAL_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
6	VCO_FREQBAND_VAL_OVRD_EN	R/W	**Description**: Enable override for rx_cdr_vco_freqband Input signals overridden by it: rx_cdr_vco_freqband **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5:4	VCO_FREQBAND_VAL_OVRD	R/W	**Description**: Override value for rx_cdr_vco_freqband. Its override enable is: RX_OVRD_IN_4.VCO_FREQBAND_VAL_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

Table 11-364 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_4 (Continued)

Bits	Name	Memory Access	Description
3	RX_CDR_VCO_TEMP_COMP_EN_OVRD_EN	R/W	**Description**: Enable override for rx_cdr_vco_temp_comp_en Input signals overridden by it: rx_cdr_vco_temp_comp_en **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_CDR_VCO_TEMP_COMP_EN_OVRD_VAL	R/W	**Description**: Override value for rx_cdr_vco_temp_comp_en. Its override enable is: RX_OVRD_IN_4.RX_CDR_VCO_TEMP_COMP_EN_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_CDR_VCO_STEP_CTRL_OVRD_EN	R/W	**Description**: Enable override for rx_cdr_vco_step_ctrl Input signals overridden by it: rx_cdr_vco_step_ctrl **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_CDR_VCO_STEP_CTRL_OVRD_VAL	R/W	**Description**: Override value for rx_cdr_vco_step_ctrl. Its override enable is: RX_OVRD_IN_4.RX_CDR_VCO_STEP_CTRL_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.361 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_5

- **Description:** rx adaptation mode override register
- **Size:** 16 bits
- **Offset:** 0x3019
- **Exists:** Always

Table 11-365 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_5

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	ADAPT_REQ_OVRD_EN	R/W	**Description**: Enable override values for rx_adapt_req Input signals overridden by it: rx_adapt_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	ADAPT_REQ	R/W	**Description**: Override value for rx_adapt_req. Its override enable is: RX_OVRD_IN_5.ADAPT_REQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	RX_ADAPT_IN_PROG_OVRD_EN	R/W	**Description**: Enable override for RX adaptation in progress Input signals overridden by it: rx_adapt_in_prog **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	RX_ADAPT_IN_PROG_OVRD	R/W	**Description**: Overridden value for the RX adaptation in progress. Its override enable is RX_OVRD_IN_5.RX_ADAPT_IN_PROG_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-365 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_5 (Continued)

Bits	Name	Memory Access	Description
10	RX_CDR_PPM_MAX_OVRD_EN	R/W	**Description**: Enable overidde for the RX CDR PPM Max Input signals overridden by it: rx_cdr_ppm_max **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9:5	RX_CDR_PPM_MAX_OVRD	R/W	**Description**: Overidden value for RX CDR PPM Max. Its override enable is: RX_OVRD_IN_5.RX_CDR_PPM_MAX_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
4	RX_ADAPT_MODE_OVRD_EN	R/W	**Description**: Enable override for the RX adaptation mode Input signals overridden by it: rx_adapt_mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:2	RX_ADAPT_MODE	R/W	**Description**: Overridden value for RX adaptation mode. Its override enable is: RX_OVRD_IN_5.RX_ADAPT_MODE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
1	RX_ADAPT_SEL_OVRD_EN	R/W	**Description**: Enable override the RX adapation setting Input signals overridden by it: rx_adapt_sel **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ADAPT_SEL_OVRD	R/W	**Description**: Overridden value for the RX adapation setting. Its override enable is: RX_OVRD_IN_5.RX_ADAPT_SEL_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.362 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5

- **Description:** Current values for incoming RX controls from PCS, register #5
- **Size:** 16 bits
- **Offset:** 0x301a
- **Exists:** Always

Table 11-366 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	RX_ADAPT_IN_PROG	R	**Description**: Value from ASIC for rx_adapt_in_prog (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13:7	RX_MARGIN_IQ	R	**Description**: Value from ASIC for rx_margin_iq (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
6:3	RX_DELTA_IQ	R	**Description**: Value from ASIC for rx_delta_iq (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
2	RX_ADAPT_SEL	R	**Description**: Value from ASIC for rx_adapt_sel (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-366 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5 (Continued)

Bits	Name	Memory Access	Description
1:0	ADAPT_MODE	R	**Description**: Value from ASIC for rx_adapt_mode (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.363 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6

- **Description:** override register for rx IQ setting
- **Size:** 16 bits
- **Offset:** 0x301b
- **Exists:** Always

Table 11-367 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12	RX_MARGIN_DELTA_OVRD_EN	R/W	**Description**: Enable Override the rx margin IQ setting Input signals overridden by it: rx_margin_iq **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11:5	RX_MARGIN_DELTA_OVRD	R/W	**Description**: Override the rx margin IQ setting. Its override enable is: RX_OVRD_IN_6.RX_MARGIN_DELTA_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f
4	RX_DELTA_IQ_OVRD_EN	R/W	**Description**: Enable Override the rx Delta IQ setting Input signals overridden by it: rx_delta_iq **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3:0	RX_DELTA_IQ_OVRD	R/W	**Description**: Override the rx Delta IQ seeting. Its override enable is: RX_OVRD_IN_6.RX_DELTA_IQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.364 RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1

- **Description:** Override values for incoming RX EQ controls from PCS, register #1
- **Size:** 16 bits
- **Offset:** 0x301c
- **Exists:** Always

Table 11-368 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	RX_EQ_OVRD_EN	R/W	**Description**: Enable override values for all RX EQ settings Input signals overridden by it: rx_eq_att_lvl rx_eq_vga2_gain rx_eq_vga1_gain rx_eq_ctle_pole rx_eq_ctle_boost rx_eq_dfe_tap1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:6	RX_EQ_ATT_LVL_OVRD_VAL	R/W	**Description**: Override value for rx_eq_att_lvl[2:0]. Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
5:3	RX_EQ_VGA2_GAIN_OVRD_VAL	R/W	**Description**: Override value for rx_eq_vga2_gain[3:0]. Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

Table 11-368 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1 (Continued)

Bits	Name	Memory Access	Description
2:0	RX_EQ_VGA1_GAIN_OVRD_VAL	R/W	**Description**: Override value for rx_eq_vga1_gain[3:0]. Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.365 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6

- **Description:** Current values for incoming RX controls from PCS, register #6
- **Size:** 16 bits
- **Offset:** 0x301d
- **Exists:** Always

Table 11-369 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:5	RX_MISC	R	**Description**: Value from ASIC fro rx_misc (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4:0	RX_CDR_PPM_MAX	R	**Description**: Value from ASIC for rx_cdr_ppm_max (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.366 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1

- **Description:** ATE Override input to control top-level inputs, register #2
- **Size:** 16 bits
- **Offset:** 0x301e
- **Exists:** Always

Table 11-370 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11	RX_ADAPT_SEL_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_adapt_sel input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	RX_ADAPT_SEL_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_adapt_sel input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
9	RX_DELTA_IQ_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_delta_iq input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8:5	RX_DELTA_IQ_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_delta_iq input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
4	RX_LPD_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_lpd input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-370 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1 (Continued)

Bits	Name	Memory Access	Description
3	RX_LPD_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_lpd input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_PSTATE_ATE_OVRD_EN	R/W	**Description**: Enable ATE override value for rx_pstate input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1:0	RX_PSTATE_ATE_OVRD_VAL	R/W	**Description**: ATE Override value for top-level rx_pstate input **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.367 RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_2

- **Description:** Override values for incoming RX EQ controls from PCS, register #2
- **Size:** 16 bits
- **Offset:** 0x301f
- **Exists:** Always

Table 11-371 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14:13	RX_EQ_CTLE_POLE_OVRD_VAL	R/W	**Description**: Override value for rx_eq_ctle_pole[2:0]. Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
12:8	RX_EQ_CTLE_BOOST_OVRD_VAL	R/W	**Description**: Override value for rx_eq_ctle_boost[4:0]. Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f
7:0	RX_EQ_DFE_TAP1_OVRD_VAL	R/W	**Description**: Override value for rx_eq_dfe_tap1[7:0] Its override enable is: RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.368 RAWLANEN_DIG_FSM_FSM_OVRD_CTL

- **Description:** FSM override control register
- **Size:** 16 bits
- **Offset:** 0x3020
- **Exists:** Always

Table 11-372 Fields for Register: RAWLANEN_DIG_FSM_FSM_OVRD_CTL

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	FSM_OVRD_EN	R/W	**Description**: Enable overriding the FSM execution of commands Must be asserted to use FSM_CMD_START and FSM_JMP_EN features (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
13	FSM_CMD_START	R/W	**Description**: Execute the current command or resume from breakpoint This is a self-clearing bit (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
12	FSM_JMP_EN	R/W	**Description**: Force the FSM to jump to FSM_JMP_ADDR in the program memory Is applied when FSM_CMD_START is pulsed. (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true
11:0	FSM_JMP_ADDR	R/W	**Description**: The jump address used when FSM_JUMP_EN=1 The address is encoded as follows: [11:8] mem_lane, [7:5] bank, [4:0] register (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xfff **Volatile**: true

11.2.369 RAWLANEN_DIG_FSM_FSM_JMP_BANK

- **Description:** FSM Jump Bank Register
- **Size:** 16 bits
- **Offset:** 0x3021
- **Exists:** Always

Table 11-373 Fields for Register: RAWLANEN_DIG_FSM_FSM_JMP_BANK

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	BANK	R/W	**Description**: The jump bank used when FSM_JUMP_EN=1 This bank need to be updated accordingly along with FSM_OVRD_CTL.FSM_JMP_ADDR **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.370 RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0

- **Description:** FSM Breakpoint 0 on SRAM Address for Debugging
- **Size:** 16 bits
- **Offset:** 0x3022
- **Exists:** Always

Table 11-374 Fields for Register: RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	BREAKPOINT_EN	R/W	**Description**: Enables the Breakpoint 0 Clearing this bit will automatically resume the FSM if it is stopped at MEM_BREAKPOINT_0.BREAKPOINT_ADDR **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:0	BREAKPOINT_ADDR	R/W	**Description**: Memory Address for Breakpoint 0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3fff

11.2.371 RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1

- **Description:** FSM Breakpoint 1 on SRAM Address for Debugging
- **Size:** 16 bits
- **Offset:** 0x3023
- **Exists:** Always

Table 11-375 Fields for Register: RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	BREAKPOINT_EN	R/W	**Description**: Enables the Breakpoint 1 Clearing this bit will automatically resume the FSM if it is stopped at MEM_BREAKPOINT_1.BREAKPOINT_ADDR **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:0	BREAKPOINT_ADDR	R/W	**Description**: Memory Address for Breakpoint 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3fff

11.2.372 RAWLANEN_DIG_FSM_MEM_ADDR_MON

- **Description:** Memory Address Monitor
- **Size:** 16 bits
- **Offset:** 0x3025
- **Exists:** Always

Table 11-376 Fields for Register: RAWLANEN_DIG_FSM_MEM_ADDR_MON

Bits	Name	Memory Access	Description
15:0	MEM_ADDR	R	**Description**: Current value of memory address used in Lane FSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.373 RAWLANEN_DIG_FSM_STATUS_MON

- **Description:** FSM Status Monitor
- **Size:** 16 bits
- **Offset:** 0x3026
- **Exists:** Always

Table 11-377 Fields for Register: RAWLANEN_DIG_FSM_STATUS_MON

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
10	RDMSK_DISABLED	R	**Description**: Check if read mask is currently disabled (i.e. mask is all ones) (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
9	WRMSK_DISABLED	R	**Description**: Check if write mask is currently disabled (i.e. mask is all ones) (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
8	WAIT_CNT_EQ0	R	**Description**: Check if wait counter currently equals zero (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
7	ALU_RES_EQ0	R	**Description**: Check if ALU result register currently equals zero (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

Table 11-377 Fields for Register: RAWLANEN_DIG_FSM_STATUS_MON (Continued)

Bits	Name	Memory Access	Description
6	ALU_OVFLW	R	**Description**: Current value of ALU overflow bit (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
5	CMD_RDY	R	**Description**: New command is ready for execution (applicable when FSM_OVRD_EN=1) (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
4:0	STATE	R	**Description**: Current state of Lane FSM (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.374 RAWLANEN_DIG_FSM_FAST_RX_STARTUP_CAL

- **Description:** Status of Fast RX Start Up Calibration
- **Size:** 16 bits
- **Offset:** 0x3027
- **Exists:** Always

Table 11-378 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_STARTUP_CAL	R	**Description**: Status of fast RX start-up calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.375 RAWLANEN_DIG_FSM_FAST_RX_ADAPT

- **Description:** Status of Fast RX Adaptation
- **Size:** 16 bits
- **Offset:** 0x3028
- **Exists:** Always

Table 11-379 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_ADAPT	R	**Description**: Status of fast RX adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.376 RAWLANEN_DIG_FSM_FAST_SUP

- **Description:** Status of Fast Support block
- **Size:** 16 bits
- **Offset:** 0x3029
- **Exists:** Always

Table 11-380 Fields for Register: RAWLANEN_DIG_FSM_FAST_SUP

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_SUP	R	**Description**: Status of fast Support block (MPLL and Rtune) (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.377 RAWLANEN_DIG_FSM_FAST_TX_CMN_MODE

- **Description:** Status of Fast TX Common-mode Charge-up
- **Size:** 16 bits
- **Offset:** 0x302a
- **Exists:** Always

Table 11-381 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_CMN_MODE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_TX_CMN_MODE	R	**Description**: Status of fast TX Common-mode Charge-up (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.378 RAWLANEN_DIG_FSM_FAST_TX_RXDET

- **Description:** Status of Fast TX detect RX
- **Size:** 16 bits
- **Offset:** 0x302b
- **Exists:** Always

Table 11-382 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_RXDET

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_TX_RXDET	R	**Description**: Status of fast TX detect RX (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.379 RAWLANEN_DIG_FSM_FAST_RX_PWRUP

- **Description:** Status of Fast RX Power-up
- **Size:** 16 bits
- **Offset:** 0x302c
- **Exists:** Always

Table 11-383 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_PWRUP

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_PWRUP	R	**Description**: Status of fast RX Power-up (LOS, VREG/AFE and DCC) (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.380 RAWLANEN_DIG_FSM_FAST_RX_VCO_WAIT

- **Description:** Status of Fast RX VCO Wait Times
- **Size:** 16 bits
- **Offset:** 0x302d
- **Exists:** Always

Table 11-384 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_VCO_WAIT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_VCO_WAIT	R	**Description**: Status of fast RX VCO wait times (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.381 RAWLANEN_DIG_FSM_FAST_RX_VCO_CAL

- **Description:** Status of Fast RX VCO Calibration
- **Size:** 16 bits
- **Offset:** 0x302e
- **Exists:** Always

Table 11-385 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_VCO_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_VCO_CAL	R	**Description**: Status of fast RX VCO Calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.382 RAWLANEN_DIG_FSM_FAST_RX_CONT_CAL_ADAPT

- **Description:** Status of Fast RX Continuous Calibration/Adaptation
- **Size:** 16 bits
- **Offset:** 0x302f
- **Exists:** Always

Table 11-386 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_CONT_CAL_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_CONT_CAL_ADAPT	R	**Description**: Status of fast RX continuous calibration/adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.383 RAWLANEN_DIG_FSM_FAST_RX_CONT_ADAPT

- **Description:** Status of Fast RX Continuous Adaptation
- **Size:** 16 bits
- **Offset:** 0x3030
- **Exists:** Always

Table 11-387 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_CONT_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	FAST_RX_CONT_ADAPT	R	**Description**: Status of fast RX continuous adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.384 RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_CAL

- **Description:** Status of RX AFE Startup Calibration
- **Size:** 16 bits
- **Offset:** 0x3031
- **Exists:** Always

Table 11-388 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_AFE_STARTUP_CAL	R	**Description**: Status of RX AFE start-up calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.385 RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_CAL

- **Description:** Status of RX DFE Startup Calibration
- **Size:** 16 bits
- **Offset:** 0x3032
- **Exists:** Always

Table 11-389 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_DFE_STARTUP_CAL	R	**Description**: Status of RX DFE Startup Calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.386 RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT_STARTUP_CAL

- **Description:** Status of DFE extended start-up Calibration
- **Size:** 16 bits
- **Offset:** 0x3033
- **Exists:** Always

Table 11-390 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_DFE_EXT_STARTUP_CAL	R	**Description**: Status of DFE extended start-up Calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.387 RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELTA_STARTUP

- **Description:** Status of RX IQ Fixed Offset
- **Size:** 16 bits
- **Offset:** 0x3034
- **Exists:** Always

Table 11-391 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELTA_STARTUP

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_IQ_DELTA_STARTUP	R	**Description**: Status of RX IQ Fixed Offset (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.388 RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_CAL

- **Description:** Status of RX IQ start-up calibration
- **Size:** 16 bits
- **Offset:** 0x3035
- **Exists:** Always

Table 11-392 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_IQ_STARTUP_CAL	R	**Description**: Status of RX IQ start-up calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.389 RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT

- **Description:** Status of RX AFE start-up Adaptation
- **Size:** 16 bits
- **Offset:** 0x3036
- **Exists:** Always

Table 11-393 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_AFE_STARTUP_ADAPT	R	**Description**: Status of RX AFE DAC start-up adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.390 RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT

- **Description:** Status of RX DFE start-up Adaptation
- **Size:** 16 bits
- **Offset:** 0x3037
- **Exists:** Always

Table 11-394 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_DFE_STARTUP_ADAPT	R	**Description**: Status of RX DFE DAC start-up adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.391 RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT

- **Description:** Status of RX IO Start-up Adaptation
- **Size:** 16 bits
- **Offset:** 0x3038
- **Exists:** Always

Table 11-395 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_IQ_STARTUP_ADAPT	R	**Description**: Status of RX IQ start-up adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.392 RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL

- **Description:** Status of RX Continuous Phase Calibration
- **Size:** 16 bits
- **Offset:** 0x3039
- **Exists:** Always

Table 11-396 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_CONT_PHASE_CAL	R	**Description**: Status of RX continuous phase calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.393 RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL

- **Description:** Status of RX AFE Continuous Calibration
- **Size:** 16 bits
- **Offset:** 0x303a
- **Exists:** Always

Table 11-397 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_AFE_CONT_CAL	R	**Description**: Status of RX AFE continuous calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.394 RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT

- **Description:** Status of RX Reference Level Continuous Adaptation
- **Size:** 16 bits
- **Offset:** 0x303b
- **Exists:** Always

Table 11-398 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_REFLVL_CONT_ADAPT	R	**Description**: Status of RX Reference Level continuous adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.395 RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT

- **Description:** Status of RX VGA Continuous Adaptation
- **Size:** 16 bits
- **Offset:** 0x303c
- **Exists:** Always

Table 11-399 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_VGA_CONT_ADAPT	R	**Description**: Status of RX VGA continuous adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.396 RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL

- **Description:** Status of RX Phase Start-up Calibration
- **Size:** 16 bits
- **Offset:** 0x303d
- **Exists:** Always

Table 11-400 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_PHS_STARTUP_CAL	R	**Description**: Status of RX Phase start-up calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.397 RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL

- **Description:** Status of RX Phase Extended Start-up Calibration
- **Size:** 16 bits
- **Offset:** 0x303e
- **Exists:** Always

Table 11-401 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	SKIP_RX_PHS_EXT_STARTUP_CAL	R	**Description**: Status of RX Phase extended start-up calibration (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.398 RAWLANEN_DIG_FSM_FW_STATES_1

- **Description:** Stores various states for Firmware - State #2
- **Size:** 16 bits
- **Offset:** 0x303f
- **Exists:** Always

Table 11-402 Fields for Register: RAWLANEN_DIG_FSM_FW_STATES_1

Bits	Name	Memory Access	Description
15:0	SPARE_STATE_0	R/W	**Description**: Spare fields to be used by Firmware **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.399 RAWLANEN_DIG_AON_AFE_ATT_IDAC_OFST

- **Description:** Offset value for RX AFE ATT iDAC
- **Size:** 16 bits
- **Offset:** 0x3040
- **Exists:** Always

Table 11-403 Fields for Register: RAWLANEN_DIG_AON_AFE_ATT_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	AFE_ATT_IDAC_OFST	R/W	**Description**: Offset value for AFE ATT iDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.400 RAWLANEN_DIG_AON_AFE_CTLE_IDAC_OFST

- **Description:** Offset value for RX AFE CTLE iDAC
- **Size:** 16 bits
- **Offset:** 0x3041
- **Exists:** Always

Table 11-404 Fields for Register: RAWLANEN_DIG_AON_AFE_CTLE_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	AFE_CTLE_IDAC_OFST	R/W	**Description**: Offset value for AFE CTLE iDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.401 RAWLANEN_DIG_AON_AFE_VGA1_IDAC_OFST

- **Description:** Offset values for RX AFE VGA1 iDAC
- **Size:** 16 bits
- **Offset:** 0x3042
- **Exists:** Always

Table 11-405 Fields for Register: RAWLANEN_DIG_AON_AFE_VGA1_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	AFE_VGA1_IDAC_OFST	R/W	**Description**: Offset value for AFE VGA1 iDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.402 RAWLANEN_DIG_AON_DFE_PHASE_EVEN_VDAC_OFST

- **Description:** Offset values for RX DFE Phase Even vDAC
- **Size:** 16 bits
- **Offset:** 0x3043
- **Exists:** Always

Table 11-406 Fields for Register: RAWLANEN_DIG_AON_DFE_PHASE_EVEN_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_PHASE_EVEN_VDAC_OFST	R/W	**Description**: Offset value for DFE Phase Even vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.403 RAWLANEN_DIG_AON_DFE_PHASE_ODD_VDAC_OFST

- **Description:** Offset values for RX DFE Phase Odd vDAC
- **Size:** 16 bits
- **Offset:** 0x3044
- **Exists:** Always

Table 11-407 Fields for Register: RAWLANEN_DIG_AON_DFE_PHASE_ODD_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_PHASE_ODD_VDAC_OFST	R/W	**Description**: Offset value for DFE Phase Odd vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.404 RAWLANEN_DIG_AON_DFE_PHASE_EVEN_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Phase Even Low vDAC
- **Size:** 16 bits
- **Offset:** 0x3045
- **Exists:** Always

Table 11-408 Fields for Register: RAWLANEN_DIG_AON_DFE_PHASE_EVEN_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_PHASE_EVEN_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Phase Even Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.405 RAWLANEN_DIG_AON_DFE_PHASE_ODD_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Phase Odd Low vDAC
- **Size:** 16 bits
- **Offset:** 0x3046
- **Exists:** Always

Table 11-409 Fields for Register: RAWLANEN_DIG_AON_DFE_PHASE_ODD_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_PHASE_ODD_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Phase Odd Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.406 RAWLANEN_DIG_AON_RX_PHSADJ_LIN

- **Description:** RX Phase Adjust Linear Value
- **Size:** 16 bits
- **Offset:** 0x3047
- **Exists:** Always

Table 11-410 Fields for Register: RAWLANEN_DIG_AON_RX_PHSADJ_LIN

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_PHSADJ_LIN	R/W	**Description**: Linear value for RX phase adjust **Value After Reset**: 0x17 **Exists**: Always **Reset Mask**: 0x3f

11.2.407 RAWLANEN_DIG_AON_DFE_DATA_EVEN_HIGH_VDAC_OFST

- **Description:** Offset values for RX DFE Data Even High vDAC
- **Size:** 16 bits
- **Offset:** 0x3048
- **Exists:** Always

Table 11-411 Fields for Register: RAWLANEN_DIG_AON_DFE_DATA_EVEN_HIGH_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_EVEN_HIGH_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Even High vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.408 RAWLANEN_DIG_AON_DFE_DATA_EVEN_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Data Even Low vDAC
- **Size:** 16 bits
- **Offset:** 0x3049
- **Exists:** Always

Table 11-412 Fields for Register: RAWLANEN_DIG_AON_DFE_DATA_EVEN_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_EVEN_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Even Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.409 RAWLANEN_DIG_AON_DFE_DATA_ODD_HIGH_VDAC_OFST

- **Description:** Offset values for RX DFE Data Odd High vDAC
- **Size:** 16 bits
- **Offset:** 0x304a
- **Exists:** Always

Table 11-413 Fields for Register: RAWLANEN_DIG_AON_DFE_DATA_ODD_HIGH_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_ODD_HIGH_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Odd High vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.410 RAWLANEN_DIG_AON_DFE_DATA_ODD_LOW_VDAC_OFST

- **Description:** Offset values for RX DFE Data Odd Low vDAC
- **Size:** 16 bits
- **Offset:** 0x304b
- **Exists:** Always

Table 11-414 Fields for Register: RAWLANEN_DIG_AON_DFE_DATA_ODD_LOW_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_DATA_ODD_LOW_VDAC_OFST	R/W	**Description**: Offset value for DFE Data Odd Low vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.411 RAWLANEN_DIG_AON_DFE_BYPASS_EVEN_VDAC_OFST

- **Description:** Offset values for RX DFE By-Pass Even vDAC
- **Size:** 16 bits
- **Offset:** 0x304c
- **Exists:** Always

Table 11-415 Fields for Register: RAWLANEN_DIG_AON_DFE_BYPASS_EVEN_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_BYPASS_EVEN_VDAC_OFST	R/W	**Description**: Offset value for DFE By-Pass Even vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.412 RAWLANEN_DIG_AON_DFE_BYPASS_ODD_VDAC_OFST

- **Description:** Offset values for RX DFE By-Pass Odd vDAC
- **Size:** 16 bits
- **Offset:** 0x304d
- **Exists:** Always

Table 11-416 Fields for Register: RAWLANEN_DIG_AON_DFE_BYPASS_ODD_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_BYPASS_ODD_VDAC_OFST	R/W	**Description**: Offset value for DFE By-Pass Odd vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.413 RAWLANEN_DIG_AON_DFE_ERROR_EVEN_VDAC_OFST

- **Description:** Offset values for RX DFE Error Even vDAC
- **Size:** 16 bits
- **Offset:** 0x304e
- **Exists:** Always

Table 11-417 Fields for Register: RAWLANEN_DIG_AON_DFE_ERROR_EVEN_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_ERROR_EVEN_VDAC_OFST	R/W	**Description**: Offset value for DFE Error Even vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.414 RAWLANEN_DIG_AON_DFE_ERROR_ODD_VDAC_OFST

- **Description:** Offset values for RX DFE Error Odd vDAC
- **Size:** 16 bits
- **Offset:** 0x304f
- **Exists:** Always

Table 11-418 Fields for Register: RAWLANEN_DIG_AON_DFE_ERROR_ODD_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	DFE_ERROR_ODD_VDAC_OFST	R/W	**Description**: Offset value for DFE Error Odd vDAC **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.415 RAWLANEN_DIG_AON_RX_CAL_IQ

- **Description:** Value for RX Calibrated IQ Phase
- **Size:** 16 bits
- **Offset:** 0x3050
- **Exists:** Always

Table 11-419 Fields for Register: RAWLANEN_DIG_AON_RX_CAL_IQ

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	RX_CAL_IQ	R/W	**Description**: Value for RX Calibrated IQ Phase **Value After Reset**: 0x17 **Exists**: Always **Reset Mask**: 0x7f

11.2.416 RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL

- **Description:** Set to Skip Firmware startup algorithms
- **Size:** 16 bits
- **Offset:** 0x3051
- **Exists:** Always

Table 11-420 Fields for Register: RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	SKIP_RX_VGA_EXT_STARTUP	R/W	**Description**: Skip RX start-up VGA calibration in extended mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13	SKIP_RX_REF_CAL_EXT_STARTUP	R/W	**Description**: Skip RX start-up reference level calibration in extended mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
12	SKIP_RX_DFE_ADAPT_STARTUP	R/W	**Description**: Skip RX DFE DAC start-up adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
11	SKIP_RX_DFE_CAL_STARTUP	R/W	**Description**: Skip RX DFE slicer start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
10	SKIP_RX_DFE_CAL_EXT_STARTUP	R/W	**Description**: Skip RX DFE slicer extended start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-420 Fields for Register: RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL (Continued)

Bits	Name	Memory Access	Description
9	SKIP_RX_AFE_ADAPT_STARTUP	R/W	**Description**: Skip RX AFE DAC start-up adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	SKIP_RX_AFE_CAL_STARTUP	R/W	**Description**: Skip RX AFE DAC start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	SKIP_RX_PHS_CAL_STARTUP	R/W	**Description**: Skip RX phase sliicer startup calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	SKIP_RX_PHS_CAL_EXT_STARTUP	R/W	**Description**: Skip RX extended phase slicer startup calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	SKIP_RX_FOM_STARTUP	R/W	**Description**: Skip RX start-up FOM measurement **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	SKIP_RX_TX_INC_DEC_STARTUP	R/W	**Description**: Skip RX continuous phase calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	SKIP_RX_IQ_CAL_STARTUP	R/W	**Description**: Skip RX IQ start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	SKIP_RX_IQ_DELTA_STARTUP	R/W	**Description**: Skip RX IQ fixed offset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	SKIP_RX_IQ_ADAPT_STARTUP	R/W	**Description**: Skip RX IQ start-up adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-420 Fields for Register: RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL (Continued)

Bits	Name	Memory Access	Description
0	SKIP_RX_REF_CAL_STARTUP	R/W	**Description**: Skip RX start-up reference level calibration **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.417 RAWLANEN_DIG_AON_RX_CONT_ALGO_CTL

- **Description:** Set to Skip Firmware mission-mode algorithms
- **Size:** 16 bits
- **Offset:** 0x3052
- **Exists:** Always

Table 11-421 Fields for Register: RAWLANEN_DIG_AON_RX_CONT_ALGO_CTL

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8	SKIP_RX_CTLE_BOOST_CONT	R/W	**Description**: Skip RX CTLE Boost continuous adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
7	SKIP_RX_DFE_ADAPT1_CONT	R/W	**Description**: Skip RX DFE DAC continuous adaptation for tap1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
6	SKIP_RX_DFE_ADAPT25_CONT	R/W	**Description**: Skip RX DFE DAC continuous adaptation for tap2 - tap5 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	SKIP_RX_DFE_CAL_CONT	R/W	**Description**: Skip RX DFE slicer continuous calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	SKIP_RX_AFE_CAL_CONT	R/W	**Description**: Skip RX AFE DAC continuous calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-421 Fields for Register: RAWLANEN_DIG_AON_RX_CONT_ALGO_CTL (Continued)

Bits	Name	Memory Access	Description
3	SKIP_RX_PHS_CAL_CONT	R/W	**Description**: Skip RX phase sliicer offset calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	SKIP_RX_VGA_ADAPT_CONT	R/W	**Description**: Skip RX VGA continuous adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	SKIP_RX_REFLVL_ADAPT_CONT	R/W	**Description**: Skip RX REF continuous adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	SKIP_RX_FOM_CONT	R/W	**Description**: Skip RX continuous FOM measurement **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.418 RAWLANEN_DIG_AON_INIT_PWRUP_DONE

- **Description:** Initial Power-Up Done Status
- **Size:** 16 bits
- **Offset:** 0x3053
- **Exists:** Always

Table 11-422 Fields for Register: RAWLANEN_DIG_AON_INIT_PWRUP_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	INIT_PWRUP_DONE	R/W	**Description**: Indicates whether initial power-up has completed or not. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.419 RAWLANEN_DIG_AON_RX_ADPT_ATT

- **Description:** RX Adapted value of ATT
- **Size:** 16 bits
- **Offset:** 0x3054
- **Exists:** Always

Table 11-423 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_ATT

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	ATT_ADPT_VAL	R/W	**Description**: Stored RX adapted ATT value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.420 RAWLANEN_DIG_AON_RX_ADPT_VGA

- **Description:** RX Adapted value of VGA
- **Size:** 16 bits
- **Offset:** 0x3055
- **Exists:** Always

Table 11-424 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_VGA

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	VGA_ADPT_VAL	R/W	**Description**: Stored RX adapted VGA value. Bits 5:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.421 RAWLANEN_DIG_AON_RX_ADPT_CTLE

- **Description:** RX Adapted value of CTLE
- **Size:** 16 bits
- **Offset:** 0x3056
- **Exists:** Always

Table 11-425 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_CTLE

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:10	CTLE_POLE_ADPT_VAL	R/W	**Description**: Stored RX adapted CTLE pole value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
9:0	CTLE_BOOST_ADPT_VAL	R/W	**Description**: Stored RX adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.422 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1

- **Description:** RX Adapted value of DFE TAP1
- **Size:** 16 bits
- **Offset:** 0x3057
- **Exists:** Always

Table 11-426 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:0	DFE_TAP1_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1fff

11.2.423 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2

- **Description:** RX Adapted value of DFE TAP2
- **Size:** 16 bits
- **Offset:** 0x3058
- **Exists:** Always

Table 11-427 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP2_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.424 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3

- **Description:** RX Adapted value of DFE TAP3
- **Size:** 16 bits
- **Offset:** 0x3059
- **Exists:** Always

Table 11-428 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP3_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP3 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.425 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4

- **Description:** RX Adapted value of DFE TAP4
- **Size:** 16 bits
- **Offset:** 0x305a
- **Exists:** Always

Table 11-429 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP4_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP4 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.426 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5

- **Description:** RX Adapted value of DFE TAP5
- **Size:** 16 bits
- **Offset:** 0x305b
- **Exists:** Always

Table 11-430 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP5_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP5 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.427 RAWLANEN_DIG_AON_RX_ADPT_IQ

- **Description:** RX Adapted value of IQ for bank 0
- **Size:** 16 bits
- **Offset:** 0x305c
- **Exists:** Always

Table 11-431 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_IQ

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	RX_ADPT_IQ	R/W	**Description**: RX adapted value of IQ for bank 0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f

11.2.428 RAWLANEN_DIG_AON_RX_ADPT_REF_ERR

- **Description:** RX Adapted value of Reference Level for Error Slicer for bank 0
- **Size:** 16 bits
- **Offset:** 0x305d
- **Exists:** Always

Table 11-432 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_REF_ERR

Bits	Name	Memory Access	Description
15:8	RX_ADPT_REF_ERR_ODD	R/W	**Description**: RX Adapted value of reference level for error odd slicer for bank 0 **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff
7:0	RX_ADPT_REF_ERR_EVEN	R/W	**Description**: RX Adapted value of reference level for error even slicer for bank 0 **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.429 RAWLANEN_DIG_AON_RX_ADAPT_DONE

- **Description:** RX Adaptation Done Status
- **Size:** 16 bits
- **Offset:** 0x305e
- **Exists:** Always

Table 11-433 Fields for Register: RAWLANEN_DIG_AON_RX_ADAPT_DONE

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RX_ADPT_IQ_VLD	R/W	**Description**: Indicates if the RX adapted value of IQ for bank 0 is valid **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ADAPT_DONE	R/W	**Description**: Indicates whether RX adaptation has completed or not. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.430 RAWLANEN_DIG_AON_RX_ADPT_ATT_B1

- **Description:** RX Adapted value of ATT for bank 1
- **Size:** 16 bits
- **Offset:** 0x305f
- **Exists:** Always

Table 11-434 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_ATT_B1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	ATT_ADPT_VAL	R/W	**Description**: Stored RX adapted ATT value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.431 RAWLANEN_DIG_AON_RX_ADPT_VGA_B1

- **Description:** RX Adapted value of VGA for bank 1
- **Size:** 16 bits
- **Offset:** 0x3060
- **Exists:** Always

Table 11-435 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_VGA_B1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	VGA_ADPT_VAL	R/W	**Description**: Stored RX adapted VGA value. Bits 5:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1ff

11.2.432 RAWLANEN_DIG_AON_RX_ADPT_CTLE_B1

- **Description:** RX Adapted value of CTLE for bank 1
- **Size:** 16 bits
- **Offset:** 0x3061
- **Exists:** Always

Table 11-436 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_CTLE_B1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:10	CTLE_POLE_ADPT_VAL	R/W	**Description**: Stored RX adapted CTLE pole value **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3
9:0	CTLE_BOOST_ADPT_VAL	R/W	**Description**: Stored RX adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.433 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1_B1

- **Description:** RX Adapted value of DFE TAP1 for bank 1
- **Size:** 16 bits
- **Offset:** 0x3062
- **Exists:** Always

Table 11-437 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP1_B1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
12:0	DFE_TAP1_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1fff

11.2.434 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2_B1

- **Description:** RX Adapted value of DFE TAP2 for bank 1
- **Size:** 16 bits
- **Offset:** 0x3063
- **Exists:** Always

Table 11-438 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP2_B1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP2_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.435 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3_B1

- **Description:** RX Adapted value of DFE TAP3 for bank 1
- **Size:** 16 bits
- **Offset:** 0x3064
- **Exists:** Always

Table 11-439 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP3_B1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP3_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP3 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.436 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4_B1

- **Description:** RX Adapted value of DFE TAP4 for bank 1
- **Size:** 16 bits
- **Offset:** 0x3065
- **Exists:** Always

Table 11-440 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP4_B1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP4_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP4 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.437 RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5_B1

- **Description:** RX Adapted value of DFE TAP5 for bank 1
- **Size:** 16 bits
- **Offset:** 0x3066
- **Exists:** Always

Table 11-441 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_DFE_TAP5_B1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	DFE_TAP5_ADPT_VAL	R/W	**Description**: Stored RX adapted DFE TAP5 value. Bits 4:0 are dither bits. **Value After Reset**: 0x800 **Exists**: Always **Reset Mask**: 0xfff

11.2.438 RAWLANEN_DIG_AON_RX_ADPT_IQ_B1

- **Description:** RX Adapted value of IQ for bank 1
- **Size:** 16 bits
- **Offset:** 0x3067
- **Exists:** Always

Table 11-442 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_IQ_B1

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:0	RX_ADPT_IQ_B1	R/W	**Description**: RX adapted value of IQ for bank 1 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7f

11.2.439 RAWLANEN_DIG_AON_RX_ADPT_REF_ERR_B1

- **Description:** RX Adapted value of Reference Level for Error Slicer for bank 1
- **Size:** 16 bits
- **Offset:** 0x3068
- **Exists:** Always

Table 11-443 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_REF_ERR_B1

Bits	Name	Memory Access	Description
15:8	RX_ADPT_REF_ERR_ODD_B1	R/W	**Description**: RX Adapted value of reference level for error odd slicer for bank 1 **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff
7:0	RX_ADPT_REF_ERR_EVEN_B1	R/W	**Description**: RX Adapted value of reference level for error even slicer for bank 1 **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.440 RAWLANEN_DIG_AON_RX_ADAPT_DONE_B1

- **Description:** RX Adaptation Done Status for bank 1
- **Size:** 16 bits
- **Offset:** 0x3069
- **Exists:** Always

Table 11-444 Fields for Register: RAWLANEN_DIG_AON_RX_ADAPT_DONE_B1

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RX_ADPT_IQ_VLD_B1	R/W	**Description**: Indicates if the RX adapted value of IQ for bank 1 is valid **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ADAPT_DONE_B1	R/W	**Description**: Indicates whether Bank 1 RX adaptation has completed or not. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.441 RAWLANEN_DIG_AON_FAST_FLAGS

- **Description:** Fast flags for simulation only
- **Size:** 16 bits
- **Offset:** 0x306a
- **Exists:** Always

Table 11-445 Fields for Register: RAWLANEN_DIG_AON_FAST_FLAGS

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9	FAST_RX_VCO_CAL	R/W	**Description**: Enable fast RX VCO Calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
8	FAST_RX_VCO_WAIT	R/W	**Description**: Enable fast RX VCO wait times **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
7	FAST_RX_PWRUP	R/W	**Description**: Enable fast RX power-up (LOS, VREG/AFE and DCC) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	FAST_TX_RXDET	R/W	**Description**: Enable fast TX Detect RX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	FAST_TX_CMN_MODE	R/W	**Description**: Enable fast TX Common Mode Charge-up **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	FAST_SUP	R/W	**Description**: Enable fast Support block (MPLL and Rtune) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-445 Fields for Register: RAWLANEN_DIG_AON_FAST_FLAGS (Continued)

Bits	Name	Memory Access	Description
3	FAST_RX_CONT_CAL_ADAPT	R/W	**Description**: Enables fast RX continuous calibration/adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	FAST_RX_CONT_ADAPT	R/W	**Description**: Enables fast RX continuous adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	FAST_RX_ADAPT	R/W	**Description**: Enables fast RX adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	FAST_RX_STARTUP_CAL	R/W	**Description**: Enables fast RX start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.442 RAWLANEN_DIG_AON_FW_STATES_0

- **Description:** Stores various states for Firmware - State #1
- **Size:** 16 bits
- **Offset:** 0x306b
- **Exists:** Always

Table 11-446 Fields for Register: RAWLANEN_DIG_AON_FW_STATES_0

Bits	Name	Memory Access	Description
15:0	SPARE_STATE_0	R/W	**Description**: Spare fields to be used by Firmware **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0xffff

11.2.443 RAWLANEN_DIG_AON_TXRX_OVRD_IN

- **Description:** Override values for incoming AON TX/RX controls from PCS
- **Size:** 16 bits
- **Offset:** 0x306c
- **Exists:** Always

Table 11-447 Fields for Register: RAWLANEN_DIG_AON_TXRX_OVRD_IN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	TX_DISABLE_OVRD_EN	R/W	**Description**: Override enable for tx_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_DISABLE_OVRD_VAL	R/W	**Description**: Override value for tx_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_DISABLE_OVRD_EN	R/W	**Description**: Override enable for rx_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_DISABLE_OVRD_VAL	R/W	**Description**: Override value for rx_disable **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.444 RAWLANEN_DIG_AON_TX_EQ_DIR_POLARITY_CTL

- **Description:** Controls the equalization direction polarity setting in PCS RAW
- **Size:** 16 bits
- **Offset:** 0x306d
- **Exists:** Always

Table 11-448 Fields for Register: RAWLANEN_DIG_AON_TX_EQ_DIR_POLARITY_CTL

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	TXPOST_DIR_INV	R/W	**Description**: Invert the txpost_dir polarity **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TXPRE_DIR_INV	R/W	**Description**: Invert the txpre_dir polarity **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	TXMAIN_DIR_INV	R/W	**Description**: Invert teh txmain_dir polarity **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.445 RAWLANEN_DIG_AON_TX_PRE_DIV

- **Description:** TX pre threshold div
- **Size:** 16 bits
- **Offset:** 0x306e
- **Exists:** Always

Table 11-449 Fields for Register: RAWLANEN_DIG_AON_TX_PRE_DIV

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:5	TX_PRE_THRESH_DIV	R/W	**Description**: TX pre threshhold div **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0xf
4:0	TX_PRE_OFF_MULT	R/W	**Description**: TX pre offset mult **Value After Reset**: 0x8 **Exists**: Always **Reset Mask**: 0x1f

11.2.446 RAWLANEN_DIG_AON_TX_MAIN_ATT_THRESHOLD

- **Description:** TX main ATT high and low threshold
- **Size:** 16 bits
- **Offset:** 0x306f
- **Exists:** Always

Table 11-450 Fields for Register: RAWLANEN_DIG_AON_TX_MAIN_ATT_THRESHOLD

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:3	TX_MAIN_ATT_HIGH_THRESHOLD	R/W	**Description**: TX main ATT high threshold **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x7
2:0	TX_MAIN_ATT_LOW_THRESHOLD	R/W	**Description**: TX main ATT low threshold **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7

11.2.447 RAWLANEN_DIG_AON_TX_MAIN_VGA_THRESHOLD

- **Description:** TX main vga high and low threshold
- **Size:** 16 bits
- **Offset:** 0x3070
- **Exists:** Always

Table 11-451 Fields for Register: RAWLANEN_DIG_AON_TX_MAIN_VGA_THRESHOLD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:4	TX_MAIN_VGA_HIGH_THRESHOLD	R/W	**Description**: TX main vga high threshold **Value After Reset**: 0xf **Exists**: Always **Reset Mask**: 0xf
3:0	TX_MAIN_VGA_LOW_THRESHOLD	R/W	**Description**: TX main vga low threshold **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf

11.2.448 RAWLANEN_DIG_AON_TX_POST_BOOST_THRESHOLD

- **Description:** TX post boost high and low threshold
- **Size:** 16 bits
- **Offset:** 0x3071
- **Exists:** Always

Table 11-452 Fields for Register: RAWLANEN_DIG_AON_TX_POST_BOOST_THRESHOLD

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	TX_POST_BOOST_HIGH_THRESHOLD	R/W	**Description**: TX post boost high threshold **Value After Reset**: 0x14 **Exists**: Always **Reset Mask**: 0x1f
4:0	TX_POST_BOOST_LOW_THRESHOLD	R/W	**Description**: TX post boost low threshold **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0x1f

11.2.449 RAWLANEN_DIG_AON_TX_POST_TAP1_THRESHOLD

- **Description:** TX post tap1 high and low threshold
- **Size:** 16 bits
- **Offset:** 0x3072
- **Exists:** Always

Table 11-453 Fields for Register: RAWLANEN_DIG_AON_TX_POST_TAP1_THRESHOLD

Bits	Name	Memory Access	Description
15:8	TX_POST_TAP1_HIGH_THRESHOLD	R/W	**Description**: TX post tap1 high threshold **Value After Reset**: 0x14 **Exists**: Always **Reset Mask**: 0xff
7:0	TX_POST_TAP1_LOW_THRESHOLD	R/W	**Description**: TX post tap1 low threshold **Value After Reset**: 0x2 **Exists**: Always **Reset Mask**: 0xff

11.2.450 RAWLANEN_DIG_AON_RX_CAL_REF_EE_VDAC_OFST

- **Description:** Value for reference level error even slicer
- **Size:** 16 bits
- **Offset:** 0x3073
- **Exists:** Always

Table 11-454 Fields for Register: RAWLANEN_DIG_AON_RX_CAL_REF_EE_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_CAL_REF_EE_VDAC_OFST	R/W	**Description**: Value for reference level error even slicer **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.451 RAWLANEN_DIG_AON_RX_CAL_REF_EO_VDAC_OFST

- **Description:** Value for reference level error odd slicer
- **Size:** 16 bits
- **Offset:** 0x3074
- **Exists:** Always

Table 11-455 Fields for Register: RAWLANEN_DIG_AON_RX_CAL_REF_EO_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_CAL_REF_EO_VDAC_OFST	R/W	**Description**: Value for reference level error odd slicer **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.452 RAWLANEN_DIG_AON_DFE_EE_VDAC_OFST

- **Description:** Value for DFE Error Even Low and High Sclier in Mission Mode
- **Size:** 16 bits
- **Offset:** 0x3075
- **Exists:** Always

Table 11-456 Fields for Register: RAWLANEN_DIG_AON_DFE_EE_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	DFE_EEL_VDAC_OFST	R/W	**Description**: Value for DFE error even low sclier in mission mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
7:0	DFE_EEH_VDAC_OFST	R/W	**Description**: Value for DFE error even high sclier in mission mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.453 RAWLANEN_DIG_AON_DFE_EO_VDAC_OFST

- **Description:** Value for DFE Eerror Odd Low and High Sclier in Mission Mode
- **Size:** 16 bits
- **Offset:** 0x3076
- **Exists:** Always

Table 11-457 Fields for Register: RAWLANEN_DIG_AON_DFE_EO_VDAC_OFST

Bits	Name	Memory Access	Description
15:8	DFE_EOL_VDAC_OFST	R/W	**Description**: Value for DFE error odd low sclier in mission mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff
7:0	DFE_EOH_VDAC_OFST	R/W	**Description**: Value for DFE error odd high sclier in mission mode **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.454 RAWLANEN_DIG_AON_RX_ADPT_ERR_SLC_MODE

- **Description:** RX Adaptation Error Slicer Mode for Reference-Level Calibration
- **Size:** 16 bits
- **Offset:** 0x3077
- **Exists:** Always

Table 11-458 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_ERR_SLC_MODE

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	RX_ADPT_ERR_SLC_MODE	R/W	**Description**: RX Adaptation error slicer mode for reference-level calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3

11.2.455 RAWLANEN_DIG_AON_SETUP_REF_CTLE_IDAC_OFST

- **Description:** Offset value for AFE CTLE IDAC During Reference-Level Calibration
- **Size:** 16 bits
- **Offset:** 0x3078
- **Exists:** Always

Table 11-459 Fields for Register: RAWLANEN_DIG_AON_SETUP_REF_CTLE_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	SETUP_REF_CTLE_IDAC_OFST	R/W	**Description**: Offset value for AFE CTLE iDAC during reference-level Calibration **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.456 RAWLANEN_DIG_AON_SETUP_REF_VGA1_IDAC_OFST

- **Description:** Offset value for AFE VGA1 IDAC During Reference-Level Calibration
- **Size:** 16 bits
- **Offset:** 0x3079
- **Exists:** Always

Table 11-460 Fields for Register: RAWLANEN_DIG_AON_SETUP_REF_VGA1_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	SETUP_REF_VGA1_IDAC_OFST	R/W	**Description**: Offset value for AFE CTLE VGA1 during reference-level Calibration **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.457 RAWLANEN_DIG_AON_SETUP_SLC_VGA1_IDAC_OFST

- **Description:** Offset Value for AFE VGA1 for Slicer Setup Calibration
- **Size:** 16 bits
- **Offset:** 0x307a
- **Exists:** Always

Table 11-461 Fields for Register: RAWLANEN_DIG_AON_SETUP_SLC_VGA1_IDAC_OFST

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	SETUP_SLC_VGA1_IDAC_OFST	R/W	**Description**: Offset value for AFE VGA1 for slicer setup calibration **Value After Reset**: 0x80 **Exists**: Always **Reset Mask**: 0xff

11.2.458 RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL

- **Description:** RX CDR Detector Control
- **Size:** 16 bits
- **Offset:** 0x307b
- **Exists:** Always

Table 11-462 Fields for Register: RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	DIS_IN_ADAPT	R/W	**Description**: Disable the CDR Detector state macine for during adaptation **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RX_CDR_PPM_MONITOR_MODE	R/W	**Description**: Enable the CDR Detector in Monitor Mode. To enable monitor mode only, disable detector mode and set this field **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_CDR_DETECTOR_EN	R/W	**Description**: Enable the CDR Detector in Detector Mode. In detector mode, monitor mode is also enabled **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.459 RAWLANEN_DIG_AON_RX_CDR_RECOVERY_TIME

- **Description:** RX CDR Recovery Time in reference clock for the intended PPM
- **Size:** 16 bits
- **Offset:** 0x307c
- **Exists:** Always

Table 11-463 Fields for Register: RAWLANEN_DIG_AON_RX_CDR_RECOVERY_TIME

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
11:0	RX_CDR_RECOVERY_TIME	R/W	**Description**: RX CDR Recovery Time in reference clock for intended PPM this default CDR Recovery Time is determined based on 15000 ppm drift on serial data. This value corresponds to the rxX_cdr_ppm_max input which is also calculated to detect 15000 ppm on serial data **Value After Reset**: 0x585 **Exists**: Always **Reset Mask**: 0xfff

11.2.460 RAWLANEN_DIG_AON_MEM_BREAKPOINT_2

- **Description:** FSM Breakpoint 2 on SRAM Address for Debugging
- **Size:** 16 bits
- **Offset:** 0x307d
- **Exists:** Always

Table 11-464 Fields for Register: RAWLANEN_DIG_AON_MEM_BREAKPOINT_2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	BREAKPOINT_EN	R/W	**Description**: Enables the Breakpoint 2 Clearing this bit will automatically resume the FSM if it is stopped at MEM_BREAKPOINT_2.BREAKPOINT_ADDR This control register resides in AON domain which can be utilized to stop/resume the power gating routine performed by FW. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
13:0	BREAKPOINT_ADDR	R/W	**Description**: Memory Address for Breakpoint 2 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3fff

11.2.461 RAWLANEN_DIG_AON_ADPT_CTL_0

- **Description:** Adaptation Control register #0
- **Size:** 16 bits
- **Offset:** 0x307e
- **Exists:** Always

Table 11-465 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_0

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.462 RAWLANEN_DIG_AON_ADPT_CTL_1

- **Description:** Adaptation Control register #1
- **Size:** 16 bits
- **Offset:** 0x307f
- **Exists:** Always

Table 11-466 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_1

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.463 RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ

- **Description:** Reset routine request
- **Size:** 16 bits
- **Offset:** 0x3080
- **Exists:** Always

Table 11-467 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RESET_RTN_REQ	R/W	**Description**: Reset routine request **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.464 RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ

- **Description:** RX reset interrupt
- **Size:** 16 bits
- **Offset:** 0x3081
- **Exists:** Always

Table 11-468 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_RESET	R	**Description**: RX reset interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.465 RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ

- **Description:** RX request interrupt
- **Size:** 16 bits
- **Offset:** 0x3082
- **Exists:** Always

Table 11-469 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_REQ	R	**Description**: RX request interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.466 RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ

- **Description:** RX rate change interrupt request
- **Size:** 16 bits
- **Offset:** 0x3083
- **Exists:** Always

Table 11-470 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_RATE_IRQ	R	**Description**: RX rate change interrupt request (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.467 RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ

- **Description:** RX pstate change interrupt request
- **Size:** 16 bits
- **Offset:** 0x3084
- **Exists:** Always

Table 11-471 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_PSTATE_IRQ	R	**Description**: RX pstate change interrupt request (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.468 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ

- **Description:** RX adaptation request interrupt
- **Size:** 16 bits
- **Offset:** 0x3085
- **Exists:** Always

Table 11-472 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_REQ_IRQ	R	**Description**: RX adaptation request interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.469 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ

- **Description:** RX adaptation disable interrupt
- **Size:** 16 bits
- **Offset:** 0x3086
- **Exists:** Always

Table 11-473 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_DIS_IRQ	R	**Description**: RX adaptation disable interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.470 RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR

- **Description:** RX reset interrupt clear
- **Size:** 16 bits
- **Offset:** 0x3087
- **Exists:** Always

Table 11-474 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_RESET_IRQ_CLR	R/W	**Description**: RX reset interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.471 RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR

- **Description:** RX request interrupt clear
- **Size:** 16 bits
- **Offset:** 0x3088
- **Exists:** Always

Table 11-475 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_REQ_IRQ_CLR	R/W	**Description**: RX request interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.472 RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR

- **Description:** RX rate change interrupt clear
- **Size:** 16 bits
- **Offset:** 0x3089
- **Exists:** Always

Table 11-476 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_RATE_IRQ_CLR	R/W	**Description**: RX rate change interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.473 RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR

- **Description:** RX pstate change interrupt clear
- **Size:** 16 bits
- **Offset:** 0x308a
- **Exists:** Always

Table 11-477 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_PSTATE_IRQ_CLR	R/W	**Description**: RX pstate change interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.474 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR

- **Description:** RX adaptation request interrupt clear
- **Size:** 16 bits
- **Offset:** 0x308b
- **Exists:** Always

Table 11-478 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_REQ_IRQ_CLR	R/W	**Description**: RX adaptation request interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.475 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR

- **Description:** RX adaptation disable interrupt clear
- **Size:** 16 bits
- **Offset:** 0x308c
- **Exists:** Always

Table 11-479 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_DIS_IRQ_CLR	R/W	**Description**: RX adaptation disable interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.476 RAWLANEN_DIG_IRQ_CTL_IRQ_MASK

- **Description:** Interrupt Mask
- **Size:** 16 bits
- **Offset:** 0x308d
- **Exists:** Always

Table 11-480 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7	RX_IQ_IRQ_MSK	R/W	**Description**: Mask for RX IQ interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
6	MPLL_RECAL_IRQ_MSK	R/W	**Description**: Mask for MPLL re-calibration interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
5	RX_RESET_IRQ_MSK	R/W	**Description**: Mask for RX reset interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
4	RX_ADAPT_DIS_IRQ_MSK	R/W	**Description**: Mask for RX adaptation disable interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
3	RX_ADAPT_REQ_IRQ_MSK	R/W	**Description**: Mask for RX adaptation request interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

Table 11-480 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (Continued)

Bits	Name	Memory Access	Description
2	RX_PSTATE_IRQ_MSK	R/W	**Description**: Mask for RX pstate change interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_RATE_IRQ_MSK	R/W	**Description**: Mask for RX rate change interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_REQ_IRQ_MSK	R/W	**Description**: Mask for RX request interrupt (0 = cannot interrupt) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.477 RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ

- **Description:** MPLLA/B Re-calibration interrupt
- **Size:** 16 bits
- **Offset:** 0x308e
- **Exists:** Always

Table 11-481 Fields for Register: RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLL_RECAL_IRQ	R	**Description**: MPLLA/B Re-calibration interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.478 RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ_CLR

- **Description:** MPLL Re-calibration interrupt clear
- **Size:** 16 bits
- **Offset:** 0x308f
- **Exists:** Always

Table 11-482 Fields for Register: RAWLANEN_DIG_IRQ_CTL_MPLL_RECAL_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	MPLL_RECAL_IRQ_CLR	R/W	**Description**: MPLL Re-Calibration interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.479 RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ

- **Description:** RX IQ interrupt
- **Size:** 16 bits
- **Offset:** 0x3090
- **Exists:** Always

Table 11-483 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_IQ_IRQ	R	**Description**: RX IQ interrupt (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.480 RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ_CLR

- **Description:** RX IQ interrupt clear
- **Size:** 16 bits
- **Offset:** 0x3091
- **Exists:** Always

Table 11-484 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_IQ_IRQ_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_IQ_IRQ_CLR	R/W	**Description**: RX IQ interrupt clear (self-clearing) (volatile) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1 **Volatile**: true

11.2.481 RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN

- **Description:** Override values for incoming LANE controls
- **Size:** 16 bits
- **Offset:** 0x30a0
- **Exists:** Always

Table 11-485 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	LANE_OVRD_EN	R/W	**Description**: Override enable for signals in this register Input Sigals overridden by it: lane_mplla_en_in lane_mpllb_en_in **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	LANE_MPLLB_EN_IN	R/W	**Description**: Override value for lane_mpllb_en_in. Its override enable is: LANE_OVRD_IN.LANE_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	LANE_MPLLA_EN_IN	R/W	**Description**: Override value for lane_mplla_en_in. Its override enable is: LANE_OVRD_IN.LANE_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.482 RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT

- **Description:** Override values for outgoing LANE controls
- **Size:** 16 bits
- **Offset:** 0x30a1
- **Exists:** Always

Table 11-486 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	LANE_OVRD_EN	R/W	**Description**: Override enable for signals in this register Output Signals overridden by it: lane_mplla_en_out lane_mpllb_en_out **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	LANE_MPLLB_EN_OUT	R/W	**Description**: Override value for lane_mpllb_en_out. Its override enable is: LANE_OVRD_OUT.LANE_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	LANE_MPLLA_EN_OUT	R/W	**Description**: Override value for lane_mplla_en_out. Its override enable is: LANE_OVRD_OUT.LANE_OVRD_EN **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.483 RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN

- **Description:** Override values for incoming SUP controls from PMA
- **Size:** 16 bits
- **Offset:** 0x30a2
- **Exists:** Always

Table 11-487 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	SUP_STATE_OVRD_EN	R/W	**Description**: Override enable for signals in this register Input Signals overridden by it: mplla_state_crclk (synchronized version of mplla_state, coming from CMN) mpllb_state_crclk (synchronized version of mpllb_state, coming from CMN) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	MPLLB_STATE	R/W	**Description**: Override value for mpllb_state. Its override enable is: SUP_OVRD_IN.SUP_STATE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	MPLLA_STATE	R/W	**Description**: Override value for mplla_state. Its override enable is: SUP_OVRD_IN.SUP_STATE_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.484 RAWLANEN_DIG_PMA_XF_SUP_PMA_IN

- **Description:** Current values for coming SUP status controls from PMA
- **Size:** 16 bits
- **Offset:** 0x30a3
- **Exists:** Always

Table 11-488 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_PMA_IN

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	RTUNE_ACK	R	**Description**: Value from PMA for rtune_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
1	MPLLB_STATE	R	**Description**: Value from PMA for mpllb_state (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
0	MPLLA_STATE	R	**Description**: Value from PMA for mplla_state (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.485 RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT

- **Description:** Override values for outgoing TX controls to PMA
- **Size:** 16 bits
- **Offset:** 0x30a4
- **Exists:** Always

Table 11-489 Fields for Register: RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	TX_RESET_OVRD_EN	R/W	**Description**: Override enable for tx_reset Output Signals to PMA overridden by it: tx_reset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_RESET_OVRD_VAL	R/W	**Description**: Override value for tx_reset. Its override enable is: TX_OVRD_OUT.TX_RESET_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TX_REQ_OVRD_EN	R/W	**Description**: Override enable for tx_req Output Signals to PMA overridden by it: tx_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	TX_REQ_OVRD_VAL	R/W	**Description**: Override value for tx_req. Its override enable is: TX_OVRD_OUT.TX_REQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.486 RAWLANEN_DIG_PMA_XF_TX_PMA_IN

- **Description:** Current values for coming TX status controls from PMA
- **Size:** 16 bits
- **Offset:** 0x30a5
- **Exists:** Always

Table 11-490 Fields for Register: RAWLANEN_DIG_PMA_XF_TX_PMA_IN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ACK	R	**Description**: Value from PMA for tx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.487 RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT

- **Description:** Override values for outgoing RX controls to PMA
- **Size:** 16 bits
- **Offset:** 0x30a6
- **Exists:** Always

Table 11-491 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	RX_RESET_OVRD_EN	R/W	**Description**: Override enable for rx_reset Output Signals to PMA overridden by it: rx_reset **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_RESET_OVRD_VAL	R/W	**Description**: Override value for rx_reset. Its override enable is: RX_OVRD_OUT.RX_RESET_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_REQ_OVRD_EN	R/W	**Description**: Override enable for rx_req Output Signals to PMA overridden by it: rx_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_REQ_OVRD_VAL	R/W	**Description**: Override value for rx_req. Its override enable is: RX_OVRD_OUT.RX_REQ_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.488 RAWLANEN_DIG_PMA_XF_RX_PMA_IN

- **Description:** Current values for coming RX status controls from PMA
- **Size:** 16 bits
- **Offset:** 0x30a7
- **Exists:** Always

Table 11-492 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_PMA_IN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ACK	R	**Description**: Value from PMA for rx_ack (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.489 RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL

- **Description:** Lane Rtune Controls
- **Size:** 16 bits
- **Offset:** 0x30a8
- **Exists:** Always

Table 11-493 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	LANE_RTUNE_REQ	R/W	**Description**: Lane value for rtune_req **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.490 RAWLANEN_DIG_PMA_XF_RX_OVRD_IN

- **Description:** Override values for incoming RX controls from PMA
- **Size:** 16 bits
- **Offset:** 0x30a9
- **Exists:** Always

Table 11-494 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_OVRD_IN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RX_ACK_OVRD_EN	R/W	**Description**: Override enable for rx_ack Input signal from PMA overridden by it: rx_ack **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ACK_OVRD_VAL	R/W	**Description**: Override value for rx_ack Its override enable is: RX_OVRD_IN.RX_ACK_OVRD_EN **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.491 RAWLANEN_DIG_PMA_XF_SRAM_REC_EN

- **Description:** SRAM Record Enable
- **Size:** 16 bits
- **Offset:** 0x30aa
- **Exists:** Always

Table 11-495 Fields for Register: RAWLANEN_DIG_PMA_XF_SRAM_REC_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	REC_EN	R	**Description**: Enable the SRAM Recording (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.492 RAWLANEN_DIG_TX_CTL_TX_FSM_CTL

- **Description:** TX FSM Control
- **Size:** 16 bits
- **Offset:** 0x30c0
- **Exists:** Always

Table 11-496 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_FSM_CTL

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	TX_P0_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P0 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	TX_P0S_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P0S **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	TX_P1_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P1 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
0	TX_P2_ALLOW_RXDET	R/W	**Description**: If asserted, then rxdet request is allowed in P2 **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.493 RAWLANEN_DIG_TX_CTL_TX_CLK_CTL

- **Description:** Select clock to act as TX input clock
- **Size:** 16 bits
- **Offset:** 0x30c1
- **Exists:** Always

Table 11-497 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CLK_CTL

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
4:1	TX_CLK_SEL	R/W	**Description**: Select clock source for tx_pma_clk 0 - tx_pcs_clk (input clock from pcs) 1 - mplla_word_clk 2 - mplla_dword_clk 3 - mplla_qword_clk 4 - mplla_oword_clk 5 - mplla_div16p5_clk 6 - mplla_div33_clk 7 - mplla_div66_clk 8 - mplla_div_clk 9 - mpllb_word_clk 10 - mpllb_dword_clk 11 - mpllb_qword_clk 12 - mpllb_oword_clk 13 - mpllb_div_clk 14 - rx_clk (from pma) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xf
0	TX_CLK_EN	R/W	**Description**: Enable the tx_clk to pma tx lane TX_CLK_EN must be deasserted when switching TX_CLK_SEL **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1

11.2.494 RAWLANEN_DIG_RX_CTL_RX_FSM_CTL

- **Description:** RX FSM control register
- **Size:** 16 bits
- **Offset:** 0x30c2
- **Exists:** Always

Table 11-498 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_FSM_CTL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1	RATE_CHG_IN_P1	R/W	**Description**: When asserted, then a rate change in P0/P0s will be sequenced such that the RX is put in P1, the rate change is applied, and then the RX is returned to P0/P0s. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	EN_RX_CTL_FSM	R/W	**Description**: Enable the RX control FSM in the Raw PCS If enabled, then when FSM detects a rate change, it moves the RX to P1, does the rate change then goes back to P0/P0s. If not enabled, then FSM is by-passed. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.495 RAWLANEN_DIG_RX_CTL_RX_LOS_MASK_CTL

- **Description:** RX LOS Mask Control
- **Size:** 16 bits
- **Offset:** 0x30c3
- **Exists:** Always

Table 11-499 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_LOS_MASK_CTL

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
8:0	RX_LOS_MASK_CNT	R/W	**Description**: Number of cycles (ref_range_clk) to mask out the rx_los output, from the time the los is powered-on. Default set for minimum 1.1us. **Value After Reset**: 0x1d **Exists**: Always **Reset Mask**: 0x1ff

11.2.496 RAWLANEN_DIG_RX_CTL_RX_DATA_EN_OVRD_CTL

- **Description:** RX Data Enable Override Control
- **Size:** 16 bits
- **Offset:** 0x30c4
- **Exists:** Always

Table 11-500 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DATA_EN_OVRD_CTL

Bits	Name	Memory Access	Description
15:5	INT_REF_TRCK_CNT	R/W	**Description**: Number of ref_range cycles to wait for integral reference tracking to settle. **Value After Reset**: 0x3e8 **Exists**: Always **Reset Mask**: 0x7ff
4:0	RX_DATA_EN_OVRD_CNT	R/W	**Description**: Number of ref_range cycles to override rx_data_en to 1. **Value After Reset**: 0x9 **Exists**: Always **Reset Mask**: 0x1f

11.2.497 RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS

- **Description:** RX continuous offset cancellation status
- **Size:** 16 bits
- **Offset:** 0x30c5
- **Exists:** Always

Table 11-501 Fields for Register: RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ENABLE	R	**Description**: Enable status for rx continuous offset cancellation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.498 RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS

- **Description:** RX continuous adaptation status
- **Size:** 16 bits
- **Offset:** 0x30c6
- **Exists:** Always

Table 11-502 Fields for Register: RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	ENABLE	R	**Description**: Enable status for rx continuous adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.499 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE

- **Description:** RX adaptation mode
- **Size:** 16 bits
- **Offset:** 0x30c7
- **Exists:** Always

Table 11-503 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
1:0	ADAPT_MODE	R	**Description**: RX adaptation mode (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.500 RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL

- **Description:** Select Between Two Banks Adaptation Settings for RX Adaptation
- **Size:** 16 bits
- **Offset:** 0x30c8
- **Exists:** Always

Table 11-504 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_SEL	R	**Description**: Select Between Two Banks Adaptation Settings for RX Adaptation (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.501 RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT

- **Description:** RX CDR PPM Drift on rx clock
- **Size:** 16 bits
- **Offset:** 0x30c9
- **Exists:** Always

Table 11-505 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
14	RX_PPM_DRIFT_VLD	R	**Description**: Indicates if the PPM Drift is valid. (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true
13:0	RX_PPM_DRIFT	R	**Description**: Indicates the amount of drift on rx clock when RX_PPM_DRIFT_VLD is asserted (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.502 RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS

- **Description:** RX CDR Detector Status Register
- **Size:** 16 bits
- **Offset:** 0x30ca
- **Exists:** Always

Table 11-506 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2:0	CDR_DET_STATE	R	**Description**: Indicate the current state for the state machine in CDR Detector (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.503 RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL

- **Description:** Values for outgoing RX controls to PMA
- **Size:** 16 bits
- **Offset:** 0x30cb
- **Exists:** Always

Table 11-507 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	RX_DFE_TAP1_ADAPT_OVRD_EN	R/W	**Description**: Enable override the DFE Tap1 to zero during RX Adaptation **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_CDR_TRACK_EN	R/W	**Description**: Value for rx_cdr_track_en to PMA **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	RX_ALIGH_EN	R/W	**Description**: Value for rx_align_en to PMA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_CLK_SHIFT	R/W	**Description**: Value for rx_clk_shift to PMA **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.504 RAWLANEN_DIG_RX_CTL_RX_PMA_EQ_IQ

- **Description:** RX PMA Equalization IQ Phase value
- **Size:** 16 bits
- **Offset:** 0x30cc
- **Exists:** Always

Table 11-508 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PMA_EQ_IQ

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_PMA_EQ_IQ	R	**Description**: RX PMA equalization IQ phase value (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.505 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN

- **Description:** Enable RX Adapt Mode Override (using Custom Setting for RX Adaptation)
- **Size:** 16 bits
- **Offset:** 0x30cd
- **Exists:** Always

Table 11-509 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
3	RX_ADAPT_MODE_3_OVRD_EN	R/W	**Description**: Enables RX adapt mode ovrride when adapt_mode is 2'b11 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
2	RX_ADAPT_MODE_2_OVRD_EN	R/W	**Description**: Enables RX adapt mode ovrride when adapt_mode is 2'b10 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	RX_ADAPT_MODE_1_OVRD_EN	R/W	**Description**: Enables RX adapt mode ovrride when adapt_mode is 2'b01 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	RX_ADAPT_MODE_0_OVRD_EN	R/W	**Description**: Enables RX adapt mode ovrride when adapt_mode is 2'b00 **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.506 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN

- **Description:** Enable Using Four Different RX Adaptation Modes
- **Size:** 16 bits
- **Offset:** 0x30ce
- **Exists:** Always

Table 11-510 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	RX_ADAPT_MODE_EN	R	**Description**: Enable using four different RX adaptation modes (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0 **Volatile**: true

11.2.507 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM

- **Description:** RX Adaptation Mission-Mode Figure of Merit
- **Size:** 16 bits
- **Offset:** 0x30cf
- **Exists:** Always

Table 11-511 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ADAPT_MM_FOM	R/W	**Description**: RX Adaptation Mission-Mode Figure of Merit **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.508 RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM

- **Description:** RX Adaptation Start-up Figure of Merit (Base Mode or Extended Mode)
- **Size:** 16 bits
- **Offset:** 0x30d0
- **Exists:** Always

Table 11-512 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ADAPT_STARTUP_FOM	R/W	**Description**: RX Adaptation Start-up Figure of Merit (Base Mode or Extended Mode) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xff

11.2.509 RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN

- **Description:** RX Adapted value of Reference Level for Error Even Slicer
- **Size:** 16 bits
- **Offset:** 0x30d1
- **Exists:** Always

Table 11-513 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ADPT_REF_ERR_EVEN	R/W	**Description**: RX Adapted value of reference level for error even slicer for bank 0 **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.510 RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD

- **Description:** RX Adapted value of Reference Level for Error Odd Slicer
- **Size:** 16 bits
- **Offset:** 0x30d2
- **Exists:** Always

Table 11-514 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
7:0	RX_ADPT_REF_ERR_ODD	R/W	**Description**: RX Adapted value of reference level for error odd slicer **Value After Reset**: 0x3a **Exists**: Always **Reset Mask**: 0xff

11.2.511 RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP

- **Description:** RX Phase Adjust Mapped Value
- **Size:** 16 bits
- **Offset:** 0x30d3
- **Exists:** Always

Table 11-515 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
5:0	RX_PHSADJ_MAP	R	**Description**: Mapped value for RX phase adjust (actual reset value is 0xX) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0

11.2.512 RAWLANEN_DIG_RX_CTL_RX_DAC_IN_USE

- **Description:** Indicates state of the DAC Interface
- **Size:** 16 bits
- **Offset:** 0x30d4
- **Exists:** Always

Table 11-516 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DAC_IN_USE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
0	DAC_IN_USE	R/W	**Description**: Indicates state of the DAC interface This is used by firmware only **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.513 RAWLANEN_DIG_AON_ADPT_CTL_2

- **Description:** Adaptation Control register #2
- **Size:** 16 bits
- **Offset:** 0x30e0
- **Exists:** Always

Table 11-517 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_2

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.514 RAWLANEN_DIG_AON_ADPT_CTL_3

- **Description:** Adaptation Control register #3
- **Size:** 16 bits
- **Offset:** 0x30e1
- **Exists:** Always

Table 11-518 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_3

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.515 RAWLANEN_DIG_AON_ADPT_CTL_4

- **Description:** Adaptation Control register #4
- **Size:** 16 bits
- **Offset:** 0x30e2
- **Exists:** Always

Table 11-519 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_4

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.516 RAWLANEN_DIG_AON_ADPT_CTL_5

- **Description:** Adaptation Control register #5
- **Size:** 16 bits
- **Offset:** 0x30e3
- **Exists:** Always

Table 11-520 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_5

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.517 RAWLANEN_DIG_AON_ADPT_CTL_6

- **Description:** Adaptation Control register #6
- **Size:** 16 bits
- **Offset:** 0x30e4
- **Exists:** Always

Table 11-521 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_6

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.518 RAWLANEN_DIG_AON_ADPT_CTL_7

- **Description:** Adaptation Control register #7
- **Size:** 16 bits
- **Offset:** 0x30e5
- **Exists:** Always

Table 11-522 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_7

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.519 RAWLANEN_DIG_AON_ADPT_CTL_8

- **Description:** Adaptation Control register #8
- **Size:** 16 bits
- **Offset:** 0x30e6
- **Exists:** Always

Table 11-523 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_8

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.520 RAWLANEN_DIG_AON_ADPT_CTL_9

- **Description:** Adaptation Control register #9
- **Size:** 16 bits
- **Offset:** 0x30e7
- **Exists:** Always

Table 11-524 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_9

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.521 RAWLANEN_DIG_AON_ADPT_CTL_10

- **Description:** Adaptation Control register #10
- **Size:** 16 bits
- **Offset:** 0x30e8
- **Exists:** Always

Table 11-525 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_10

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.522 RAWLANEN_DIG_AON_ADPT_CTL_11

- **Description:** Adaptation Control register #11
- **Size:** 16 bits
- **Offset:** 0x30e9
- **Exists:** Always

Table 11-526 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_11

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.523 RAWLANEN_DIG_AON_ADPT_CTL_12

- **Description:** Adaptation Control register #12
- **Size:** 16 bits
- **Offset:** 0x30ea
- **Exists:** Always

Table 11-527 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_12

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.524 RAWLANEN_DIG_AON_ADPT_CTL_13

- **Description:** Adaptation Control register #13
- **Size:** 16 bits
- **Offset:** 0x30eb
- **Exists:** Always

Table 11-528 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_13

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.525 RAWLANEN_DIG_AON_ADPT_CTL_14

- **Description:** Adaptation Control register #14
- **Size:** 16 bits
- **Offset:** 0x30ec
- **Exists:** Always

Table 11-529 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_14

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.526 RAWLANEN_DIG_AON_ADPT_CTL_15

- **Description:** Adaptation Control register #15
- **Size:** 16 bits
- **Offset:** 0x30ed
- **Exists:** Always

Table 11-530 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_15

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.527 RAWLANEN_DIG_AON_ADPT_CTL_16

- **Description:** Adaptation Control register #16
- **Size:** 16 bits
- **Offset:** 0x30ee
- **Exists:** Always

Table 11-531 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_16

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.528 RAWLANEN_DIG_AON_ADPT_CTL_17

- **Description:** Adaptation Control register #17
- **Size:** 16 bits
- **Offset:** 0x30ef
- **Exists:** Always

Table 11-532 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_17

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.529 RAWLANEN_DIG_AON_ADPT_CTL_18

- **Description:** Adaptation Control register #18
- **Size:** 16 bits
- **Offset:** 0x30f0
- **Exists:** Always

Table 11-533 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_18

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.530 RAWLANEN_DIG_AON_ADPT_CTL_19

- **Description:** Adaptation Control register #19
- **Size:** 16 bits
- **Offset:** 0x30f1
- **Exists:** Always

Table 11-534 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_19

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.531 RAWLANEN_DIG_AON_ADPT_CTL_20

- **Description:** Adaptation Control register #20
- **Size:** 16 bits
- **Offset:** 0x30f2
- **Exists:** Always

Table 11-535 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_20

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.532 RAWLANEN_DIG_AON_ADPT_CTL_21

- **Description:** Adaptation Control register #21
- **Size:** 16 bits
- **Offset:** 0x30f3
- **Exists:** Always

Table 11-536 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_21

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.533 RAWLANEN_DIG_AON_ADPT_CTL_22

- **Description:** Adaptation Control register #22
- **Size:** 16 bits
- **Offset:** 0x30f4
- **Exists:** Always

Table 11-537 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_22

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.534 RAWLANEN_DIG_AON_ADPT_CTL_23

- **Description:** Adaptation Control register #23
- **Size:** 16 bits
- **Offset:** 0x30f5
- **Exists:** Always

Table 11-538 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_23

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.535 RAWLANEN_DIG_AON_ADPT_CTL_24

- **Description:** Adaptation Control register #24
- **Size:** 16 bits
- **Offset:** 0x30f6
- **Exists:** Always

Table 11-539 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_24

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.536 RAWLANEN_DIG_AON_ADPT_CTL_25

- **Description:** Adaptation Control register #25
- **Size:** 16 bits
- **Offset:** 0x30f7
- **Exists:** Always

Table 11-540 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_25

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.537 RAWLANEN_DIG_AON_ADPT_CTL_26

- **Description:** Adaptation Control register #26
- **Size:** 16 bits
- **Offset:** 0x30f8
- **Exists:** Always

Table 11-541 Fields for Register: RAWLANEN_DIG_AON_ADPT_CTL_26

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Description**: Value of adaptation control **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.538 RAWLANEN_DIG_AON_SRAM_REC_CTRL

- **Description:** SRAM Record Control
- **Size:** 16 bits
- **Offset:** 0x30f9
- **Exists:** Always

Table 11-542 Fields for Register: RAWLANEN_DIG_AON_SRAM_REC_CTRL

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	REC_EN	R/W	**Description**: Enable the SRAM Recording **Value After Reset**: 0x1 **Exists**: Always **Reset Mask**: 0x1
1	REC_MODE	R/W	**Description**: The SRAM Recording Mode If REC_MODE = 1'b0, only the first 16 records will be stored in SRAM If REC_MODE = 1'b1, the most recent 16 records (if there are) will be stored in SRAM **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	REC_INIT_DONE	R/W	**Description**: Indicates if the SRAM_REC_ADDR and SRAM_REC_ITER have been initialized. Write 1'b0 to this field if they need be re-initialized. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.539 RAWLANEN_DIG_AON_SRAM_REC_ADDR

- **Description:** Current SRAM Recording Address
- **Size:** 16 bits
- **Offset:** 0x30fa
- **Exists:** Always

Table 11-543 Fields for Register: RAWLANEN_DIG_AON_SRAM_REC_ADDR

Bits	Name	Memory Access	Description
15:0	SRAM_REC_ADDR	R/W	**Description**: Current SRAM Recording Address **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.540 RAWLANEN_DIG_AON_SRAM_REC_ITER

- **Description:** Current Iteration Count for SRAM Recording
- **Size:** 16 bits
- **Offset:** 0x30fb
- **Exists:** Always

Table 11-544 Fields for Register: RAWLANEN_DIG_AON_SRAM_REC_ITER

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:0	SRAM_REC_ITER	R/W	**Description**: Current Iteration Count for SRAM Recording **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x3ff

11.2.541 RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL_1

- **Description:** Set to Skip Firmware startup/rate-change algorithms
- **Size:** 16 bits
- **Offset:** 0x30fc
- **Exists:** Always

Table 11-545 Fields for Register: RAWLANEN_DIG_AON_RX_STARTUP_ALGO_CTL_1

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
2	SKIP_RX_VGA_SLICER_CAL_STARTUP	R/W	**Description**: Skip RX VGA slicer start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
1	SKIP_RX_BYPASS_CAL_STARTUP	R/W	**Description**: Skip RX bypass slicer start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1
0	SKIP_RX_ERROR_CAL_STARTUP	R/W	**Description**: Skip RX error slicer start-up calibration **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.542 RAWLANEN_DIG_AON_RX_IQ_CTL

- **Description:** RX IQ Adapatation Control Register #1
- **Size:** 16 bits
- **Offset:** 0x30fd
- **Exists:** Always

Table 11-546 Fields for Register: RAWLANEN_DIG_AON_RX_IQ_CTL

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
6:4	IQ_ADPT_RESERVED	R/W	**Description**: reserved **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x7
3:0	IQ_ADPT_OFST	R/W	**Description**: Inital IQ offset for IQ adpatation This is the additional IQ offset that will be added before doing the IQ adaptation **Value After Reset**: 0x3 **Exists**: Always **Reset Mask**: 0xf

11.2.543 RAWLANEN_DIG_AON_RX_IQ_CTL_1

- **Description:** RX IQ Adapatation Control Register #2
- **Size:** 16 bits
- **Offset:** 0x30fe
- **Exists:** Always

Table 11-547 Fields for Register: RAWLANEN_DIG_AON_RX_IQ_CTL_1

Bits	Name	Memory Access	Description
15:1	DPLL_THRESH	R/W	**Description**: DPLL frequency threshold **Value After Reset**: 0x1000 **Exists**: Always **Reset Mask**: 0x7fff
0	DPLL_THRESH_CENTER	R/W	**Description**: Indicates the center of the DPLL threshold value 0 - Center is relative to the current value of DPLL frequency 1 - Absolute center (for example 'h8192) **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1

11.2.544 RAWLANEN_DIG_AON_RX_ADPT_IQ_LIMIT

- **Description:** RX IQ Adaptation Offset Limit
- **Size:** 16 bits
- **Offset:** 0x30ff
- **Exists:** Always

Table 11-548 Fields for Register: RAWLANEN_DIG_AON_RX_ADPT_IQ_LIMIT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	**Description**: Reserved for Future use and actual reset value is 0xX **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x0
9:5	RIGHT_LIMIT	R/W	**Description**: The max IQ offset from Calibrated IQ for right-side IQ sweep This is a signed number in sign and magnitude format; the bit 4 is the sign bit, and the bits [3:0] is its magnitude. **Value After Reset**: 0x7 **Exists**: Always **Reset Mask**: 0x1f
4:0	LEFT_LIMIT	R/W	**Description**: The max IQ offset from Calibrated IQ for left-side IQ sweep This is a signed number in sign and magnitude format; the bit 4 is the sign bit, and the bits [3:0] is its magnitude. **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0x1f

11.2.545 RAWMEM_DIG_ROM_CMN0_B0_R0

- **Description:** This is the first ROM memory address
- **Size:** 16 bits
- **Offset:** 0x4000
- **Exists:** Always

Table 11-549 Fields for Register: RAWMEM_DIG_ROM_CMN0_B0_R0

Bits	Name	Memory Access	Description
15:0	DATA	R	**Description**: Memory data **Value After Reset**: 0x81d **Exists**: Always **Reset Mask**: 0xffff

11.2.546 RAWMEM_DIG_ROM_CMN63_B7_R31

- **Description:** This is the last ROM memory address
- **Size:** 16 bits
- **Offset:** 0x7fff
- **Exists:** Always

Table 11-550 Fields for Register: RAWMEM_DIG_ROM_CMN63_B7_R31

Bits	Name	Memory Access	Description
15:0	DATA	R	**Description**: Memory data **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

11.2.547 RAWMEM_DIG_RAM_CMN0_B0_R0

- **Description:** This is the first RAM memory address
- **Size:** 16 bits
- **Offset:** 0xc000
- **Exists:** Always

Table 11-551 Fields for Register: RAWMEM_DIG_RAM_CMN0_B0_R0

Bits	Name	Memory Access	Description
15:0	DATA	R	**Description**: Memory data **Value After Reset**: 0x81d **Exists**: Always **Reset Mask**: 0xffff

11.2.548 RAWMEM_DIG_RAM_CMN63_B7_R31

- **Description:** This is the last RAM memory address
- **Size:** 16 bits
- **Offset:** 0xffff
- **Exists:** Always

Table 11-552 Fields for Register: RAWMEM_DIG_RAM_CMN63_B7_R31

Bits	Name	Memory Access	Description
15:0	DATA	R	**Description**: Memory data **Value After Reset**: 0x0 **Exists**: Always **Reset Mask**: 0xffff

A
Support Configuration Settings

This section contains support configuration values for configuring the PHY for your application.

Information/Primary Inputs		PCIe										
		Gen 1 SRIS/SRNS Refclk										Gen1 Common Refclk
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	20	25	40	50	62.5	78.125	100	125	156.25	200	100
	MPLL VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	20	25	40	50	62.5	78.125	100	62.5	78.125	100	100
	divide ratio determined by mpll{a,b}_multiplier	125	100	125	50	40	32	25	20	32	25	25
	mpll{a,b}_word_clk frequency (MHz)	500	500	500	500	500	500	500	500	500	500	500
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
Primary Inputs	ref_range[2:0]	0	0	1	1	2	3	3	4	3	3	3
	ref_clk_div2_en	0	0	0	0	0	0	0	0	1	1	0
	ref_clk_mpll{a,b}_div2_en	0	0	1	0	0	0	0	0	0	0	0
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	102	102	102	102	102	102	102	102	102	102	102
	mpll{a,b}_ssc_freq_cnt_init[11:0]	37	37	37	37	37	37	37	37	37	37	37
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	125	100	125	50	40	32	25	20	32	25	25
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41039	41039	41039	41031	41031	41031	41029	41029	41031	41029	41356
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	1	1	3	3	3	3	3	3
	rxX_ref_ld_val[6:0]	6	7	11	14	17	21	27	34	21	27	27
	rxX_vco_ld_val[12:0]	1500	1400	1375	1400	1360	1344	1350	1360	1344	1350	1350
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	20	19	18	19	18	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		PCIe										
		Gen 2 SRIS/SRNS Refclk										Gen2 Common Refclk
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	20	25	40	50	62.5	78.125	100	125	156.25	200	100
	MPLL VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	20	25	40	50	62.5	78.125	100	62.5	78.125	100	100
	divide ratio determined by mpll{a,b}_multiplier	125	100	125	50	40	32	25	20	32	25	25
	mpll{a,b}_word_clk frequency (MHz)	500	500	500	500	500	500	500	500	500	500	500
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
Primary Inputs	ref_range[2:0]	0	0	1	1	2	3	3	4	3	3	3
	ref_clk_div2_en	0	0	0	0	0	0	0	0	1	1	0
	ref_clk_mpll{a,b}_div2_en	0	0	1	0	0	0	0	0	0	0	0
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	102	102	102	102	102	102	102	102	102	102	102
	mpll{a,b}_ssc_freq_cnt_init[11:0]	37	37	37	37	37	37	37	37	37	37	37
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	125	100	125	50	40	32	25	20	32	25	25
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41039	41039	41039	41031	41031	41031	41029	41029	41031	41029	41356
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	1	1	3	3	3	3	3	3
	rxX_ref_ld_val[6:0]	6	7	11	14	17	21	27	34	21	27	27
	rxX_vco_ld_val[12:0]	1500	1400	1375	1400	1360	1344	1350	1360	1344	1350	1350
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	20	19	18	19	18	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		PCIe								
		Gen3 SRIS/SRNS Refclk								Gen3 Common Refclk
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	20	25	40	50	62.5	100	125	200	100
	MPLL VCO frequency (GHz)	8	8	8	8	8	8	8	8	8
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	20	25	40	50	62.5	100	62.5	100	100
	divide ratio determined by mpll{a,b}_multiplier	200	160	100	80	64	80	64	80	80
	mpll{a,b}_word_clk frequency (MHz)	1000	1000	1000	1000	1000	1000	1000	1000	1000
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	1	1	1	1	1	1	1	1	1
	RX VCO frequency (GHz)	4	4	4	4	4	4	4	4	4
Primary Inputs	ref_range[2:0]	0	0	1	1	2	3	4	3	3
	ref_clk_div2_en	0	0	0	0	0	0	0	1	0
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	0	1	1	1	1
	mpll{a,b}_div8_clk_en	1	1	1	1	1	1	1	1	1
	mpll{a,b}_div10_clk_en	0	0	0	0	0	0	0	0	0
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	81	81	81	81	81	81	81	81	81
	mpll{a,b}_ssc_freq_cnt_init[11:0]	96	96	96	96	96	96	96	96	96
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	100	80	100	80	64	80	64	80	80
	mpll{a,b}_multiplier[7]	1	1	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41038	41036	41034	41014	41014	41014	41014	41014	41094
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	1	1	3	3	3	3
	rxX_ref_ld_val[6:0]	7	9	14	17	21	34	42	34	34
	rxX_vco_ld_val[12:0]	1400	1440	1400	1360	1344	1360	1344	1360	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	19	19	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		PCIe								
		Gen4 SRIS/SRNS Refclk								Gen4 Common Refclk
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	20	25	40	50	62.5	100	125	200	100
	MPLL VCO frequency (GHz)	8	8	8	8	8	8	8	8	8
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	20	25	40	50	62.5	100	62.5	100	100
	divide ratio determined by mpll{a,b}_multiplier	200	160	100	80	64	80	64	80	80
	mpll{a,b}_word_clk frequency (MHz)	1000	1000	1000	1000	1000	1000	1000	1000	1000
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	1	1	1	1	1	1	1	1	1
	RX VCO frequency (GHz)	8	8	8	8	8	8	8	8	8
Primary Inputs	ref_range[2:0]	0	0	1	1	2	3	4	3	3
	ref_clk_div2_en	0	0	0	0	0	0	0	1	0
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	0	1	1	1	1
	mpll{a,b}_div8_clk_en	1	1	1	1	1	1	1	1	1
	mpll{a,b}_div10_clk_en	0	0	0	0	0	0	0	0	0
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	81	81	81	81	81	81	81	81	81
	mpll{a,b}_ssc_freq_cnt_init[11:0]	96	96	96	96	96	96	96	96	96
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	100	80	100	80	64	80	64	80	80
	mpll{a,b}_multiplier[7]	1	1	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41038	41036	41034	41014	41014	41014	41014	41014	41094
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	1	1	3	3	3	3
	rxX_ref_ld_val[6:0]	4	5	7	9	11	17	21	17	17
	rxX_vco_ld_val[12:0]	1600	1600	1400	1440	1408	1360	1344	1360	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	22	22	19	19	19	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		SATA										
		Gen1/2/3										
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	24	25	30	40	48	50	60	80	96	100	120
	MPLL VCO frequency (GHz)	6	6	6	6	6	6	6	6	6	6	6
	Max negative frequency offset for SSC on RX data (PPM)	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000	5000
	phy_ref_dig_clk frequency (MHz)	24	25	30	40	48	50	60	80	48	50	60
	divide ratio determined by mpll{a,b}_multiplier	125	120	100	75	125	120	100	75	125	120	100
	mpll{a,b}_word_clk frequency (MHz)	600	600	600	600	600	600	600	600	600	600	600
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	6	6	6	6	6	6	6	6	6	6	6
Primary Inputs	ref_range[2:0]	0	0	1	1	1	1	2	3	1	1	2
	ref_clk_div2_en	0	0	0	0	0	0	0	0	1	1	1
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	1	1	1	1	1	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	102	102	102	102	102	102	102	102	102	102	102
	mpll{a,b}_ssc_freq_cnt_init[11:0]	45	45	45	45	45	45	45	45	45	45	45
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2
	bs_rx_bigswing	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3
	mpll{a,b}_multiplier[6:0]	125	120	100	75	125	120	100	75	125	120	100
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41084	41084	41067	41050	41084	41084	41067	41050	41084	41084	41067
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	0	1	0	0	0	1	0	0	0
	rxX_ref_ld_val[6:0]	6	6	7	9	11	12	14	18	11	12	14
	rxX_vco_ld_val[12:0]	1500	1440	1400	1350	1375	1440	1400	1350	1375	1440	1400
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	20	19	19	18	18	19	19	18	18	19	19

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		Ethernet										
		1000BASE-KX/SGMII/QSGMII										
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	20	25	40	50	62.5	78.125	100	125	156.25	200	312.5
	MPLL VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	20	25	40	50	62.5	78.125	100	62.5	78.125	100	78.125
	divide ratio determined by mpll{a,b}_multiplier	125	100	125	50	40	32	50	40	32	50	32
	mpll{a,b}_word_clk frequency (MHz)	500	500	500	500	500	500	500	500	500	500	500
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	5	5	5	5	5	5	5	5	5	5	5
Primary Inputs	ref_range[2:0]	0	0	1	1	2	3	3	4	6	3	6
	ref_clk_div2_en	0	0	0	0	0	0	0	0	0	1	1
	ref_clk_mpll{a,b}_div2_en	0	0	1	0	0	0	1	1	1	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	125	100	125	50	40	32	50	40	32	50	32
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41039	41039	41039	41031	41031	41031	41031	41031	41031	41031	41031
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	0	1	1	3	1	1	3	1	3
	rxX_ref_ld_val[6:0]	6	7	11	14	17	21	27	34	42	27	42
	rxX_vco_ld_val[12:0]	1500	1400	1375	1400	1360	1344	1350	1360	1344	1350	1344
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	20	19	18	19	18	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		Ethernet						
		1000BASE-KX/SGMII (Bifurcation using 7.5GHz)						
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	25	50	62.5	78.125	125	156.25	312.5
	MPLL VCO frequency (GHz)	7.5	7.5	7.5	7.5	7.5	7.5	7.5
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	25	50	62.5	78.125	62.5	78.125	78.125
	divide ratio determined by mpll{a,b}_multiplier	150	150	60	48	60	48	48
	mpll{a,b}_word_clk frequency (MHz)	750	750	750	750	750	750	750
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	1	1	1	1	1	1	1
	RX VCO frequency (GHz)	5	5	5	5	5	5	5
Primary Inputs	ref_range[2:0]	0	1	2	3	4	6	6
	ref_clk_div2_en	0	0	0	0	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	1	0	0	1	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	1	1	1	1	1	1	1
	mpll{a,b}_div_multiplier[6:0]	30	30	30	30	30	30	30
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	75	75	60	48	60	48	48
	mpll{a,b}_multiplier[7]	1	1	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41100	41100	41095	41047	41095	41047	41047
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	3	1	3	3
	rxX_ref_ld_val[6:0]	7	14	17	21	34	42	42
	rxX_vco_ld_val[12:0]	1400	1400	1360	1344	1360	1344	1344
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	19	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		Ethernet									
		XAUI/10GBASE-KX4					RXAUI				
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	62.5	78.125	125	156.25	312.5	62.5	78.125	125	156.25	312.5
	MPLL VCO frequency (GHz)	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	62.5	78.125	62.5	78.125	78.125	62.5	78.125	62.5	78.125	78.125
	divide ratio determined by mpll{a,b}_multiplier	50	40	50	40	40	50	40	50	40	40
	mpll{a,b}_word_clk frequency (MHz)	625	625	625	625	625	625	625	625	625	625
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25
Primary Inputs	ref_range[2:0]	2	3	4	6	6	2	3	4	6	6
	ref_clk_div2_en	0	0	0	0	1	0	0	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	0	1	1	1	0	0	1	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	50	40	50	40	40	50	40	50	40	40
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41031	41031	41031	41031	41031	41031	41031	41031	41031	41031
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	1	3	1	3	3	1	3	1	3	3
	rxX_ref_ld_val[6:0]	14	17	27	34	34	14	17	27	34	34
	rxX_vco_ld_val[12:0]	1400	1360	1350	1360	1360	1400	1360	1350	1360	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	18	18	18	18	19	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		Ethernet					
		10GBASE-KR@10.3125G for 16-bit			10GBASE-KR@10.3125G for 20-bit		
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	78.125	156.25	312.5	78.125	156.25	312.5
	MPLL VCO frequency (GHz)	5.15625	5.15625	5.15625	5.15625	5.15625	5.15625
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	78.125	78.125	78.125	78.125	78.125	78.125
	divide ratio determined by mpll{a,b}_multiplier	33	33	33	33	33	33
	mpll{a,b}_word_clk frequency (MHz)	644.53125	644.53125	644.53125	515.625	515.625	515.625
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0
	RX VCO frequency (GHz)	5.15625	5.15625	5.15625	5.15625	5.15625	5.15625
Primary Inputs	ref_range[2:0]	3	6	6	3	6	6
	ref_clk_div2_en	0	0	1	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	1	1	0	1	1
	mpll{a,b}_div8_clk_en	1	1	1	0	0	0
	mpll{a,b}_div10_clk_en	0	0	0	1	1	1
	mplla_div16p5_clk_en	1	1	1	1	1	1
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	33	33	33	33	33	33
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41046	41046	41046	41046	41046	41046
	rx_vref_ctrl[4:0]	17	17	17	17	17	17
	sup_misc[7:0]	3	3	3	3	3	3
	rxX_ref_ld_val[6:0]	21	41	41	21	41	41
	rxX_vco_ld_val[12:0]	1386	1353	1353	1386	1353	1353
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		CEI									
		CEI-6G@6.25G							CEI-11G@10.3125G		
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	25	50	62.5	78.125	125	156.25	312.5	78.125	156.25	312.5
	MPLL VCO frequency (GHz)	6.25	6.25	6.25	6.25	6.25	6.25	6.25	5.15625	5.15625	5.15625
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	25	50	62.5	78.125	62.5	78.125	78.125	78.125	78.125	78.125
	divide ratio determined by mpll{a,b}_multiplier	125	125	50	40	50	40	40	33	33	33
	mpll{a,b}_word_clk frequency (MHz)	625	625	625	625	625	625	625	515.625	515.625	515.625
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	6.25	6.25	6.25	6.25	6.25	6.25	6.25	5.15625	5.15625	5.15625
Primary Inputs	ref_range[2:0]	0	1	2	3	4	6	6	3	6	6
	ref_clk_div2_en	0	0	0	0	0	0	1	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	1	0	0	1	1	1	0	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	1	1	1
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2
	bs_rx_bigswing	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3
	mpll{a,b}_multiplier[6:0]	125	125	50	40	50	40	40	33	33	33
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41085	41085	41031	41031	41031	41031	41031	41046	41046	41046
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	0	1	3	1	3	3	3	3	3
	rxX_ref_ld_val[6:0]	6	11	14	17	27	34	34	21	41	41
	rxX_vco_ld_val[12:0]	1500	1375	1400	1360	1350	1360	1360	1386	1353	1353
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	20	18	19	18	18	18	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		CPRI/OBSAI											
		CPRI/OBSAI @ 2.4576G						CPRI/OBSAI @ 3.072G					
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	30.72	38.4	61.44	76.8	153.6	307.2	30.72	38.4	61.44	76.8	153.6	307.2
	MPLL VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152	6.144	6.144	6.144	6.144	6.144	6.144
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	30.72	38.4	61.44	76.8	76.8	76.8	30.72	38.4	61.44	76.8	76.8	76.8
	divide ratio determined by mpll{a,b}_multiplier	80	64	40	32	32	32	100	80	50	40	40	40
	mpll{a,b}_word_clk frequency (MHz)	491.52	491.52	491.52	491.52	491.52	491.52	614.4	614.4	614.4	614.4	614.4	614.4
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152	6.144	6.144	6.144	6.144	6.144	6.144
Primary Inputs	ref_range[2:0]	1	1	2	2	5	5	1	1	2	2	5	5
	ref_clk_div2_en	0	0	0	0	0	1	0	0	0	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	1	1	0	0	0	0	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	80	64	40	32	32	32	100	80	50	40	40	40
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41150	41150	41143	41143	41143	41143	41054	41054	41063	41047	41047	41047
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	1	1	3	3	3	0	1	1	3	3	3
	rxX_ref_ld_val[6:0]	9	11	17	21	42	42	7	9	14	17	34	34
	rxX_vco_ld_val[12:0]	1440	1408	1360	1344	1344	1344	1400	1440	1400	1360	1360	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	19	18	18	18	18	19	19	19	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		CPRI/OBSAI											
		CPRI/OBSAI @ 4.9152G						CPRI/OBSAI @ 6.144G					
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	30.72	38.4	61.44	76.8	153.6	307.2	30.72	38.4	61.44	76.8	153.6	307.2
	MPLL VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152	6.144	6.144	6.144	6.144	6.144	6.144
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	30.72	38.4	61.44	76.8	76.8	76.8	30.72	38.4	61.44	76.8	76.8	76.8
	divide ratio determined by mpll{a,b}_multiplier	80	64	40	32	32	32	100	80	50	40	40	40
	mpll{a,b}_word_clk frequency (MHz)	491.52	491.52	491.52	491.52	491.52	491.52	614.4	614.4	614.4	614.4	614.4	614.4
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152	6.144	6.144	6.144	6.144	6.144	6.144
Primary Inputs	ref_range[2:0]	1	1	2	2	5	5	1	1	2	2	5	5
	ref_clk_div2_en	0	0	0	0	0	1	0	0	0	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	1	1	0	0	0	0	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	80	64	40	32	32	32	100	80	50	40	40	40
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41150	41150	41143	41143	41143	41143	41054	41054	41063	41047	41047	41047
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17	17	17	17	17
	sup_misc[7:0]	0	1	1	3	3	3	0	1	1	3	3	3
	rxX_ref_ld_val[6:0]	9	11	17	21	42	42	7	9	14	17	34	34
	rxX_vco_ld_val[12:0]	1440	1408	1360	1344	1344	1344	1400	1440	1400	1360	1360	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	19	18	18	18	18	19	19	19	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		CPRI/OBSAI					
		CPRI/OBSAI @ 9.8304G					
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	30.72	38.4	61.44	76.8	153.6	307.2
	MPLL VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	30.72	38.4	61.44	76.8	76.8	76.8
	divide ratio determined by mpll{a,b}_multiplier	80	64	40	32	32	32
	mpll{a,b}_word_clk frequency (MHz)	491.52	491.52	491.52	491.52	491.52	491.52
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0
	RX VCO frequency (GHz)	4.9152	4.9152	4.9152	4.9152	4.9152	4.9152
Primary Inputs	ref_range[2:0]	1	1	2	2	5	5
	ref_clk_div2_en	0	0	0	0	0	1
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0
	bs_tx_lowswing	0	0	0	0	0	0
	bs_rx_bigswing	1	1	1	1	1	1
	mpll{a,b}_multiplier[6:0]	80	64	40	32	32	32
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41150	41150	41143	41143	41143	41143
	rx_vref_ctrl[4:0]	17	17	17	17	17	17
	sup_misc[7:0]	0	1	1	3	3	3
	rxX_ref_ld_val[6:0]	9	11	17	21	42	42
	rxX_vco_ld_val[12:0]	1440	1408	1360	1344	1344	1344
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	19	19	18	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		JESD204B				
		JESD204B @1.25G	JESD204B @3.125G	JESD204B @6.25G	JESD204B @10.3125G	JESD204B @12.5G
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	62.5	62.5	62.5	78.125	62.5
	MPLL VCO frequency (GHz)	5	6.25	6.25	5.15625	6.25
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	62.5	62.5	62.5	78.125	62.5
	divide ratio determined by mpll{a,b}_multiplier	40	50	50	33	50
	mpll{a,b}_word_clk frequency (MHz)	500	625	625	515.625	625
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0
	RX VCO frequency (GHz)	5	6.25	6.25	5.15625	6.25
Primary Inputs	ref_range[2:0]	2	2	2	3	2
	ref_clk_div2_en	0	0	0	0	0
	ref_clk_mpll{a,b}_div2_en	0	0	0	0	0
	mpll{a,b}_div8_clk_en	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0
	bs_tx_lowswing	1	1	1	1	1
	bs_rx_bigswing	0	0	0	0	0
	mpll{a,b}_multiplier[6:0]	40	50	50	33	50
	mpll{a,b}_multiplier[7]	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41031	41031	41031	41046	41031
	rx_vref_ctrl[4:0]	17	17	17	17	17
	sup_misc[7:0]	1	1	1	3	1
	rxX_ref_ld_val[6:0]	17	14	14	21	14
	rxX_vco_ld_val[12:0]	1360	1400	1400	1386	1400
	bs_rx_level[4:0] (DC)	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	18	19	19	18	19

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

Information/Primary Inputs		SRIO							
		SRIO Level II SR/MR/LR @ 5G or SRIO Level I @ 2.5G or SRIO Level I @ 1.25G				SRIO Level II SR/MR/LR @6.25G or SRIO Level I @ 3.125G			
Clock Information	ref_alt_clk_p/m (ref_pad_clk_p/m) frequency (MHz)	62.5	78.125	125	156.25	62.5	78.125	125	156.25
	MPLL VCO frequency (GHz)	5	5	5	5	6.25	6.25	6.25	6.25
	Max negative frequency offset for SSC on RX data (PPM)	0	0	0	0	0	0	0	0
	phy_ref_dig_clk frequency (MHz)	62.5	78.125	62.5	78.125	62.5	78.125	62.5	78.125
	divide ratio determined by mpll{a,b}_multiplier	40	32	40	32	50	40	50	40
	mpll{a,b}_word_clk frequency (MHz)	500	500	500	500	625	625	625	625
	MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0
	RX VCO frequency (GHz)	5	5	5	5	6.25	6.25	6.25	6.25
Primary Inputs	ref_range[2:0]	2	3	4	6	2	3	4	6
	ref_clk_div2_en	0	0	0	0	0	0	0	0
	ref_clk_mpll{a,b}_div2_en	0	0	1	1	0	0	1	1
	mpll{a,b}_div8_clk_en	0	0	0	0	0	0	0	0
	mpll{a,b}_div10_clk_en	1	1	1	1	1	1	1	1
	mplla_div16p5_clk_en	0	0	0	0	0	0	0	0
	mpll{a,b}_tx_clk_div[1:0]	0	0	0	0	0	0	0	0
	mpll{a,b}_div_clk_en	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_div_multiplier[6:0]	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1	Note 1
	mpll{a,b}_ssc_en	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_clk_sel	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_peak[7:0]	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_freq_cnt_init[11:0]	0	0	0	0	0	0	0	0
	mpll{a,b}_ssc_up_spread	0	0	0	0	0	0	0	0
	mpll{a,b}_fracn_ctrl[10:0]	0	0	0	0	0	0	0	0
	bs_tx_lowswing	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2	Note 2
	bs_rx_bigswing	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3	Note 3
	mpll{a,b}_multiplier[6:0]	40	32	40	32	50	40	50	40
	mpll{a,b}_multiplier[7]	0	0	0	0	0	0	0	0
	tx_vboost_lvl[2:0]	5	5	5	5	5	5	5	5
	mpll{a,b}_bandwidth[15:0]	41031	41031	41031	41031	41031	41031	41031	41031
	rx_vref_ctrl[4:0]	17	17	17	17	17	17	17	17
	sup_misc[7:0]	1	3	1	3	1	3	1	3
	rxX_ref_ld_val[6:0]	17	21	34	42	14	17	27	34
	rxX_vco_ld_val[12:0]	1360	1344	1360	1344	1400	1360	1350	1360
	bs_rx_level[4:0] (DC)	12	12	12	12	12	12	12	12
	bs_rx_level[4:0] (AC)	11	11	11	11	11	11	11	11
	rxX_cdr_ppm_max[4:0]	18	18	18	18	19	18	18	18

Note 1: User specific setting.

Note 2: For CEI-6/11G SR, the bs_tx_lowswing needs to set to 1 and for CEI-6/11G LR it needs to set to 0.
For SATA Gen1/2, the bs_tx_lowswing needs to set to 1 and for SATA Gen3 it needs to set to 0.
For SRIO II SR, the bs_tx_lowswing needs to set to 1 and for SRIO II MR/LR it needs to set to 0.

Note 3: For CEI-6/11G SR, the bs_rx_bigswing needs to set to 0 and for CEI-6/11G LR it needs to set to 1.
For SATA Gen1/2, the bs_rx_bigswing needs to set to 0 and for SATA Gen 3 it needs to set to 1.
For SRIO II SR, the bs_rx_bigswing needs to set to 0 and for SRIO II MR/LR it needs to set to 1.

Note 4: For Bifurcated 1000BASE-KX/SGMII, use the mpll_div_clk output.

Note 5: **Sup_misc[2:1]**
00 (0.8V mode)
01 (0.85V mode)
10 (0.9V mode)
11 (1.0V mode)

B
Lane Configuration Settings

This section contains lane configuration values for configuring the PHY for your application.

Information/Primary Inputs		PCIe			
		Gen1	Gen2	Gen3	Gen4
Data Width	Tx Path (bits)	20	20	16	16
	Rx Path (bits)	20	20	16	16
Clock Information	Tx Baud rate (Gbps)	2.5	5	8	16
	Rx Baud rate (Gbps)	2.5	5	8	16
	txX_clk frequency (MHz)	125	250	500	1000
	rxX_clk frequency (MHz)	125	250	500	1000
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	125	250	500	1000
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	1	1
	Corresponding MPLL VCO Frequency (GHz)	5	5	8	8
	Corresponding MPLL Output Clock Frequency (GHz)	2.5	2.5	8	8
	Rx VCO Frequency (GHz)	5	5	4	8
	txX_misc[7:0]	0	0	0	0
	txX_rate[2:0]	1	0	1	0
	txX_mpllb_sel	0	0	1	1
Tx Primary Inputs	txX_width[1:0]	3	3	2	2
	txX_vboost_en	1	1	1	1
	txX_iboost_lvl[3:0]	15	15	15	15
	txX_eq_pre[5:0]	0	0	NOTE 1	NOTE 1
	txX_eq_post[5:0]	28	40	NOTE 1	NOTE 1
	txX_eq_main[5:0]	33	30	NOTE 1	NOTE 1
Rx Primary Inputs	rxX_rate[1:0]	2	1	0	0
	rxX_eq_ctle_pole[1:0]	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	14	12	16
	rxX_los_threshold[2:0]	1	1	1	1
	rxX_cdr_vco_step_ctrl	1	1	1	0
	rxX_cdr_vco_temp_comp_en	1	1	1	1
	rxX_misc[7:0]	6	4	0	1
	rxX_width[1:0]	3	3	2	2
	rxX_div16p5_clk_en	0	0	0	0
	rxX_cdr_ssc_en	0/1	0/1	0/1	0/1
	rxX_los_lfps_en	0	0	0	0
	rxX_dfe_bypass	1	1	0	0
	rxX_eq_att_lvl[2:0]	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	5	5
	rxX_eq_vga2_gain[2:0]	4	4	5	5
	rxX_eq_dfe_tap1[7:0]	0	0	0	10
	rxX_cdr_vco_freqband[1:0]	1	1	1	2
	rxX_term_acdc	1	1	1	1
	rxX_delta_iq[3:0]	0	0	0	3
	rxX_adapt_sel	0	0	0	1
	rxX_adapt_mode[1:0]	0	0	1	0

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		SATA				
		Gen1		Gen2		Gen3
Data Width	Tx Path (bits)	10	20	10	20	20
	Rx Path (bits)	10	20	10	20	20
Clock Information	Tx Baud rate (Gbps)	1.5	1.5	3	3	6
	Rx Baud rate (Gbps)	1.5	1.5	3	3	6
	txX_clk frequency (MHz)	600	300	600	300	300
	rxX_clk frequency (MHz)	150	75	300	150	300
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	600	300	600	300	300
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0
	Corresponding MPLL VCO Frequency (GHz)	6	6	6	6	6
	Corresponding MPLL Output Clock Frequency (GHz)	3	3	3	3	3
	Rx VCO Frequency (GHz)	6	6	6	6	6
	txX_misc[7:0]	0	0	0	0	0
	txX_rate[2:0]	0	0	0	0	0
	txX_mpllb_sel	0	0	0	0	0
Tx Primary Inputs	txX_width[1:0]	1	3	1	3	3
	txX_vboost_en	0	0	0	0	0
	txX_iboost_lvl[3:0]	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	0	0	0
	txX_eq_post[5:0]	0	0	0	0	18
	txX_eq_main[5:0]	23	23	27	27	34
Rx Primary Inputs	rxX_rate[1:0]	3	3	2	2	1
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	6	6	14
	rxX_los_threshold[2:0]	3	3	3	3	3
	rxX_cdr_vco_step_ctrl	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1
	rxX_misc[7:0]	7	7	7	7	5
	rxX_width[1:0]	1	3	1	3	3
	rxX_div16p5_clk_en	0	0	0	0	0
	rxX_cdr_ssc_en	1	1	1	1	1
	rxX_los_lfps_en	0	0	0	0	0
	rxX_dfe_bypass	1	1	1	1	1
	rxX_eq_att_lvl[2:0]	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	4
	rxX_eq_vga2_gain[2:0]	4	4	4	4	4
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	2	2	2	2	2
	rxX_term_acdc	1	1	1	1	1
	rxX_delta_iq[3:0]	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	0	0	0

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

1

Information/Primary Inputs		Ethernet									
		1000BASE-KX		SGMII		QSGMII		1000BASE-KX Bifurcation using 7.5GHz		SGMII Bifurcation using 7.5GHz	
Data Width	Tx Path (bits)	10	20	10	20	10	20	10	20	10	20
	Rx Path (bits)	10	20	10	20	10	20	10	20	10	20
Clock Information	Tx Baud rate (Gbps)	1.25	1.25	1.25	1.25	5	5	1.25	1.25	1.25	1.25
	Rx Baud rate (Gbps)	1.25	1.25	1.25	1.25	5	5	1.25	1.25	1.25	1.25
	txX_clk frequency (MHz)	125	62.5	125	62.5	500	250	125	62.5	125	62.5
	rxX_clk frequency (MHz)	125	62.5	125	62.5	500	250	125	62.5	125	62.5
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	125	62.5	125	62.5	500	250	125	62.5	125	62.5
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	1	1	1	1
	Corresponding MPLL VCO Frequency (GHz)	5	5	5	5	5	5	7.5	7.5	7.5	7.5
	Corresponding MPLL Output Clock Frequency (GHz)	5	5	5	5	5	5	7.5	7.5	7.5	7.5
	Rx VCO Frequency (GHz)	5	5	5	5	5	5	5	5	5	5
	txX_misc[7:0]	0	0	0	0	0	0	0	0	0	0
	txX_rate[2:0]	3	3	3	3	1	1	6	6	6	6
	txX_mpllb_sel	0	0	0	0	0	0	1	1	1	1
Tx Primary Inputs	txX_width[1:0]	1	3	1	3	1	3	1	3	1	3
	txX_vboost_en	1	1	0	0	0	0	1	1	0	0
	txX_iboost_lvl[3:0]	15	15	15	15	15	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	0	0	0	0	0	0	0	0
	txX_eq_post[5:0]	0	0	0	0	0	0	0	0	0	0
	txX_eq_main[5:0]	40	40	20	20	36	36	40	40	20	20
Rx Primary Inputs	rxX_rate[1:0]	3	3	3	3	1	1	3	3	3	3
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	6	6	14	14	6	6	6	6
	rxX_los_threshold[2:0]	3	3	3	3	3	3	3	3	3	3
	rxX_cdr_vco_step_ctrl	1	1	1	1	1	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1	1	1	1	1	1
	rxX_misc[7:0]	6	6	6	6	4	4	6	6	6	6
	rxX_width[1:0]	1	3	1	3	1	3	1	3	1	3
	rxX_div16p5_clk_en	0	0	0	0	0	0	0	0	0	0
	rxX_cdr_ssc_en	0	0	0	0	0	0	0	0	0	0
	rxX_los_lfps_en	0	0	0	0	0	0	0	0	0	0
	rxX_dfe_bypass	1	1	1	1	0	0	1	1	1	1
	rxX_eq_att_lvl[2:0]	0	0	0	0	0	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	4	4	4	4	4	4
	rxX_eq_vga2_gain[2:0]	4	4	4	4	4	4	4	4	4	4
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	1	1	1	1	1	1	1	1	1	1
	rxX_term_acdc	1	1	1	1	0/1	0/1	1	1	1	1
	rxX_delta_iq[3:0]	0	0	0	0	0	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	0	0	3	3	0	0	0	0

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		Ethernet					
		XAUI/10GBASE-KX4		RXAUI		10GBASE-KR@10.3125G	
Data Width	Tx Path (bits)	10	20	10	20	20	16
	Rx Path (bits)	10	20	10	20	20	16
Clock Information	Tx Baud rate (Gbps)	3.125	3.125	6.25	6.25	10.3125	10.3125
	Rx Baud rate (Gbps)	3.125	3.125	6.25	6.25	10.3125	10.3125
	txX_clk frequency (MHz)	312.5	156.25	625	312.5	515.625	644.53125
	rxX_clk frequency (MHz)	312.5	156.25	625	312.5	515.625	644.53125
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	312.5	312.5
	txX_word_clk frequency (MHz)	312.5	156.25	625	312.5	515.625	644.53125
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0
	Corresponding MPLL VCO Frequency (GHz)	6.25	6.25	6.25	6.25	5.15625	5.15625
	Corresponding MPLL Output Clock Frequency (GHz)	6.25	6.25	6.25	6.25	5.15625	5.15625
	Rx VCO Frequency (GHz)	6.25	6.25	6.25	6.25	5.15625	5.15625
	txX_misc[7:0]	0	0	0	0	0	0
	txX_rate[2:0]	2	2	1	1	0	0
	txX_mpllb_sel	0	0	0	0	0	0
Tx Primary Inputs	txX_width[1:0]	1	3	1	3	3	2
	txX_vboost_en	1	1	1	1	1	1
	txX_iboost_lvl[3:0]	15	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	0	0	NOTE 1	NOTE 1
	txX_eq_post[5:0]	32	32	40	40	NOTE 1	NOTE 1
	txX_eq_main[5:0]	32	32	30	30	NOTE 1	NOTE 1
Rx Primary Inputs	rxX_rate[1:0]	2	2	1	1	0	0
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	14	14	16	16
	rxX_los_threshold[2:0]	3	3	3	3	7	7
	rxX_cdr_vco_step_ctrl	1	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1	1
	rxX_misc[7:0]	7	7	5	5	0	0
	rxX_width[1:0]	1	3	1	3	3	2
	rxX_div16p5_clk_en	0	0	0	0	1	1
	rxX_cdr_ssc_en	0	0	0	0	0	0
	rxX_los_lfps_en	0	0	0	0	0	0
	rxX_dfe_bypass	1	1	0	0	0	0
	rxX_eq_att_lvl[2:0]	0	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	5	5
	rxX_eq_vga2_gain[2:0]	4	4	4	4	5	5
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	2	2	2	2	1	1
	rxX_term_acdc	1	1	0/1	0/1	1	1
	rxX_delta_iq[3:0]	0	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	3	3	3	3

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		CEI	
		CEI-6G-SR @6.25G	CEI-11G-SR @10.3125G
Data Width	Tx Path (bits)	20	20
	Rx Path (bits)	20	20
Clock Information	Tx Baud rate (Gbps)	6.25	10.3125
	Rx Baud rate (Gbps)	6.25	10.3125
	txX_clk frequency (MHz)	312.5	515.625
	rxX_clk frequency (MHz)	312.5	515.625
	rxX_div16p5_clk frequency (MHz)	NA	NA
	txX_word_clk frequency (MHz)	312.5	515.625
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0
	Corresponding MPLL VCO Frequency (GHz)	6.25	5.15625
	Corresponding MPLL Output Clock Frequency (GHz)	6.25	5.15625
	Rx VCO Frequency (GHz)	6.25	5.15625
	txX_misc[7:0]	0	0
	txX_rate[2:0]	1	0
	txX_mpllb_sel	0	0
Tx Primary Inputs	txX_width[1:0]	3	3
	txX_vboost_en	0	0
	txX_iboost_lvl[3:0]	15	15
	txX_eq_pre[5:0]	0	0
	txX_eq_post[5:0]	0	32
	txX_eq_main[5:0]	30	22
Rx Primary Inputs	rxX_rate[1:0]	1	0
	rxX_eq_ctle_pole[1:0]	0	0
	rxX_eq_ctle_boost[4:0]	14	12
	rxX_los_threshold[2:0]	3	3
	rxX_cdr_vco_step_ctrl	1	1
	rxX_cdr_vco_temp_comp_en	1	1
	rxX_misc[7:0]	5	0
	rxX_width[1:0]	3	3
	rxX_div16p5_clk_en	0	0
	rxX_cdr_ssc_en	0	0
	rxX_los_lfps_en	0	0
	rxX_dfe_bypass	0	0
	rxX_eq_att_lvl[2:0]	0	0
	rxX_eq_vga1_gain[2:0]	4	5
	rxX_eq_vga2_gain[2:0]	4	5
	rxX_eq_dfe_tap1[7:0]	0	0
	rxX_cdr_vco_freqband[1:0]	2	1
	rxX_term_acdc	0/1	0/1
	rxX_delta_iq[3:0]	0	0
	rxX_adapt_sel	0	0
	rxX_adapt_mode[1:0]	3	3

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		CPRI/OBSAI					
		CPRI/OBSAI @2.4576G	CPRI/OBSAI @3.072G	CPRI/OBSAI@4.9152G		CPRI/OBSAI @6.144G	CPRI/OBSAI @9.8304G
Data Width	Tx Path (bits)	20	20	10	20	20	20
	Rx Path (bits)	20	20	10	20	20	20
Clock Information	Tx Baud rate (Gbps)	2.4576	3.072	4.9152	4.9152	6.144	9.8304
	Rx Baud rate (Gbps)	2.4576	3.072	4.9152	4.9152	6.144	9.8304
	txX_clk frequency (MHz)	122.88	153.6	491.52	245.76	307.2	491.52
	rxX_clk frequency (MHz)	122.88	153.6	491.52	245.76	307.2	491.52
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	122.88	153.6	491.52	245.76	307.2	491.52
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0
	Corresponding MPLL VCO Frequency (GHz)	4.9152	6.144	4.9152	4.9152	6.144	4.9152
	Corresponding MPLL Output Clock Frequency (GHz)	4.9152	6.144	4.9152	4.9152	6.144	4.9152
	Rx VCO Frequency (GHz)	4.9152	6.144	4.9152	4.9152	6.144	4.9152
	txX_misc[7:0]	0	0	0	0	0	0
	txX_rate[2:0]	2	2	1	1	1	0
	txX_mpllb_sel	0	0	0	0	0	0
Tx Primary Inputs	txX_width[1:0]	3	3	1	3	3	3
	txX_vboost_en	1	1	1	1	1	1
	txX_iboost_lvl[3:0]	15	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	NOTE 1	NOTE 1	0	NOTE 1
	txX_eq_post[5:0]	0	0	NOTE 1	NOTE 1	32	NOTE 1
	txX_eq_main[5:0]	36	36	NOTE 1	NOTE 1	22	NOTE 1
Rx Primary Inputs	rxX_rate[1:0]	2	2	1	1	1	0
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	14	14	14	12
	rxX_los_threshold[2:0]	3	3	3	3	3	3
	rxX_cdr_vco_step_ctrl	1	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1	1
	rxX_misc[7:0]	6	7	4	4	5	0
	rxX_width[1:0]	3	3	1	3	3	3
	rxX_div16p5_clk_en	0	0	0	0	0	0
	rxX_cdr_ssc_en	0	0	0	0	0	0
	rxX_los_lfps_en	0	0	0	0	0	0
	rxX_dfe_bypass	1	1	0	0	0	0
	rxX_eq_att_lvl[2:0]	0	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	4	5
	rxX_eq_vga2_gain[2:0]	4	4	4	4	4	5
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	1	2	1	1	2	1
	rxX_term_acdc	0/1	0/1	0/1	0/1	0/1	0/1
	rxX_delta_iq[3:0]	0	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	3	3	3	3

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		JESD204B (related to CEI-11G-SR)							
		JESD204B @1.2288G	JESD204B @1.25G		JESD204B @3.125G		JESD204B @6.25G	JESD204B @10.3125G	JESD204B @12.5G
Data Width	Tx Path (bits)	20	10	20	10	20	20	20	20
	Rx Path (bits)	20	10	20	10	20	20	20	20
Clock Information	Tx Baud rate (Gbps)	1.2288	1.25	1.25	3.125	3.125	6.25	10.3125	12.5
	Rx Baud rate (Gbps)	1.2288	1.25	1.25	3.125	3.125	6.25	10.3125	12.5
	txX_clk frequency (MHz)	61.44	125	62.5	312.5	156.25	312.5	515.625	625
	rxX_clk frequency (MHz)	61.44	125	62.5	312.5	156.25	312.5	515.625	625
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	61.44	125	62.5	312.5	156.25	312.5	515.625	625
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0
	Corresponding MPLL VCO Frequency (GHz)	4.9152	5	5	6.25	6.25	6.25	5.15625	6.25
	Corresponding MPLL Output Clock Frequency (GHz)	4.9152	5	5	6.25	6.25	6.25	5.15625	6.25
	Rx VCO Frequency (GHz)	4.9152	5	5	6.25	6.25	6.25	5.15625	6.25
	txX_misc[7:0]	0	0	0	0	0	0	0	0
	txX_rate[2:0]	3	3	3	2	2	1	0	0
	txX_mpllb_sel	0	0	0	0	0	0	0	0
Tx Primary Inputs	txX_width[1:0]	3	1	3	1	3	3	3	3
	txX_vboost_en	0	0	0	0	0	0	0	0
	txX_iboost_lvl[3:0]	15	15	15	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	0	0	0	0	0	0
	txX_eq_post[5:0]	0	0	0	0	0	0	20	20
	txX_eq_main[5:0]	30	30	30	30	30	30	25	25
Rx Primary Inputs	rxX_rate[1:0]	3	3	3	2	2	1	0	0
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	6	6	6	14	12	12
	rxX_los_threshold[2:0]	3	3	3	3	3	3	3	3
	rxX_cdr_vco_step_ctrl	1	1	1	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1	1	1	1
	rxX_misc[7:0]	6	6	6	7	7	5	2	1
	rxX_width[1:0]	3	1	3	1	3	3	3	3
	rxX_div16p5_clk_en	0	0	0	0	0	0	0	0
	rxX_cdr_ssc_en	0	0	0	0	0	0	0	0
	rxX_los_lfps_en	0	0	0	0	0	0	0	0
	rxX_dfe_bypass	1	1	1	1	1	0	0	0
	rxX_eq_att_lvl[2:0]	0	0	0	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	4	4	4	4
	rxX_eq_vga2_gain[2:0]	4	4	4	4	4	4	4	4
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	1	1	1	2	2	2	1	2
	rxX_term_acdc	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
	rxX_delta_iq[3:0]	0	0	0	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	0	0	0	3	3	3

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.

Information/Primary Inputs		SRIO Level I			SRIO Level II					
		SRIO Level I @1.25G	SRIO Level I @2.5G	SRIO Level I @3.125G	SRIO Level II SR@5G	SRIO Level II LR/MR@5G	SRIO Level II SR@6.25G	SRIO Level II LR/MR@6.25G	SRIO Level II SR@6.25G	SRIO Level II LR/MR@6.25G
Data Width	Tx Path (bits)	10	10	10	10	10	10	10	20	20
	Rx Path (bits)	10	10	10	10	10	10	10	20	20
Clock Information	Tx Baud rate (Gbps)	1.25	2.5	3.125	5	5	6.25	6.25	6.25	6.25
	Rx Baud rate (Gbps)	1.25	2.5	3.125	5	5	6.25	6.25	6.25	6.25
	txX_clk frequency (MHz)	125	250	312.5	500	500	625	625	312.5	312.5
	rxX_clk frequency (MHz)	125	250	312.5	500	500	625	625	312.5	312.5
	rxX_div16p5_clk frequency (MHz)	NA	NA	NA	NA	NA	NA	NA	NA	NA
	txX_word_clk frequency (MHz)	125	250	312.5	500	500	625	625	312.5	312.5
	Tx CLK sourced from MPLL#: MPLL{A,B}: 0-MPLLA, 1-MPLLB	0	0	0	0	0	0	0	0	0
	Corresponding MPLL VCO Frequency (GHz)	5	5	6.25	5	5	6.25	6.25	6.25	6.25
	Corresponding MPLL Output Clock Frequency (GHz)	5	5	6.25	5	5	6.25	6.25	6.25	6.25
	Rx VCO Frequency (GHz)	5	5	6.25	5	5	6.25	6.25	6.25	6.25
	txX_misc[7:0]	0	0	0	0	0	0	0	0	0
	txX_rate[2:0]	3	2	2	1	1	1	1	1	1
	txX_mpllb_sel	0	0	0	0	0	0	0	0	0
Tx Primary Inputs	txX_width[1:0]	1	1	1	1	1	1	1	3	3
	txX_vboost_en	1	0	0	0	1	0	1	0	1
	txX_iboost_lvl[3:0]	15	15	15	15	15	15	15	15	15
	txX_eq_pre[5:0]	0	0	0	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1
	txX_eq_post[5:0]	0	32	32	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1
	txX_eq_main[5:0]	40	32	32	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1	NOTE 1
Rx Primary Inputs	rxX_rate[1:0]	3	2	2	1	1	1	1	1	1
	rxX_eq_ctle_pole[1:0]	0	0	0	0	0	0	0	0	0
	rxX_eq_ctle_boost[4:0]	6	6	6	14	14	14	18	14	18
	rxX_los_threshold[2:0]	3	3	3	3	3	3	3	3	3
	rxX_cdr_vco_step_ctrl	1	1	1	1	1	1	1	1	1
	rxX_cdr_vco_temp_comp_en	1	1	1	1	1	1	1	1	1
	rxX_misc[7:0]	6	6	7	4	4	5	5	5	5
	rxX_width[1:0]	1	1	1	1	1	1	1	3	3
	rxX_div16p5_clk_en	0	0	0	0	0	0	0	0	0
	rxX_cdr_ssc_en	0	0	0	0	0	0	0	0	0
	rxX_los_lfps_en	0	0	0	0	0	0	0	0	0
	rxX_dfe_bypass	1	1	1	0	0	0	0	0	0
	rxX_eq_att_lvl[2:0]	0	0	0	0	0	0	0	0	0
	rxX_eq_vga1_gain[2:0]	4	4	4	4	4	4	4	4	4
	rxX_eq_vga2_gain[2:0]	4	4	4	4	4	4	4	4	4
	rxX_eq_dfe_tap1[7:0]	0	0	0	0	0	0	0	0	0
	rxX_cdr_vco_freqband[1:0]	1	1	2	1	1	2	2	2	2
	rxX_term_acdc	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
	rxX_delta_iq[3:0]	0	0	0	0	0	0	0	0	0
	rxX_adapt_sel	0	0	0	0	0	0	0	0	0
	rxX_adapt_mode[1:0]	0	0	0	3	3	3	3	3	3

Note 1: Multiple TX settings/presets are possible as the TX equalization is adapted. Refer to the TX Presets table for more details on PRE/MAIN/POST settings for specific presets.